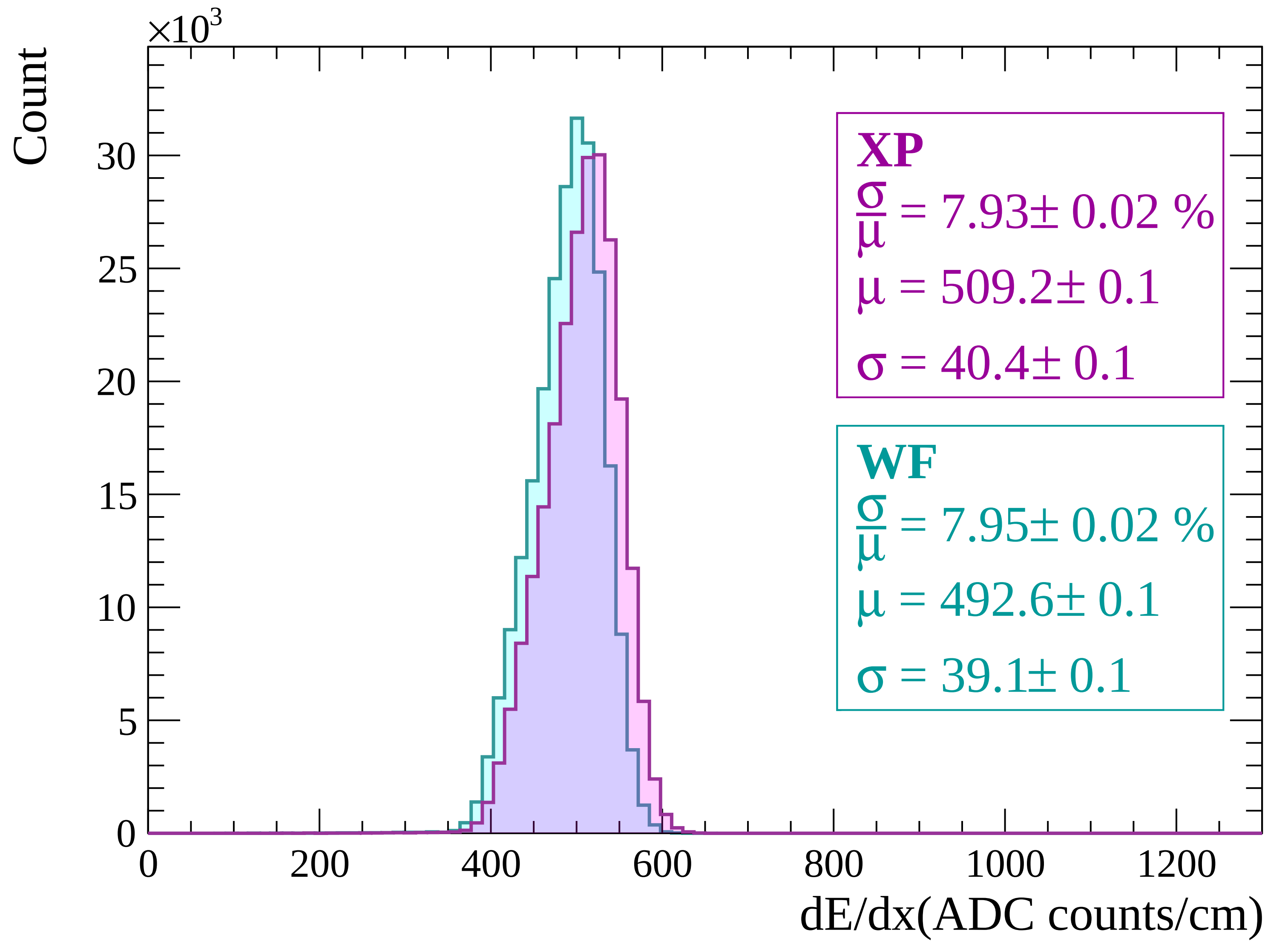
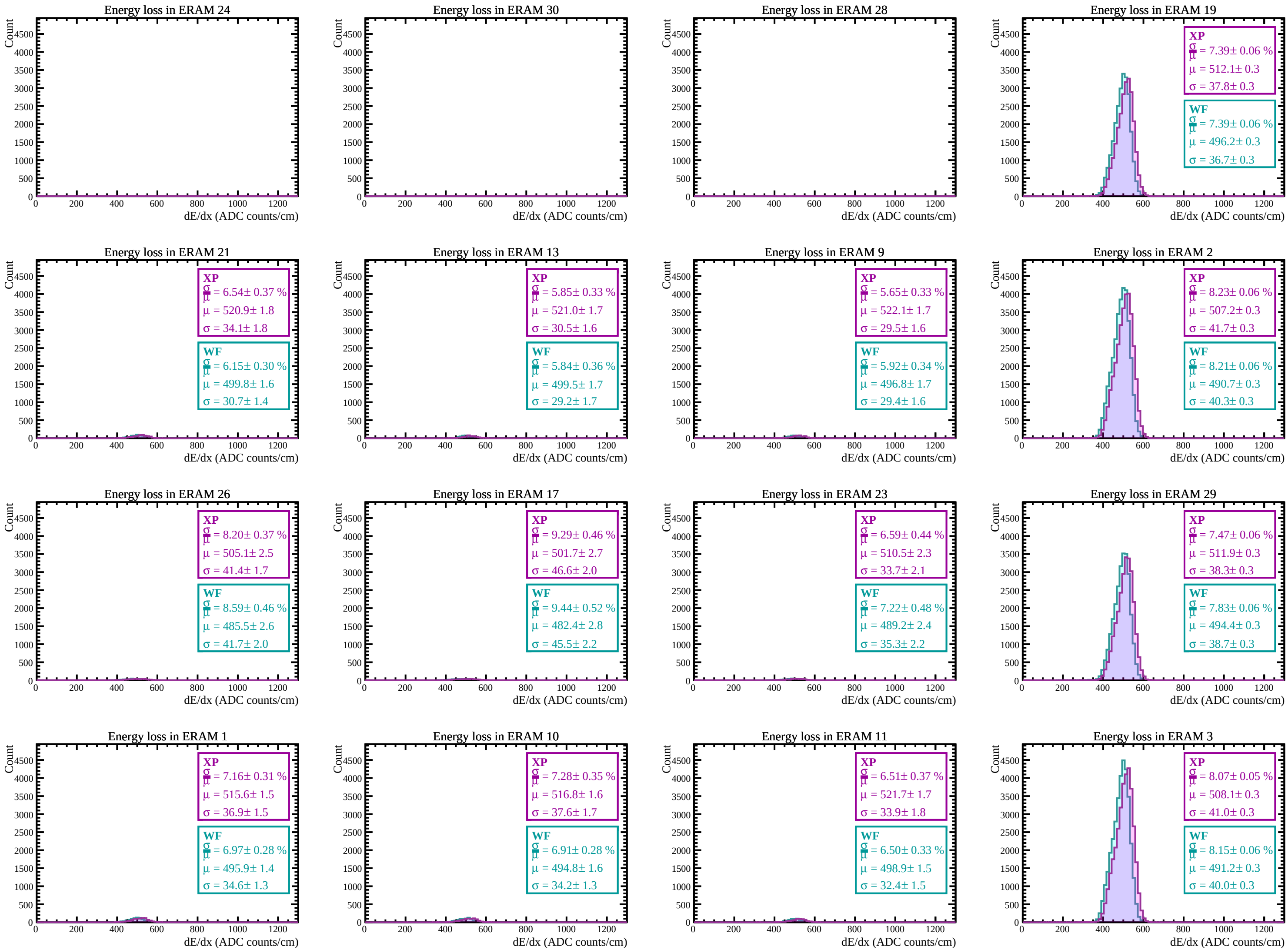
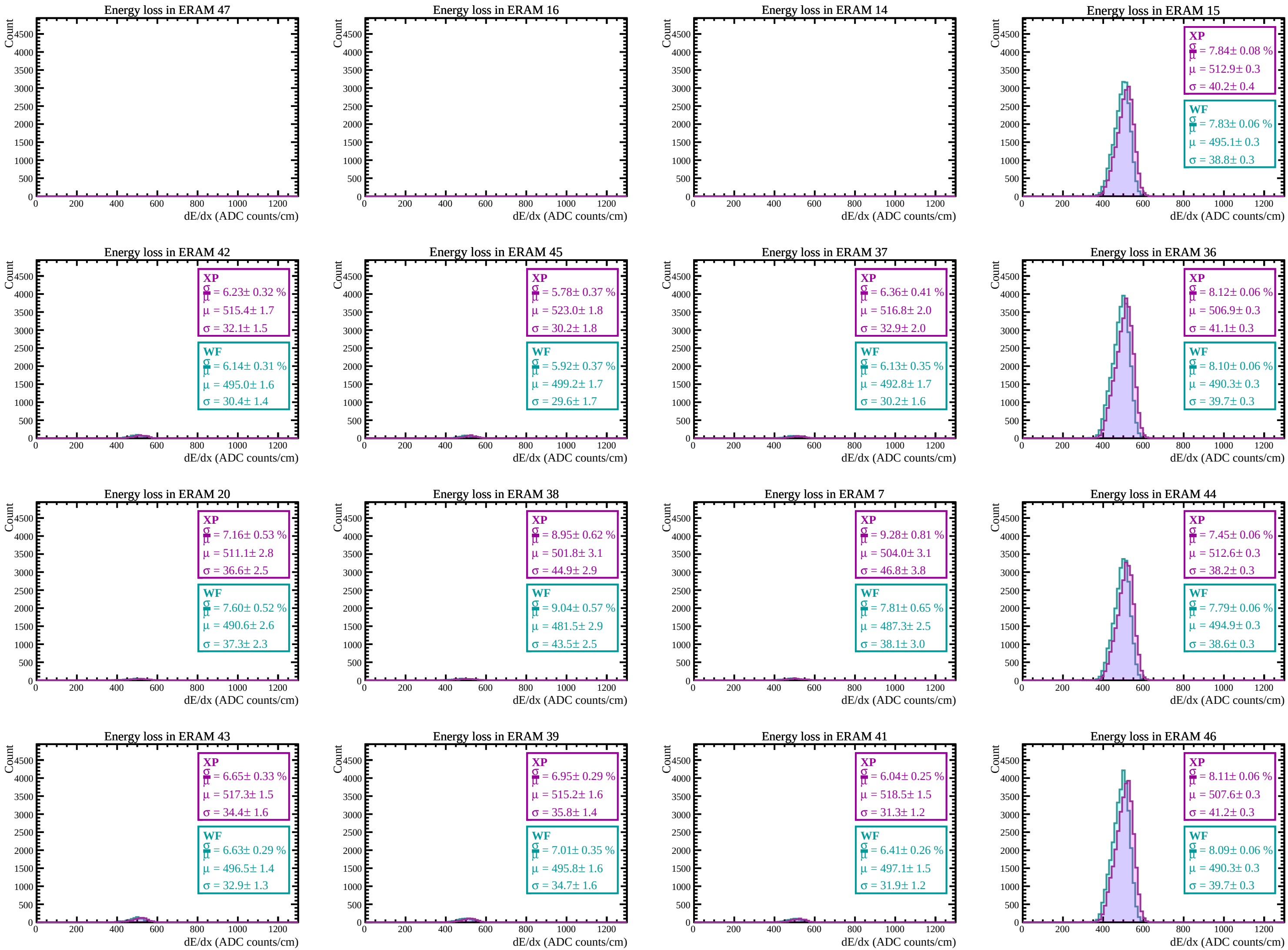
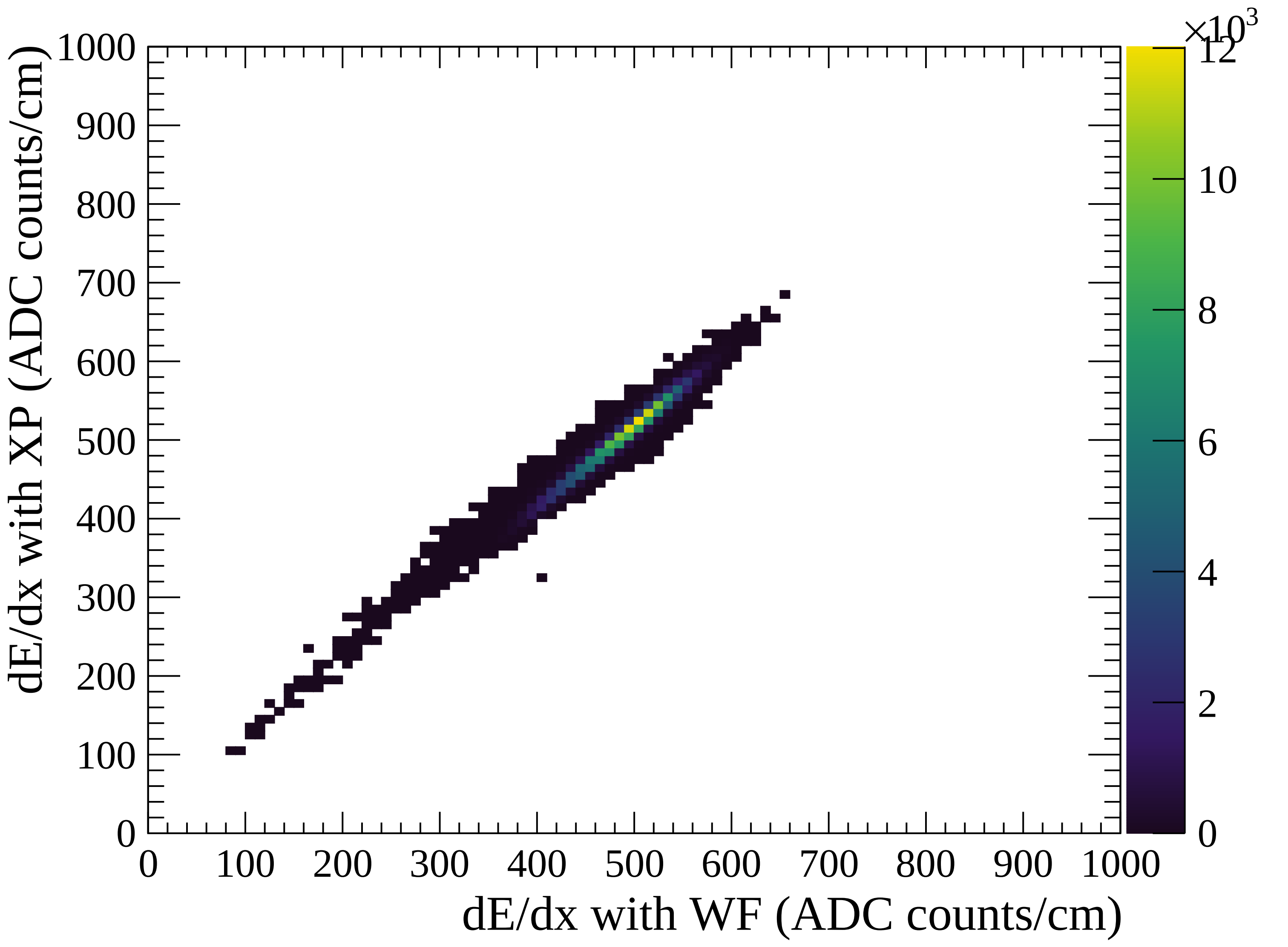


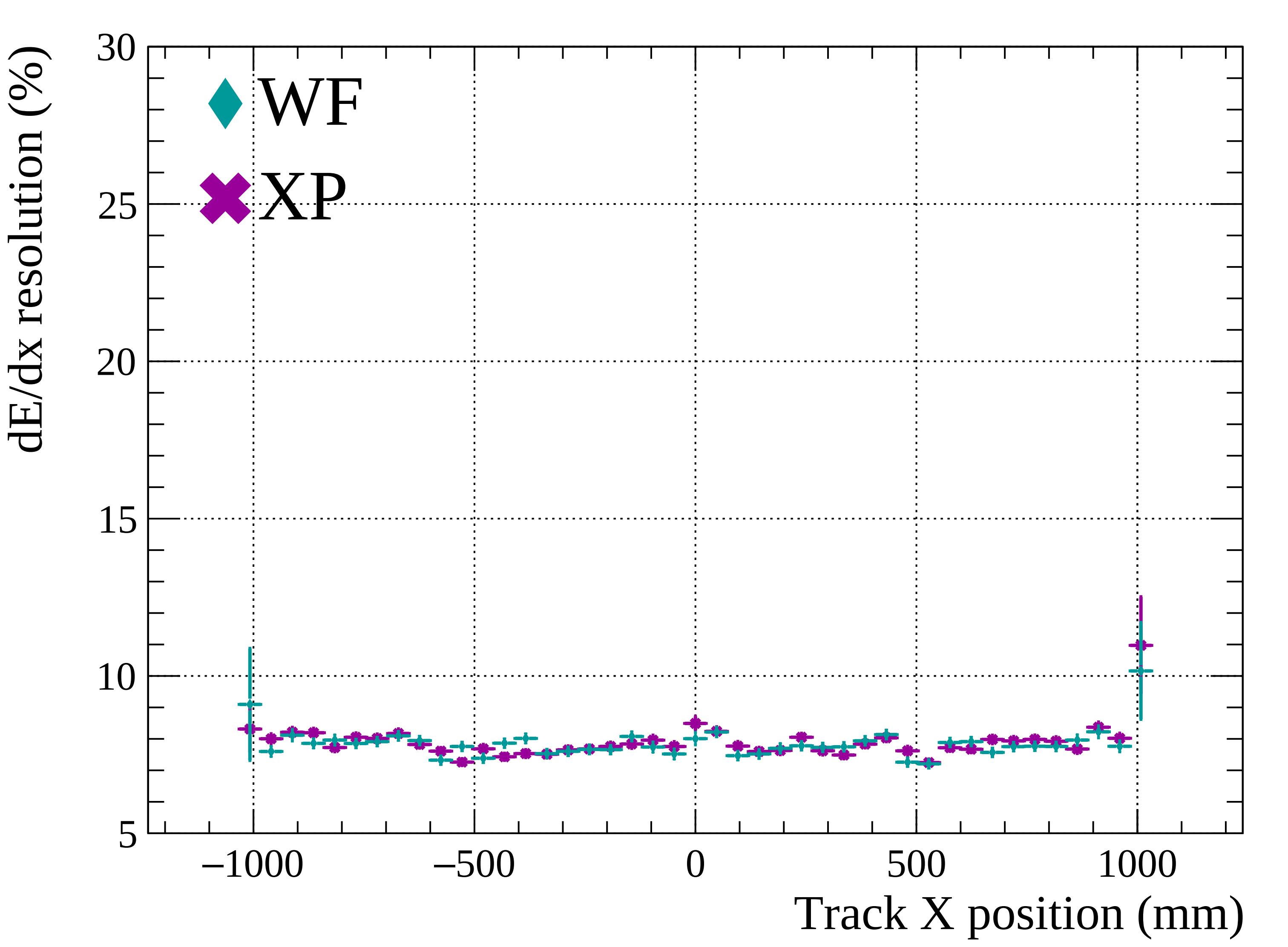
$\times 10^3$
Count
30
25
20
15
10
5
0
0
200
400
600
800
1000
1200
dE/dx(ADC counts/cm)
XP
$\frac{\sigma}{\mu} = 7.93 \pm 0.02$ %
$\mu = 509.2 \pm 0.1$
$\sigma = 40.4 \pm 0.1$
WF
$\frac{\sigma}{\mu} = 7.95 \pm 0.02$ %
$\mu = 492.6 \pm 0.1$
$\sigma = 39.1 \pm 0.1$


Energy loss in ERAM 24
Energy loss in ERAM 30
Energy loss in ERAM 28
Energy loss in ERAM 19
XP σ/μ = 7.39± 0.06 % μ = 512.1± 0.3 σ = 37.8± 0.3
WF σ/μ = 7.39± 0.06 % μ = 496.2± 0.3 σ = 36.7± 0.3
Energy loss in ERAM 21
XP σ/μ = 6.54± 0.37 % μ = 520.9± 1.8 σ = 34.1± 1.8
WF σ/μ = 6.15± 0.30 % μ = 499.8± 1.6 σ = 30.7± 1.4
Energy loss in ERAM 13
XP σ/μ = 5.85± 0.33 % μ = 521.0± 1.7 σ = 30.5± 1.6
WF σ/μ = 5.84± 0.36 % μ = 499.5± 1.7 σ = 29.2± 1.7
Energy loss in ERAM 9
XP σ/μ = 5.65± 0.33 % μ = 522.1± 1.7 σ = 29.5± 1.6
WF σ/μ = 5.92± 0.34 % μ = 496.8± 1.7 σ = 29.4± 1.6
Energy loss in ERAM 2
XP σ/μ = 8.23± 0.06 % μ = 507.2± 0.3 σ = 41.7± 0.3
WF σ/μ = 8.21± 0.06 % μ = 490.7± 0.3 σ = 40.3± 0.3
Energy loss in ERAM 26
XP σ/μ = 8.20± 0.37 % μ = 505.1± 2.5 σ = 41.4± 1.7
WF σ/μ = 8.59± 0.46 % μ = 485.5± 2.6 σ = 41.7± 2.0
Energy loss in ERAM 17
XP σ/μ = 9.29± 0.46 % μ = 501.7± 2.7 σ = 46.6± 2.0
WF σ/μ = 9.44± 0.52 % μ = 482.4± 2.8 σ = 45.5± 2.2
Energy loss in ERAM 23
XP σ/μ = 6.59± 0.44 % μ = 510.5± 2.3 σ = 33.7± 2.1
WF σ/μ = 7.22± 0.48 % μ = 489.2± 2.4 σ = 35.3± 2.2
Energy loss in ERAM 29
XP σ/μ = 7.47± 0.06 % μ = 511.9± 0.3 σ = 38.3± 0.3
WF σ/μ = 7.83± 0.06 % μ = 494.4± 0.3 σ = 38.7± 0.3
Energy loss in ERAM 1
XP σ/μ = 7.16± 0.31 % μ = 515.6± 1.5 σ = 36.9± 1.5
WF σ/μ = 6.97± 0.28 % μ = 495.9± 1.4 σ = 34.6± 1.3
Energy loss in ERAM 10
XP σ/μ = 7.28± 0.35 % μ = 516.8± 1.6 σ = 37.6± 1.7
WF σ/μ = 6.91± 0.28 % μ = 494.8± 1.6 σ = 34.2± 1.3
Energy loss in ERAM 11
XP σ/μ = 6.51± 0.37 % μ = 521.7± 1.7 σ = 33.9± 1.8
WF σ/μ = 6.50± 0.33 % μ = 498.9± 1.5 σ = 32.4± 1.5
Energy loss in ERAM 3
XP σ/μ = 8.07± 0.05 % μ = 508.1± 0.3 σ = 41.0± 0.3
WF σ/μ = 8.15± 0.06 % μ = 491.2± 0.3 σ = 40.0± 0.3
Count
dE/dx (ADC counts/cm)

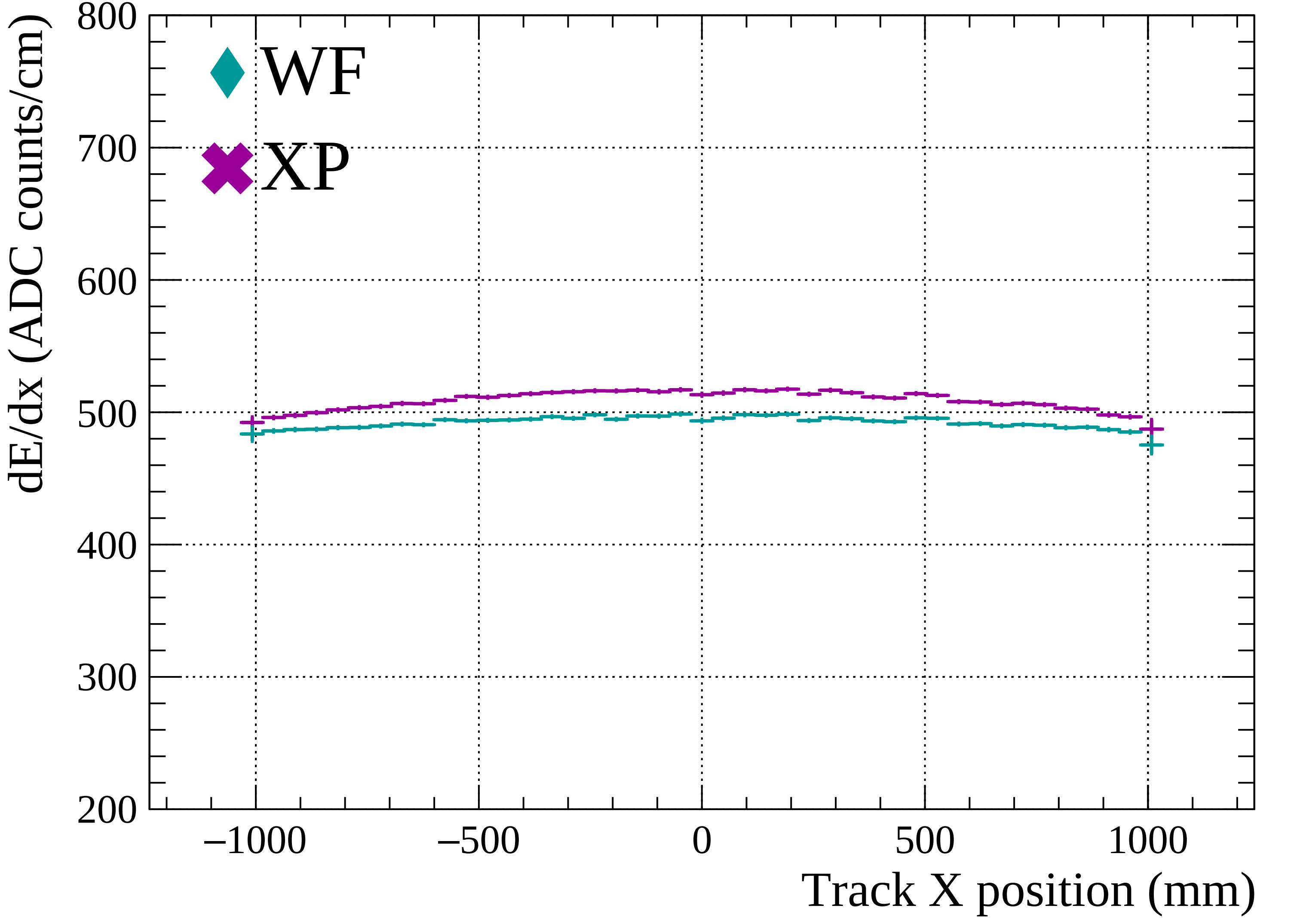

Energy loss in ERAM 47
Energy loss in ERAM 16
Energy loss in ERAM 14
Energy loss in ERAM 15
XP
σ/μ = 7.84± 0.08 %
μ = 512.9± 0.3
σ = 40.2± 0.4
WF
σ/μ = 7.83± 0.06 %
μ = 495.1± 0.3
σ = 38.8± 0.3
Energy loss in ERAM 42
XP
σ/μ = 6.23± 0.32 %
μ = 515.4± 1.7
σ = 32.1± 1.5
WF
σ/μ = 6.14± 0.31 %
μ = 495.0± 1.6
σ = 30.4± 1.4
Energy loss in ERAM 45
XP
σ/μ = 5.78± 0.37 %
μ = 523.0± 1.8
σ = 30.2± 1.8
WF
σ/μ = 5.92± 0.37 %
μ = 499.2± 1.7
σ = 29.6± 1.7
Energy loss in ERAM 37
XP
σ/μ = 6.36± 0.41 %
μ = 516.8± 2.0
σ = 32.9± 2.0
WF
σ/μ = 6.13± 0.35 %
μ = 492.8± 1.7
σ = 30.2± 1.6
Energy loss in ERAM 36
XP
σ/μ = 8.12± 0.06 %
μ = 506.9± 0.3
σ = 41.1± 0.3
WF
σ/μ = 8.10± 0.06 %
μ = 490.3± 0.3
σ = 39.7± 0.3
Energy loss in ERAM 20
XP
σ/μ = 7.16± 0.53 %
μ = 511.1± 2.8
σ = 36.6± 2.5
WF
σ/μ = 7.60± 0.52 %
μ = 490.6± 2.6
σ = 37.3± 2.3
Energy loss in ERAM 38
XP
σ/μ = 8.95± 0.62 %
μ = 501.8± 3.1
σ = 44.9± 2.9
WF
σ/μ = 9.04± 0.57 %
μ = 481.5± 2.9
σ = 43.5± 2.5
Energy loss in ERAM 7
XP
σ/μ = 9.28± 0.81 %
μ = 504.0± 3.1
σ = 46.8± 3.8
WF
σ/μ = 7.81± 0.65 %
μ = 487.3± 2.5
σ = 38.1± 3.0
Energy loss in ERAM 44
XP
σ/μ = 7.45± 0.06 %
μ = 512.6± 0.3
σ = 38.2± 0.3
WF
σ/μ = 7.79± 0.06 %
μ = 494.9± 0.3
σ = 38.6± 0.3
Energy loss in ERAM 43
XP
σ/μ = 6.65± 0.33 %
μ = 517.3± 1.5
σ = 34.4± 1.6
WF
σ/μ = 6.63± 0.29 %
μ = 496.5± 1.4
σ = 32.9± 1.3
Energy loss in ERAM 39
XP
σ/μ = 6.95± 0.29 %
μ = 515.2± 1.6
σ = 35.8± 1.4
WF
σ/μ = 7.01± 0.35 %
μ = 495.8± 1.6
σ = 34.7± 1.6
Energy loss in ERAM 41
XP
σ/μ = 6.04± 0.25 %
μ = 518.5± 1.5
σ = 31.3± 1.2
WF
σ/μ = 6.41± 0.26 %
μ = 497.1± 1.5
σ = 31.9± 1.2
Energy loss in ERAM 46
XP
σ/μ = 8.11± 0.06 %
μ = 507.6± 0.3
σ = 41.2± 0.3
WF
σ/μ = 8.09± 0.06 %
μ = 490.3± 0.3
σ = 39.7± 0.3
Count
dE/dx (ADC counts/cm)



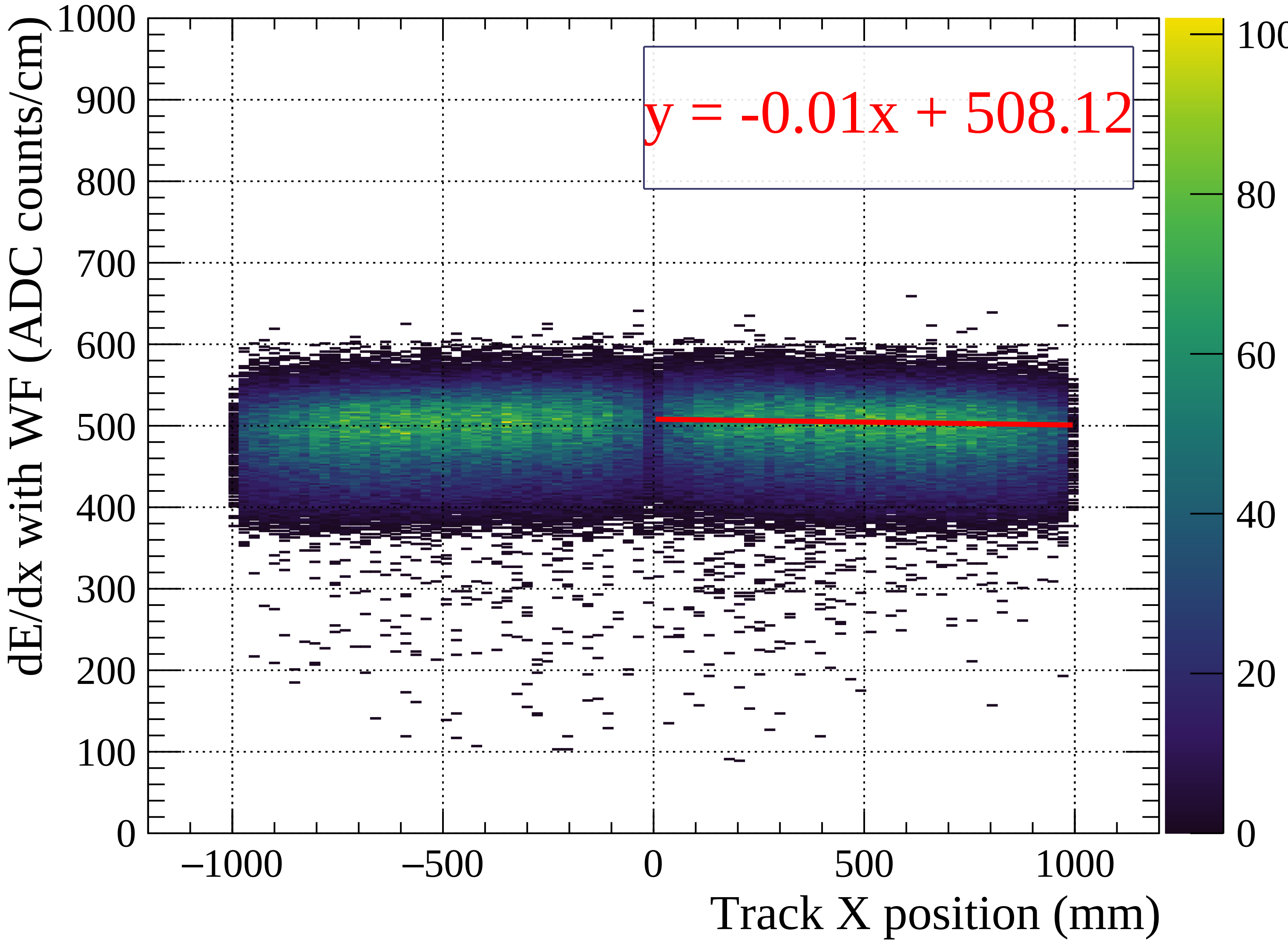
dE/dx with XP (ADC counts/cm)
dE/dx with WF (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
1000
×10^3
0
2
4
6
8
10
12

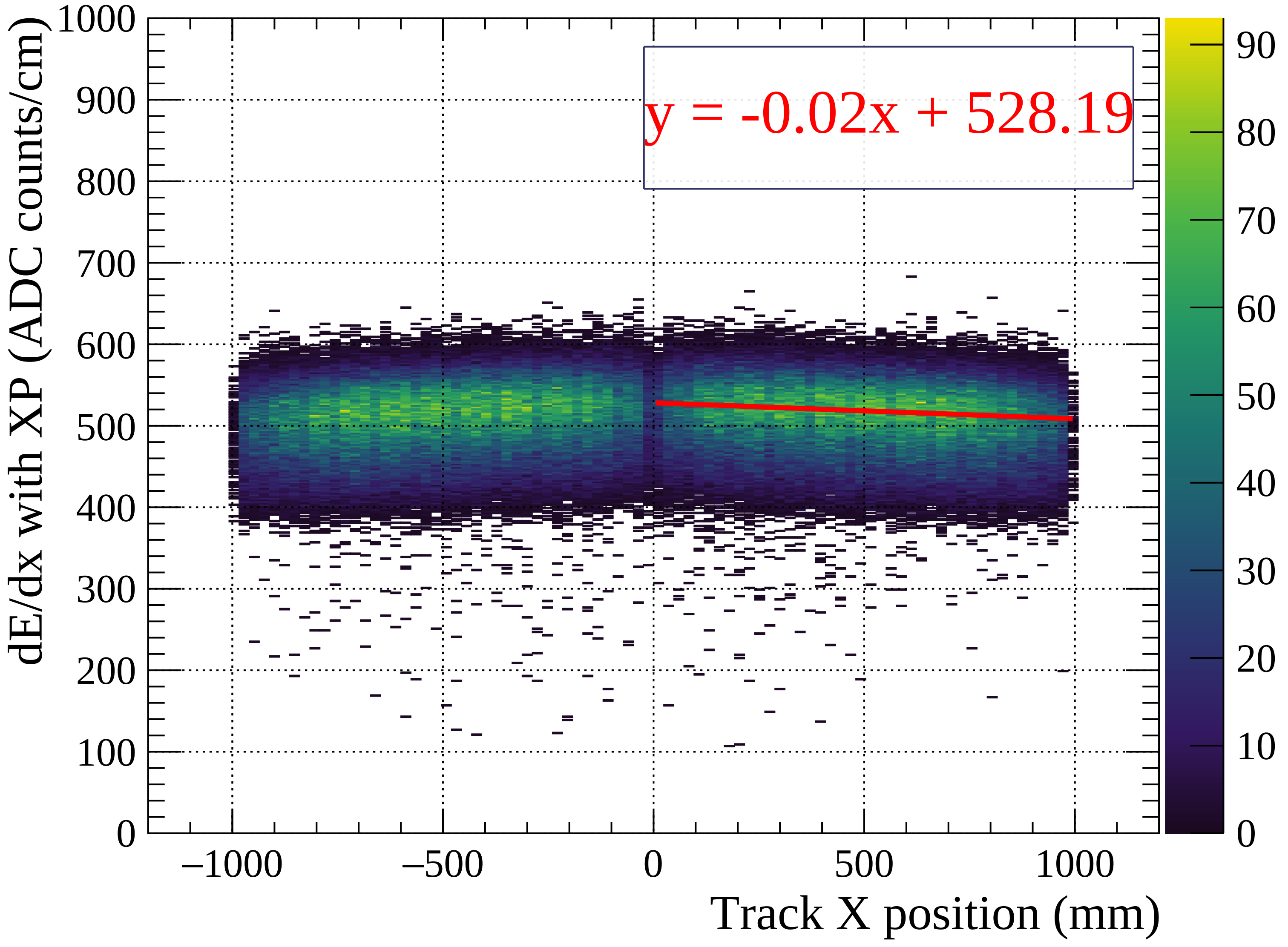


WF
XP
dE/dx resolution (%)
30
25
20
15
10
5
−1000
−500
0
500
1000
Track X position (mm)

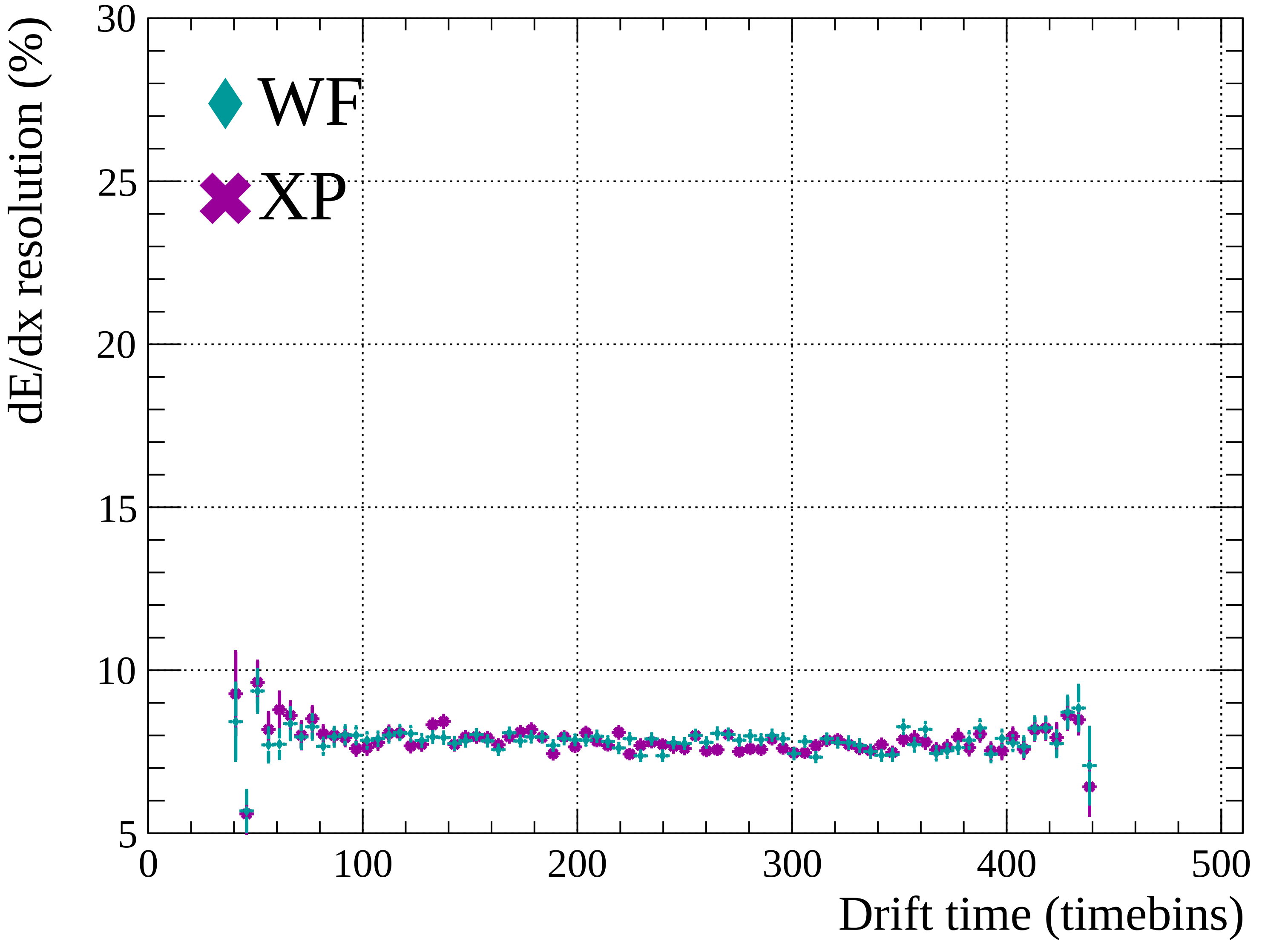


WF
XP
dE/dx (ADC counts/cm)
800
700
600
500
400
300
200
−1000
−500
0
500
1000
Track X position (mm)

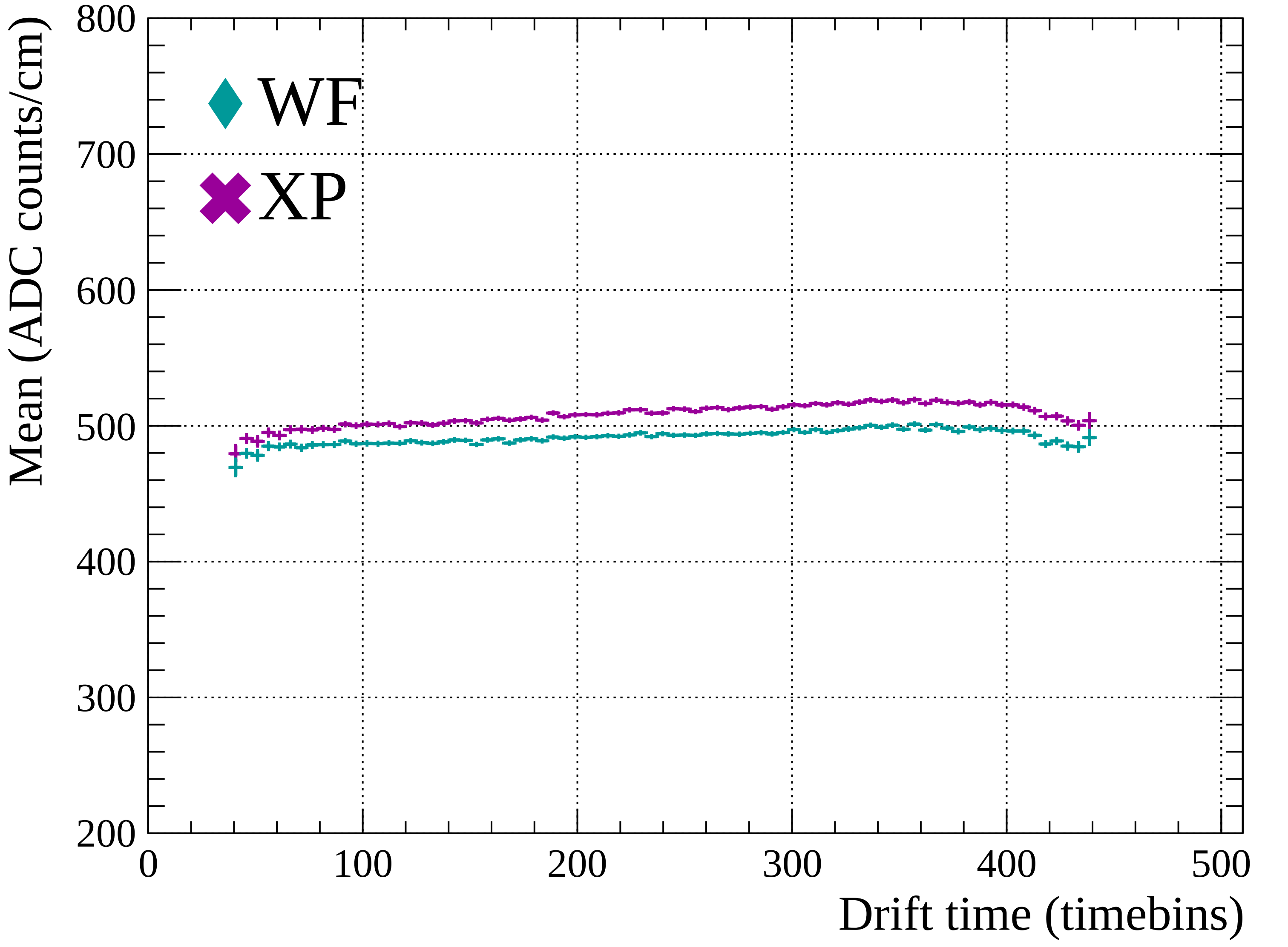

y = -0.01x + 508.12
dE/dx with WF (ADC counts/cm)
Track X position (mm)
0
100
200
300
400
500
600
700
800
900
1000
−1000
−500
0
500
1000
0
20
40
60
80



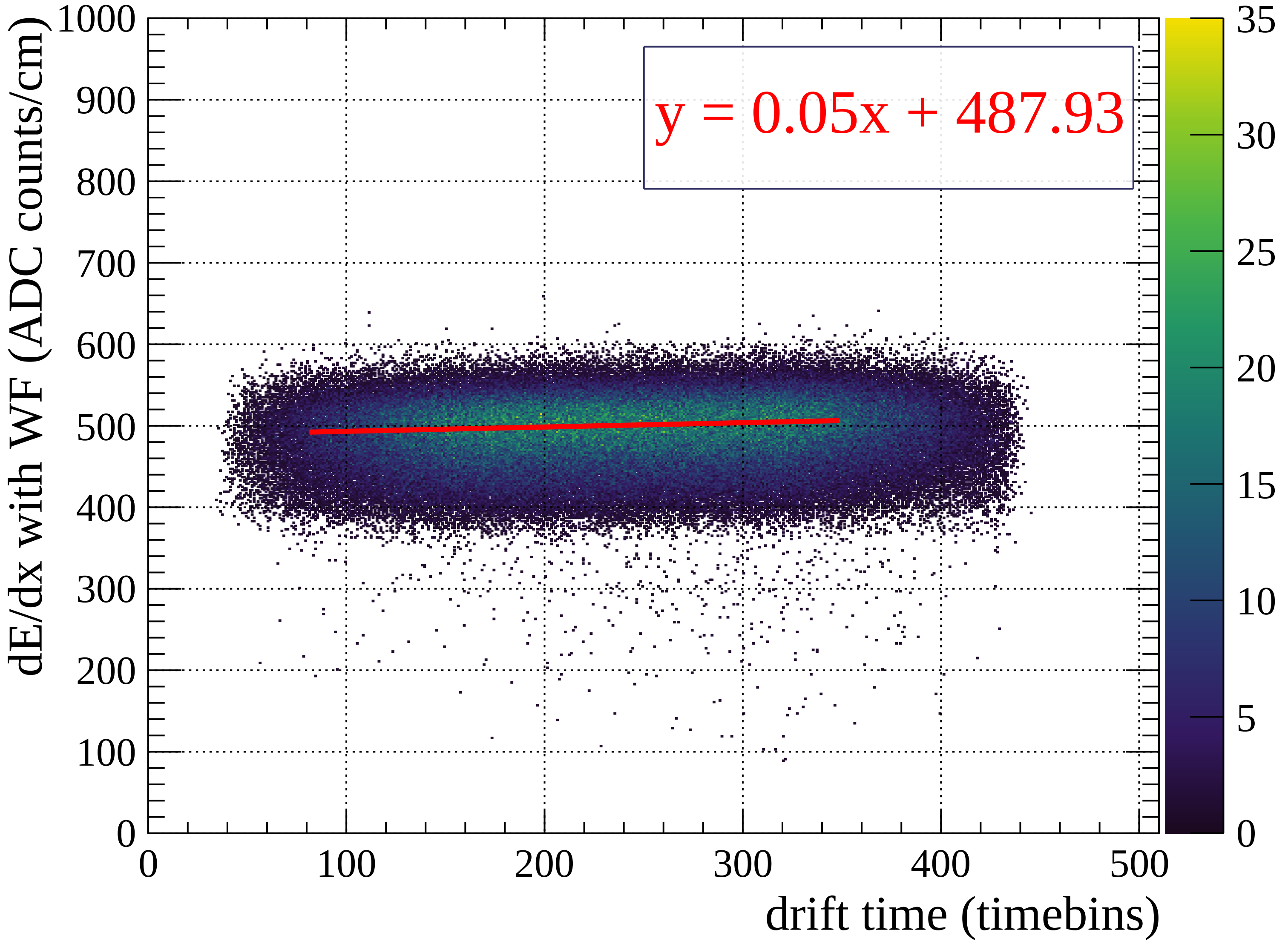
y = -0.02x + 528.19
dE/dx with XP (ADC counts/cm)
Track X position (mm)
0
100
200
300
400
500
600
700
800
900
1000
−1000
−500
0
500
1000
0
10
20
30
40
50
60
70
80
90



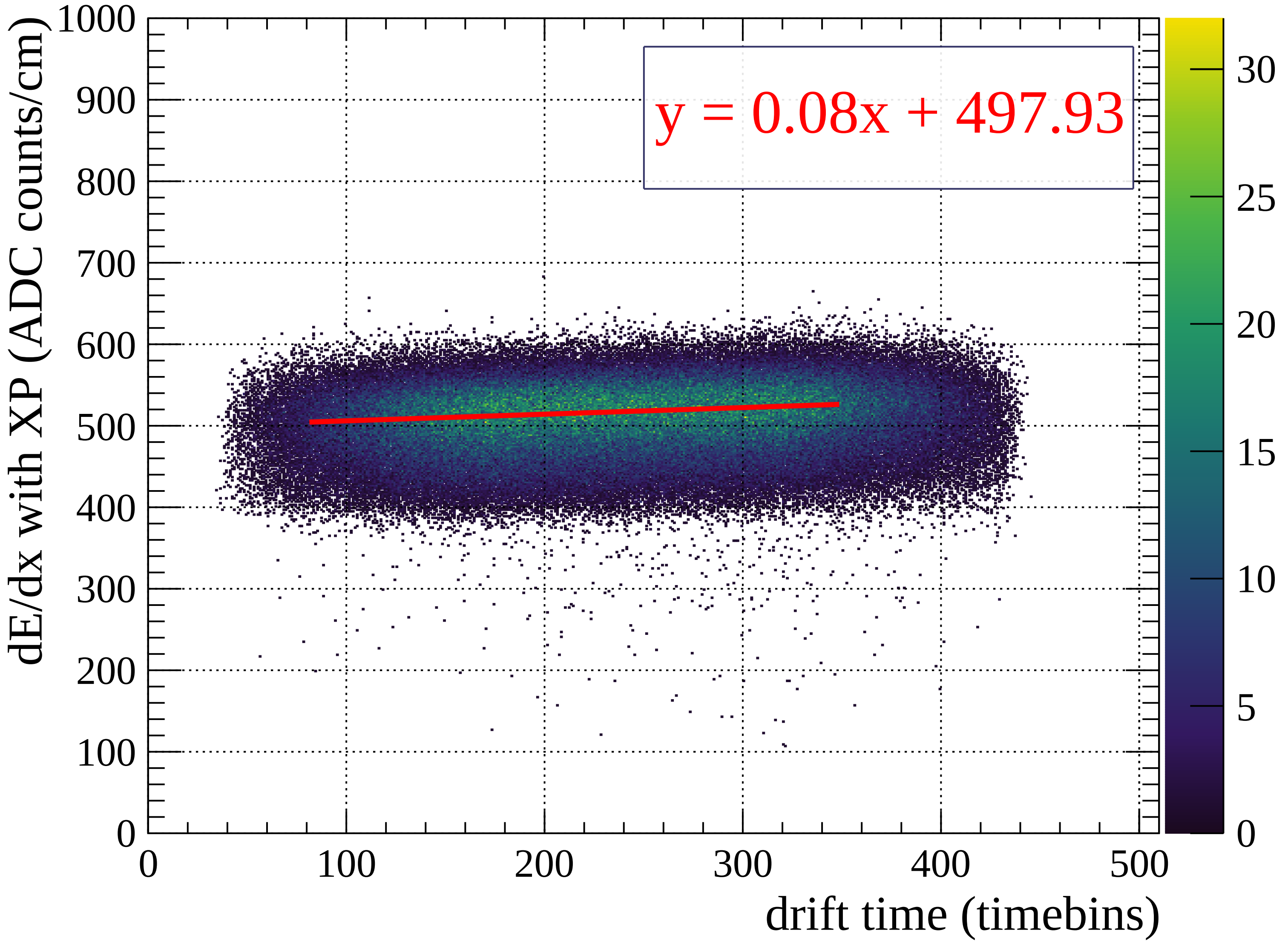
dE/dx resolution (%)
Drift time (timebins)
WF
XP
30
25
20
15
10
5
0
100
200
300
400
500



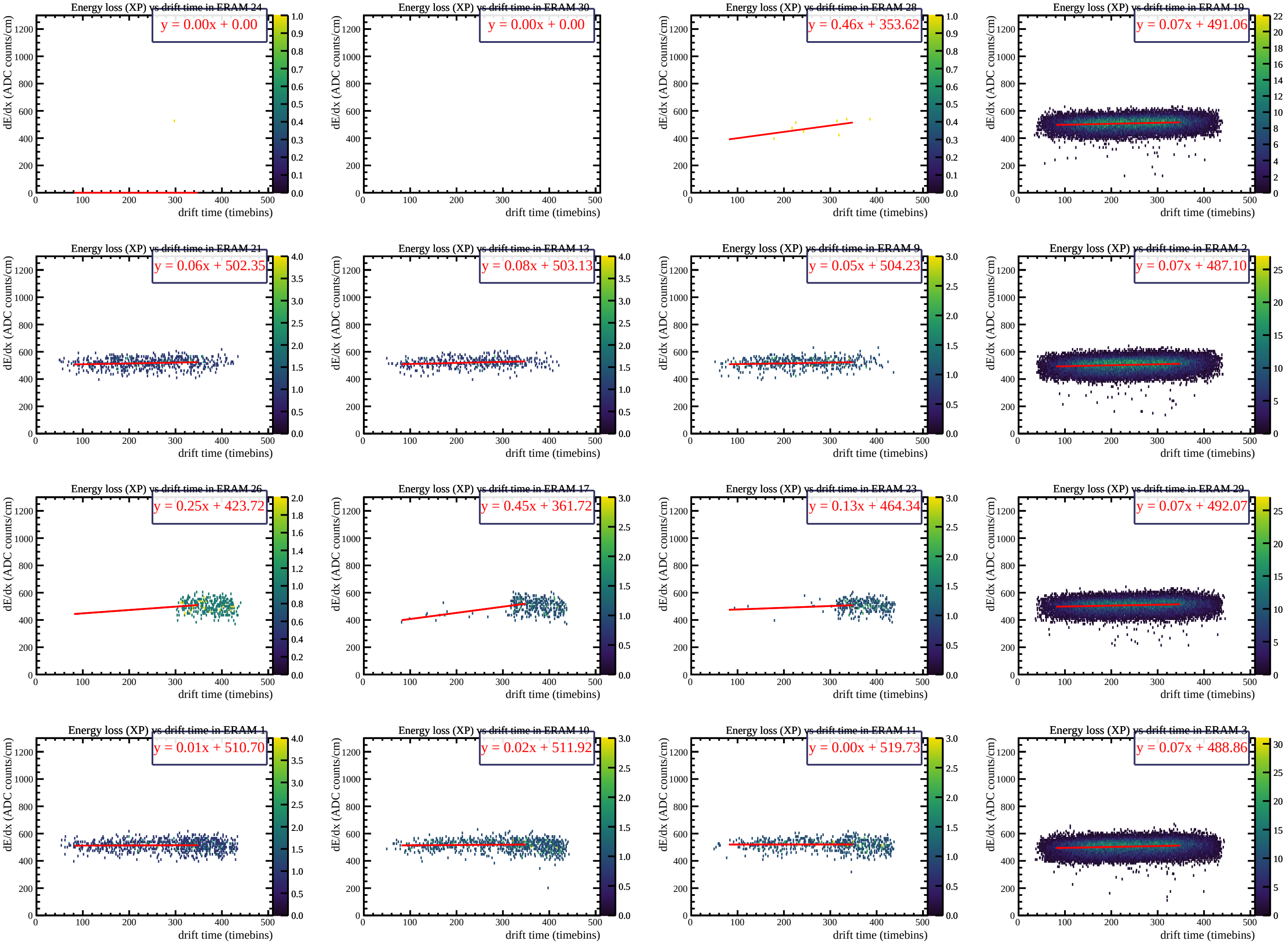
Mean (ADC counts/cm)
800
700
600
500
400
300
200
0
100
200
300
400
500
Drift time (timebins)
WF
XP



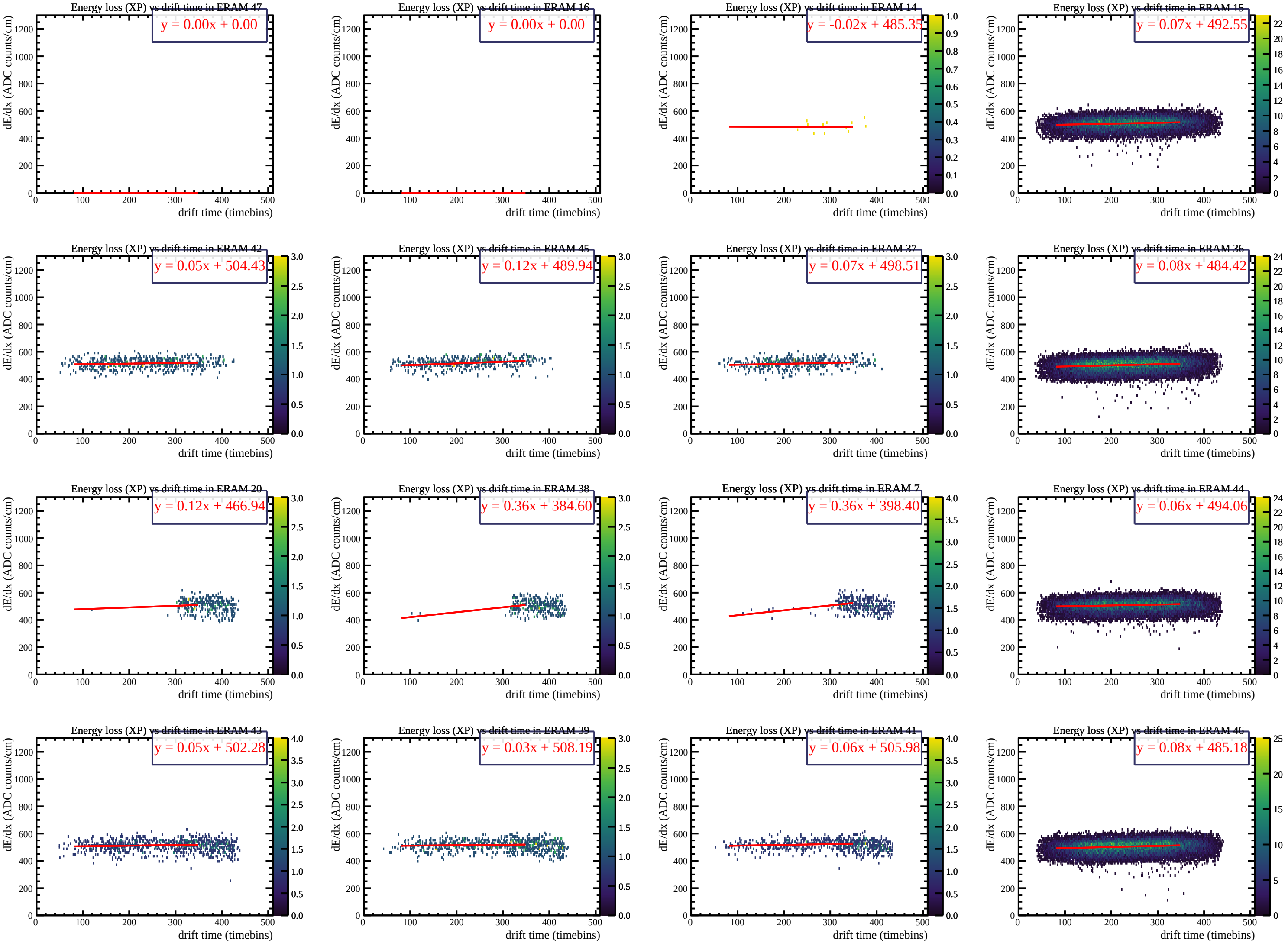
y = 0.05x + 487.93
dE/dx with WF (ADC counts/cm)
drift time (timebins)
0
100
200
300
400
500
600
700
800
900
1000
0
100
200
300
400
500
0
5
10
15
20
25
30
35



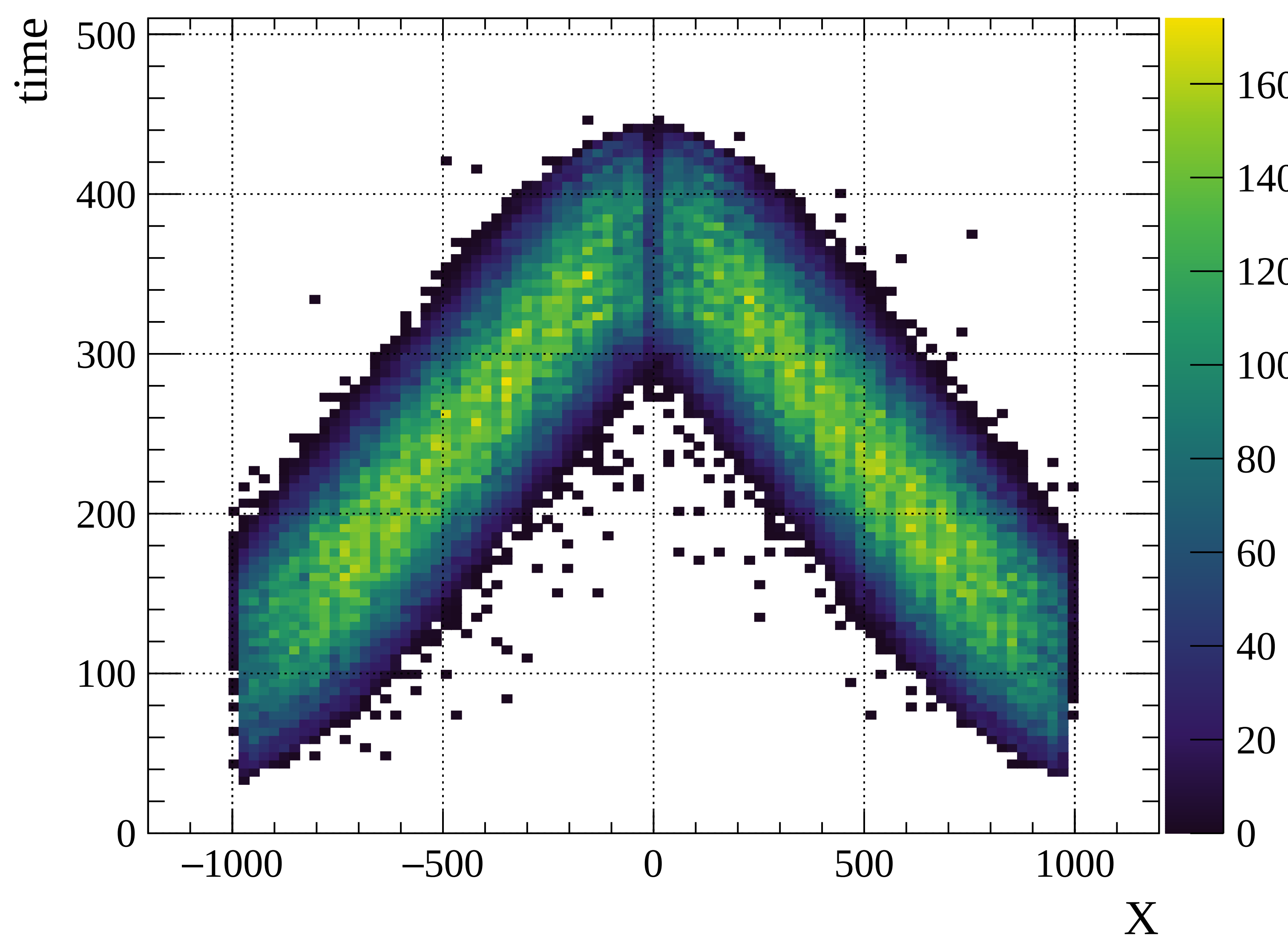
y = 0.08x + 497.93
dE/dx with XP (ADC counts/cm)
drift time (timebins)
1000
900
800
700
600
500
400
300
200
100
0
0
100
200
300
400
500
30
25
20
15
10
5
0

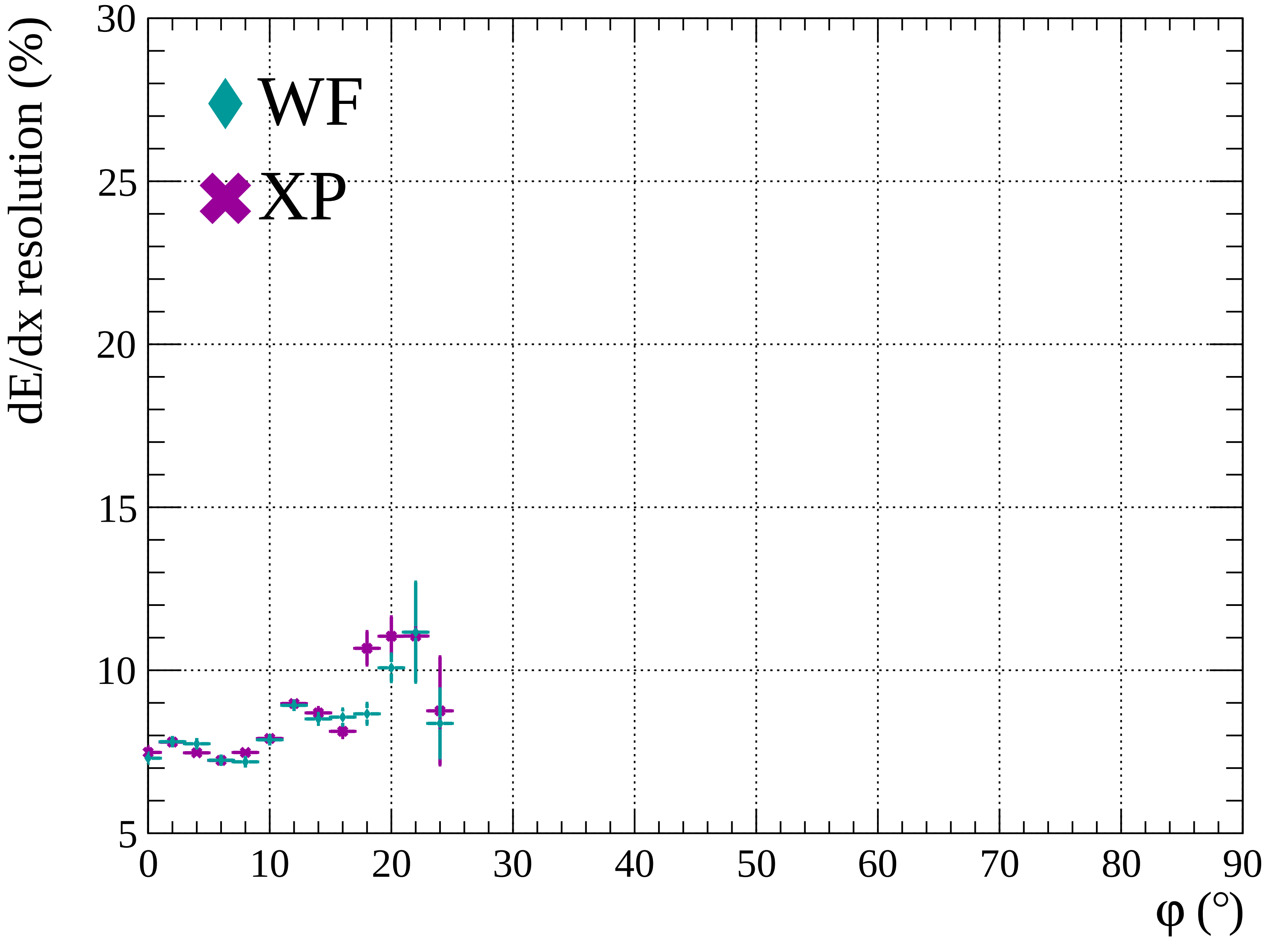


Energy loss (XP) vs drift time in ERAM 24
y = 0.00x + 0.00
Energy loss (XP) vs drift time in ERAM 30
y = 0.00x + 0.00
Energy loss (XP) vs drift time in ERAM 28
y = 0.46x + 353.62
Energy loss (XP) vs drift time in ERAM 19
y = 0.07x + 491.06
Energy loss (XP) vs drift time in ERAM 21
y = 0.06x + 502.35
Energy loss (XP) vs drift time in ERAM 13
y = 0.08x + 503.13
Energy loss (XP) vs drift time in ERAM 9
y = 0.05x + 504.23
Energy loss (XP) vs drift time in ERAM 2
y = 0.07x + 487.10
Energy loss (XP) vs drift time in ERAM 26
y = 0.25x + 423.72
Energy loss (XP) vs drift time in ERAM 17
y = 0.45x + 361.72
Energy loss (XP) vs drift time in ERAM 23
y = 0.13x + 464.34
Energy loss (XP) vs drift time in ERAM 29
y = 0.07x + 492.07
Energy loss (XP) vs drift time in ERAM 1
y = 0.01x + 510.70
Energy loss (XP) vs drift time in ERAM 10
y = 0.02x + 511.92
Energy loss (XP) vs drift time in ERAM 11
y = 0.00x + 519.73
Energy loss (XP) vs drift time in ERAM 3
y = 0.07x + 488.86
dE/dx (ADC counts/cm)
drift time (timebins)

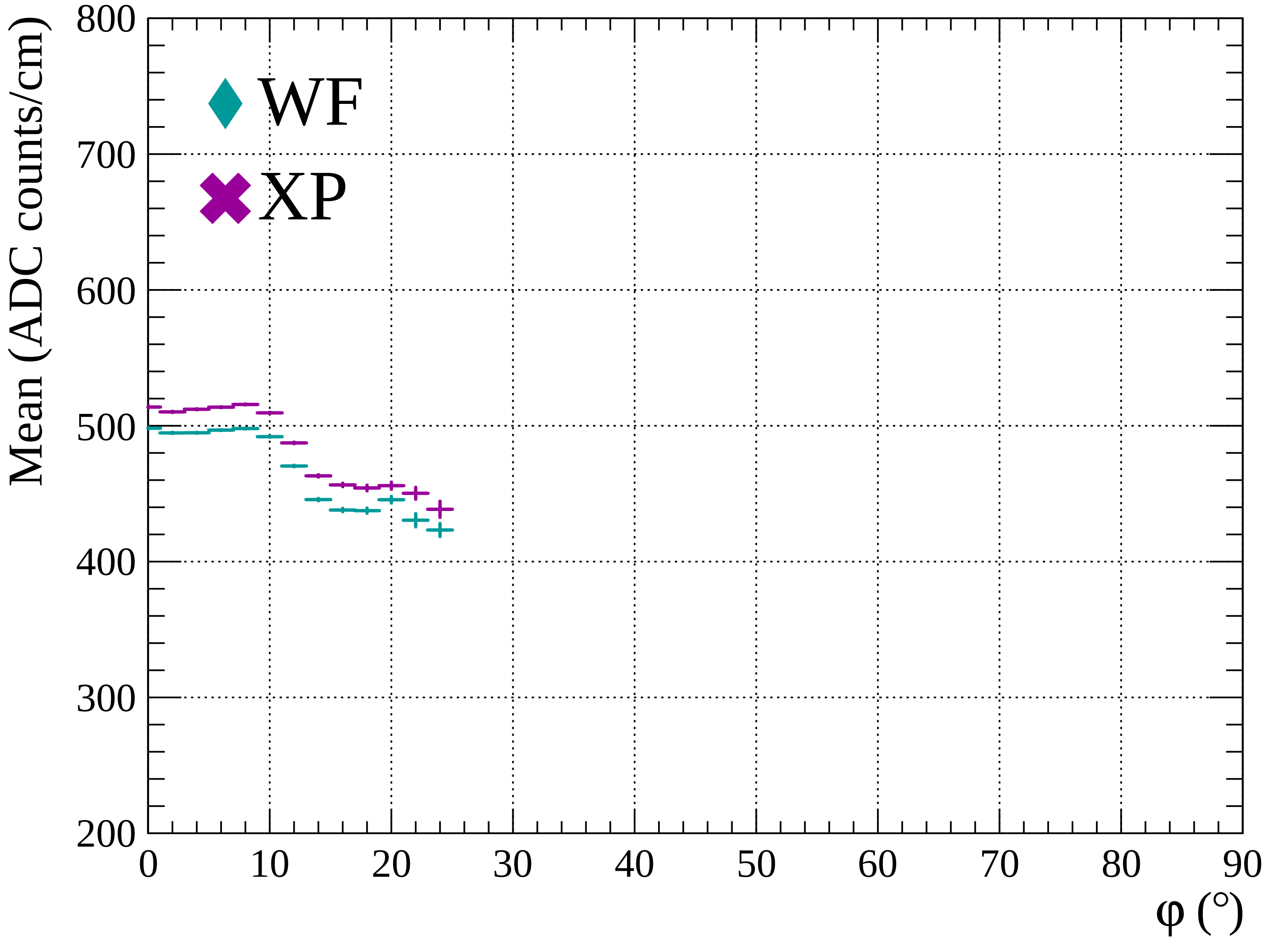


Energy loss (XP) vs drift time in ERAM 47
y = 0.00x + 0.00
Energy loss (XP) vs drift time in ERAM 16
y = 0.00x + 0.00
Energy loss (XP) vs drift time in ERAM 14
y = -0.02x + 485.35
Energy loss (XP) vs drift time in ERAM 15
y = 0.07x + 492.55
Energy loss (XP) vs drift time in ERAM 42
y = 0.05x + 504.43
Energy loss (XP) vs drift time in ERAM 45
y = 0.12x + 489.94
Energy loss (XP) vs drift time in ERAM 37
y = 0.07x + 498.51
Energy loss (XP) vs drift time in ERAM 36
y = 0.08x + 484.42
Energy loss (XP) vs drift time in ERAM 20
y = 0.12x + 466.94
Energy loss (XP) vs drift time in ERAM 38
y = 0.36x + 384.60
Energy loss (XP) vs drift time in ERAM 7
y = 0.36x + 398.40
Energy loss (XP) vs drift time in ERAM 44
y = 0.06x + 494.06
Energy loss (XP) vs drift time in ERAM 43
y = 0.05x + 502.28
Energy loss (XP) vs drift time in ERAM 39
y = 0.03x + 508.19
Energy loss (XP) vs drift time in ERAM 41
y = 0.06x + 505.98
Energy loss (XP) vs drift time in ERAM 46
y = 0.08x + 485.18
dE/dx (ADC counts/cm)
drift time (timebins)

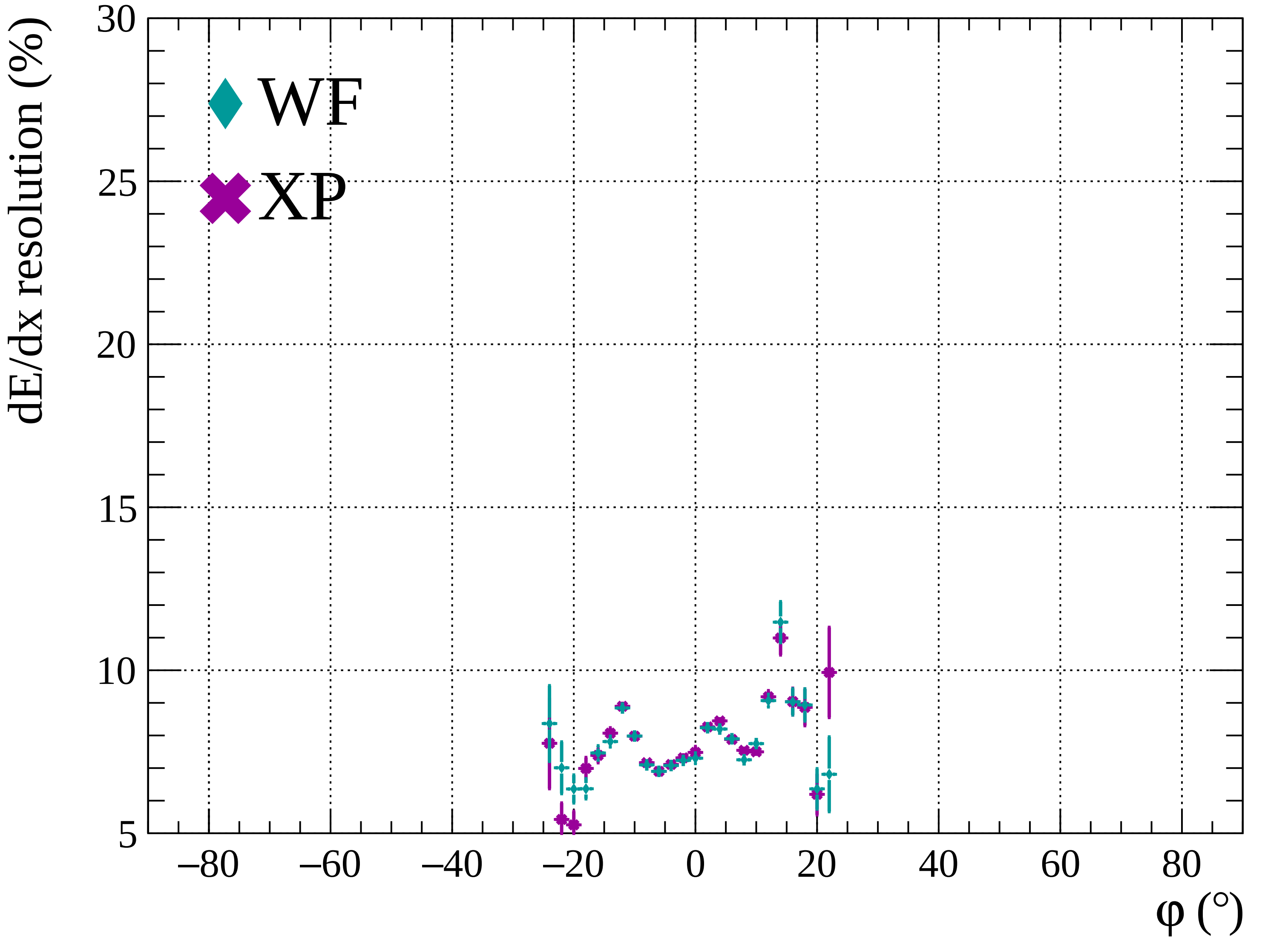

time
500
400
300
200
100
0
−1000
−500
0
500
1000
X
160
140
120
100
80
60
40
20
0



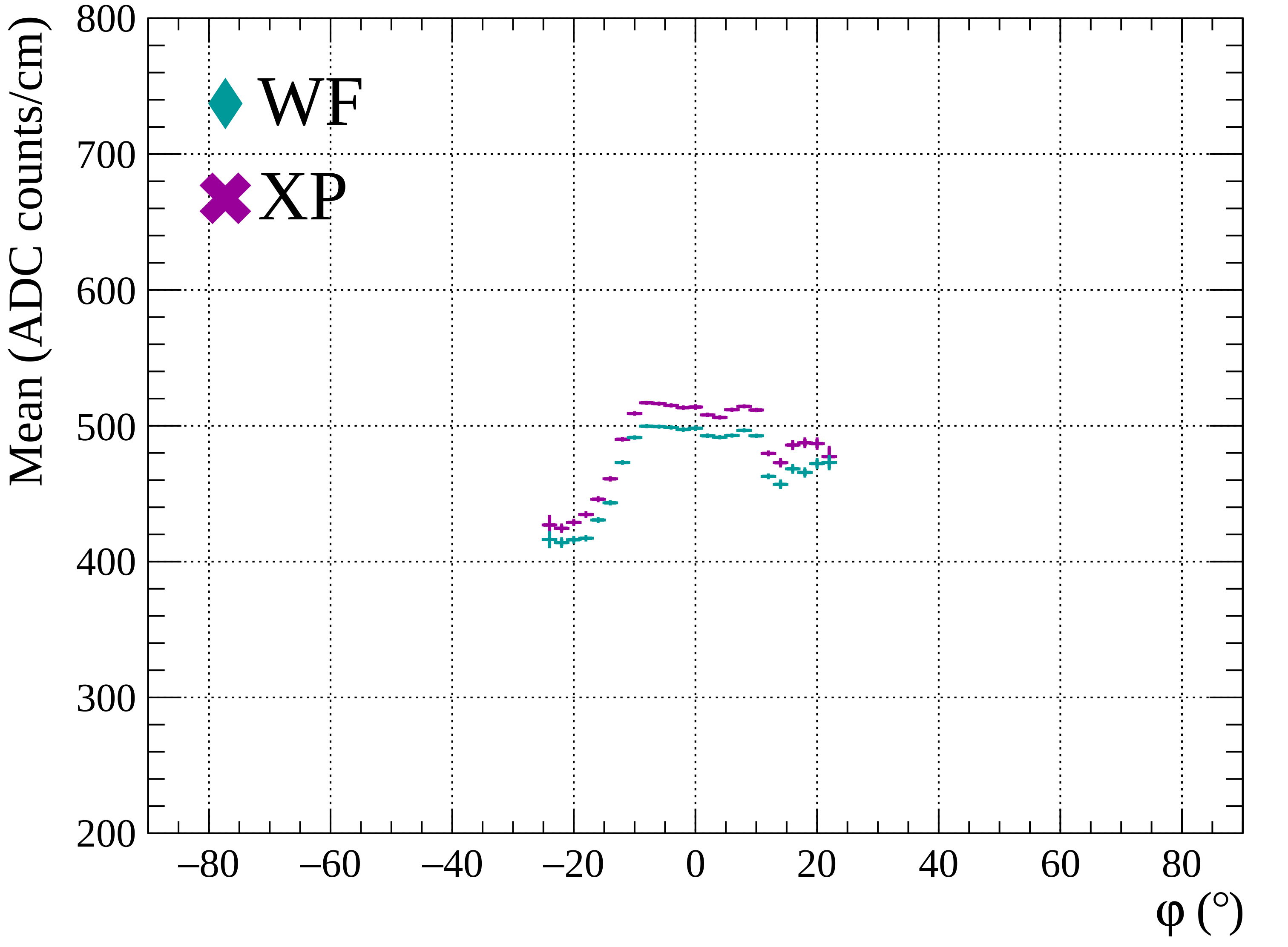
WF
XP
dE/dx resolution (%)
φ (°)
30
25
20
15
10
5
0
10
20
30
40
50
60
70
80
90

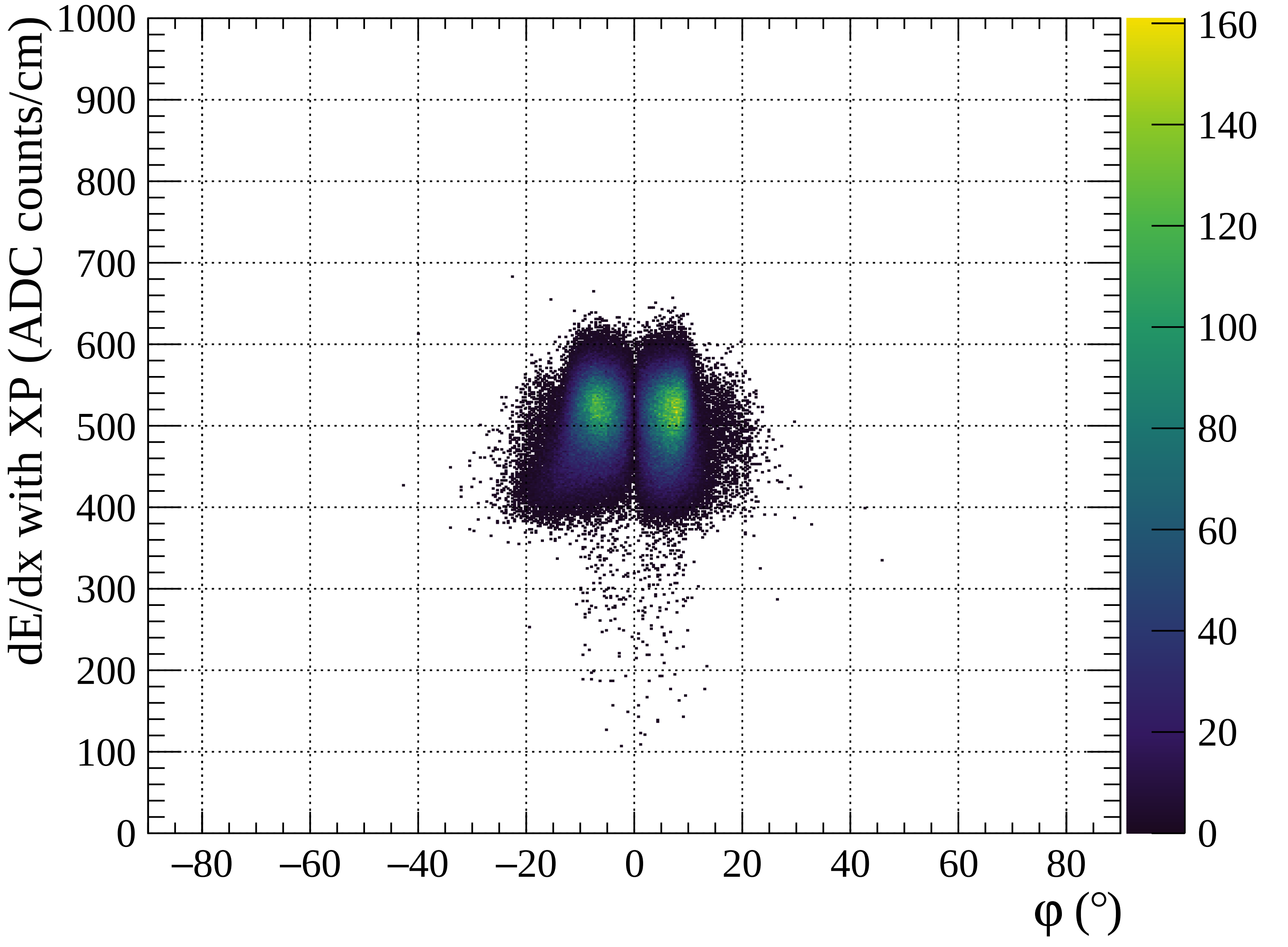


Mean (ADC counts/cm)
800
700
600
500
400
300
200
WF
XP
0
10
20
30
40
50
60
70
80
90
φ (°)

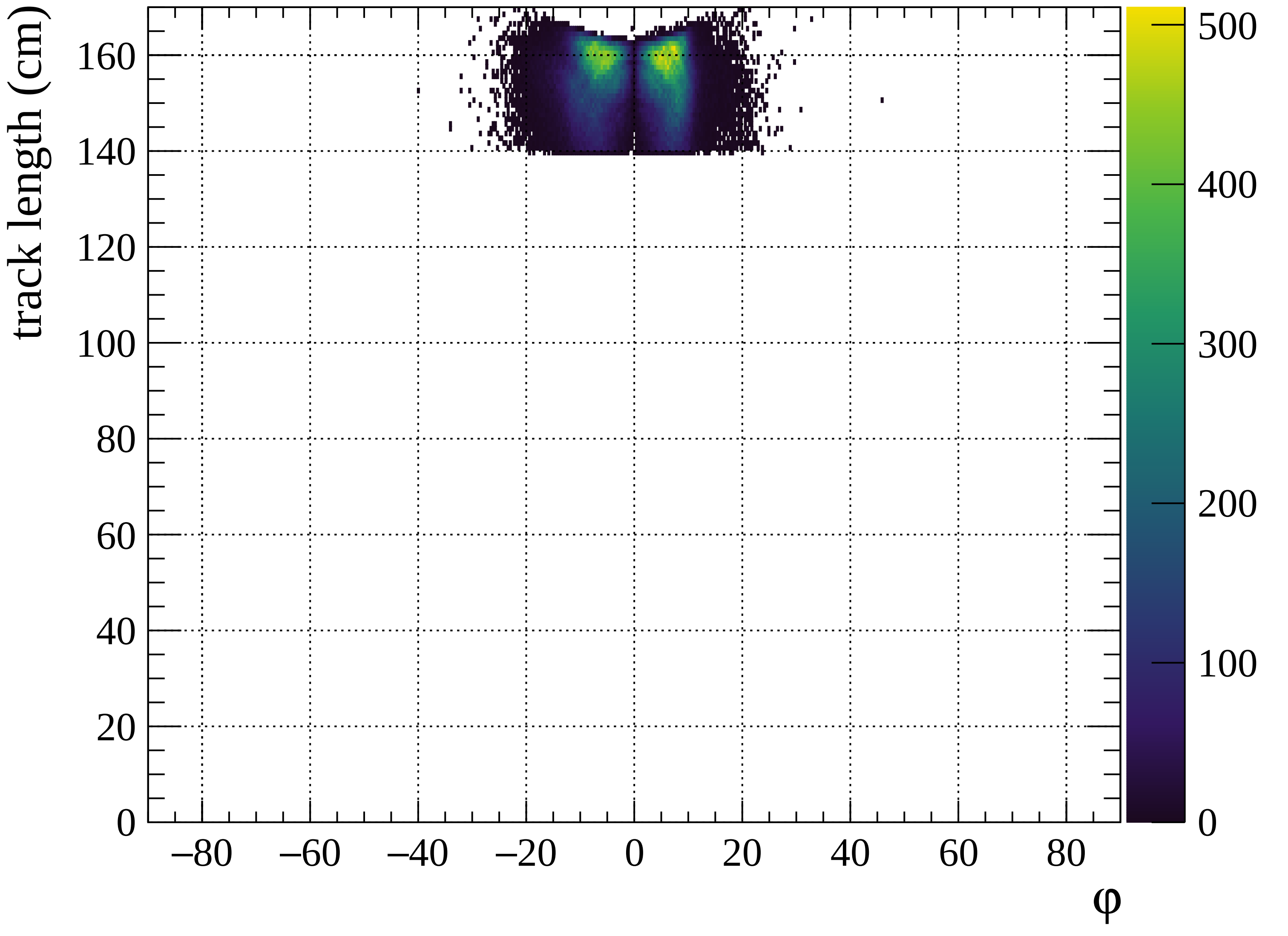


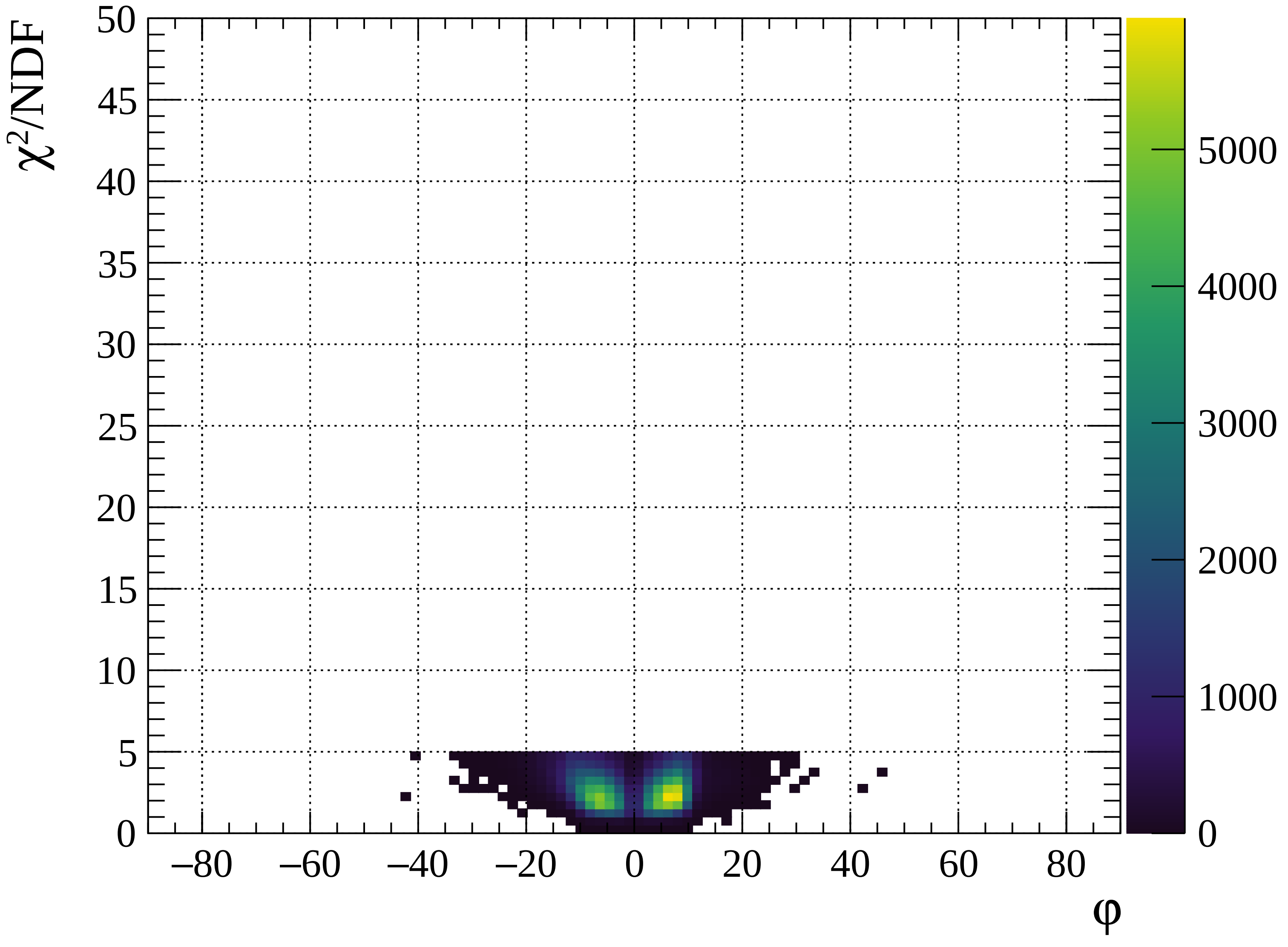
dE/dx resolution (%)
30
25
20
15
10
5
WF
XP
−80
−60
−40
−20
0
20
40
60
80
φ (°)


WF
XP
Mean (ADC counts/cm)
800
700
600
500
400
300
200
−80
−60
−40
−20
0
20
40
60
80
φ (°)

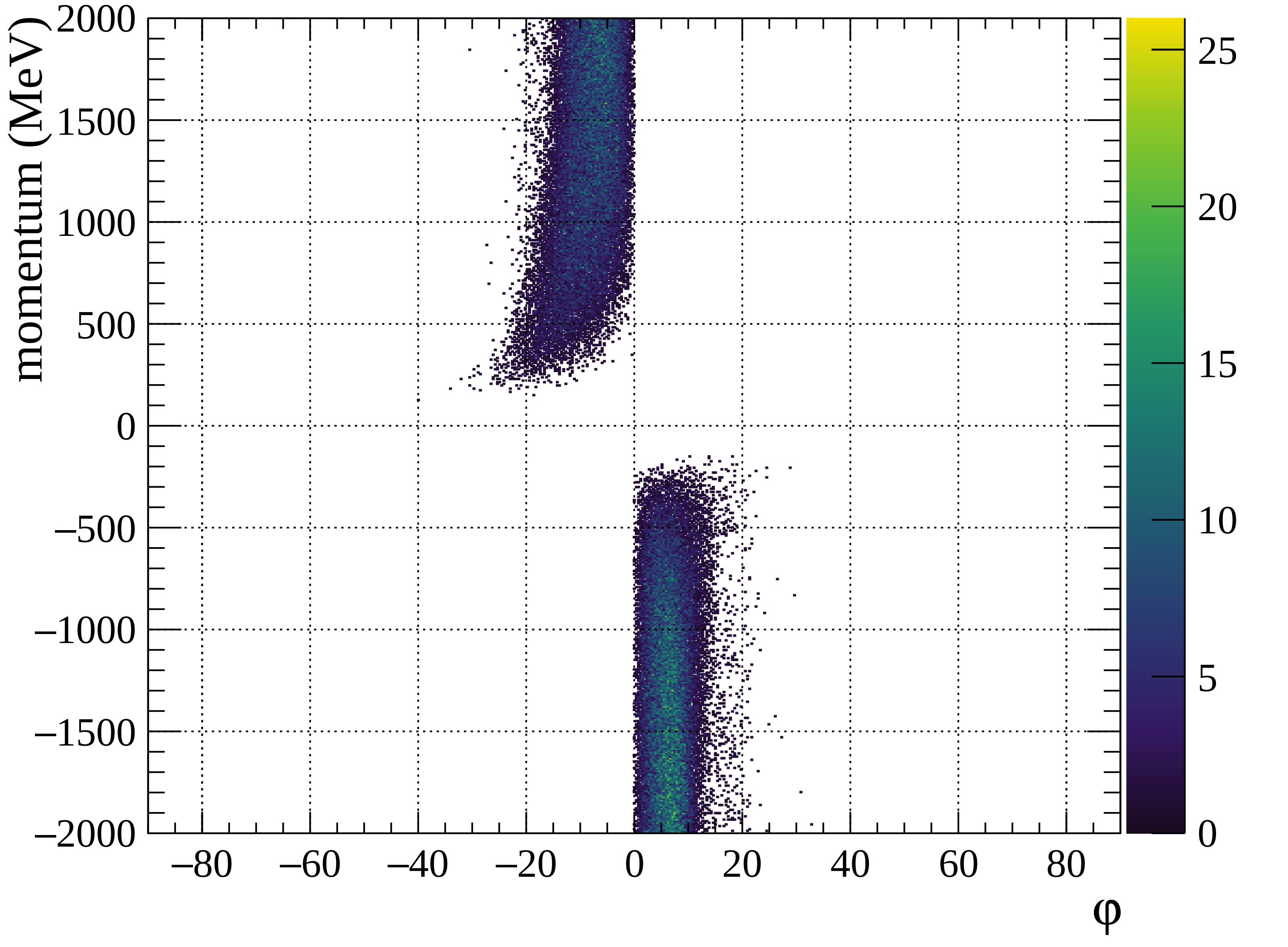


dE/dx with XP (ADC counts/cm)
1000
900
800
700
600
500
400
300
200
100
0
−80
−60
−40
−20
0
20
40
60
80
φ (°)
160
140
120
100
80
60
40
20
0

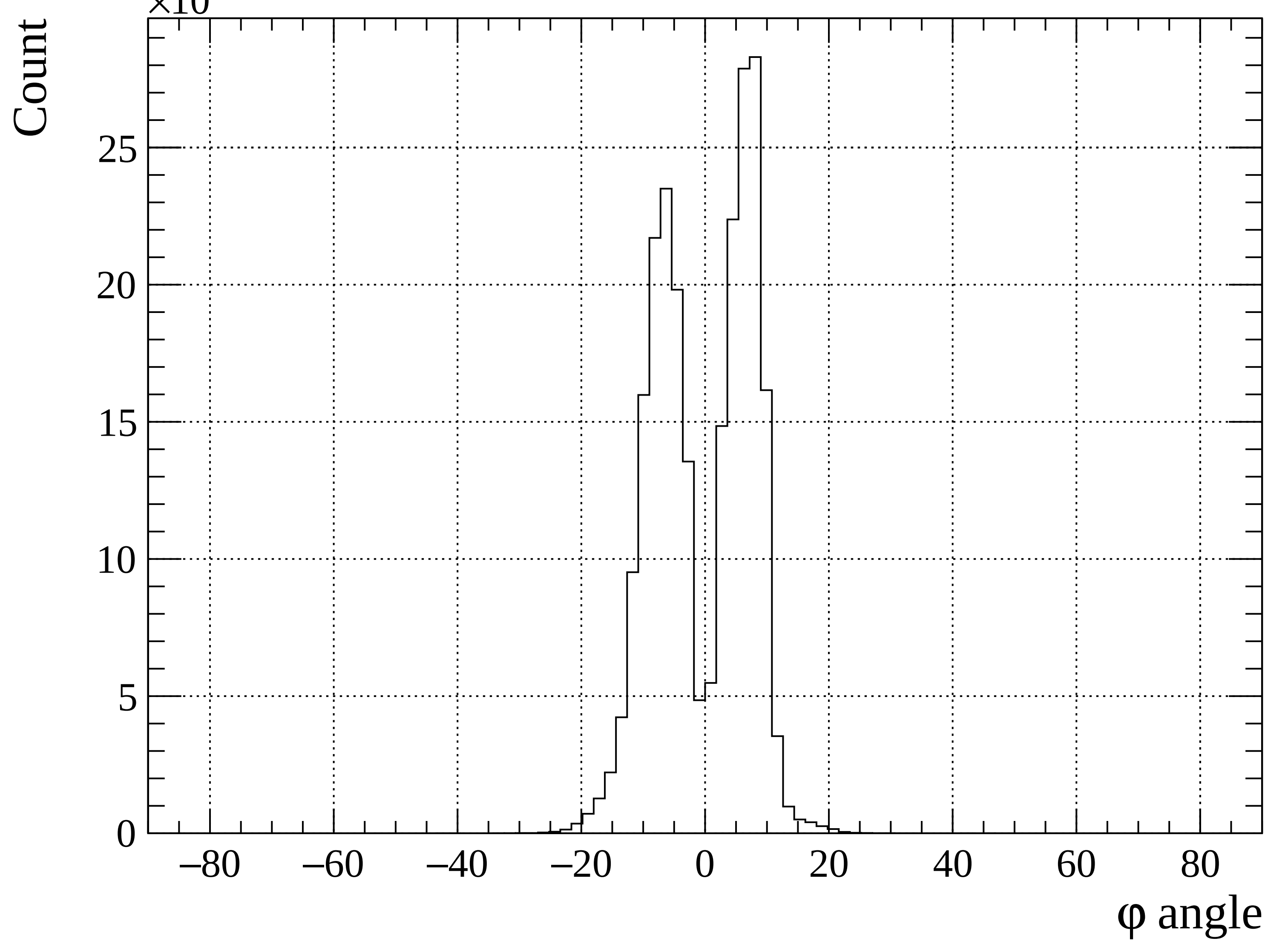

track length (cm)
160
140
120
100
80
60
40
20
0
−80
−60
−40
−20
0
20
40
60
80
φ
500
400
300
200
100
0



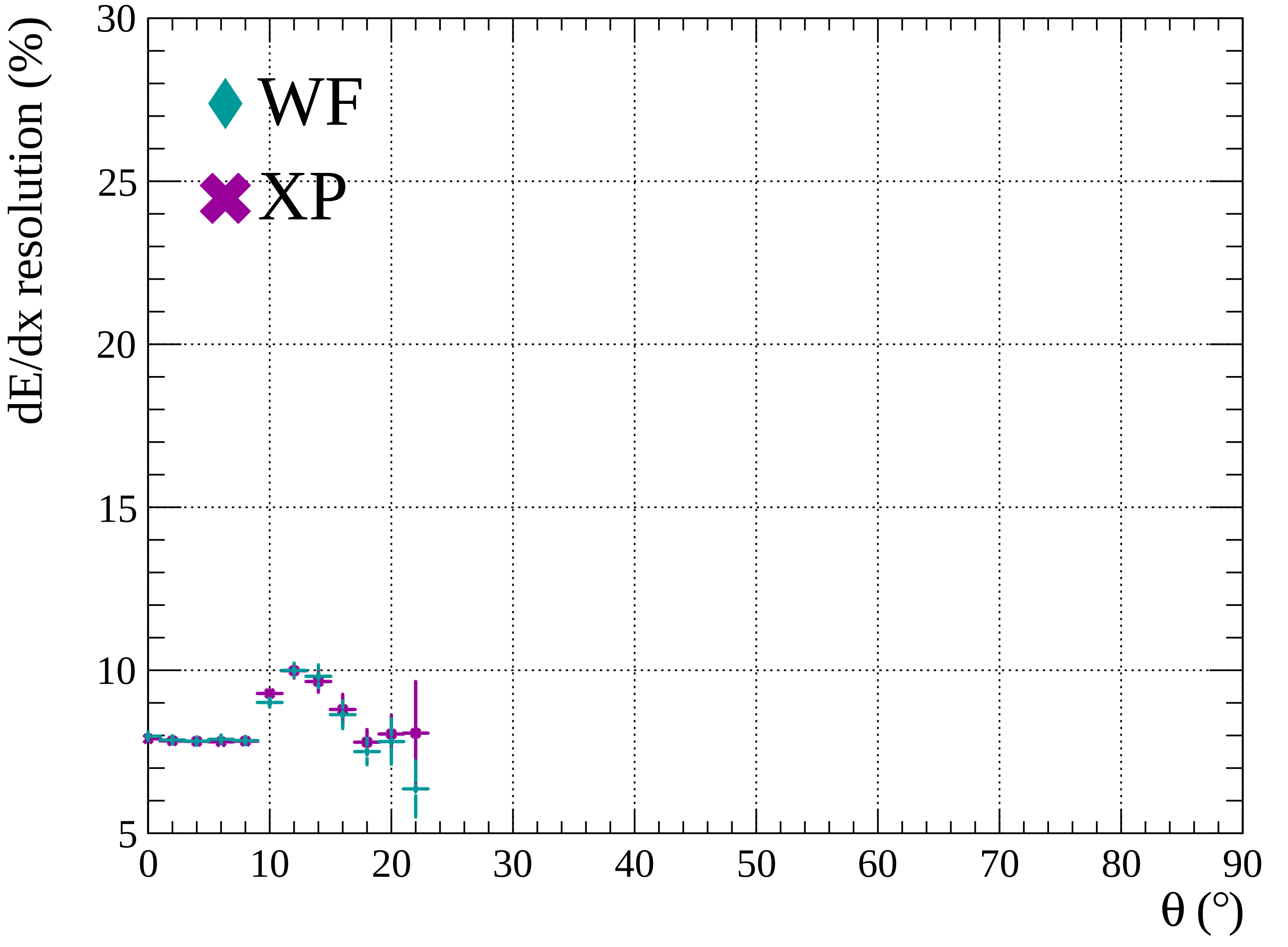
χ²/NDF
50
45
40
35
30
25
20
15
10
5
0
−80
−60
−40
−20
0
20
40
60
80
φ
5000
4000
3000
2000
1000
0



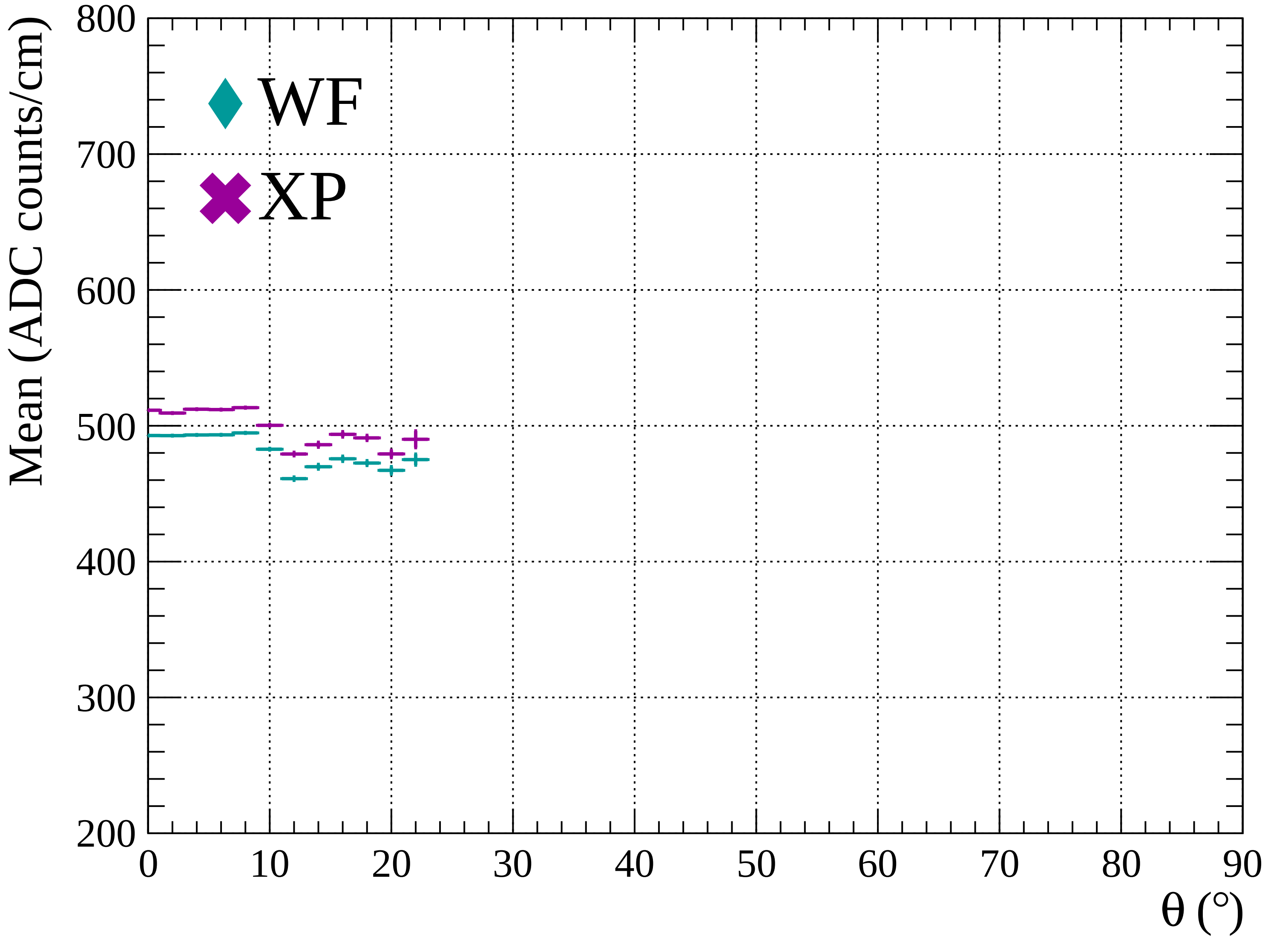
momentum (MeV)
2000
1500
1000
500
0
−500
−1000
−1500
−2000
−80
−60
−40
−20
0
20
40
60
80
φ
25
20
15
10
5
0

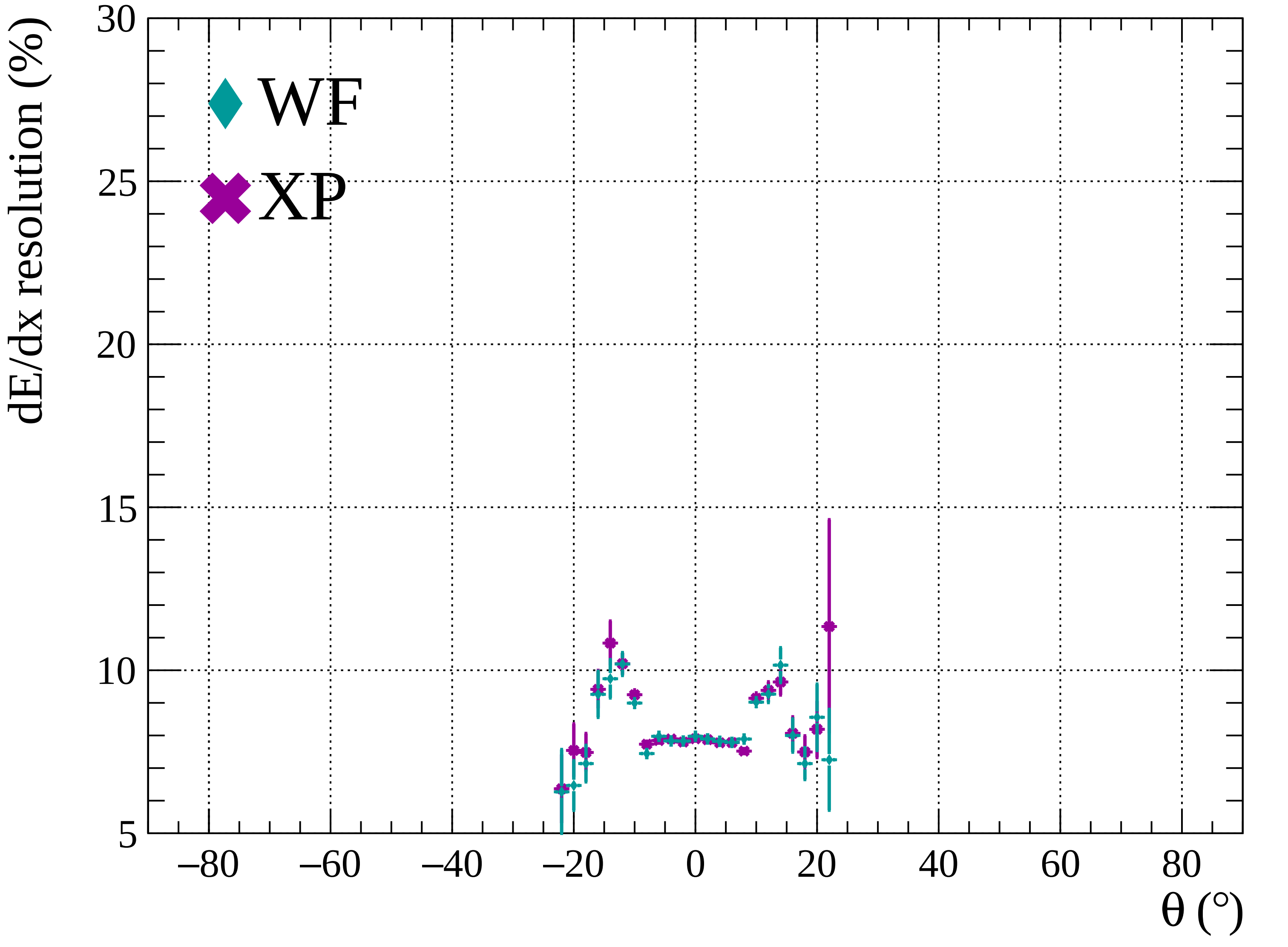


Count
×10
25
20
15
10
5
0
−80
−60
−40
−20
0
20
40
60
80
φ angle

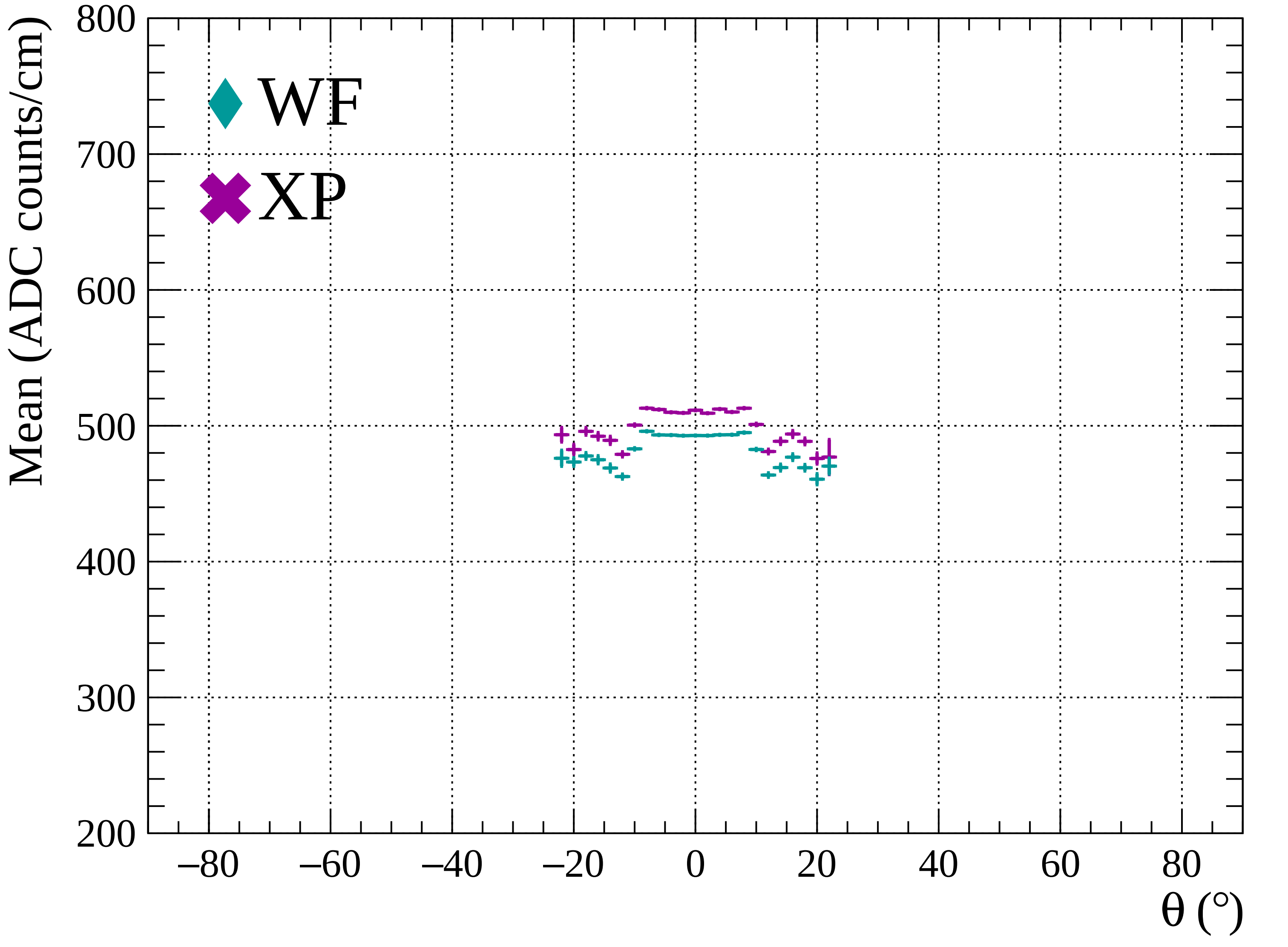


dE/dx resolution (%)
30
25
20
15
10
5
0
10
20
30
40
50
60
70
80
90
θ (°)
WF
XP

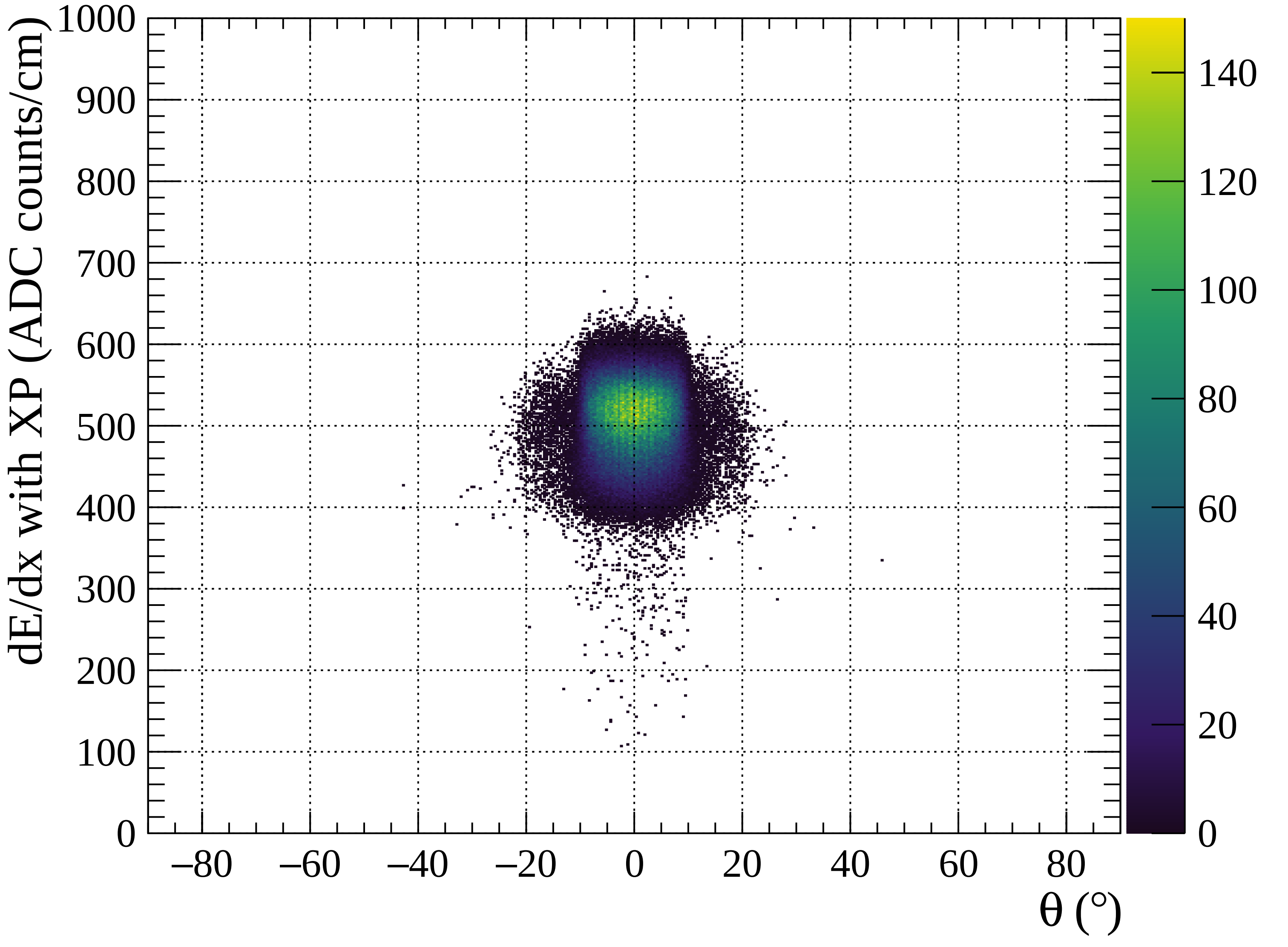

WF
XP
Mean (ADC counts/cm)
θ (°)
200
300
400
500
600
700
800
0
10
20
30
40
50
60
70
80
90



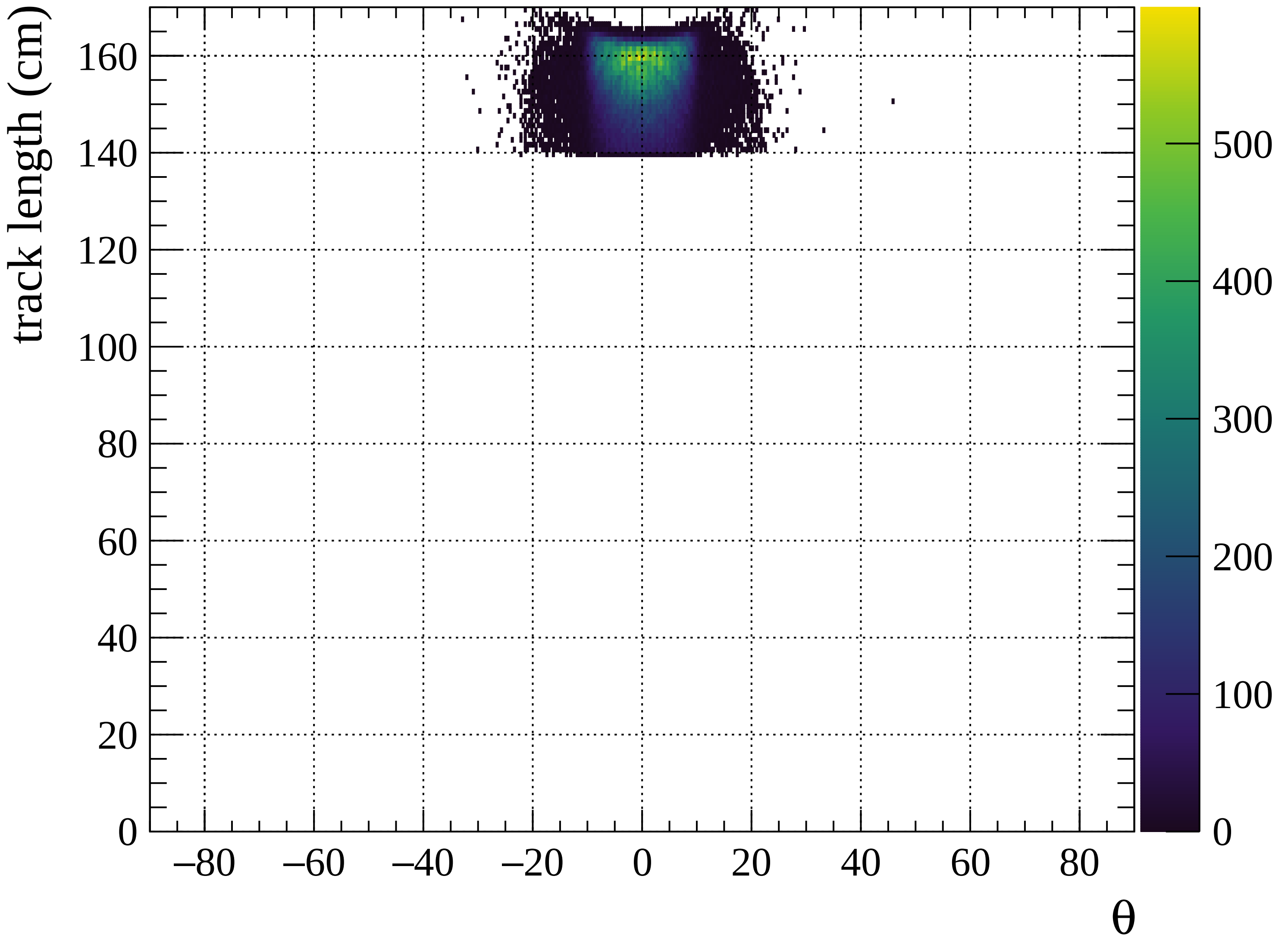
dE/dx resolution (%)
WF
XP
θ (°)
30
25
20
15
10
5
−80
−60
−40
−20
0
20
40
60
80

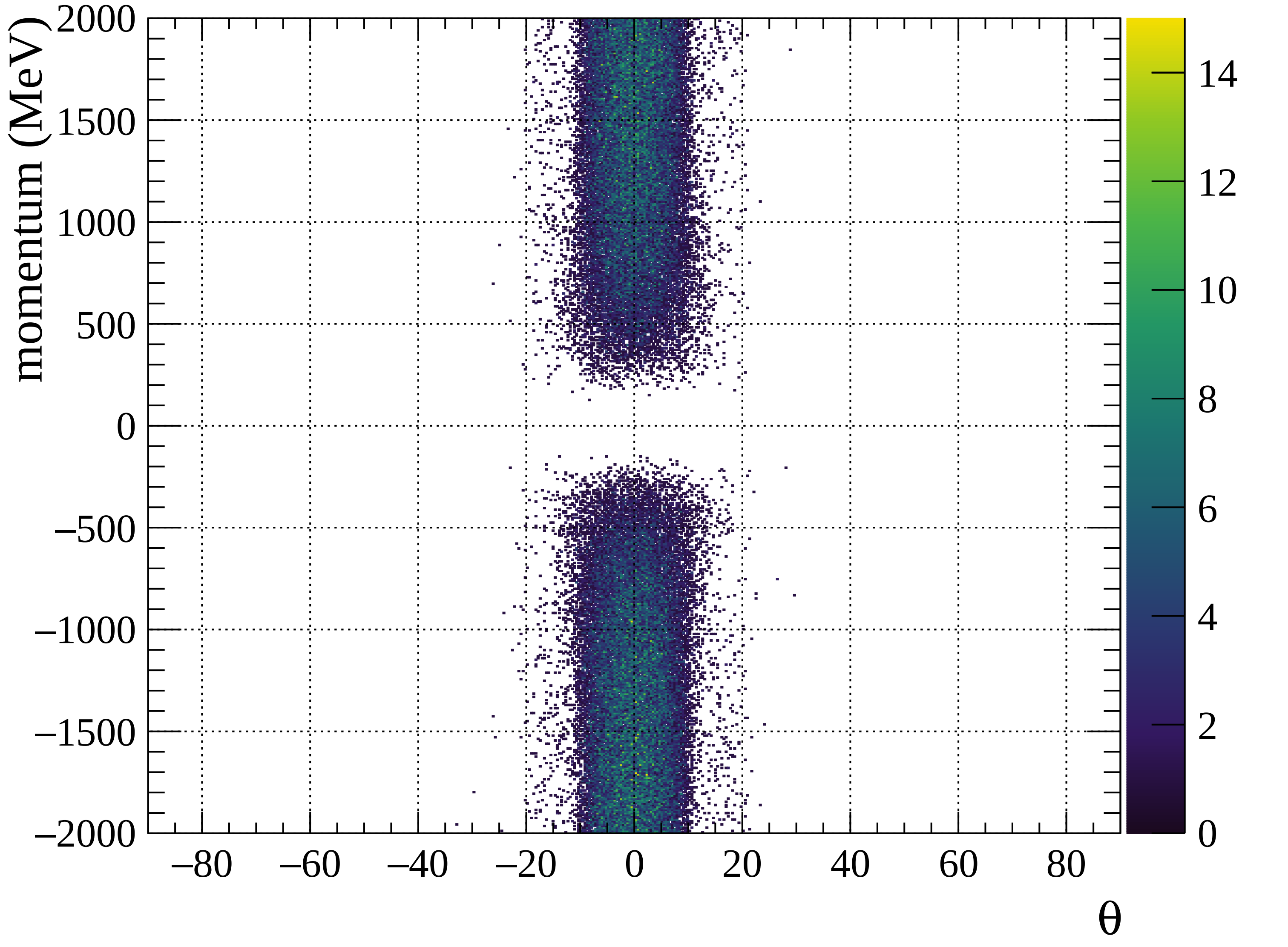


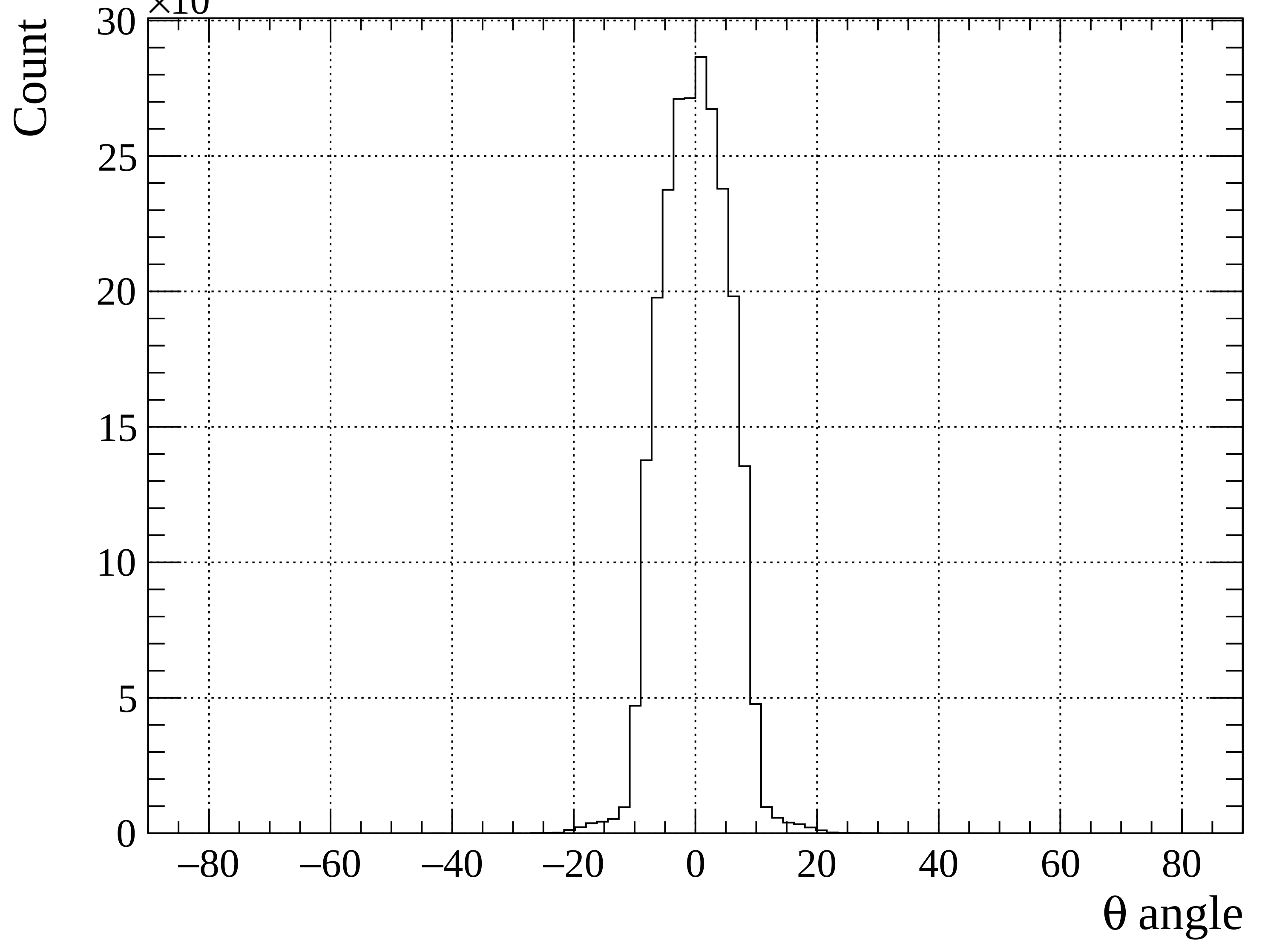
WF
XP
Mean (ADC counts/cm)
θ (°)
800
700
600
500
400
300
200
−80
−60
−40
−20
0
20
40
60
80



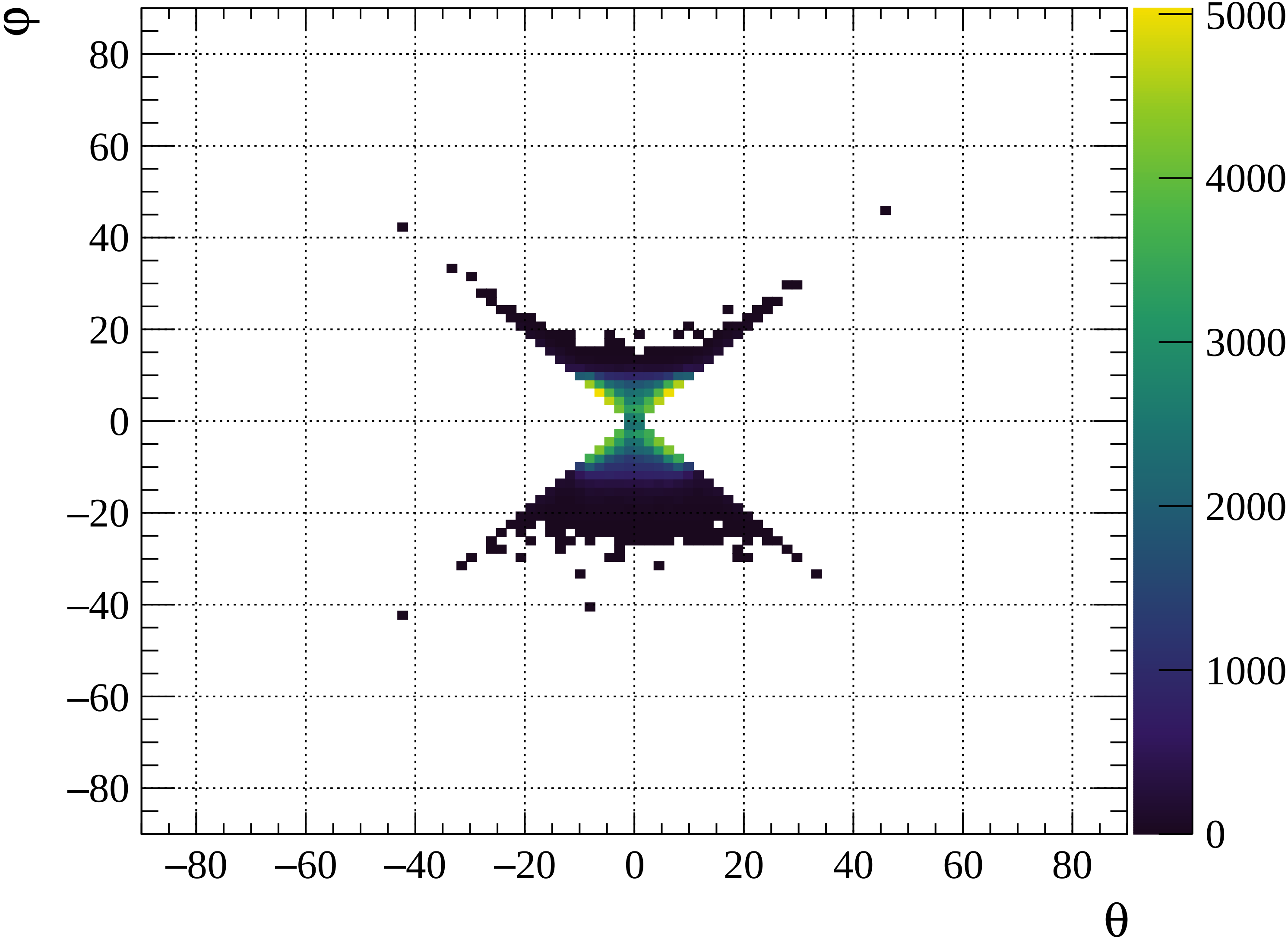
dE/dx with XP (ADC counts/cm)
1000
900
800
700
600
500
400
300
200
100
0
−80
−60
−40
−20
0
20
40
60
80
θ (°)
140
120
100
80
60
40
20
0


track length (cm)
θ
0
20
40
60
80
100
120
140
160
−80
−60
−40
−20
0
20
40
60
80
0
100
200
300
400
500

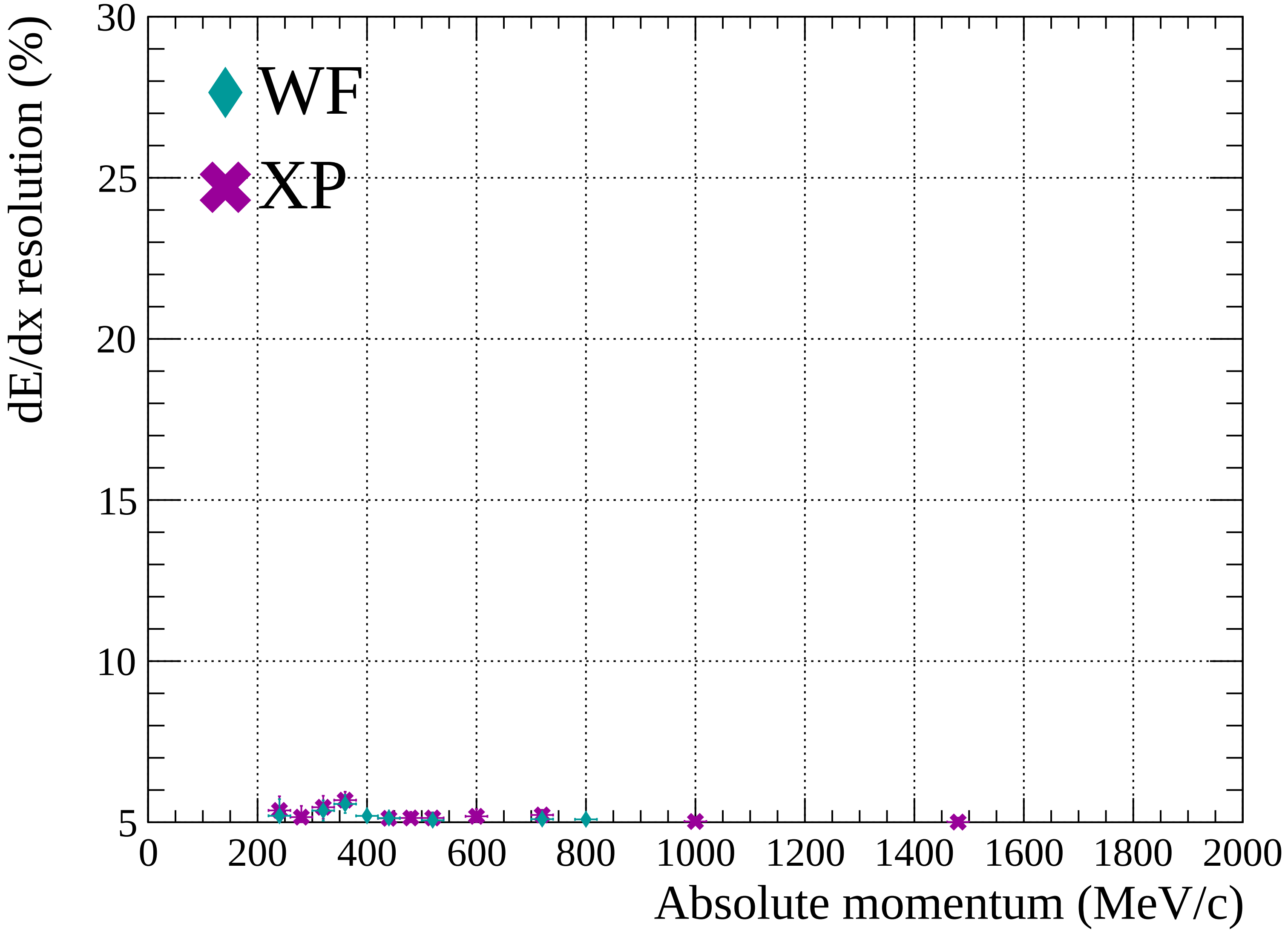

momentum (MeV)
2000
1500
1000
500
0
−500
−1000
−1500
−2000
−80
−60
−40
−20
0
20
40
60
80
θ
14
12
10
8
6
4
2
0

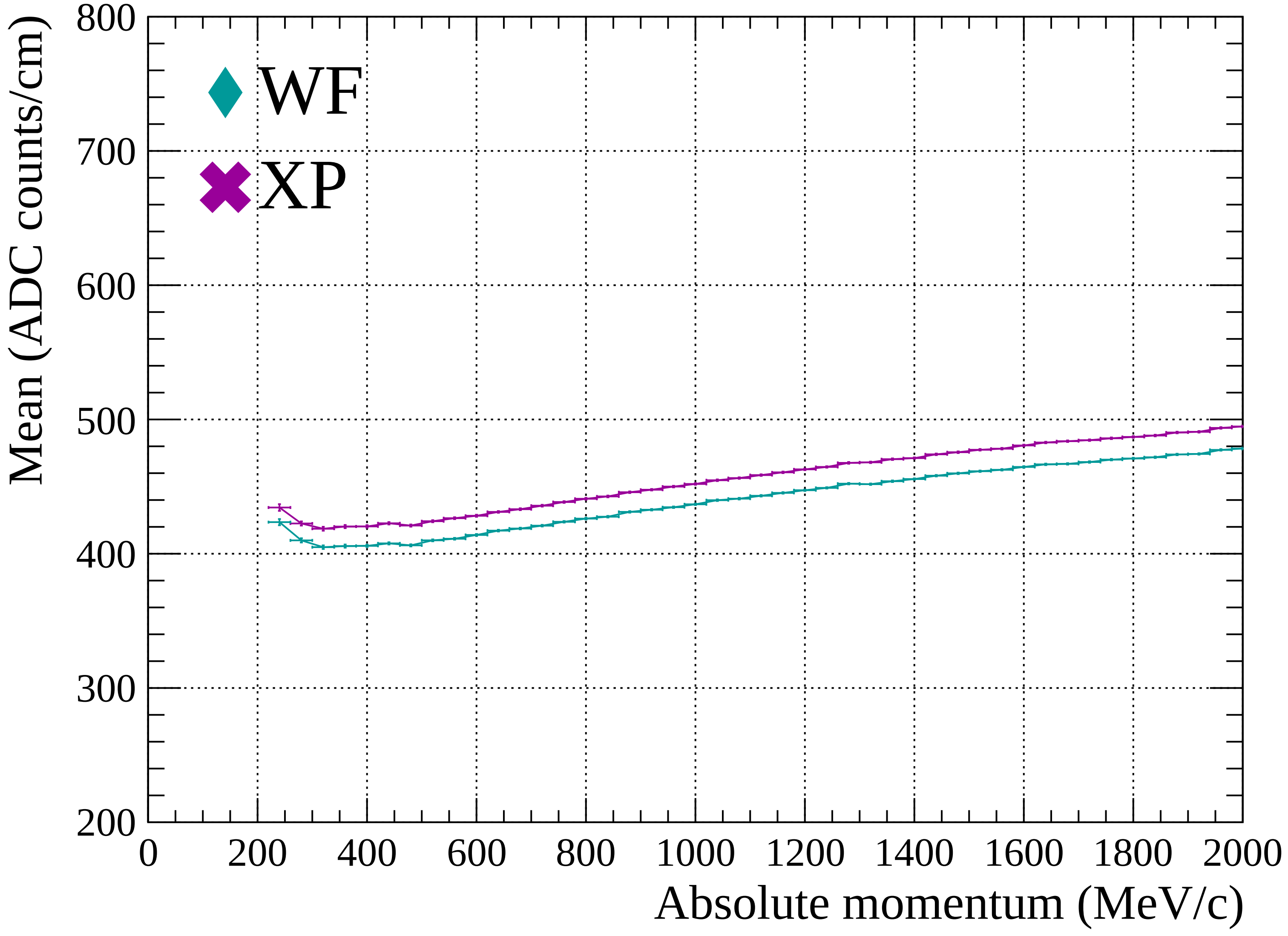


×10
Count
30
25
20
15
10
5
0
−80
−60
−40
−20
0
20
40
60
80
θ angle

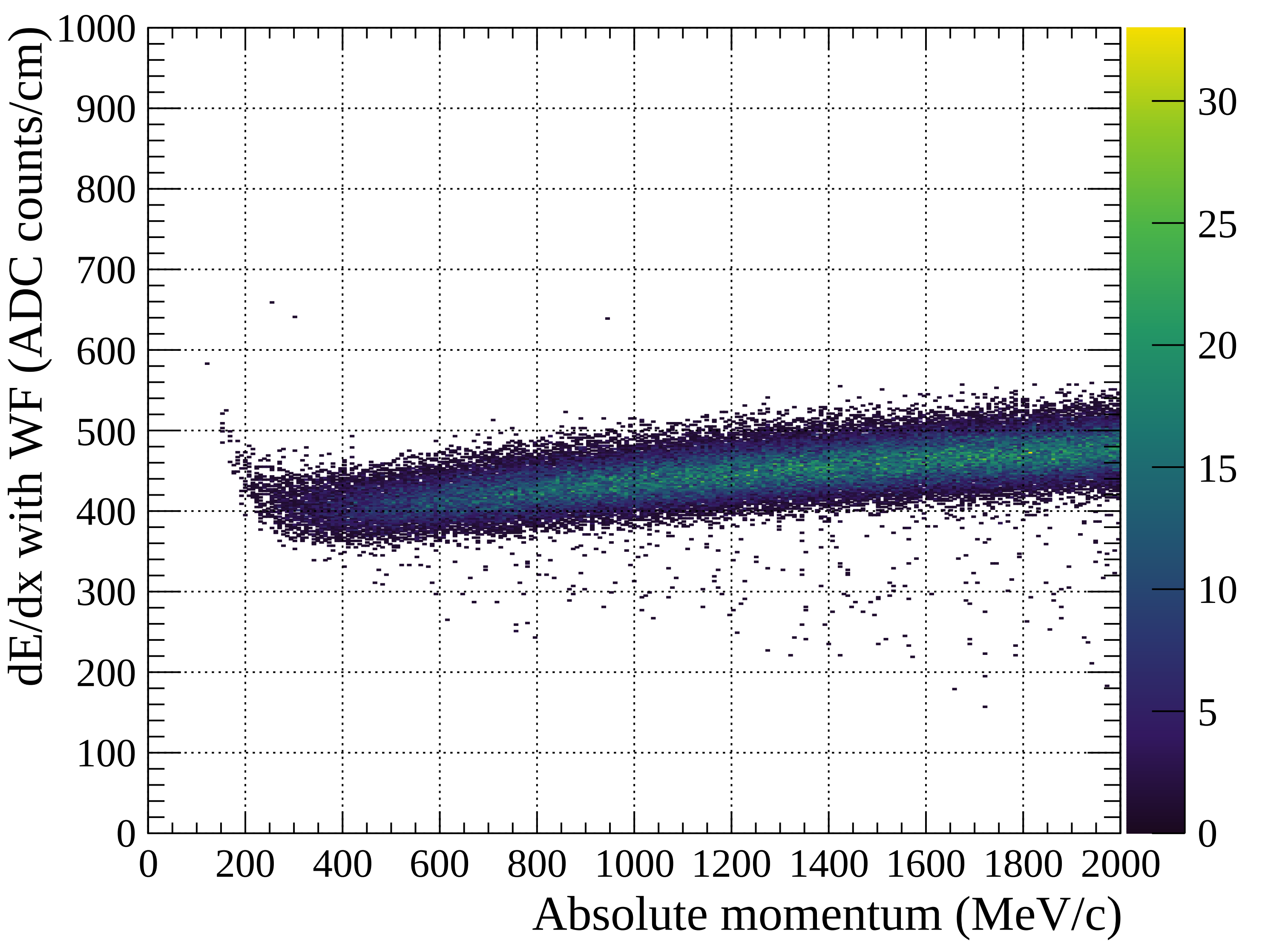


φ
θ
80
60
40
20
0
−20
−40
−60
−80
5000
4000
3000
2000
1000

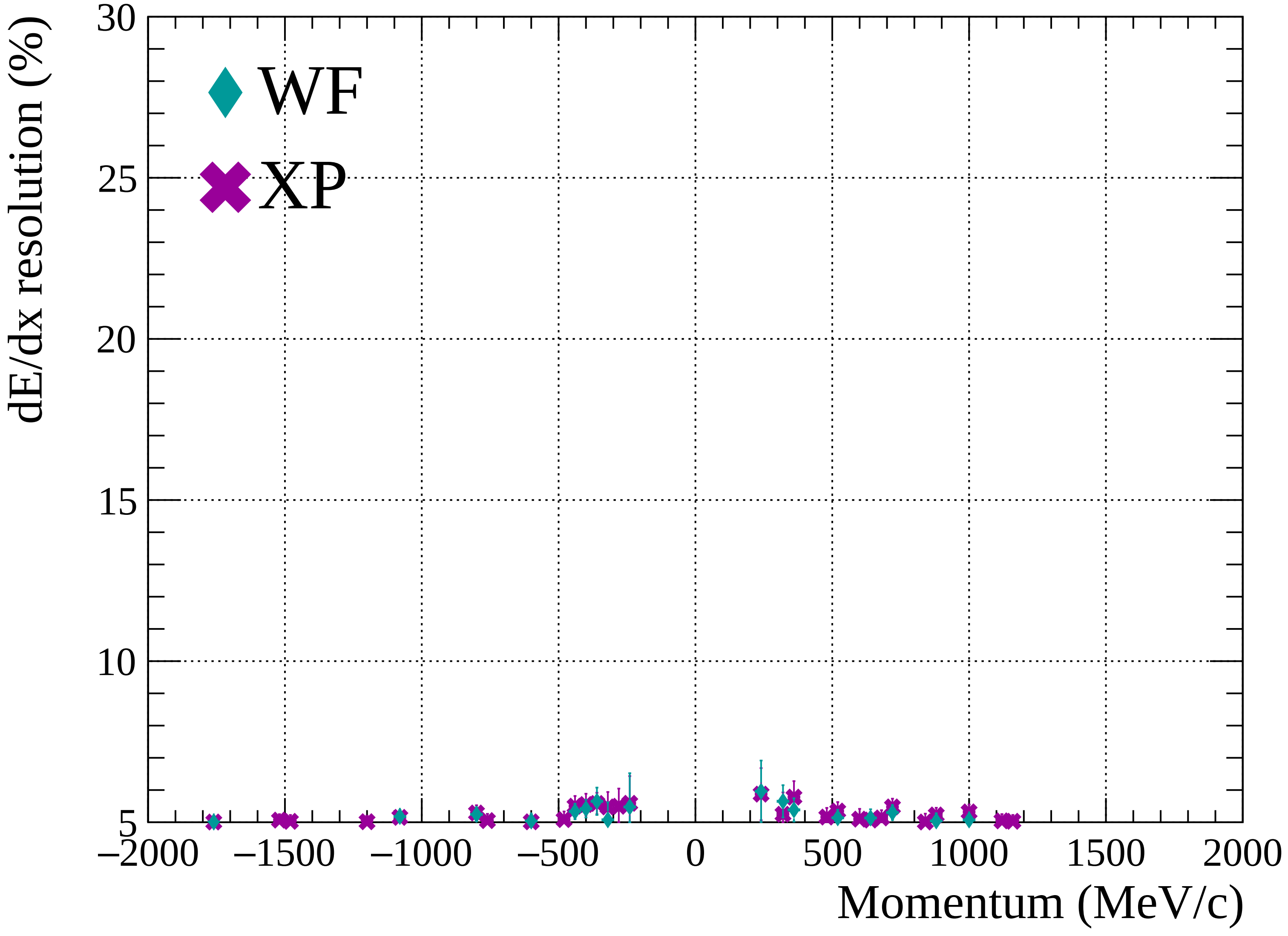

dE/dx resolution (%)
30
25
20
15
10
5
WF
XP
0
200
400
600
800
1000
1200
1400
1600
1800
2000
Absolute momentum (MeV/c)



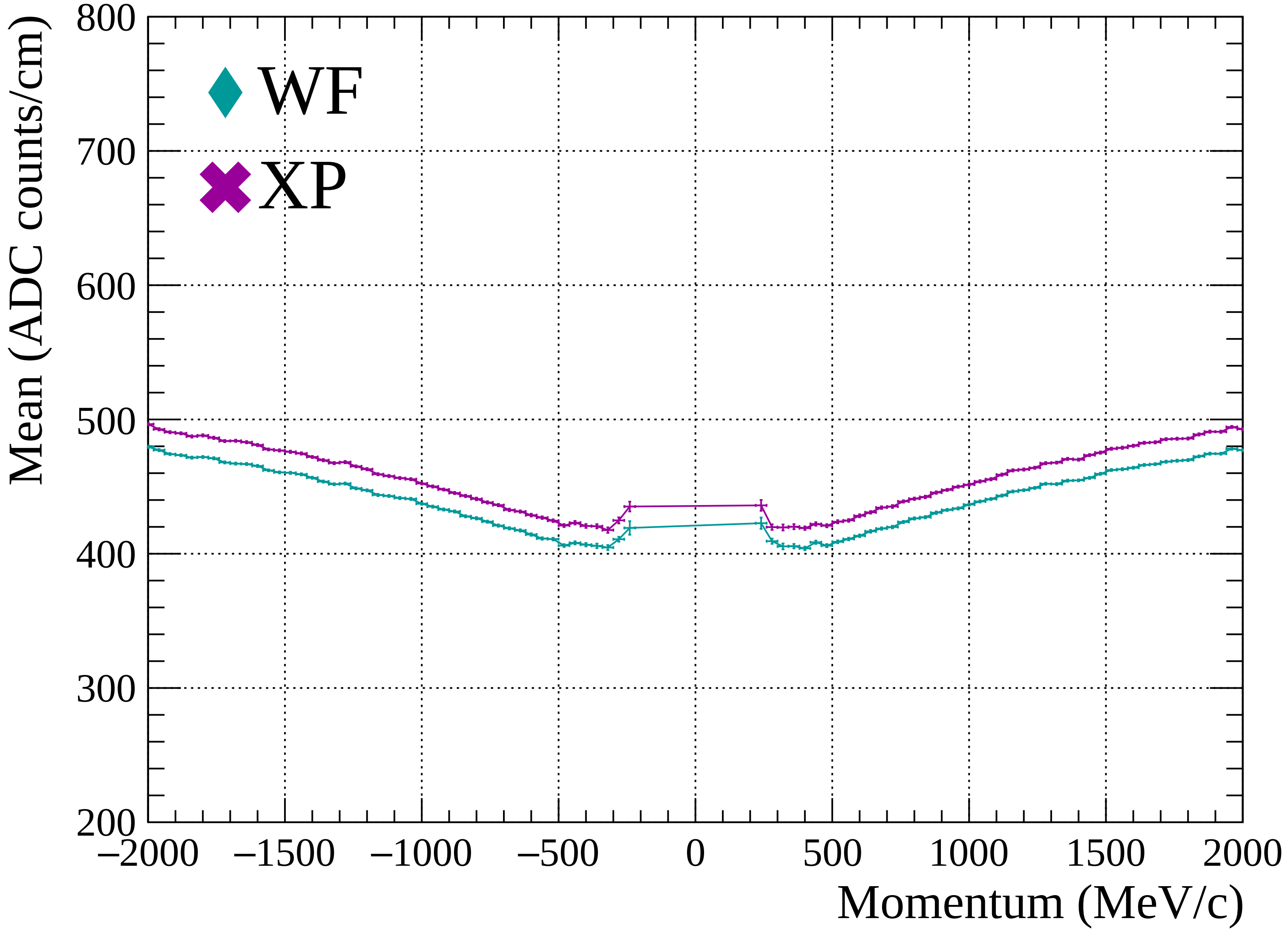
WF
XP
Mean (ADC counts/cm)
200
300
400
500
600
700
800
0
200
400
600
800
1000
1200
1400
1600
1800
2000
Absolute momentum (MeV/c)



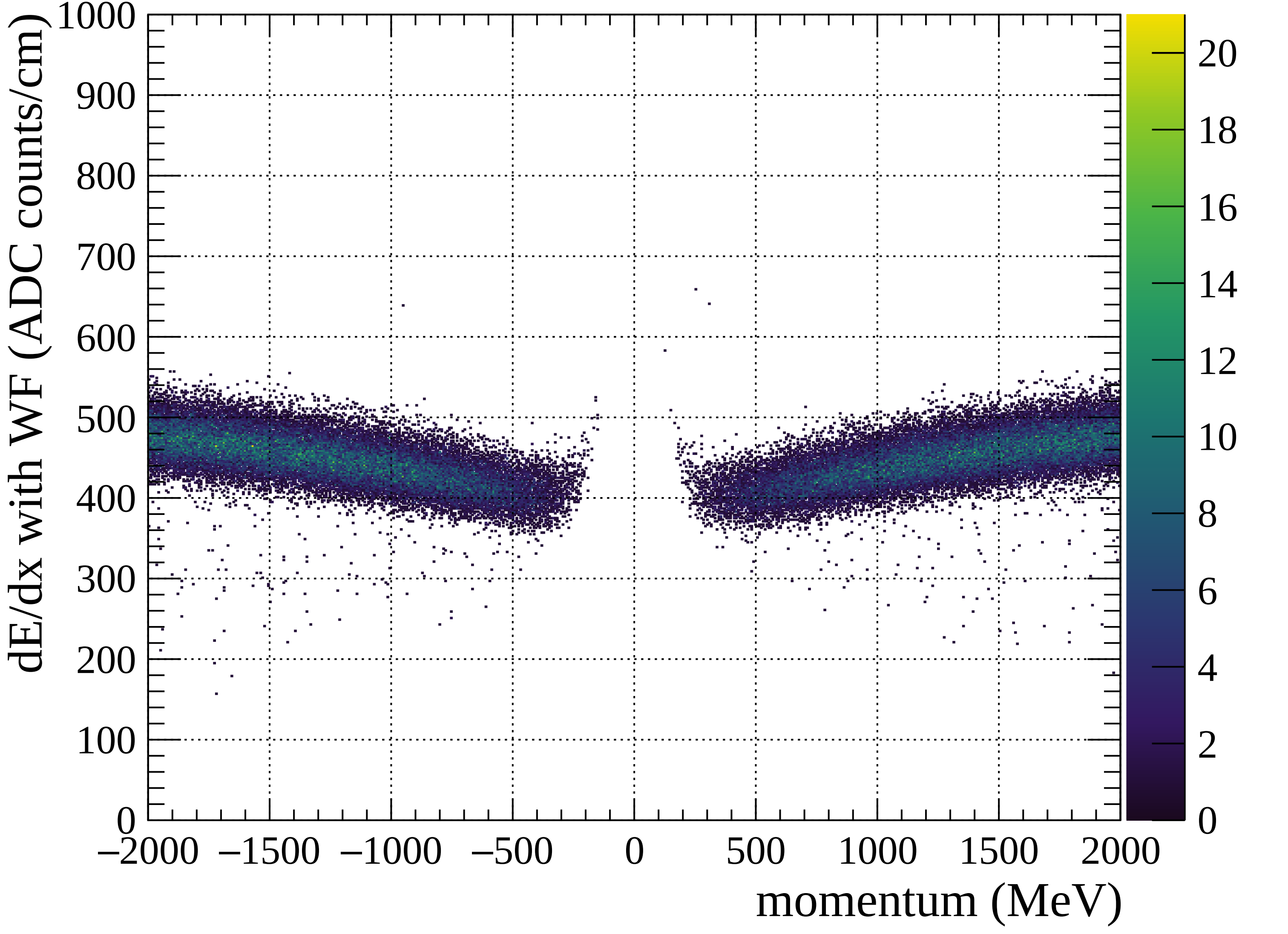
dE/dx with WF (ADC counts/cm)
Absolute momentum (MeV/c)
0
100
200
300
400
500
600
700
800
900
1000
0
200
400
600
800
1000
1200
1400
1600
1800
2000
0
5
10
15
20
25
30

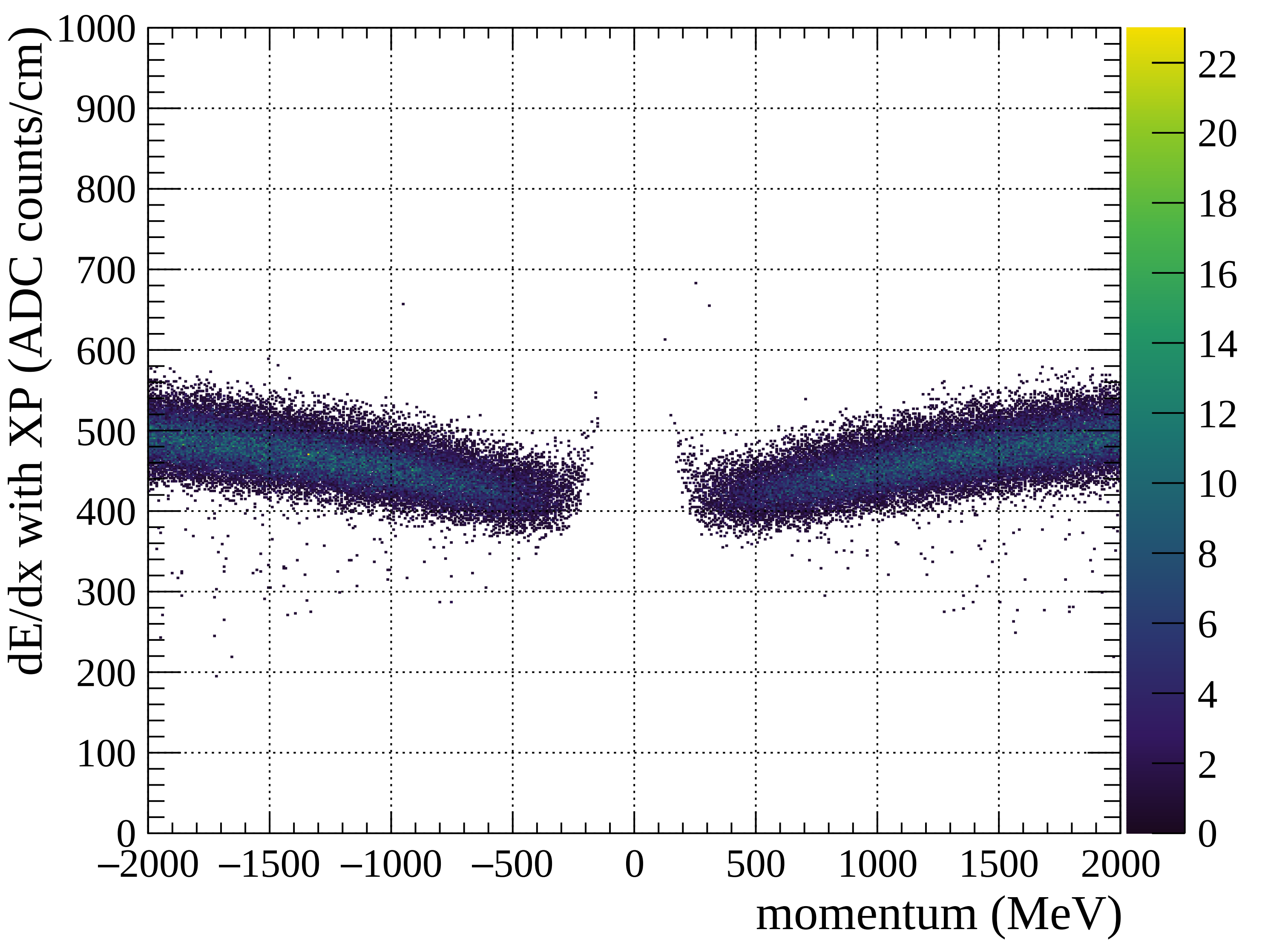


dE/dx resolution (%)
30
25
20
15
10
5
WF
XP
−2000
−1500
−1000
−500
0
500
1000
1500
2000
Momentum (MeV/c)

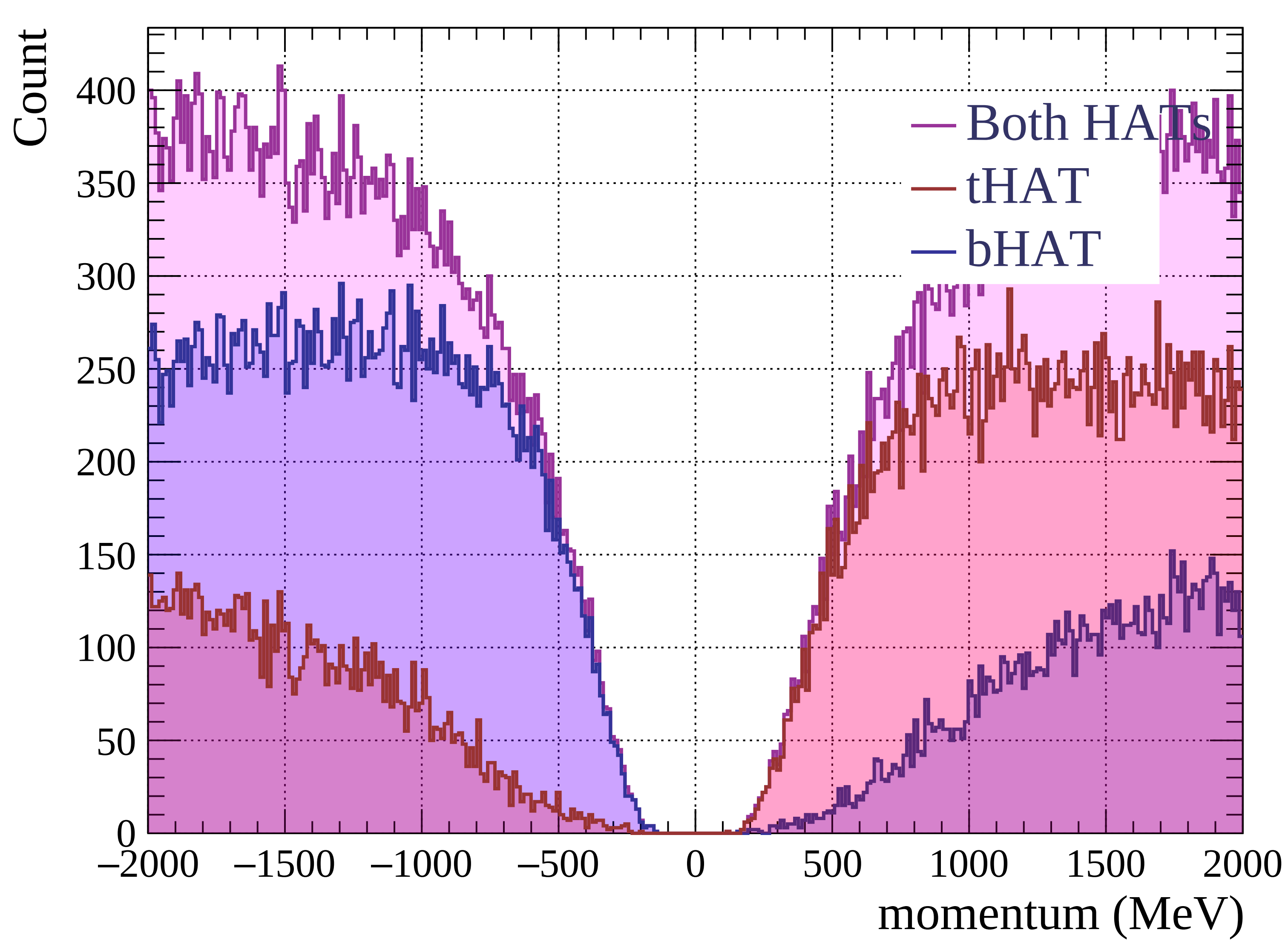


WF
XP
Mean (ADC counts/cm)
800
700
600
500
400
300
200
−2000
−1500
−1000
−500
0
500
1000
1500
2000
Momentum (MeV/c)

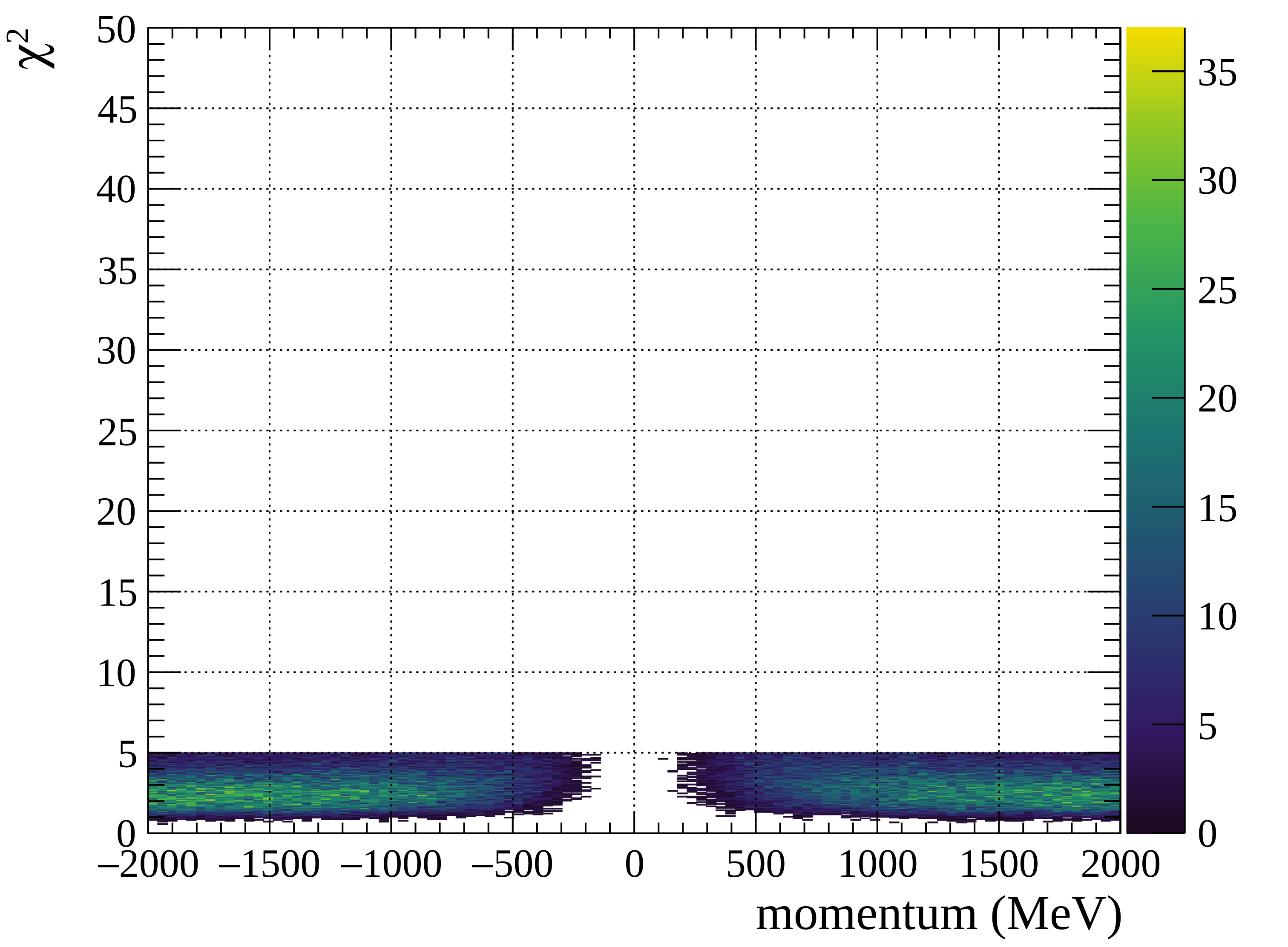

dE/dx with WF (ADC counts/cm)
momentum (MeV)
1000
900
800
700
600
500
400
300
200
100
0
−2000
−1500
−1000
−500
0
500
1000
1500
2000
20
18
16
14
12
10
8
6
4
2
0



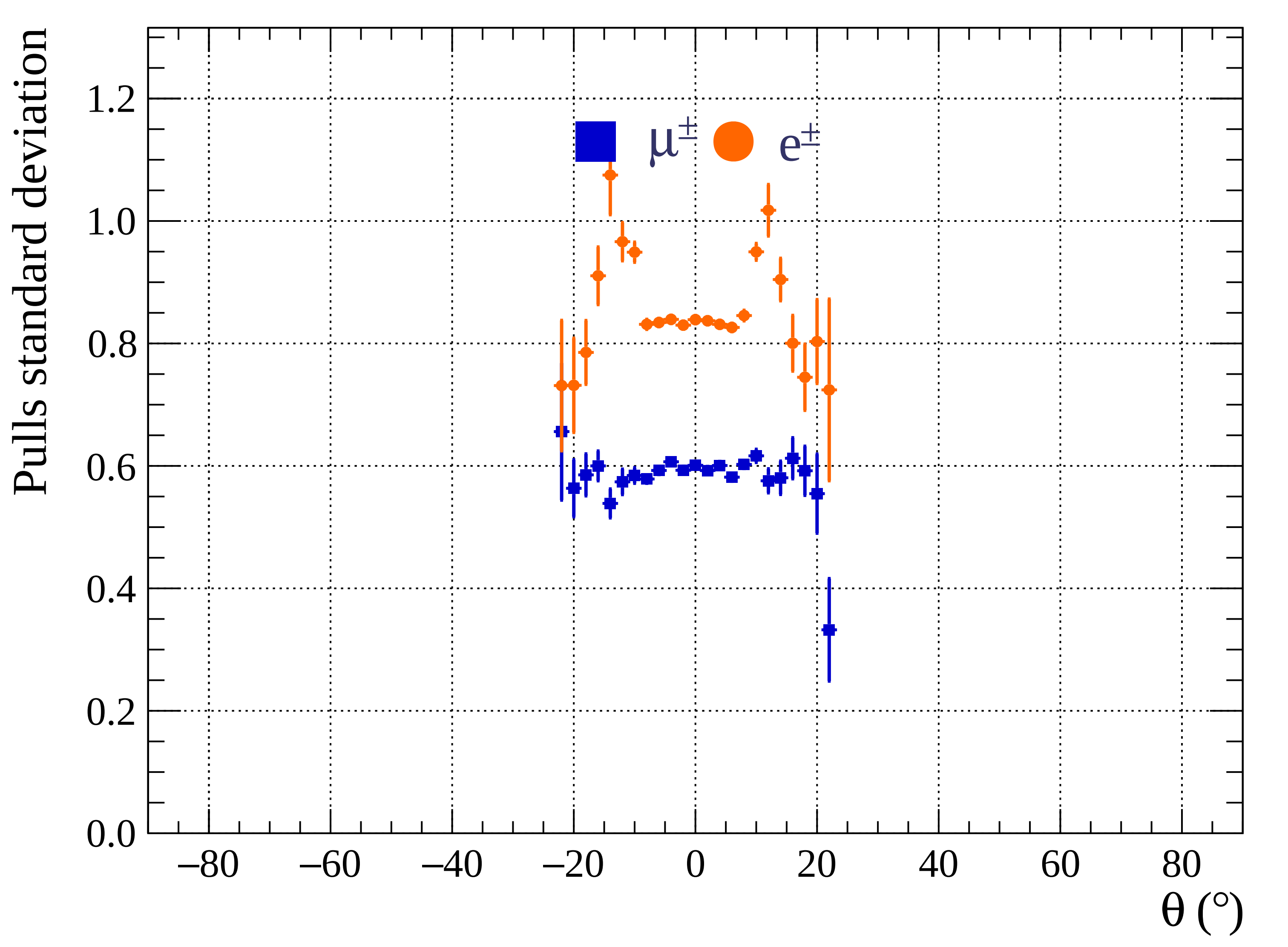
dE/dx with XP (ADC counts/cm)
momentum (MeV)
1000
900
800
700
600
500
400
300
200
100
0
−2000
−1500
−1000
−500
0
500
1000
1500
2000
22
20
18
16
14
12
10
8
6
4
2
0



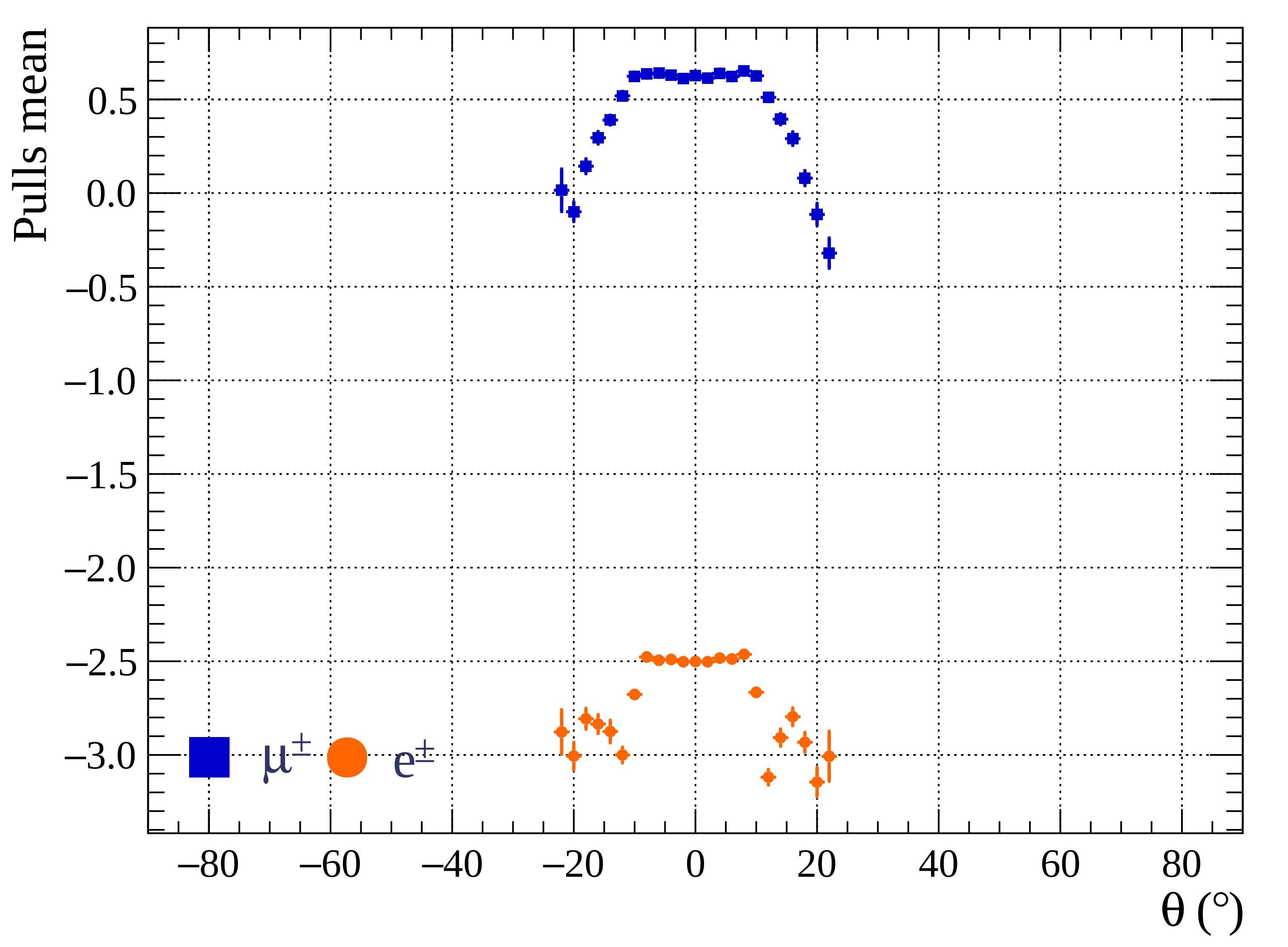
Count
400
350
300
250
200
150
100
50
0
−2000
−1500
−1000
−500
0
500
1000
1500
2000
momentum (MeV)
Both HATs
tHAT
bHAT



χ^2
50
45
40
35
30
25
20
15
10
5
0
−2000
−1500
−1000
−500
0
500
1000
1500
2000
momentum (MeV)
35
30
25
20
15
10
5
0



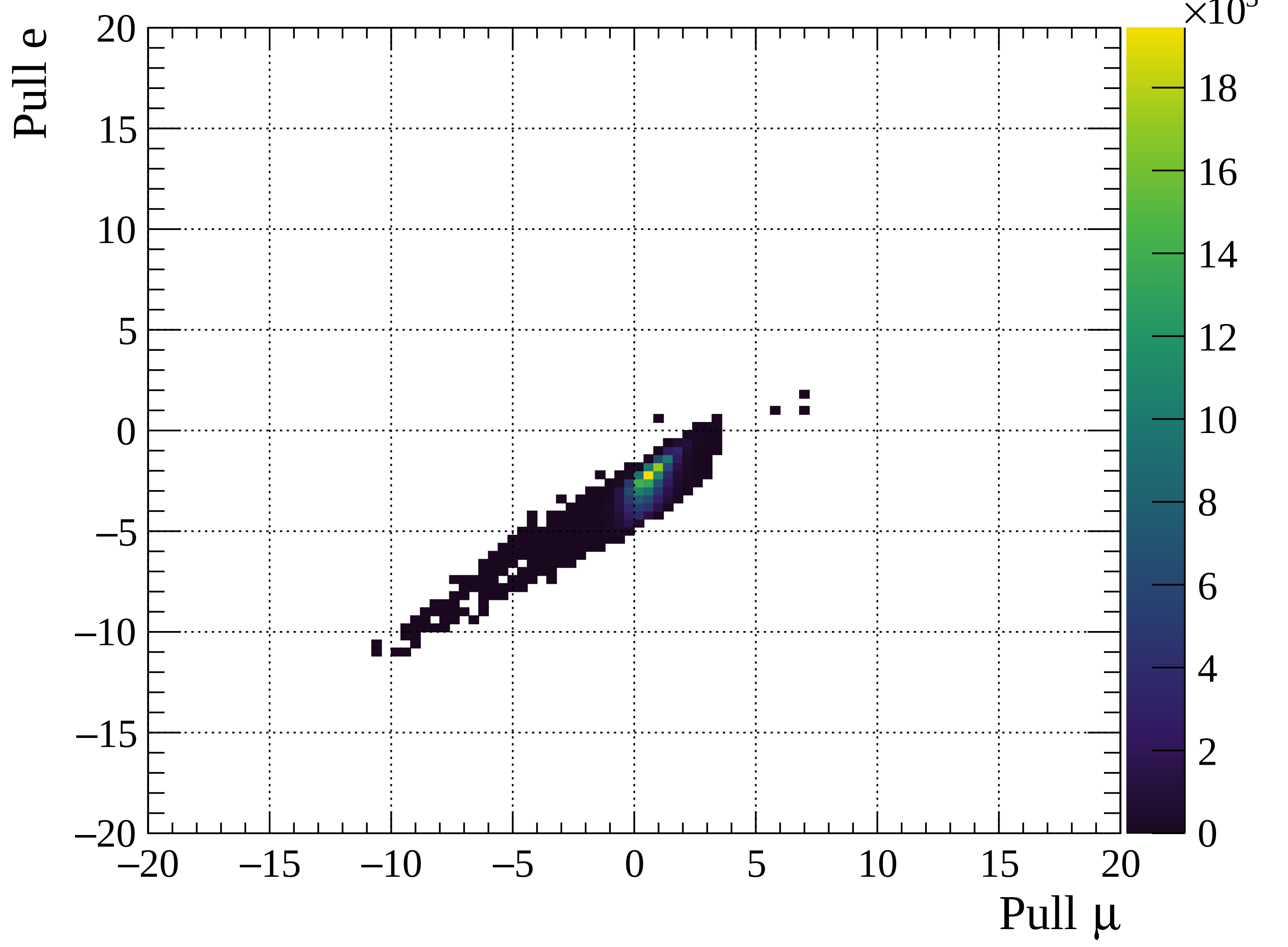
Pulls standard deviation
0.0
0.2
0.4
0.6
0.8
1.0
1.2
μ±
e±
−80
−60
−40
−20
0
20
40
60
80
θ (°)



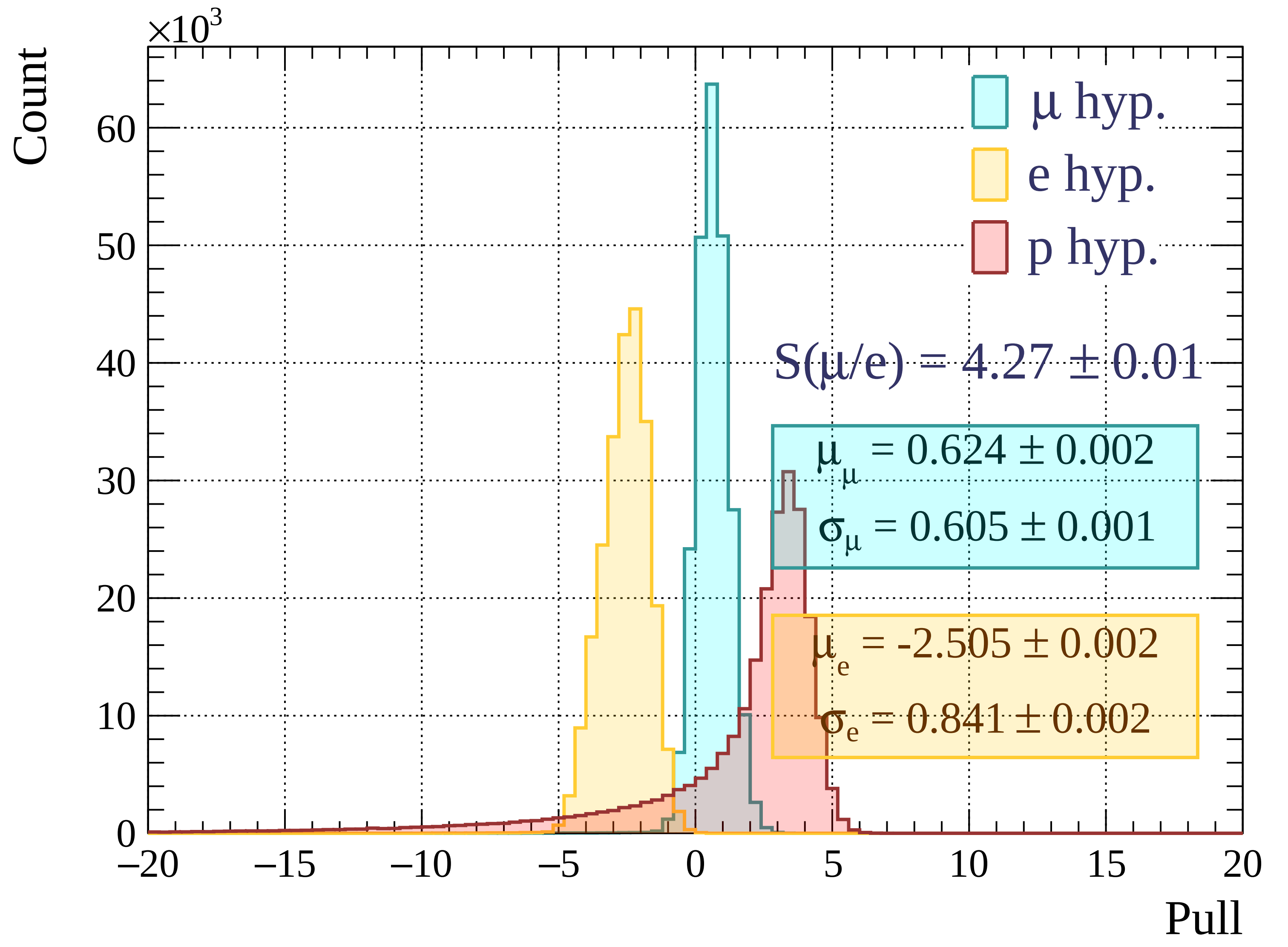
Pulls mean
0.5
0.0
−0.5
−1.0
−1.5
−2.0
−2.5
−3.0
μ±
e±
−80
−60
−40
−20
0
20
40
60
80
θ (°)

$\mu^{\pm}$ $e^{\pm}$

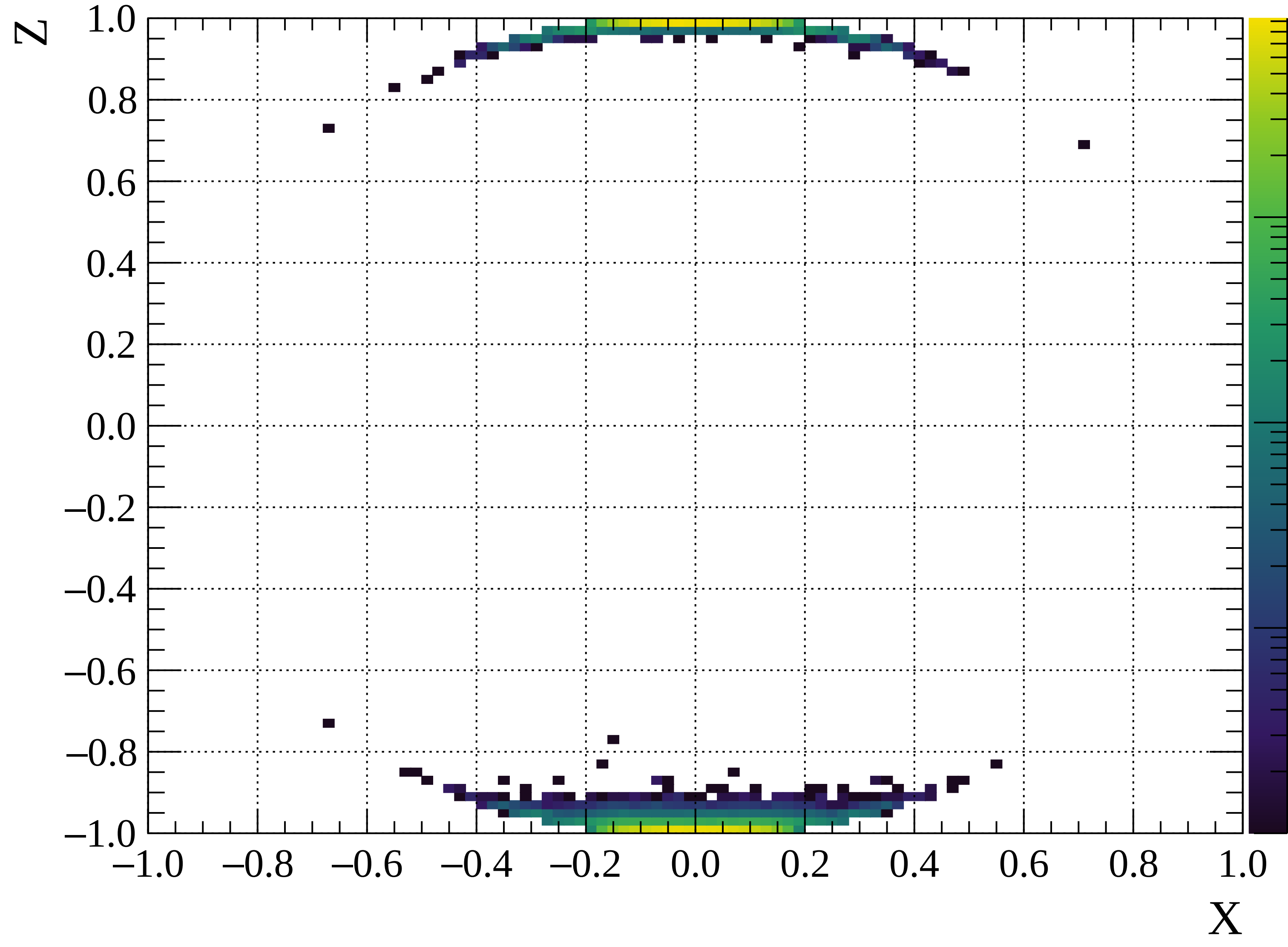
$\mu^{\pm}$
$e^{\pm}$



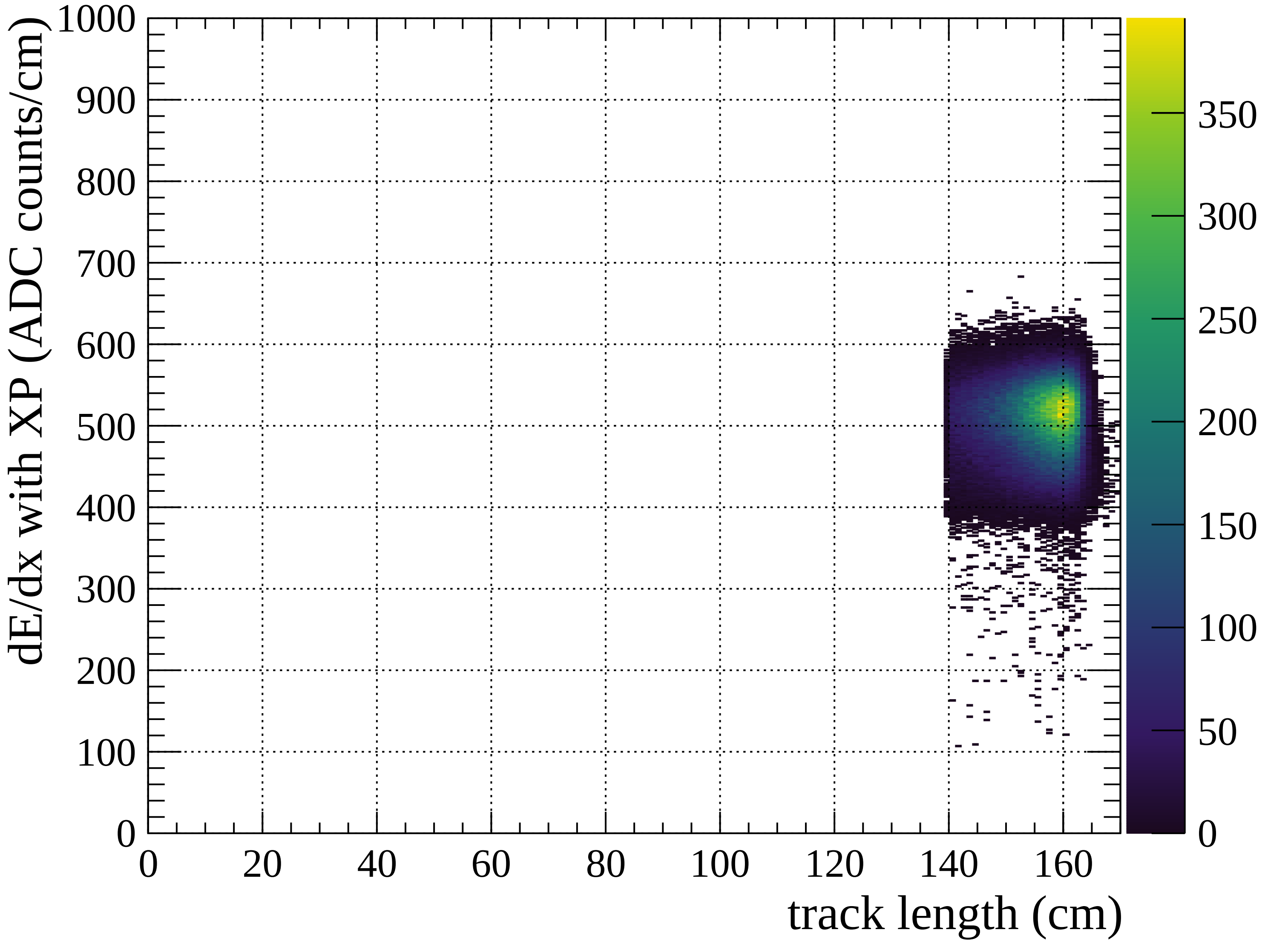
Pull e
Pull μ
×10³
20
15
10
5
0
−5
−10
−15
−20
18
16
14
12
10
8
6
4
2
0



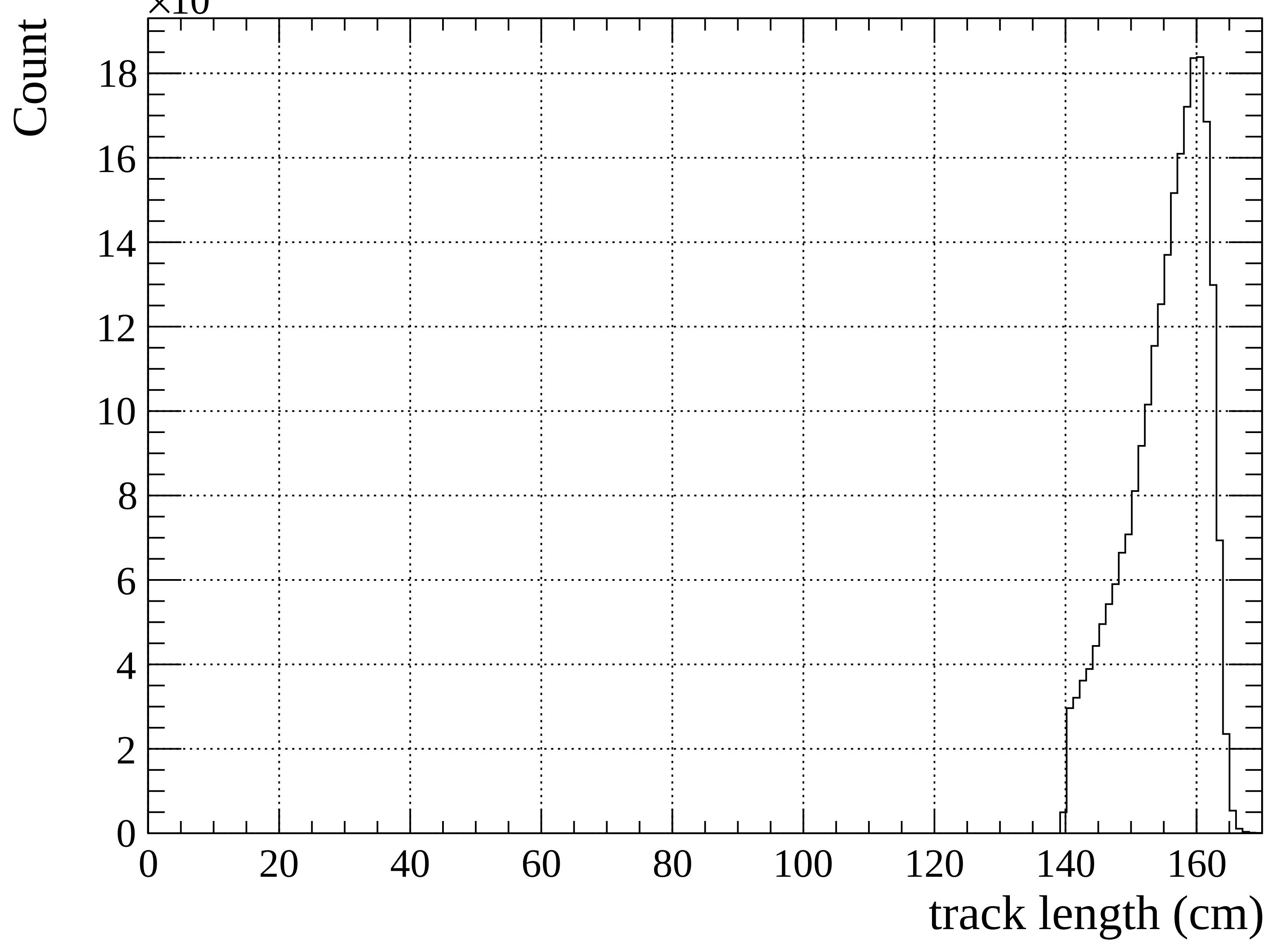
Count
×10³
Pull
μ hyp.
e hyp.
p hyp.
S(μ/e) = 4.27 ± 0.01
μ_μ = 0.624 ± 0.002
σ_μ = 0.605 ± 0.001
μ_e = -2.505 ± 0.002
σ_e = 0.841 ± 0.002

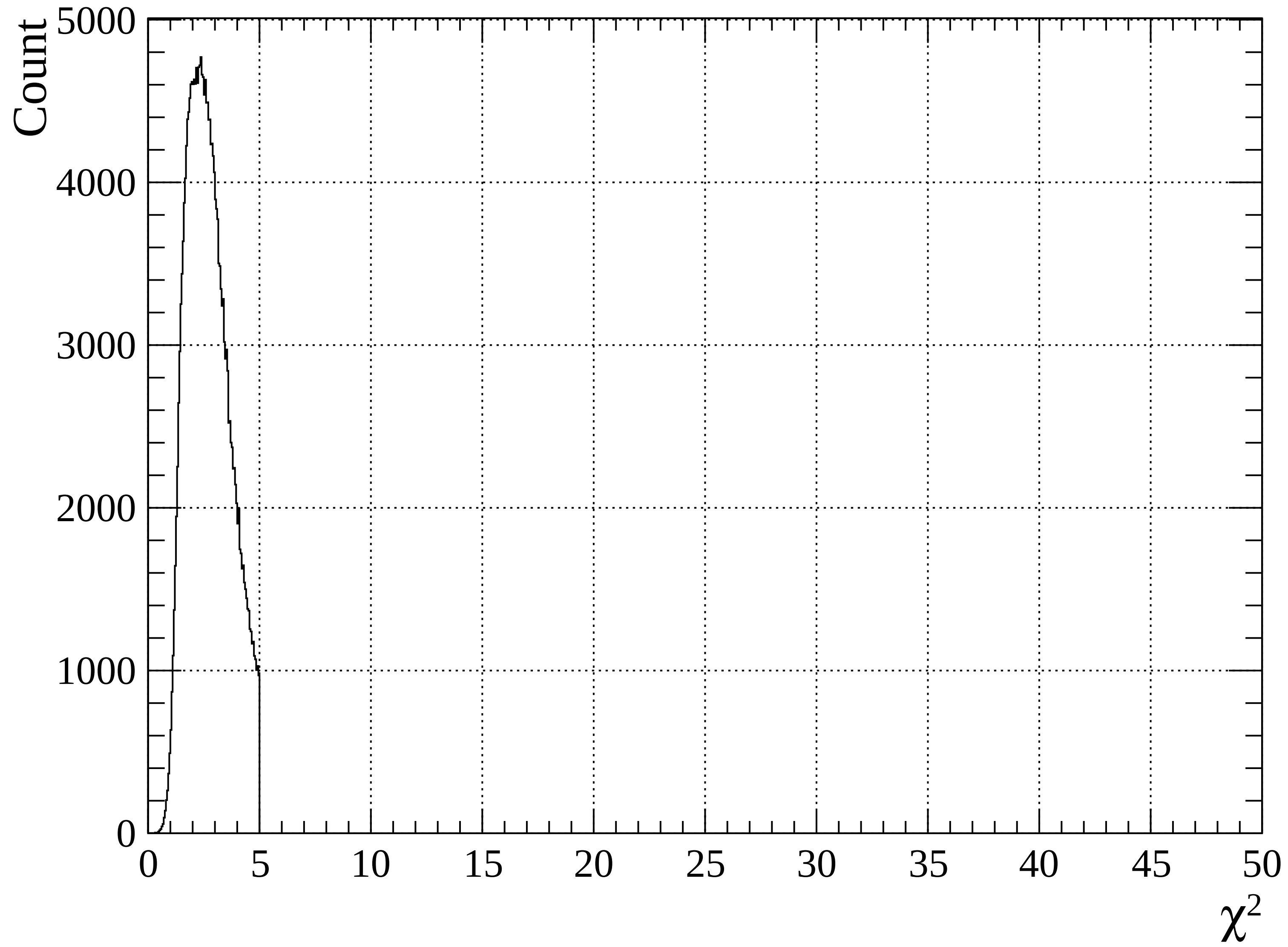


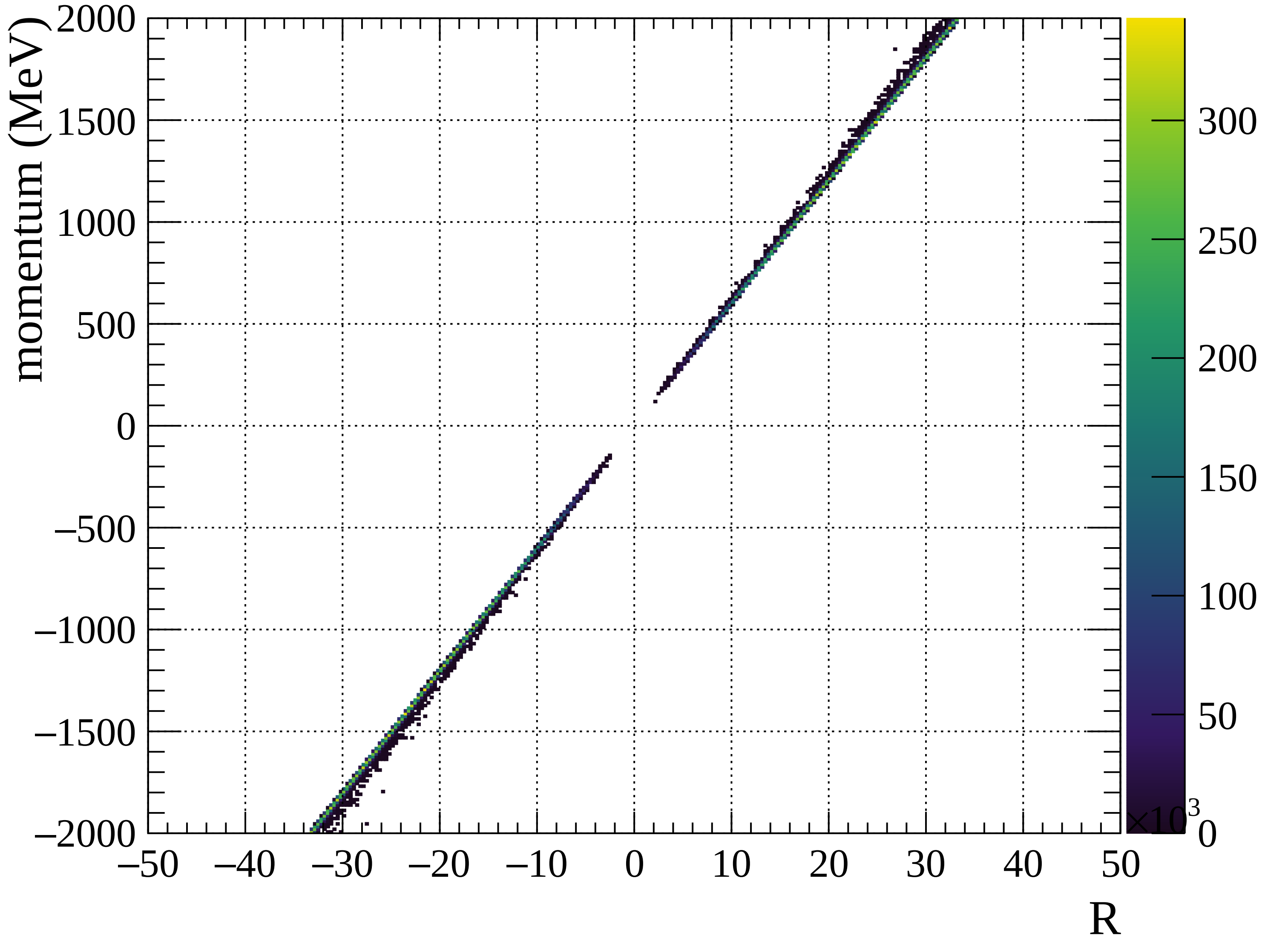
Z
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
−1.0
−0.8
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
X



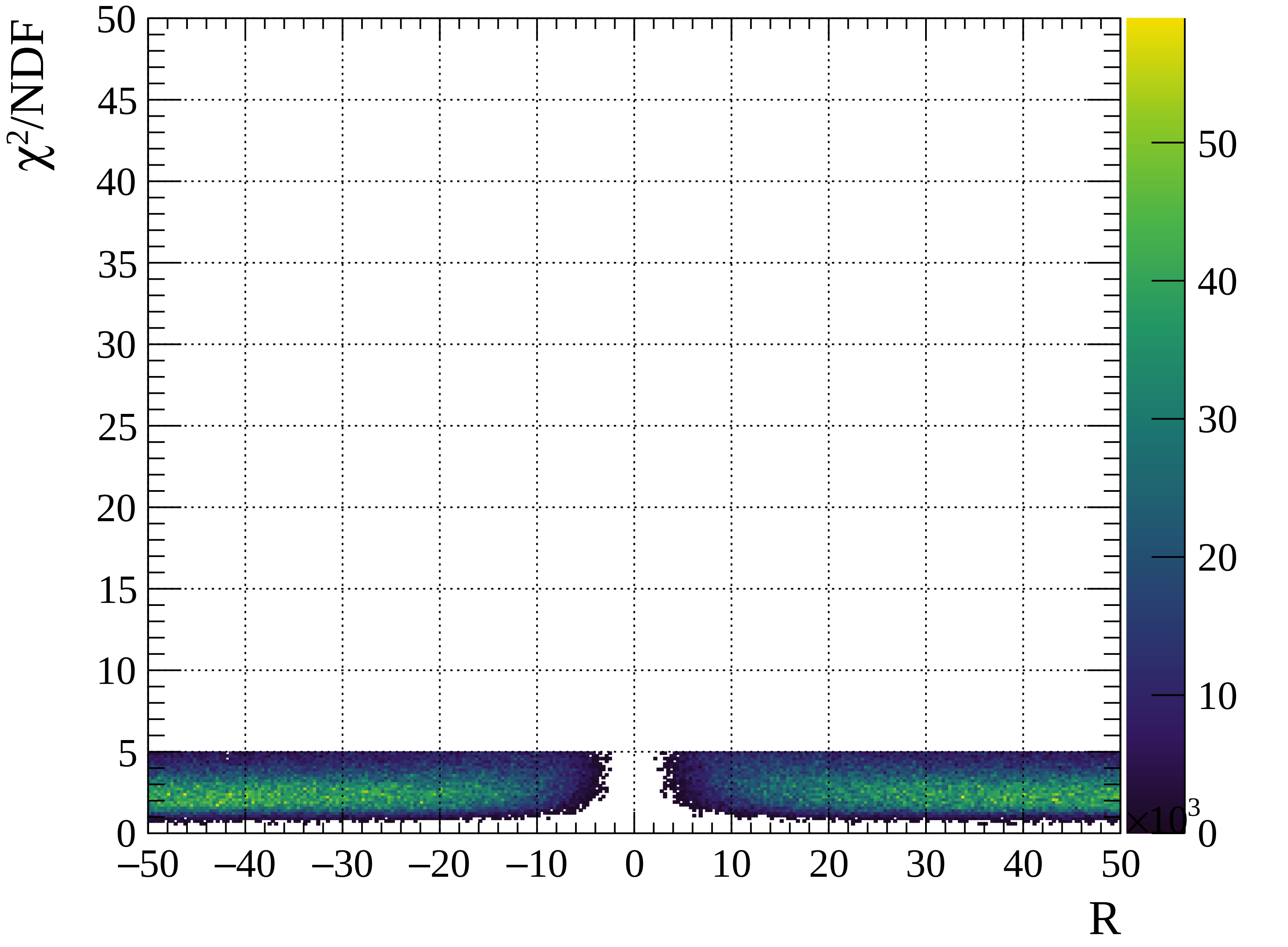
dE/dx with XP (ADC counts/cm)
track length (cm)
0
100
200
300
400
500
600
700
800
900
1000
0
20
40
60
80
100
120
140
160
0
50
100
150
200
250
300
350


Count
×10
18
16
14
12
10
8
6
4
2
0
0
20
40
60
80
100
120
140
160
track length (cm)

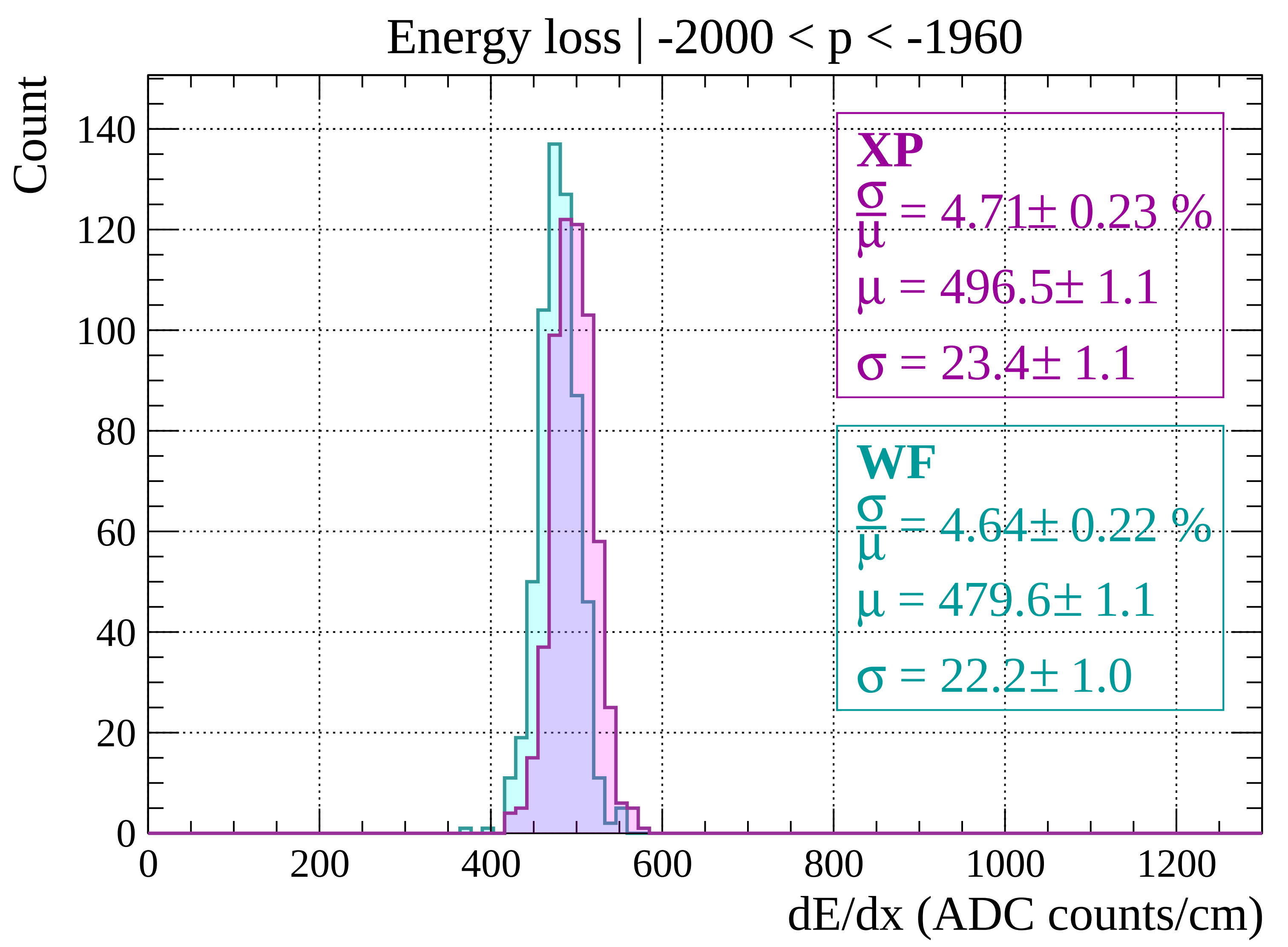

Count
5000
4000
3000
2000
1000
0
0
5
10
15
20
25
30
35
40
45
50
χ^2



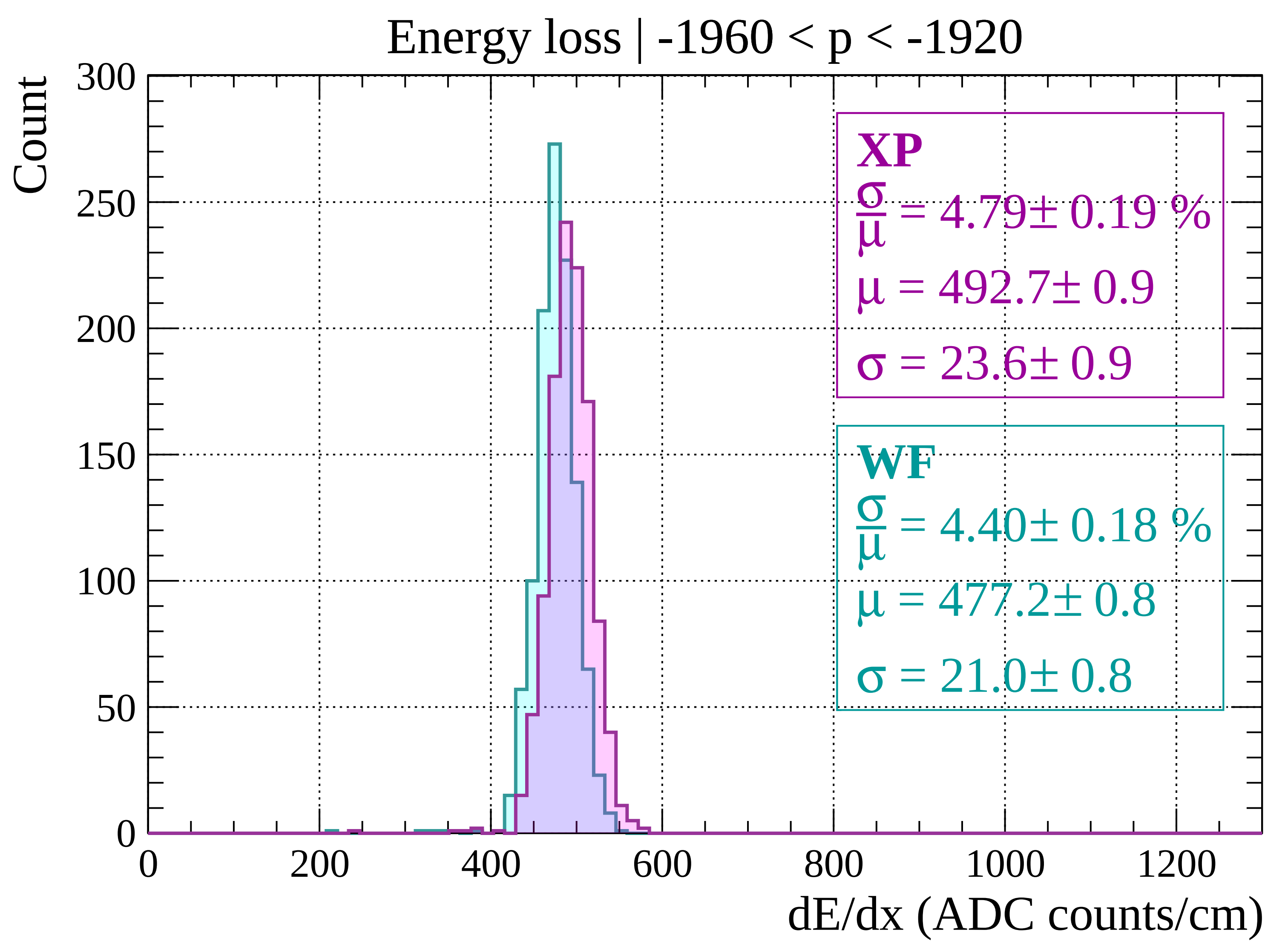
momentum (MeV)
2000
1500
1000
500
0
−500
−1000
−1500
−2000
−50
−40
−30
−20
−10
0
10
20
30
40
50
R
×10^3
300
250
200
150
100
50
0



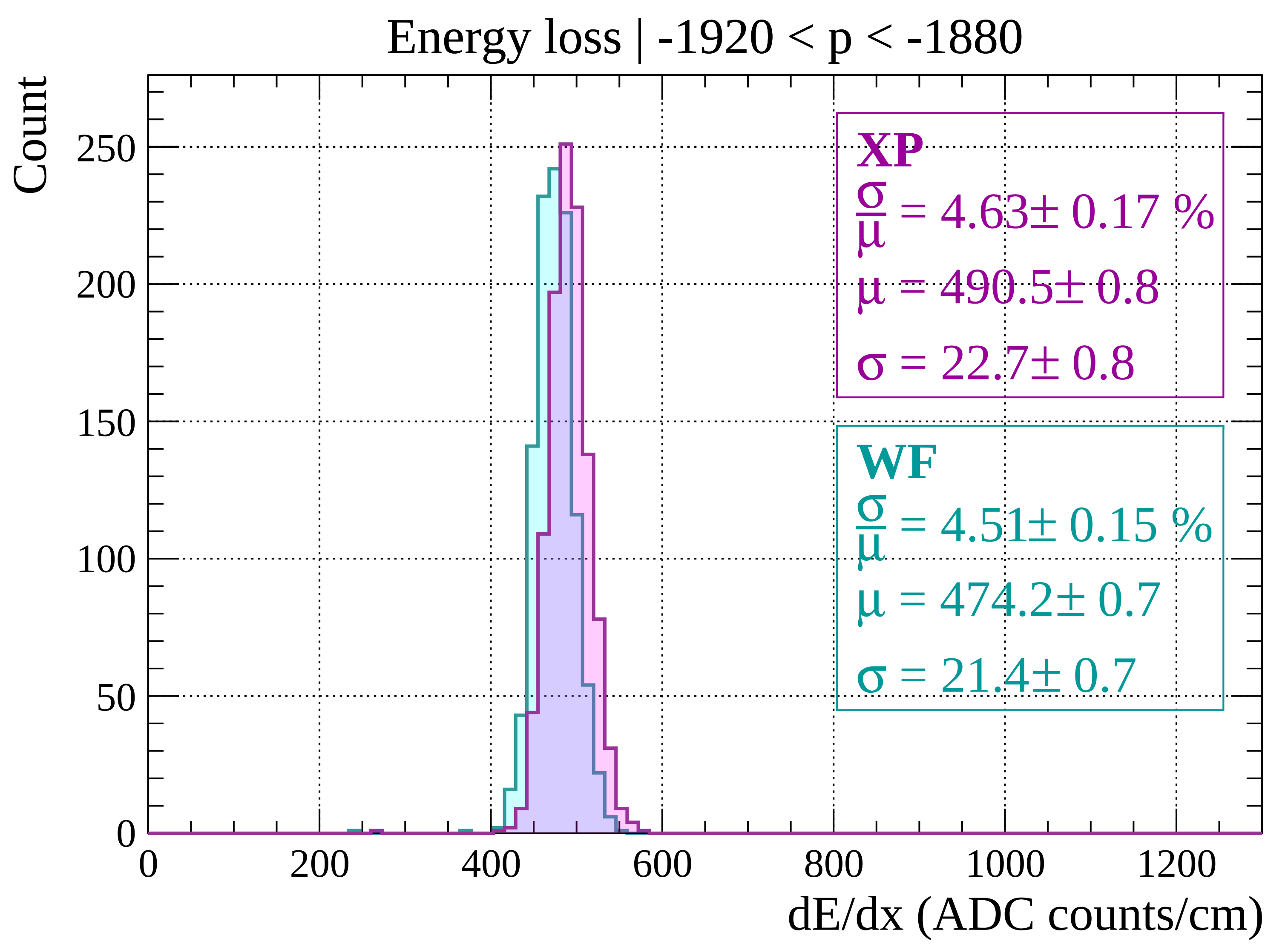
χ^2/NDF
50
45
40
35
30
25
20
15
10
5
0
−50
−40
−30
−20
−10
0
10
20
30
40
50
R
×10³
50
40
30
20
10
0



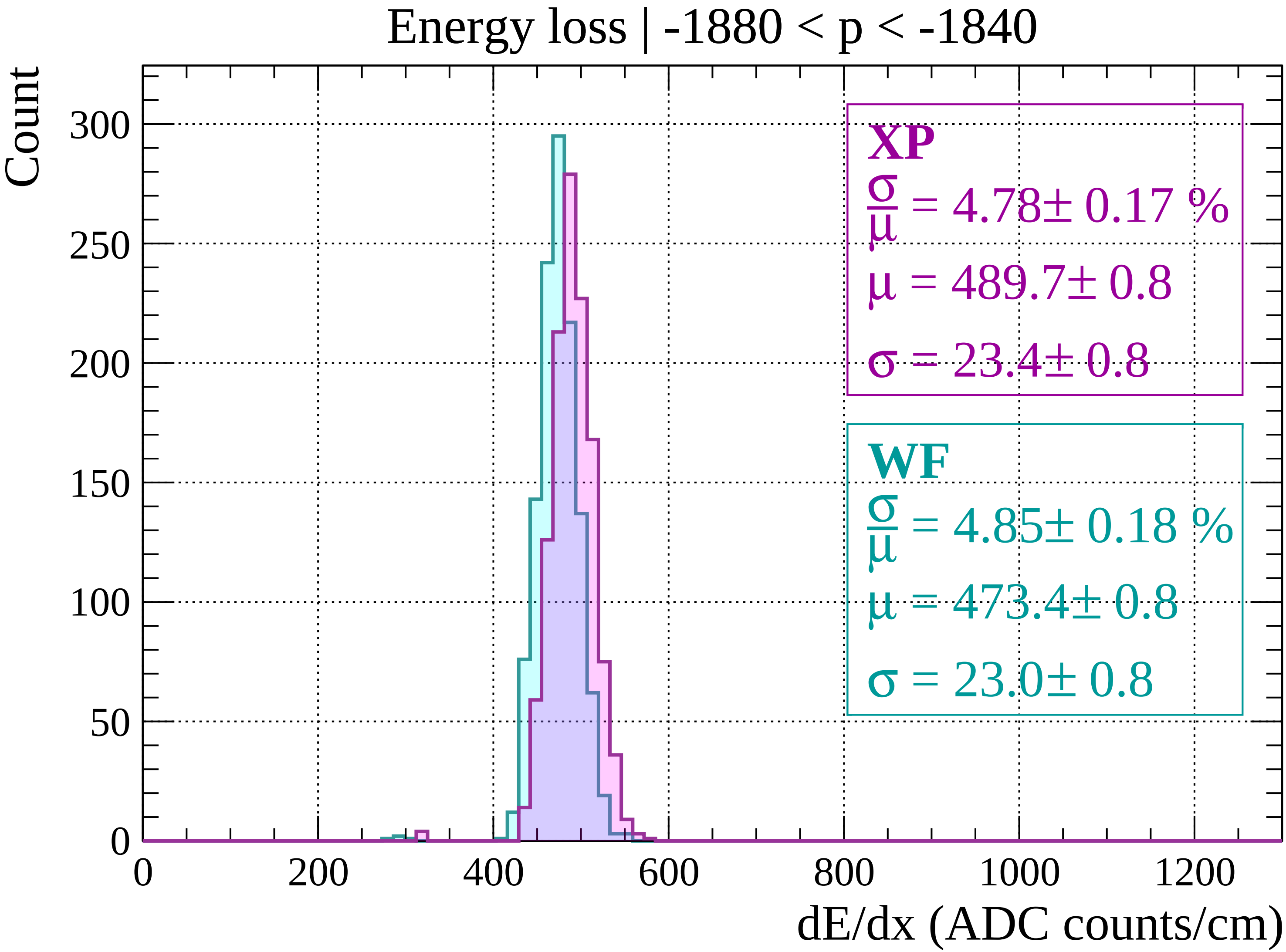
Energy loss | -2000 < p < -1960
Count
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 4.71± 0.23 %
μ = 496.5± 1.1
σ = 23.4± 1.1
WF
σ/μ = 4.64± 0.22 %
μ = 479.6± 1.1
σ = 22.2± 1.0



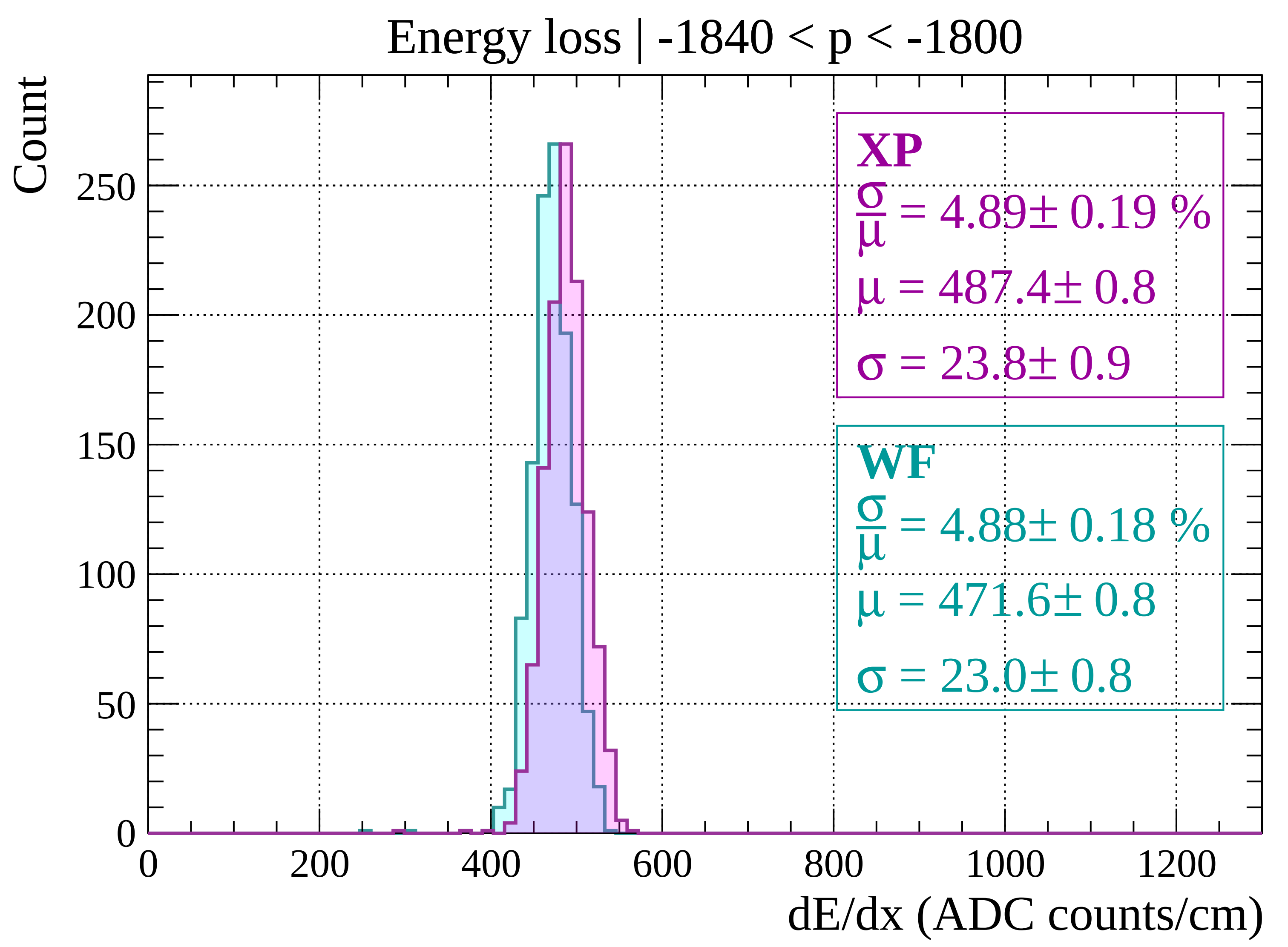
Energy loss | -1960 < p < -1920
Count
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 4.79± 0.19 %
μ = 492.7± 0.9
σ = 23.6± 0.9
WF
$\frac{\sigma}{\mu}$ = 4.40± 0.18 %
μ = 477.2± 0.8
σ = 21.0± 0.8



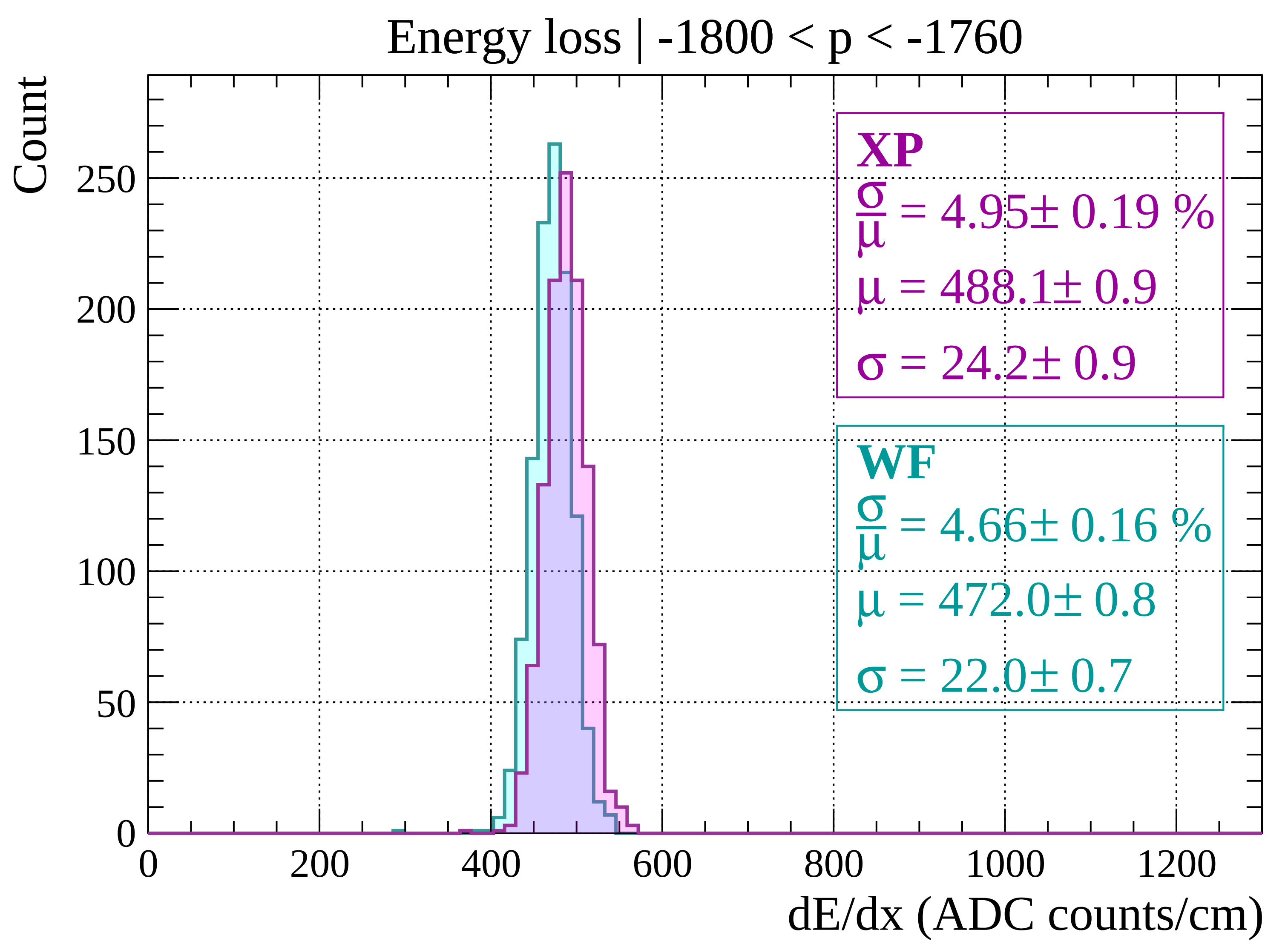
Energy loss | -1920 < p < -1880
Count
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 4.63± 0.17 %
μ = 490.5± 0.8
σ = 22.7± 0.8
WF
σ/μ = 4.51± 0.15 %
μ = 474.2± 0.7
σ = 21.4± 0.7

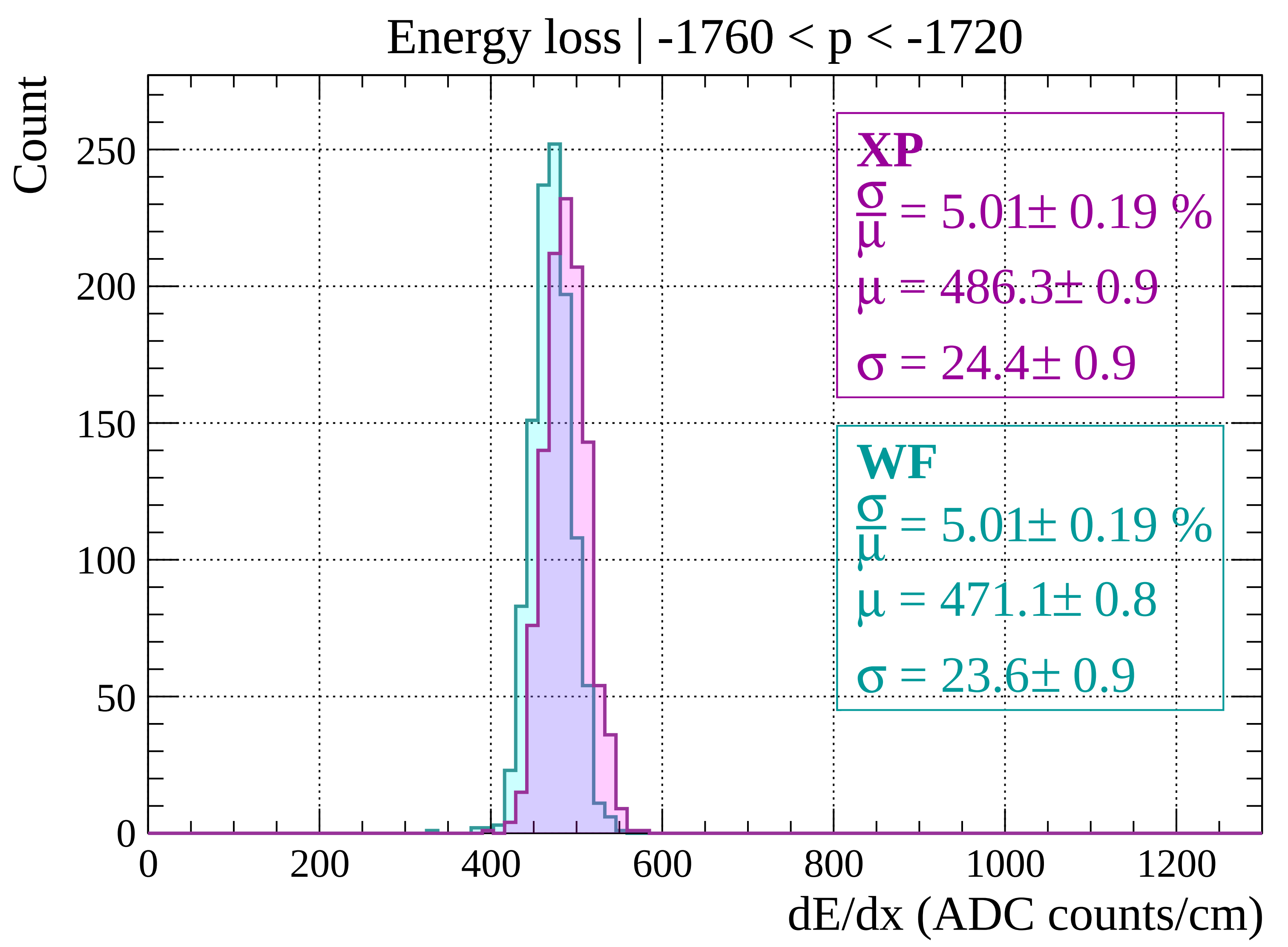


Energy loss | -1880 < p < -1840
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP
σ/μ = 4.78± 0.17 %
μ = 489.7± 0.8
σ = 23.4± 0.8
WF
σ/μ = 4.85± 0.18 %
μ = 473.4± 0.8
σ = 23.0± 0.8

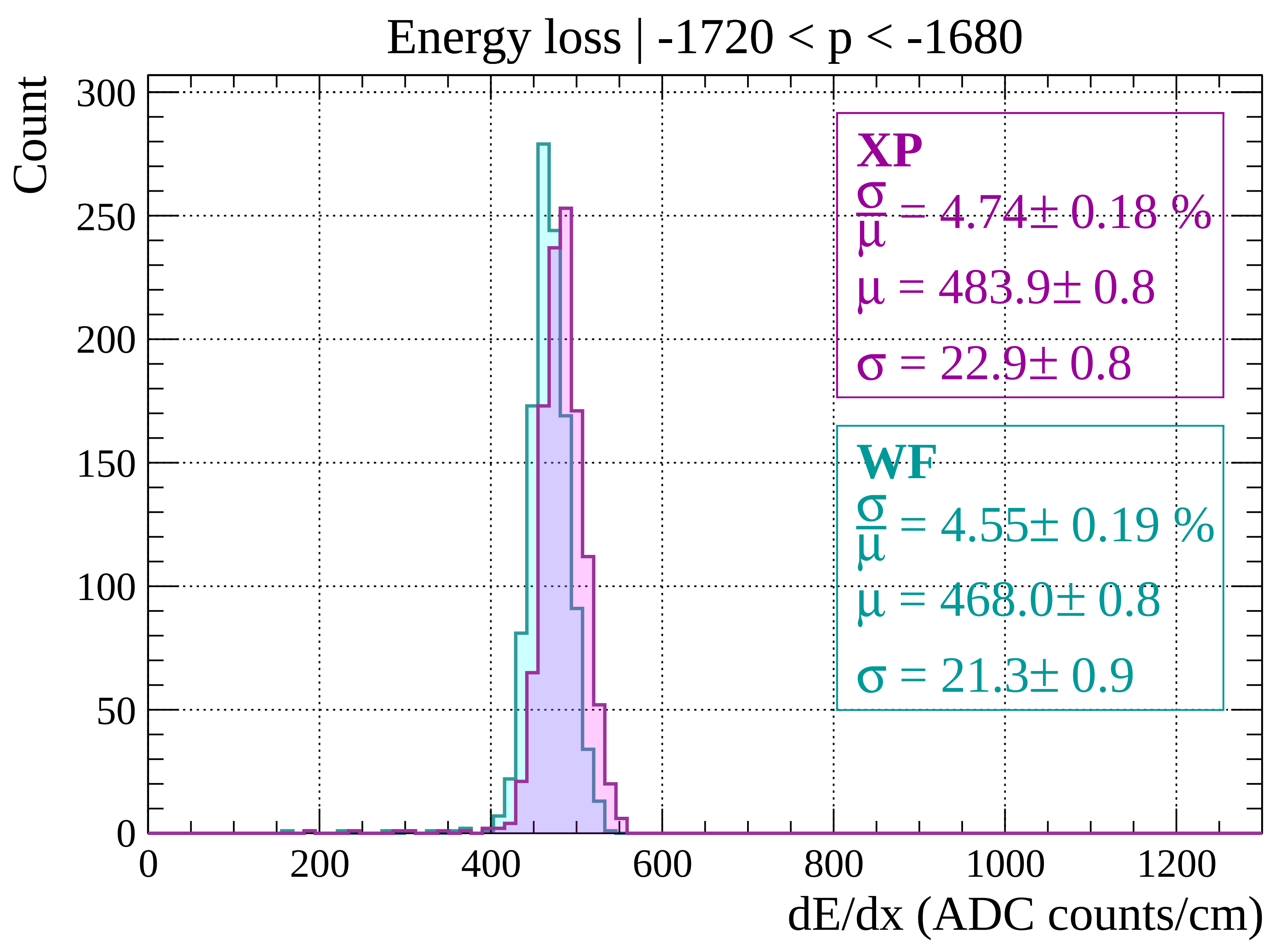


Energy loss | -1840 < p < -1800
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 4.89± 0.19 %
μ = 487.4± 0.8
σ = 23.8± 0.9
WF
$\frac{\sigma}{\mu}$ = 4.88± 0.18 %
μ = 471.6± 0.8
σ = 23.0± 0.8

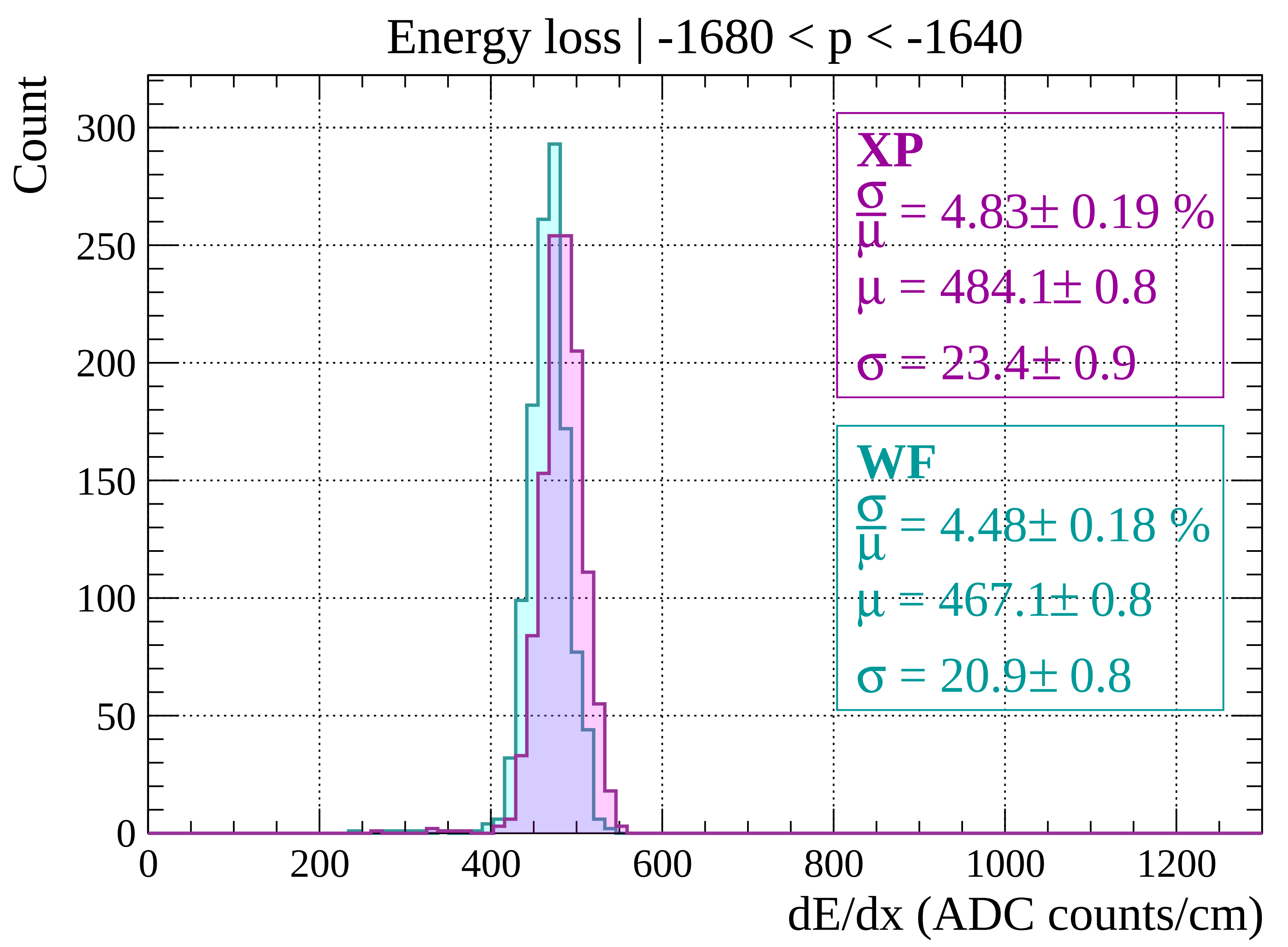

Energy loss | -1800 < p < -1760
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP
σ/μ = 4.95± 0.19 %
μ = 488.1± 0.9
σ = 24.2± 0.9
WF
σ/μ = 4.66± 0.16 %
μ = 472.0± 0.8
σ = 22.0± 0.7



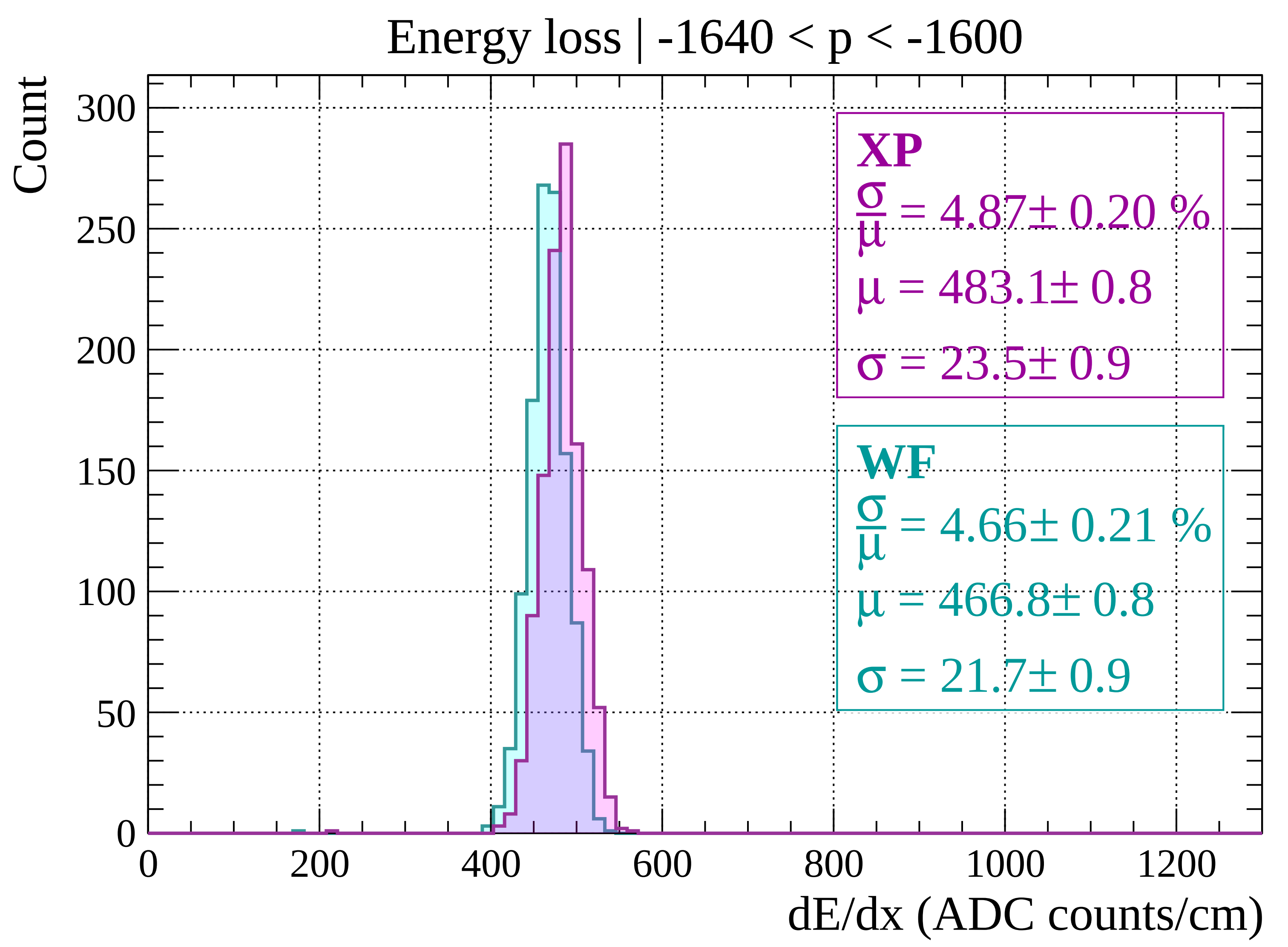
Energy loss | -1760 < p < -1720
Count
0
50
100
150
200
250
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 5.01± 0.19 %
μ = 486.3± 0.9
σ = 24.4± 0.9
WF
σ/μ = 5.01± 0.19 %
μ = 471.1± 0.8
σ = 23.6± 0.9

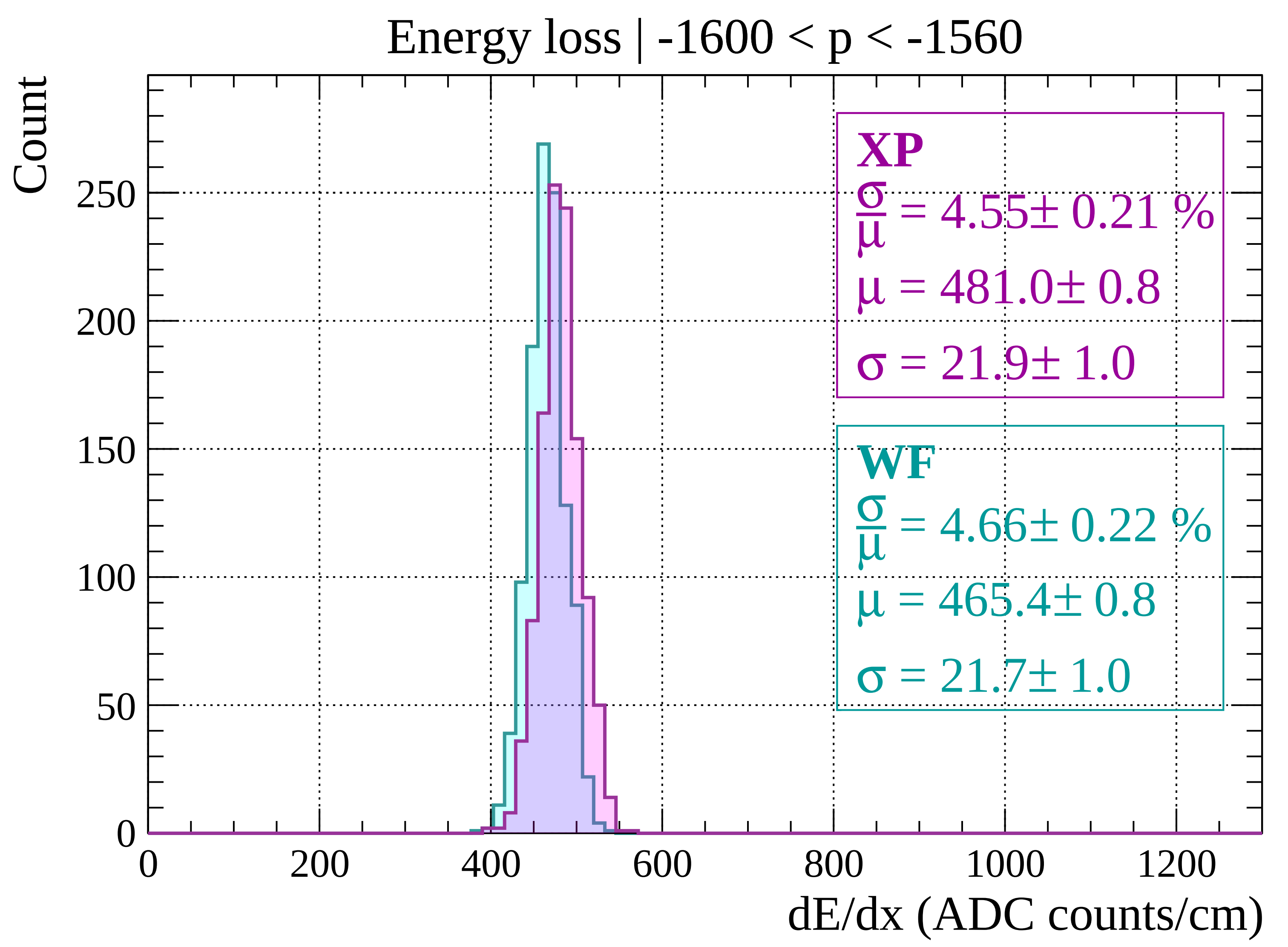


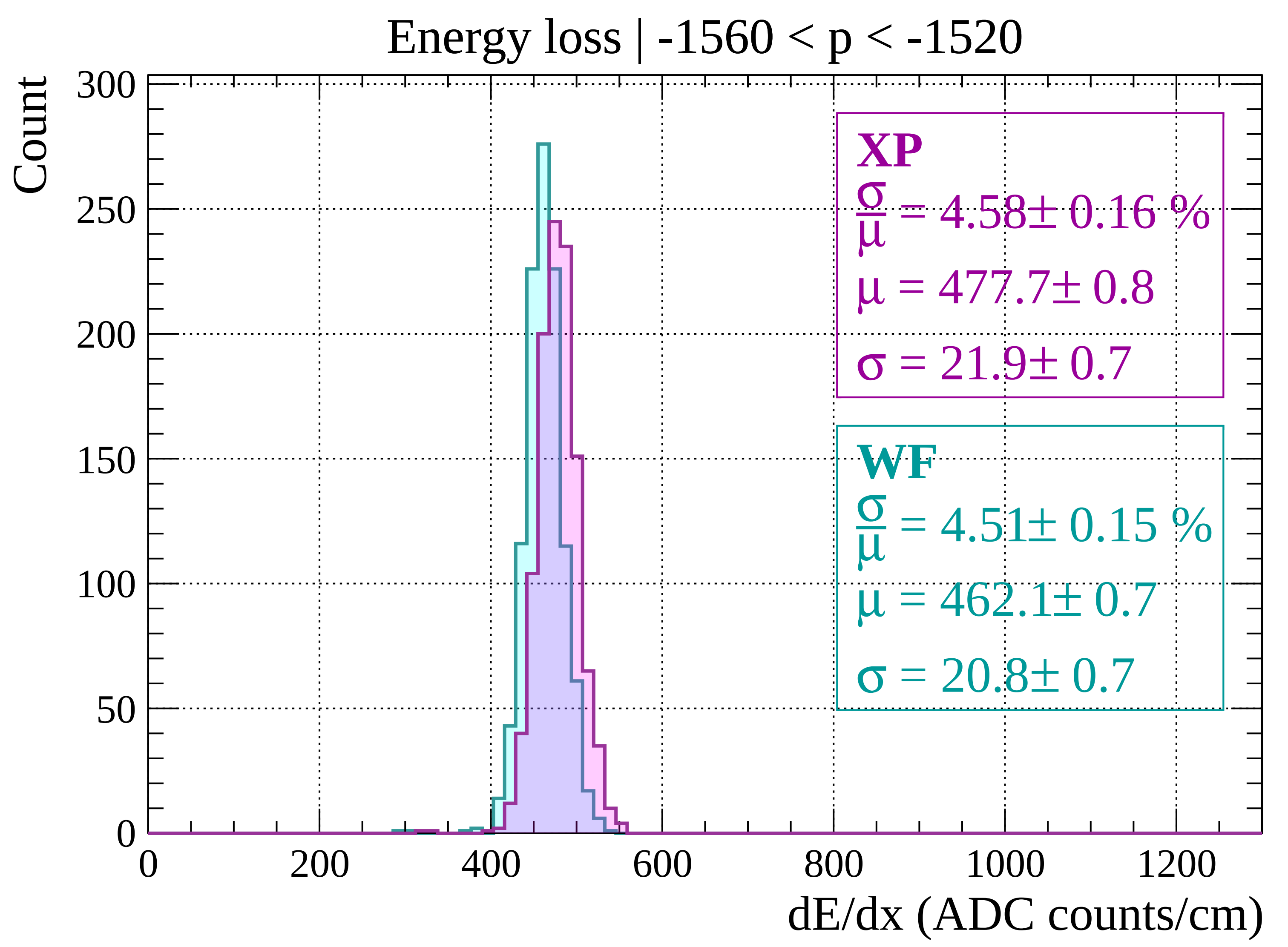
Energy loss | -1720 < p < -1680
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
50
100
150
200
250
300
XP
$\frac{\sigma}{\mu}$ = 4.74± 0.18 %
μ = 483.9± 0.8
σ = 22.9± 0.8
WF
$\frac{\sigma}{\mu}$ = 4.55± 0.19 %
μ = 468.0± 0.8
σ = 21.3± 0.9



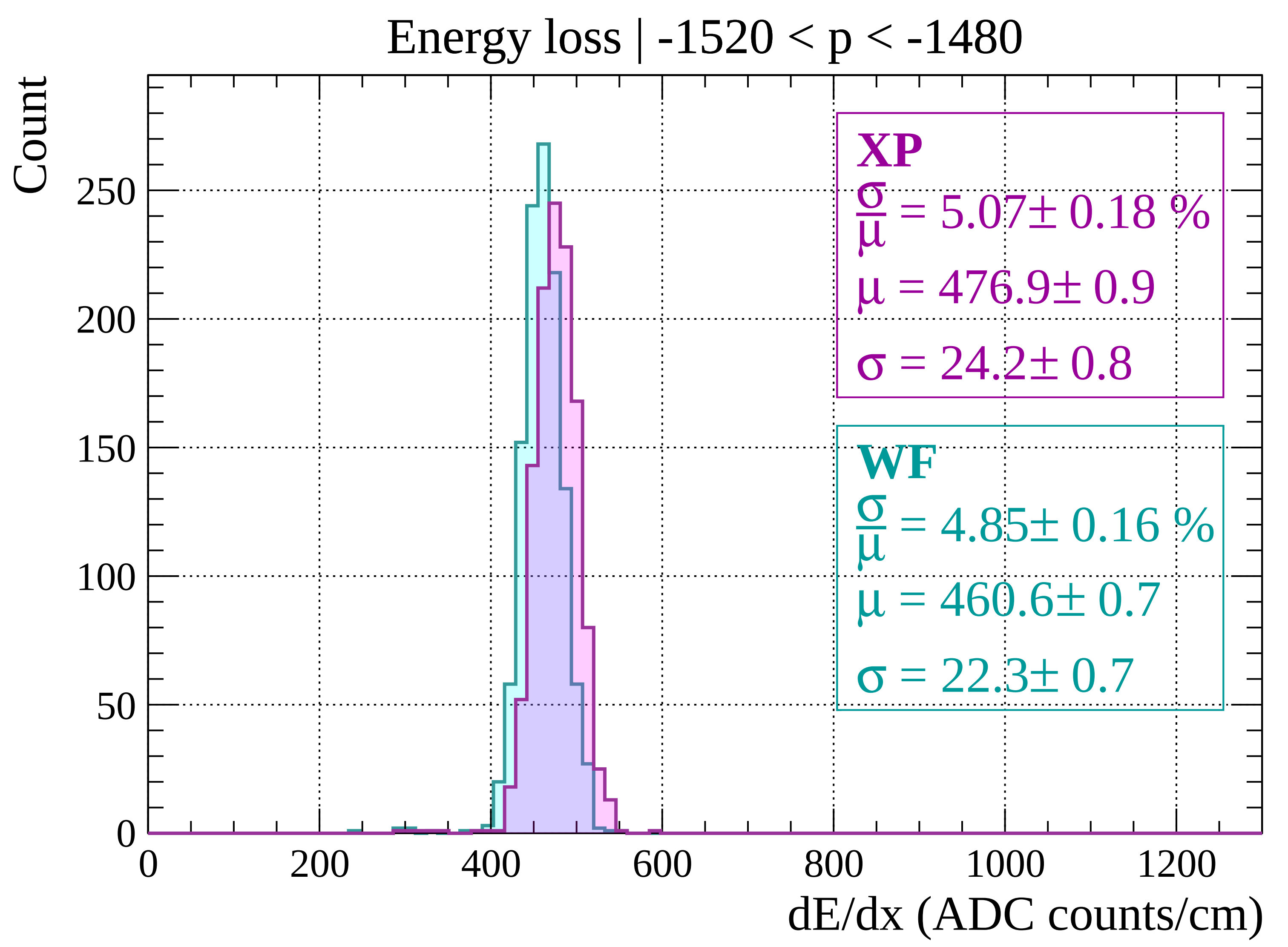
Energy loss | -1680 < p < -1640
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP
σ/μ = 4.83± 0.19 %
μ = 484.1± 0.8
σ = 23.4± 0.9
WF
σ/μ = 4.48± 0.18 %
μ = 467.1± 0.8
σ = 20.9± 0.8


Energy loss | -1640 < p < -1600
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 4.87± 0.20 %
μ = 483.1± 0.8
σ = 23.5± 0.9
WF
$\frac{\sigma}{\mu}$ = 4.66± 0.21 %
μ = 466.8± 0.8
σ = 21.7± 0.9

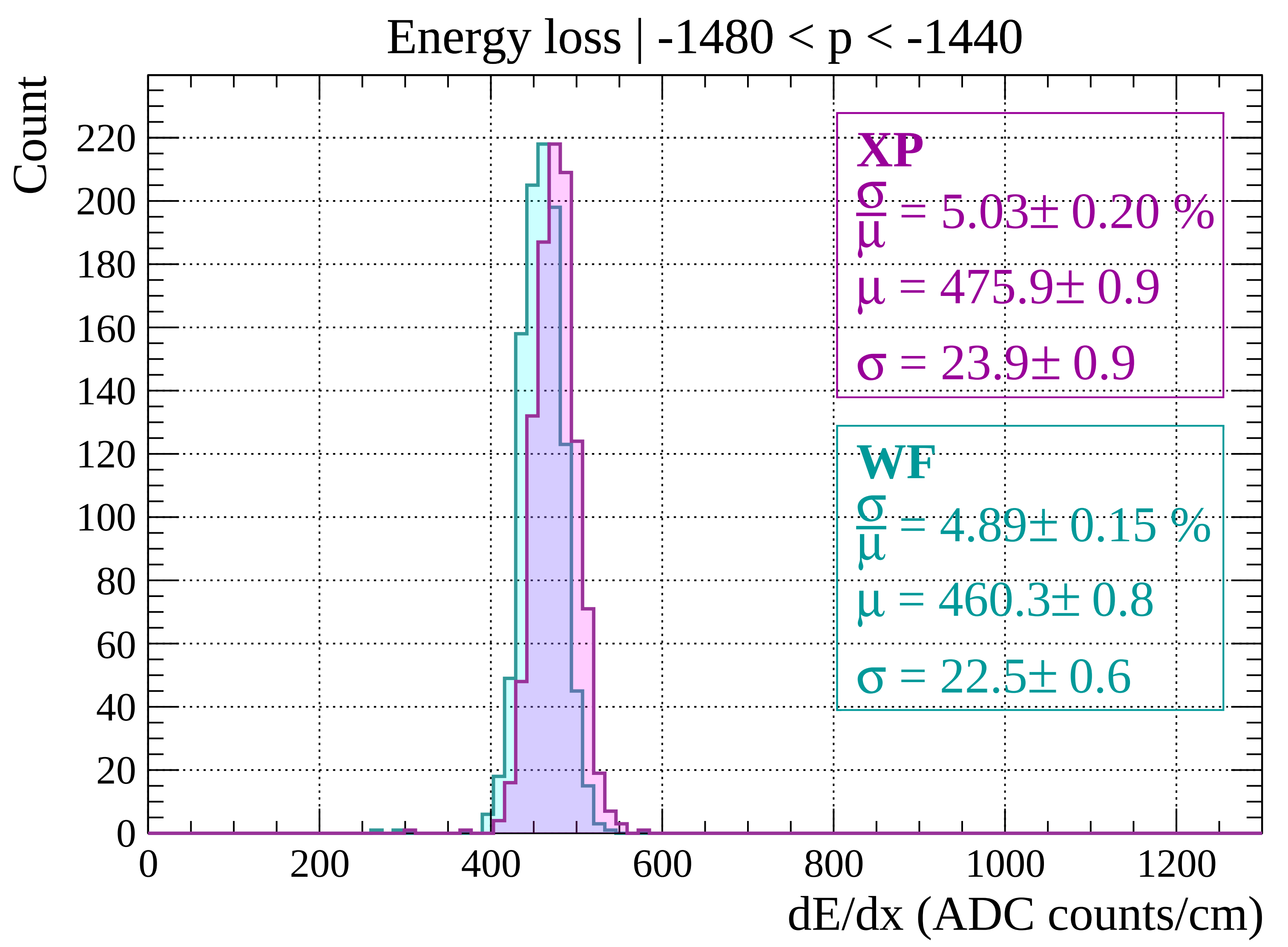

Energy loss | -1600 < p < -1560
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
50
100
150
200
250
XP
$\frac{\sigma}{\mu}$ = 4.55± 0.21 %
μ = 481.0± 0.8
σ = 21.9± 1.0
WF
$\frac{\sigma}{\mu}$ = 4.66± 0.22 %
μ = 465.4± 0.8
σ = 21.7± 1.0



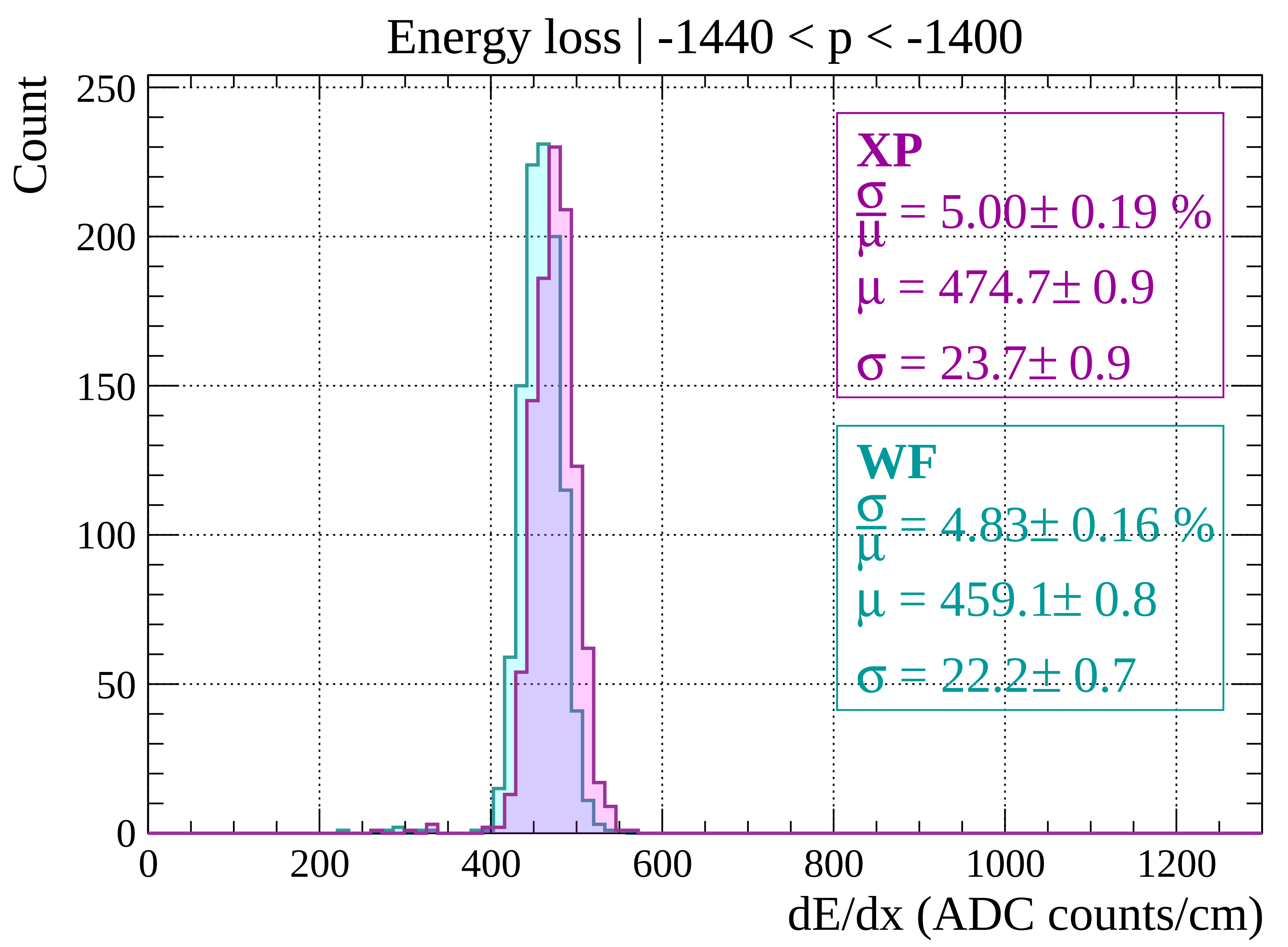
Energy loss | -1560 < p < -1520
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 4.58± 0.16 %
μ = 477.7± 0.8
σ = 21.9± 0.7
WF
$\frac{\sigma}{\mu}$ = 4.51± 0.15 %
μ = 462.1± 0.7
σ = 20.8± 0.7

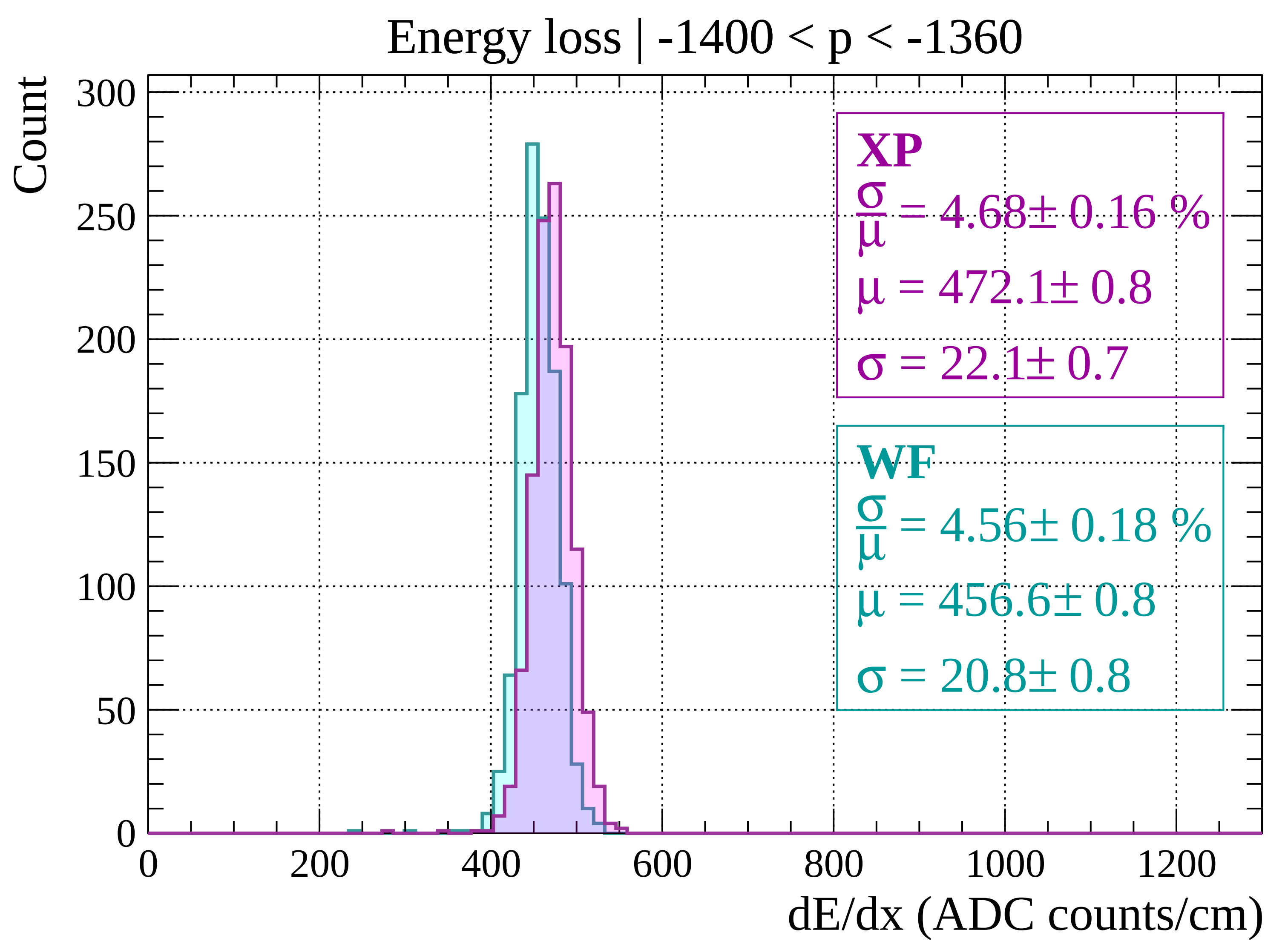


Energy loss | -1520 < p < -1480
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP
σ/μ = 5.07± 0.18 %
μ = 476.9± 0.9
σ = 24.2± 0.8
WF
σ/μ = 4.85± 0.16 %
μ = 460.6± 0.7
σ = 22.3± 0.7

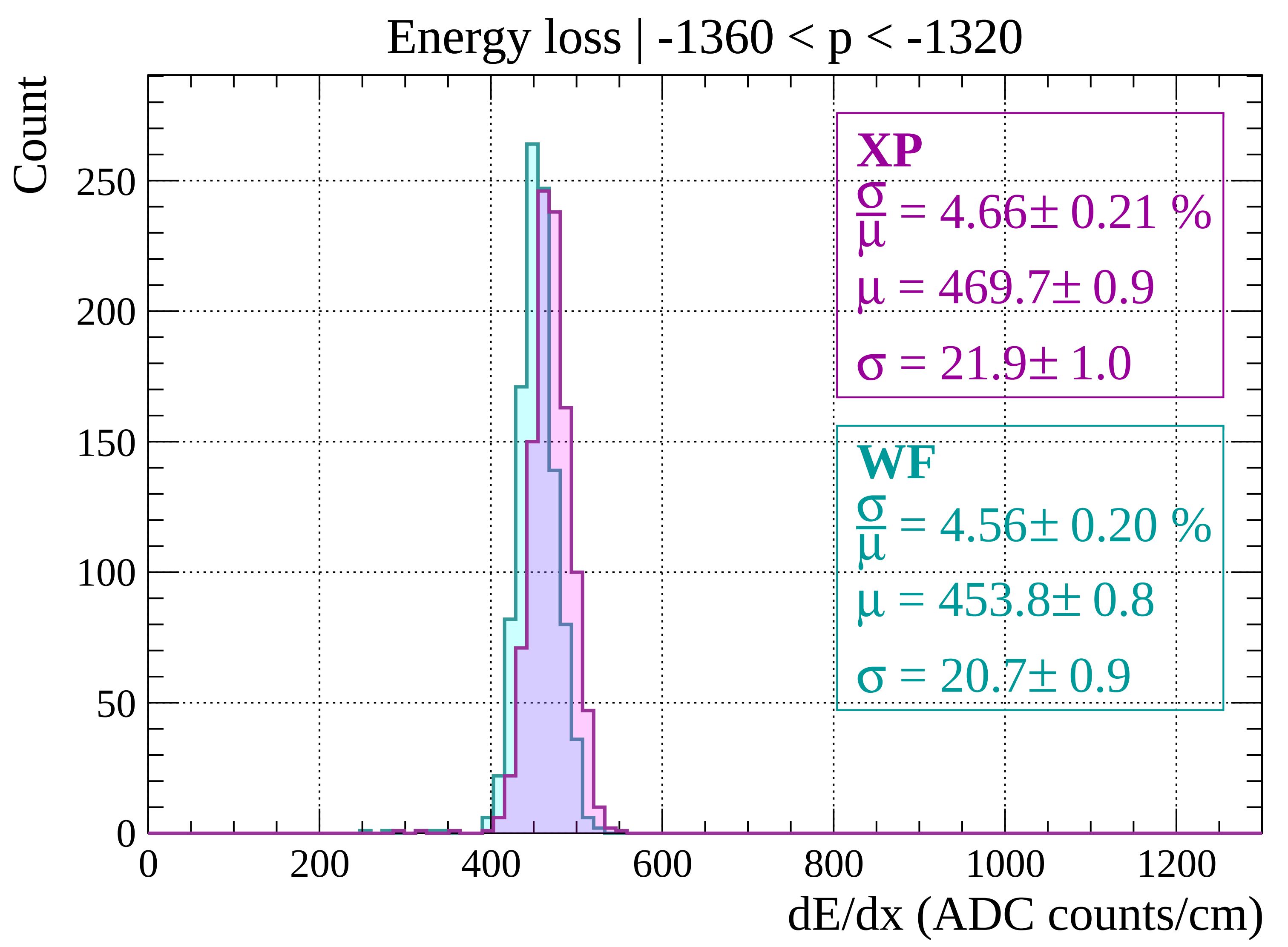


Energy loss | -1480 < p < -1440
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
200
220
XP
σ/μ = 5.03± 0.20 %
μ = 475.9± 0.9
σ = 23.9± 0.9
WF
σ/μ = 4.89± 0.15 %
μ = 460.3± 0.8
σ = 22.5± 0.6

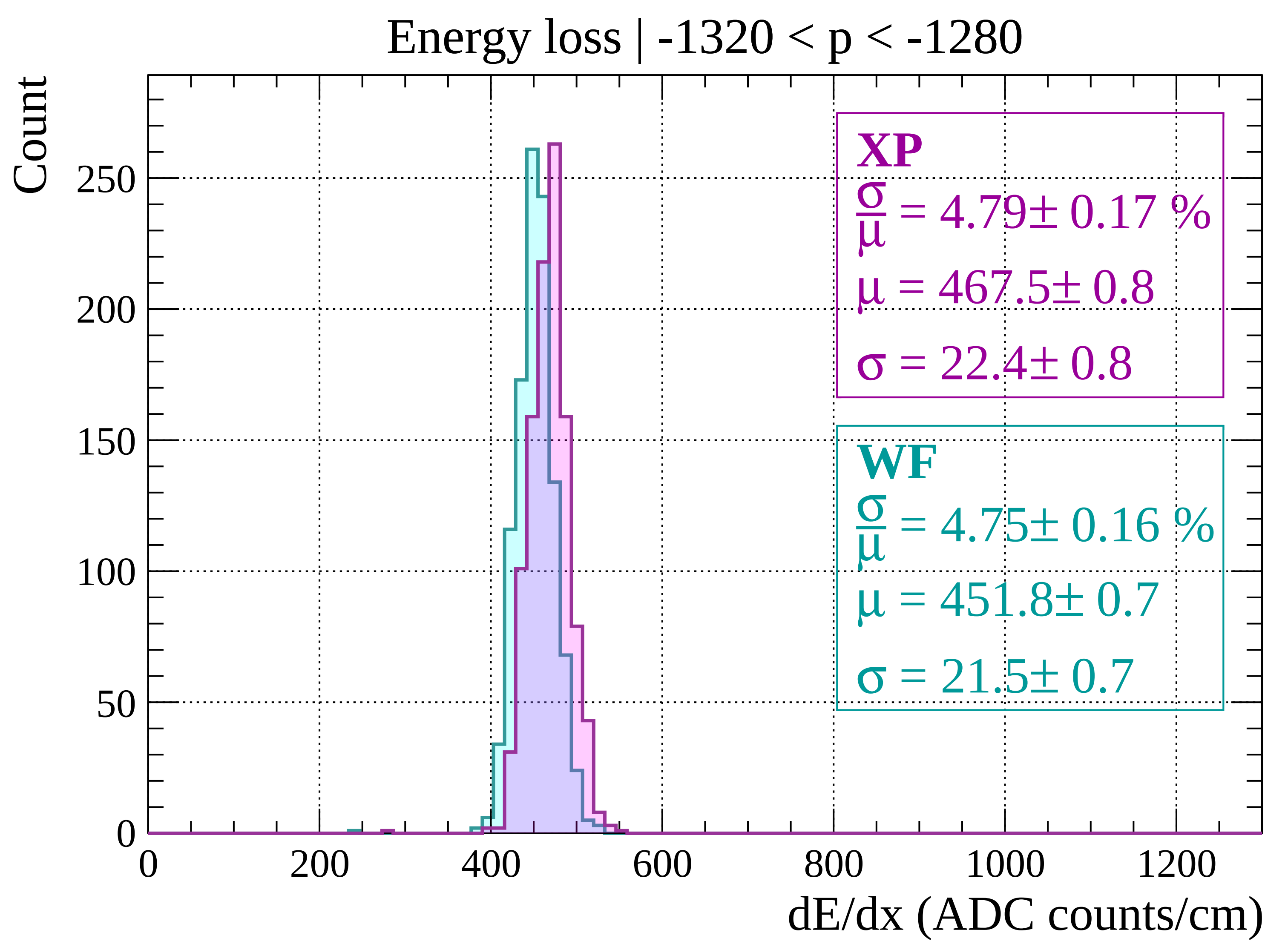

Energy loss | -1440 < p < -1400
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP
σ/μ = 5.00± 0.19 %
μ = 474.7± 0.9
σ = 23.7± 0.9
WF
σ/μ = 4.83± 0.16 %
μ = 459.1± 0.8
σ = 22.2± 0.7



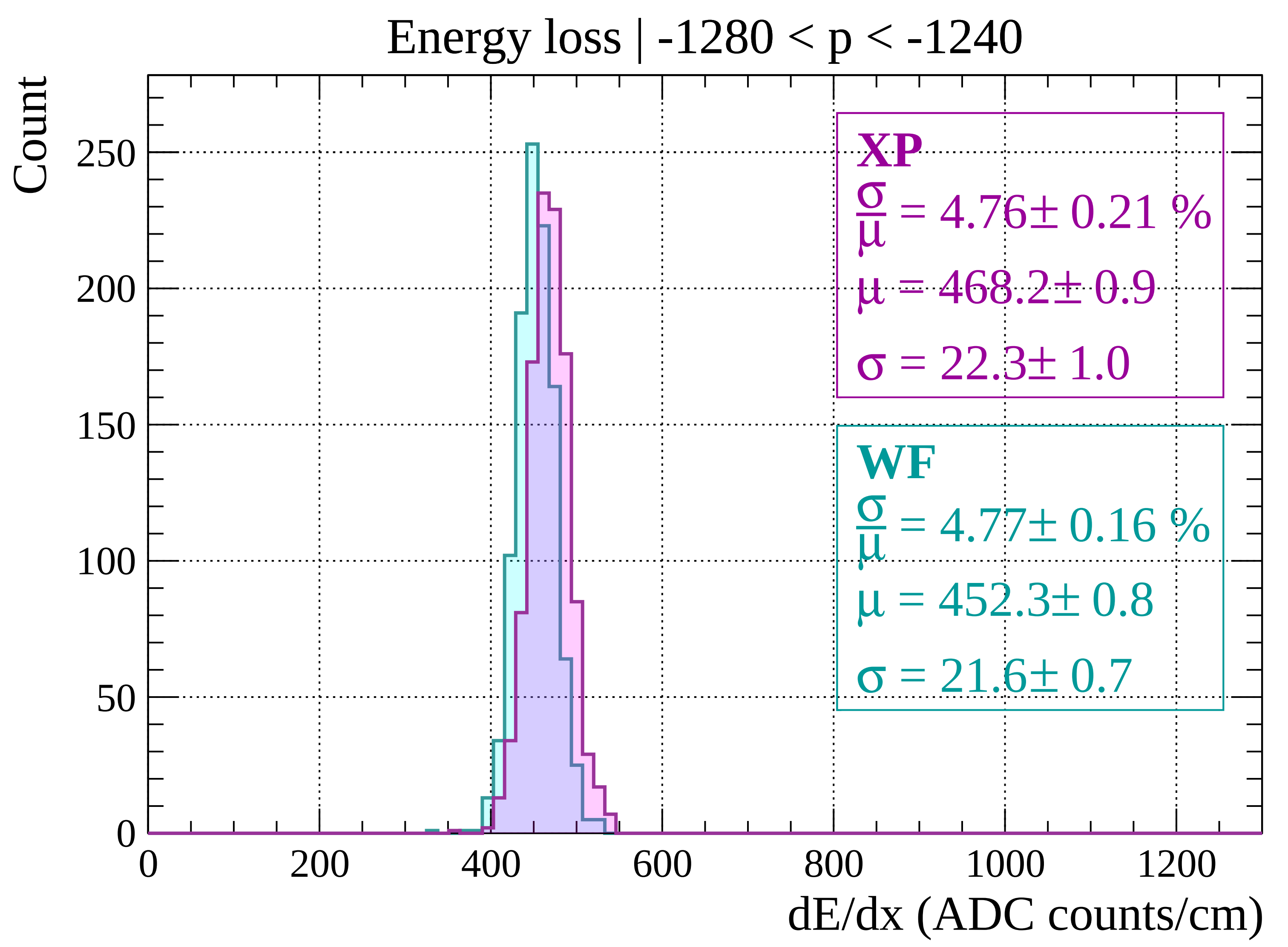
Energy loss | -1400 < p < -1360
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 4.68± 0.16 %
μ = 472.1± 0.8
σ = 22.1± 0.7
WF
$\frac{\sigma}{\mu}$ = 4.56± 0.18 %
μ = 456.6± 0.8
σ = 20.8± 0.8



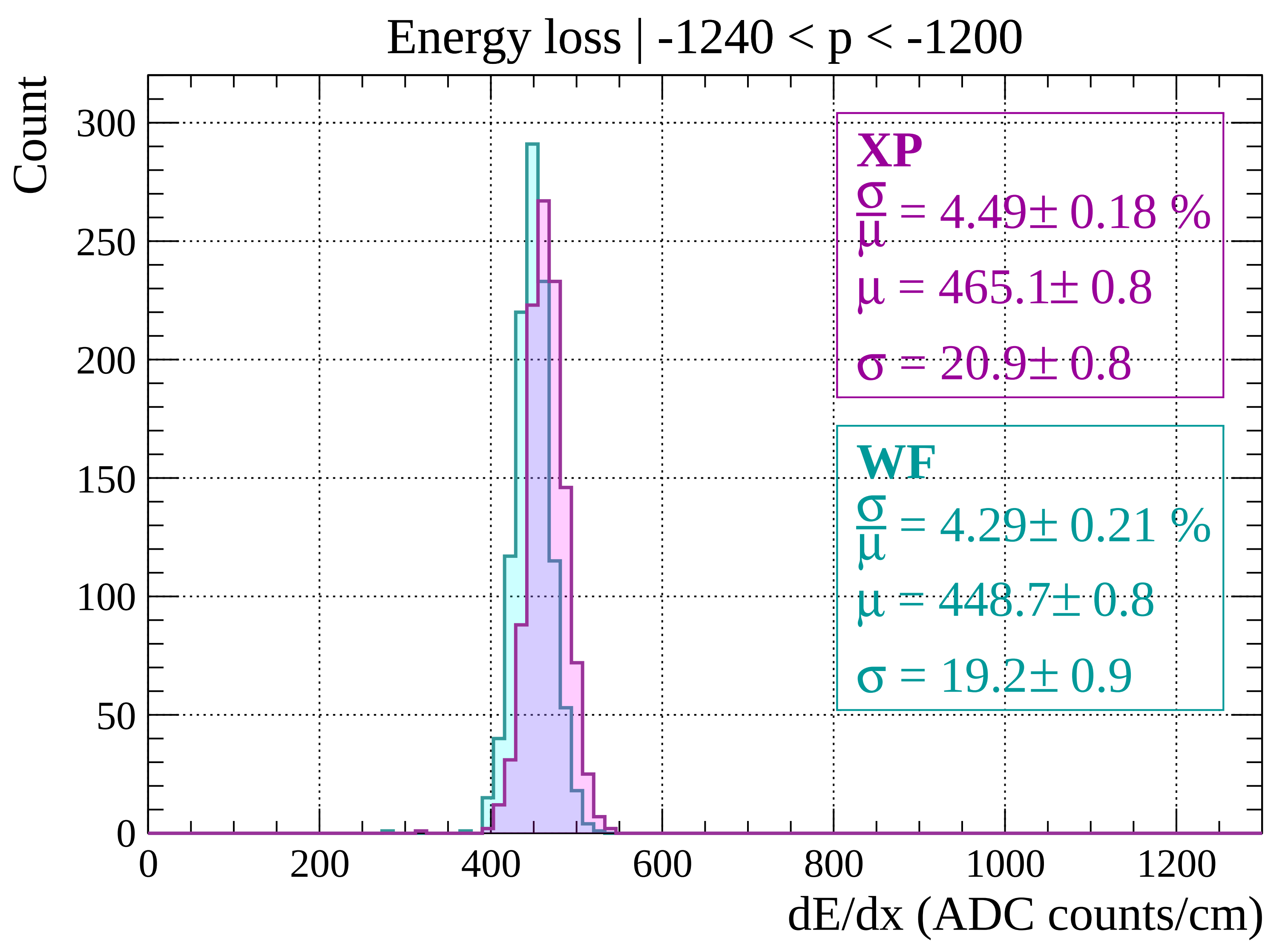
Energy loss | -1360 < p < -1320
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP
σ/μ = 4.66± 0.21 %
μ = 469.7± 0.9
σ = 21.9± 1.0
WF
σ/μ = 4.56± 0.20 %
μ = 453.8± 0.8
σ = 20.7± 0.9

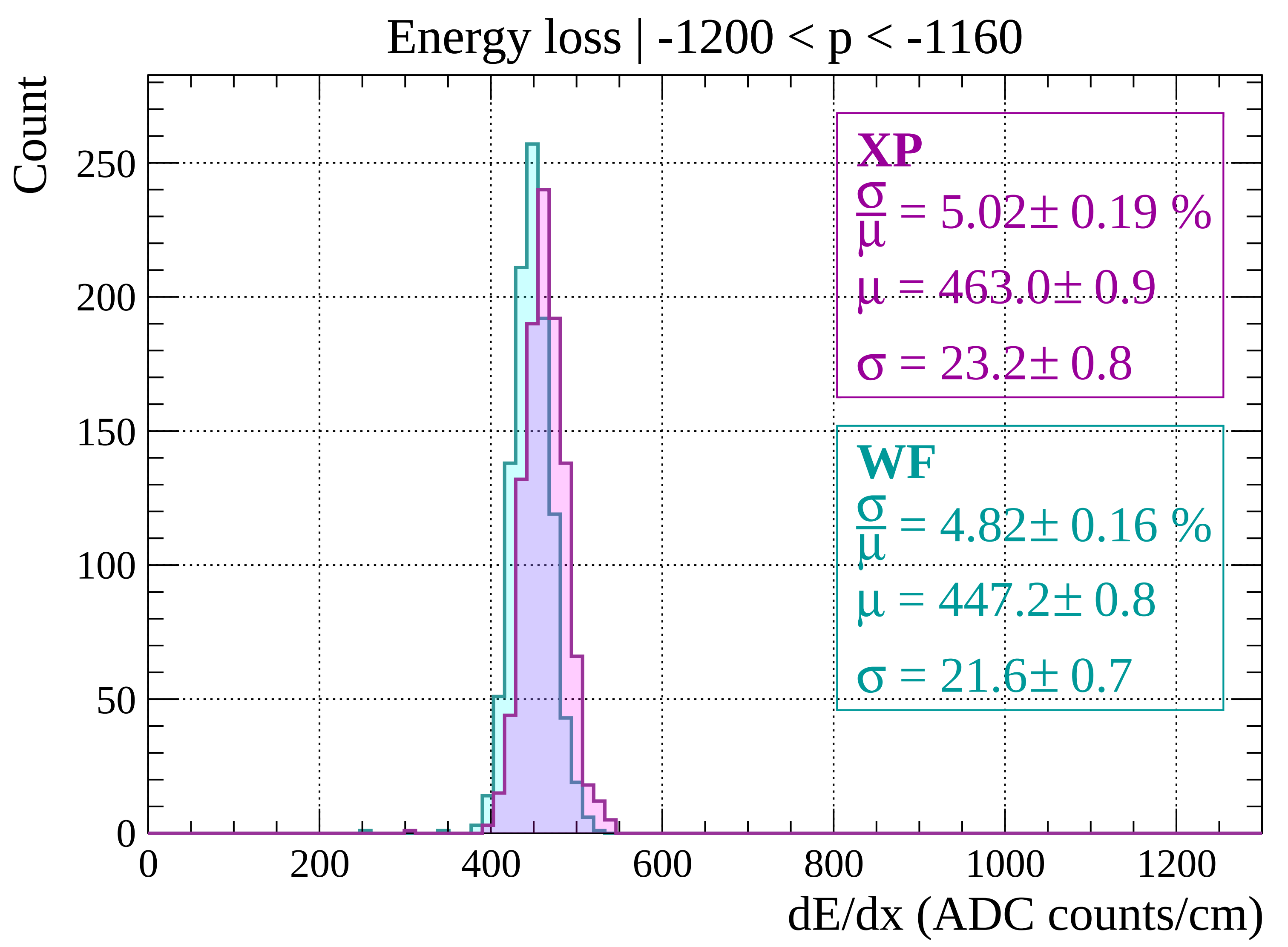


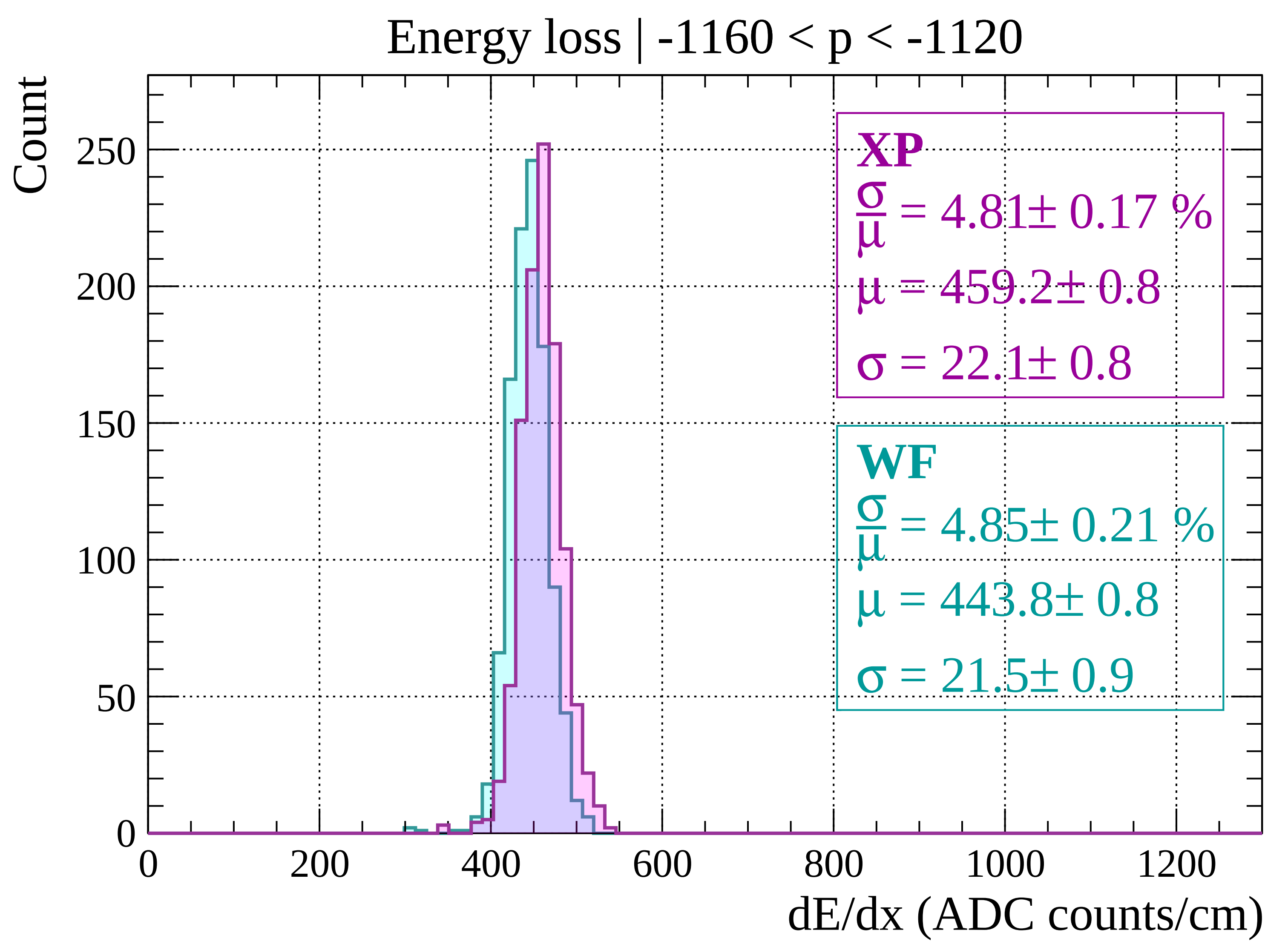
Energy loss | -1320 < p < -1280
Count
0
50
100
150
200
250
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 4.79± 0.17 %
μ = 467.5± 0.8
σ = 22.4± 0.8
WF
σ/μ = 4.75± 0.16 %
μ = 451.8± 0.7
σ = 21.5± 0.7



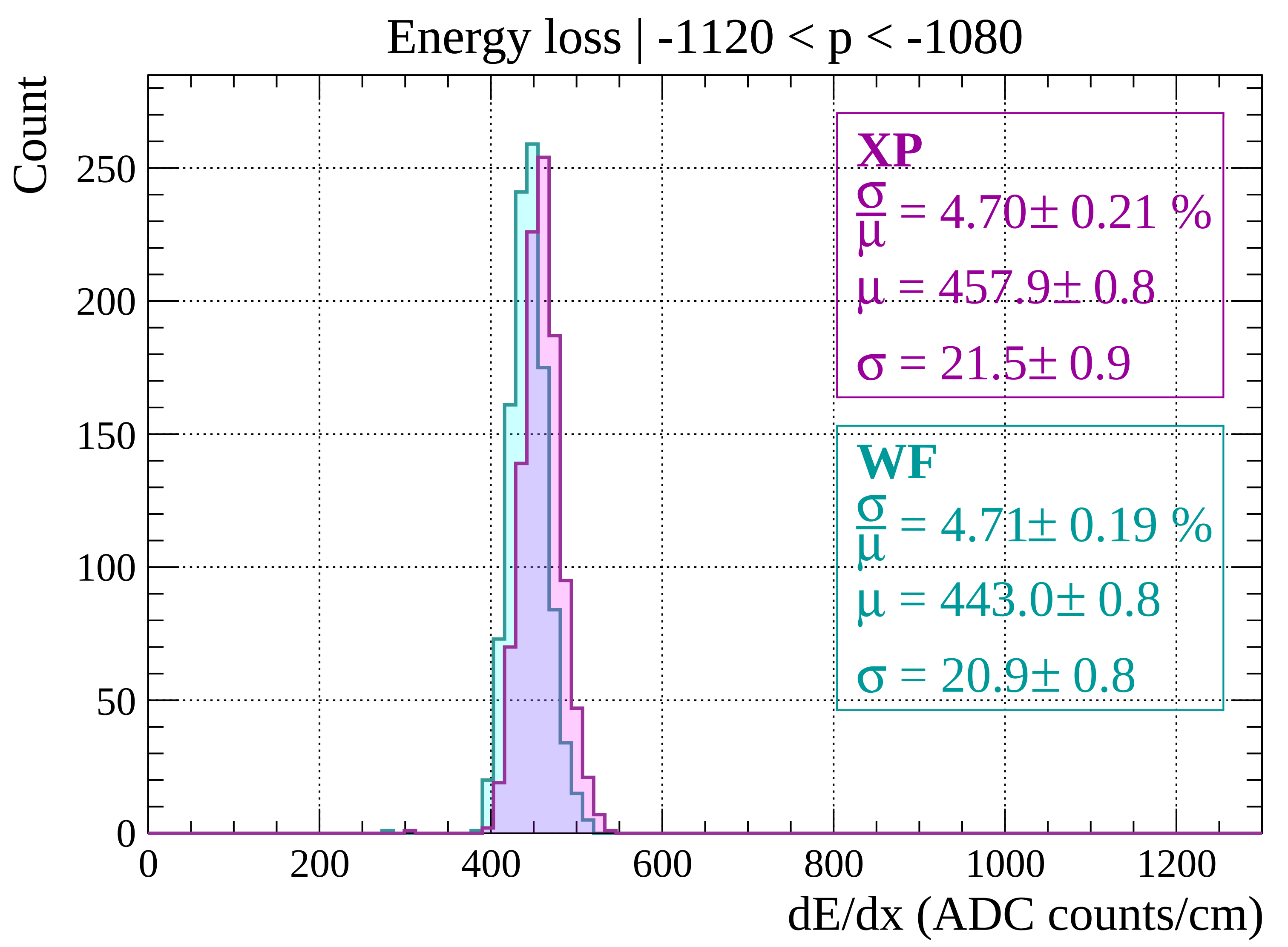
Energy loss | -1280 < p < -1240
Count
0
50
100
150
200
250
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 4.76± 0.21 %
μ = 468.2± 0.9
σ = 22.3± 1.0
WF
$\frac{\sigma}{\mu}$ = 4.77± 0.16 %
μ = 452.3± 0.8
σ = 21.6± 0.7

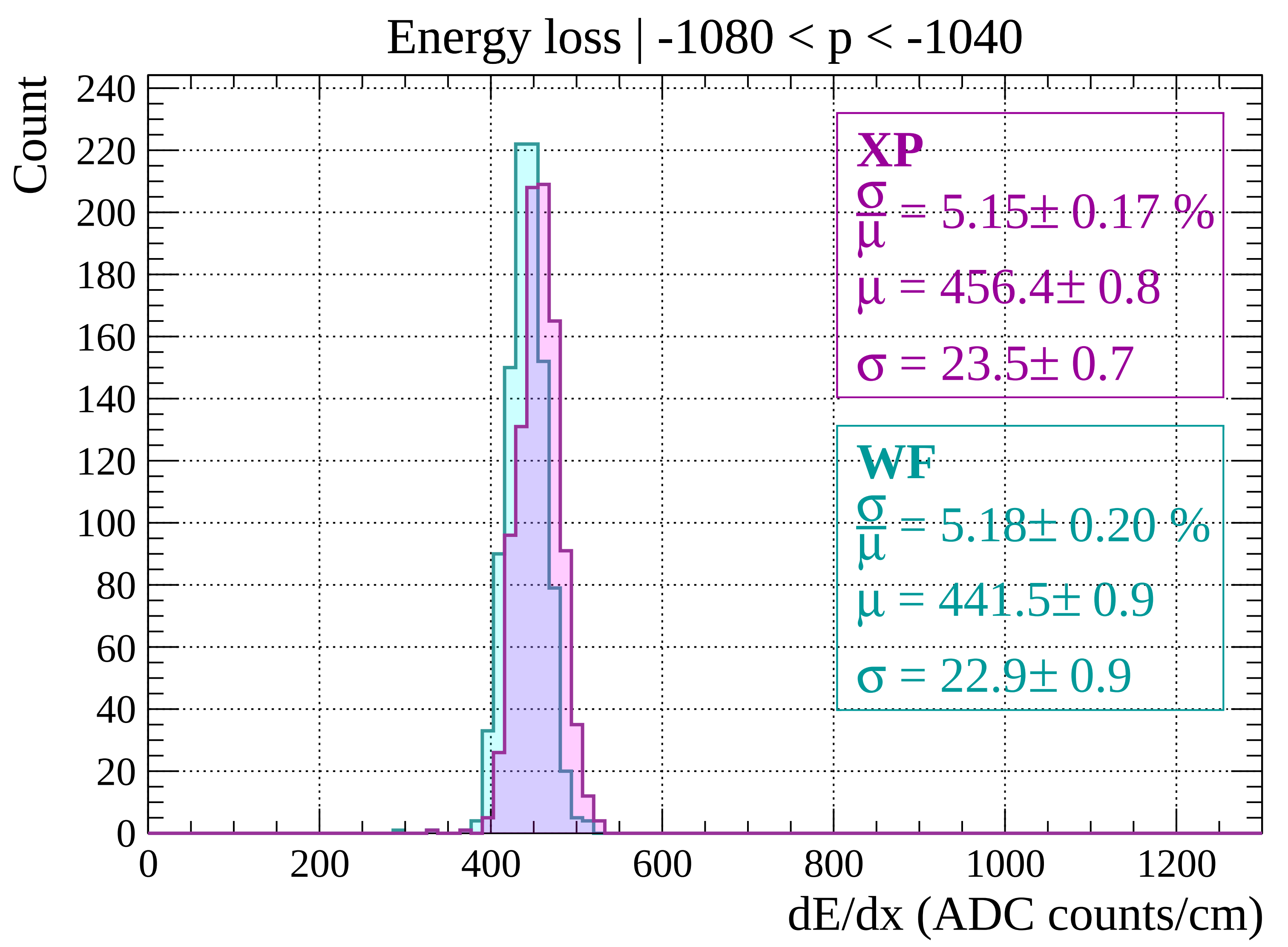

Energy loss | -1240 < p < -1200
Count
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 4.49± 0.18 %
μ = 465.1± 0.8
σ = 20.9± 0.8
WF
σ/μ = 4.29± 0.21 %
μ = 448.7± 0.8
σ = 19.2± 0.9

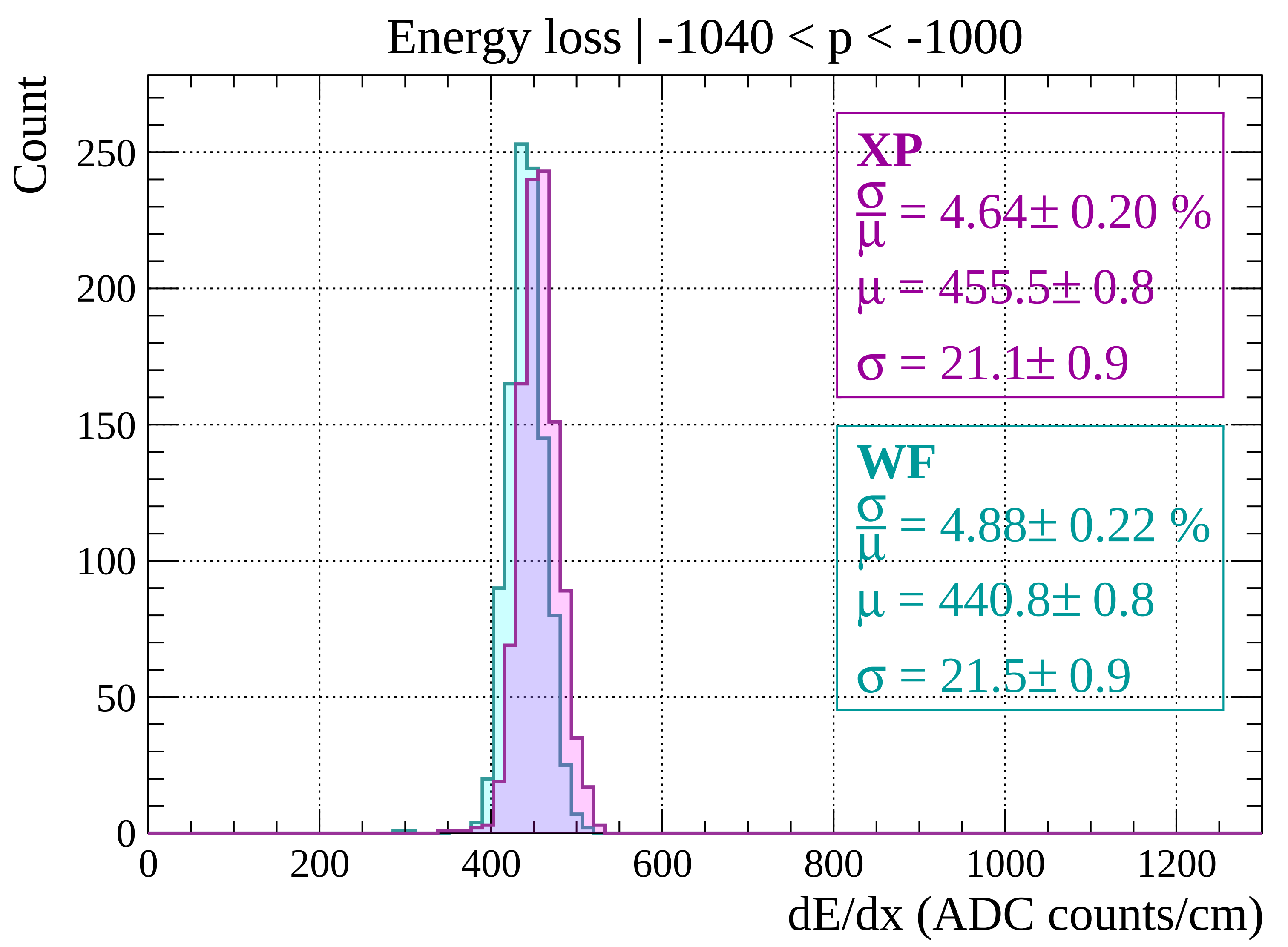

Energy loss | -1200 < p < -1160
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP
σ/μ = 5.02± 0.19 %
μ = 463.0± 0.9
σ = 23.2± 0.8
WF
σ/μ = 4.82± 0.16 %
μ = 447.2± 0.8
σ = 21.6± 0.7



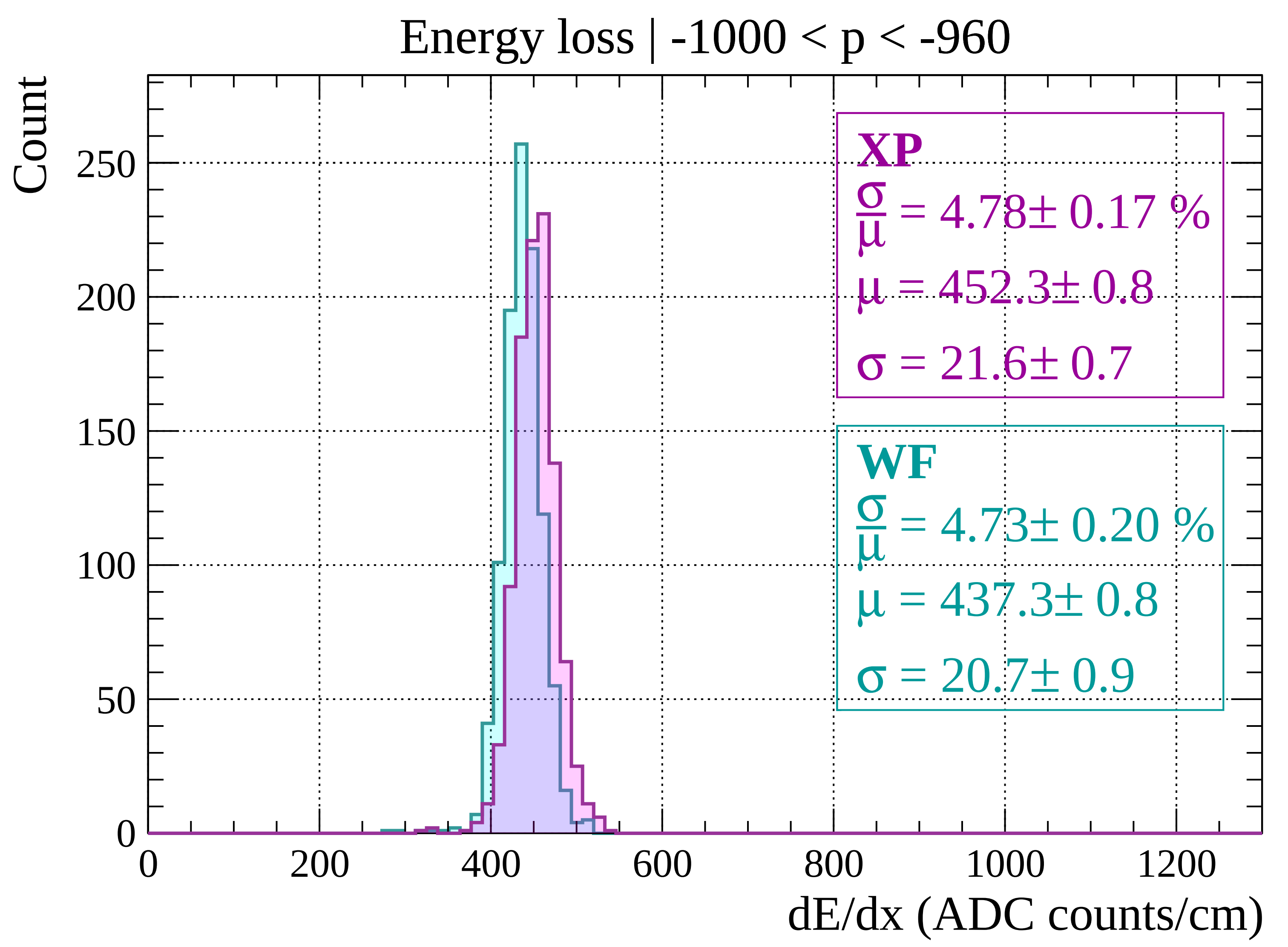
Energy loss | -1160 < p < -1120
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP
σ/μ = 4.81± 0.17 %
μ = 459.2± 0.8
σ = 22.1± 0.8
WF
σ/μ = 4.85± 0.21 %
μ = 443.8± 0.8
σ = 21.5± 0.9


Energy loss | -1120 < p < -1080
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 4.70 ± 0.21 %
μ = 457.9 ± 0.8
σ = 21.5 ± 0.9
WF
$\frac{\sigma}{\mu}$ = 4.71 ± 0.19 %
μ = 443.0 ± 0.8
σ = 20.9 ± 0.8

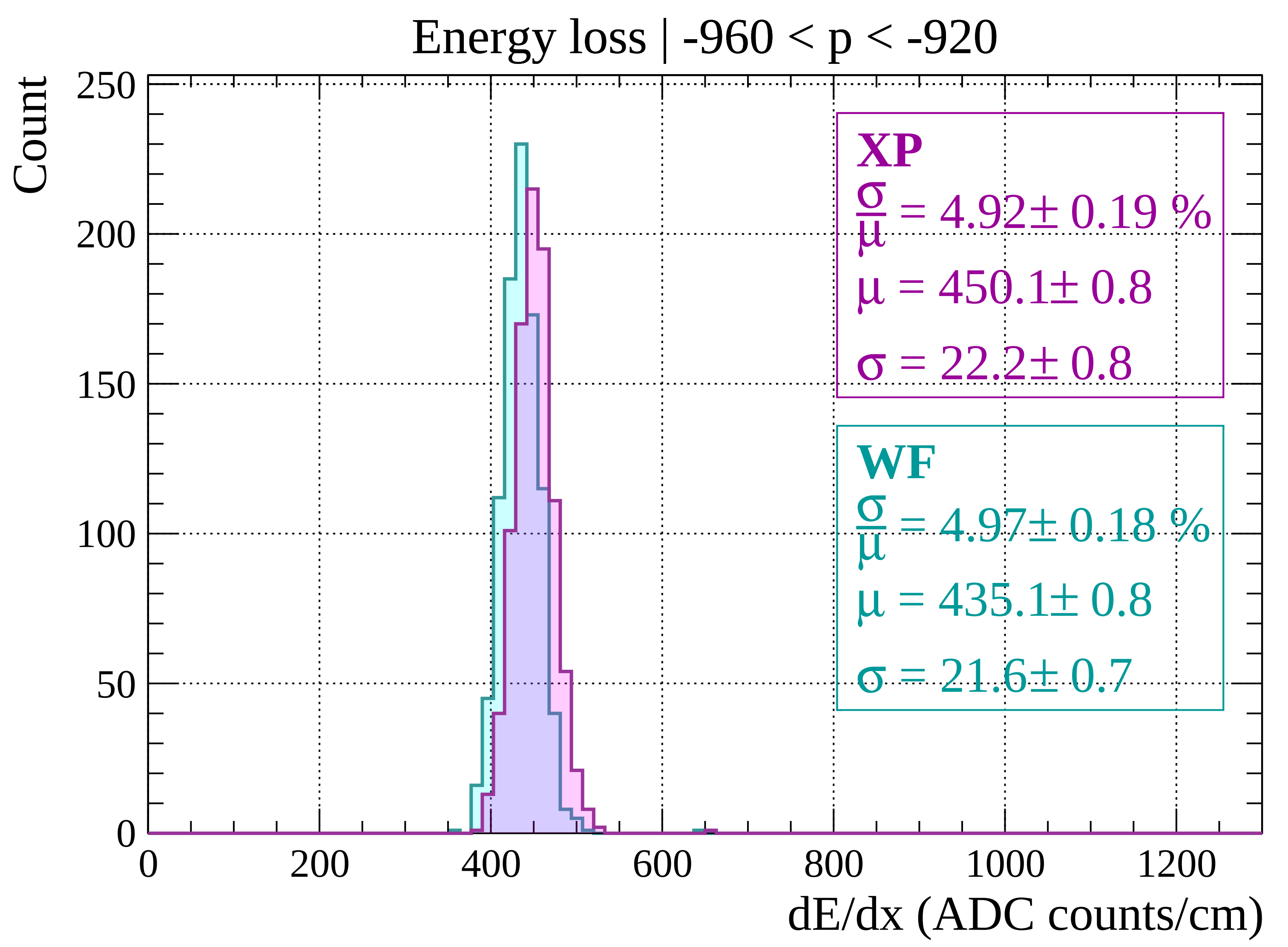

Energy loss | -1080 < p < -1040
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
200
220
240
0
200
400
600
800
1000
1200
XP
σ/μ = 5.15± 0.17 %
μ = 456.4± 0.8
σ = 23.5± 0.7
WF
σ/μ = 5.18± 0.20 %
μ = 441.5± 0.9
σ = 22.9± 0.9



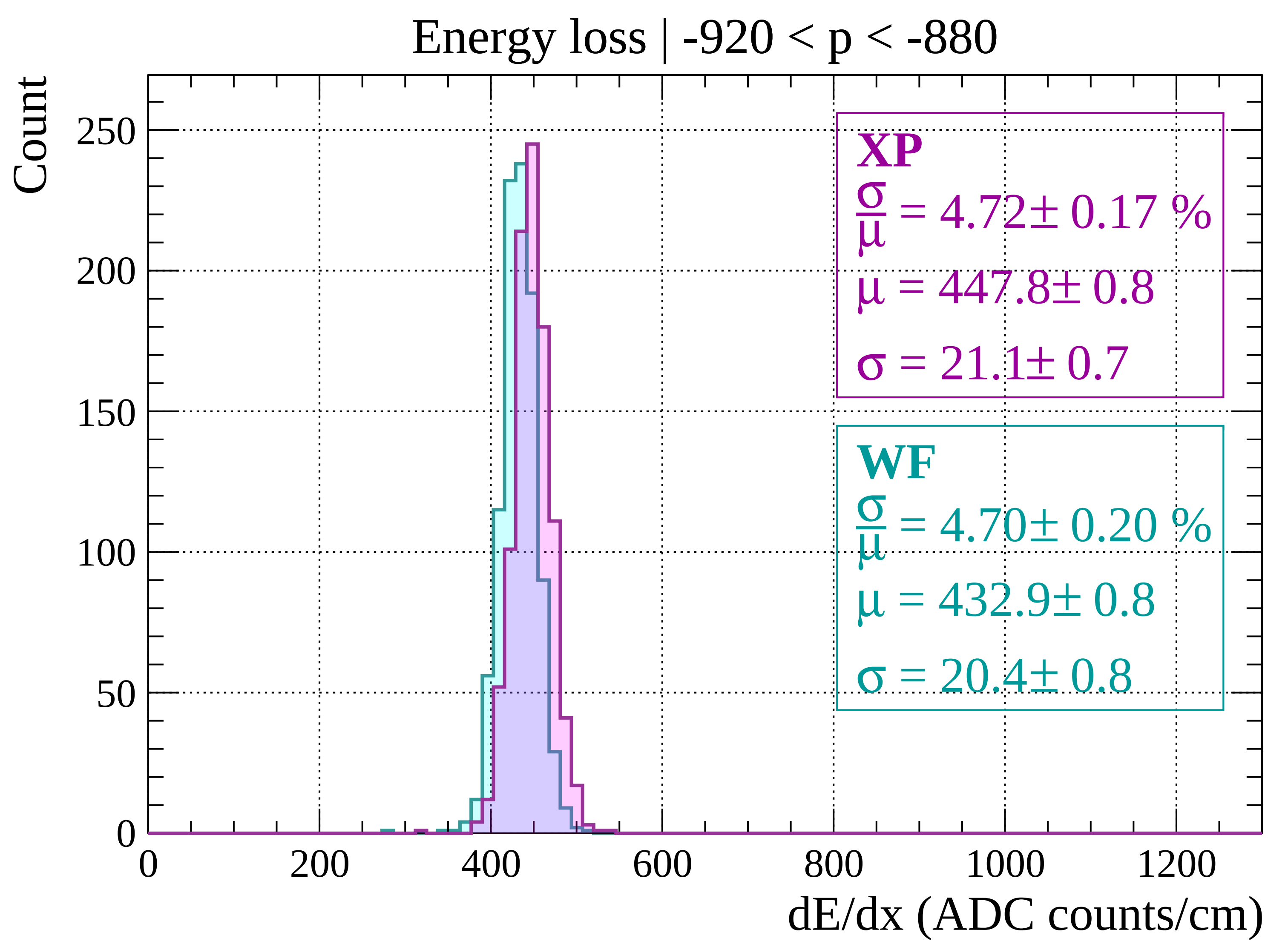
Energy loss | -1040 < p < -1000
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP
σ/μ = 4.64± 0.20 %
μ = 455.5± 0.8
σ = 21.1± 0.9
WF
σ/μ = 4.88± 0.22 %
μ = 440.8± 0.8
σ = 21.5± 0.9



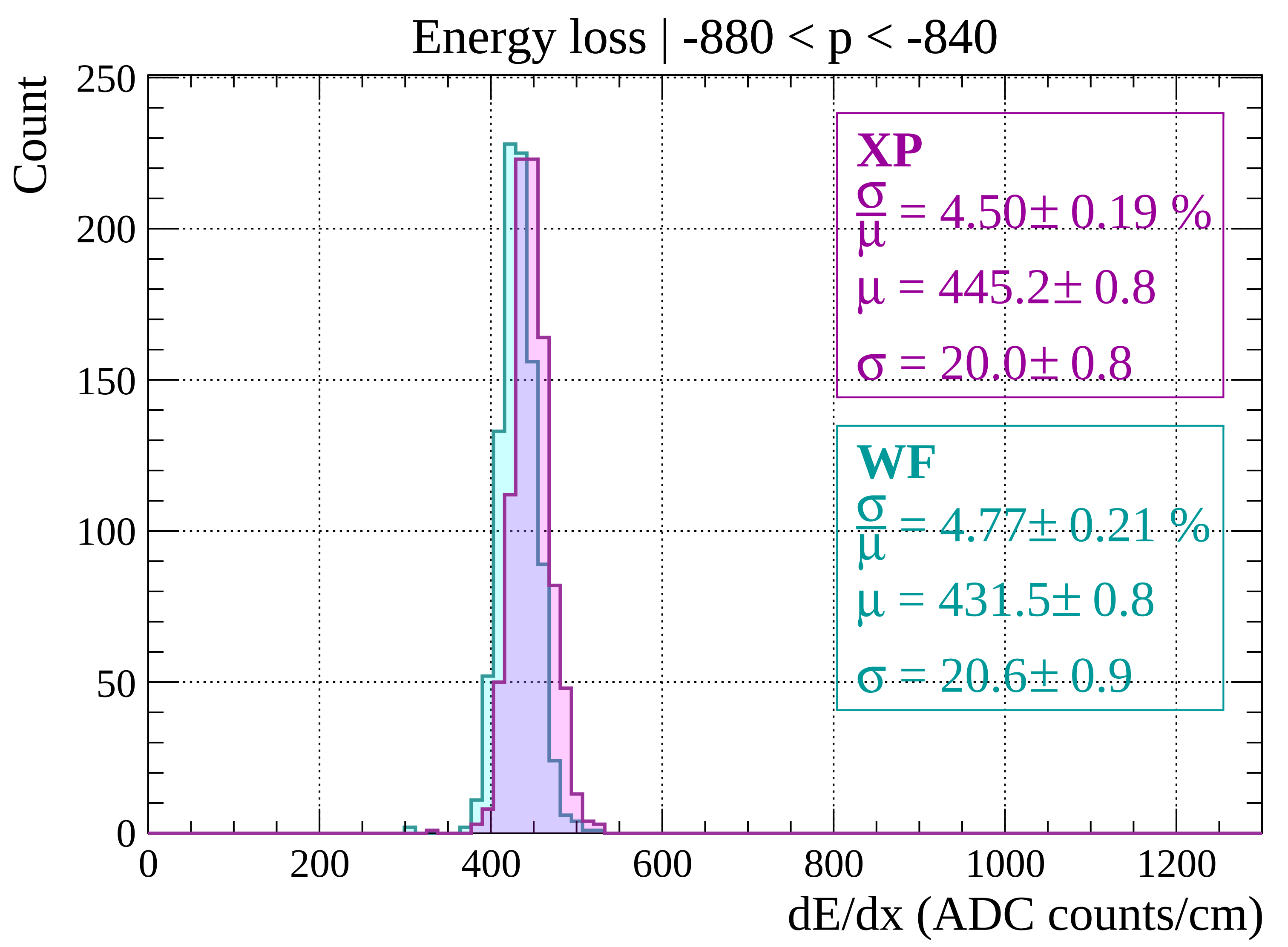
Energy loss | -1000 < p < -960
Count
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 4.78± 0.17 %
μ = 452.3± 0.8
σ = 21.6± 0.7
WF
$\frac{\sigma}{\mu}$ = 4.73± 0.20 %
μ = 437.3± 0.8
σ = 20.7± 0.9



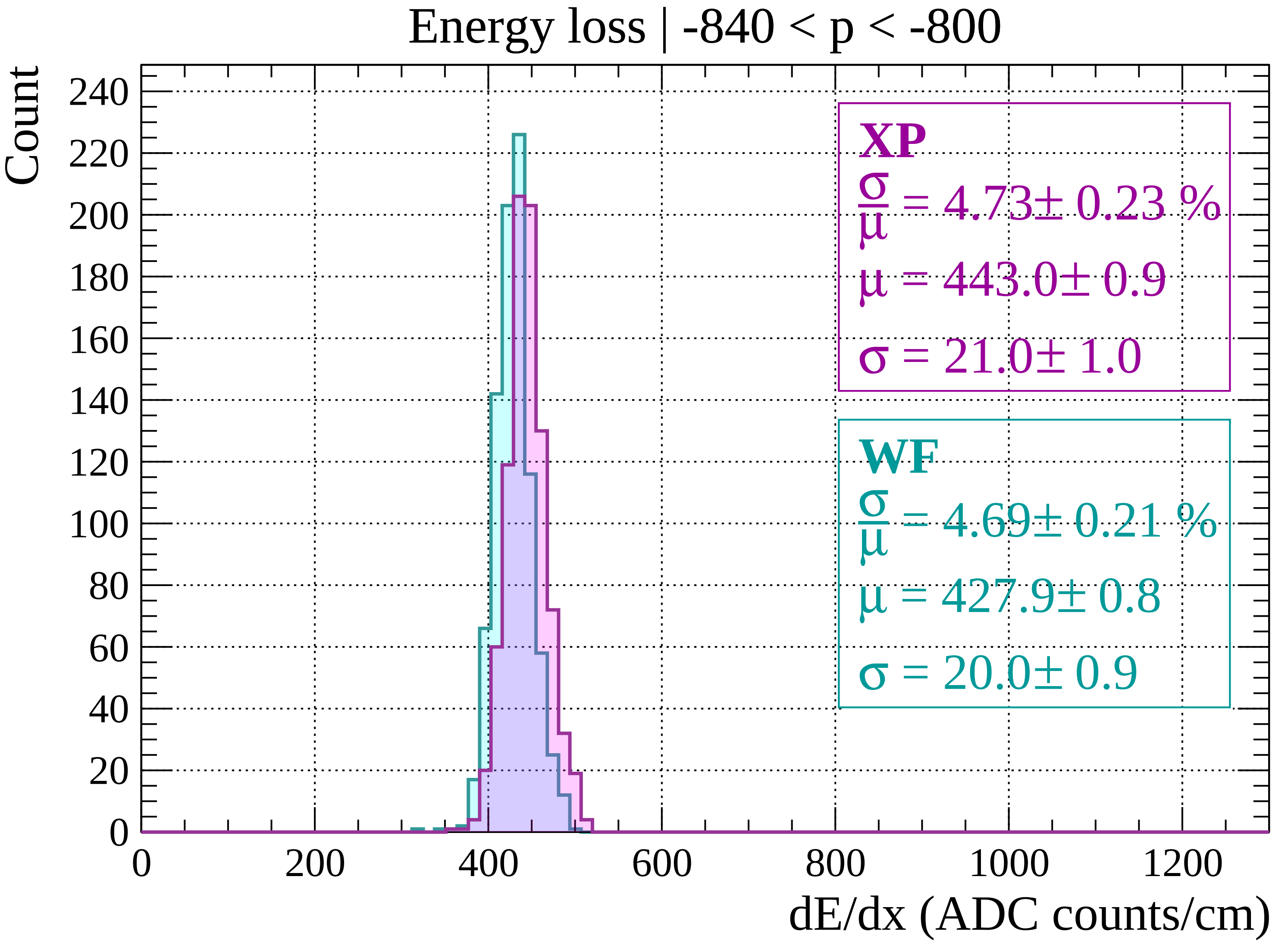
Energy loss | -960 < p < -920
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 4.92± 0.19 %
μ = 450.1± 0.8
σ = 22.2± 0.8
WF
σ/μ = 4.97± 0.18 %
μ = 435.1± 0.8
σ = 21.6± 0.7



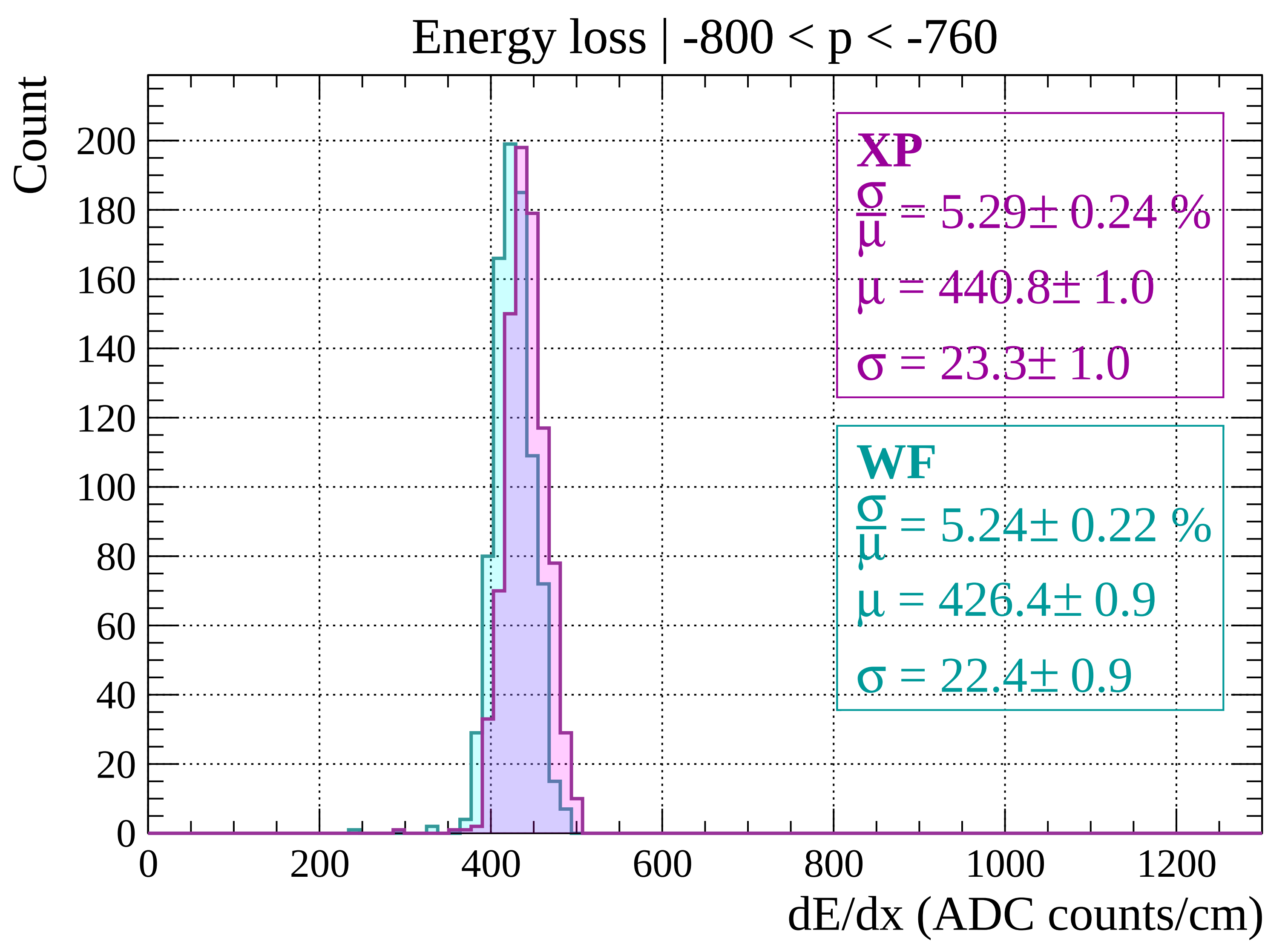
Energy loss | -920 < p < -880
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 4.72± 0.17 %
μ = 447.8± 0.8
σ = 21.1± 0.7
WF
$\frac{\sigma}{\mu}$ = 4.70± 0.20 %
μ = 432.9± 0.8
σ = 20.4± 0.8

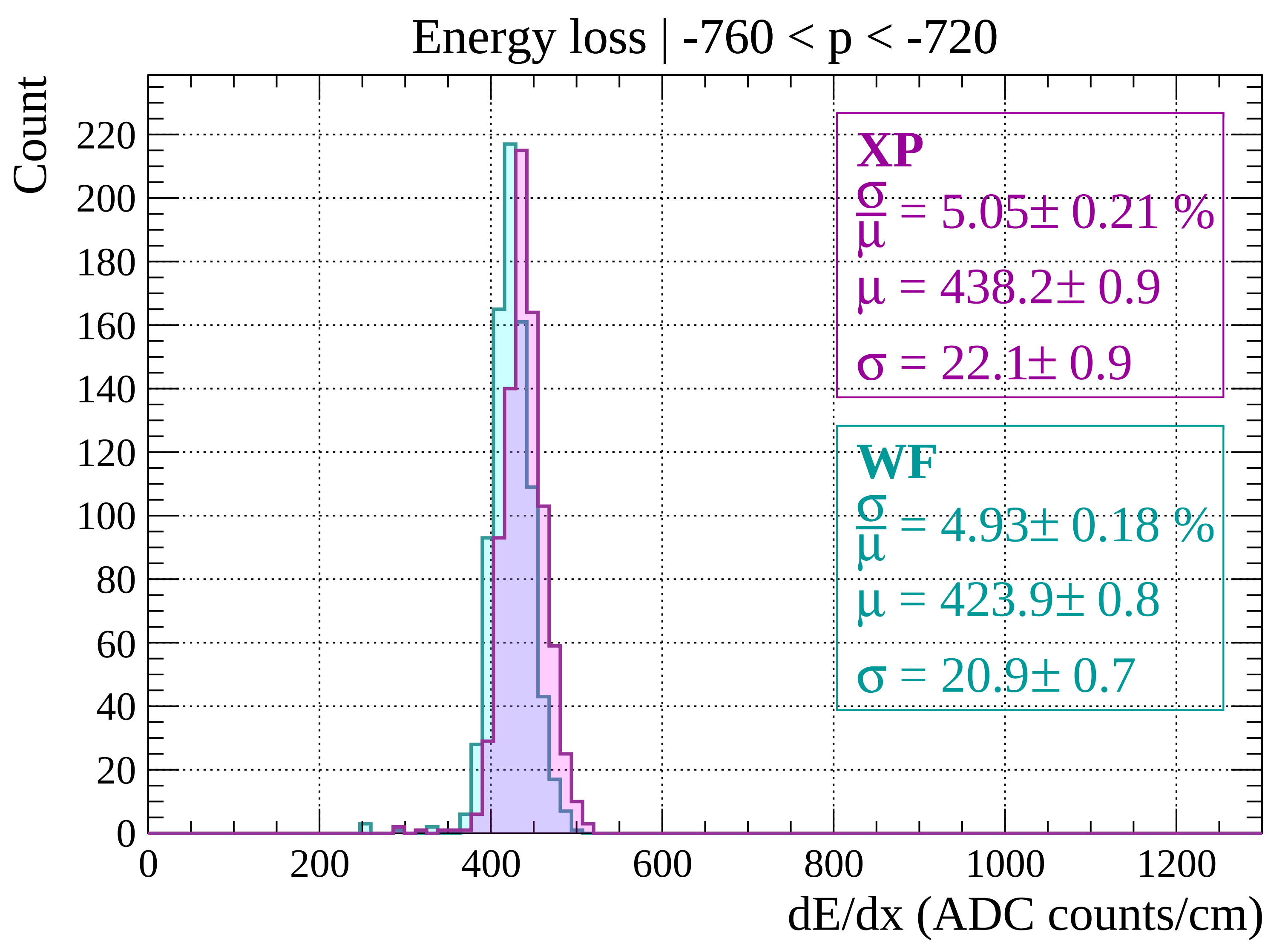


Energy loss | -880 < p < -840
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
50
100
150
200
250
XP
σ/μ = 4.50± 0.19 %
μ = 445.2± 0.8
σ = 20.0± 0.8
WF
σ/μ = 4.77± 0.21 %
μ = 431.5± 0.8
σ = 20.6± 0.9

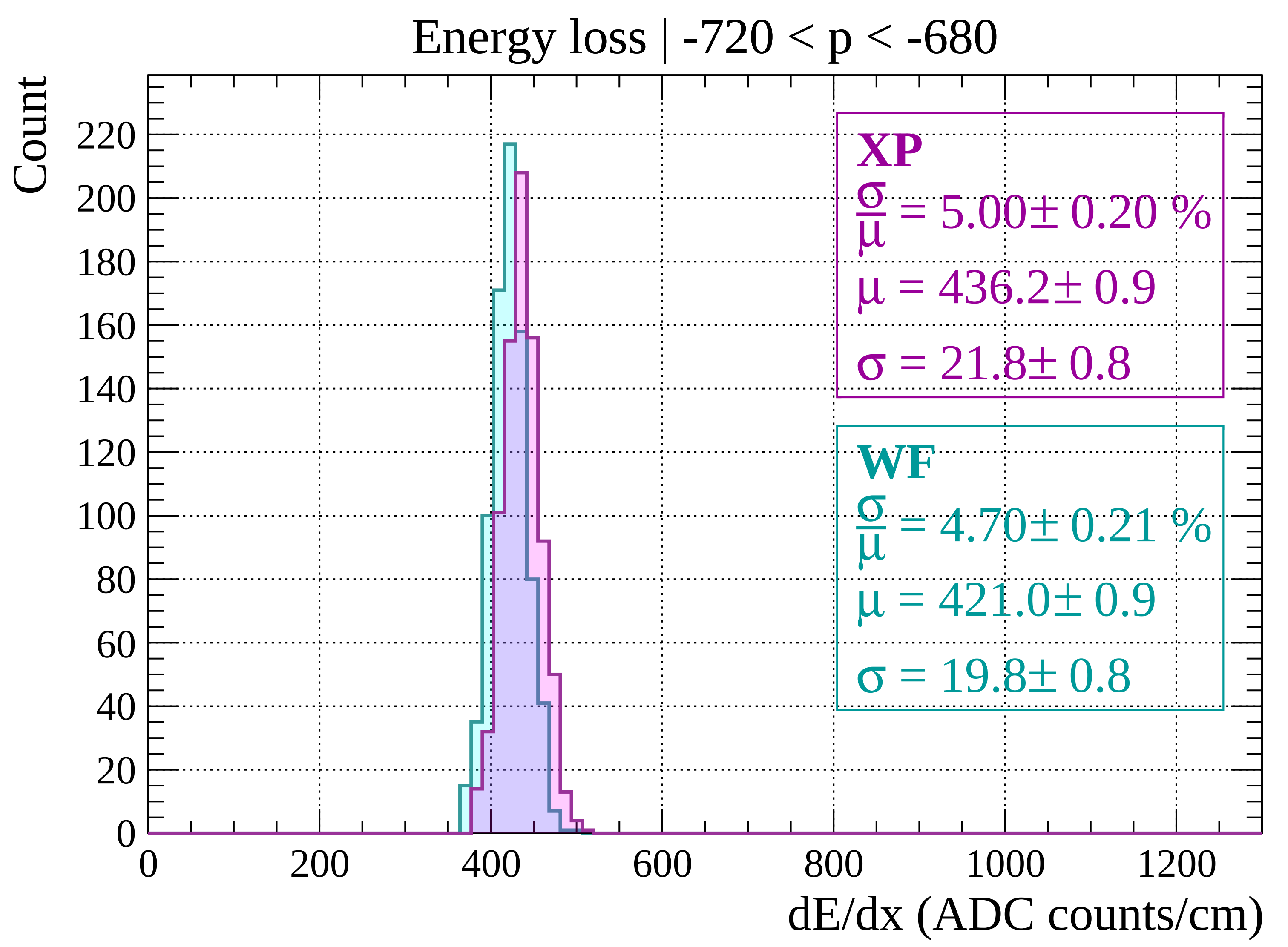


Energy loss | -840 < p < -800
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
200
220
240
0
200
400
600
800
1000
1200
XP
σ/μ = 4.73± 0.23 %
μ = 443.0± 0.9
σ = 21.0± 1.0
WF
σ/μ = 4.69± 0.21 %
μ = 427.9± 0.8
σ = 20.0± 0.9

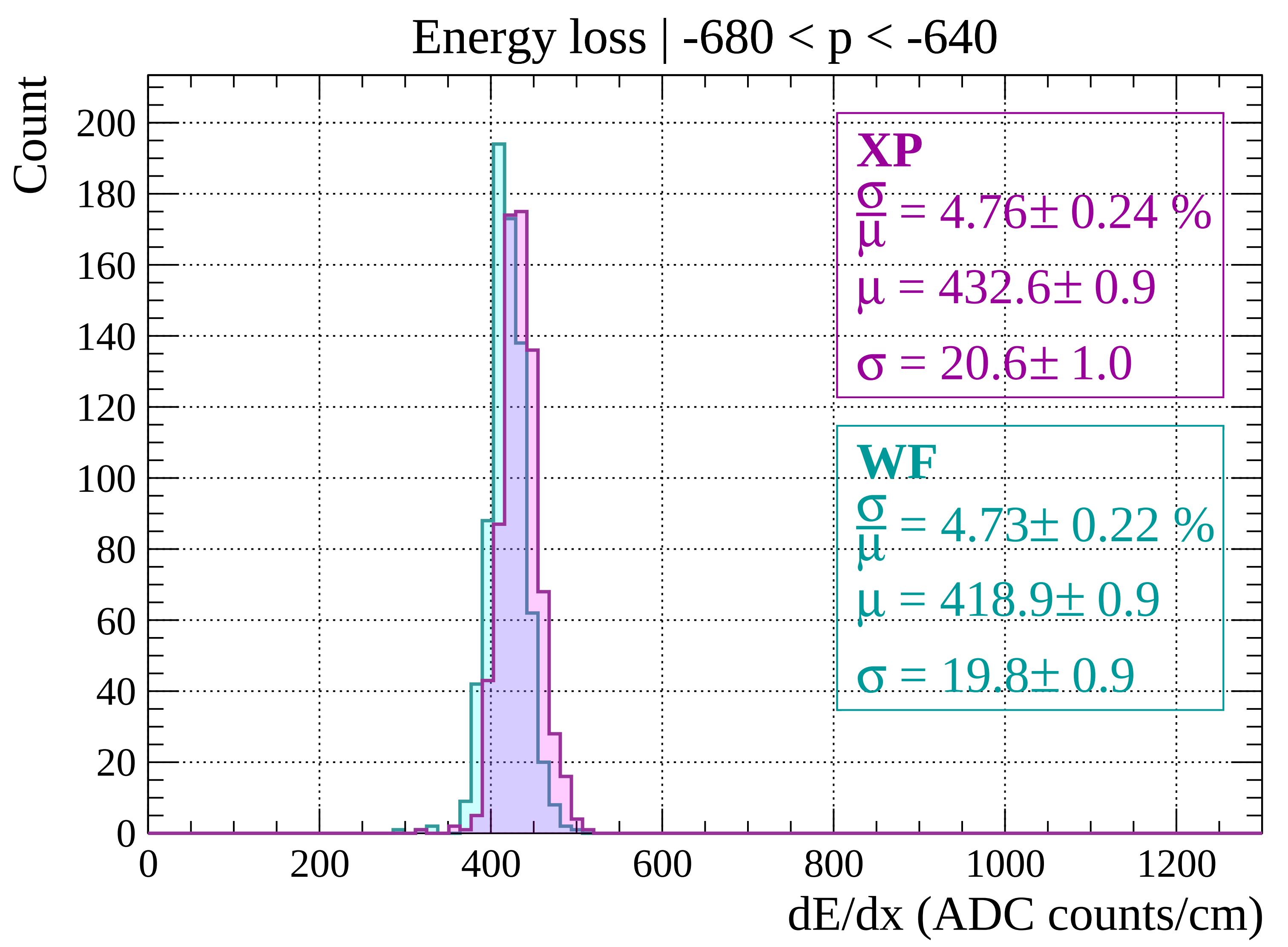

Energy loss | -800 < p < -760
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 5.29± 0.24 %
μ = 440.8± 1.0
σ = 23.3± 1.0
WF
σ/μ = 5.24± 0.22 %
μ = 426.4± 0.9
σ = 22.4± 0.9
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
200



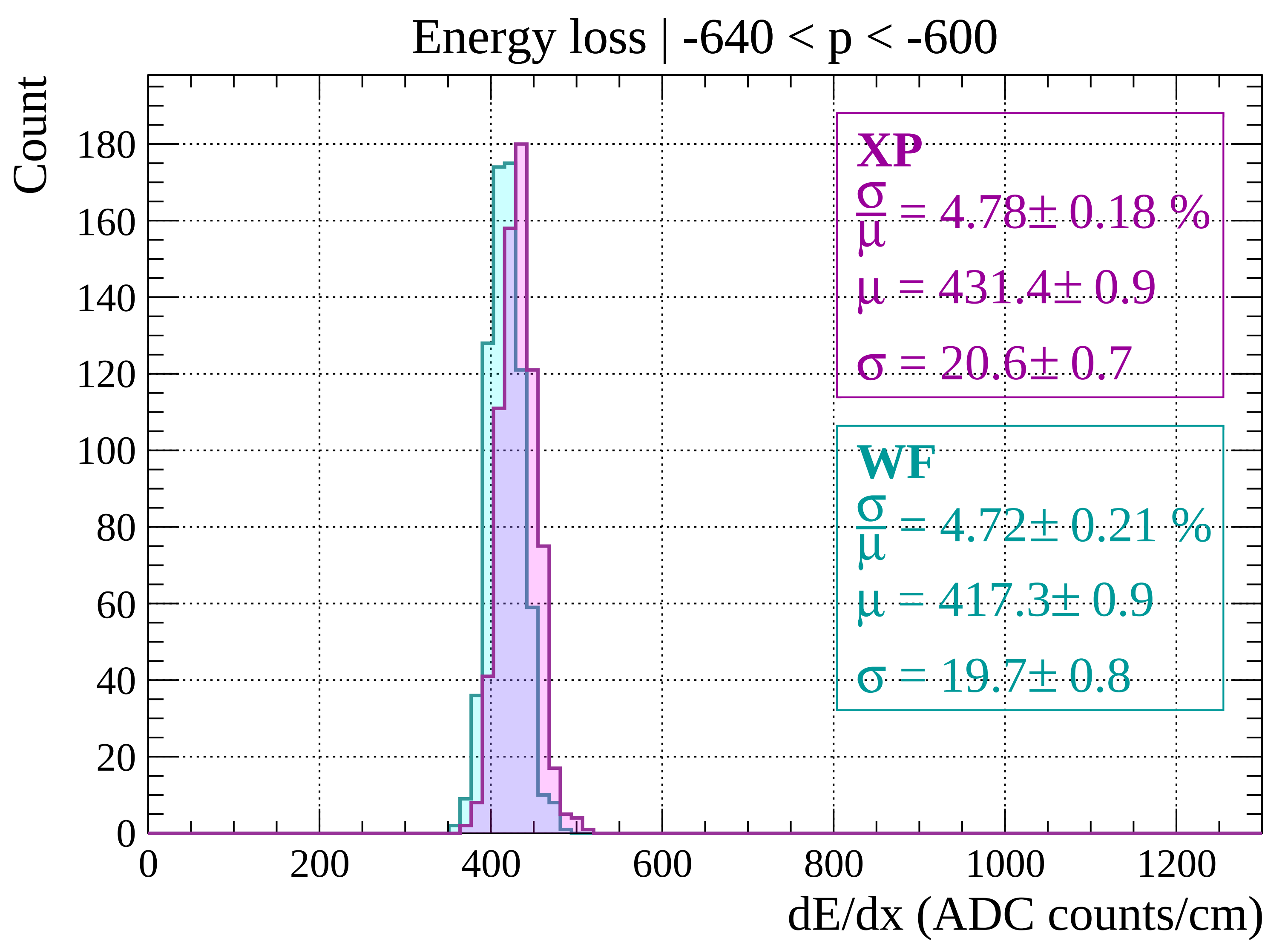
Energy loss | -760 < p < -720
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 5.05± 0.21 %
μ = 438.2± 0.9
σ = 22.1± 0.9
WF
σ/μ = 4.93± 0.18 %
μ = 423.9± 0.8
σ = 20.9± 0.7
0
20
40
60
80
100
120
140
160
180
200
220
0
200
400
600
800
1000
1200



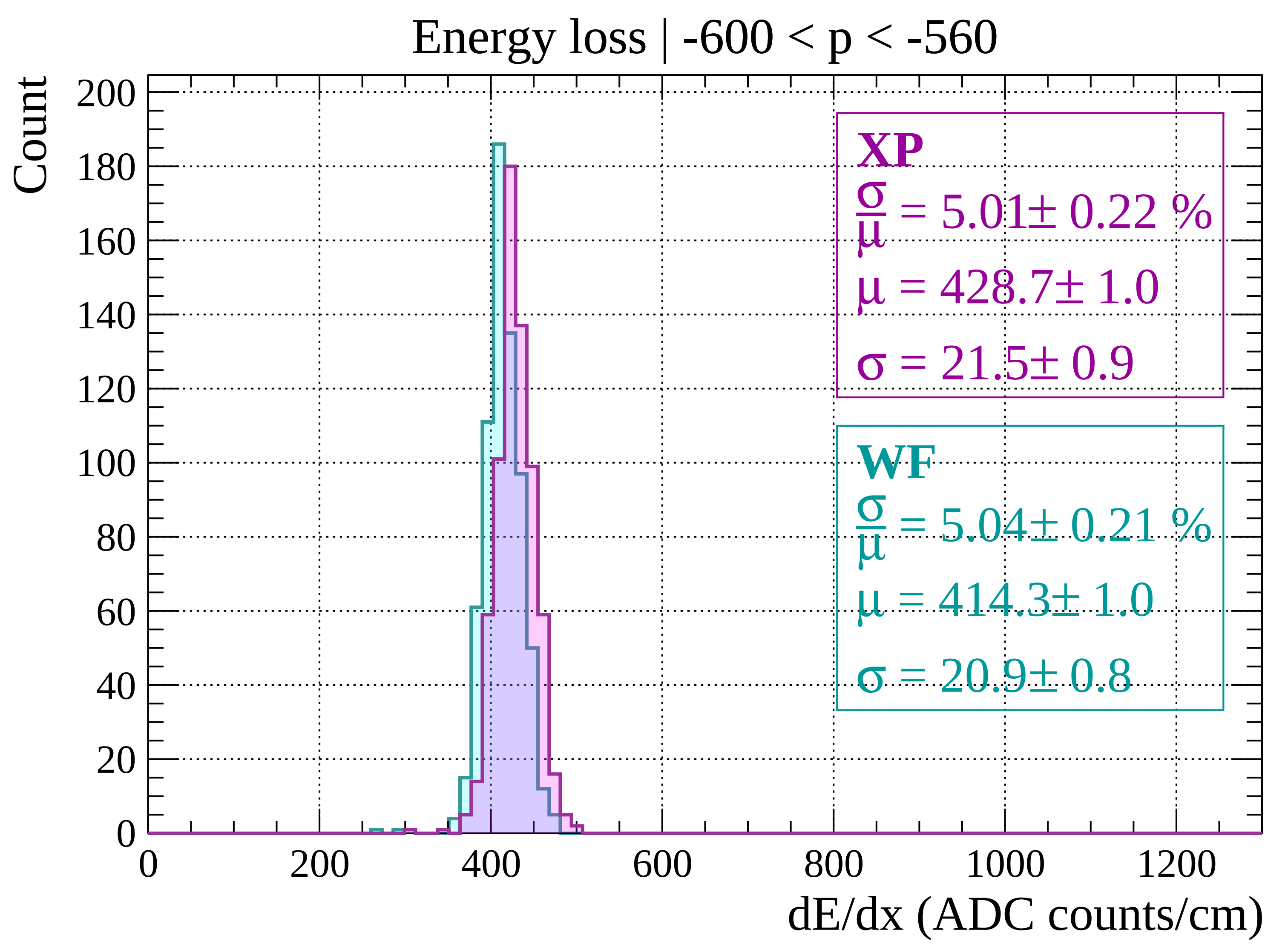
Energy loss | -720 < p < -680
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
200
220
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 5.00± 0.20 %
μ = 436.2± 0.9
σ = 21.8± 0.8
WF
$\frac{\sigma}{\mu}$ = 4.70± 0.21 %
μ = 421.0± 0.9
σ = 19.8± 0.8



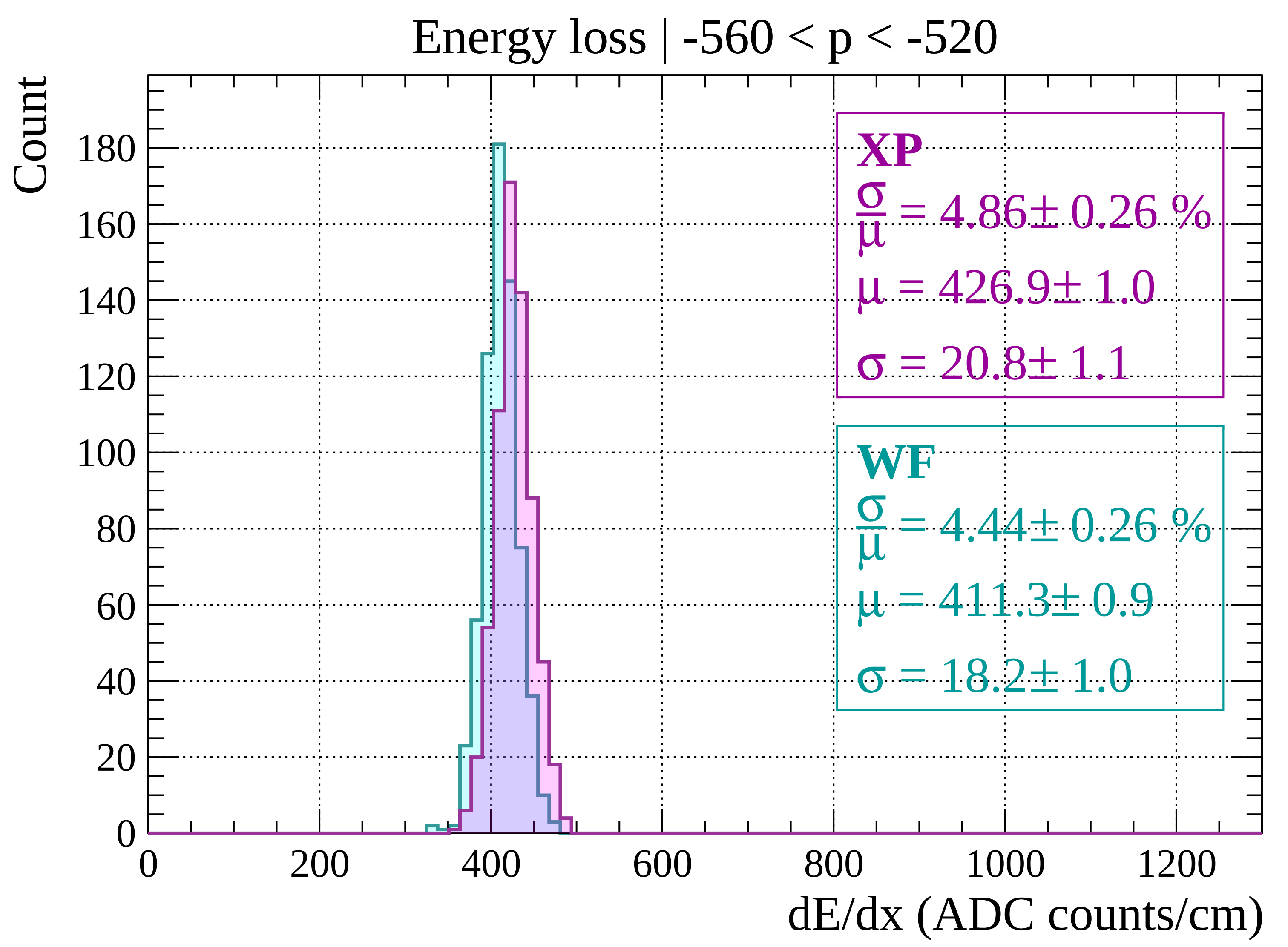
Energy loss | -680 < p < -640
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
XP
σ/μ = 4.76± 0.24 %
μ = 432.6± 0.9
σ = 20.6± 1.0
WF
σ/μ = 4.73± 0.22 %
μ = 418.9± 0.9
σ = 19.8± 0.9



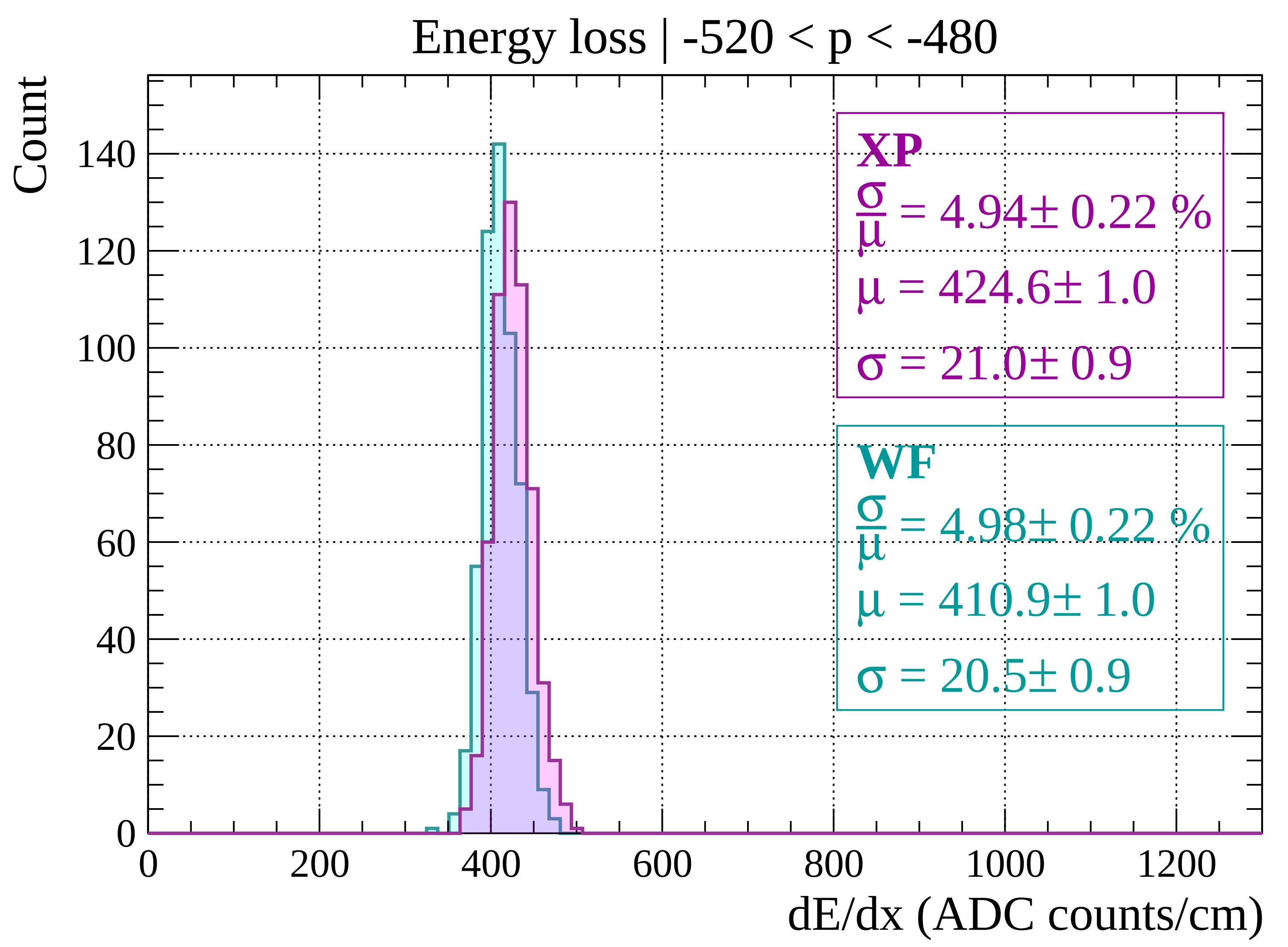
Energy loss | -640 < p < -600
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
XP
σ/μ = 4.78± 0.18 %
μ = 431.4± 0.9
σ = 20.6± 0.7
WF
σ/μ = 4.72± 0.21 %
μ = 417.3± 0.9
σ = 19.7± 0.8



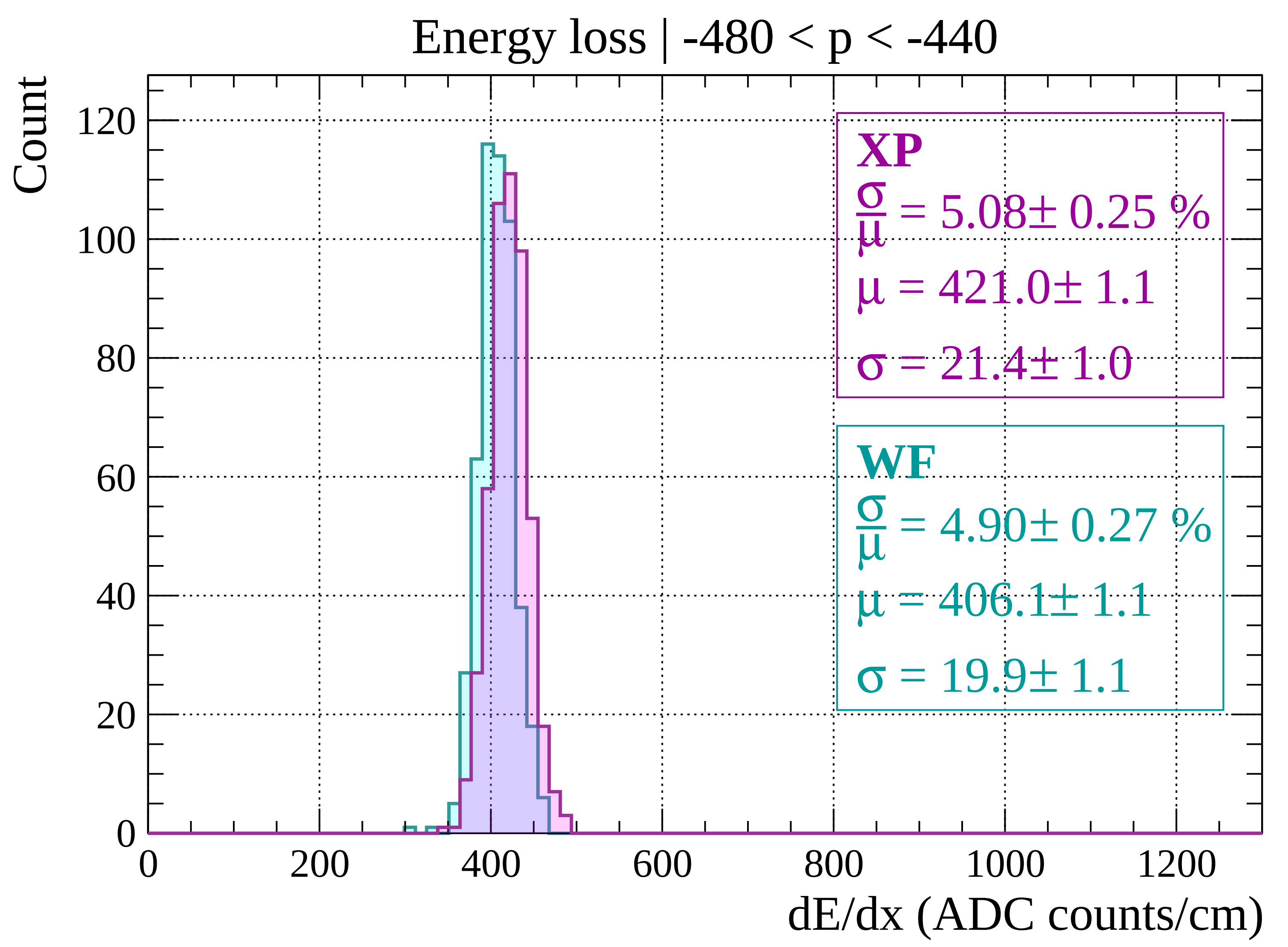
Energy loss | -600 < p < -560
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
200
XP
$\frac{\sigma}{\mu}$ = 5.01± 0.22 %
μ = 428.7± 1.0
σ = 21.5± 0.9
WF
$\frac{\sigma}{\mu}$ = 5.04± 0.21 %
μ = 414.3± 1.0
σ = 20.9± 0.8



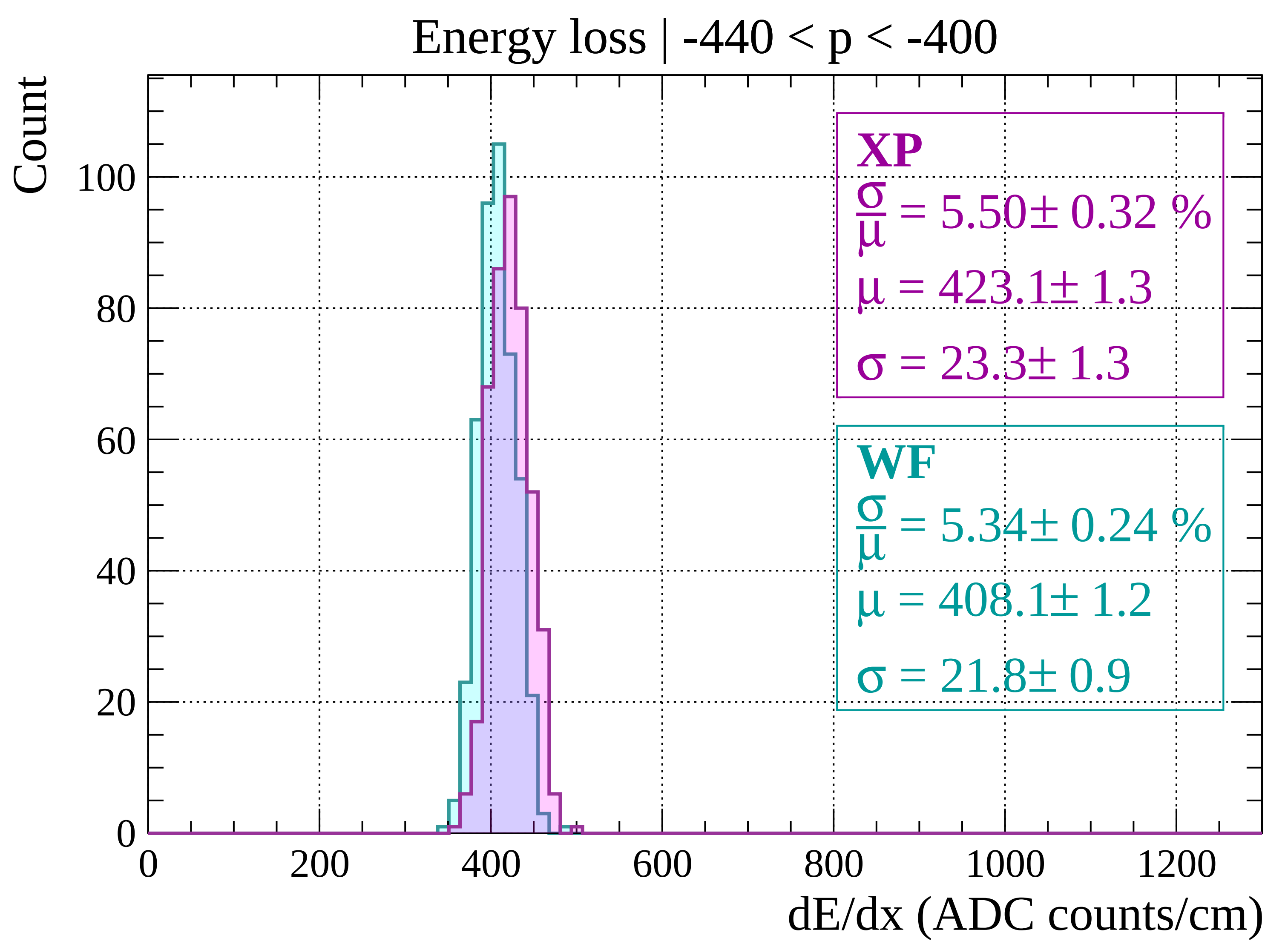
Energy loss | -560 < p < -520
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
0
200
400
600
800
1000
1200
XP
σ/μ = 4.86± 0.26 %
μ = 426.9± 1.0
σ = 20.8± 1.1
WF
σ/μ = 4.44± 0.26 %
μ = 411.3± 0.9
σ = 18.2± 1.0



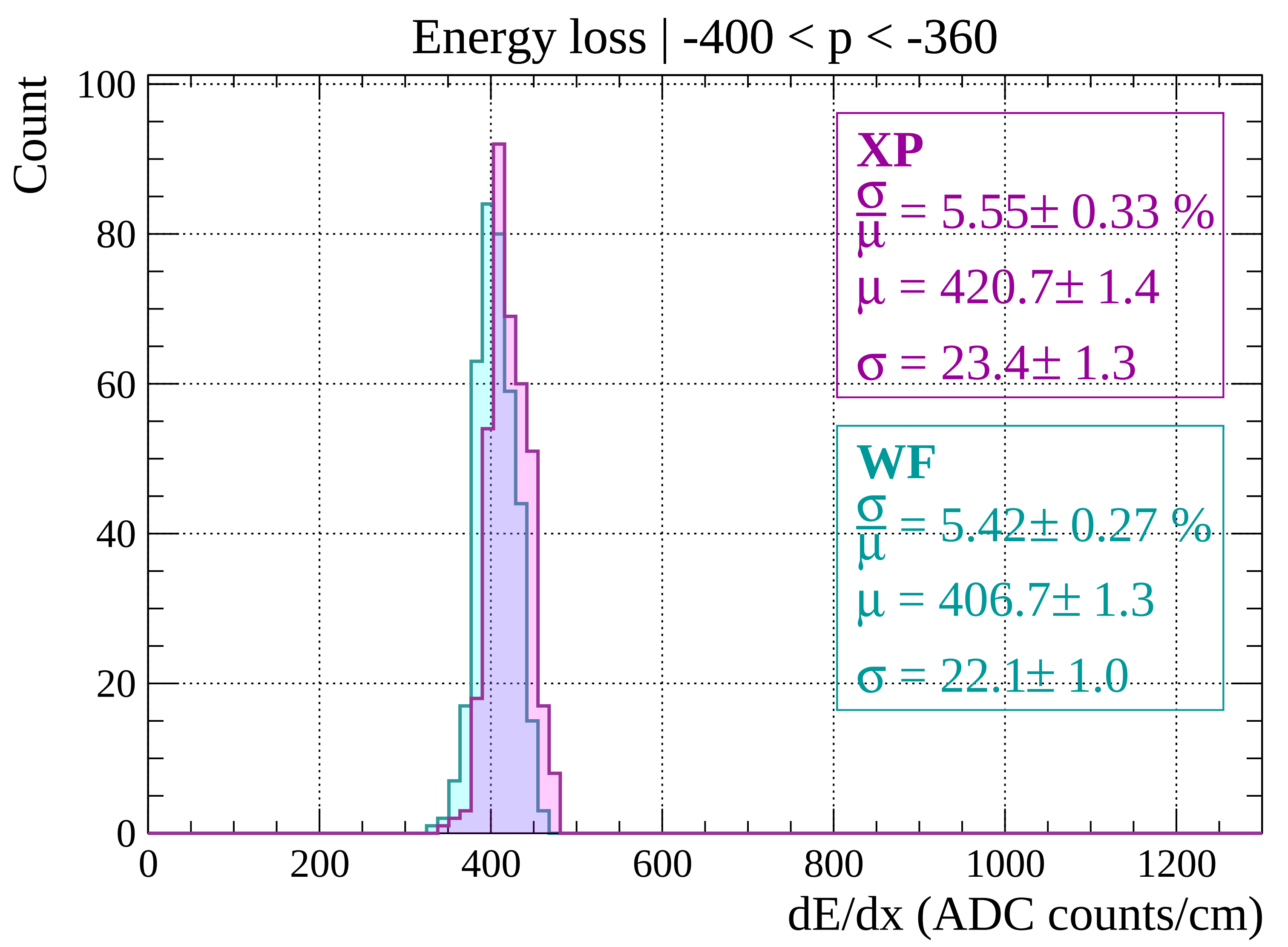
Energy loss | -520 < p < -480
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 4.94± 0.22 %
μ = 424.6± 1.0
σ = 21.0± 0.9
WF
$\frac{\sigma}{\mu}$ = 4.98± 0.22 %
μ = 410.9± 1.0
σ = 20.5± 0.9



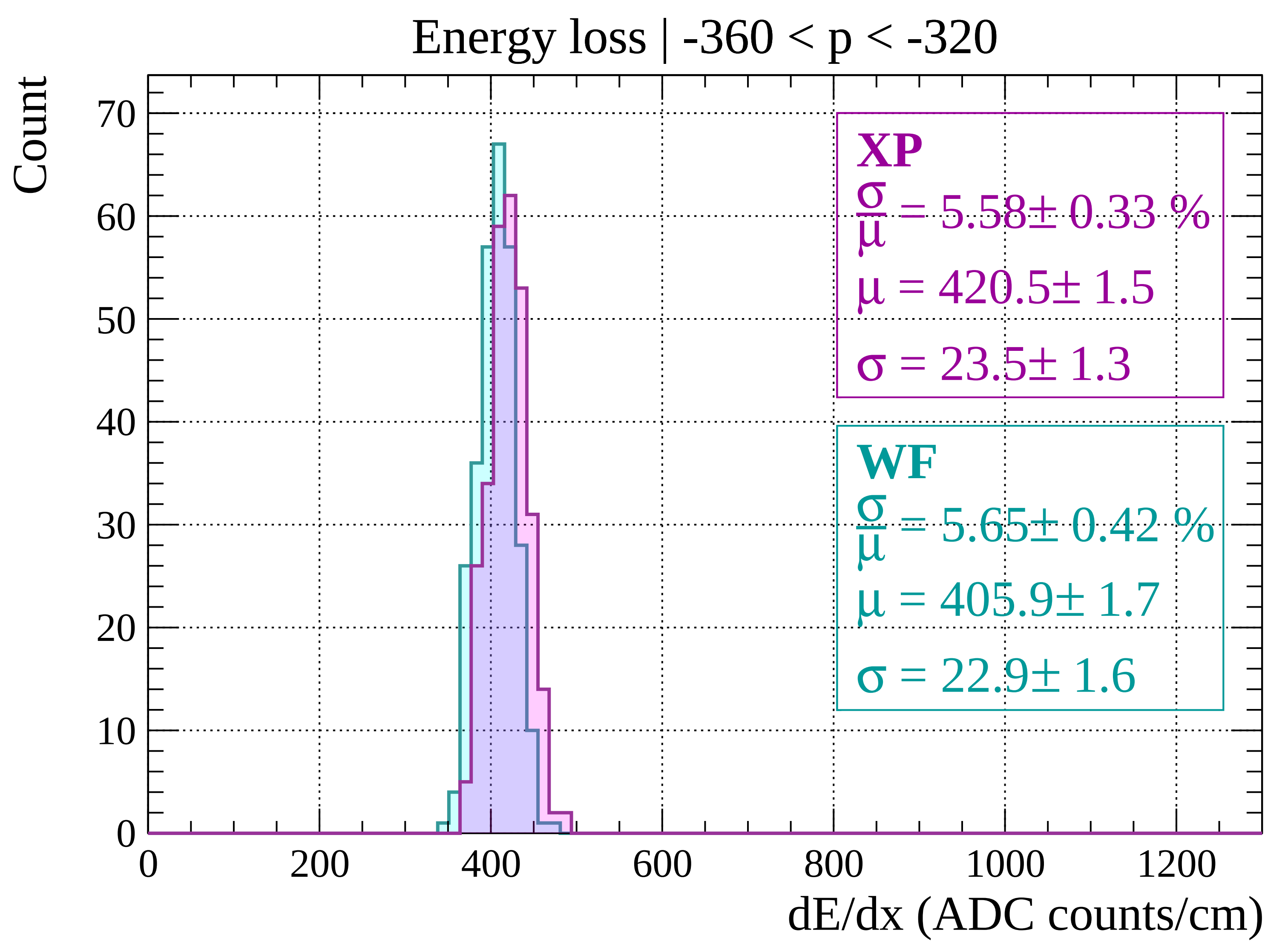
Energy loss | -480 < p < -440
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 5.08± 0.25 %
μ = 421.0± 1.1
σ = 21.4± 1.0
WF
$\frac{\sigma}{\mu}$ = 4.90± 0.27 %
μ = 406.1± 1.1
σ = 19.9± 1.1



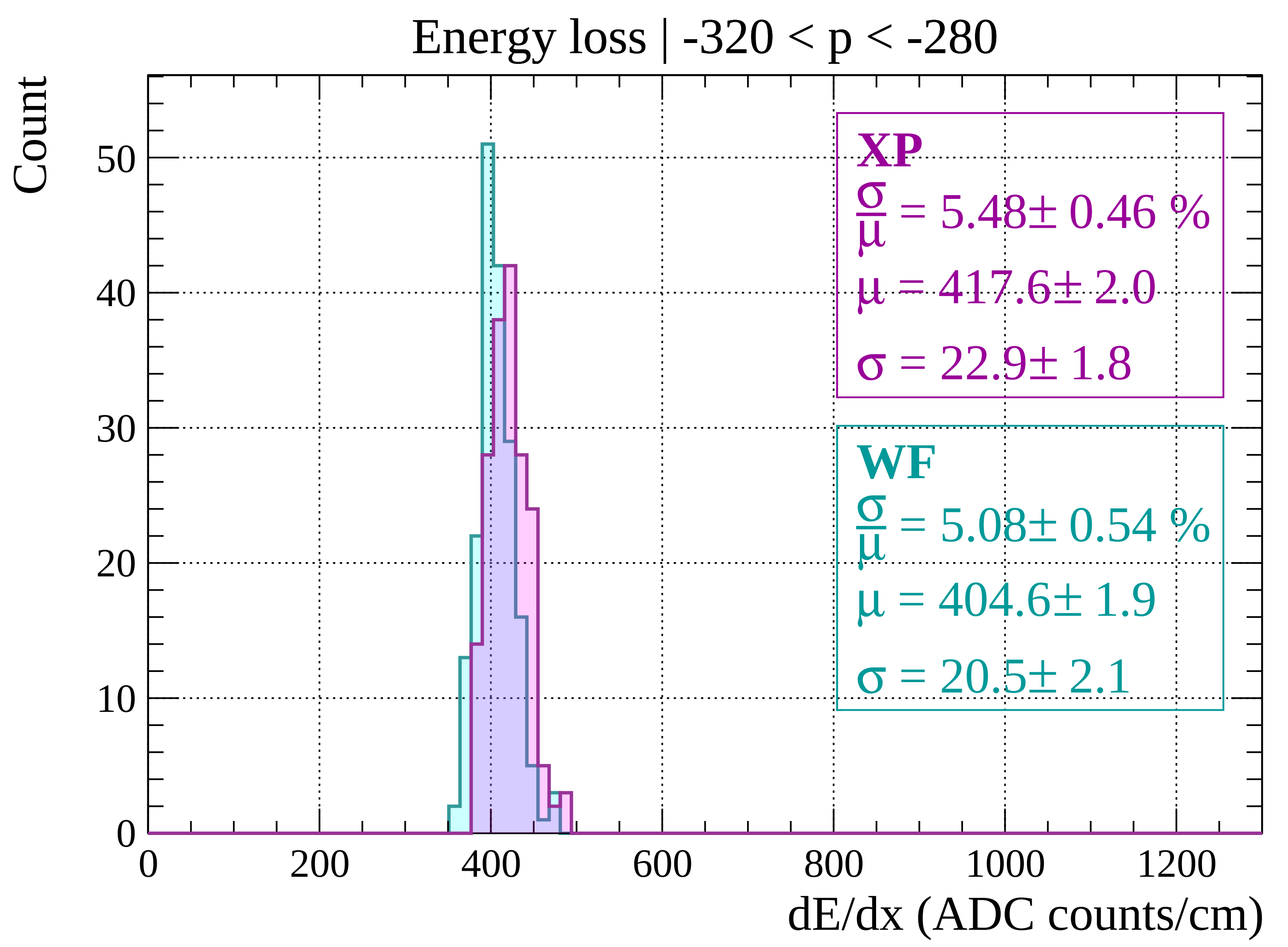
Energy loss | -440 < p < -400
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 5.50 ± 0.32 %
μ = 423.1 ± 1.3
σ = 23.3 ± 1.3
WF
$\frac{\sigma}{\mu}$ = 5.34 ± 0.24 %
μ = 408.1 ± 1.2
σ = 21.8 ± 0.9



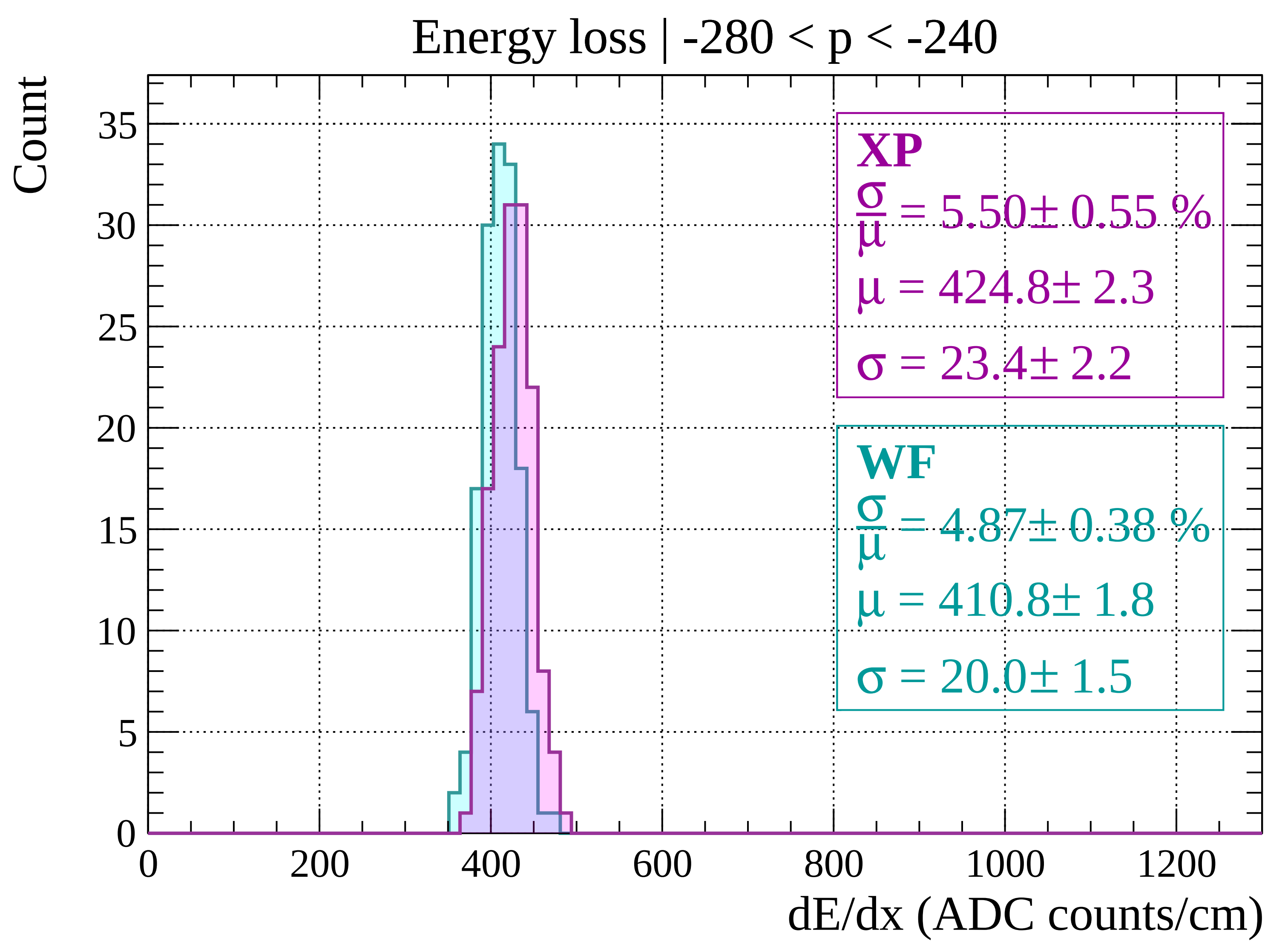
Energy loss | -400 < p < -360
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
XP
$\frac{\sigma}{\mu}$ = 5.55± 0.33 %
μ = 420.7± 1.4
σ = 23.4± 1.3
WF
$\frac{\sigma}{\mu}$ = 5.42± 0.27 %
μ = 406.7± 1.3
σ = 22.1± 1.0



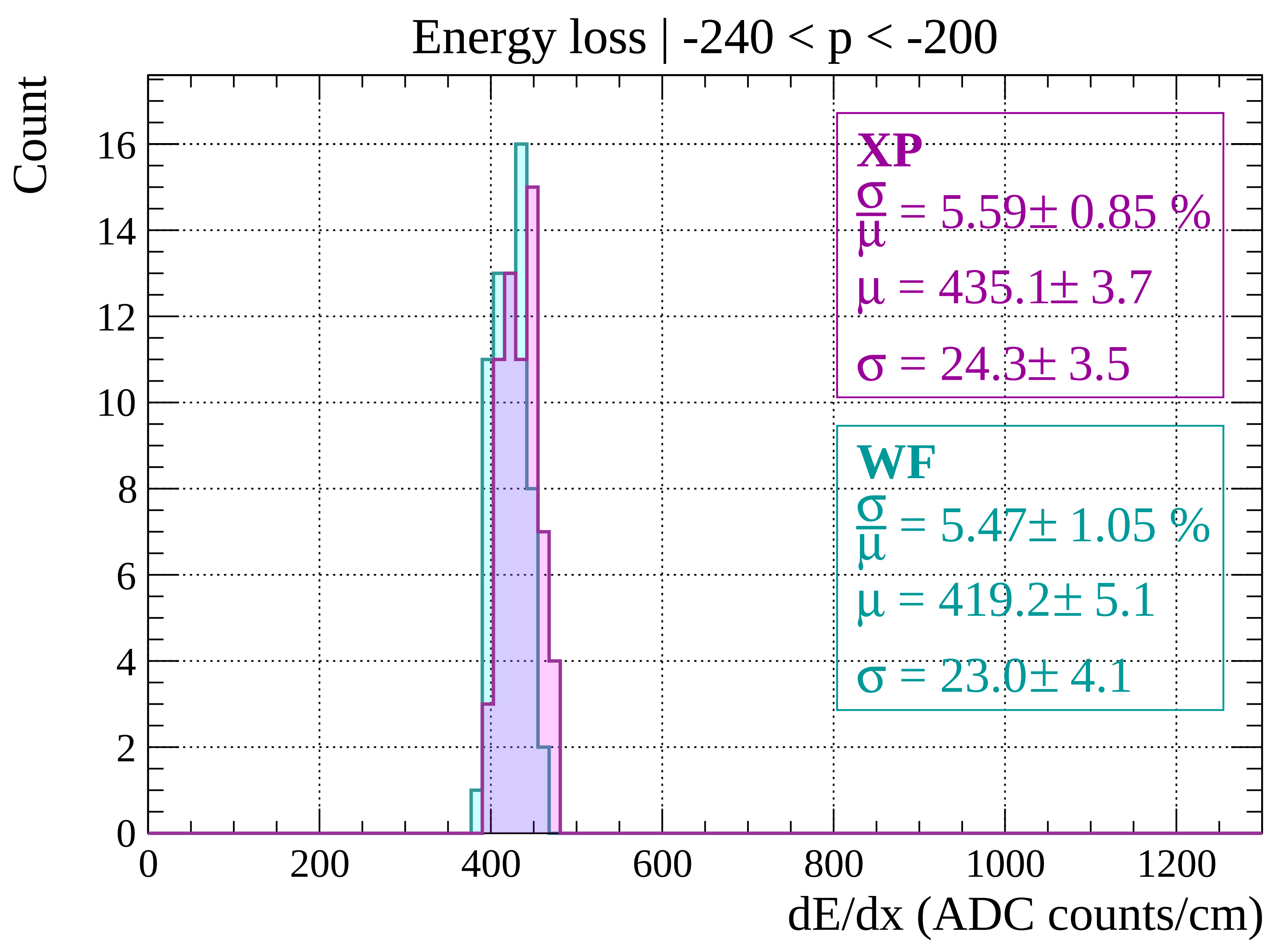
Energy loss | -360 < p < -320
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
10
20
30
40
50
60
70
XP
σ/μ = 5.58± 0.33 %
μ = 420.5± 1.5
σ = 23.5± 1.3
WF
σ/μ = 5.65± 0.42 %
μ = 405.9± 1.7
σ = 22.9± 1.6



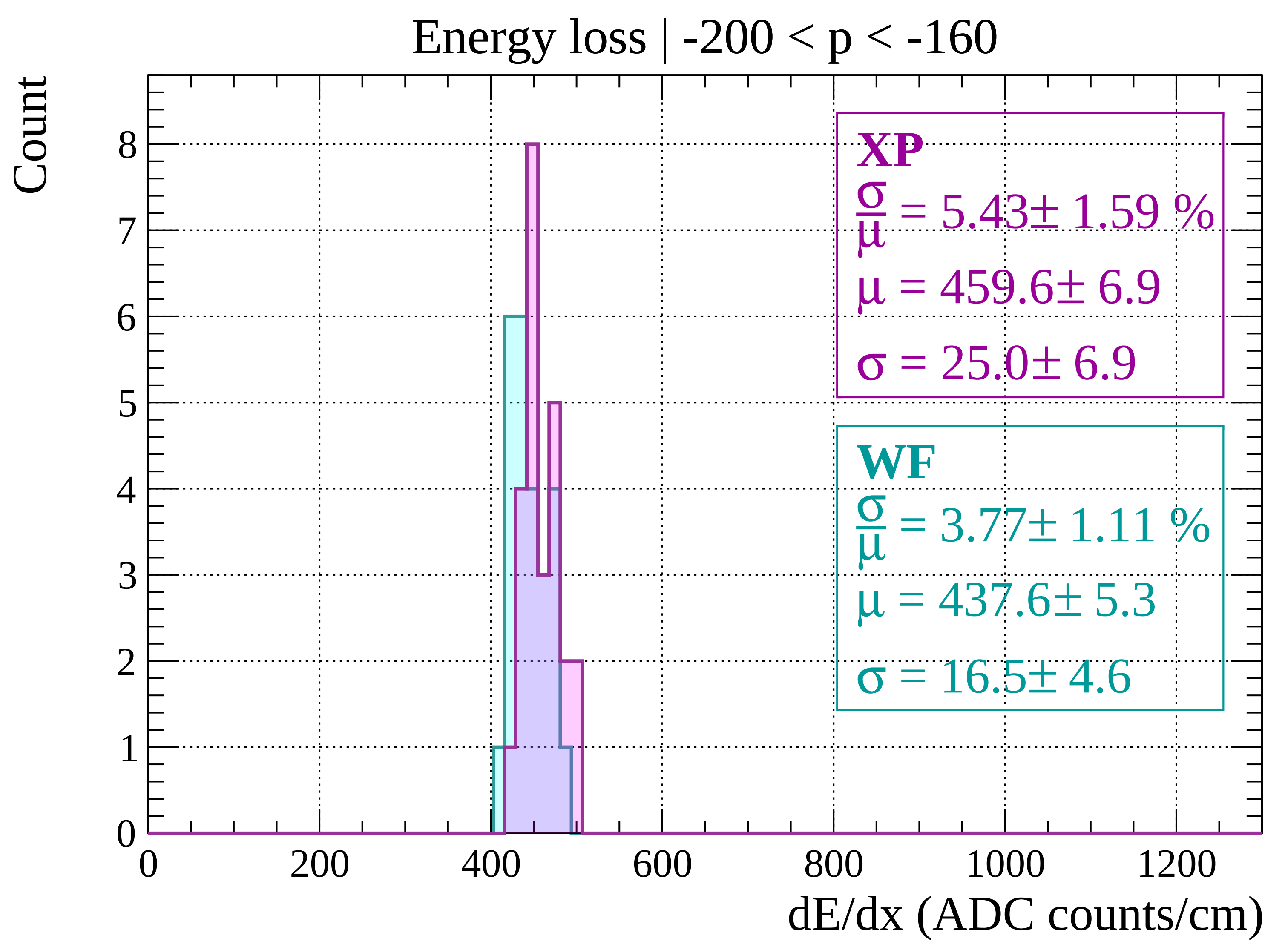
Energy loss | -320 < p < -280
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
0
200
400
600
800
1000
1200
XP
σ/μ = 5.48± 0.46 %
μ = 417.6± 2.0
σ = 22.9± 1.8
WF
σ/μ = 5.08± 0.54 %
μ = 404.6± 1.9
σ = 20.5± 2.1

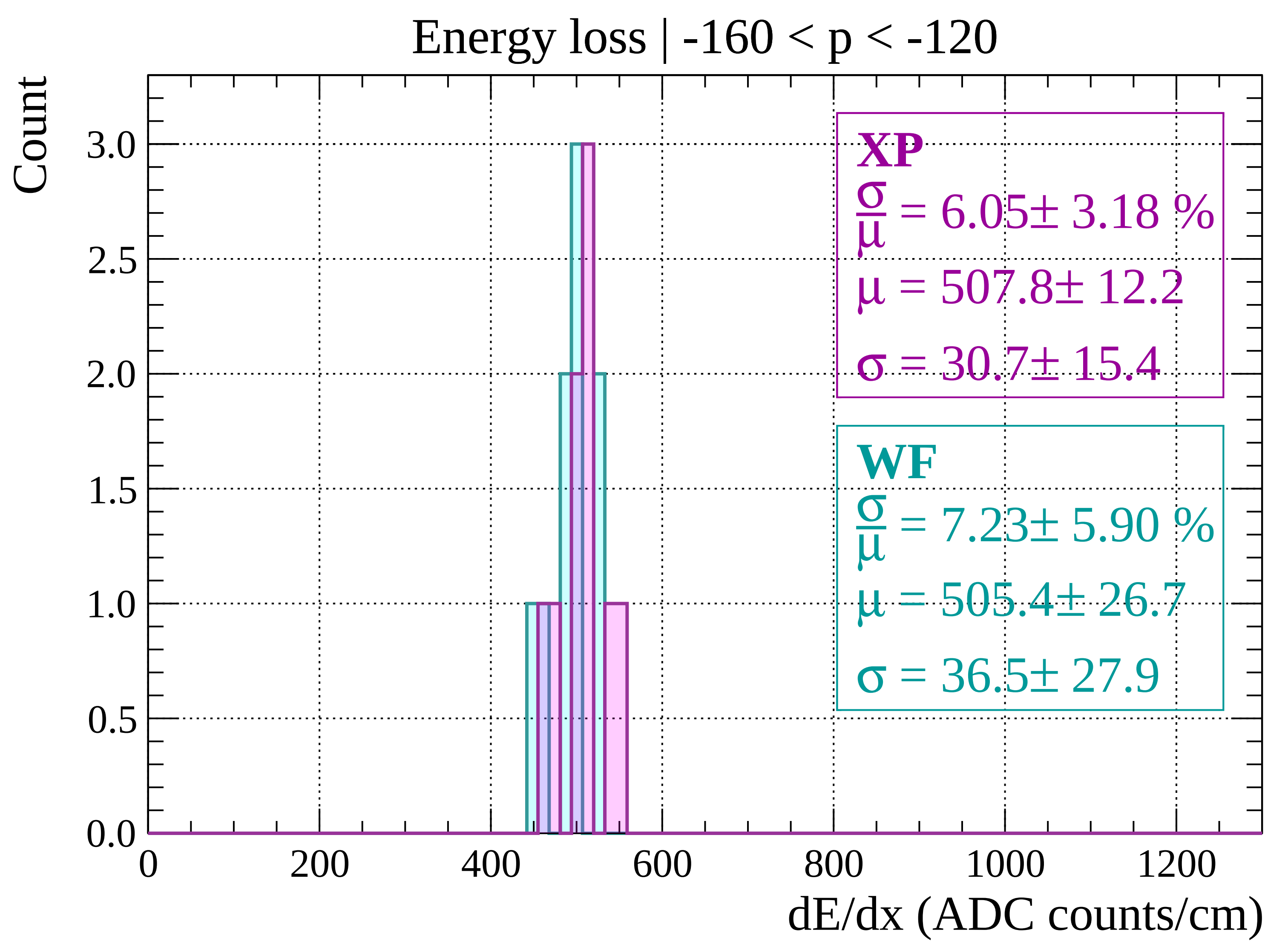


Energy loss | -280 < p < -240
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
30
35
0
200
400
600
800
1000
1200
XP
σ/μ = 5.50± 0.55 %
μ = 424.8± 2.3
σ = 23.4± 2.2
WF
σ/μ = 4.87± 0.38 %
μ = 410.8± 1.8
σ = 20.0± 1.5

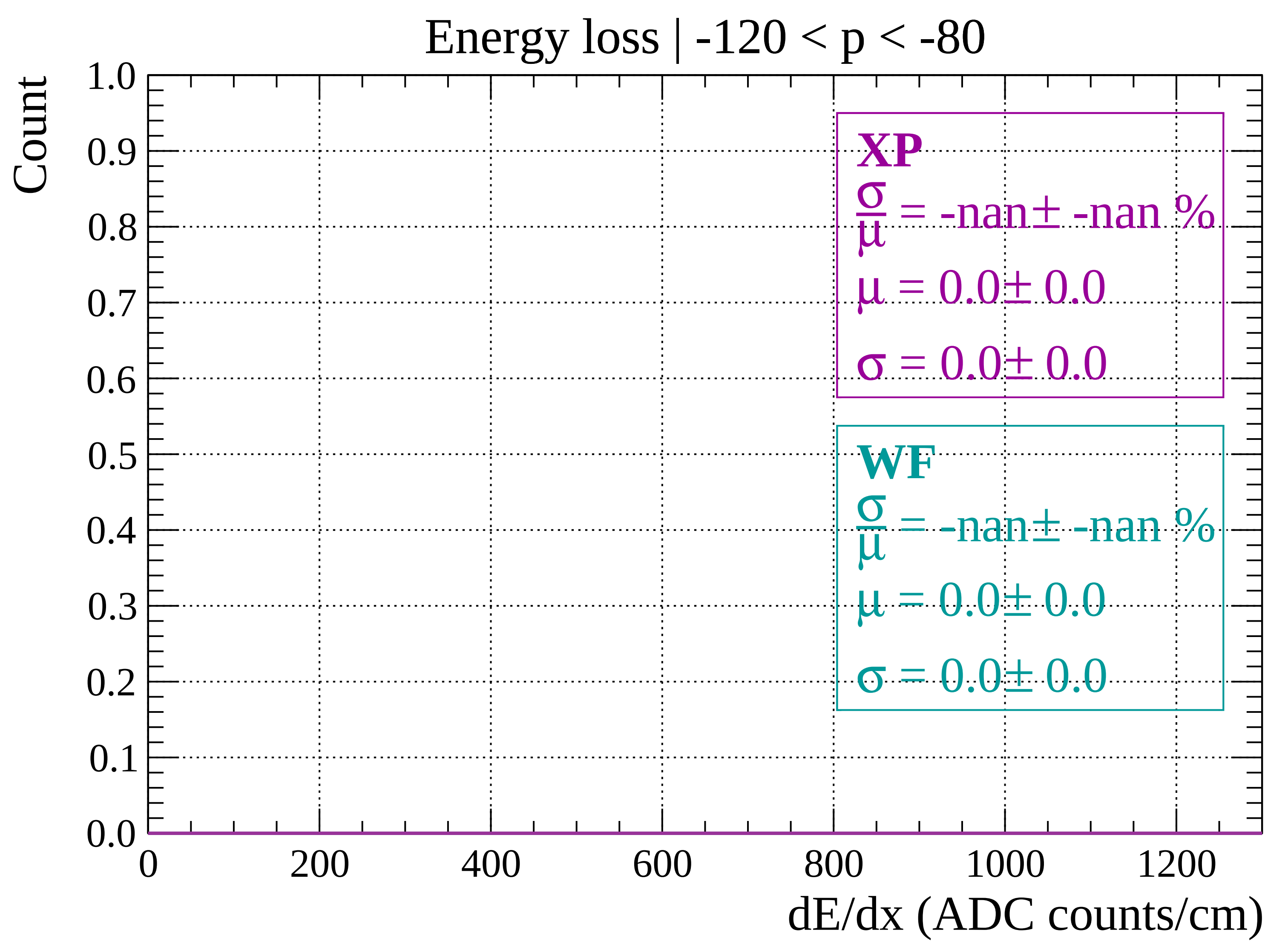


Energy loss | -240 < p < -200
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 5.59± 0.85 %
μ = 435.1± 3.7
σ = 24.3± 3.5
WF
σ/μ = 5.47± 1.05 %
μ = 419.2± 5.1
σ = 23.0± 4.1

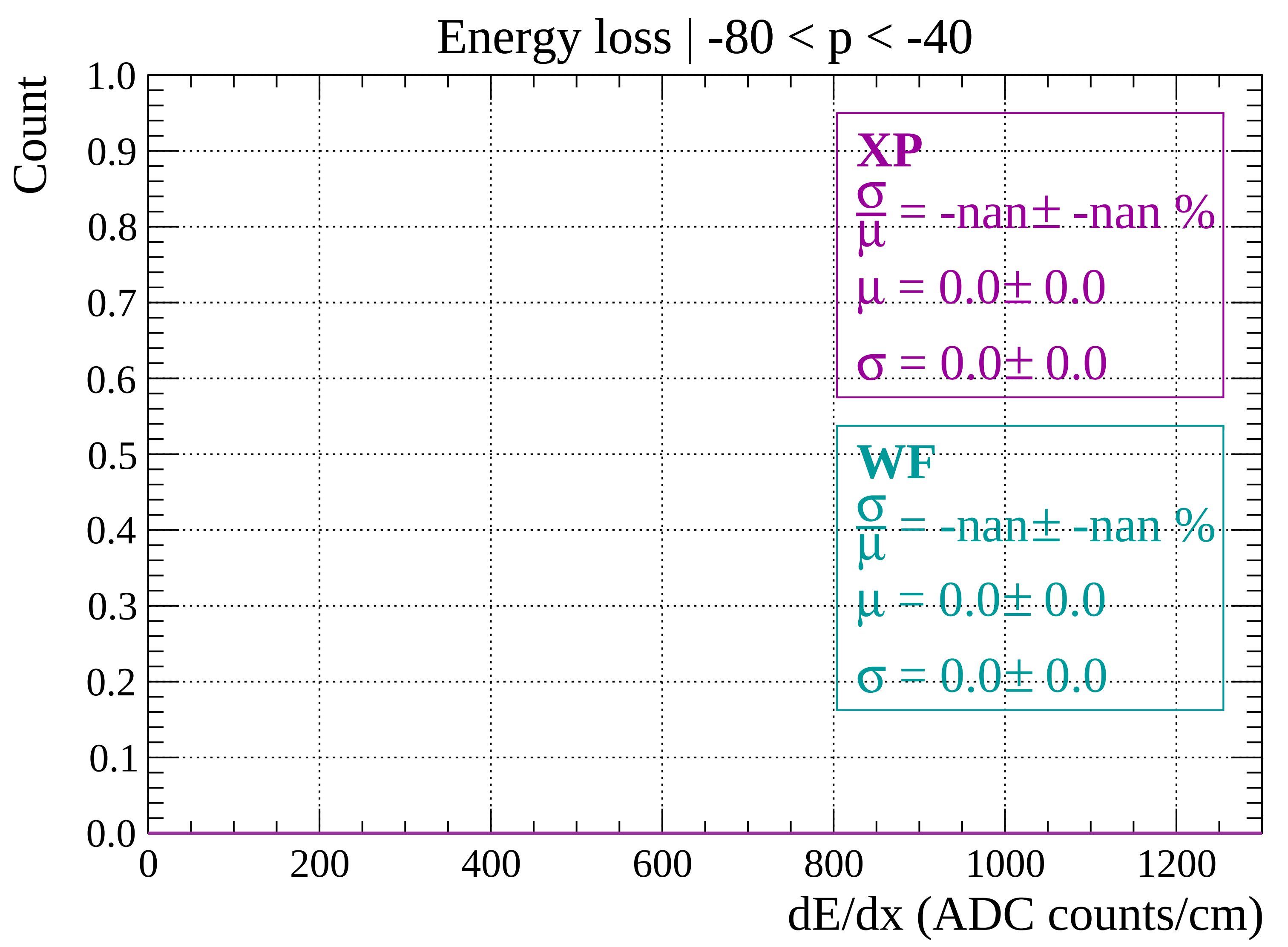

Energy loss | -200 < p < -160
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 5.43± 1.59 %
μ = 459.6± 6.9
σ = 25.0± 6.9
WF
$\frac{\sigma}{\mu}$ = 3.77± 1.11 %
μ = 437.6± 5.3
σ = 16.5± 4.6
0
1
2
3
4
5
6
7
8
0
200
400
600
800
1000
1200

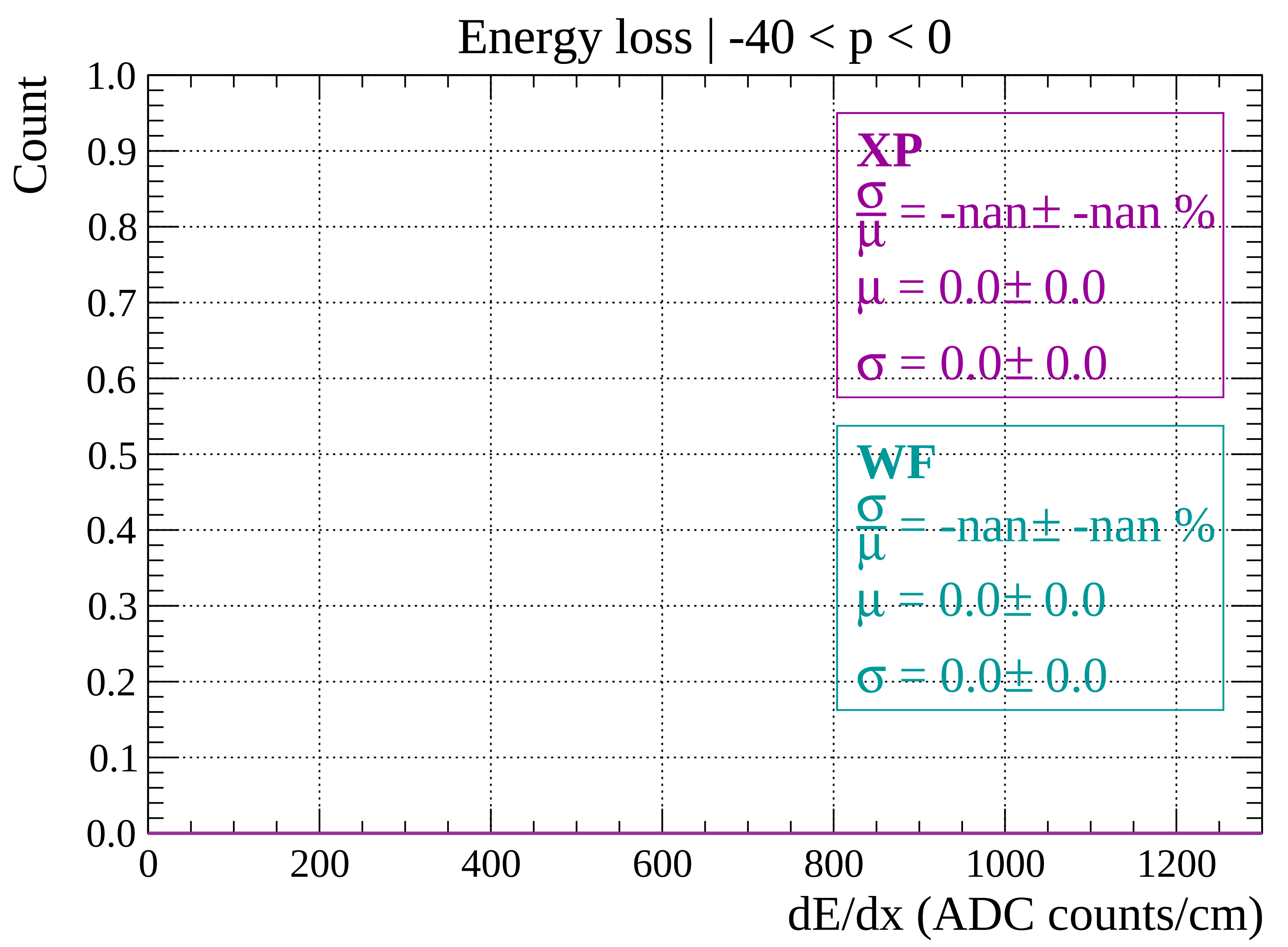


Energy loss | -160 < p < -120
Count
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 6.05± 3.18 %
μ = 507.8± 12.2
σ = 30.7± 15.4
WF
$\frac{\sigma}{\mu}$ = 7.23± 5.90 %
μ = 505.4± 26.7
σ = 36.5± 27.9

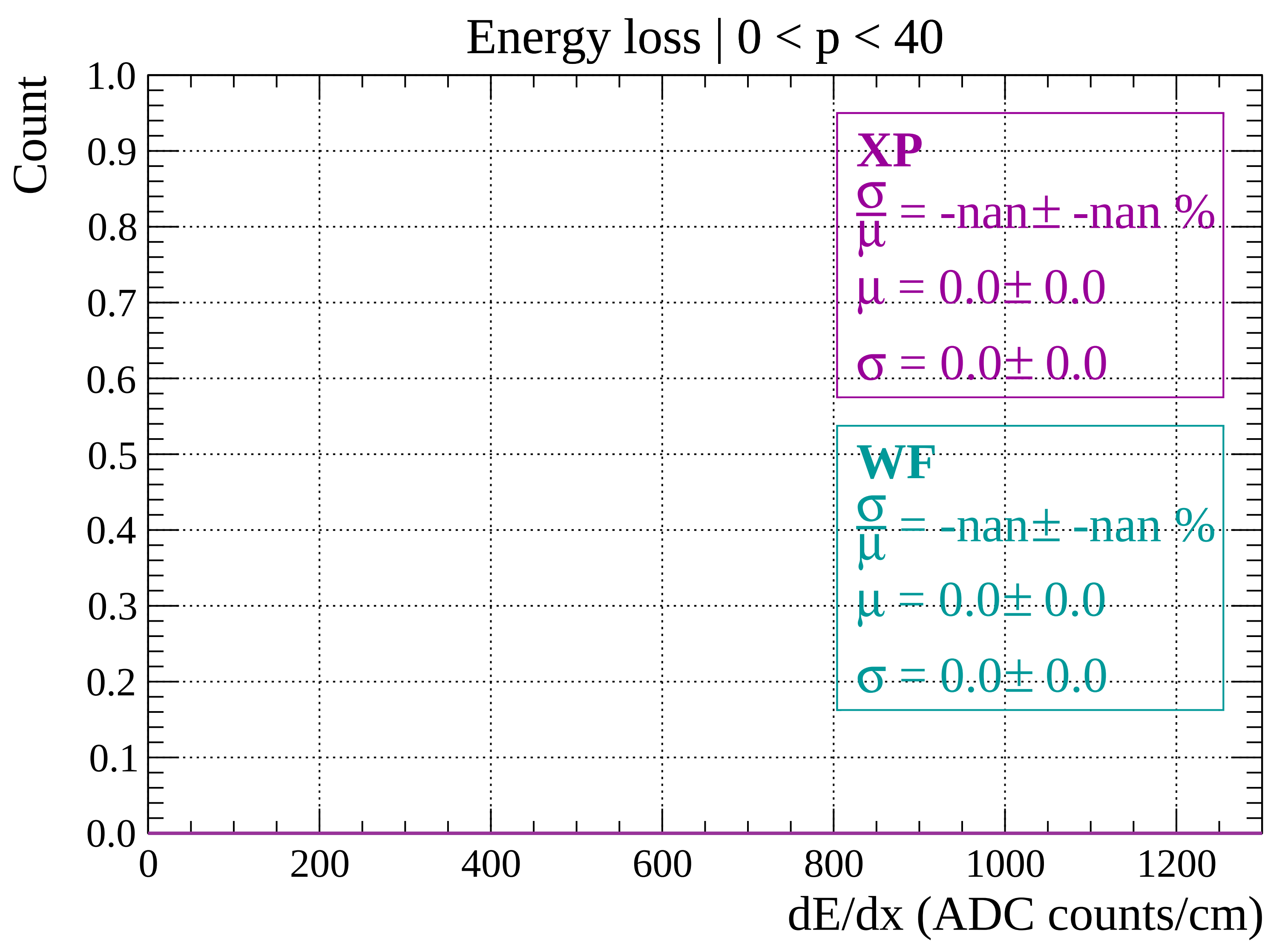


Energy loss | -120 < p < -80
Count
dE/dx (ADC counts/cm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

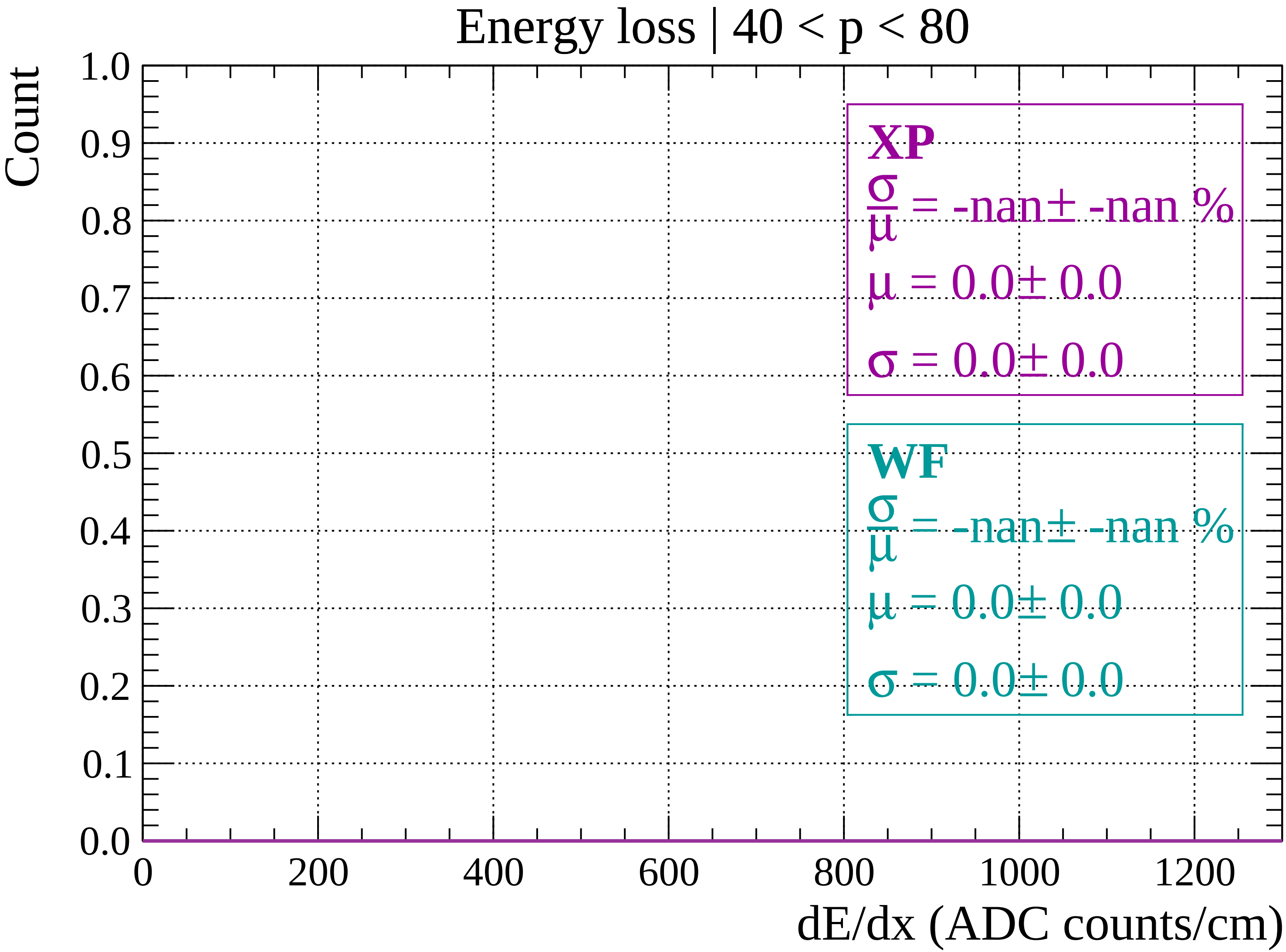

Energy loss | -80 < p < -40
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



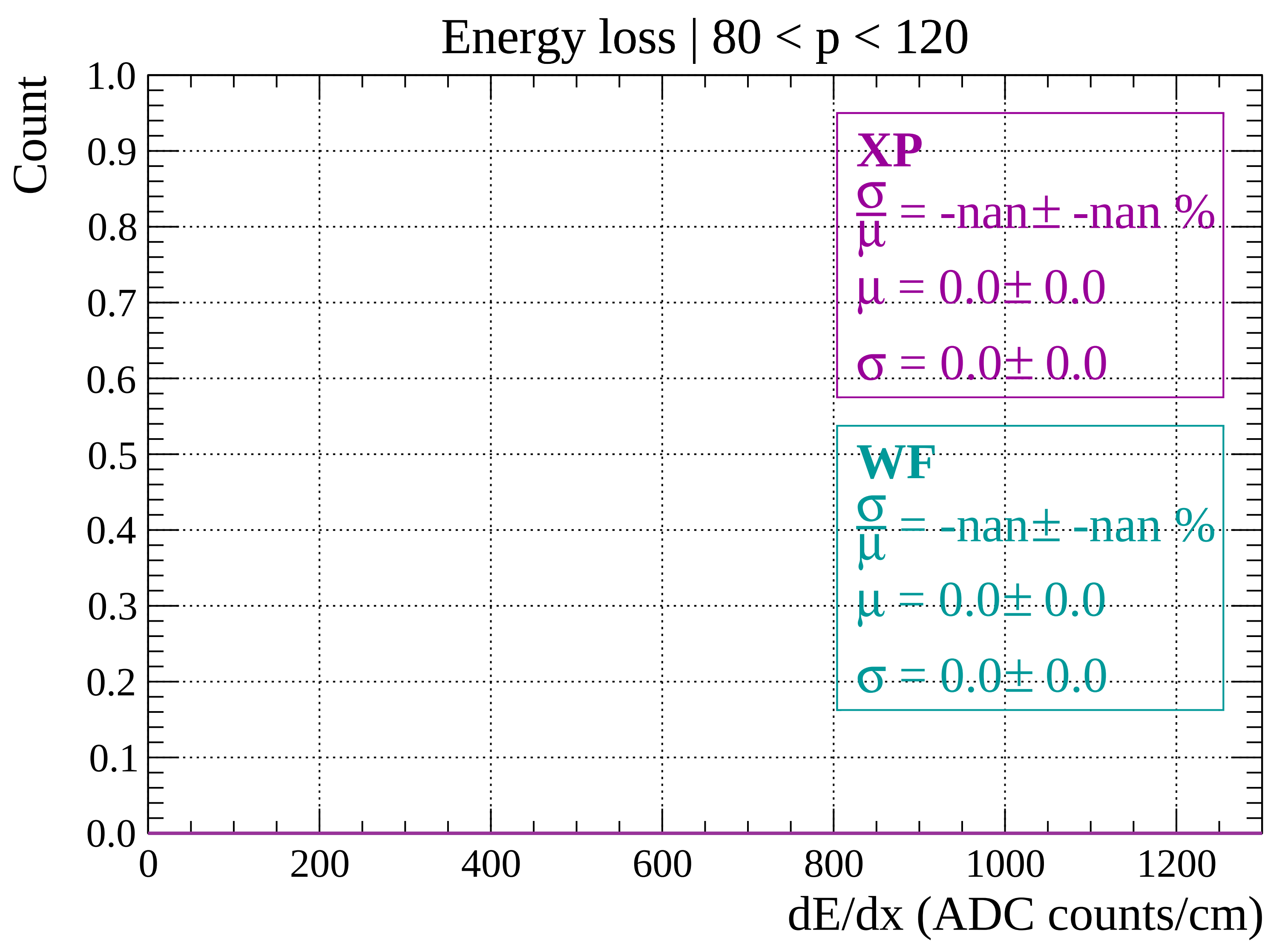
Energy loss | -40 < p < 0
Count
dE/dx (ADC counts/cm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



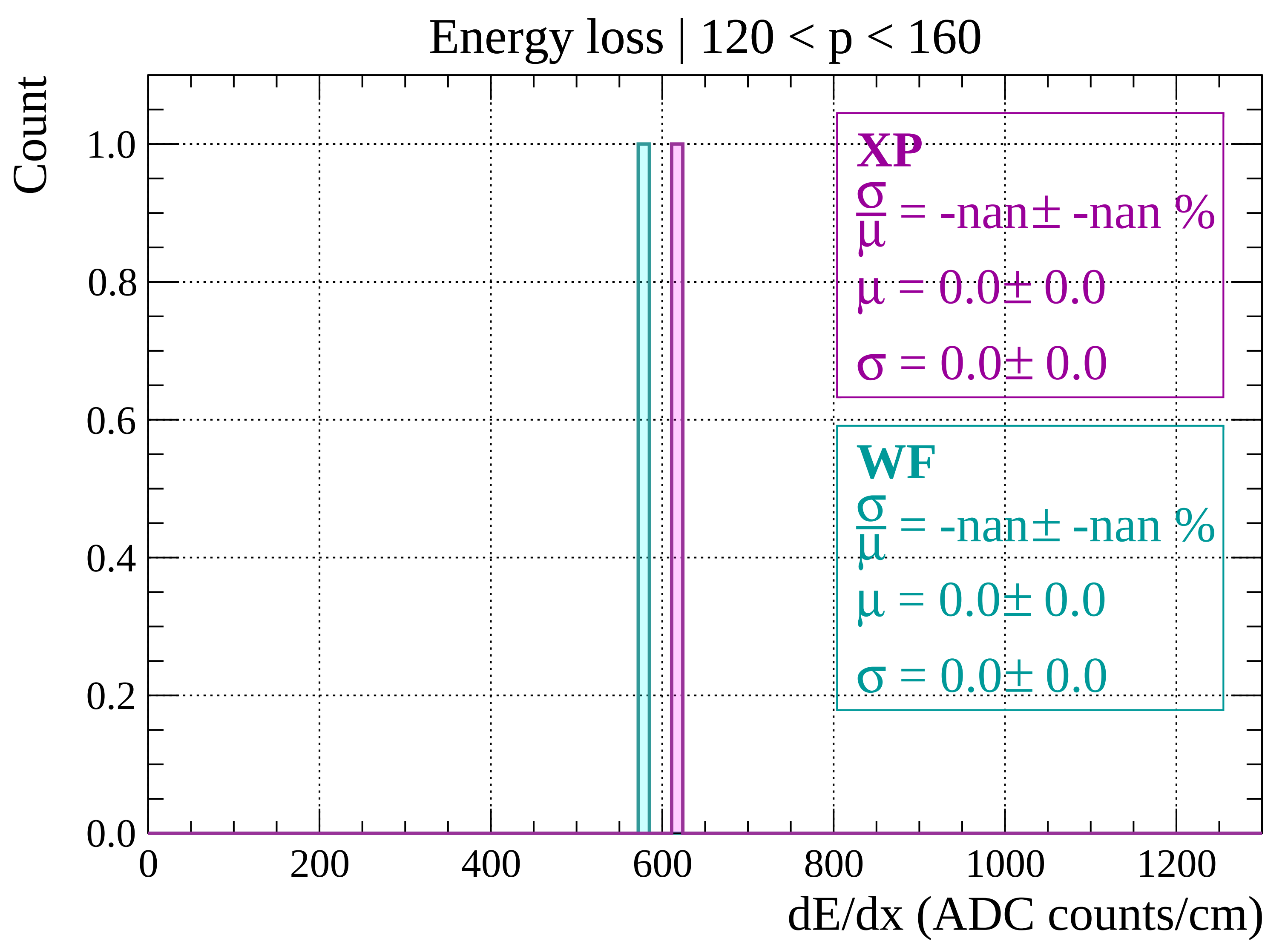
Energy loss | 0 < p < 40
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



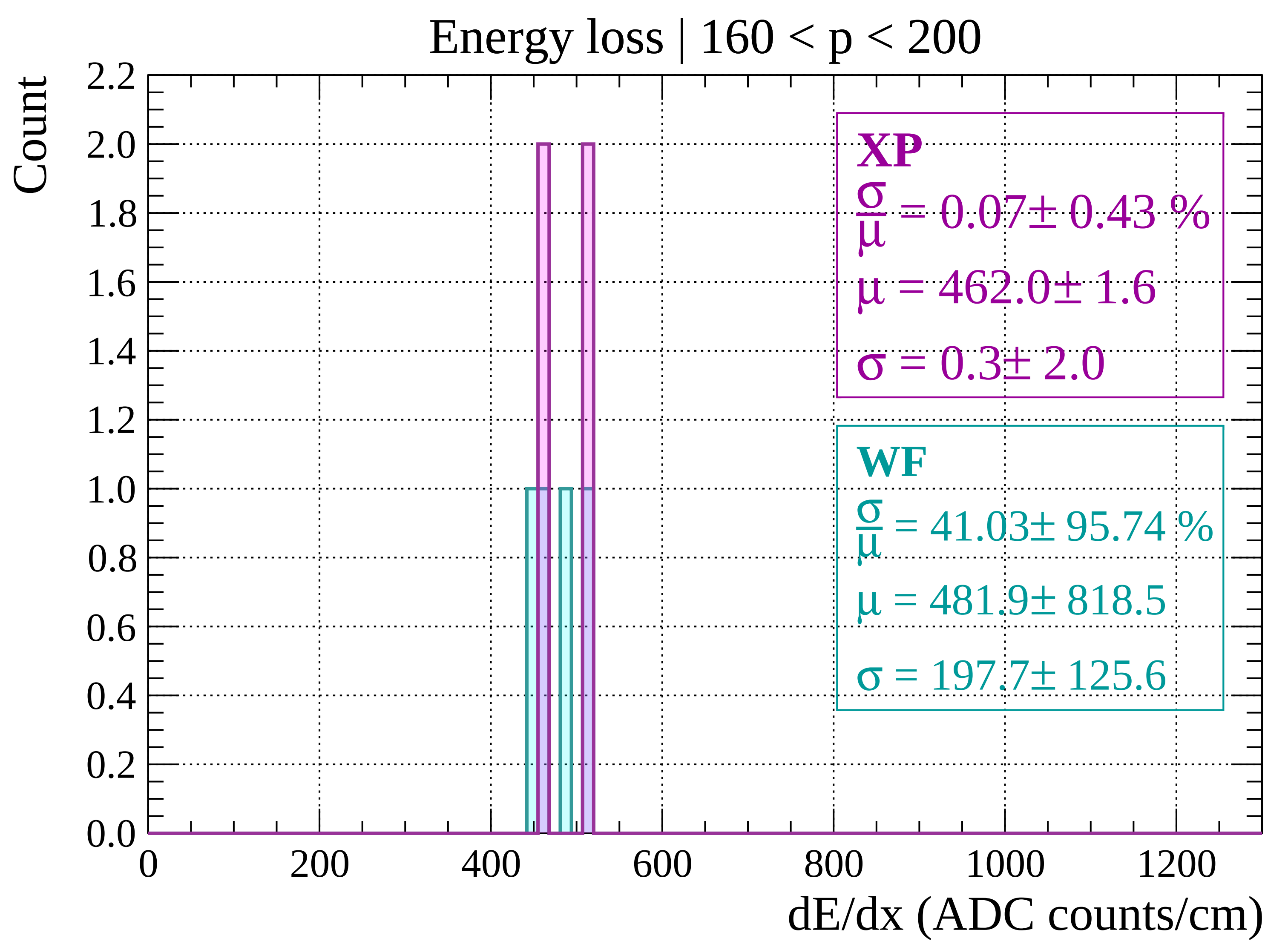
Energy loss | 40 < p < 80
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



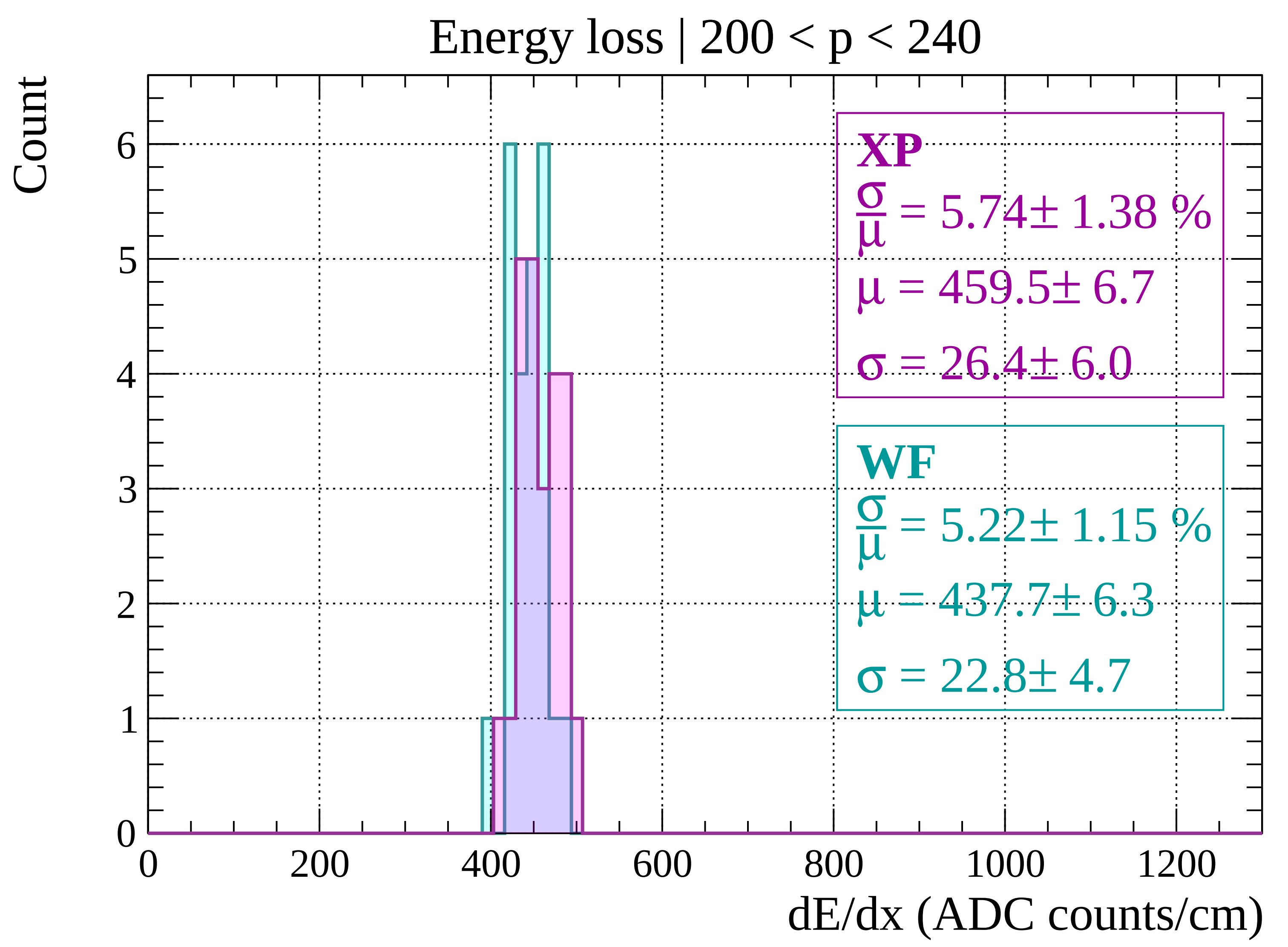
Energy loss | 80 < p < 120
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

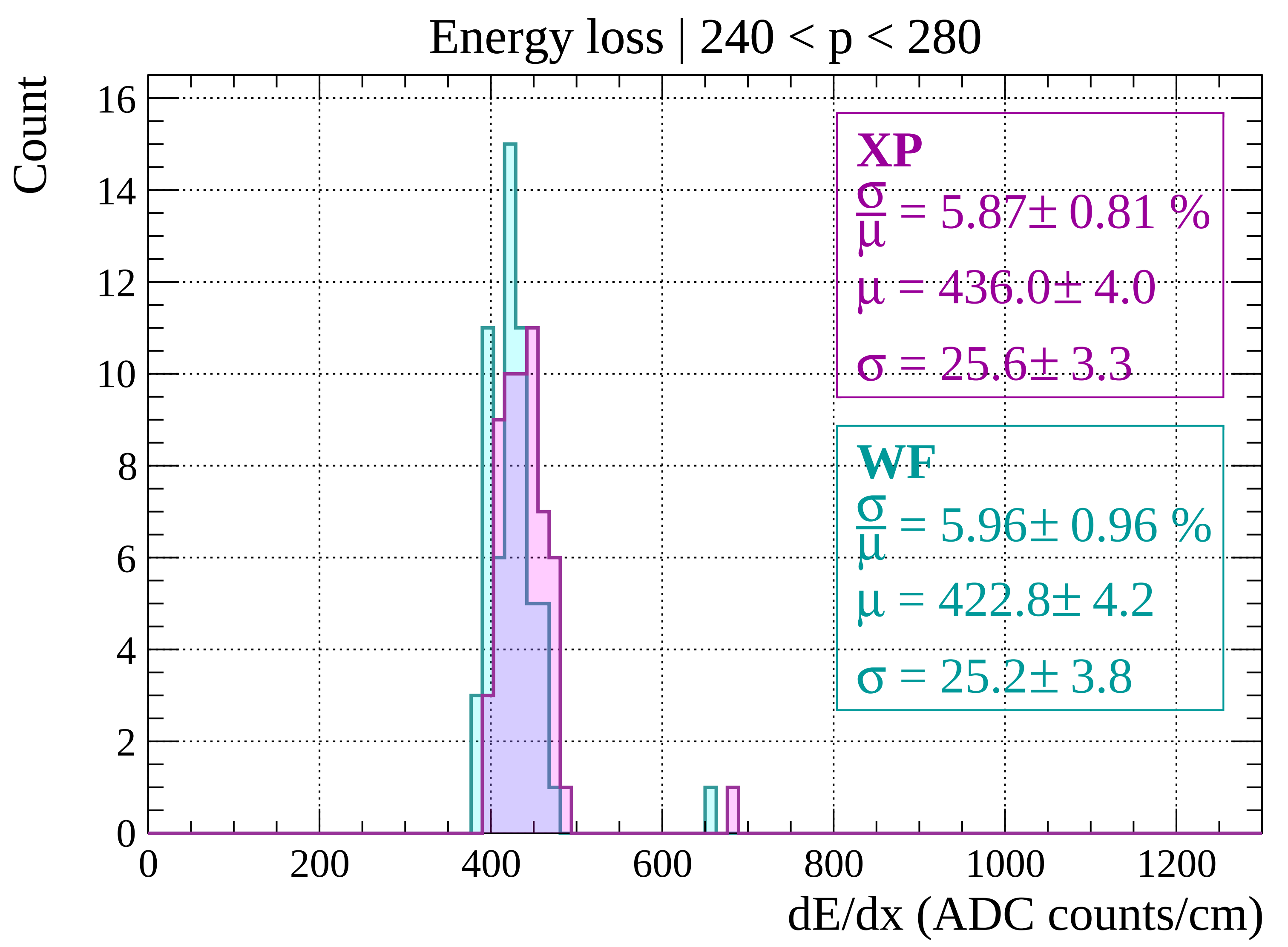


Energy loss | 120 < p < 160
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

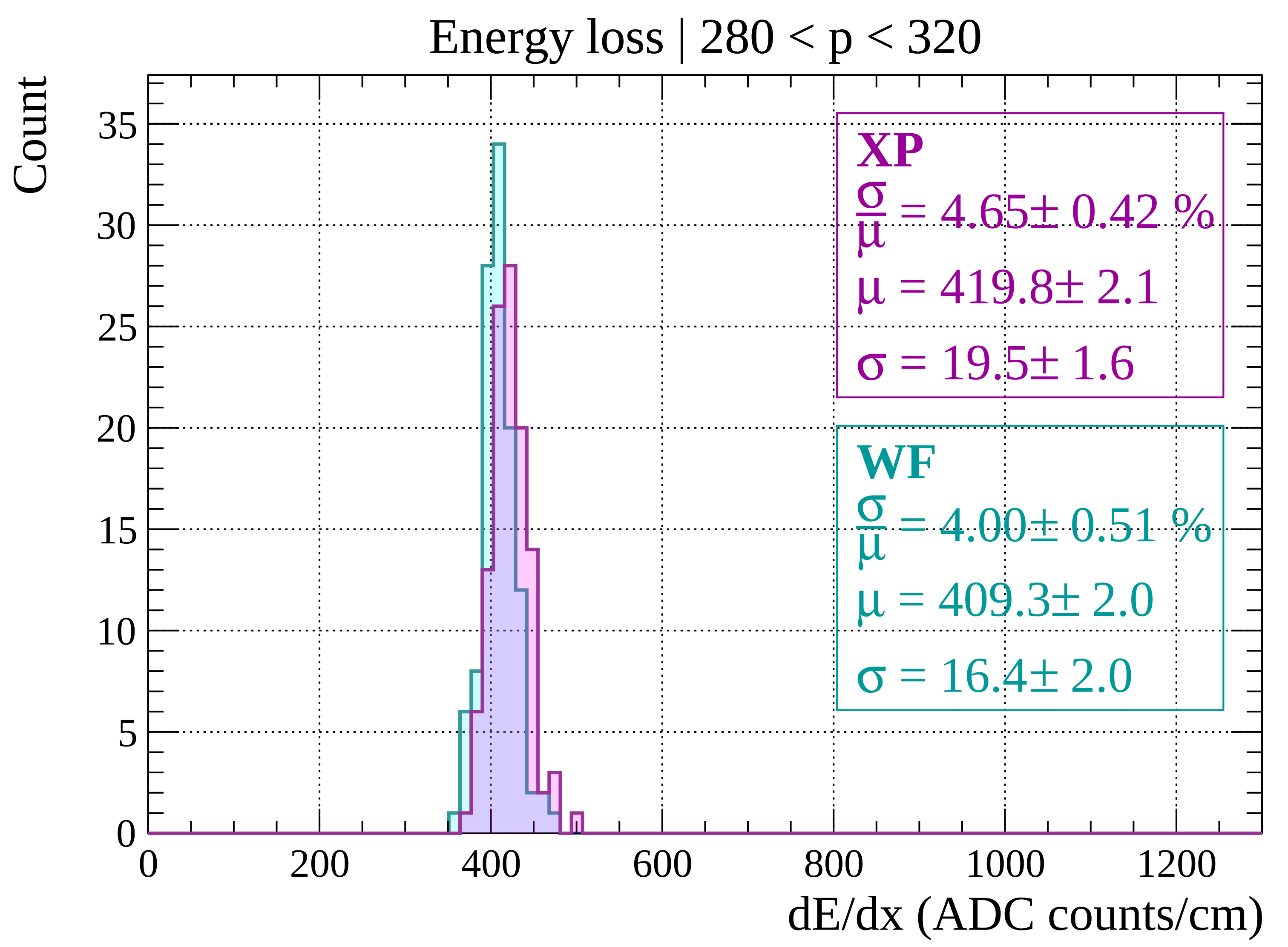


Energy loss | 160 < p < 200
Count
dE/dx (ADC counts/cm)
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
2.2
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 0.07± 0.43 %
μ = 462.0± 1.6
σ = 0.3± 2.0
WF
$\frac{\sigma}{\mu}$ = 41.03± 95.74 %
μ = 481.9± 818.5
σ = 197.7± 125.6

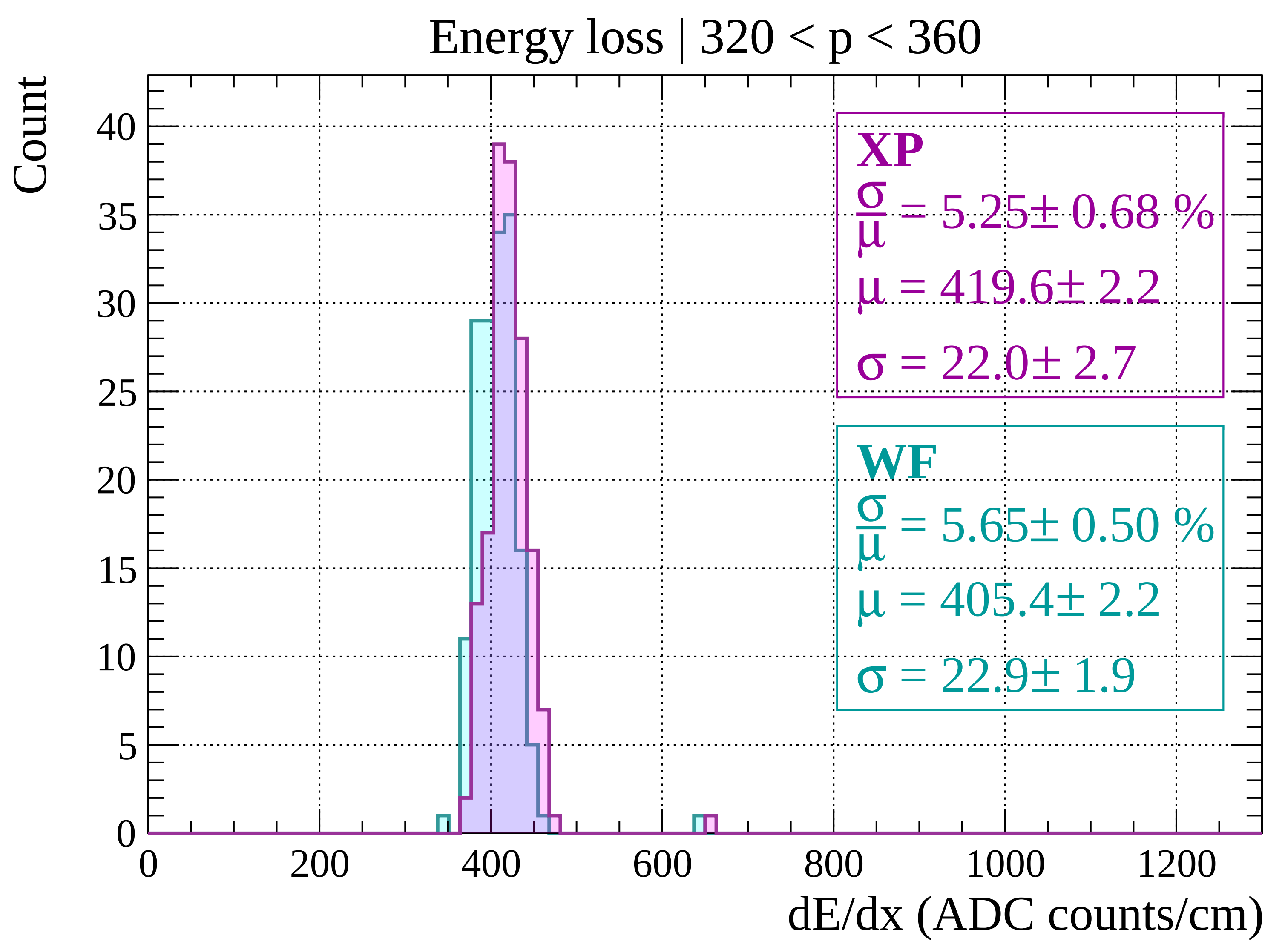

Energy loss | 200 < p < 240
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1
2
3
4
5
6
XP
σ/μ = 5.74± 1.38 %
μ = 459.5± 6.7
σ = 26.4± 6.0
WF
σ/μ = 5.22± 1.15 %
μ = 437.7± 6.3
σ = 22.8± 4.7



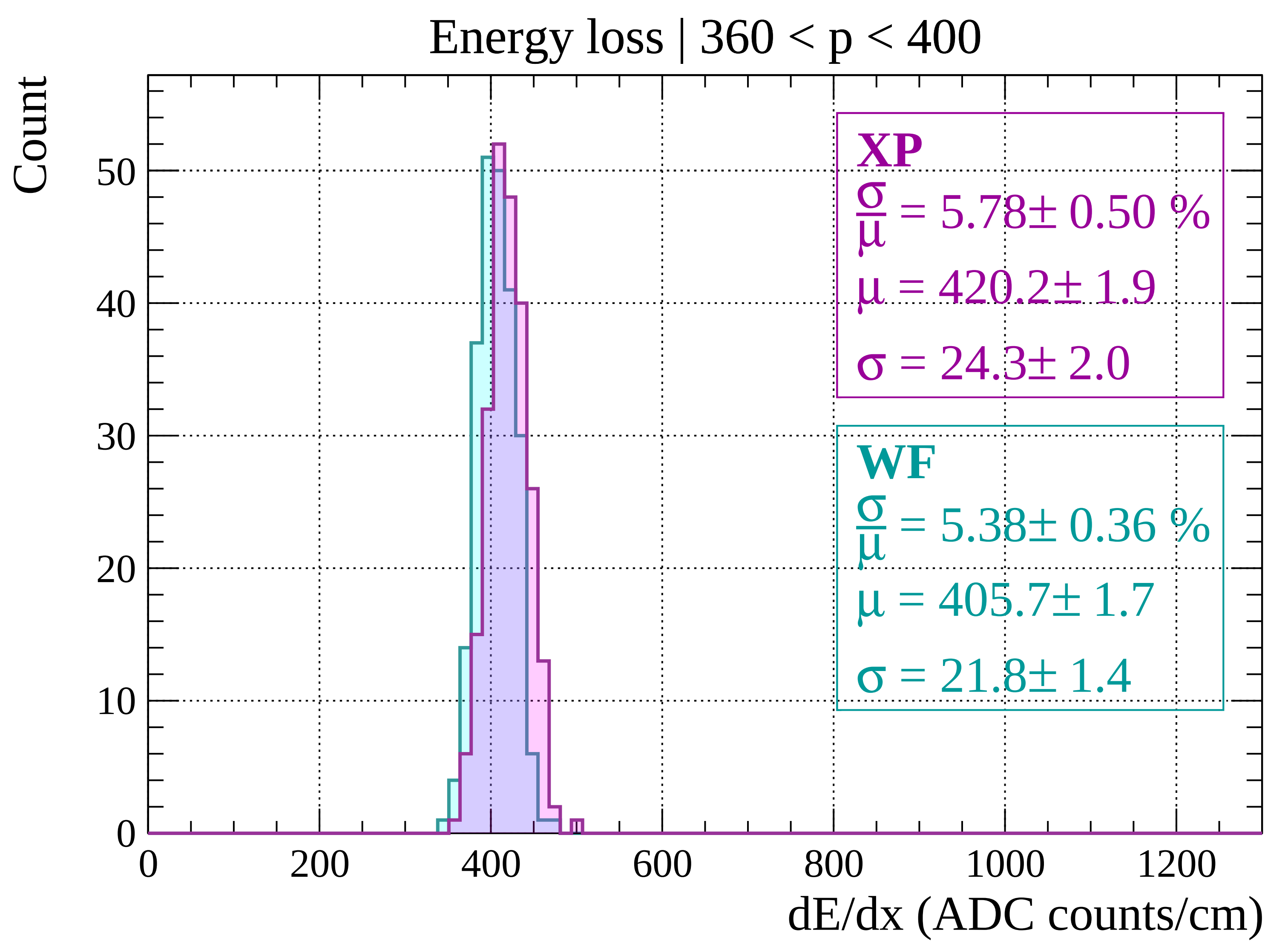
Energy loss | 240 < p < 280
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
2
4
6
8
10
12
14
16
XP
σ/μ = 5.87± 0.81 %
μ = 436.0± 4.0
σ = 25.6± 3.3
WF
σ/μ = 5.96± 0.96 %
μ = 422.8± 4.2
σ = 25.2± 3.8



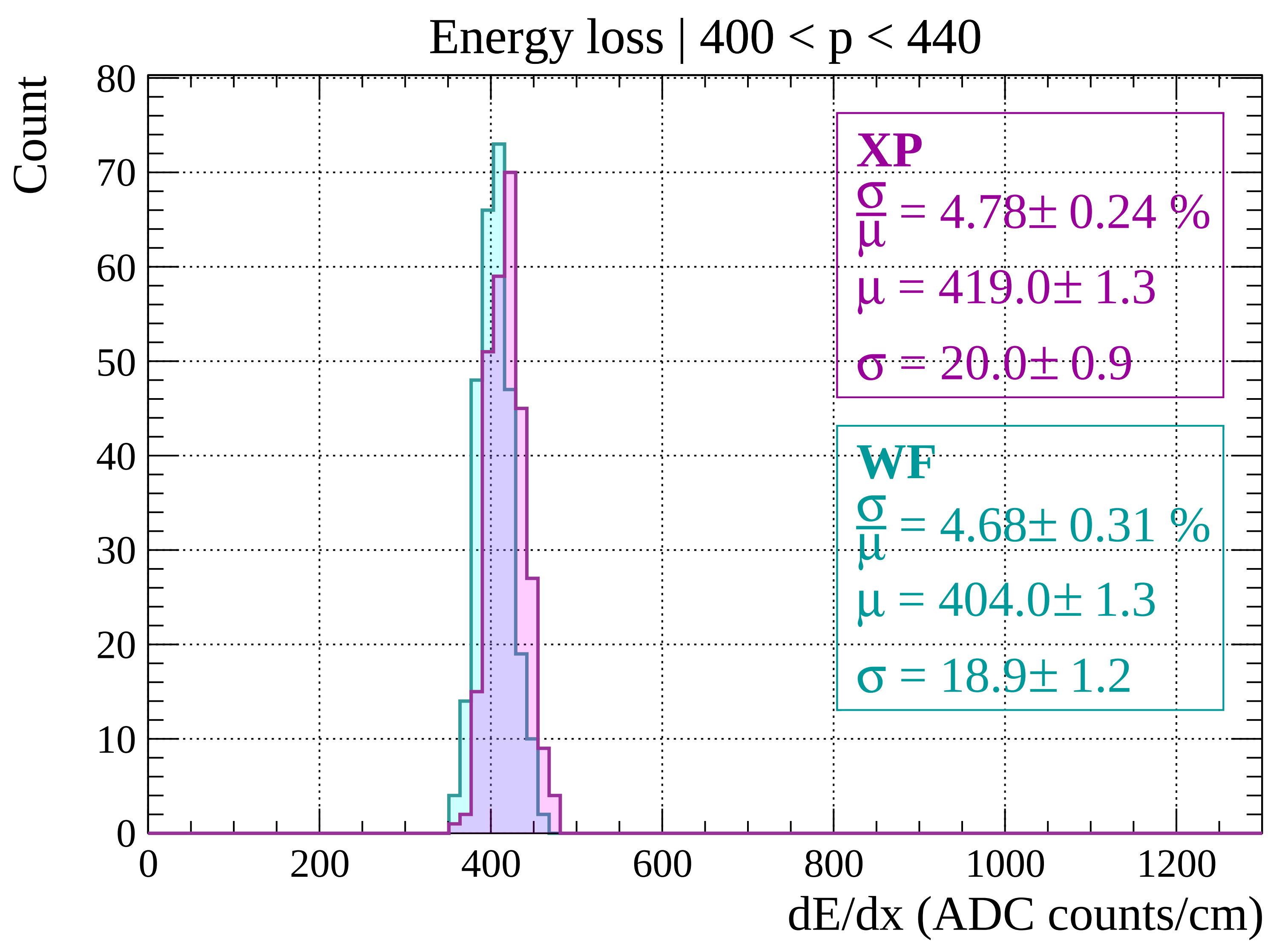
Energy loss | 280 < p < 320
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
5
10
15
20
25
30
35
XP
$\frac{\sigma}{\mu}$ = 4.65± 0.42 %
μ = 419.8± 2.1
σ = 19.5± 1.6
WF
$\frac{\sigma}{\mu}$ = 4.00± 0.51 %
μ = 409.3± 2.0
σ = 16.4± 2.0



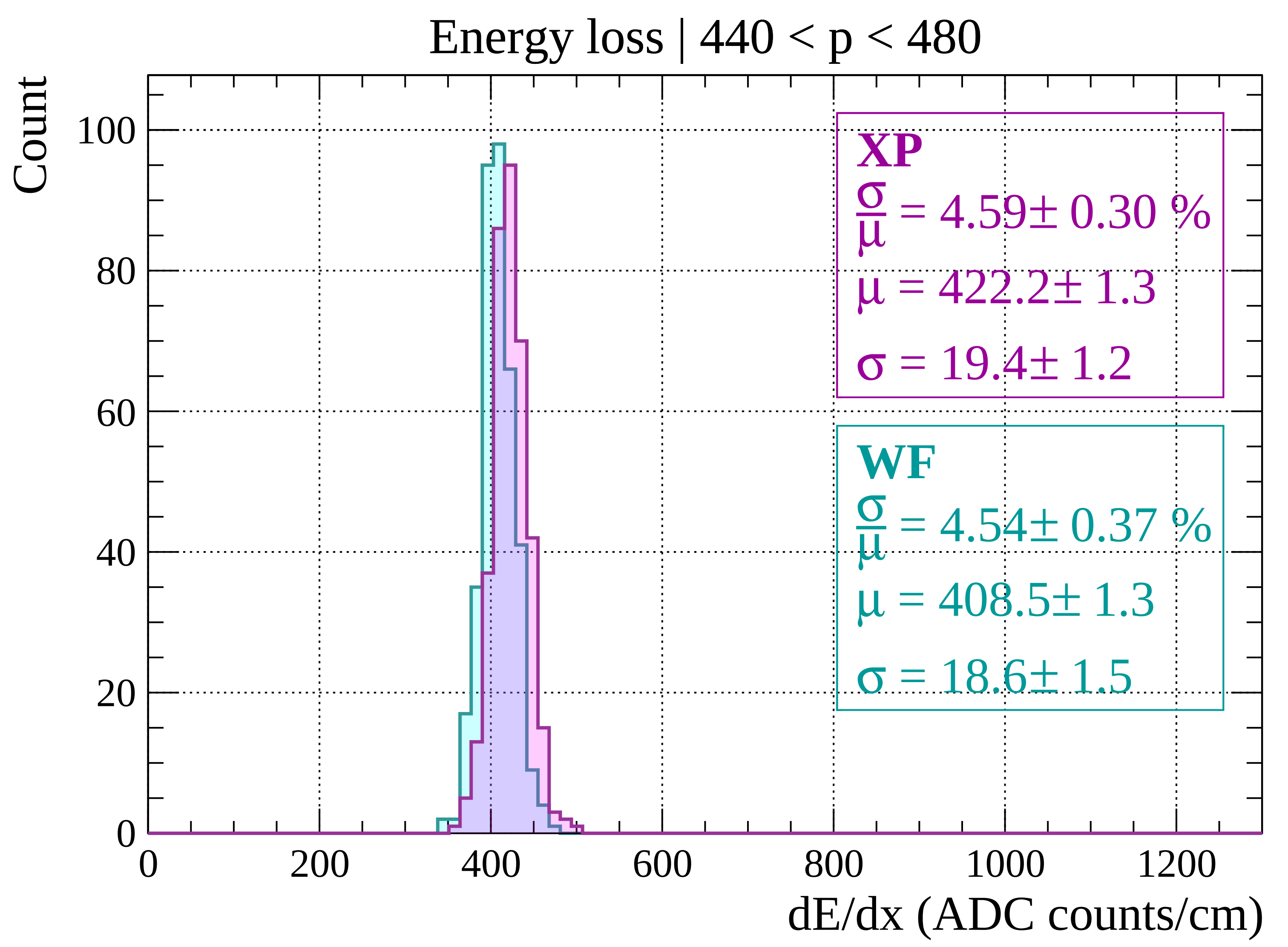
Energy loss | 320 < p < 360
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 5.25± 0.68 %
μ = 419.6± 2.2
σ = 22.0± 2.7
WF
σ/μ = 5.65± 0.50 %
μ = 405.4± 2.2
σ = 22.9± 1.9
0
5
10
15
20
25
30
35
40
0
200
400
600
800
1000
1200



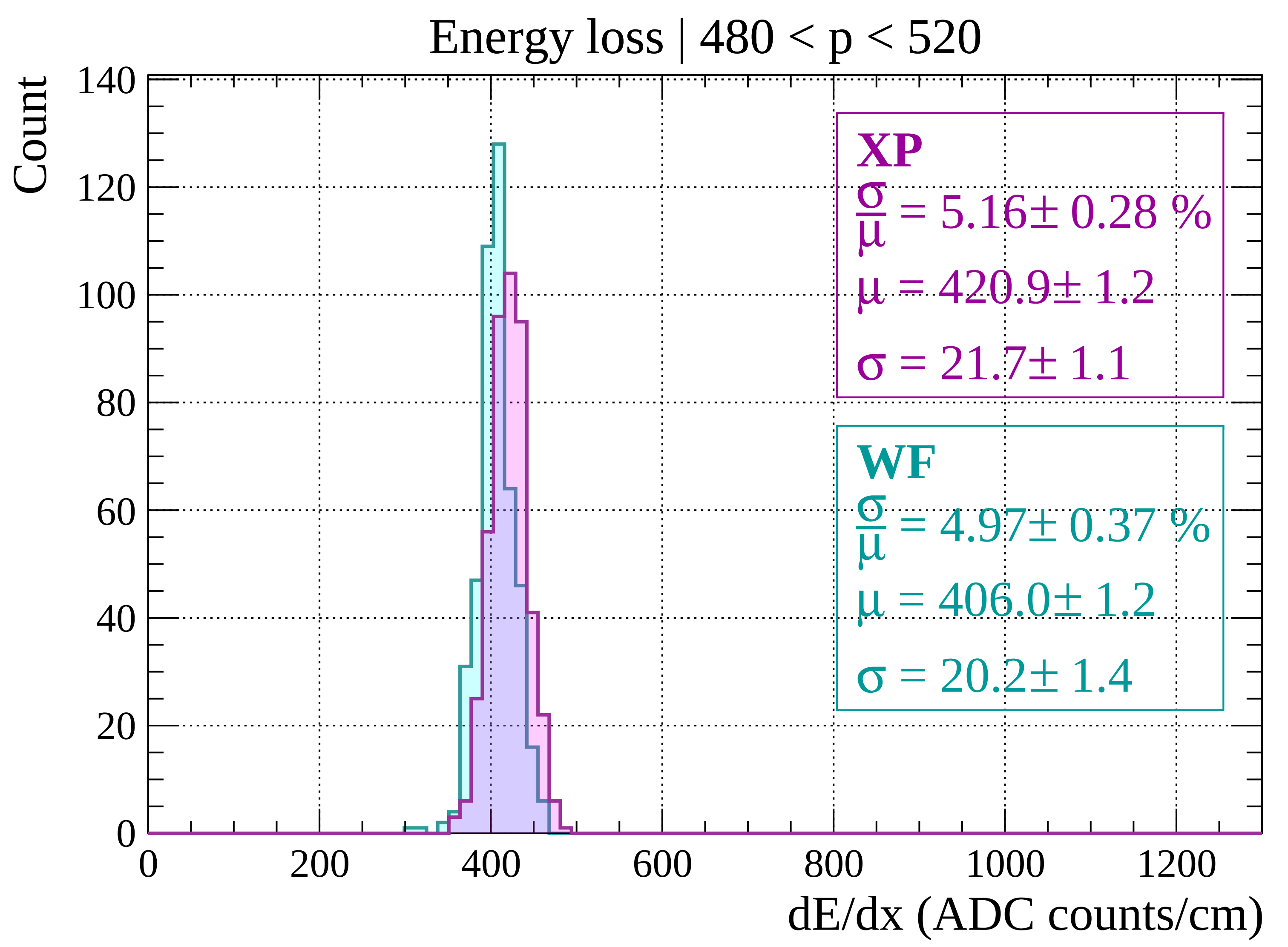
Energy loss | 360 < p < 400
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
0
200
400
600
800
1000
1200
XP
σ/μ = 5.78± 0.50 %
μ = 420.2± 1.9
σ = 24.3± 2.0
WF
σ/μ = 5.38± 0.36 %
μ = 405.7± 1.7
σ = 21.8± 1.4

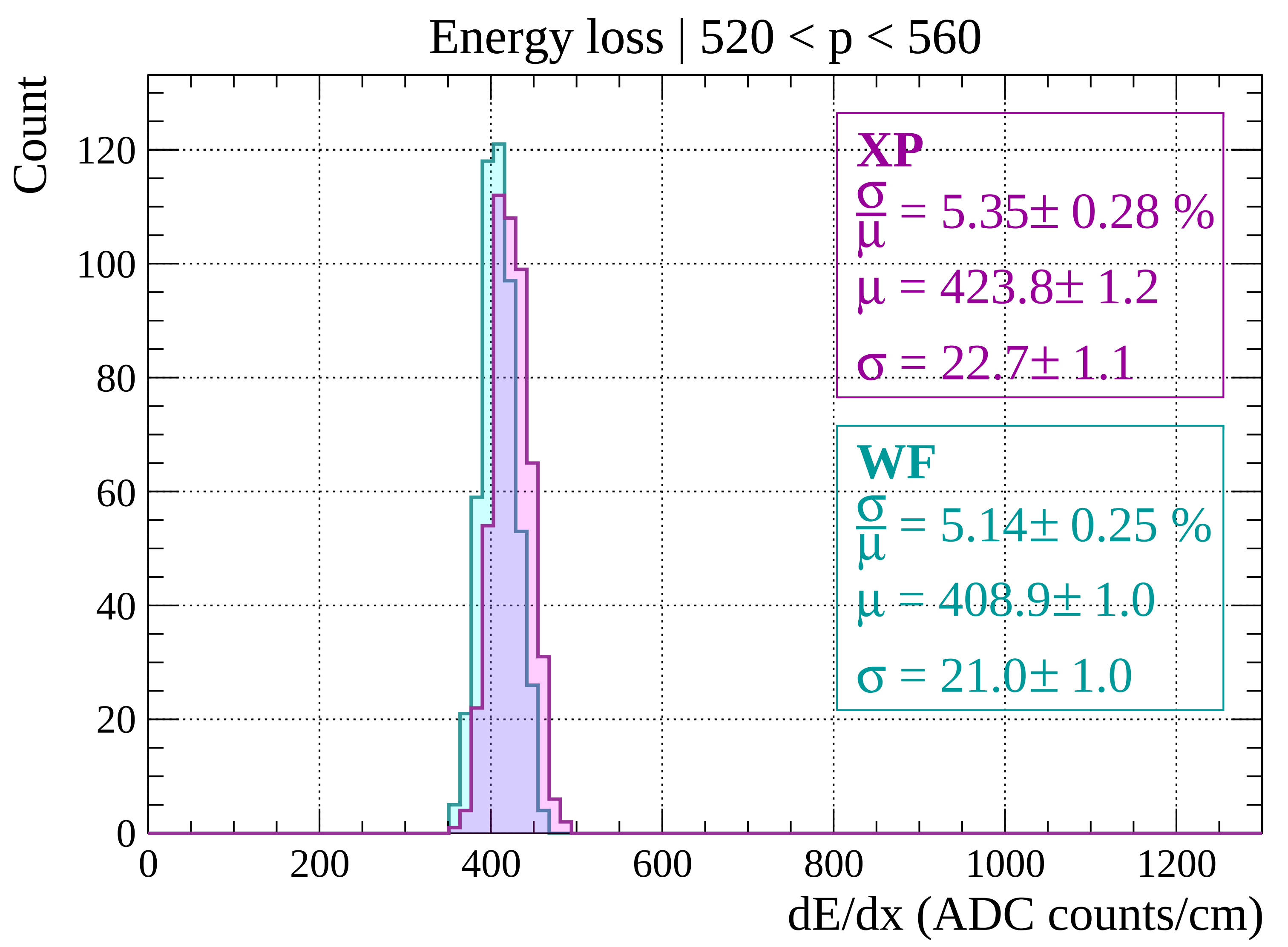


Energy loss | 400 < p < 440
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
80
0
200
400
600
800
1000
1200
XP
σ/μ = 4.78± 0.24 %
μ = 419.0± 1.3
σ = 20.0± 0.9
WF
σ/μ = 4.68± 0.31 %
μ = 404.0± 1.3
σ = 18.9± 1.2

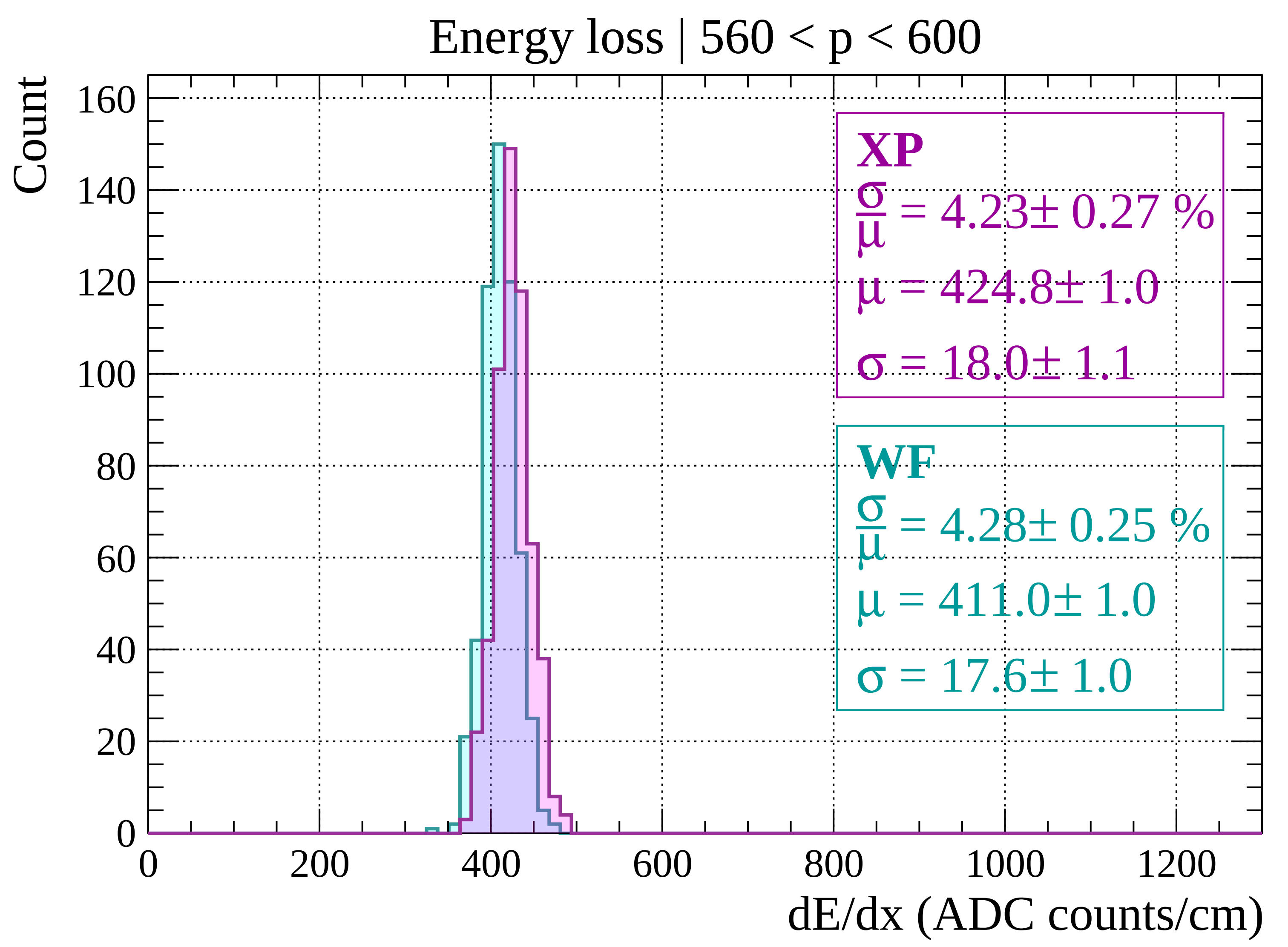


Energy loss | 440 < p < 480
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 4.59± 0.30 %
μ = 422.2± 1.3
σ = 19.4± 1.2
WF
$\frac{\sigma}{\mu}$ = 4.54± 0.37 %
μ = 408.5± 1.3
σ = 18.6± 1.5

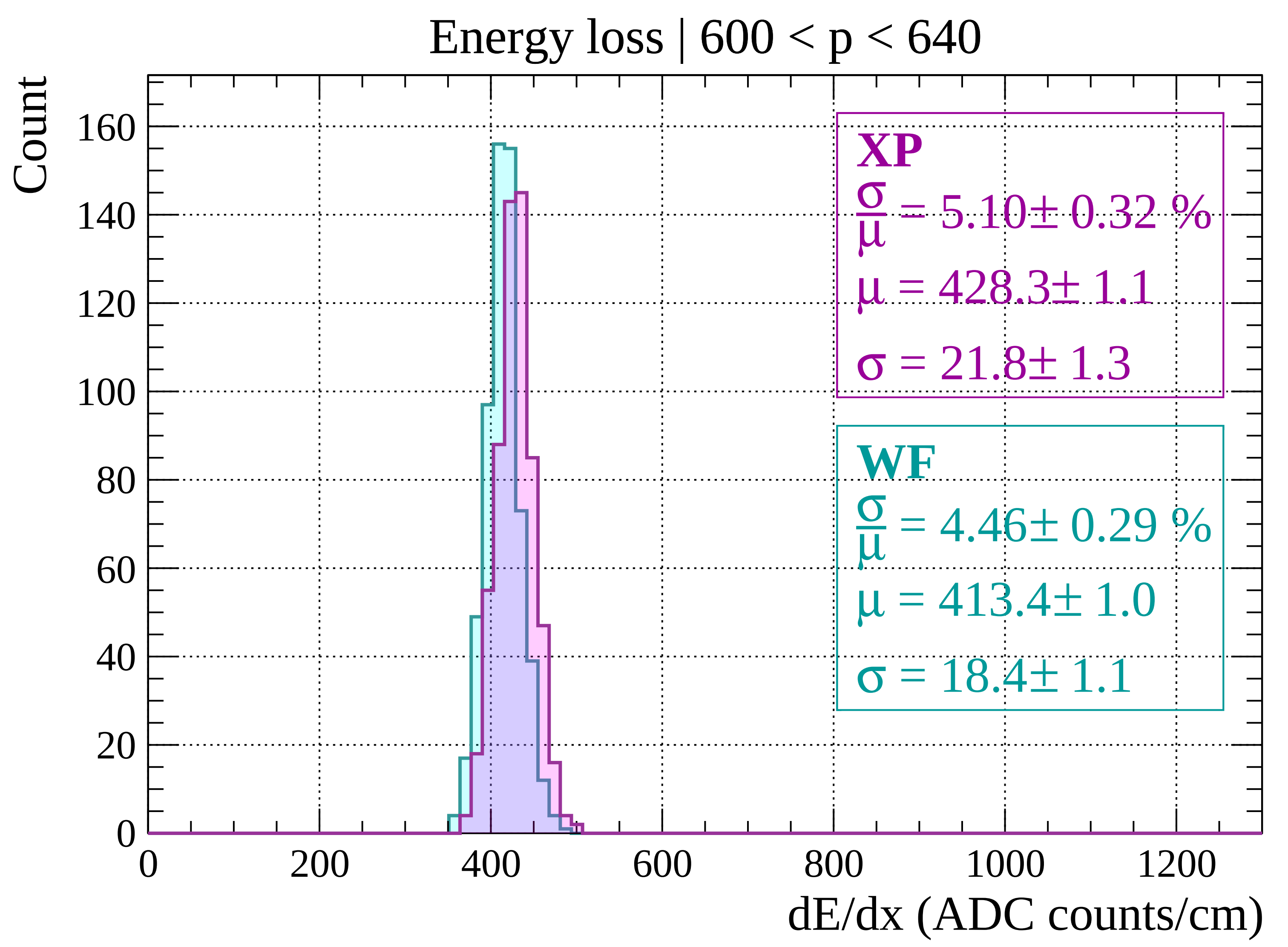

Energy loss | 480 < p < 520
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200
XP
σ/μ = 5.16± 0.28 %
μ = 420.9± 1.2
σ = 21.7± 1.1
WF
σ/μ = 4.97± 0.37 %
μ = 406.0± 1.2
σ = 20.2± 1.4



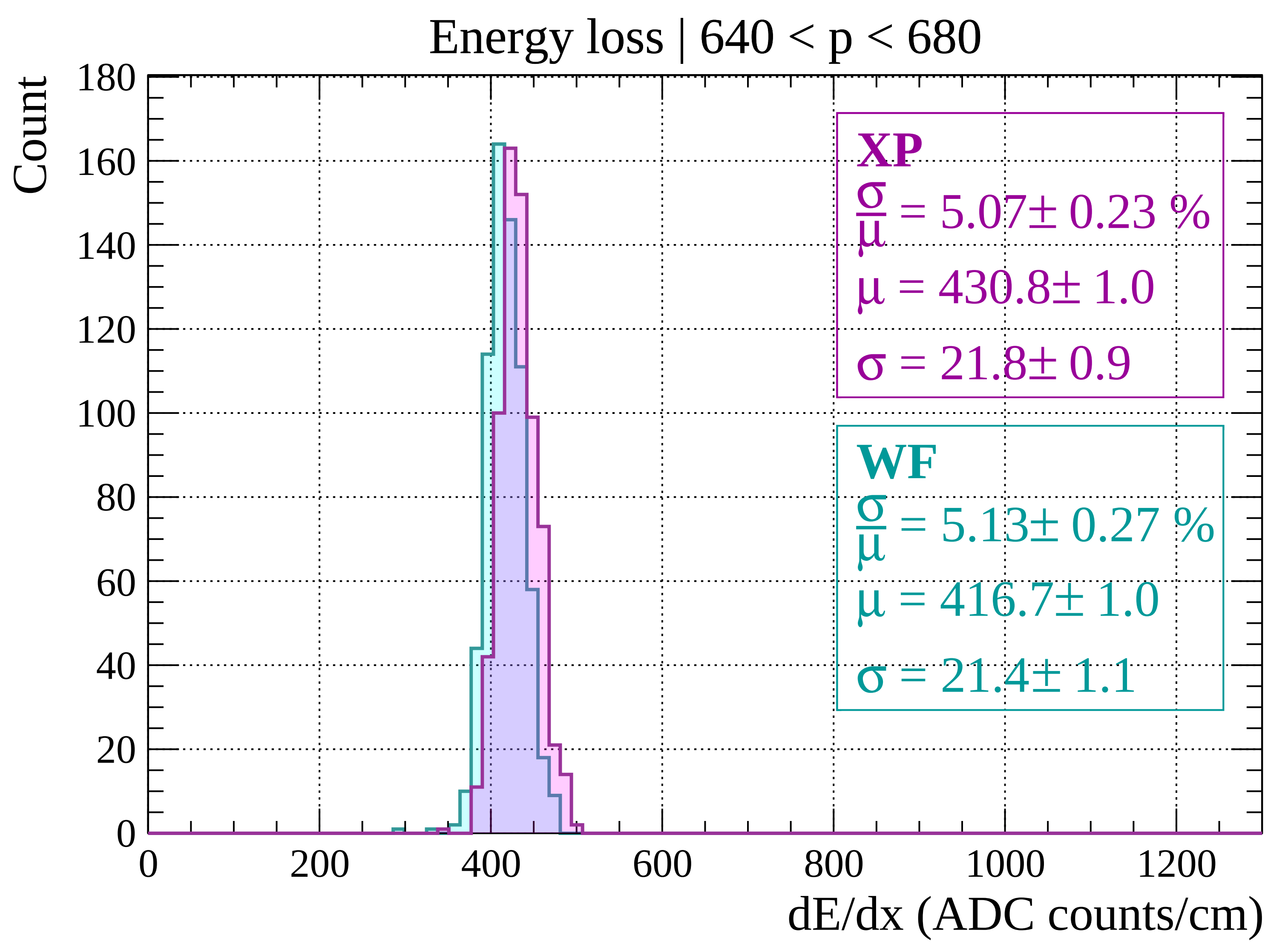
Energy loss | 520 < p < 560
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 5.35± 0.28 %
μ = 423.8± 1.2
σ = 22.7± 1.1
WF
σ/μ = 5.14± 0.25 %
μ = 408.9± 1.0
σ = 21.0± 1.0
0
200
400
600
800
1000
1200
20
40
60
80
100
120



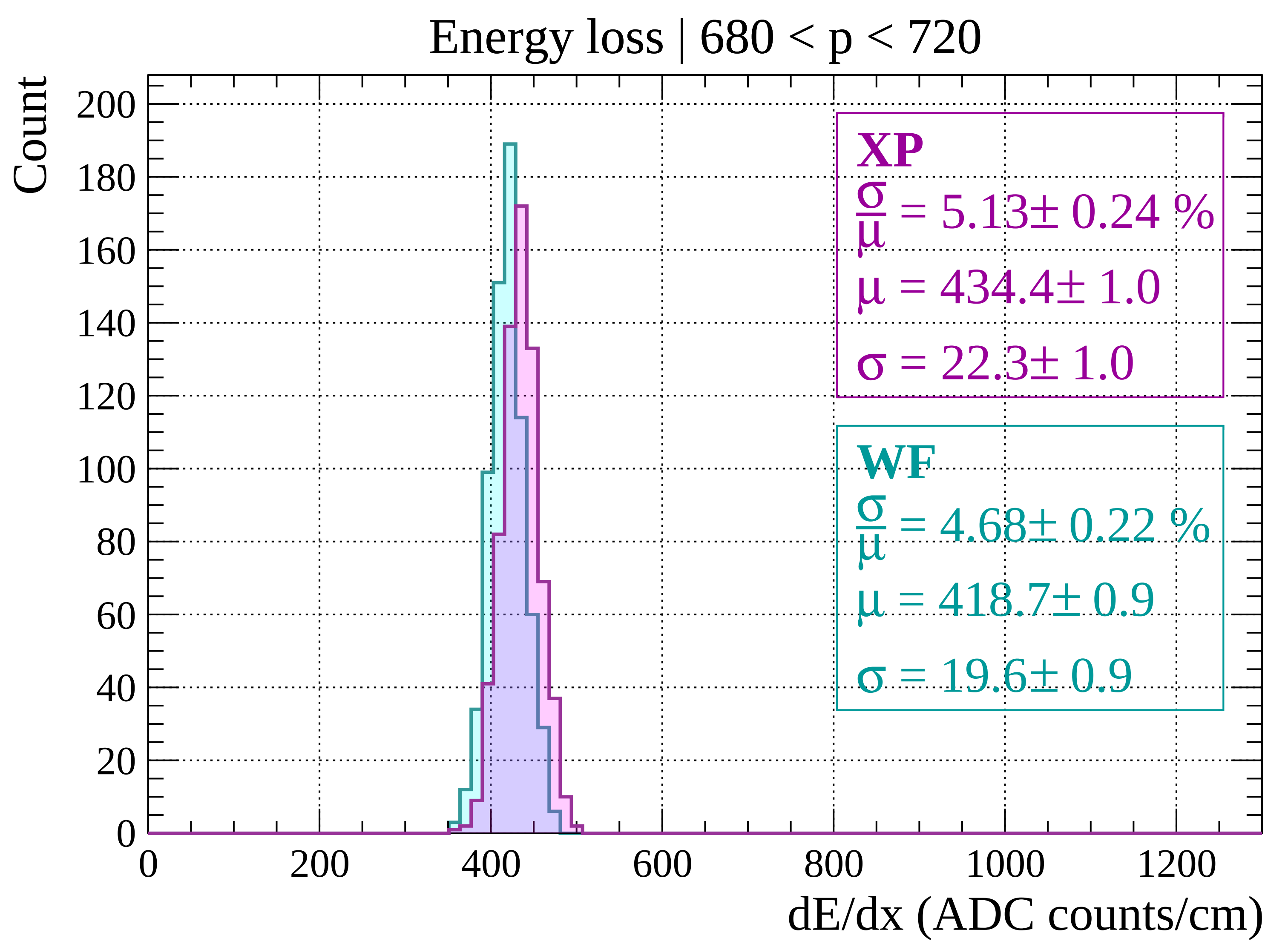
Energy loss | 560 < p < 600
Count
0
20
40
60
80
100
120
140
160
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 4.23± 0.27 %
μ = 424.8± 1.0
σ = 18.0± 1.1
WF
σ/μ = 4.28± 0.25 %
μ = 411.0± 1.0
σ = 17.6± 1.0



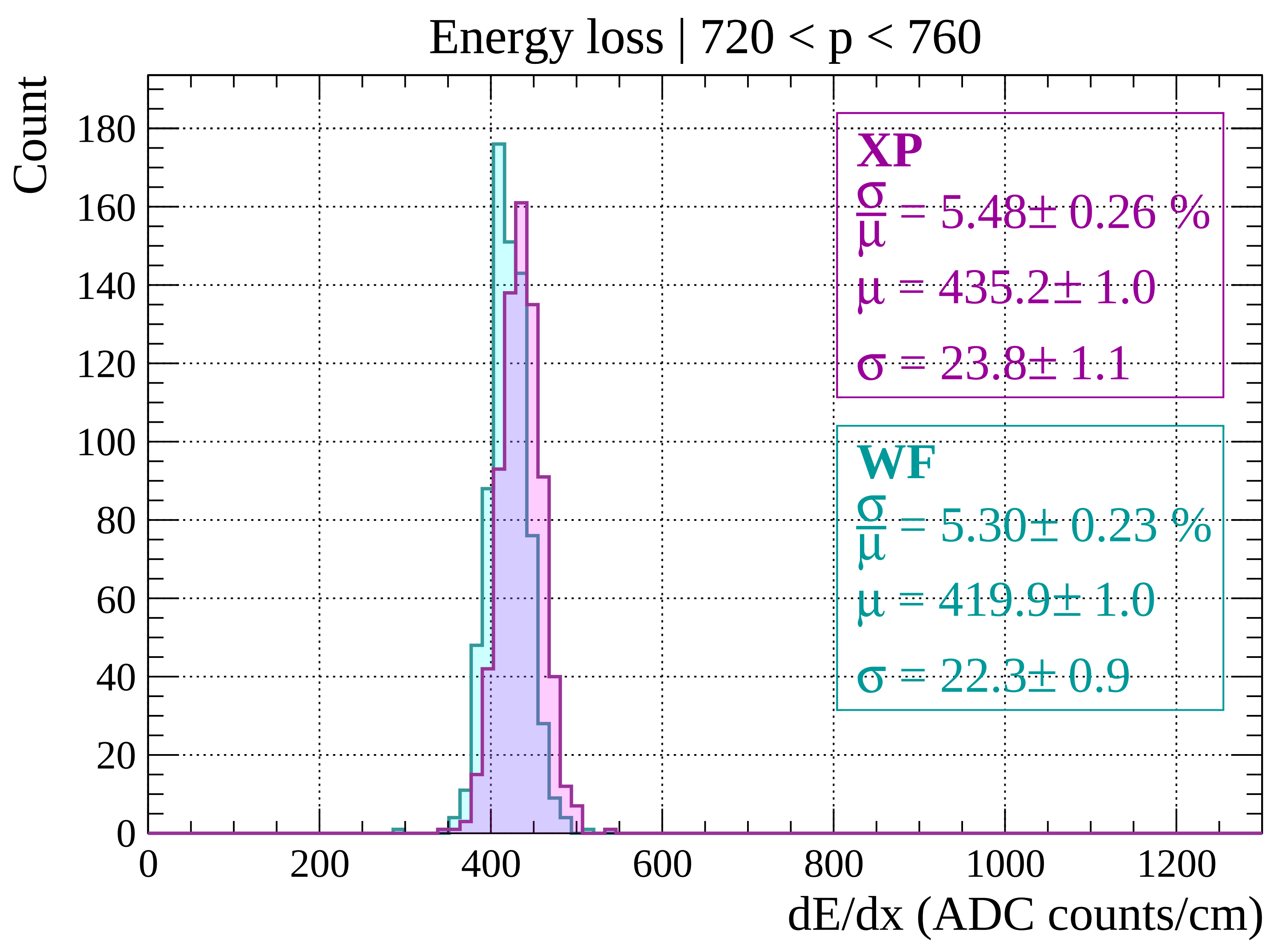
Energy loss | 600 < p < 640
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
XP
$\frac{\sigma}{\mu} = 5.10 \pm 0.32$ %
$\mu = 428.3 \pm 1.1$
$\sigma = 21.8 \pm 1.3$
WF
$\frac{\sigma}{\mu} = 4.46 \pm 0.29$ %
$\mu = 413.4 \pm 1.0$
$\sigma = 18.4 \pm 1.1$



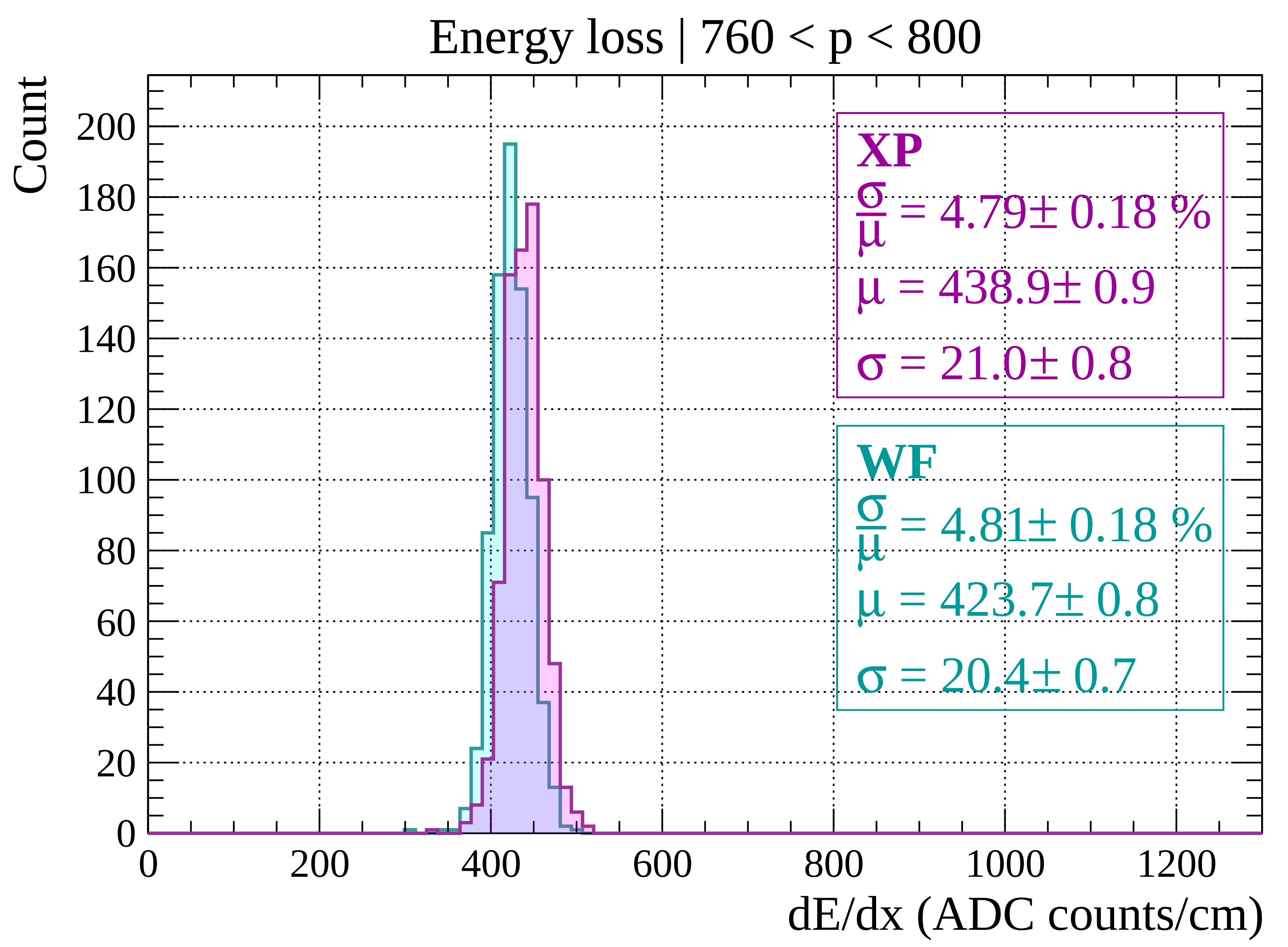
Energy loss | 640 < p < 680
Count
180
160
140
120
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 5.07± 0.23 %
μ = 430.8± 1.0
σ = 21.8± 0.9
WF
$\frac{\sigma}{\mu}$ = 5.13± 0.27 %
μ = 416.7± 1.0
σ = 21.4± 1.1



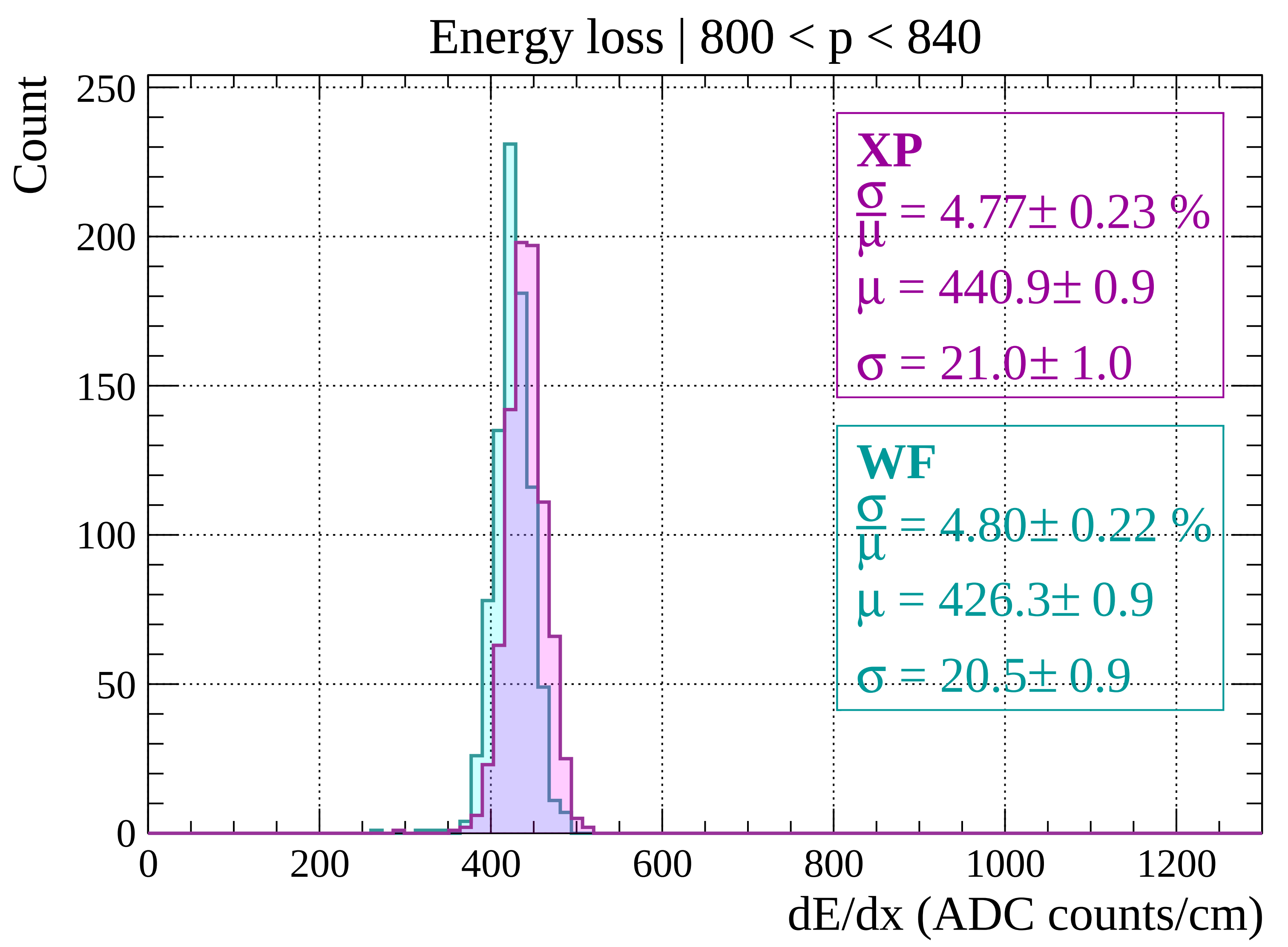
Energy loss | 680 < p < 720
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
200
XP
$\frac{\sigma}{\mu}$ = 5.13± 0.24 %
μ = 434.4± 1.0
σ = 22.3± 1.0
WF
$\frac{\sigma}{\mu}$ = 4.68± 0.22 %
μ = 418.7± 0.9
σ = 19.6± 0.9



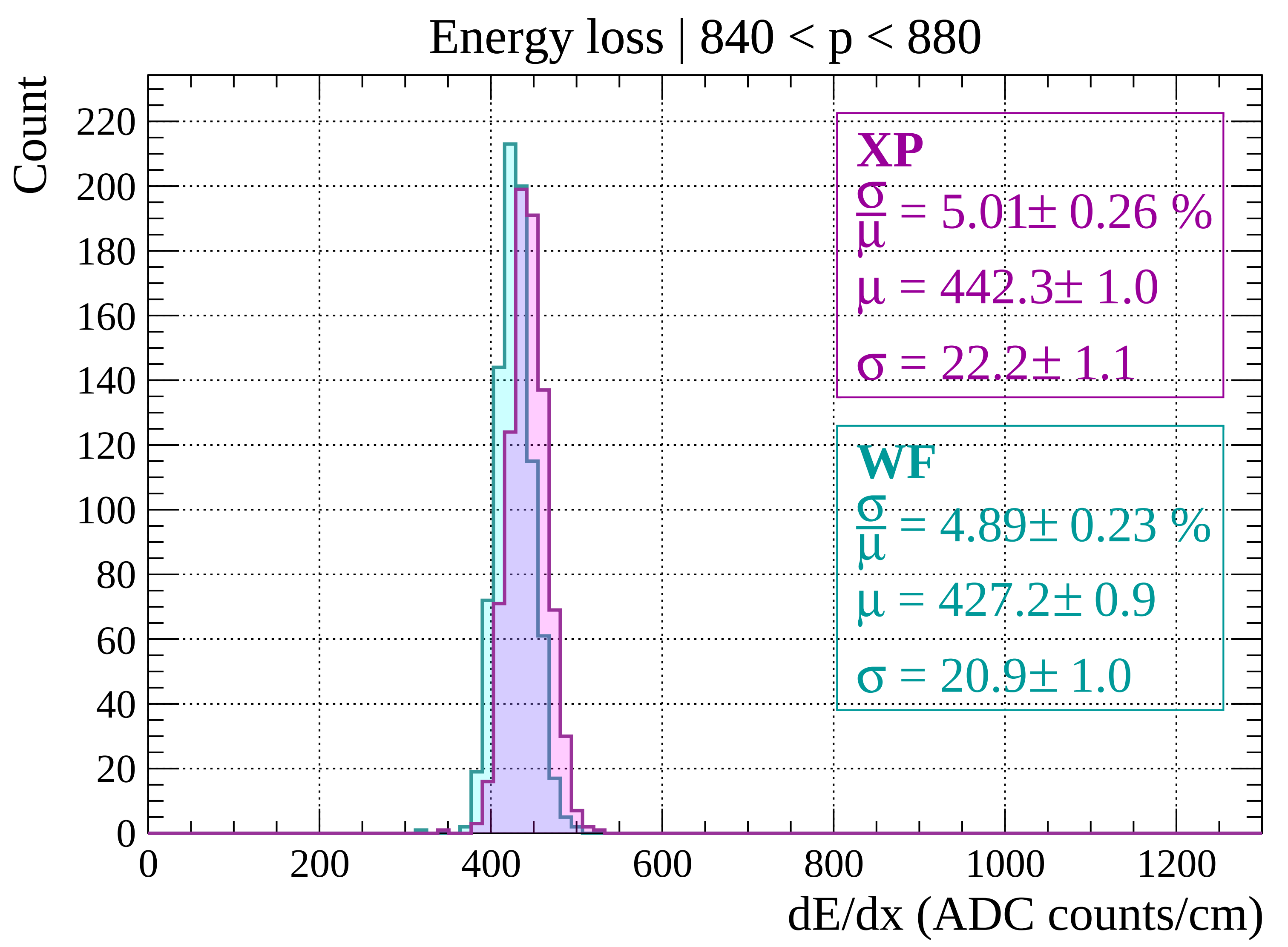
Energy loss | 720 < p < 760
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
XP
σ/μ = 5.48± 0.26 %
μ = 435.2± 1.0
σ = 23.8± 1.1
WF
σ/μ = 5.30± 0.23 %
μ = 419.9± 1.0
σ = 22.3± 0.9



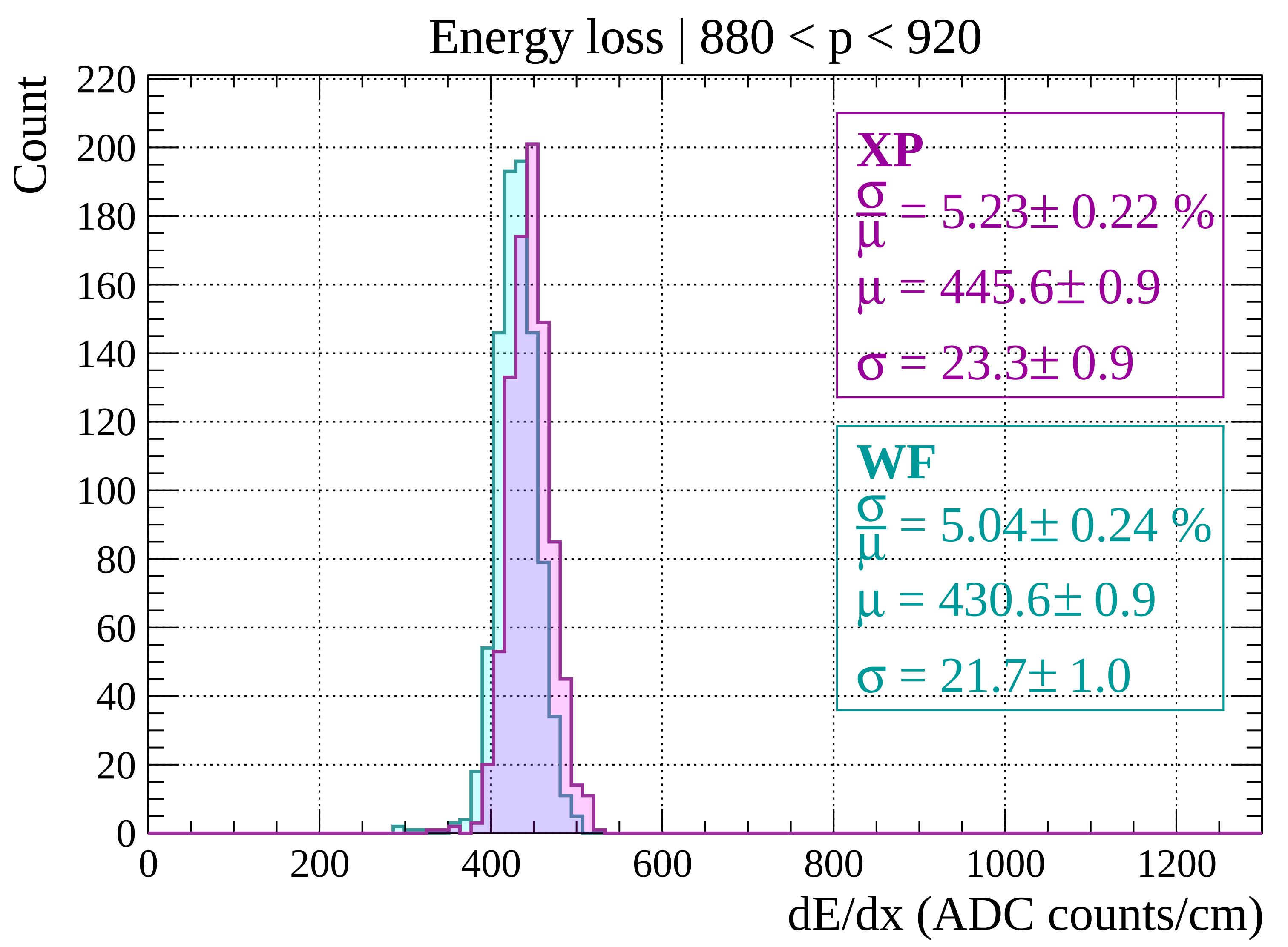
Energy loss | 760 < p < 800
Count
0
20
40
60
80
100
120
140
160
180
200
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 4.79± 0.18 %
μ = 438.9± 0.9
σ = 21.0± 0.8
WF
σ/μ = 4.81± 0.18 %
μ = 423.7± 0.8
σ = 20.4± 0.7



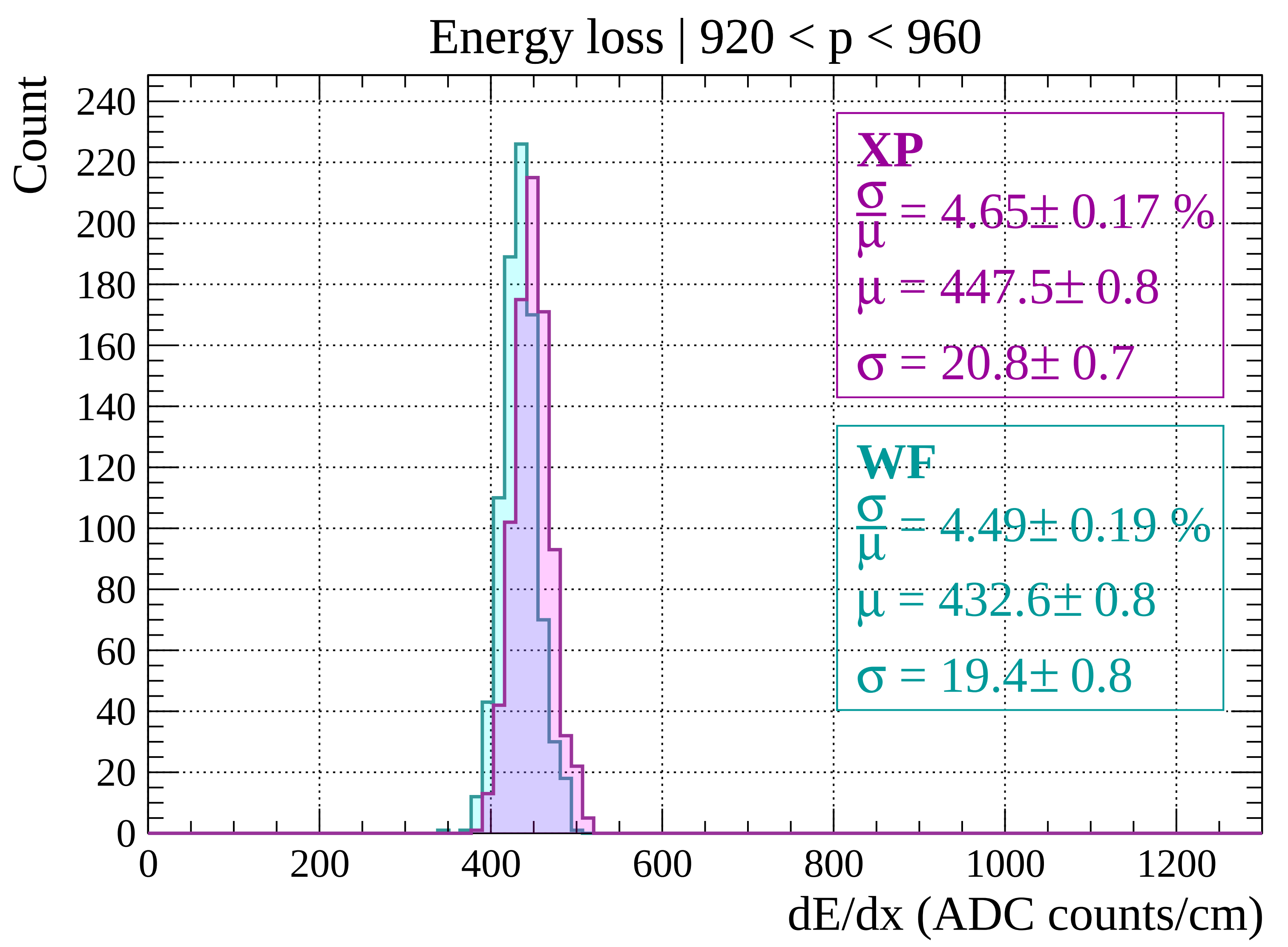
Energy loss | 800 < p < 840
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 4.77± 0.23 %
μ = 440.9± 0.9
σ = 21.0± 1.0
WF
$\frac{\sigma}{\mu}$ = 4.80± 0.22 %
μ = 426.3± 0.9
σ = 20.5± 0.9



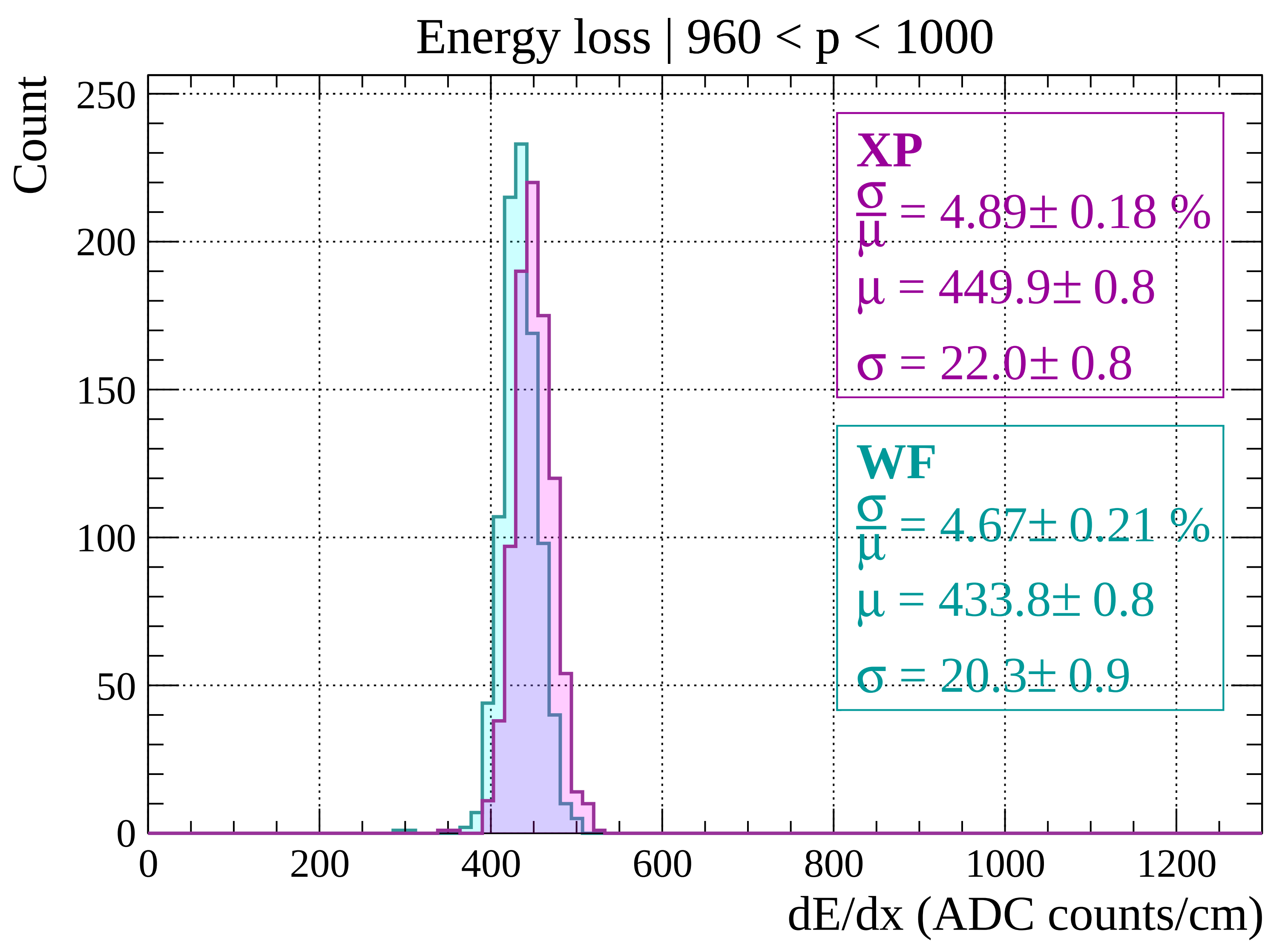
Energy loss | 840 < p < 880
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 5.01± 0.26 %
μ = 442.3± 1.0
σ = 22.2± 1.1
WF
σ/μ = 4.89± 0.23 %
μ = 427.2± 0.9
σ = 20.9± 1.0



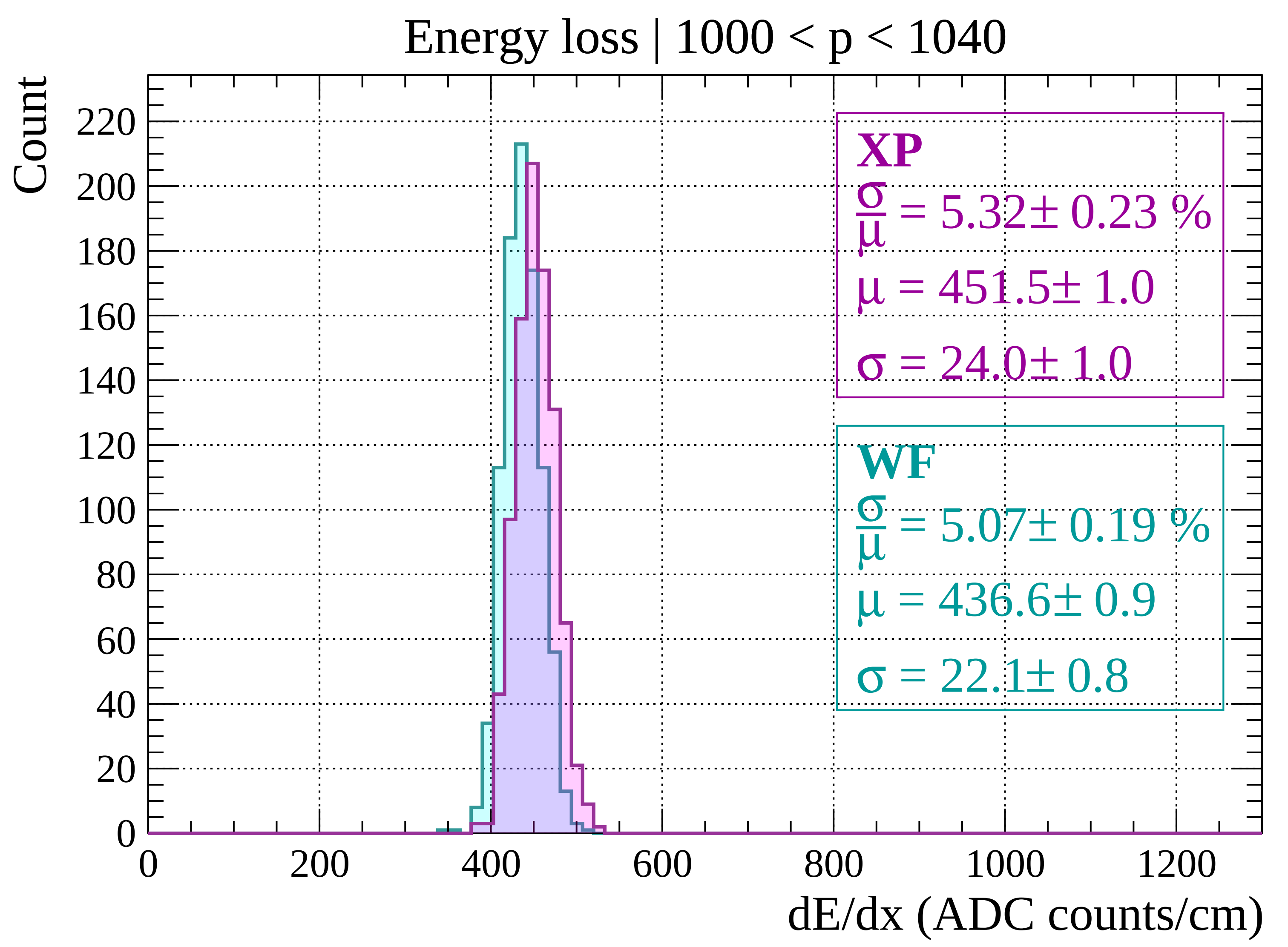
Energy loss | 880 < p < 920
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
200
220
XP
$\frac{\sigma}{\mu} = 5.23 \pm 0.22$ %
$\mu = 445.6 \pm 0.9$
$\sigma = 23.3 \pm 0.9$
WF
$\frac{\sigma}{\mu} = 5.04 \pm 0.24$ %
$\mu = 430.6 \pm 0.9$
$\sigma = 21.7 \pm 1.0$

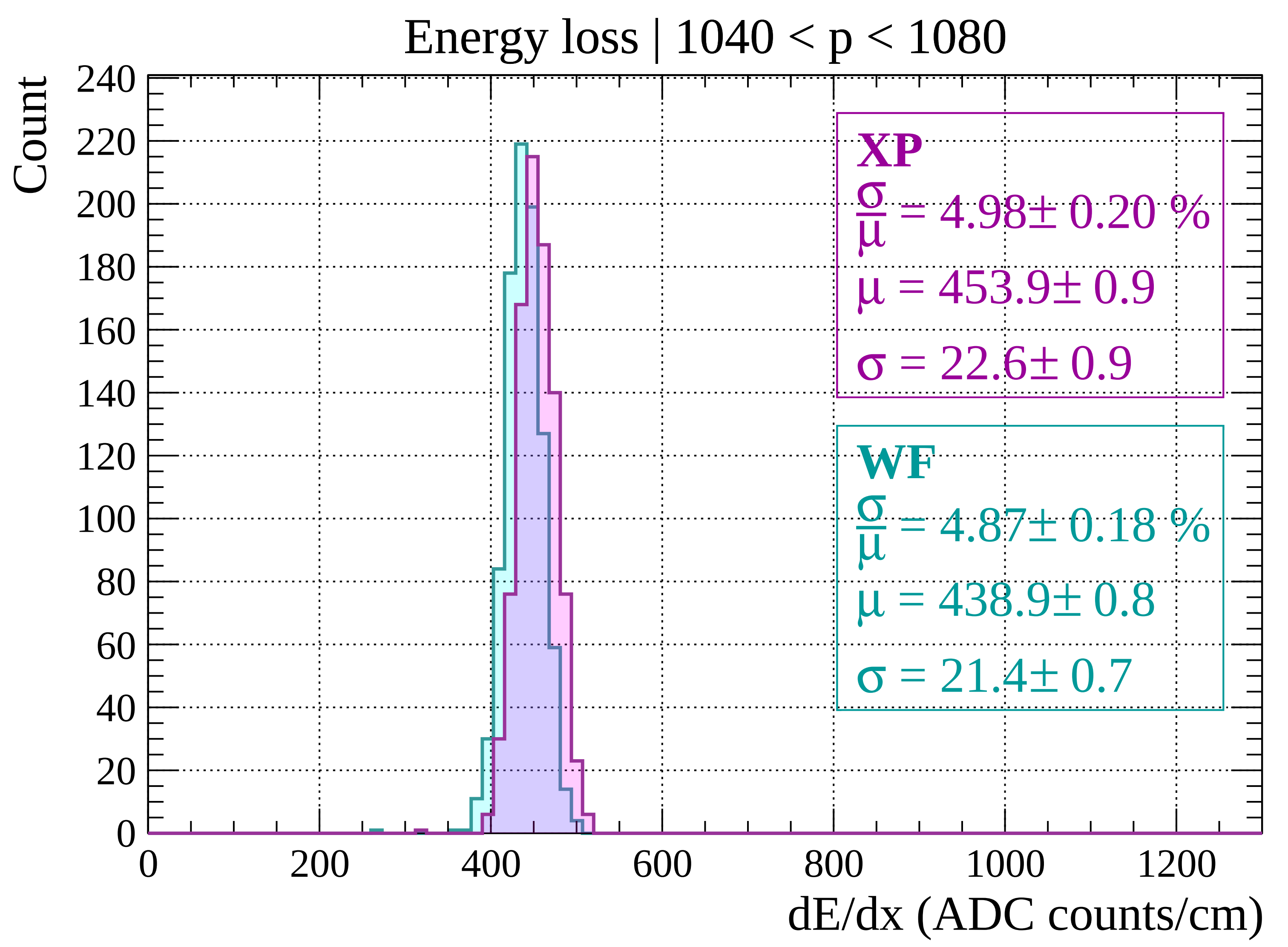


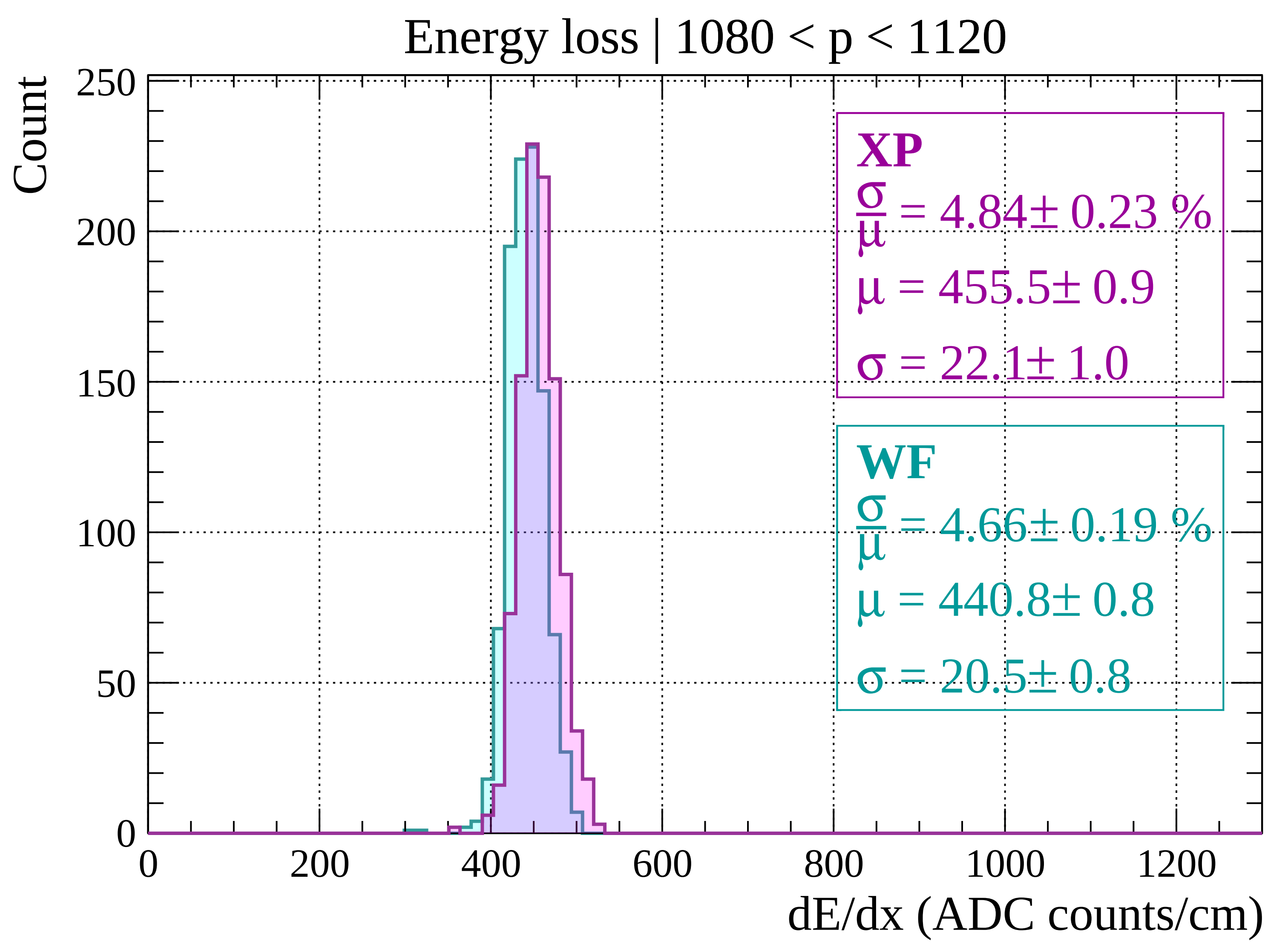
Energy loss | 920 < p < 960
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
200
220
240
XP
$\frac{\sigma}{\mu} = 4.65 \pm 0.17$ %
$\mu = 447.5 \pm 0.8$
$\sigma = 20.8 \pm 0.7$
WF
$\frac{\sigma}{\mu} = 4.49 \pm 0.19$ %
$\mu = 432.6 \pm 0.8$
$\sigma = 19.4 \pm 0.8$



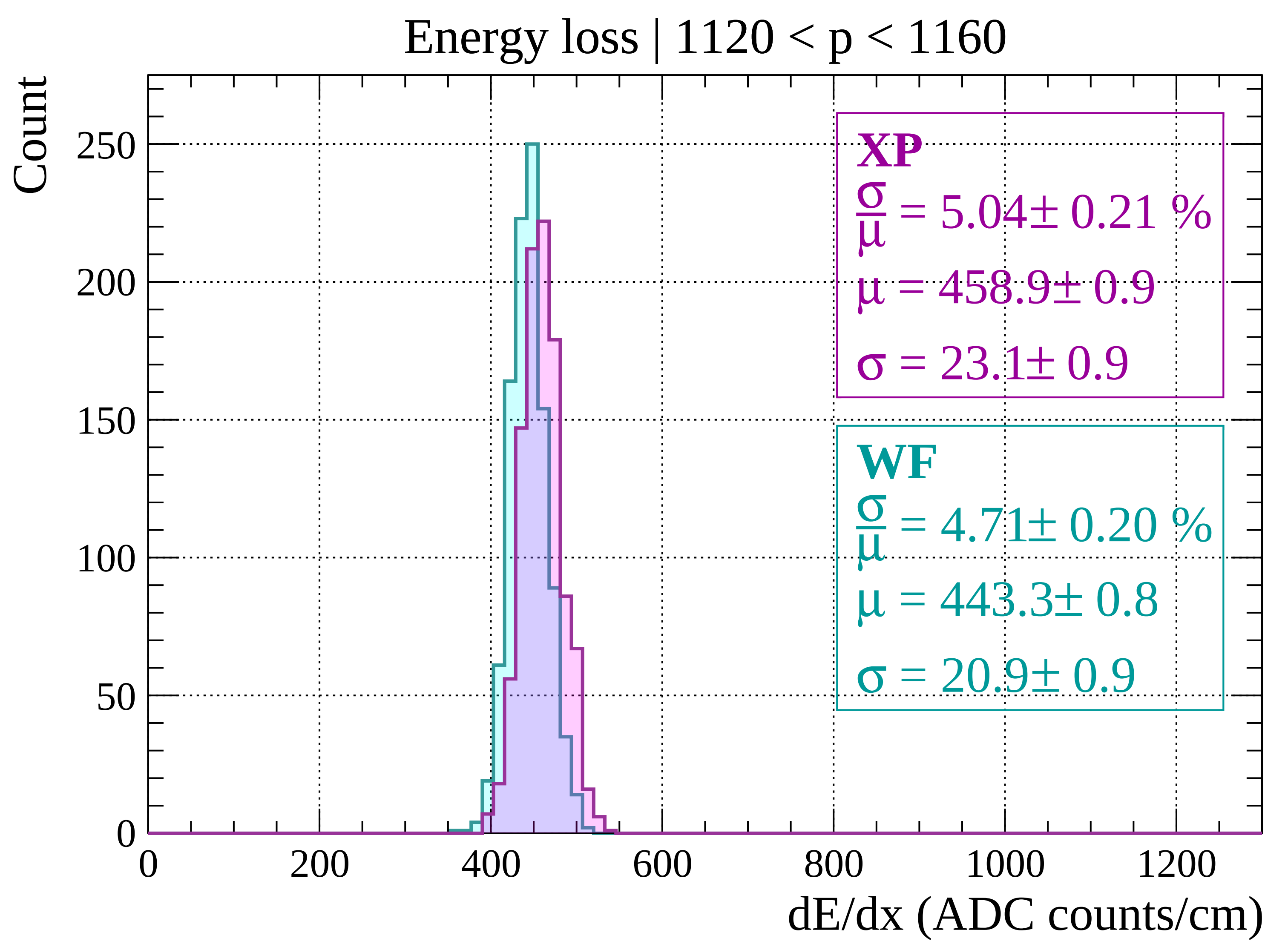
Energy loss | 960 < p < 1000
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 4.89± 0.18 %
μ = 449.9± 0.8
σ = 22.0± 0.8
WF
$\frac{\sigma}{\mu}$ = 4.67± 0.21 %
μ = 433.8± 0.8
σ = 20.3± 0.9


Energy loss | 1000 < p < 1040
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
200
220
XP
$\frac{\sigma}{\mu}$ = 5.32± 0.23 %
μ = 451.5± 1.0
σ = 24.0± 1.0
WF
$\frac{\sigma}{\mu}$ = 5.07± 0.19 %
μ = 436.6± 0.9
σ = 22.1± 0.8

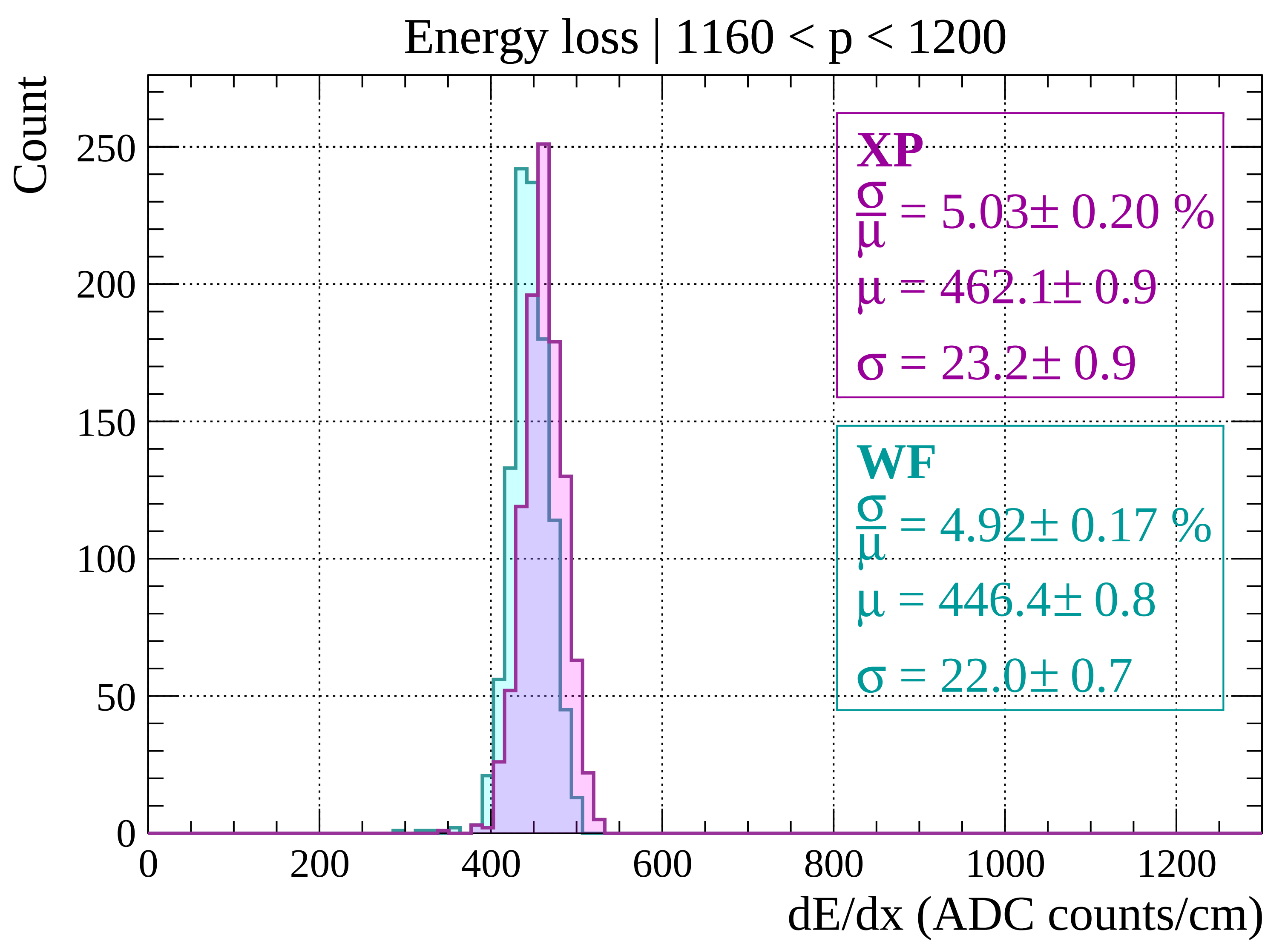

Energy loss | 1040 < p < 1080
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu} = 4.98 \pm 0.20$ %
$\mu = 453.9 \pm 0.9$
$\sigma = 22.6 \pm 0.9$
WF
$\frac{\sigma}{\mu} = 4.87 \pm 0.18$ %
$\mu = 438.9 \pm 0.8$
$\sigma = 21.4 \pm 0.7$



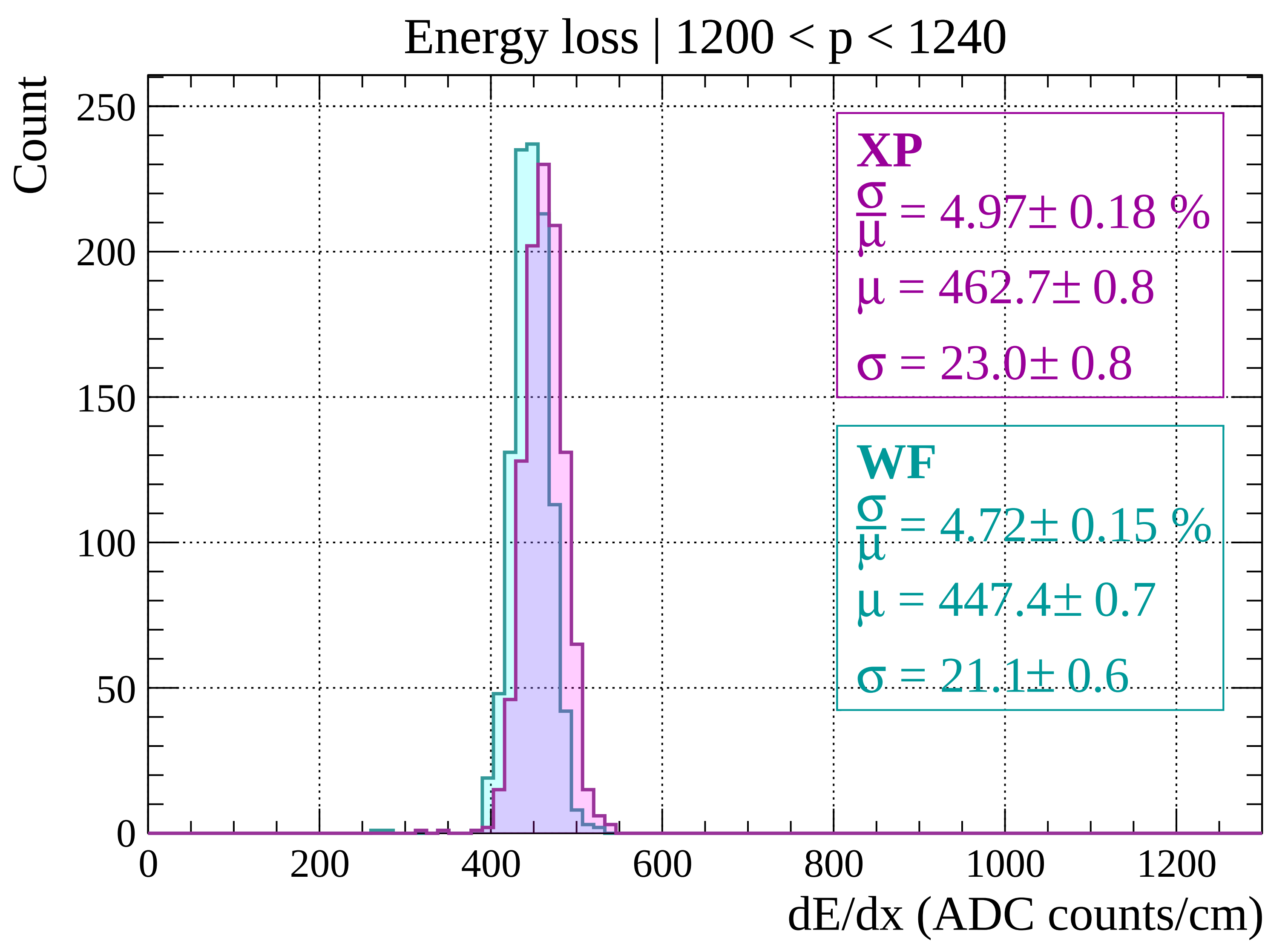
Energy loss | 1080 < p < 1120
Count
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 4.84± 0.23 %
μ = 455.5± 0.9
σ = 22.1± 1.0
WF
σ/μ = 4.66± 0.19 %
μ = 440.8± 0.8
σ = 20.5± 0.8



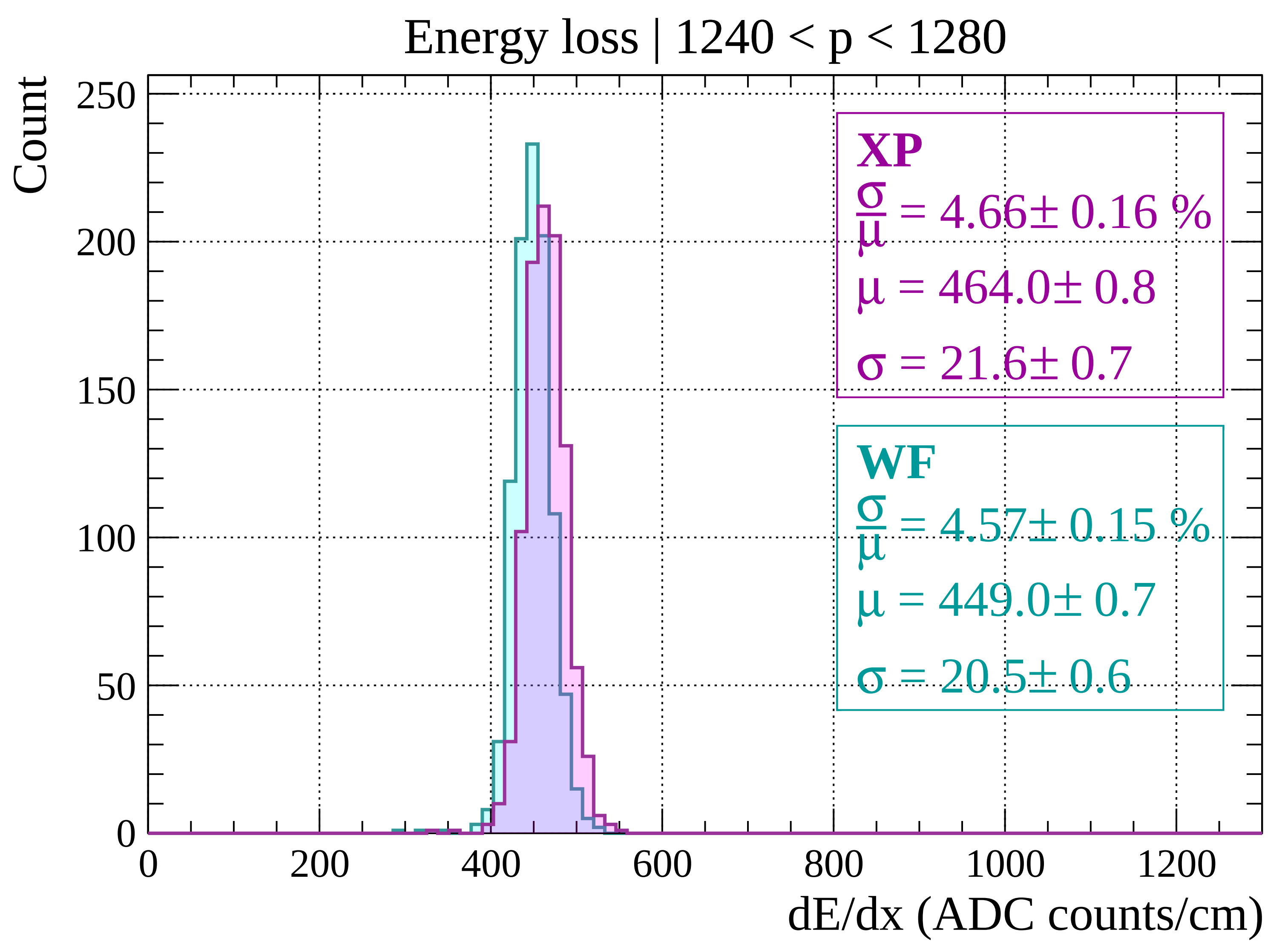
Energy loss | 1120 < p < 1160
Count
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 5.04± 0.21 %
μ = 458.9± 0.9
σ = 23.1± 0.9
WF
σ/μ = 4.71± 0.20 %
μ = 443.3± 0.8
σ = 20.9± 0.9



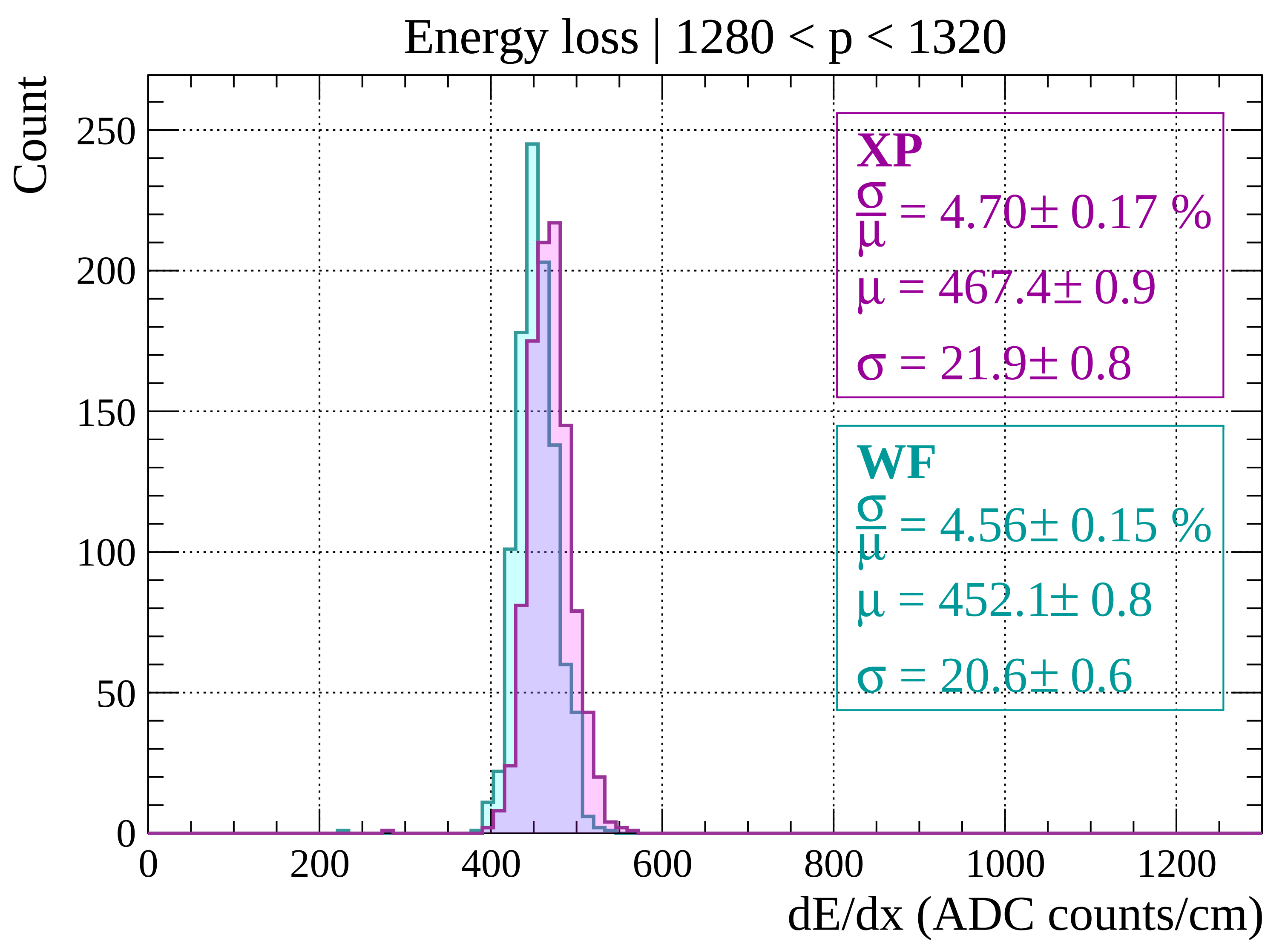
Energy loss | 1160 < p < 1200
Count
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 5.03± 0.20 %
μ = 462.1± 0.9
σ = 23.2± 0.9
WF
σ/μ = 4.92± 0.17 %
μ = 446.4± 0.8
σ = 22.0± 0.7



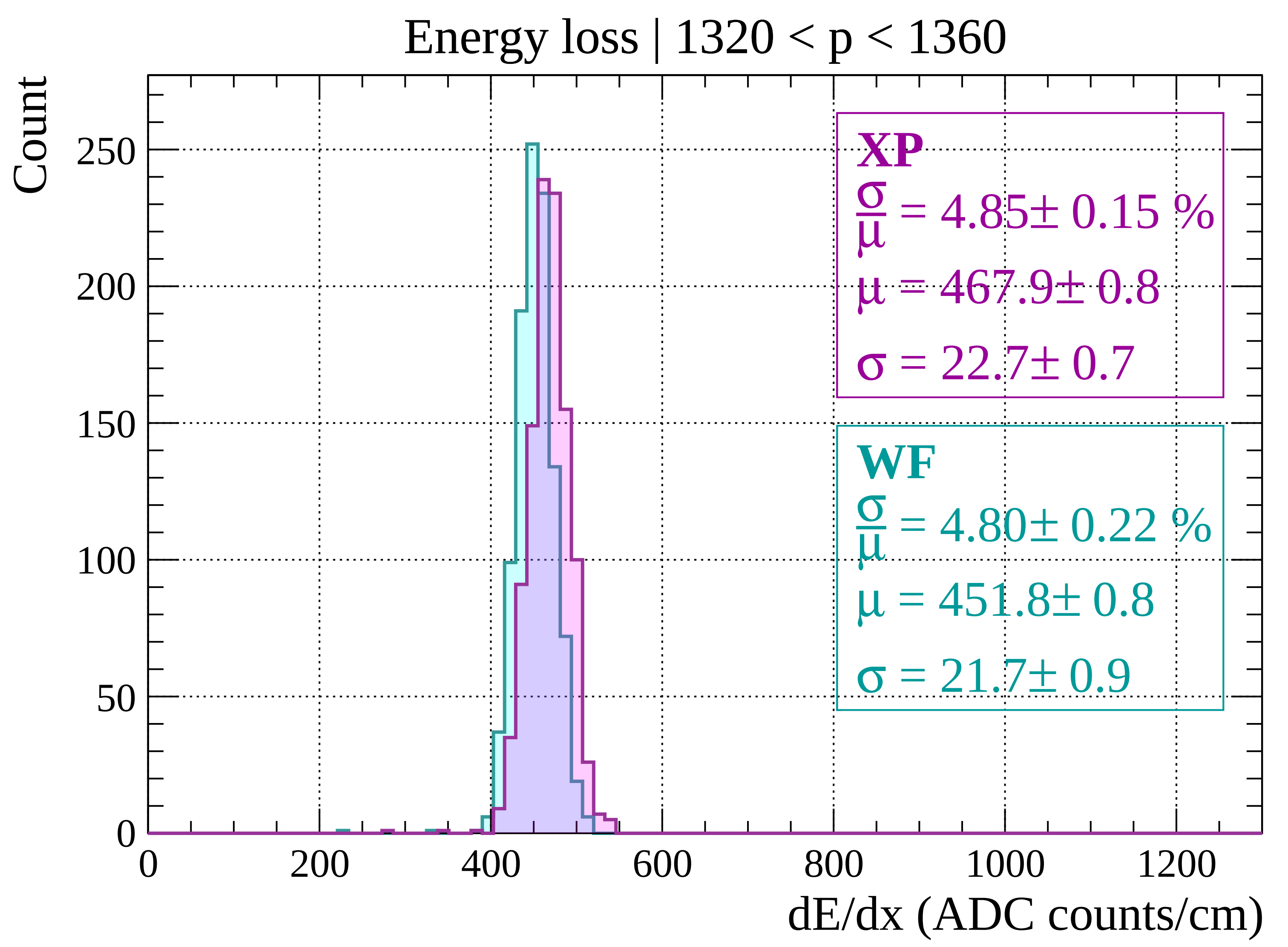
Energy loss | 1200 < p < 1240
Count
0
50
100
150
200
250
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 4.97± 0.18 %
μ = 462.7± 0.8
σ = 23.0± 0.8
WF
σ/μ = 4.72± 0.15 %
μ = 447.4± 0.7
σ = 21.1± 0.6



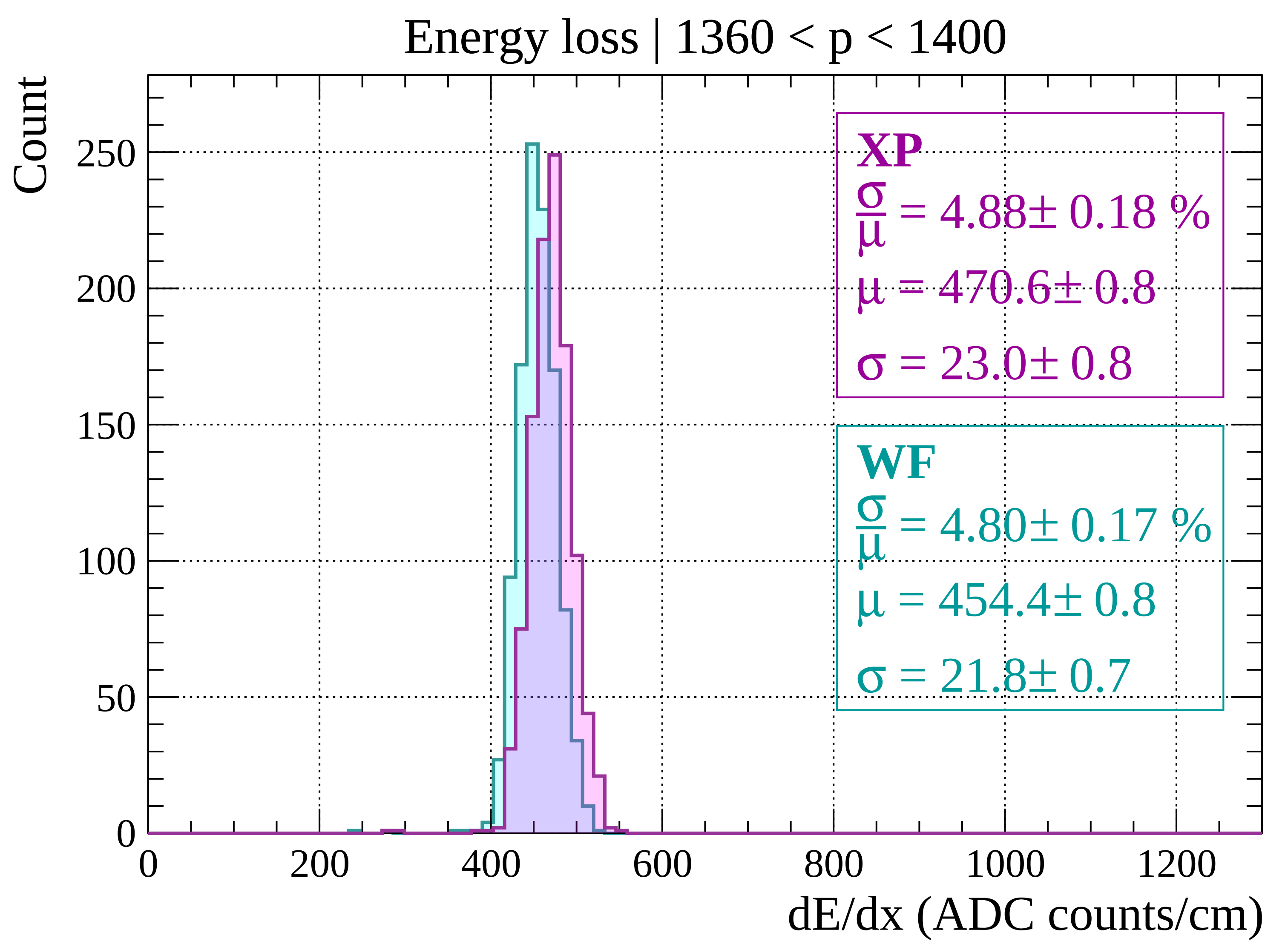
Energy loss | 1240 < p < 1280
Count
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 4.66± 0.16 %
μ = 464.0± 0.8
σ = 21.6± 0.7
WF
σ/μ = 4.57± 0.15 %
μ = 449.0± 0.7
σ = 20.5± 0.6



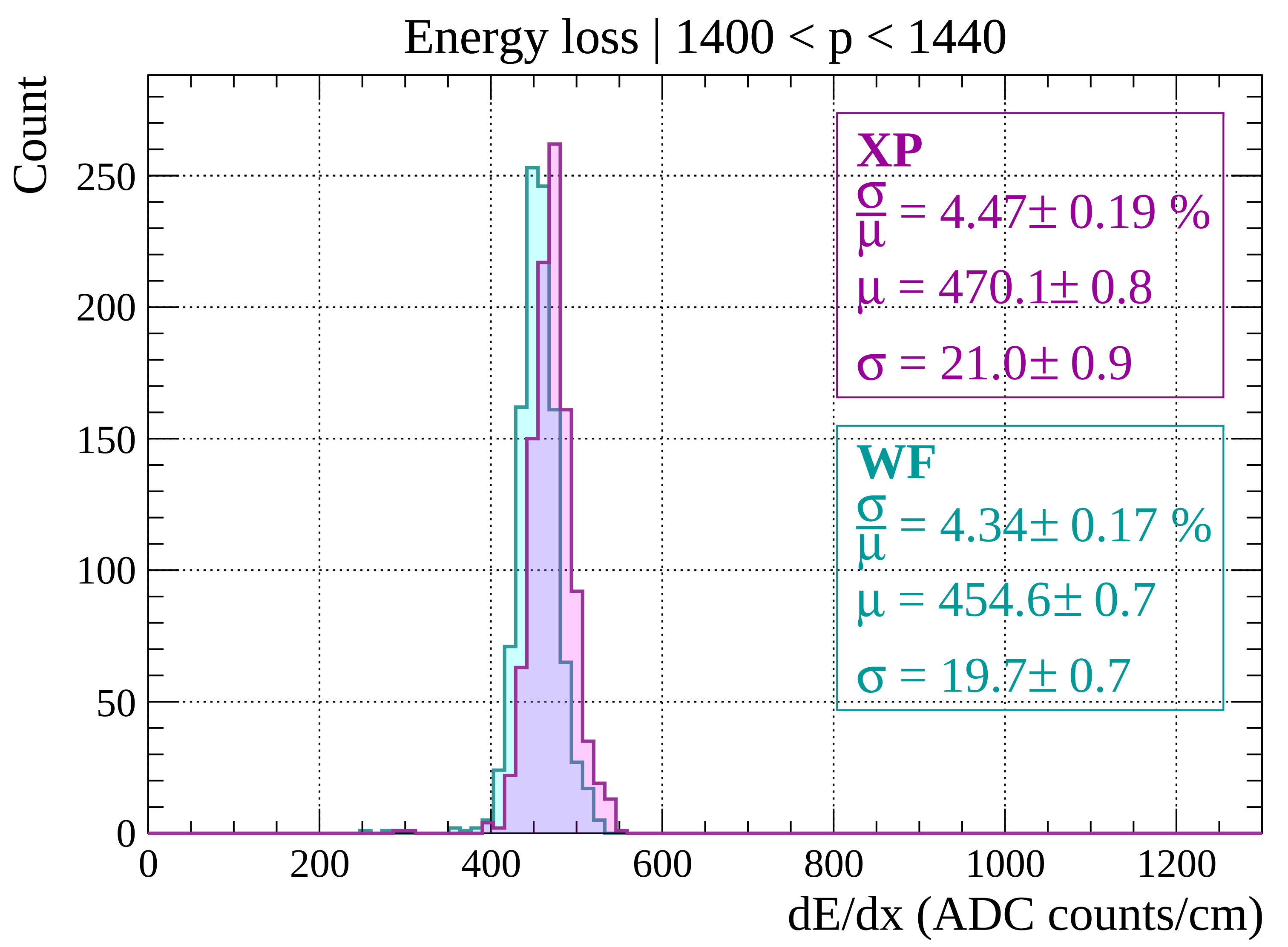
Energy loss | 1280 < p < 1320
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP
σ/μ = 4.70± 0.17 %
μ = 467.4± 0.9
σ = 21.9± 0.8
WF
σ/μ = 4.56± 0.15 %
μ = 452.1± 0.8
σ = 20.6± 0.6

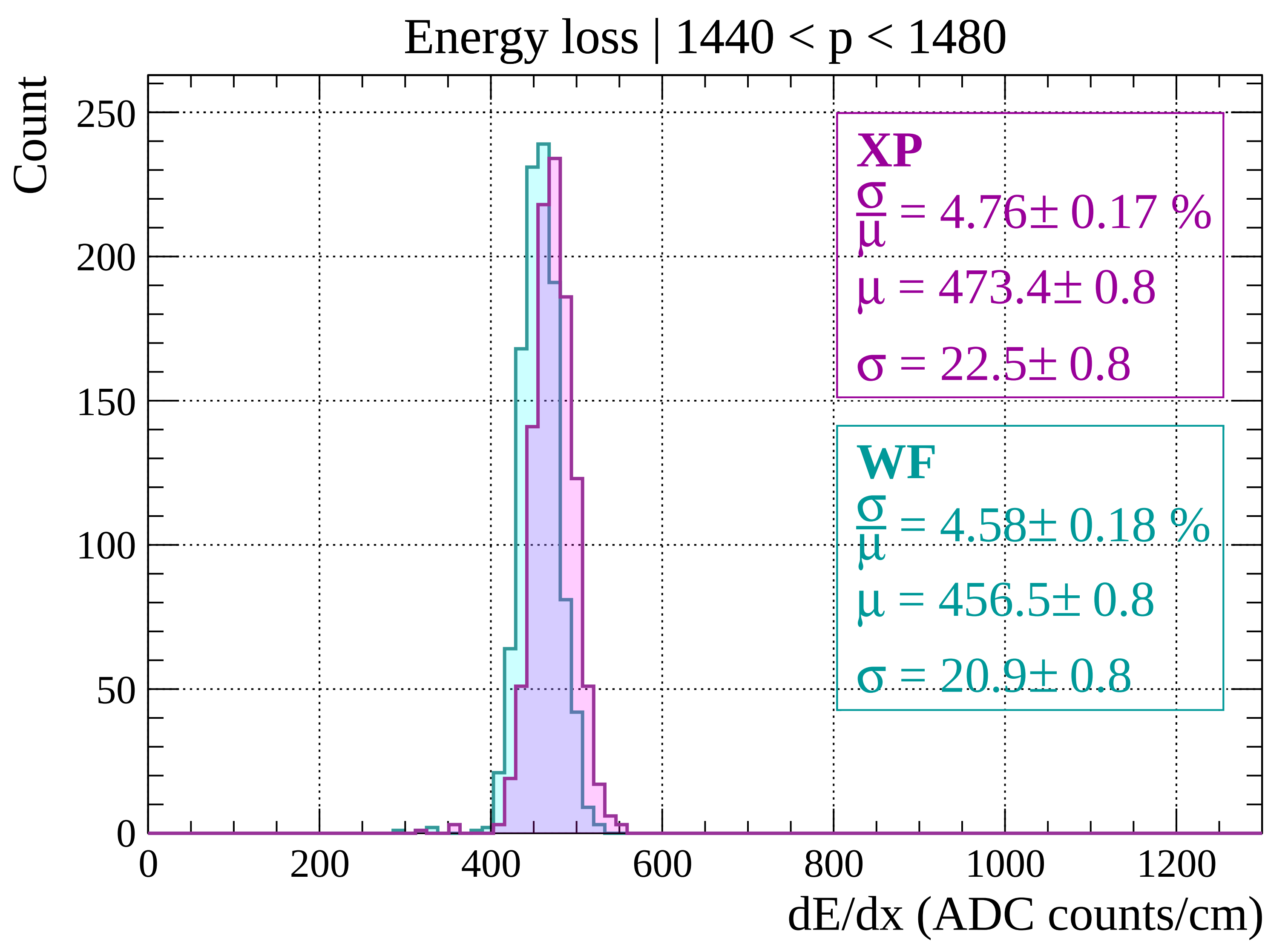


Energy loss | 1320 < p < 1360
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP
σ/μ = 4.85± 0.15 %
μ = 467.9± 0.8
σ = 22.7± 0.7
WF
σ/μ = 4.80± 0.22 %
μ = 451.8± 0.8
σ = 21.7± 0.9

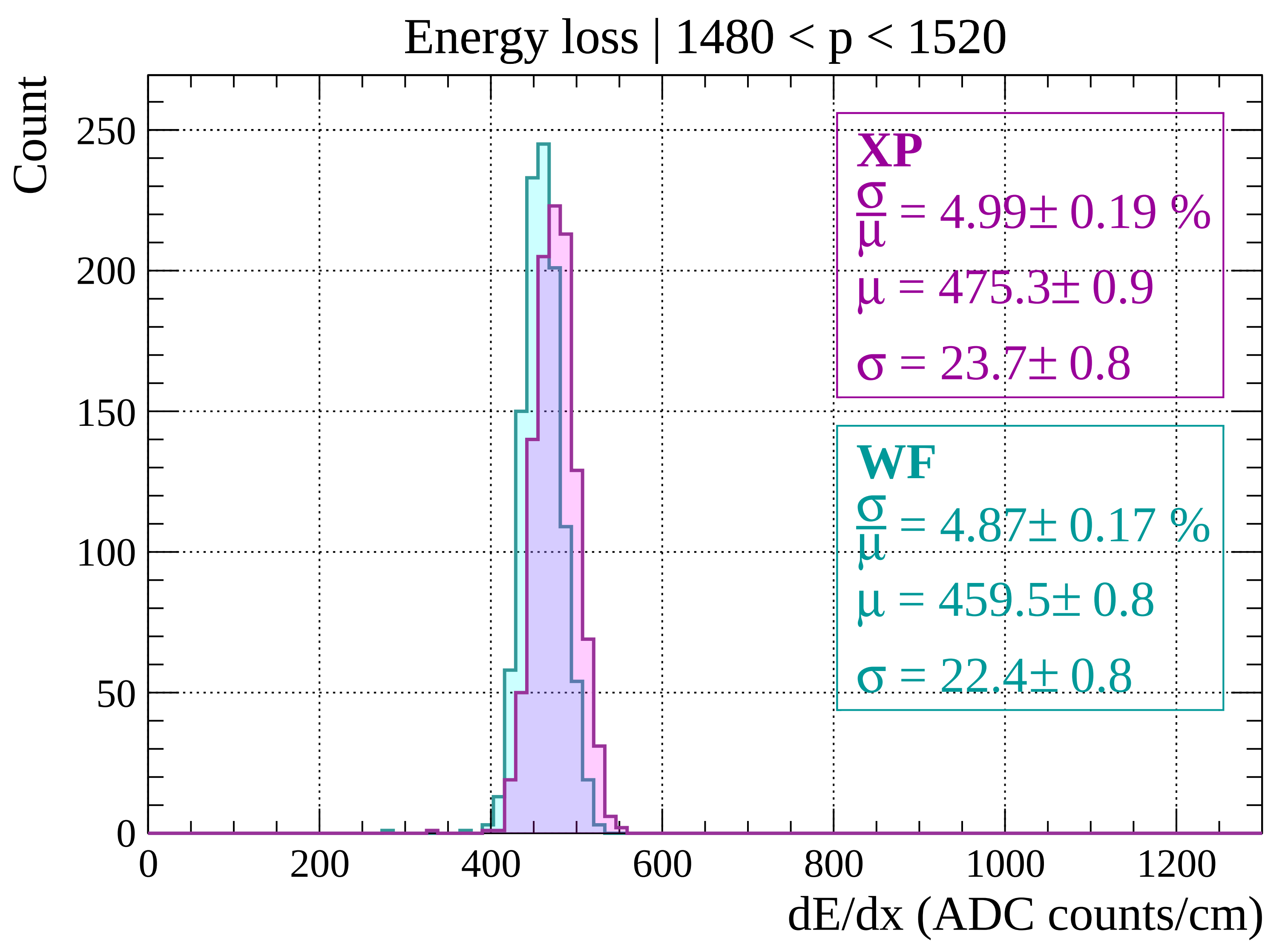


Energy loss | 1360 < p < 1400
Count
0
50
100
150
200
250
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 4.88± 0.18 %
μ = 470.6± 0.8
σ = 23.0± 0.8
WF
σ/μ = 4.80± 0.17 %
μ = 454.4± 0.8
σ = 21.8± 0.7

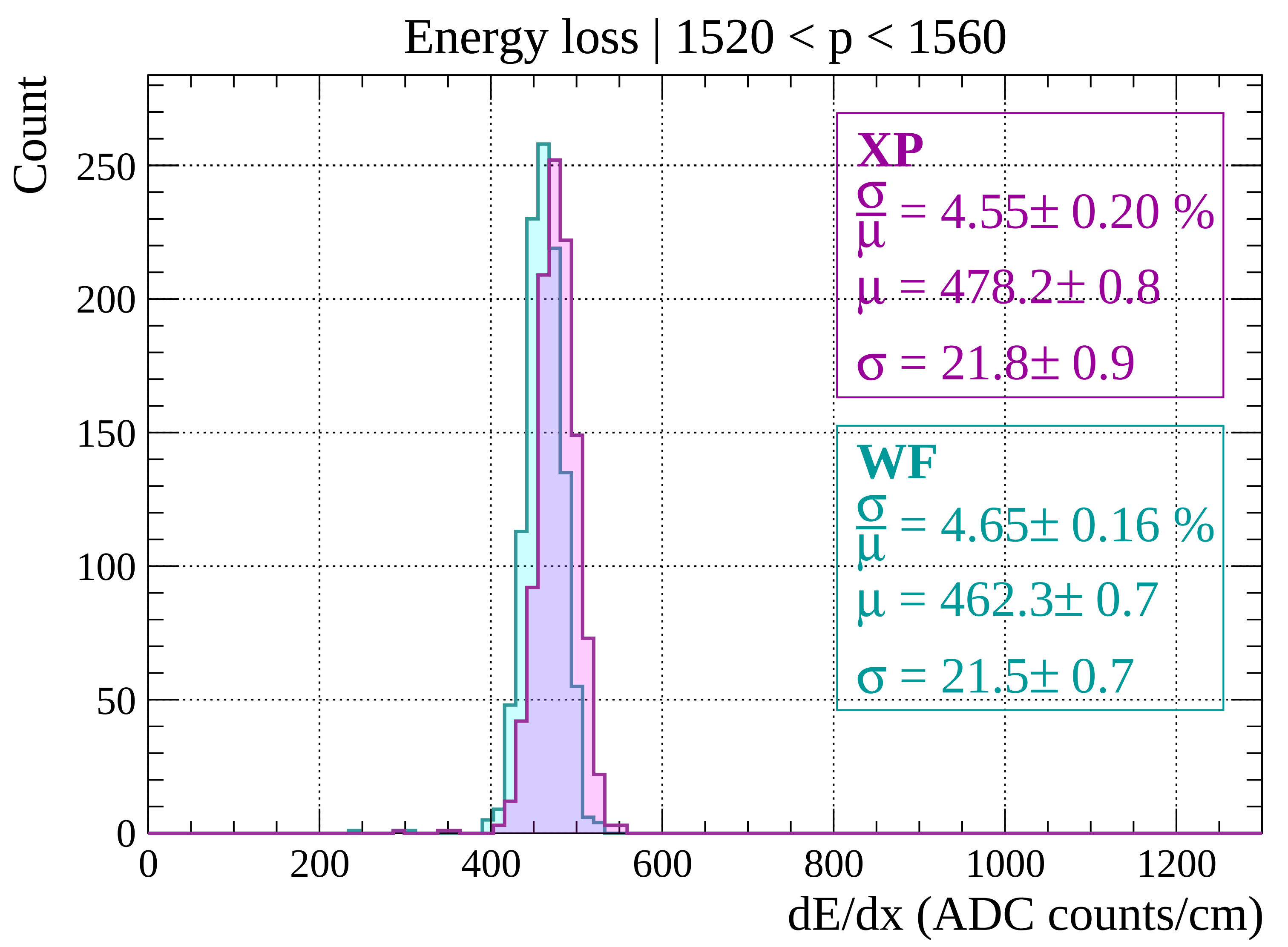

Energy loss | 1400 < p < 1440
Count
0
50
100
150
200
250
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 4.47± 0.19 %
μ = 470.1± 0.8
σ = 21.0± 0.9
WF
$\frac{\sigma}{\mu}$ = 4.34± 0.17 %
μ = 454.6± 0.7
σ = 19.7± 0.7



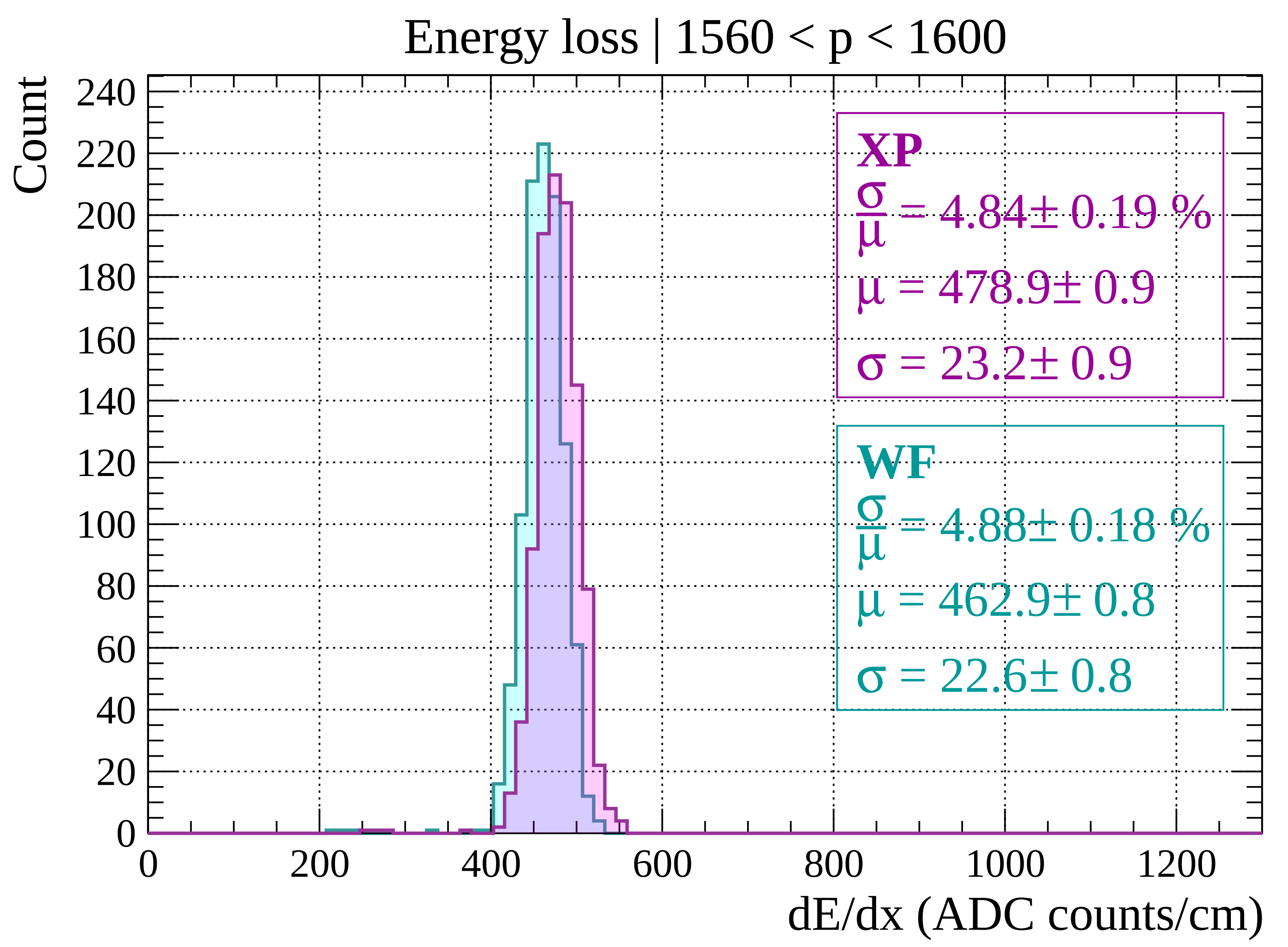
Energy loss | 1440 < p < 1480
Count
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 4.76± 0.17 %
μ = 473.4± 0.8
σ = 22.5± 0.8
WF
σ/μ = 4.58± 0.18 %
μ = 456.5± 0.8
σ = 20.9± 0.8



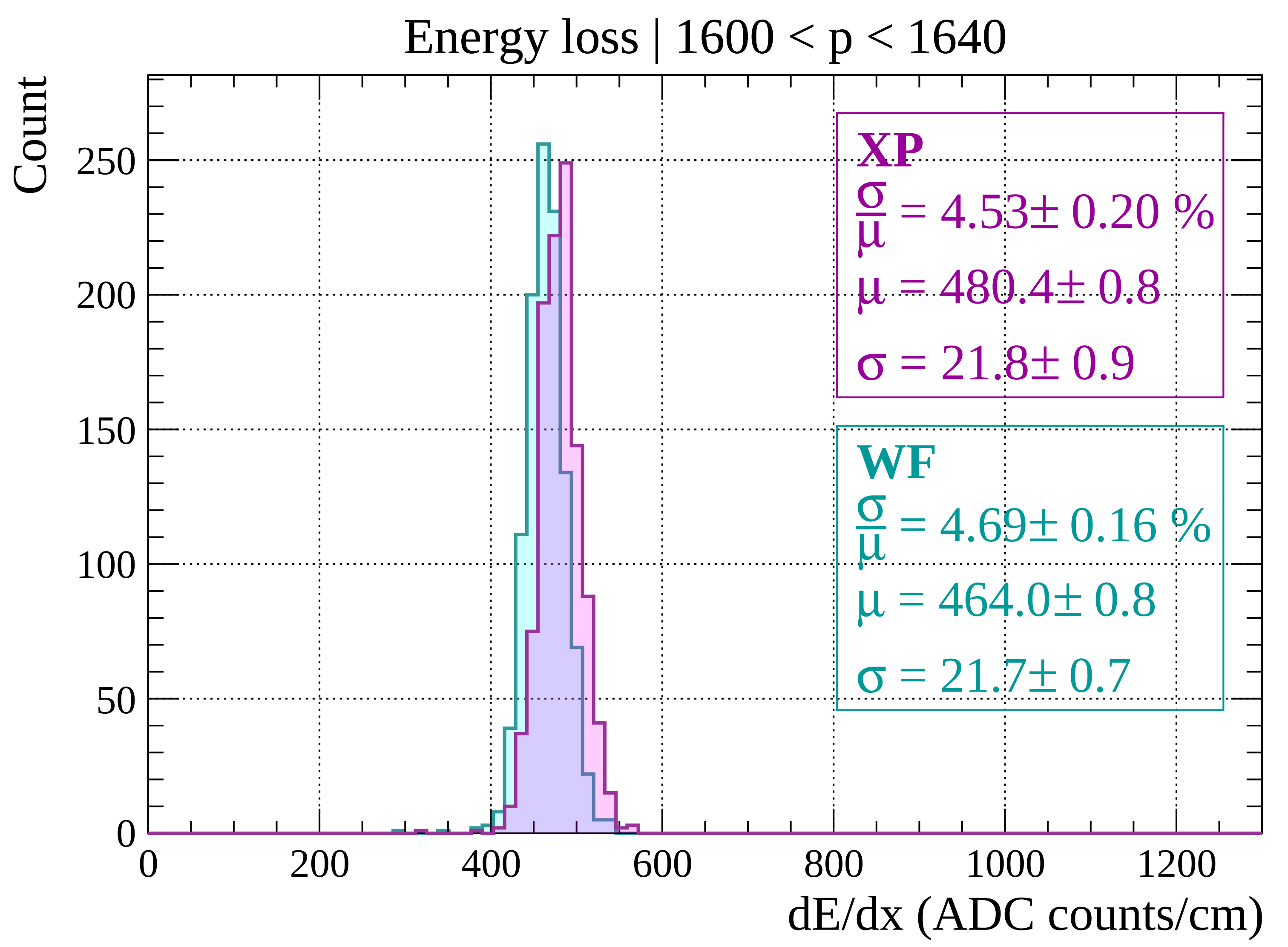
Energy loss | 1480 < p < 1520
Count
0
50
100
150
200
250
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 4.99± 0.19 %
μ = 475.3± 0.9
σ = 23.7± 0.8
WF
$\frac{\sigma}{\mu}$ = 4.87± 0.17 %
μ = 459.5± 0.8
σ = 22.4± 0.8

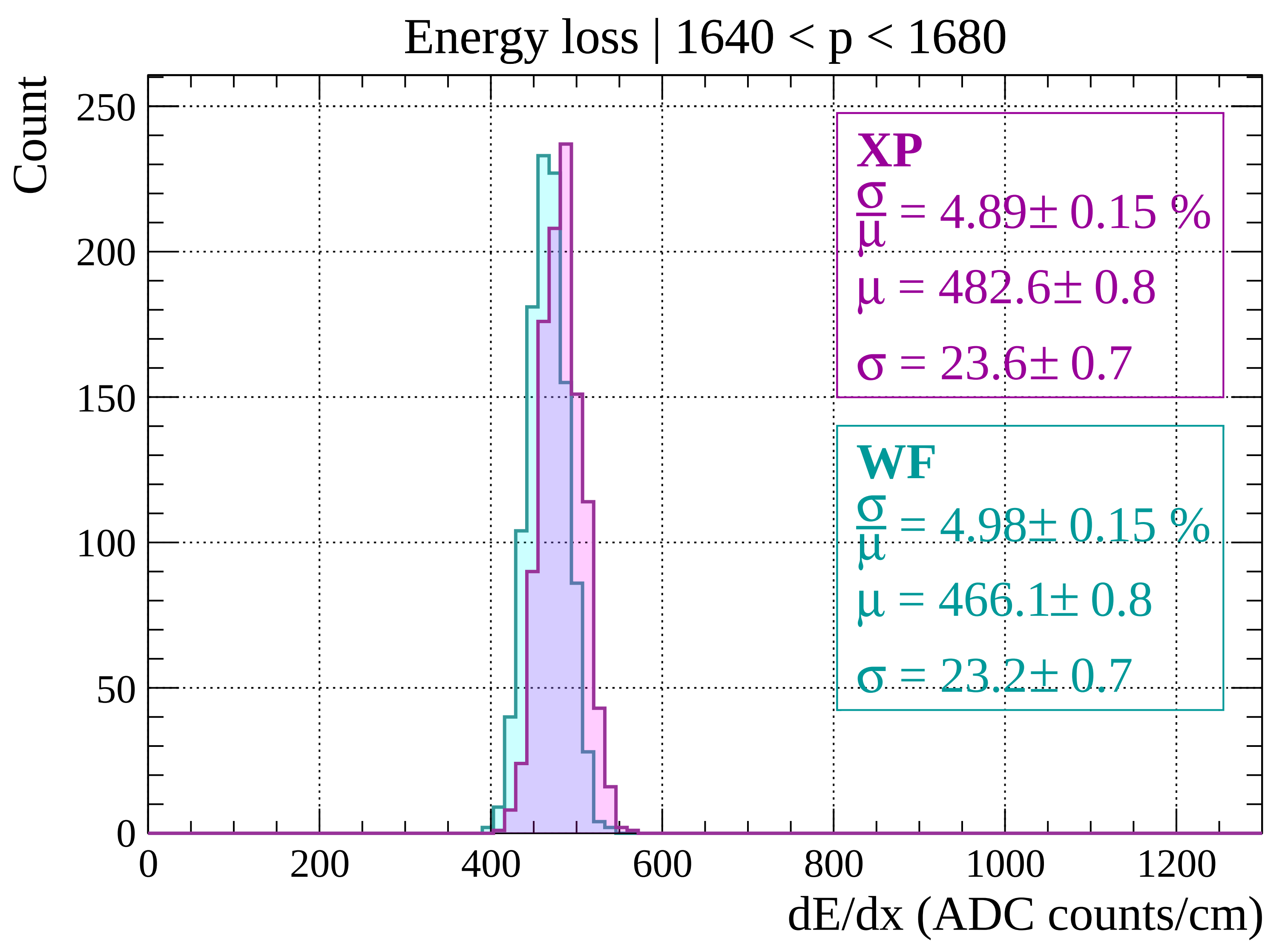


Energy loss | 1520 < p < 1560
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 4.55± 0.20 %
μ = 478.2± 0.8
σ = 21.8± 0.9
WF
$\frac{\sigma}{\mu}$ = 4.65± 0.16 %
μ = 462.3± 0.7
σ = 21.5± 0.7

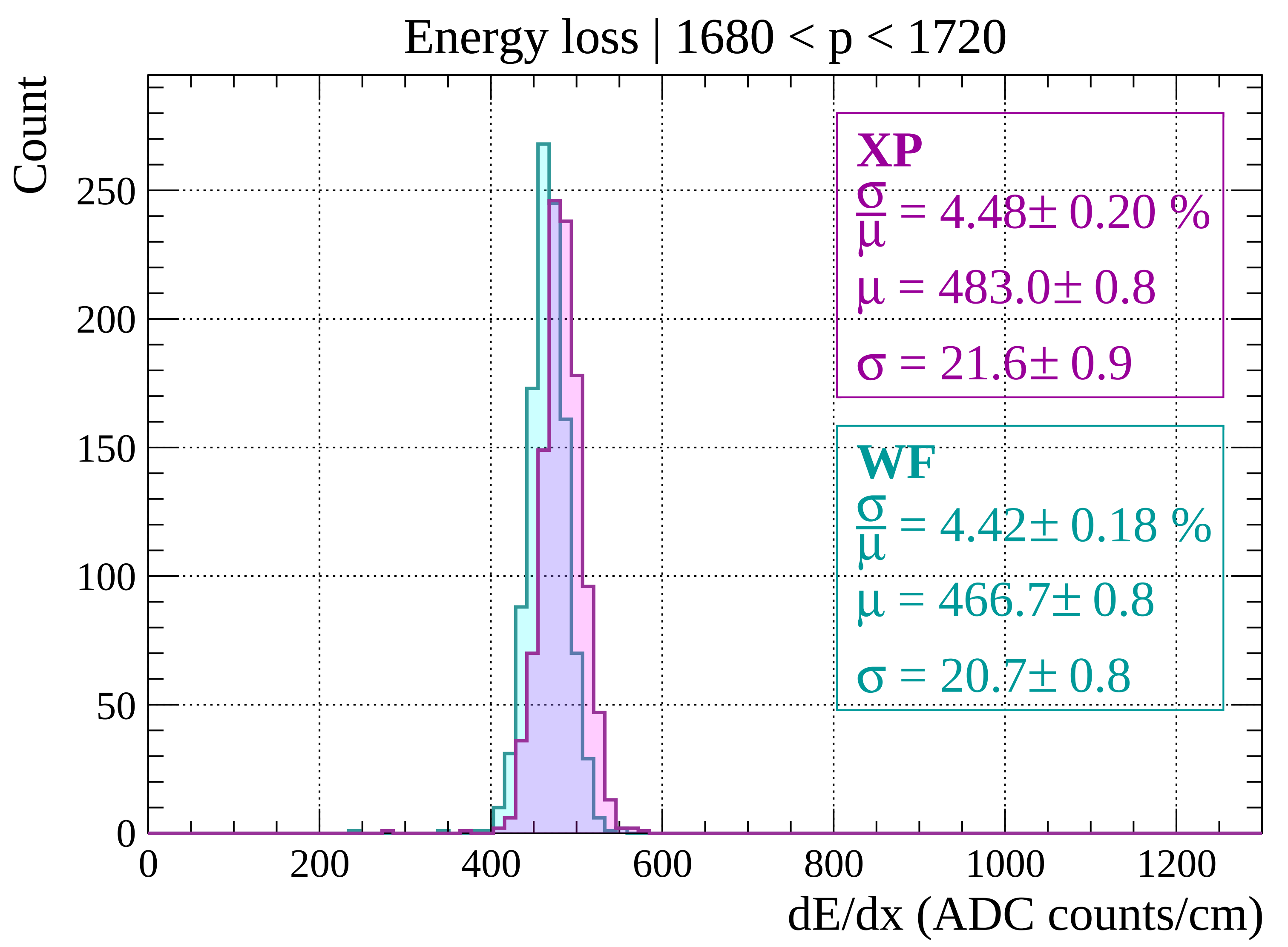


Energy loss | 1560 < p < 1600
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 4.84± 0.19 %
μ = 478.9± 0.9
σ = 23.2± 0.9
WF
σ/μ = 4.88± 0.18 %
μ = 462.9± 0.8
σ = 22.6± 0.8

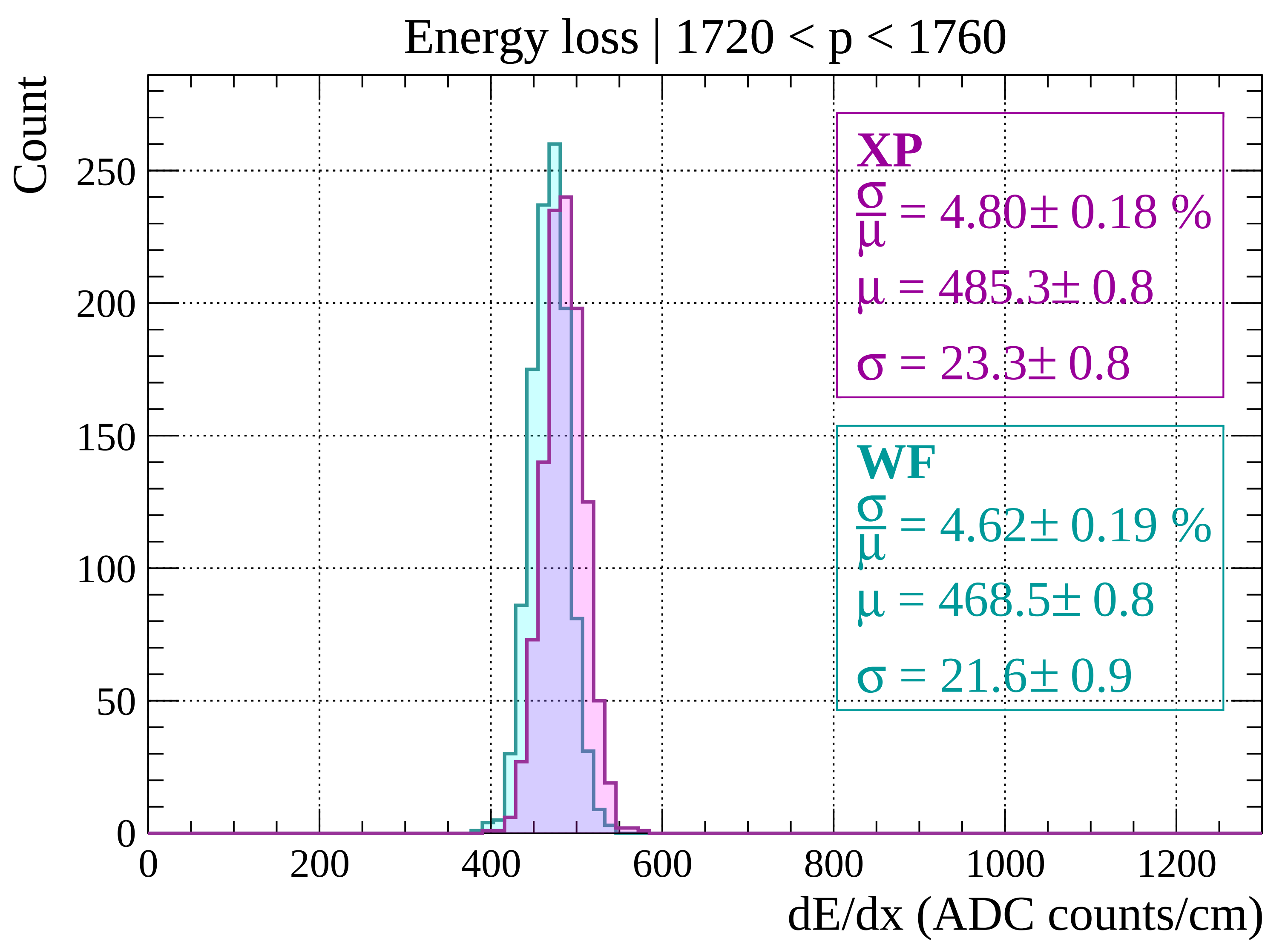

Energy loss | 1600 < p < 1640
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 4.53± 0.20 %
μ = 480.4± 0.8
σ = 21.8± 0.9
WF
σ/μ = 4.69± 0.16 %
μ = 464.0± 0.8
σ = 21.7± 0.7
0
50
100
150
200
250
0
200
400
600
800
1000
1200



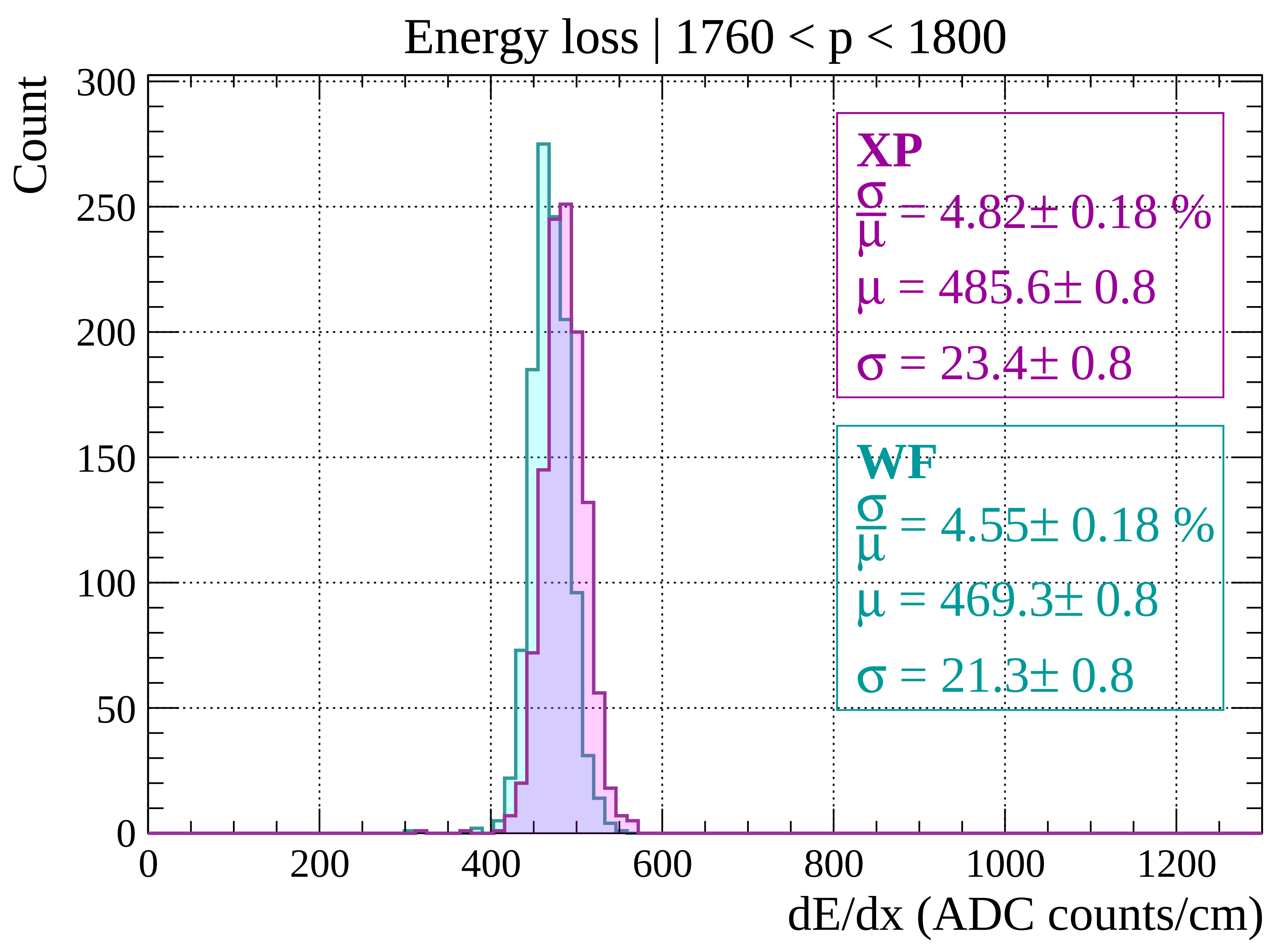
Energy loss | 1640 < p < 1680
Count
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 4.89± 0.15 %
μ = 482.6± 0.8
σ = 23.6± 0.7
WF
$\frac{\sigma}{\mu}$ = 4.98± 0.15 %
μ = 466.1± 0.8
σ = 23.2± 0.7



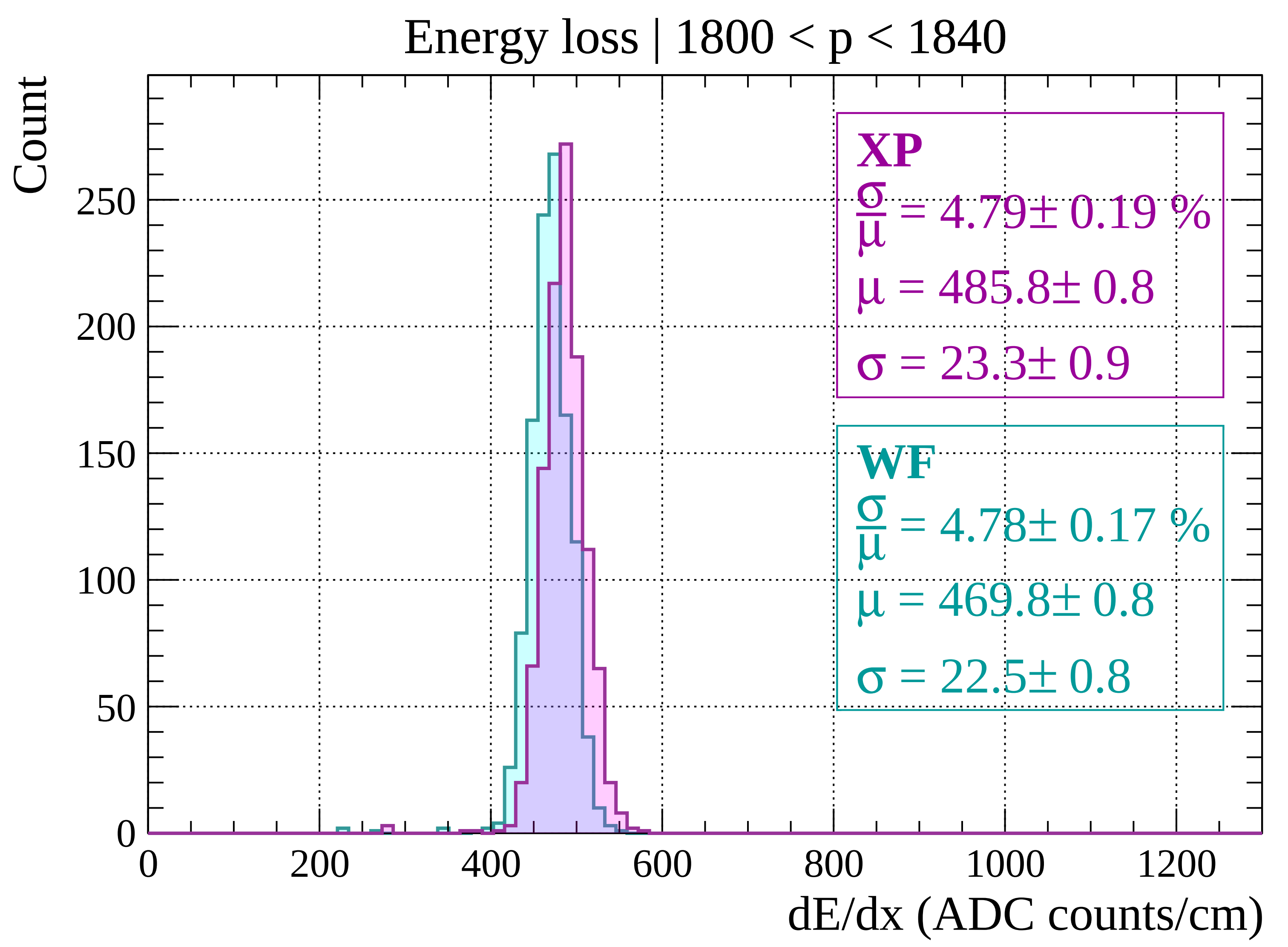
Energy loss | 1680 < p < 1720
Count
0
50
100
150
200
250
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 4.48± 0.20 %
μ = 483.0± 0.8
σ = 21.6± 0.9
WF
$\frac{\sigma}{\mu}$ = 4.42± 0.18 %
μ = 466.7± 0.8
σ = 20.7± 0.8

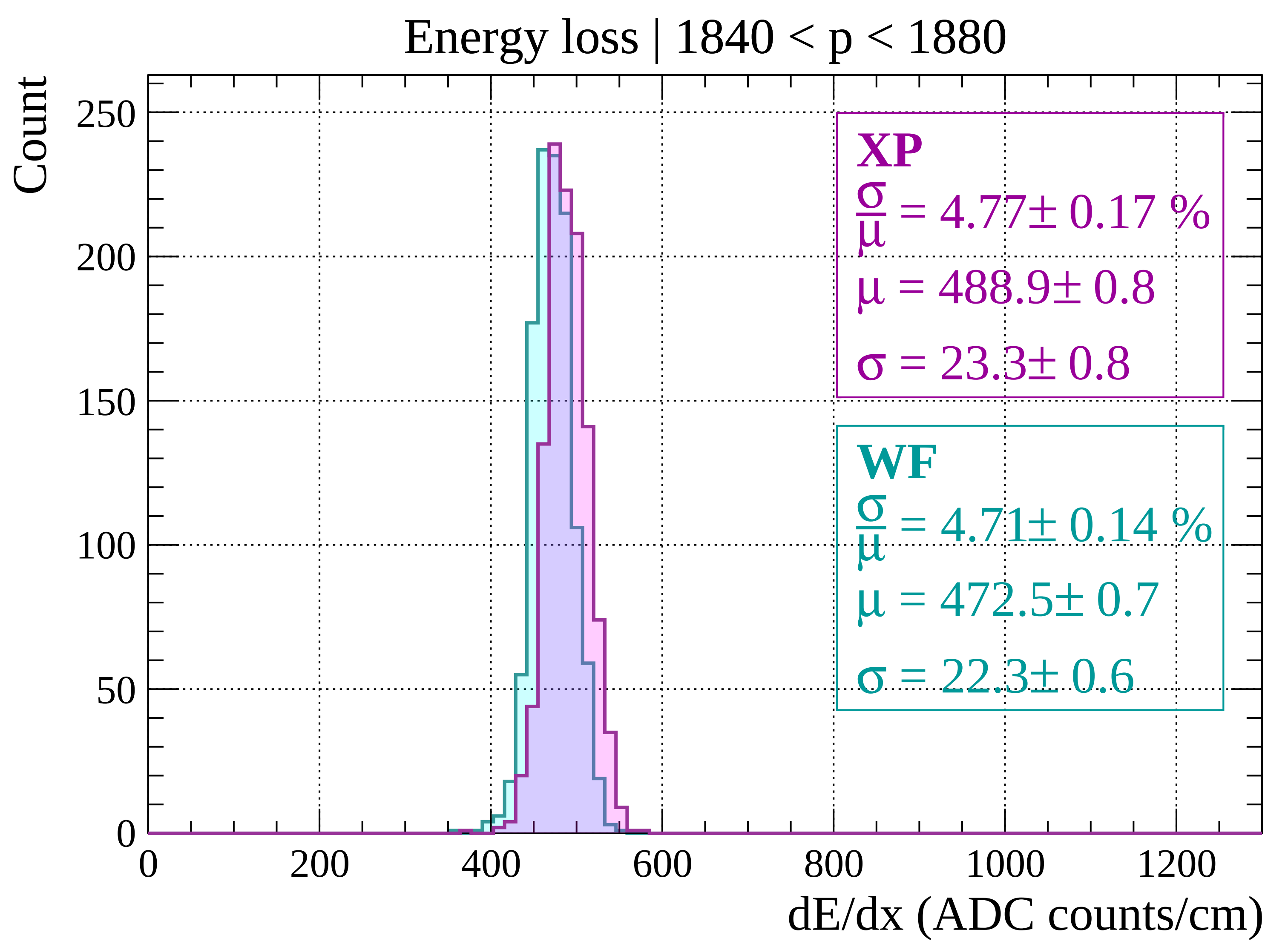


Energy loss | 1720 < p < 1760
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP
σ/μ = 4.80± 0.18 %
μ = 485.3± 0.8
σ = 23.3± 0.8
WF
σ/μ = 4.62± 0.19 %
μ = 468.5± 0.8
σ = 21.6± 0.9

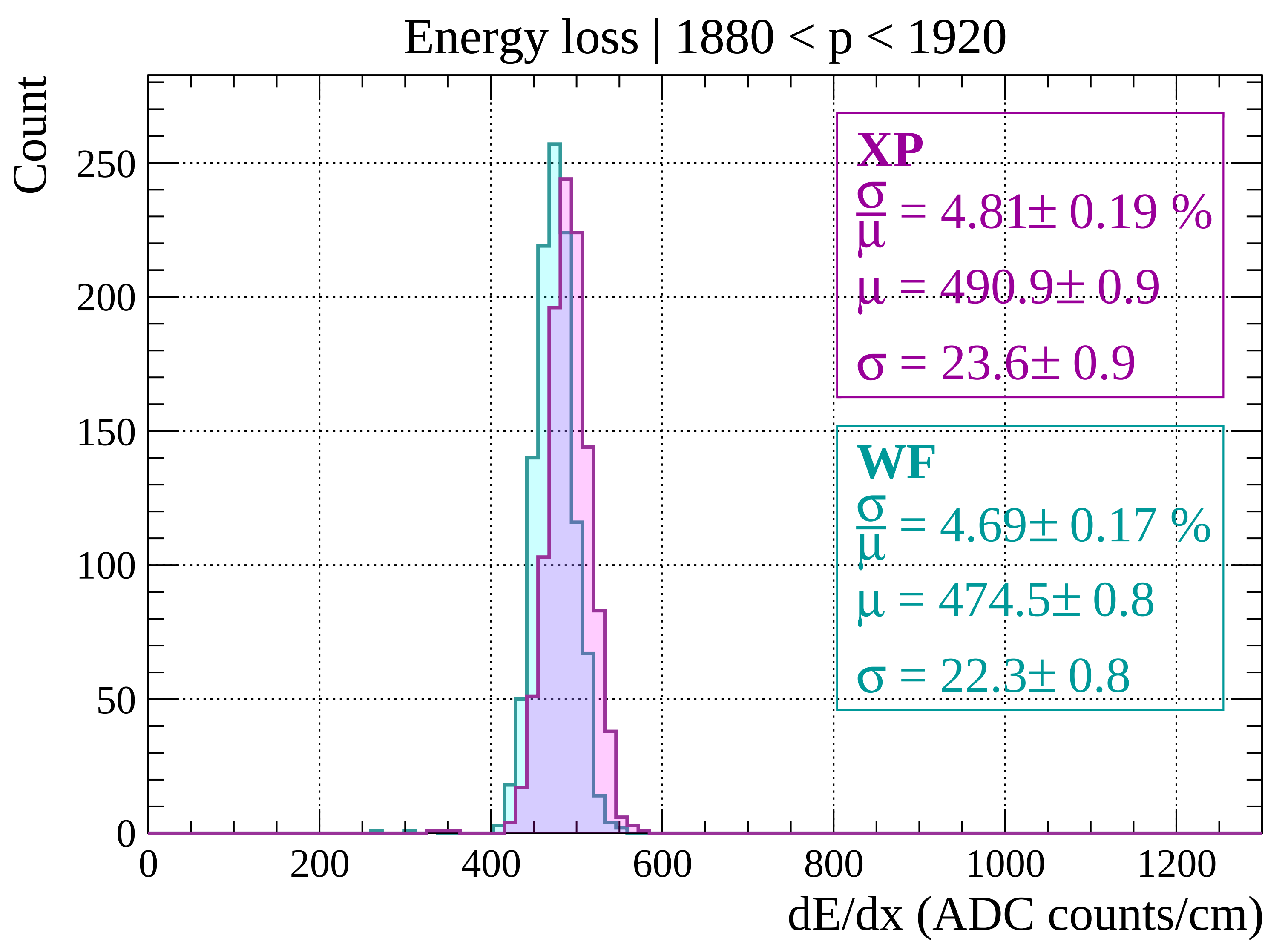


Energy loss | 1760 < p < 1800
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 4.82± 0.18 %
μ = 485.6± 0.8
σ = 23.4± 0.8
WF
$\frac{\sigma}{\mu}$ = 4.55± 0.18 %
μ = 469.3± 0.8
σ = 21.3± 0.8

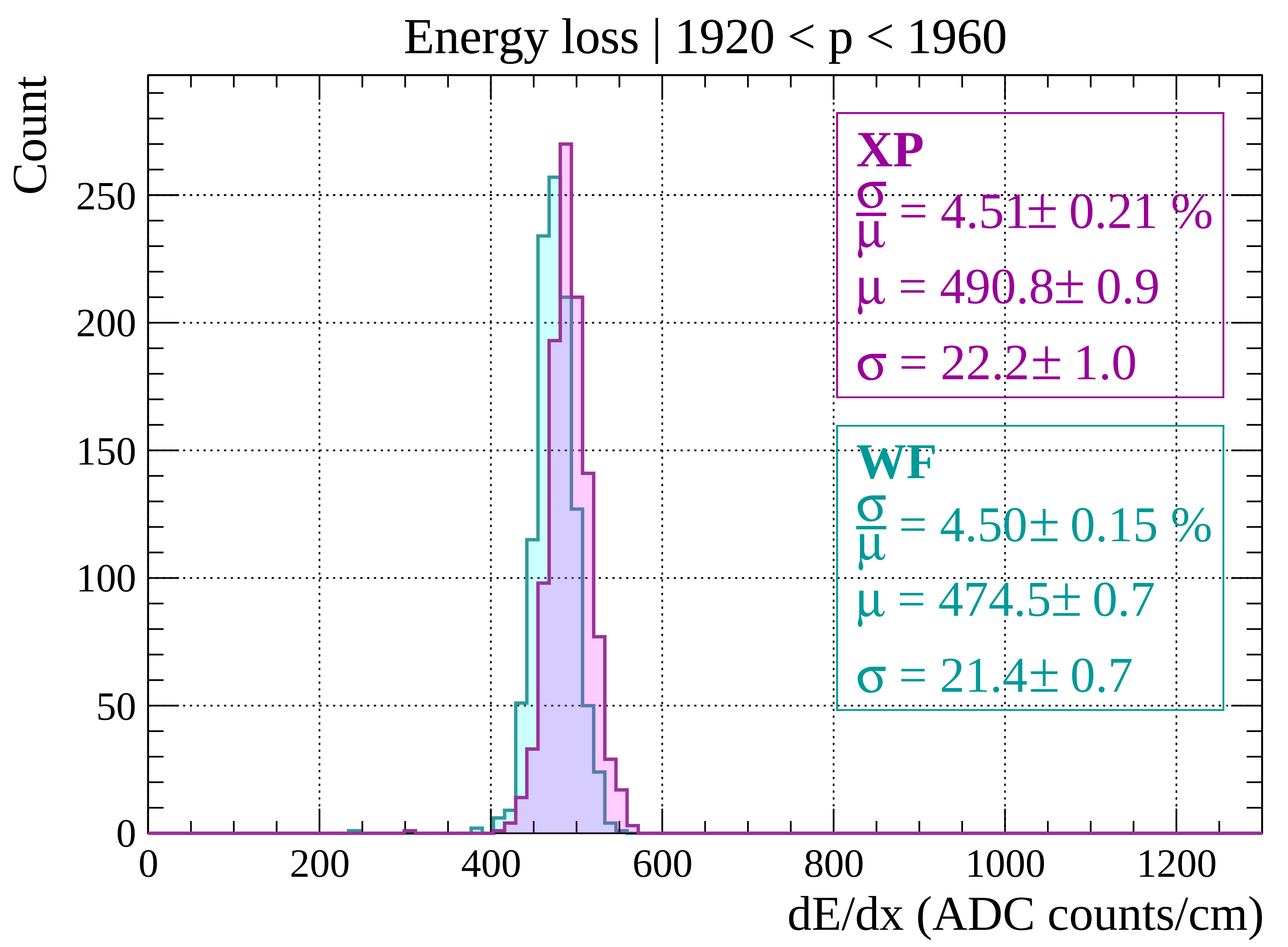

Energy loss | 1800 < p < 1840
Count
0
50
100
150
200
250
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 4.79± 0.19 %
μ = 485.8± 0.8
σ = 23.3± 0.9
WF
σ/μ = 4.78± 0.17 %
μ = 469.8± 0.8
σ = 22.5± 0.8



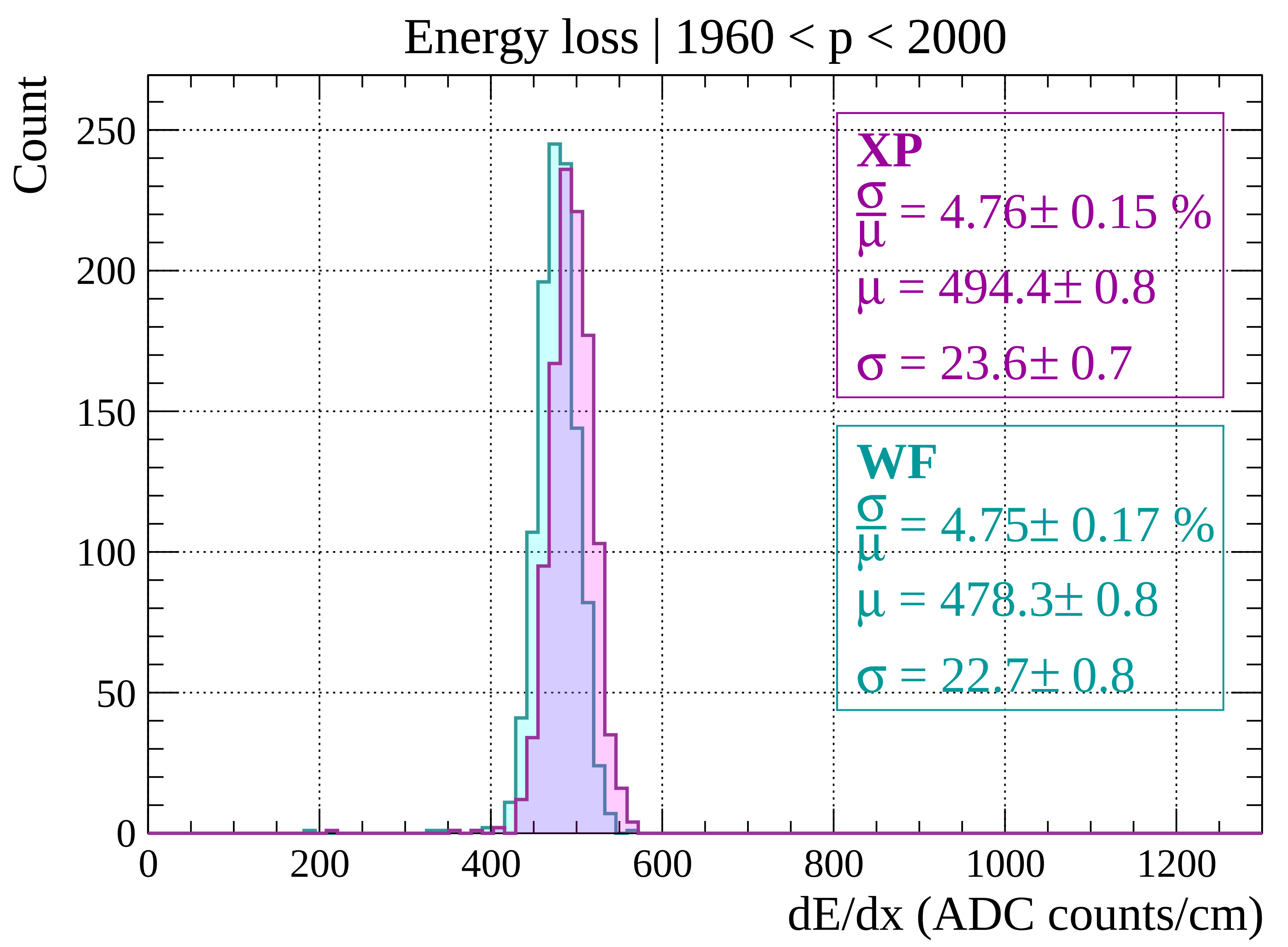
Energy loss | 1840 < p < 1880
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP
σ/μ = 4.77 ± 0.17 %
μ = 488.9 ± 0.8
σ = 23.3 ± 0.8
WF
σ/μ = 4.71 ± 0.14 %
μ = 472.5 ± 0.7
σ = 22.3 ± 0.6

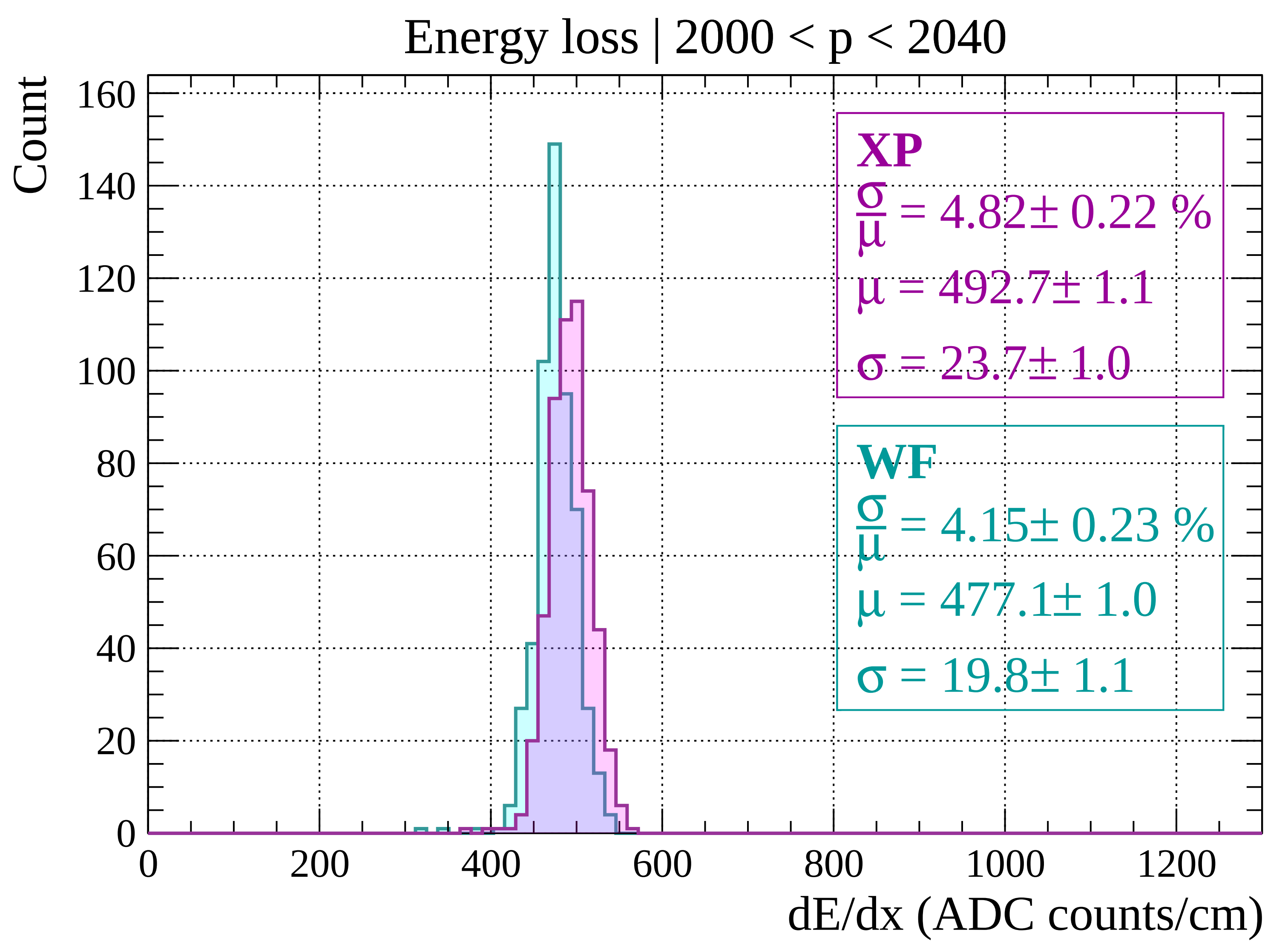


Energy loss | 1880 < p < 1920
Count
0
50
100
150
200
250
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 4.81± 0.19 %
μ = 490.9± 0.9
σ = 23.6± 0.9
WF
$\frac{\sigma}{\mu}$ = 4.69± 0.17 %
μ = 474.5± 0.8
σ = 22.3± 0.8

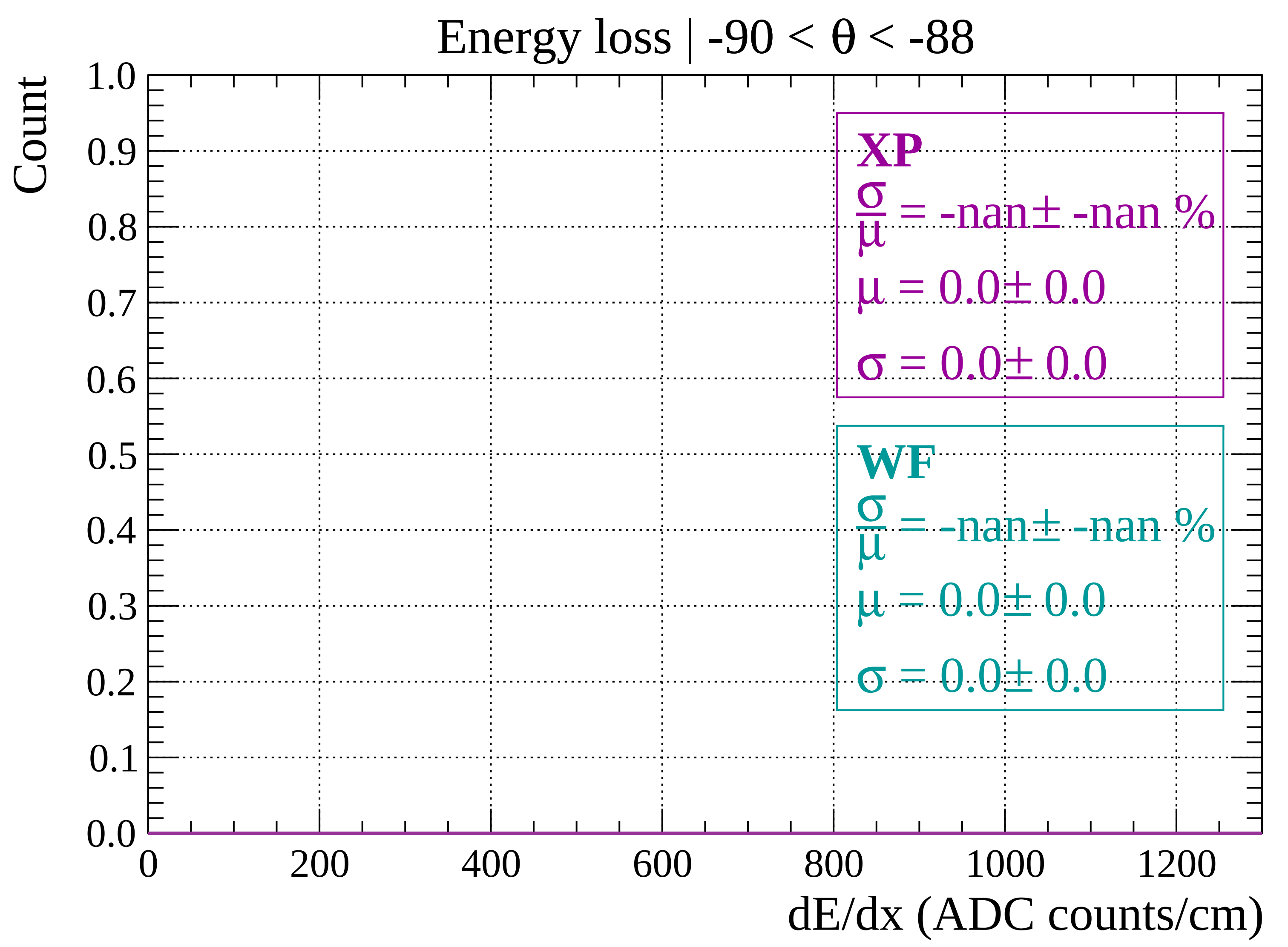


Energy loss | 1920 < p < 1960
Count
0
50
100
150
200
250
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 4.51± 0.21 %
μ = 490.8± 0.9
σ = 22.2± 1.0
WF
σ/μ = 4.50± 0.15 %
μ = 474.5± 0.7
σ = 21.4± 0.7

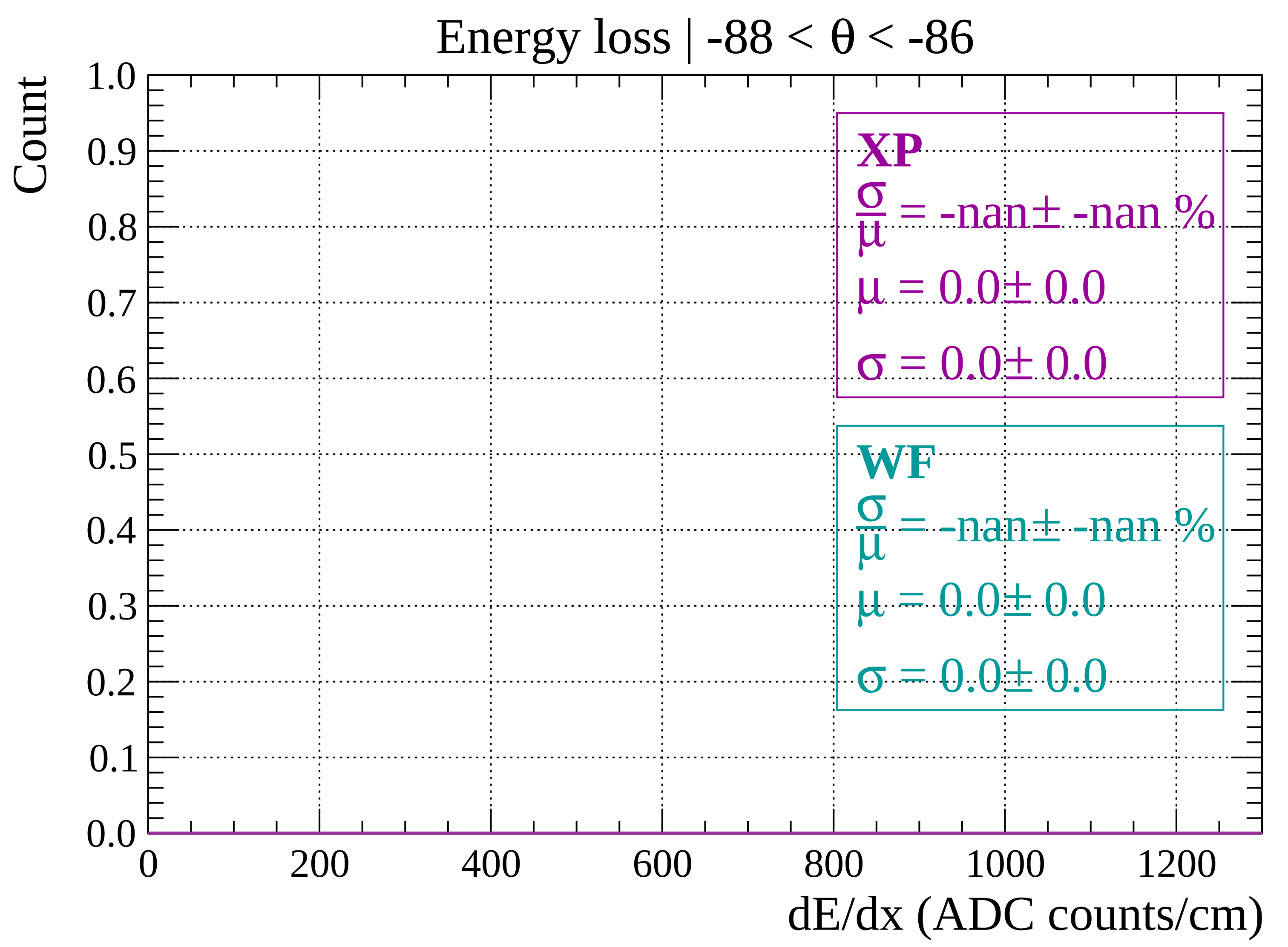

Energy loss | 1960 < p < 2000
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP
σ/μ = 4.76± 0.15 %
μ = 494.4± 0.8
σ = 23.6± 0.7
WF
σ/μ = 4.75± 0.17 %
μ = 478.3± 0.8
σ = 22.7± 0.8

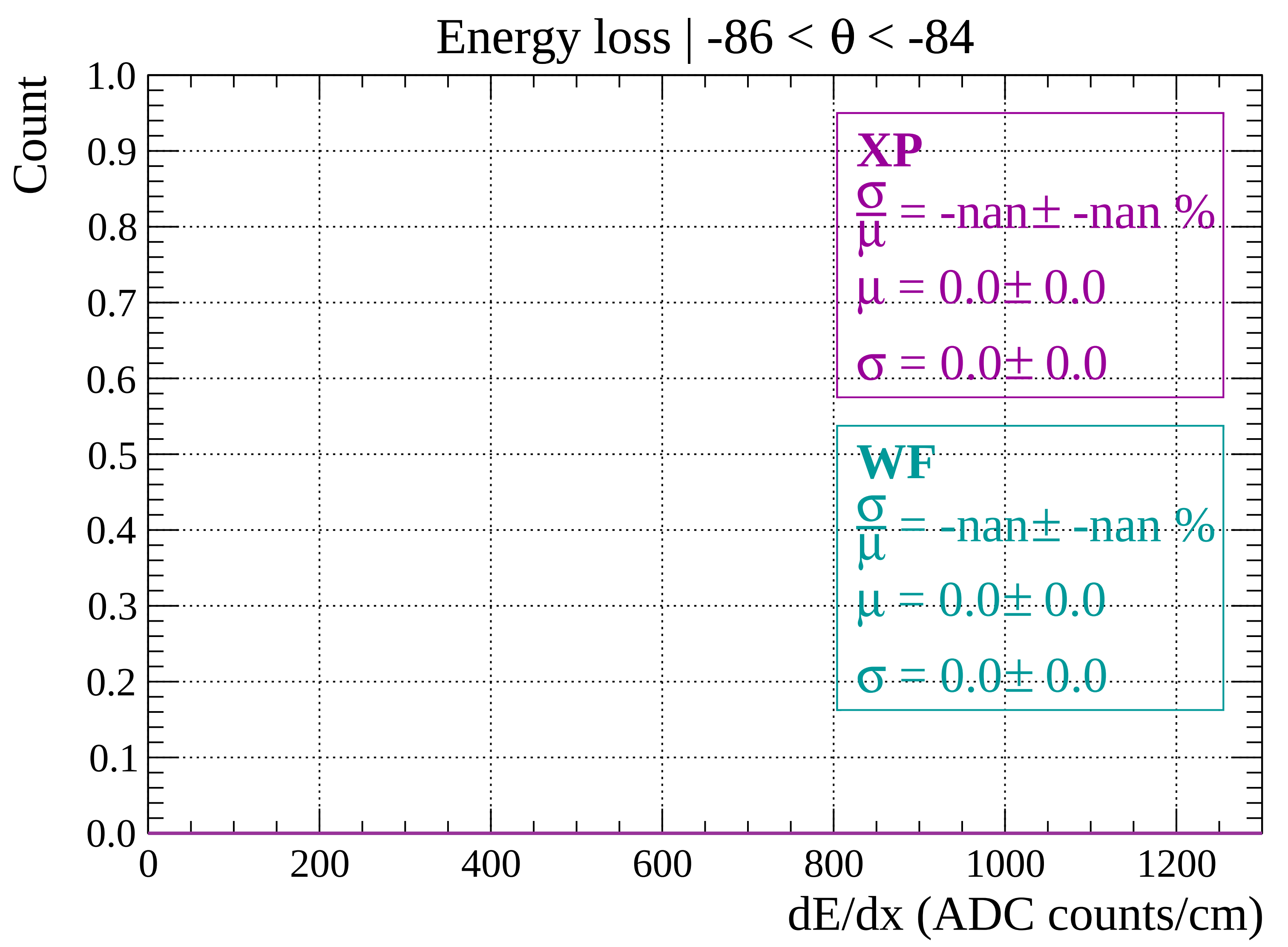


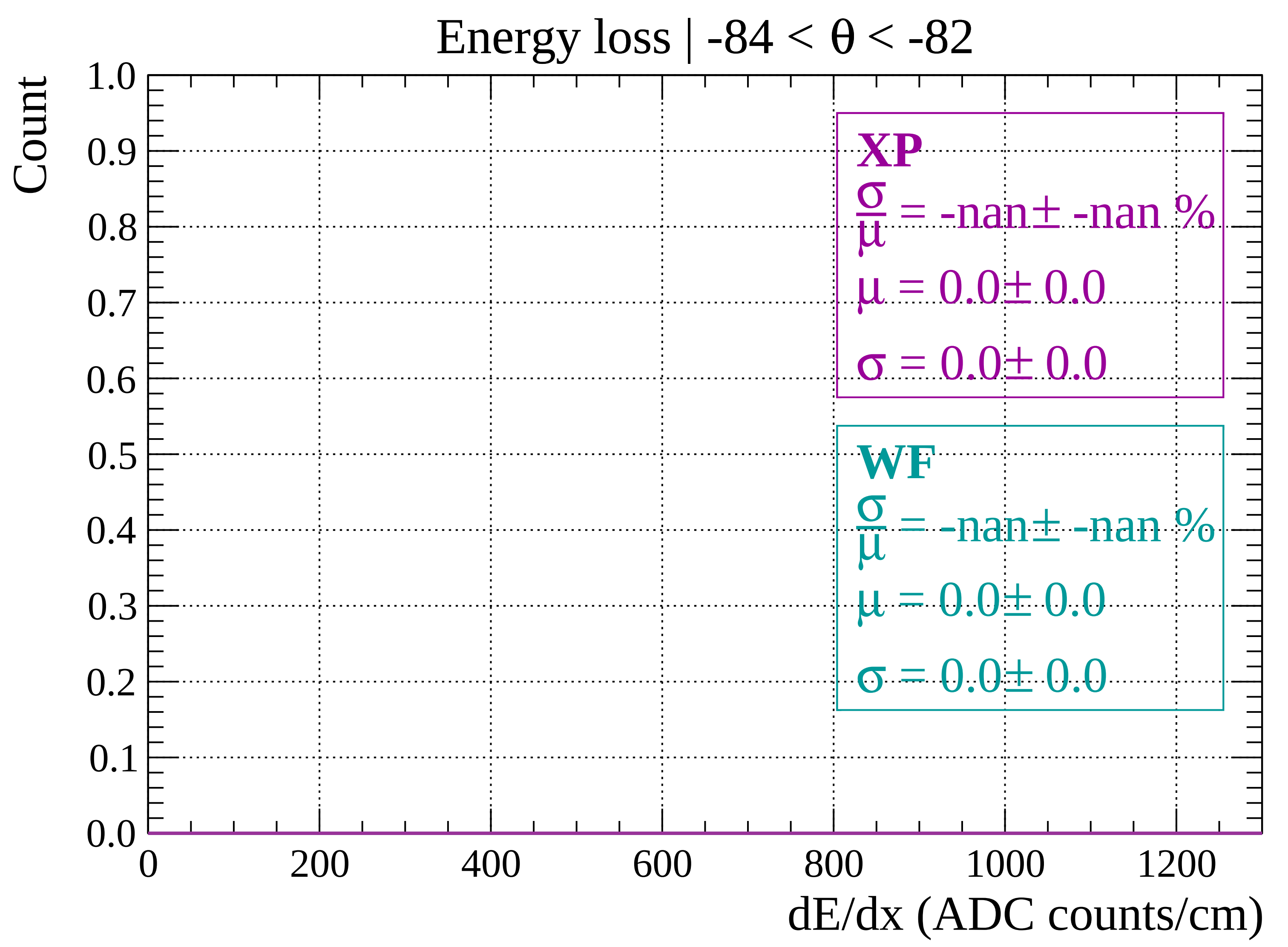
Energy loss | 2000 < p < 2040
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
XP
σ/μ = 4.82± 0.22 %
μ = 492.7± 1.1
σ = 23.7± 1.0
WF
σ/μ = 4.15± 0.23 %
μ = 477.1± 1.0
σ = 19.8± 1.1



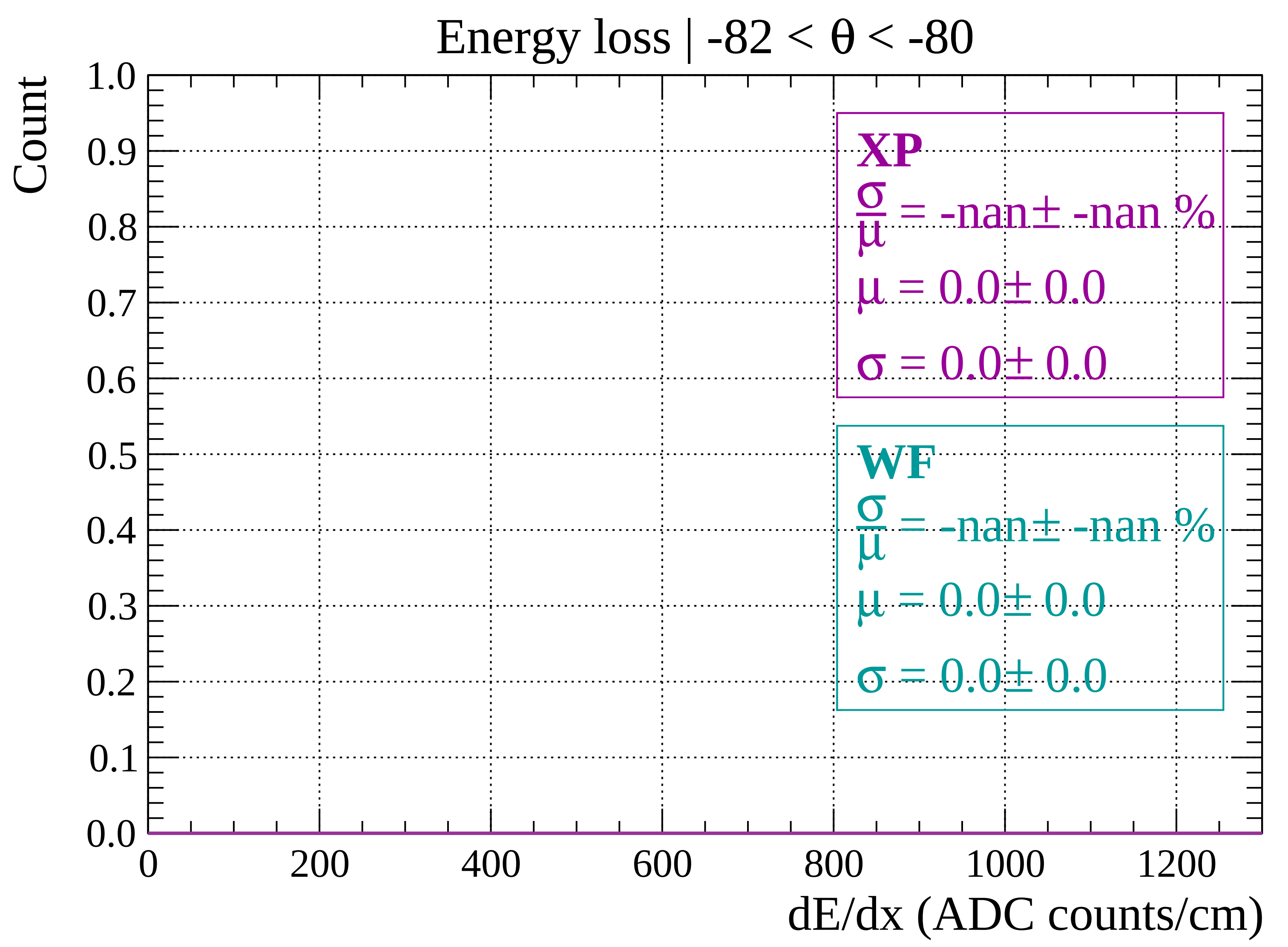
Energy loss | -90 < θ < -88
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0


Energy loss | -88 < θ < -86
Count
dE/dx (ADC counts/cm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

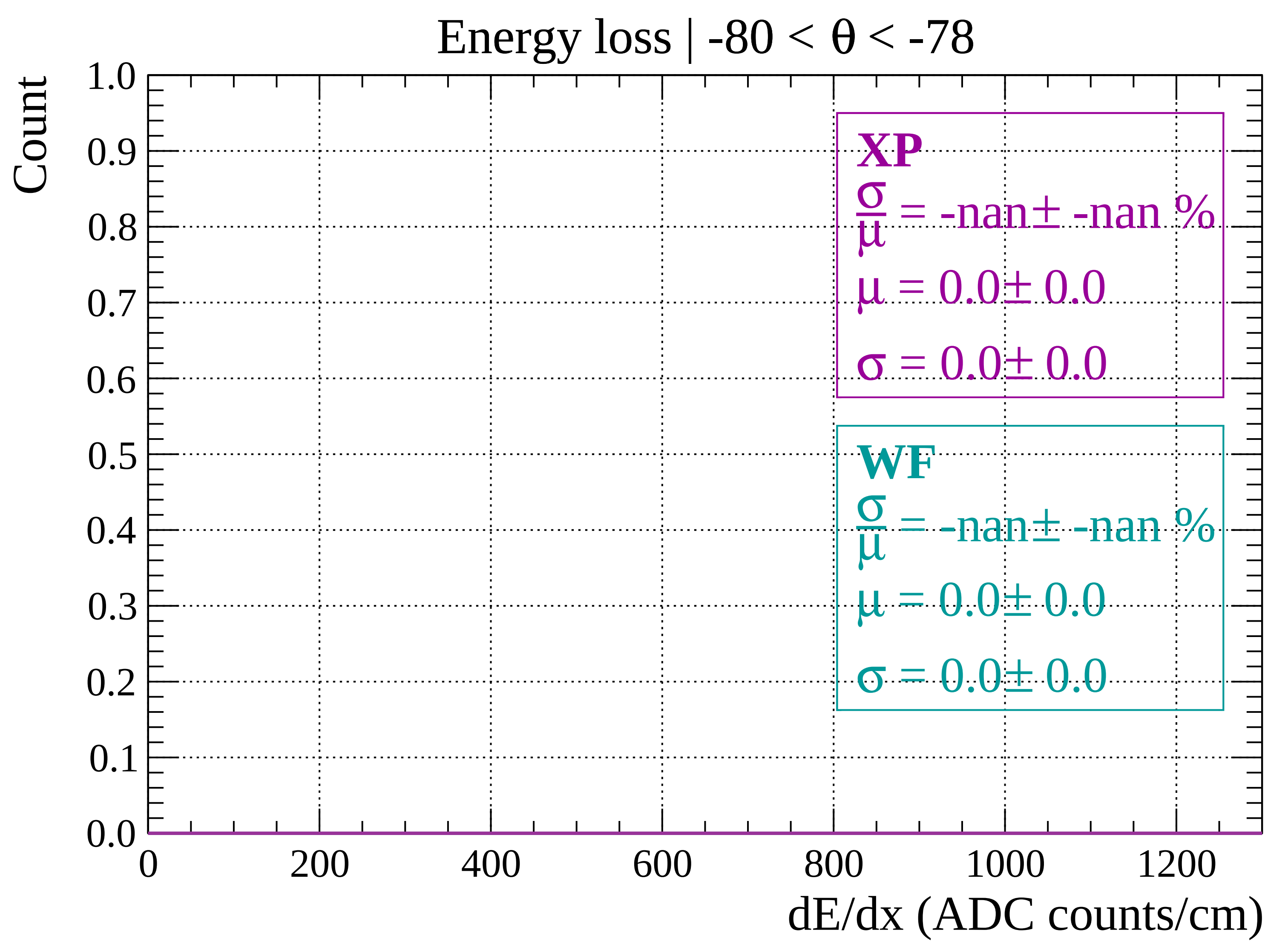

Energy loss | -86 < θ < -84
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



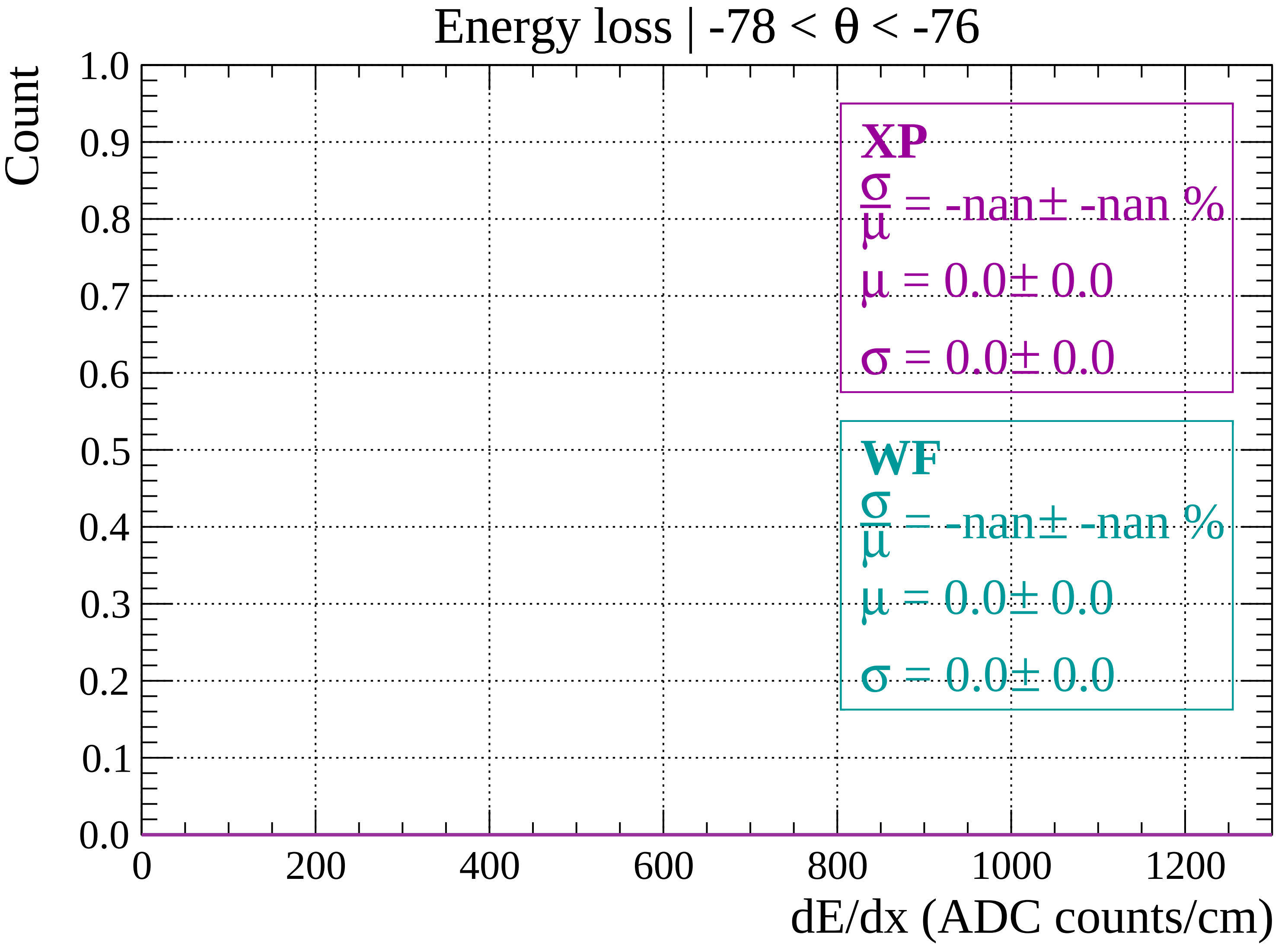
Energy loss | -84 < θ < -82
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



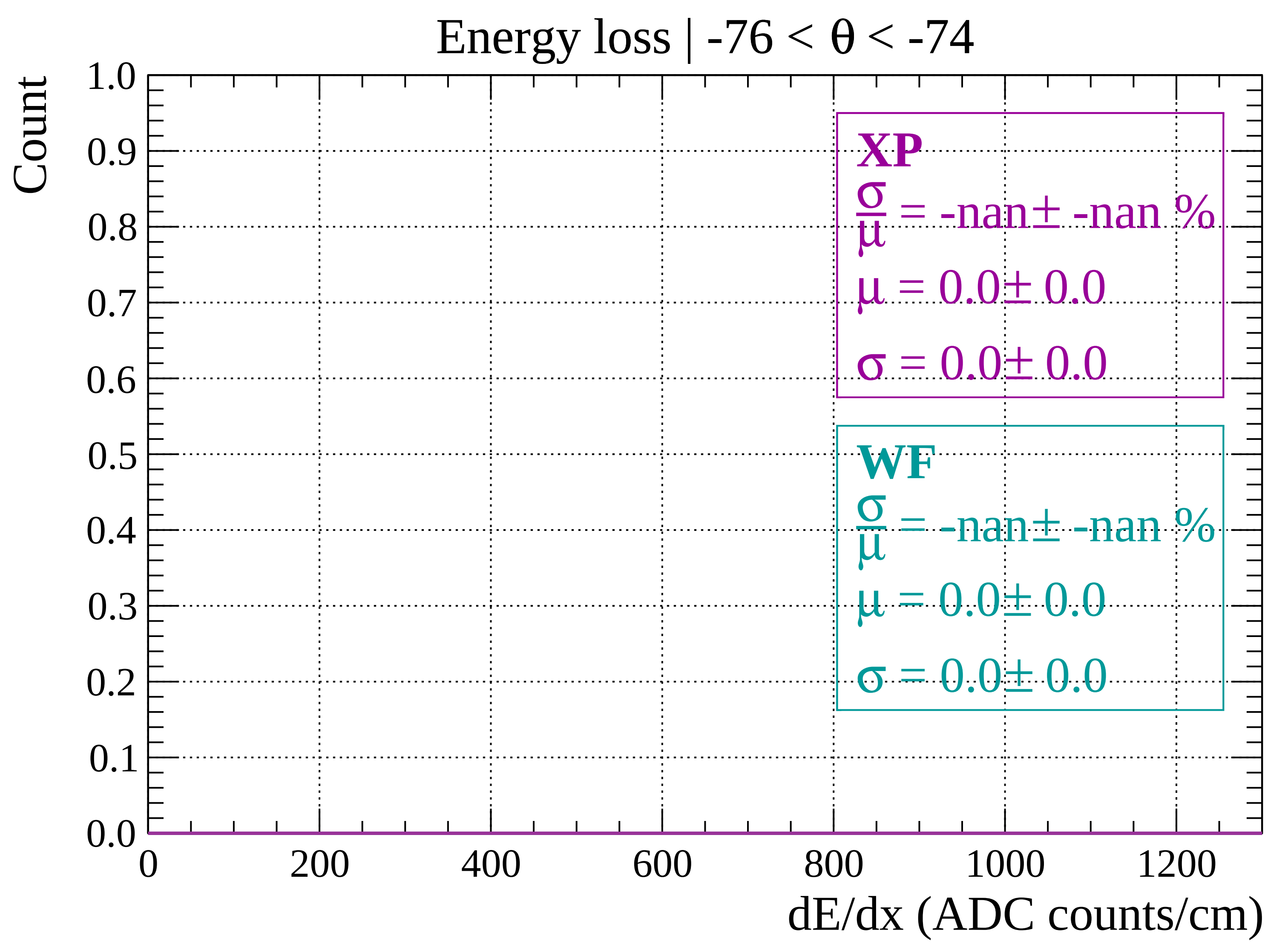
Energy loss | -82 < θ < -80
Count
dE/dx (ADC counts/cm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



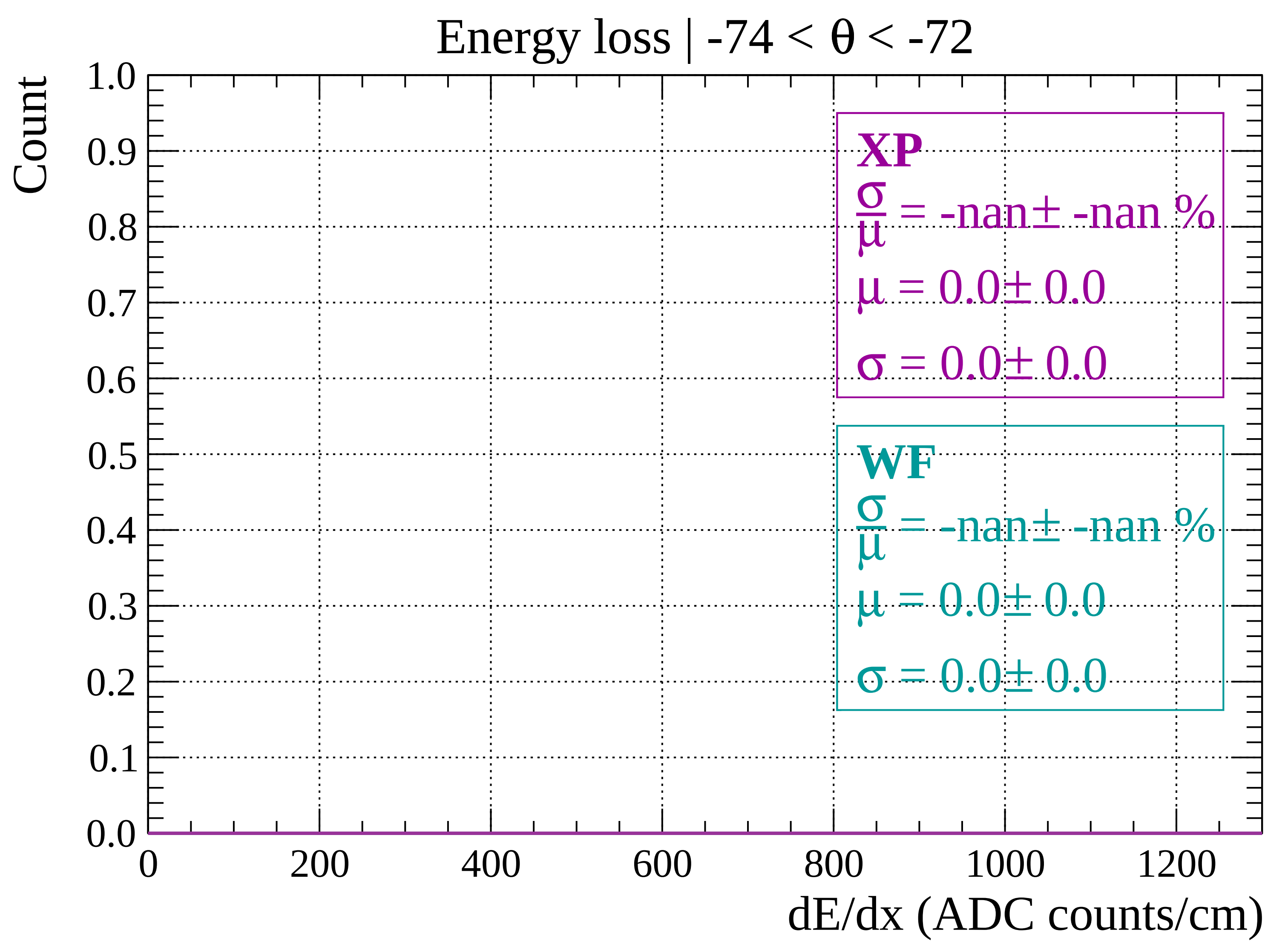
Energy loss | -80 < θ < -78
Count
dE/dx (ADC counts/cm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



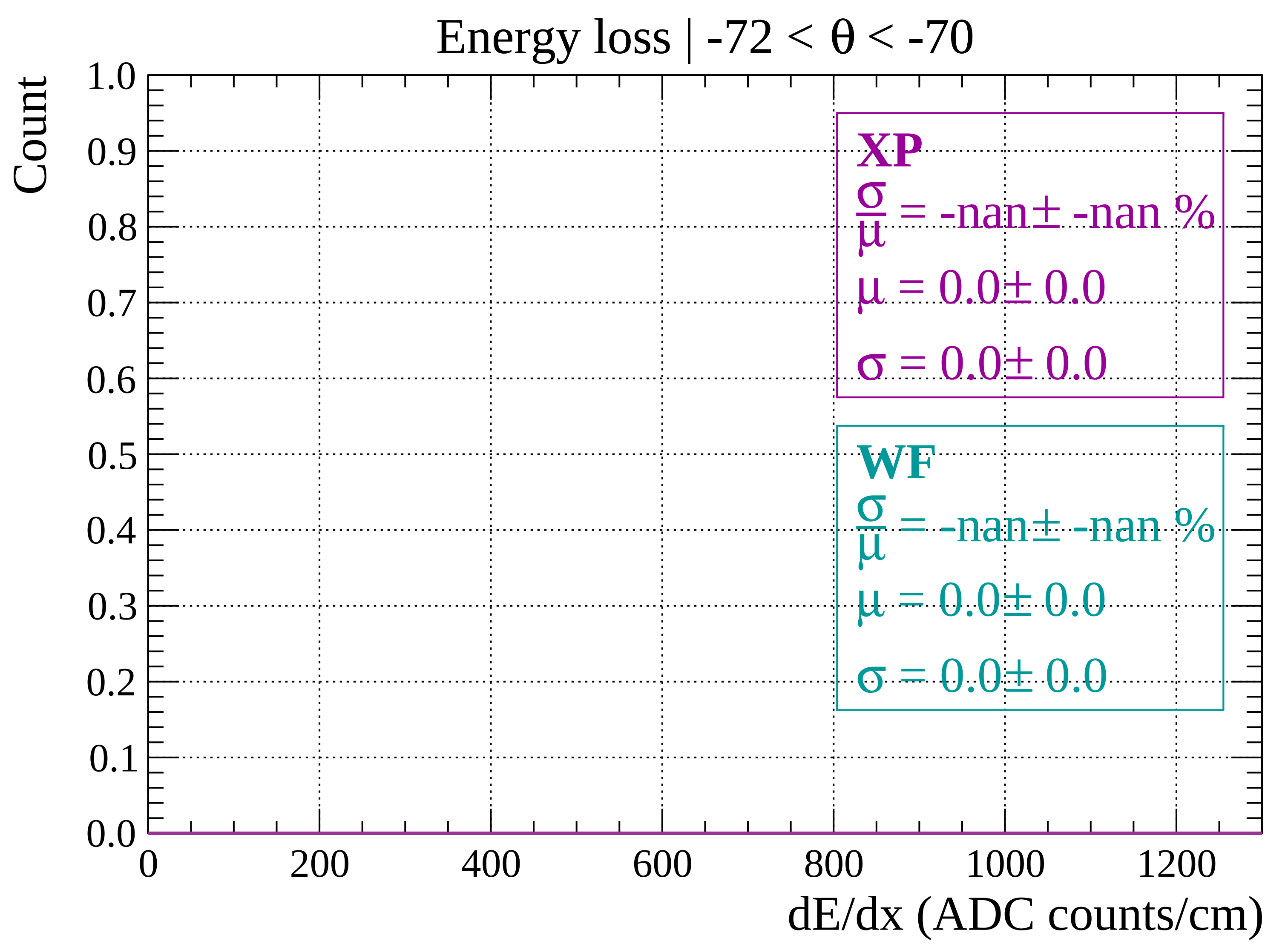
Energy loss | -78 < θ < -76
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

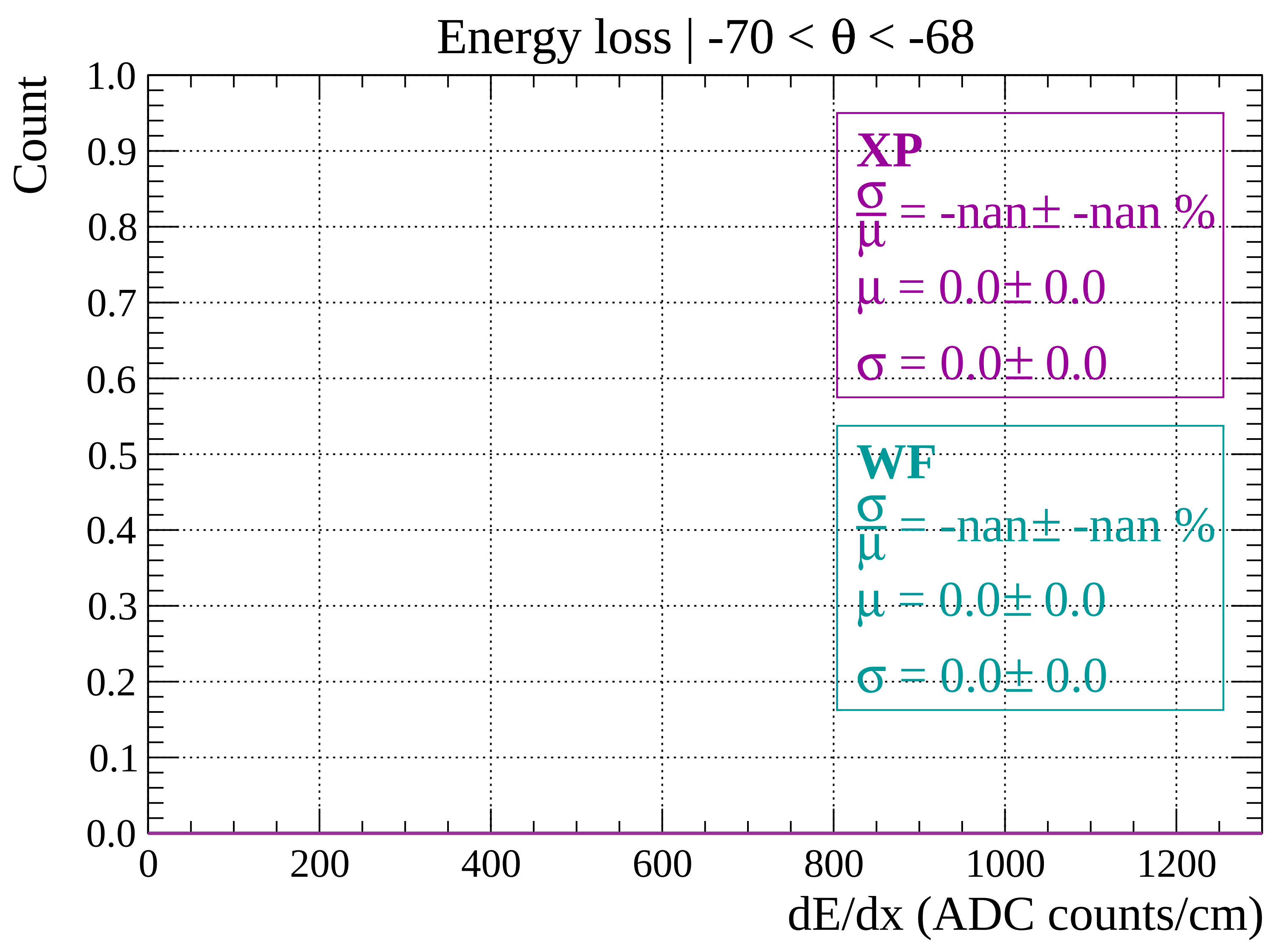


Energy loss | -76 < θ < -74
Count
dE/dx (ADC counts/cm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

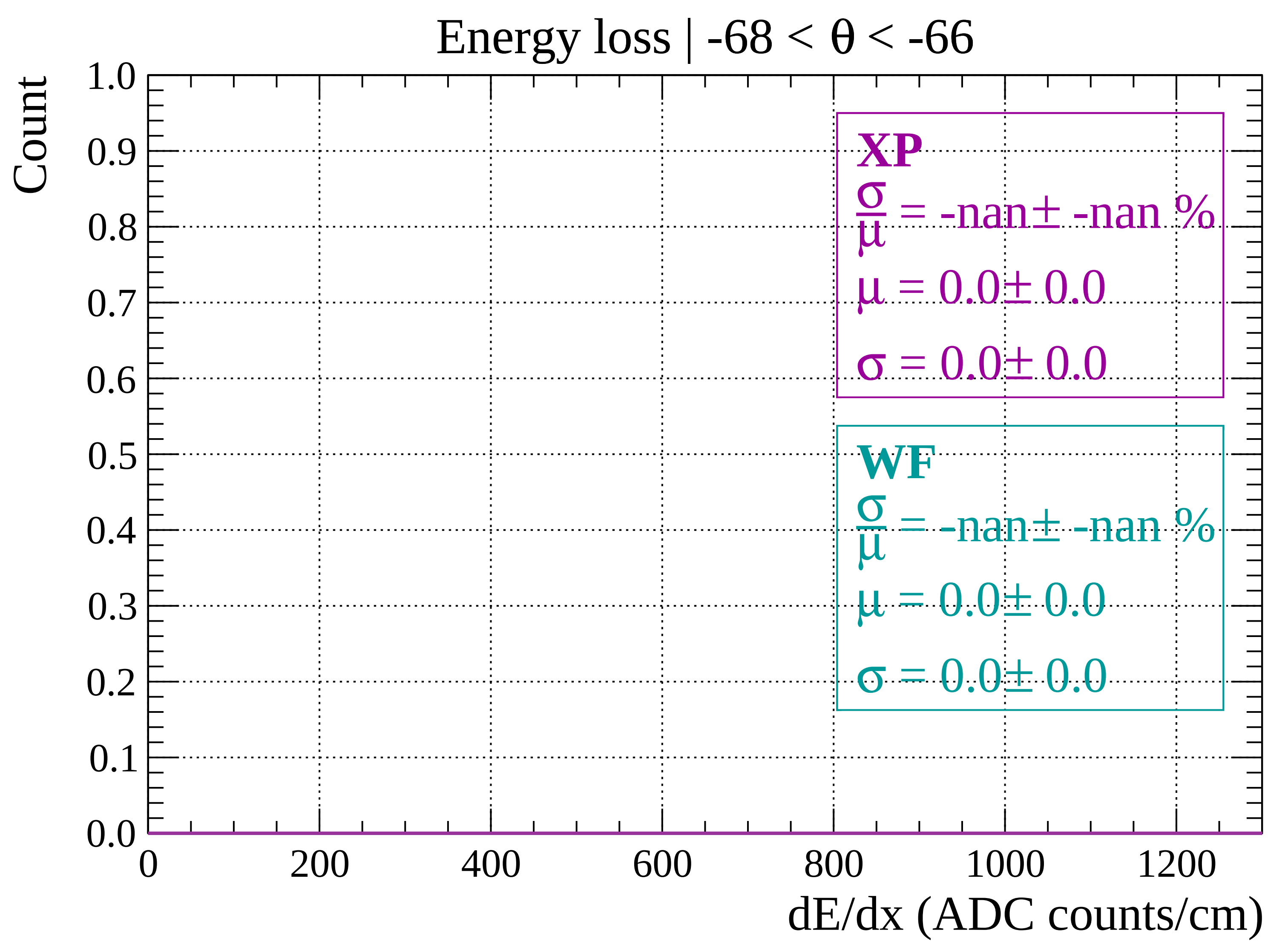


Energy loss | -74 < θ < -72
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

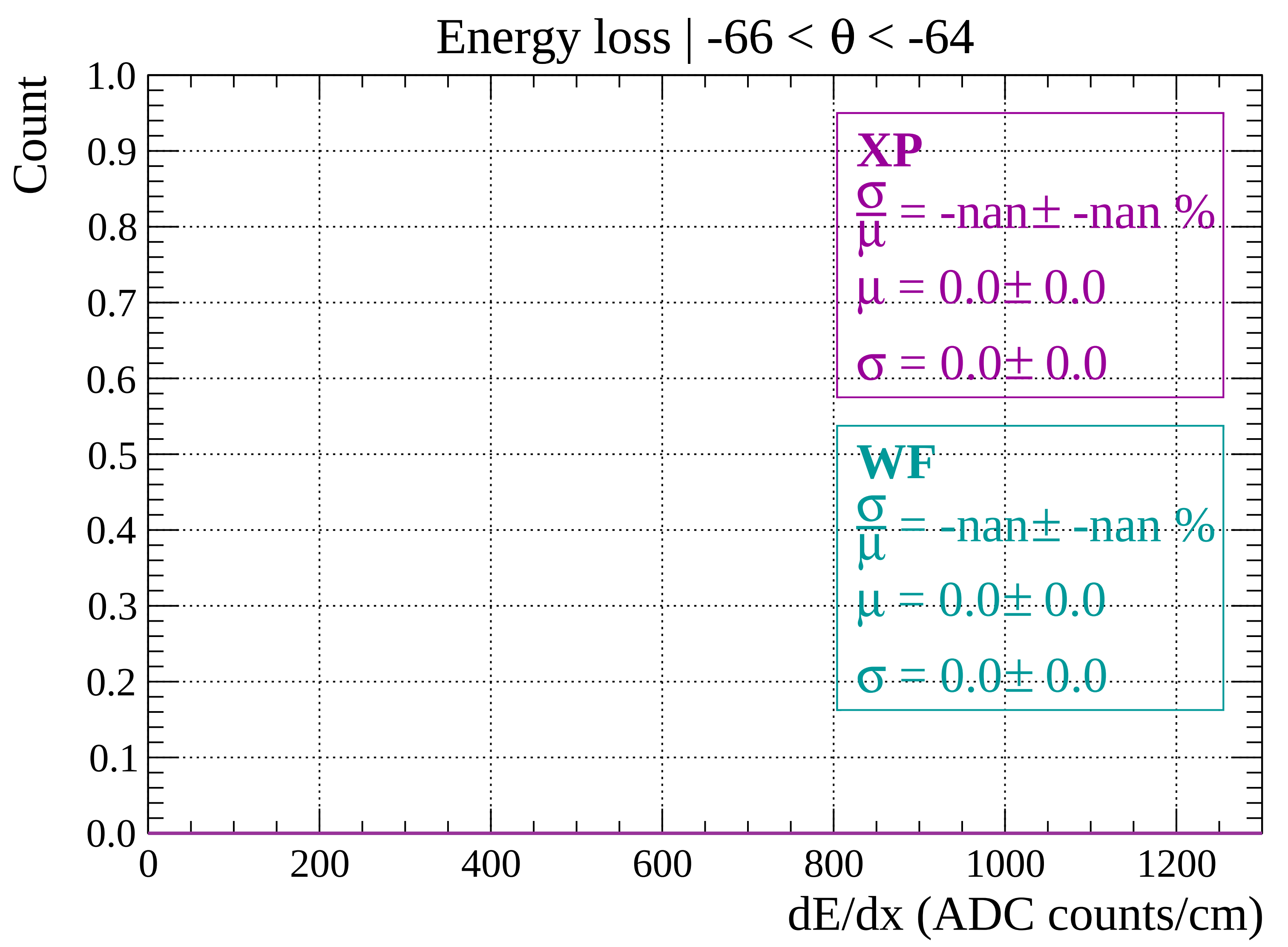

Energy loss | -72 < θ < -70
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

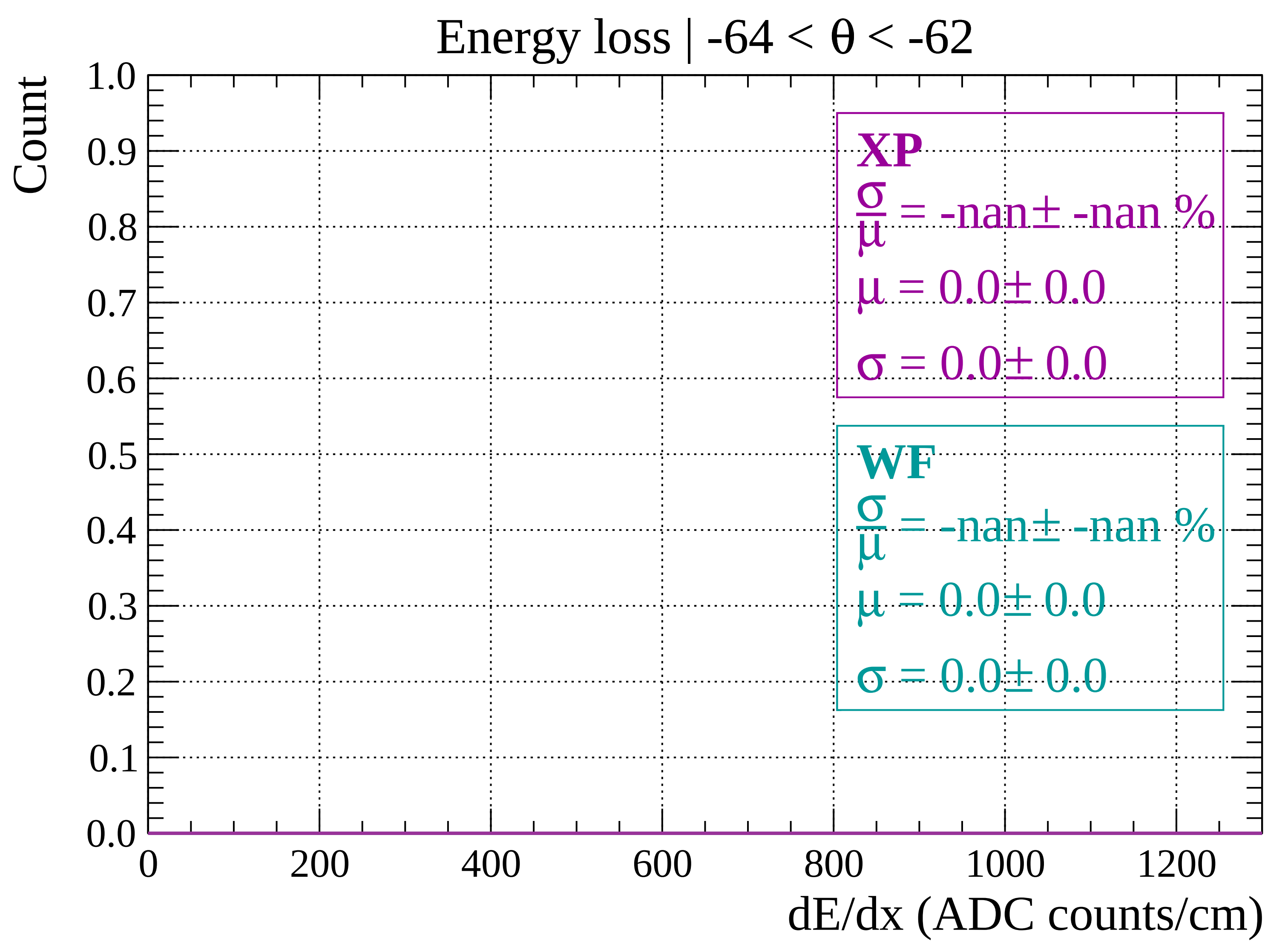


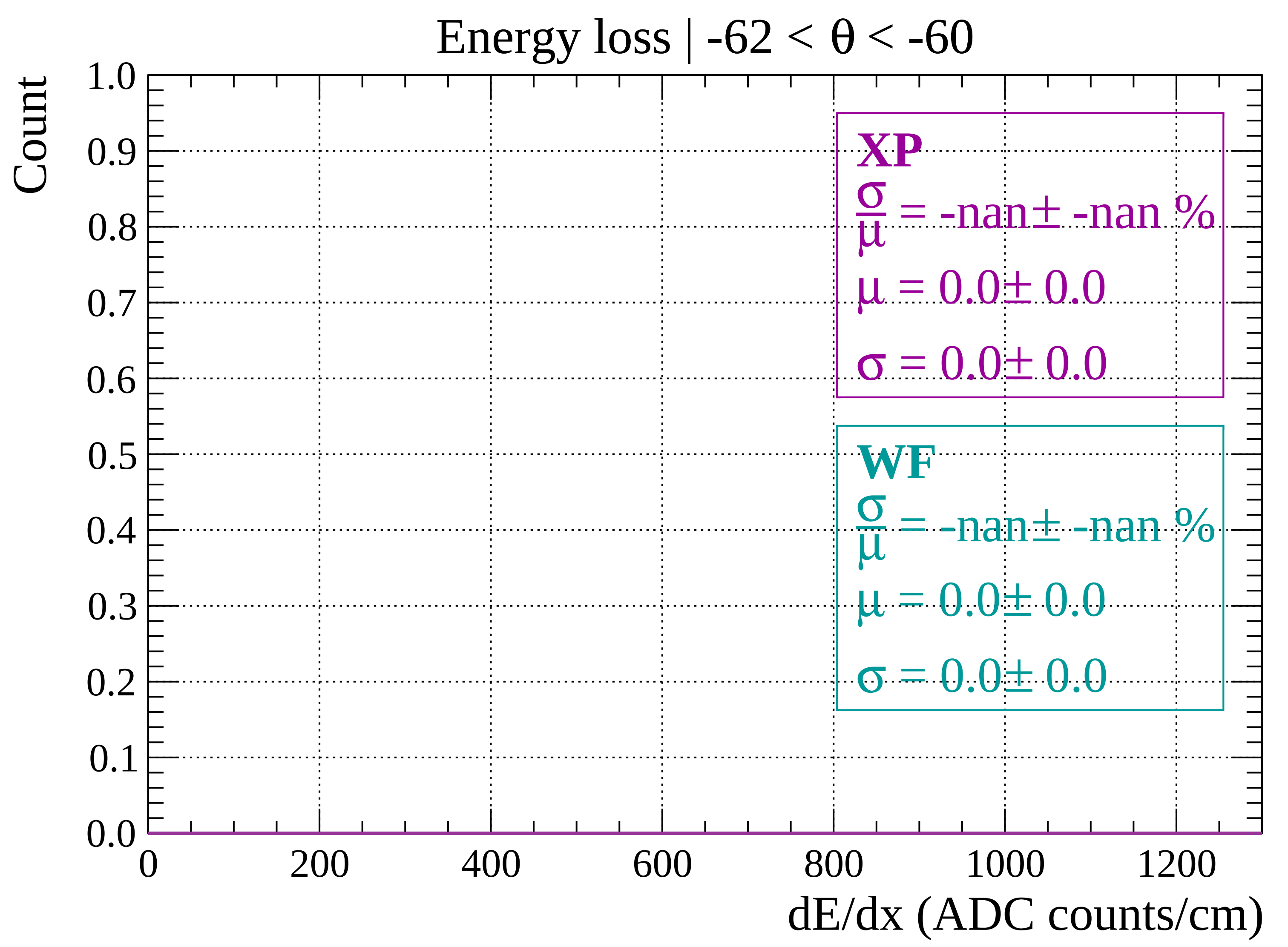
Energy loss | -70 < θ < -68
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



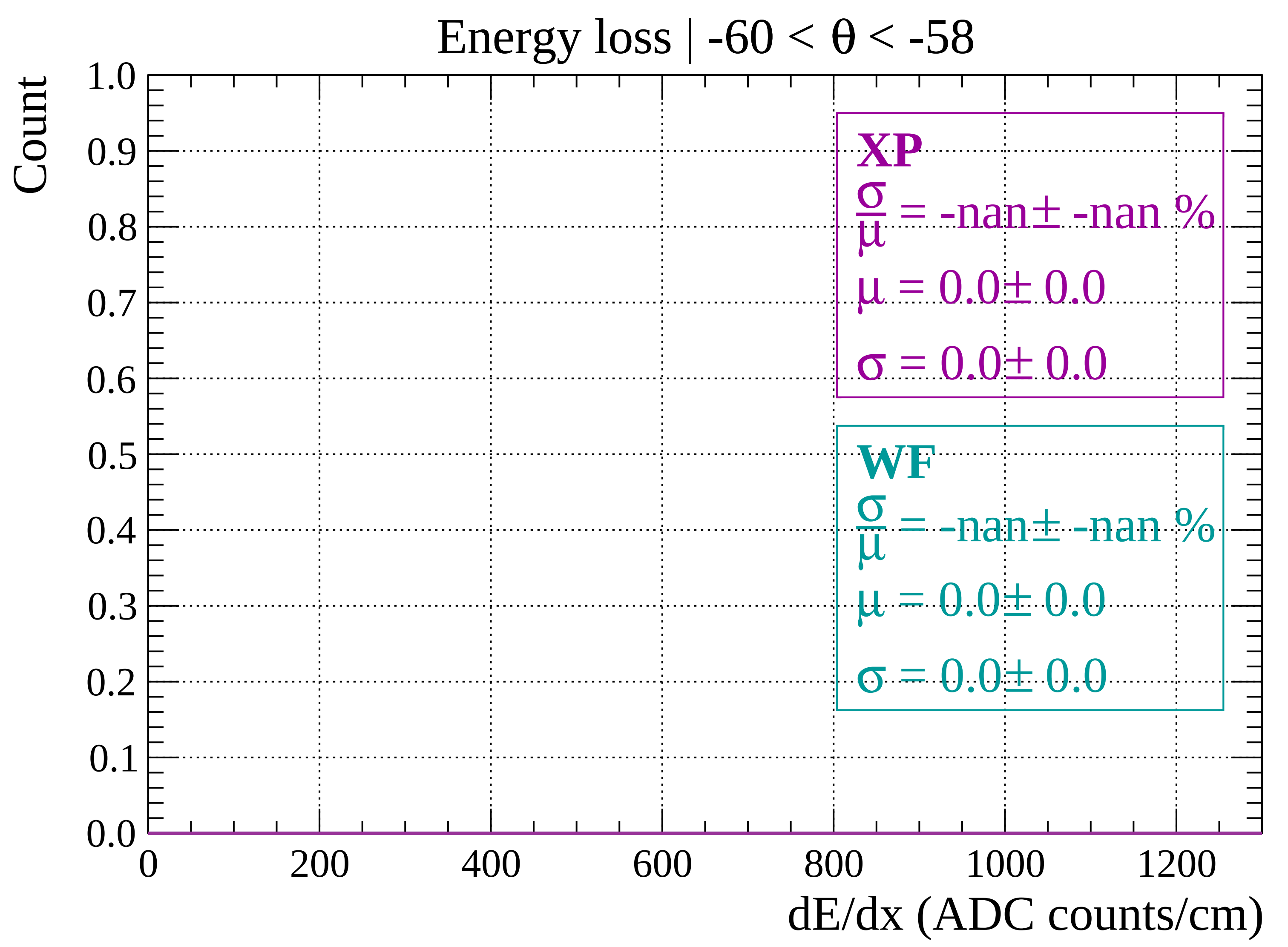
Energy loss | -68 < θ < -66
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0


Energy loss | -66 < θ < -64
Count
dE/dx (ADC counts/cm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

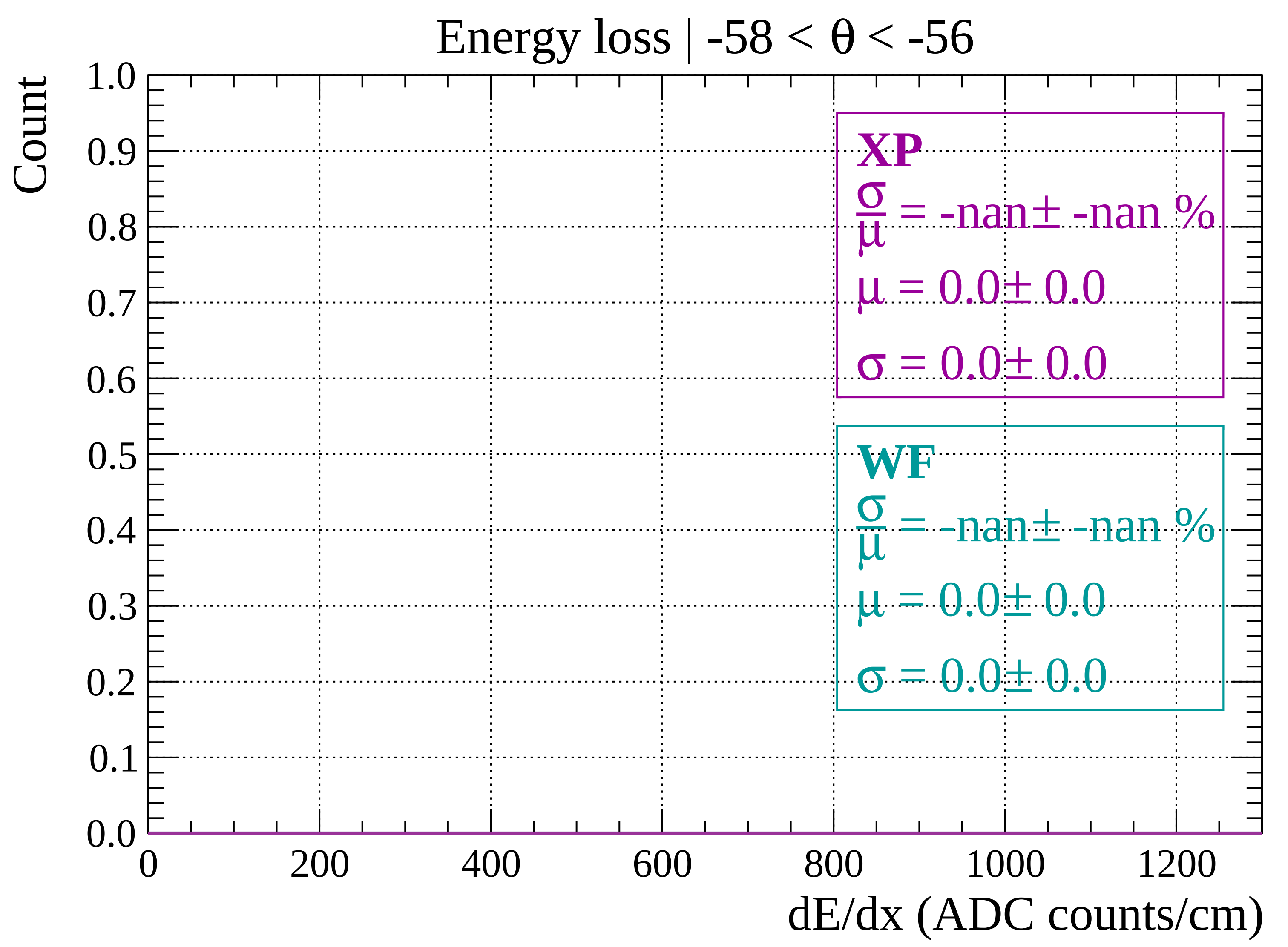

Energy loss | -64 < θ < -62
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



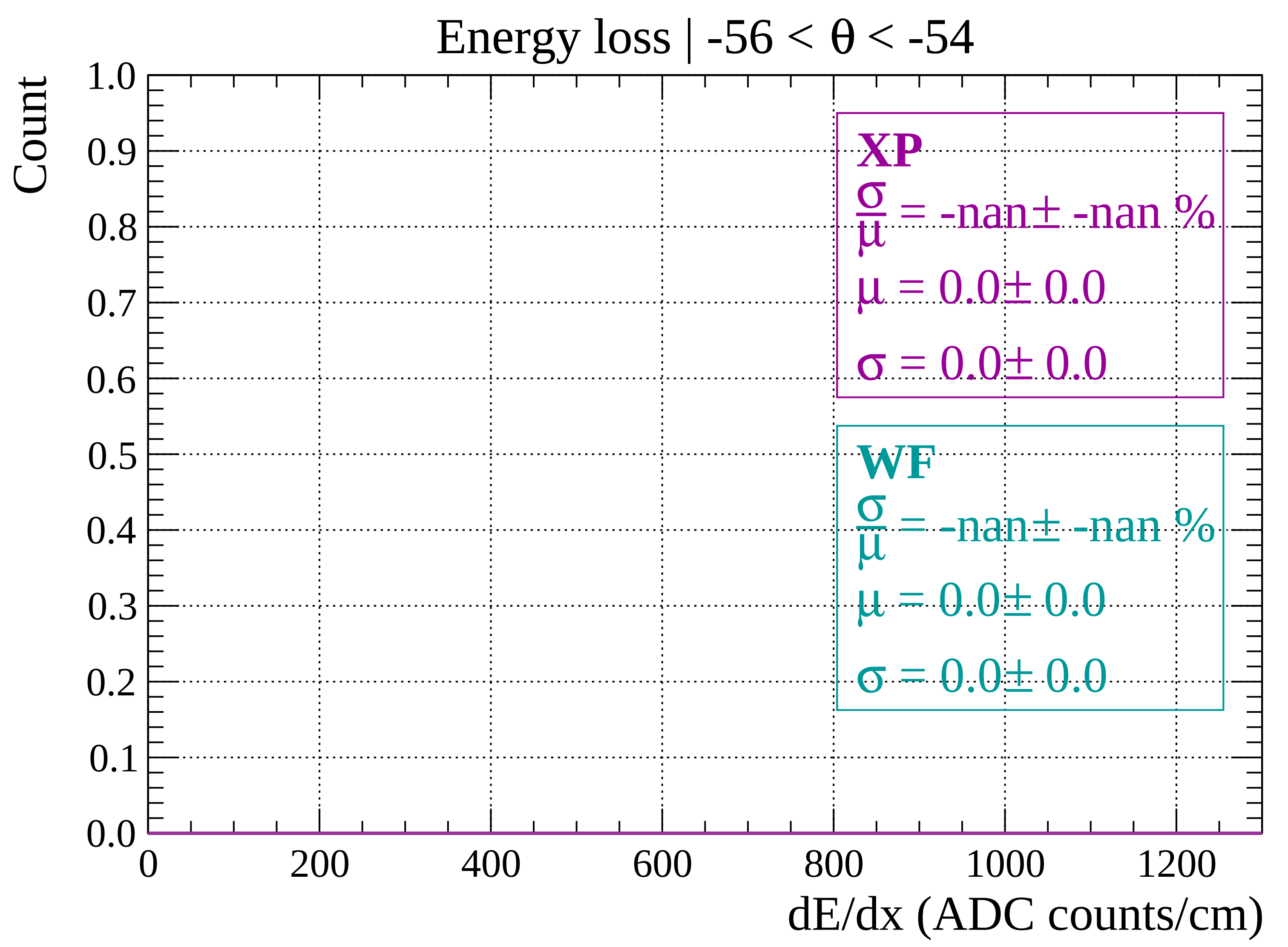
Energy loss | -62 < θ < -60
Count
dE/dx (ADC counts/cm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



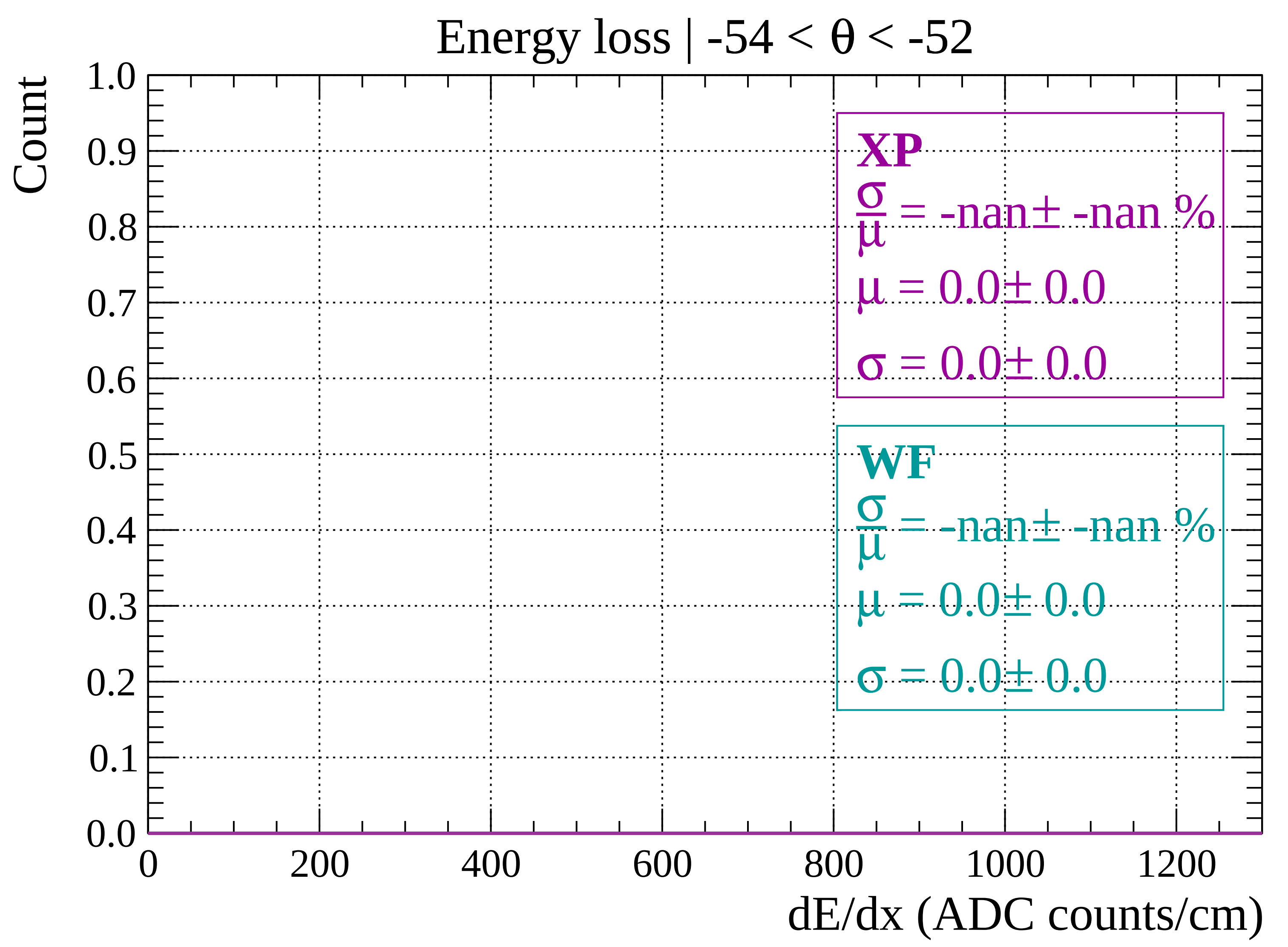
Energy loss | -60 < θ < -58
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

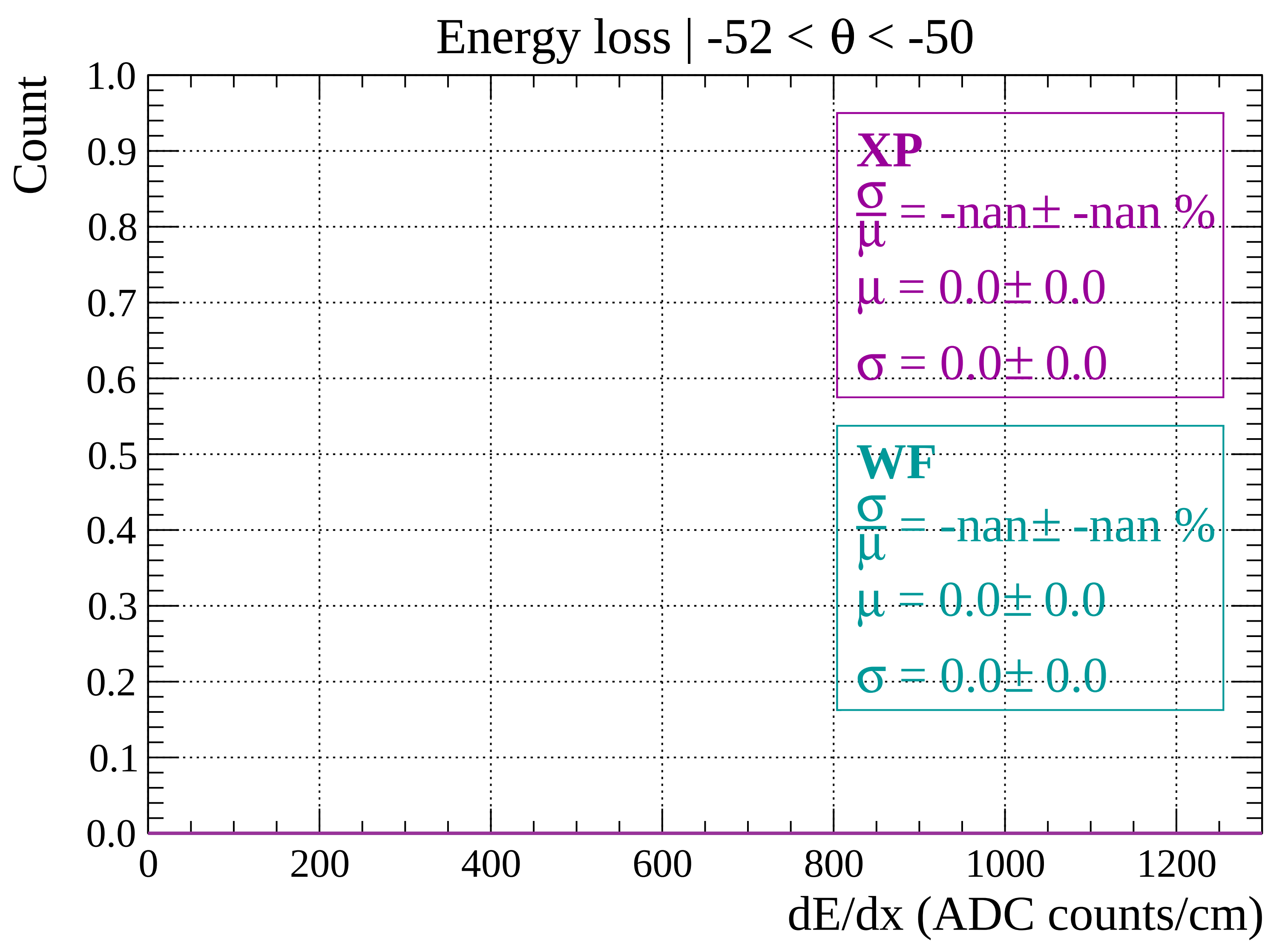


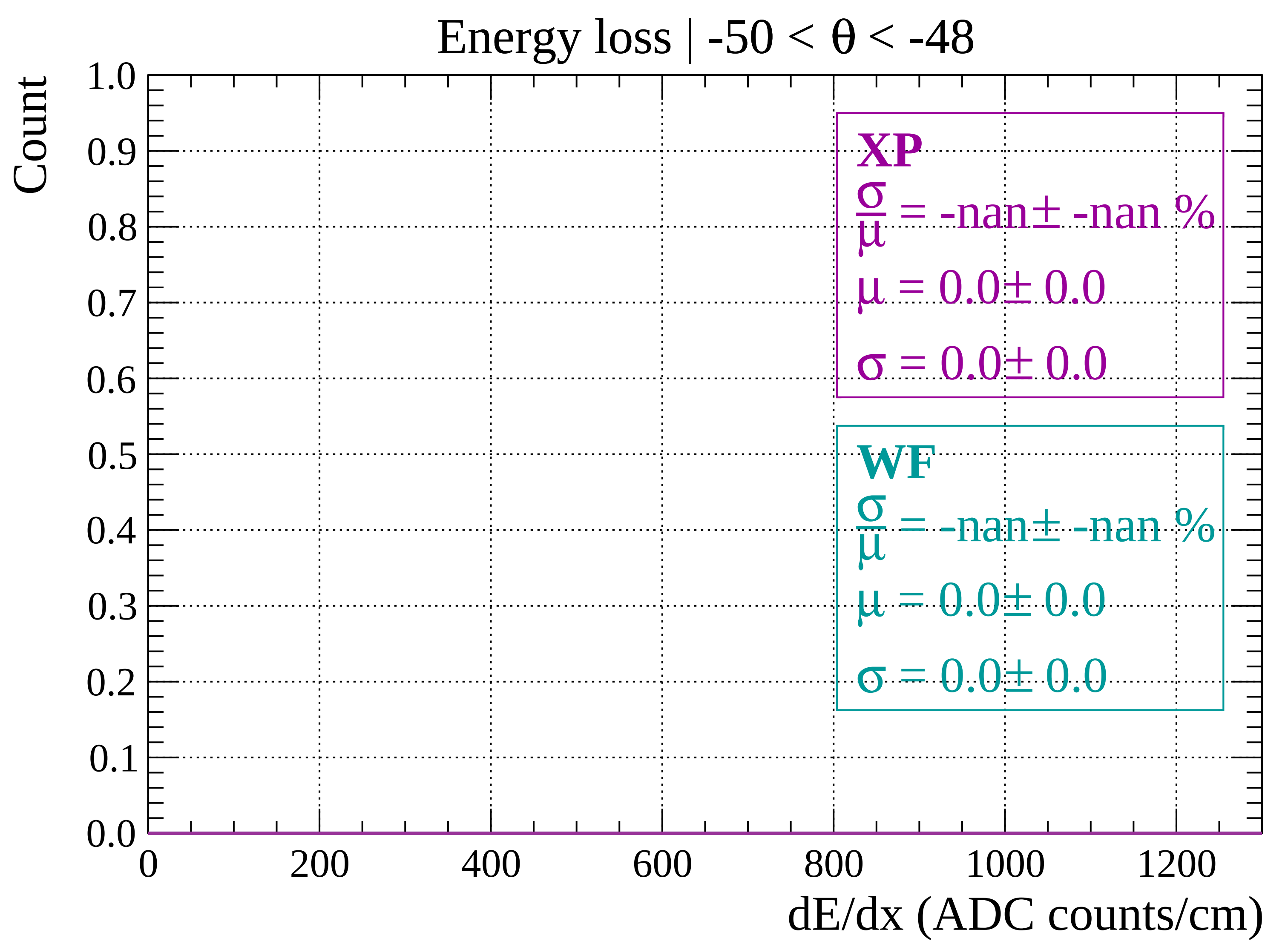
Energy loss | -58 < θ < -56
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



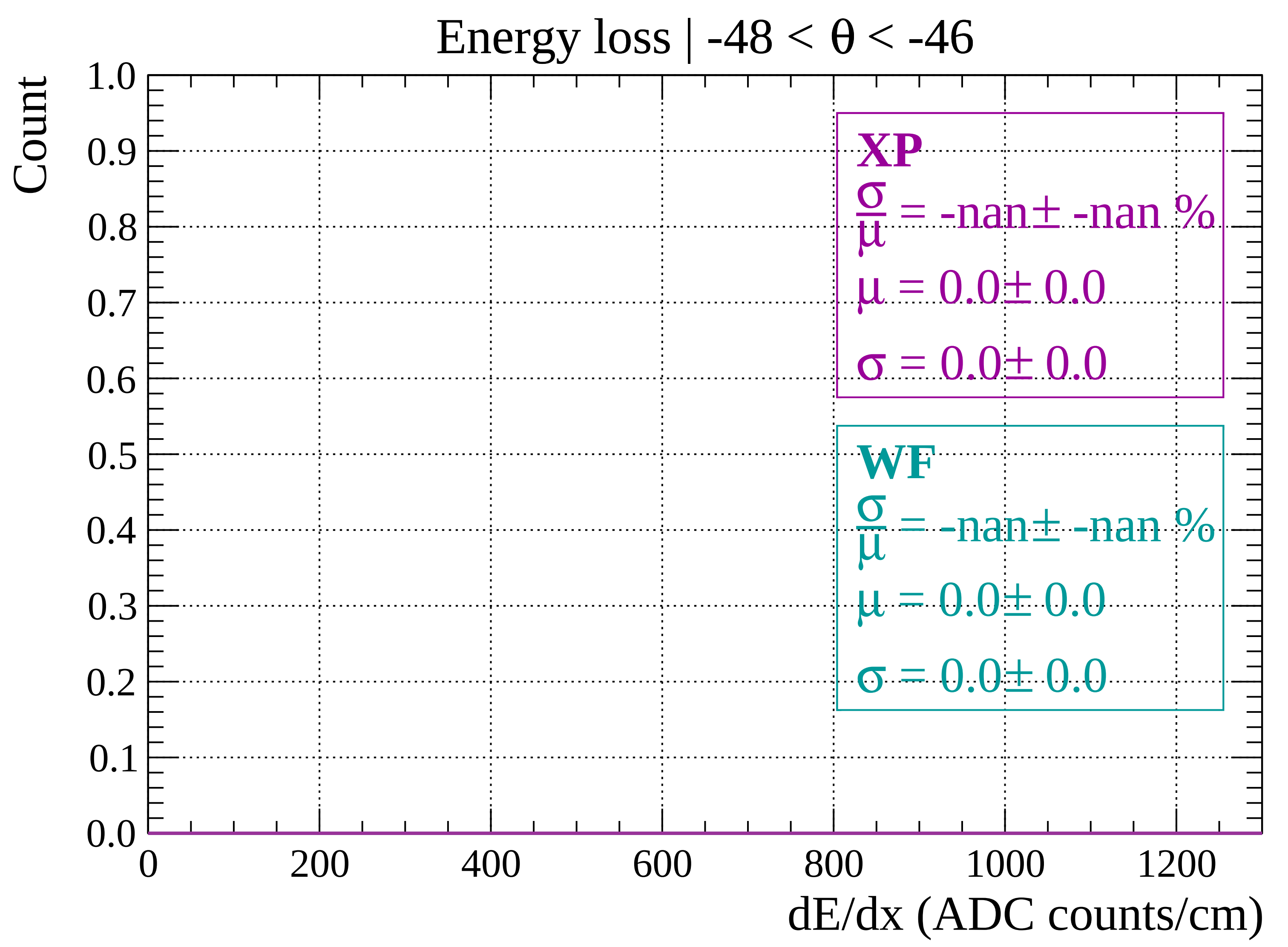
Energy loss | -56 < θ < -54
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0


Energy loss | -54 < θ < -52
Count
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
0
200
400
600
800
1000
1200
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0

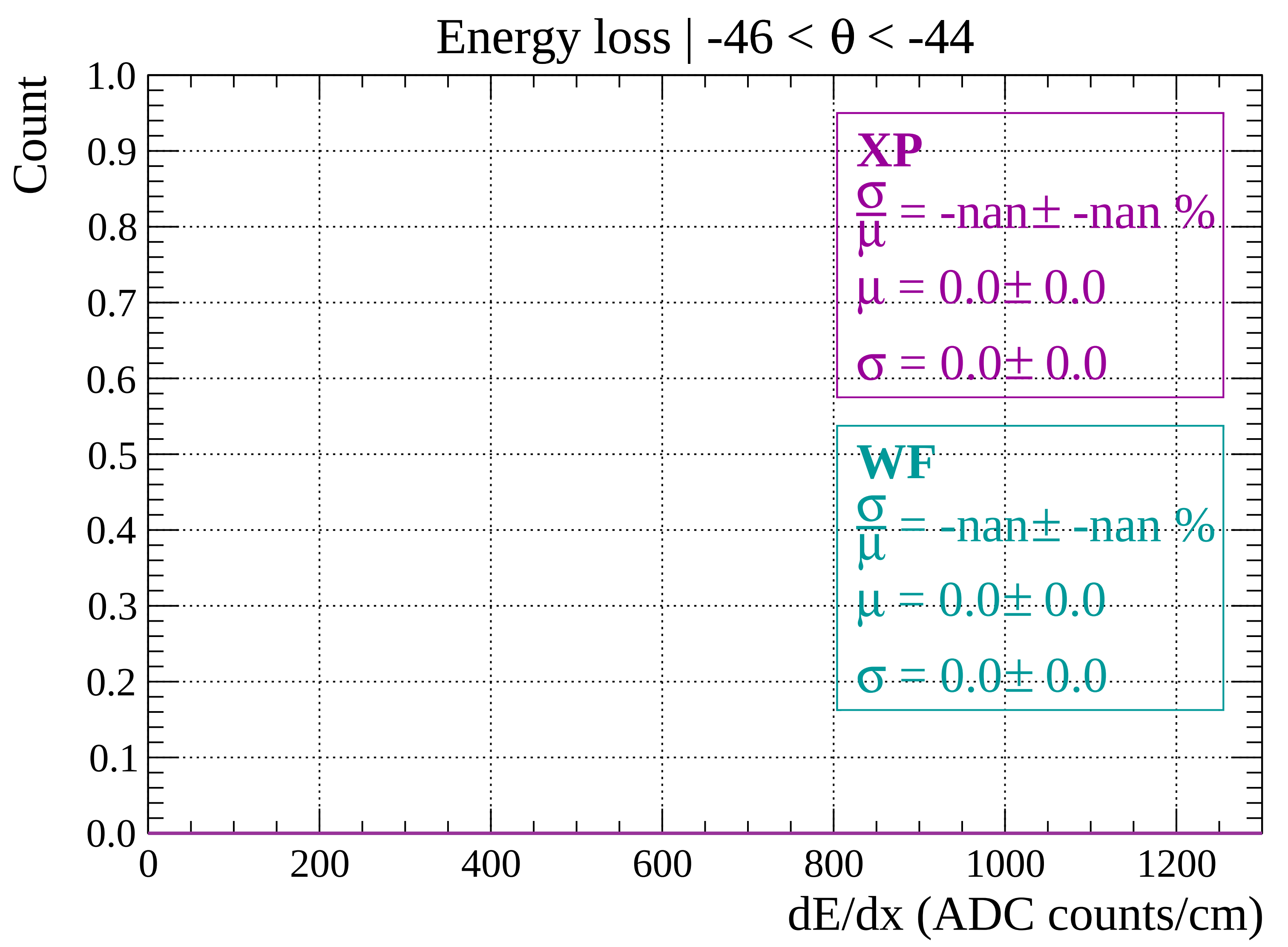

Energy loss | -52 < θ < -50
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

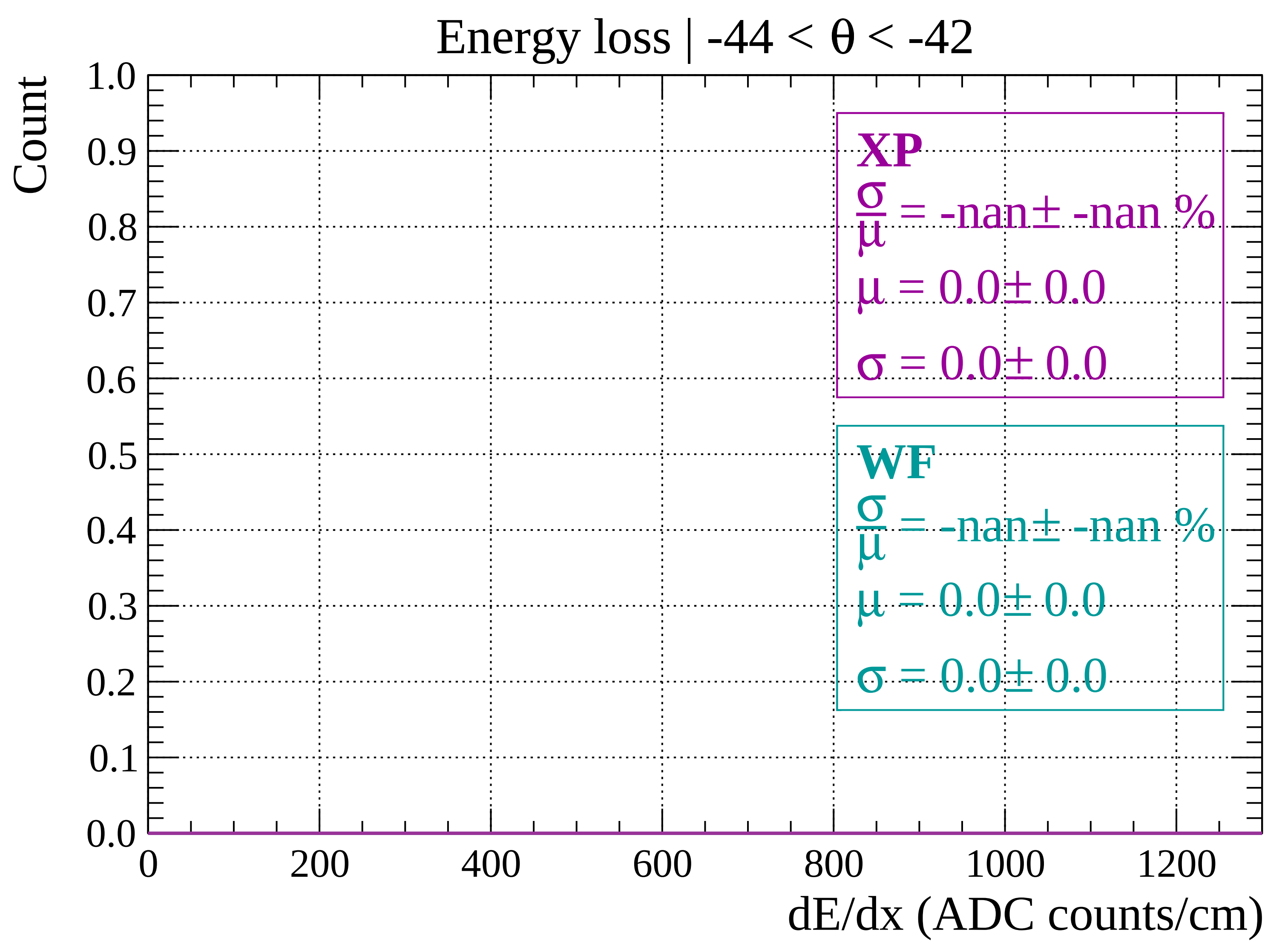


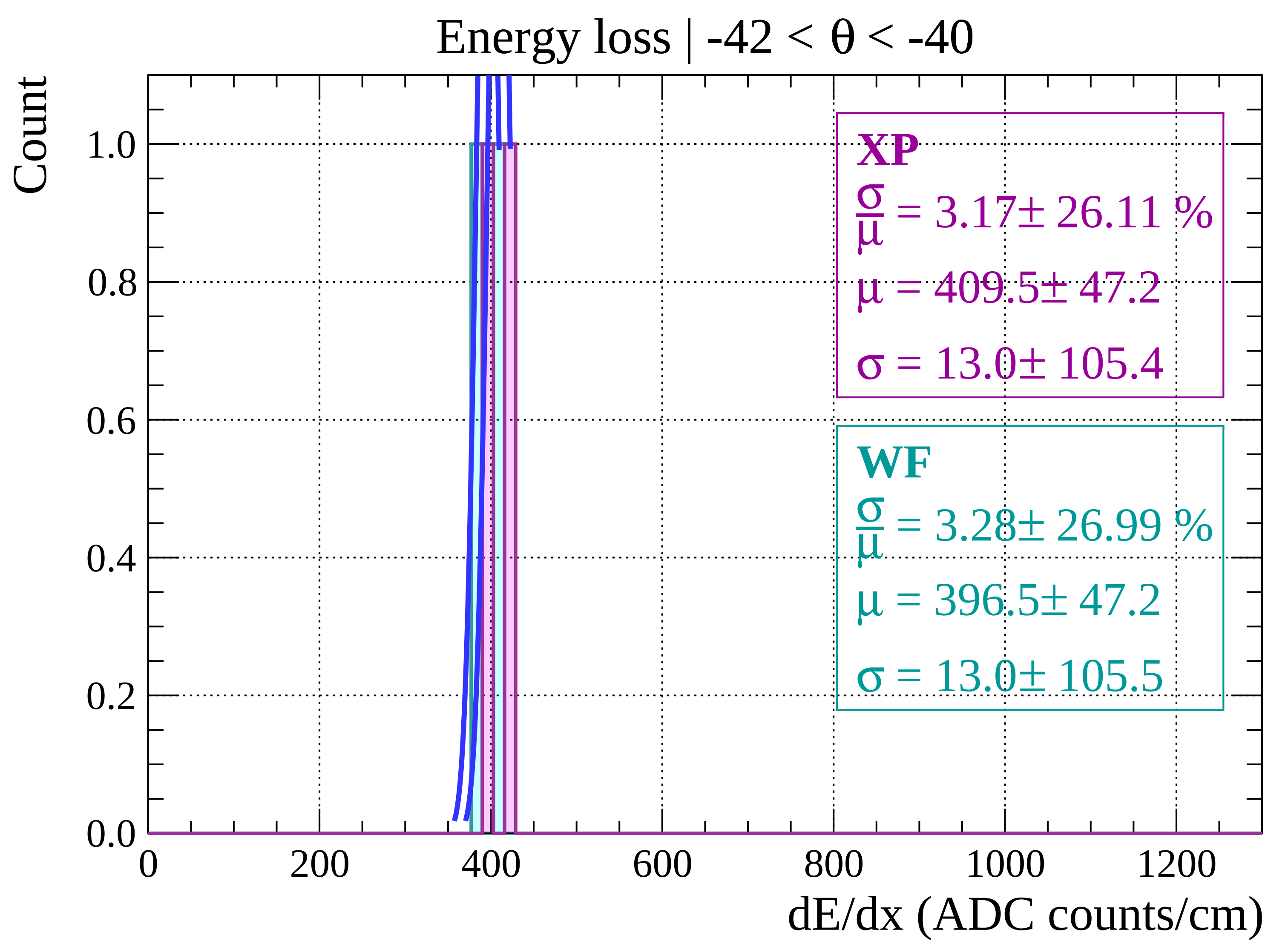
Energy loss | -50 < θ < -48
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



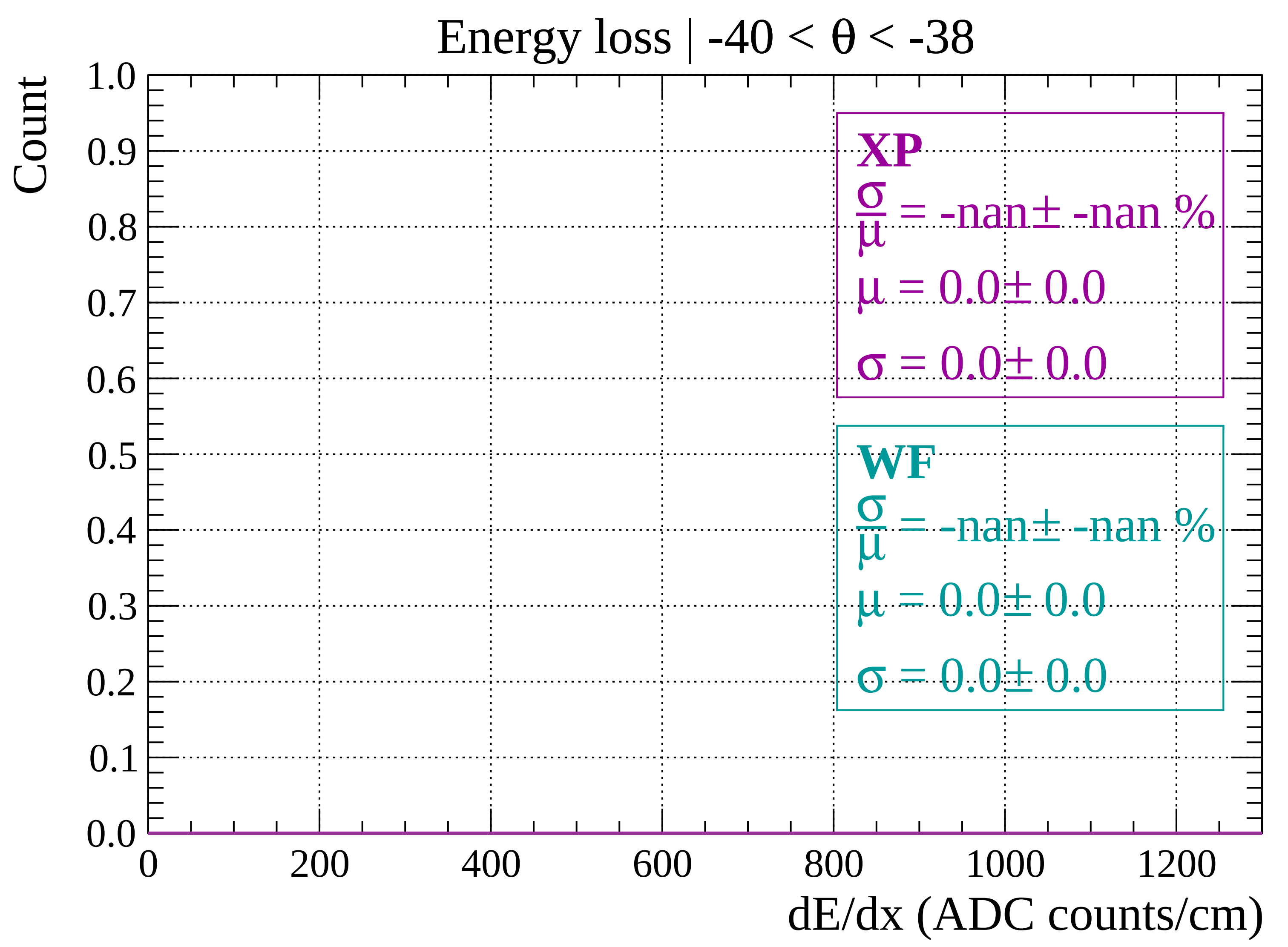
Energy loss | -48 < θ < -46
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0


Energy loss | -46 < θ < -44
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

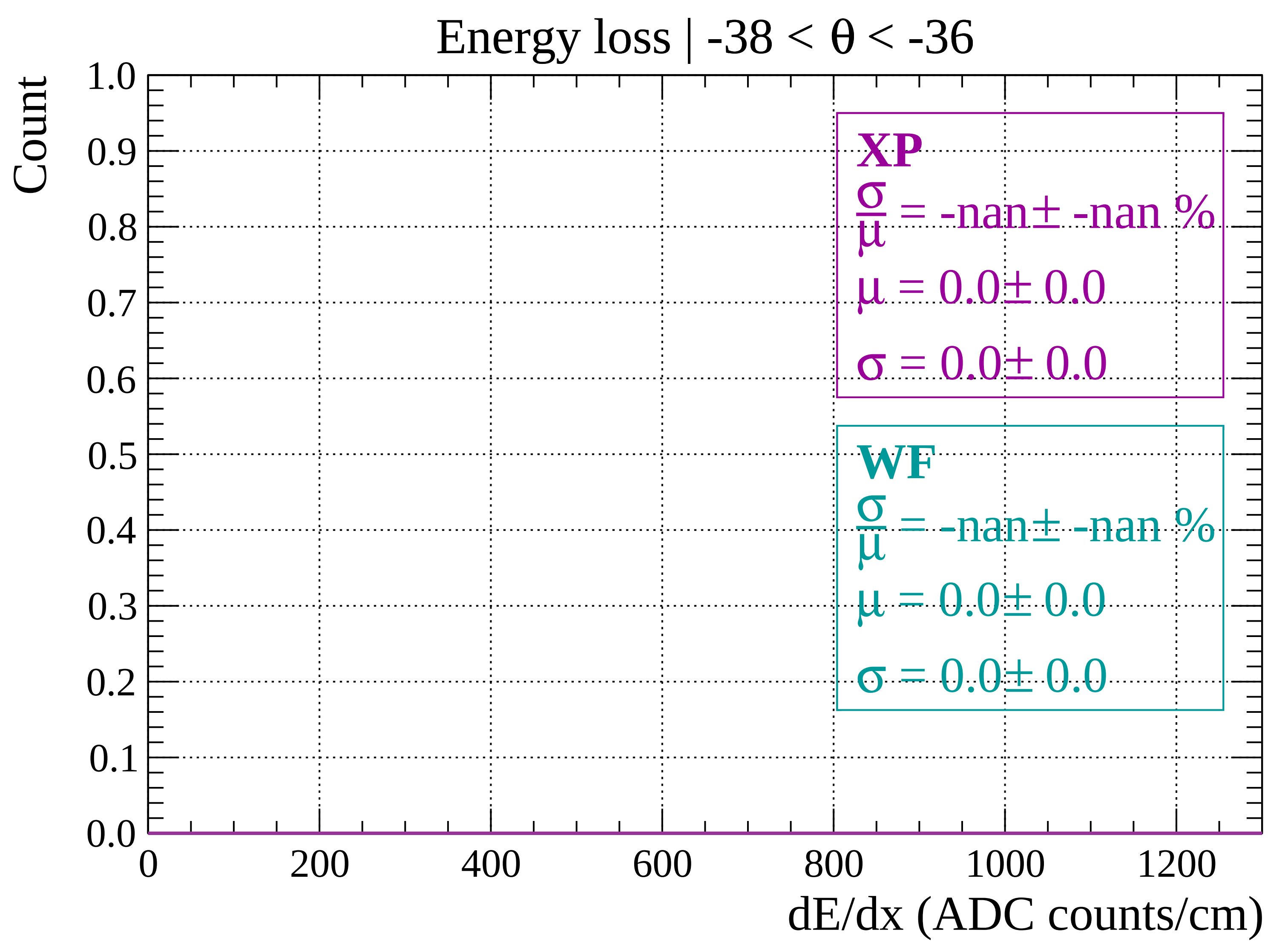

Energy loss | -44 < θ < -42
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

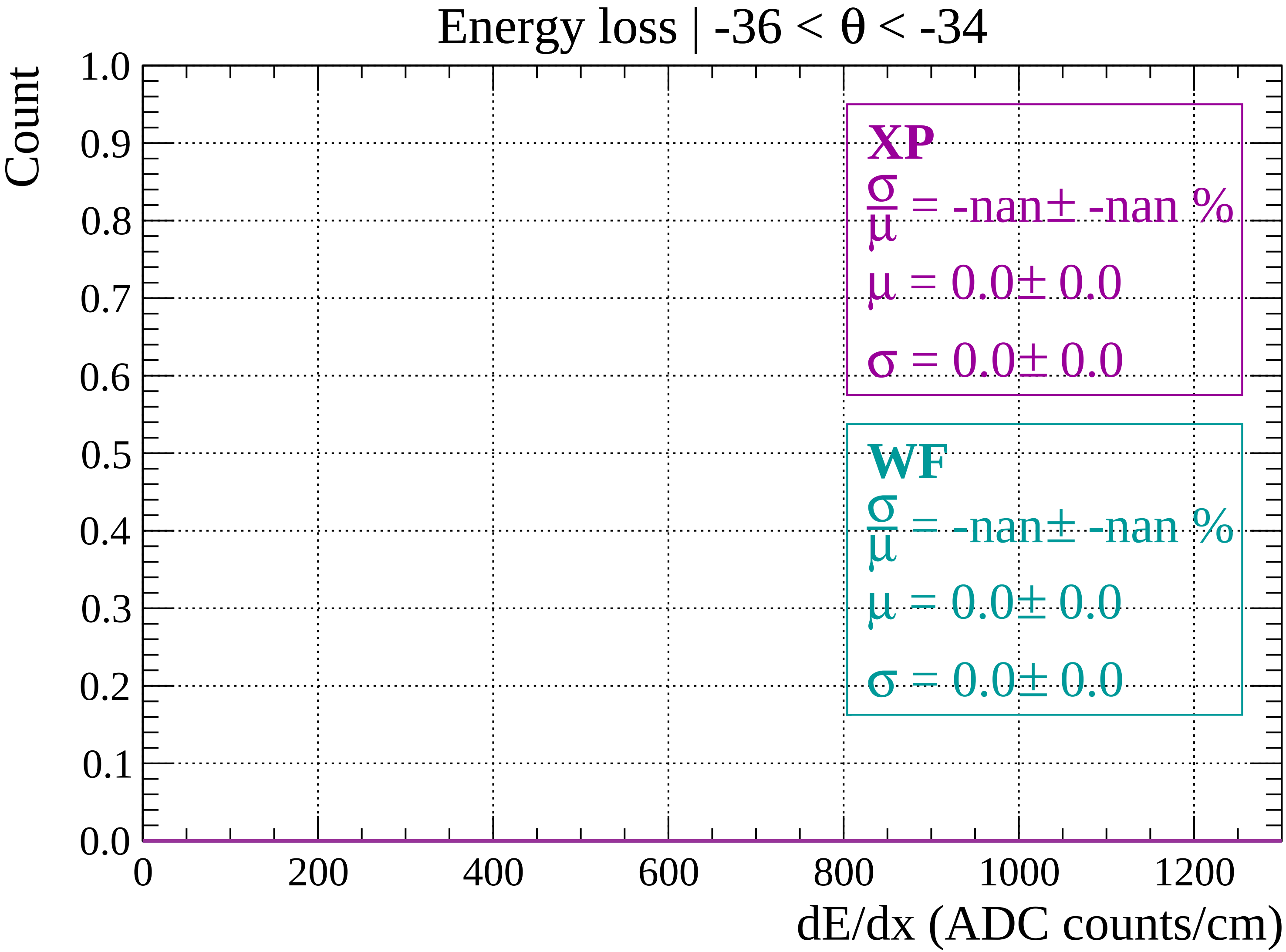


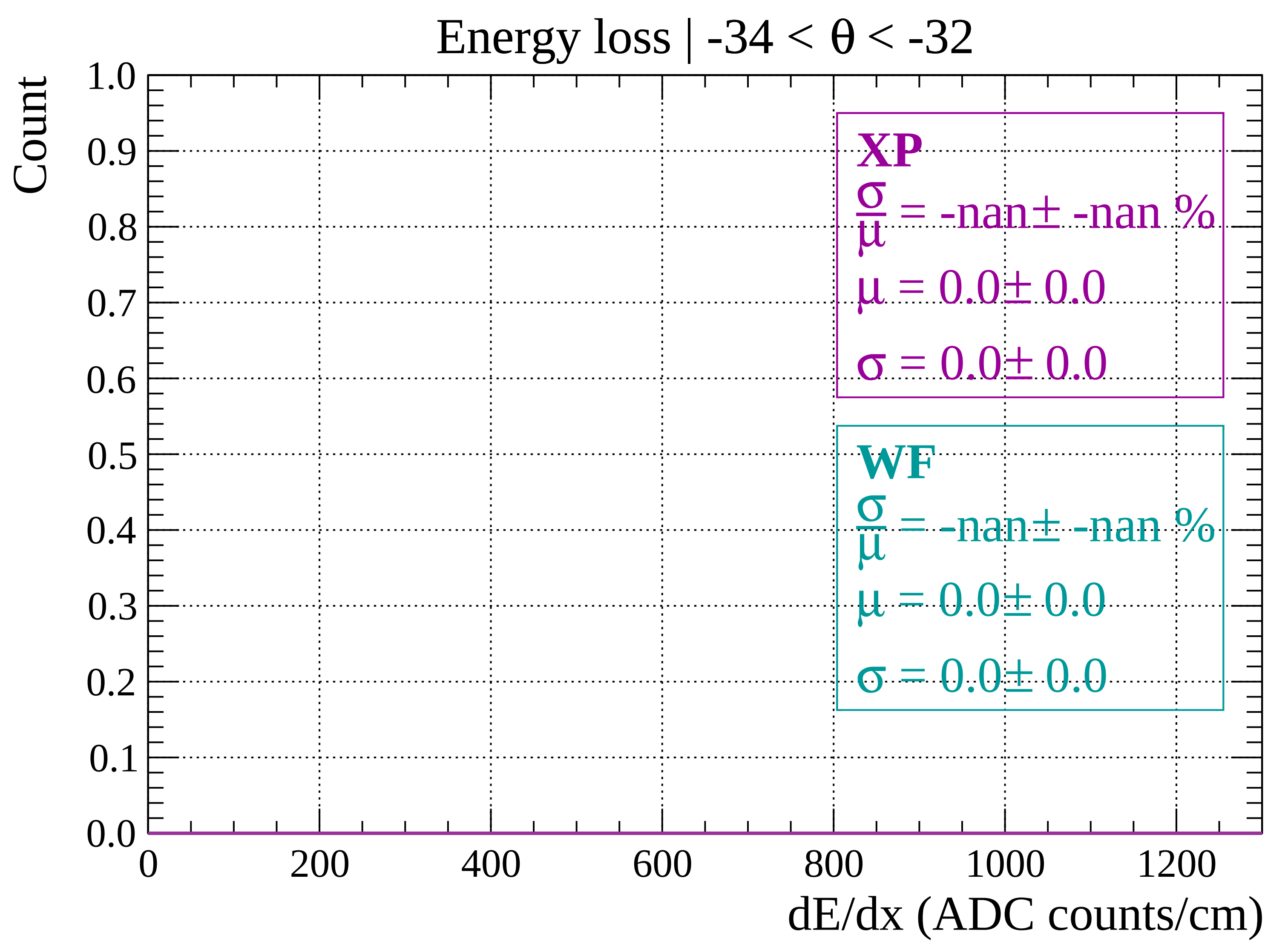
Energy loss | -42 < θ < -40
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 3.17± 26.11 %
μ = 409.5± 47.2
σ = 13.0± 105.4
WF
σ/μ = 3.28± 26.99 %
μ = 396.5± 47.2
σ = 13.0± 105.5



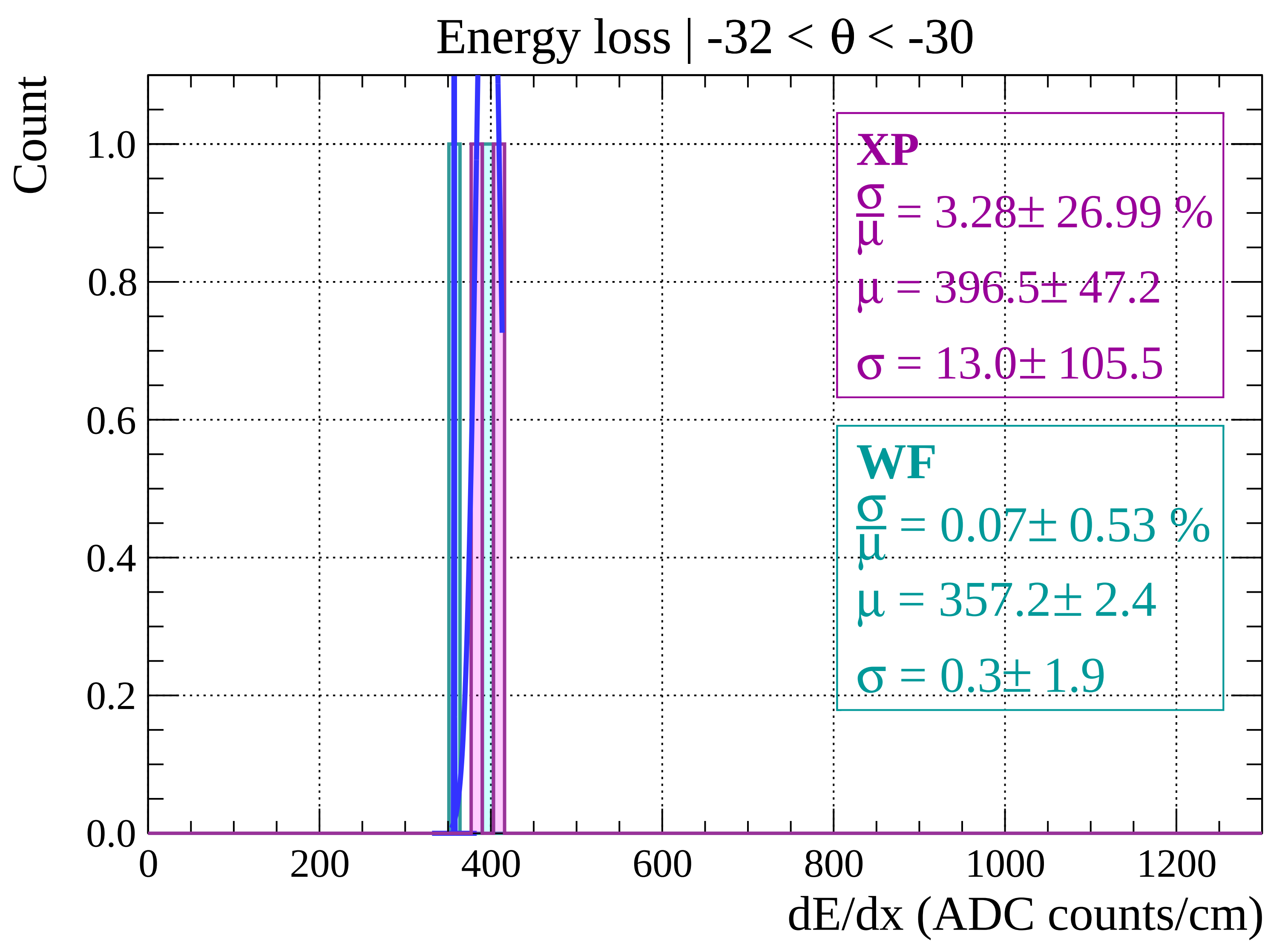
Energy loss | -40 < θ < -38
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0


Energy loss | -38 < θ < -36
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

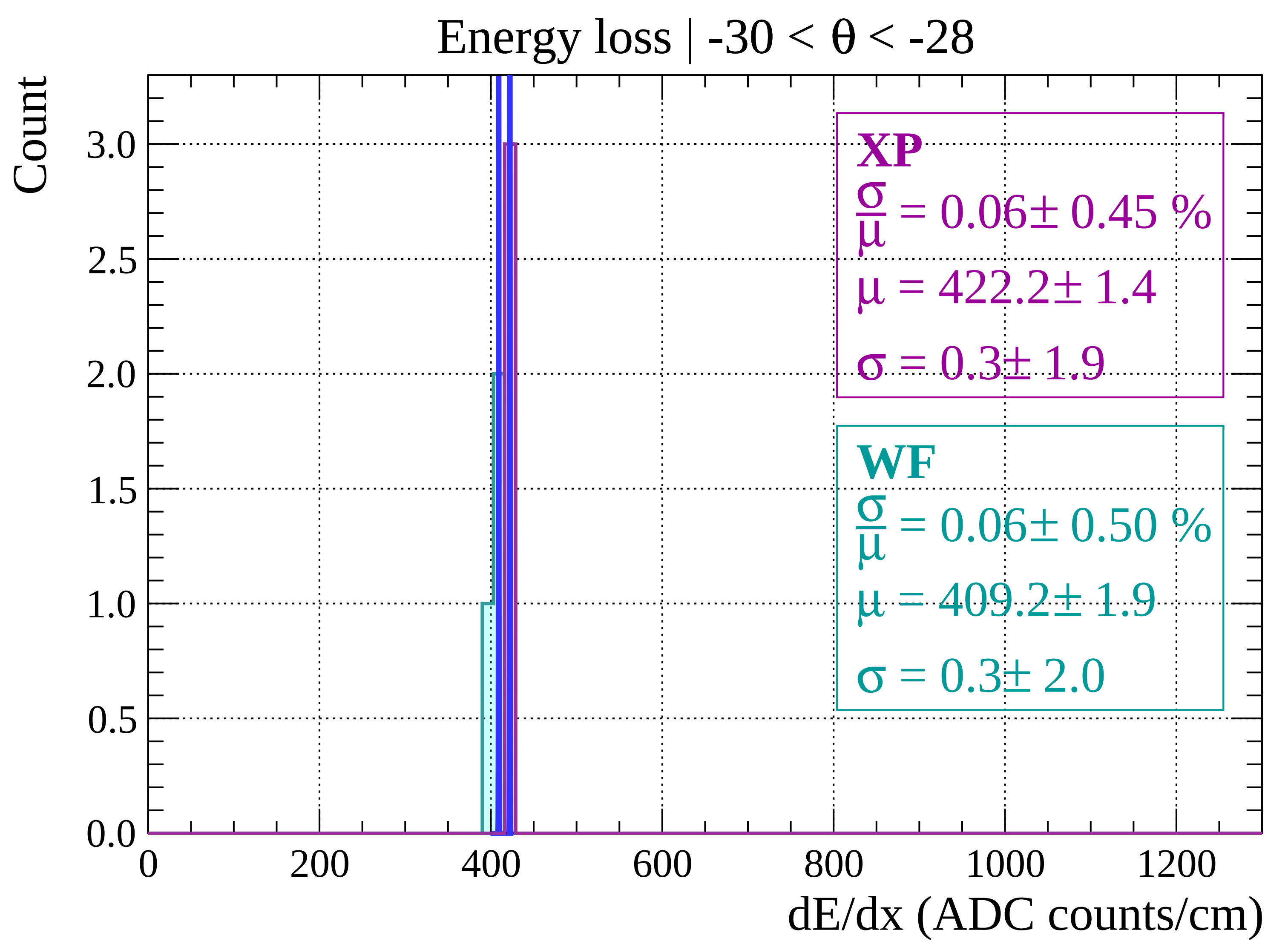

Energy loss | -36 < θ < -34
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



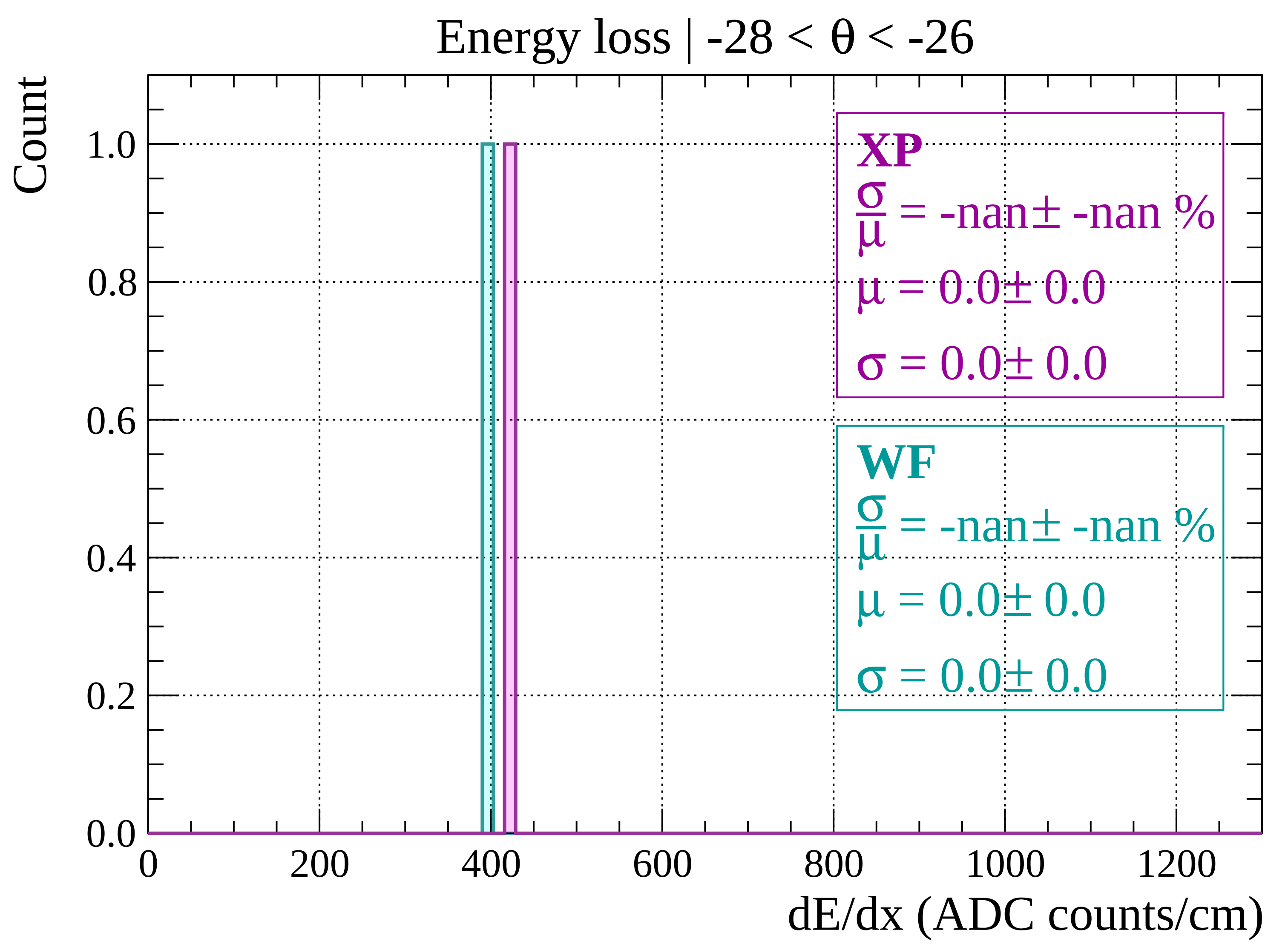
Energy loss | -34 < θ < -32
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



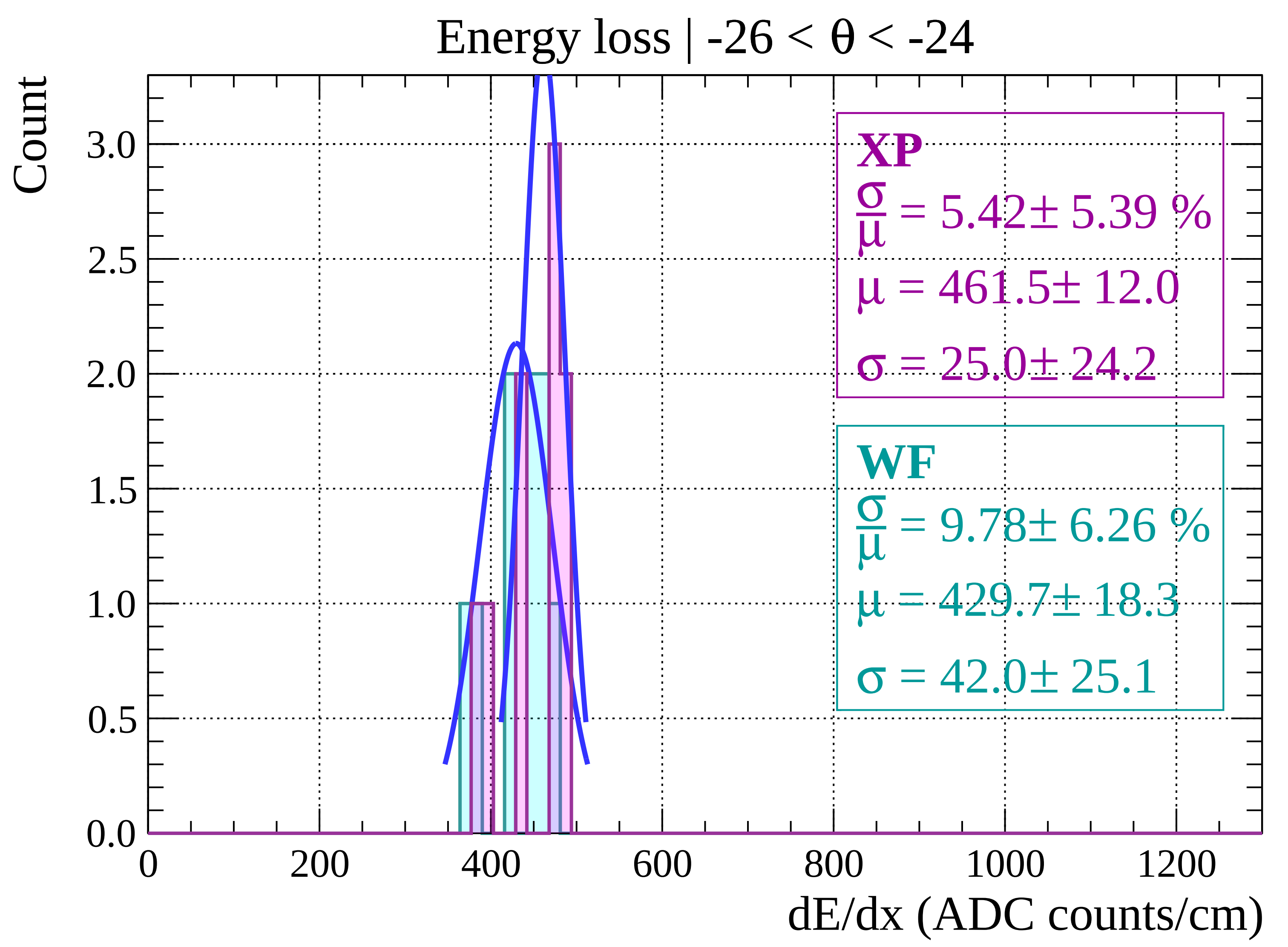
Energy loss | -32 < θ < -30
Count
dE/dx (ADC counts/cm)
0.0
0.2
0.4
0.6
0.8
1.0
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 3.28± 26.99 %
μ = 396.5± 47.2
σ = 13.0± 105.5
WF
$\frac{\sigma}{\mu}$ = 0.07± 0.53 %
μ = 357.2± 2.4
σ = 0.3± 1.9



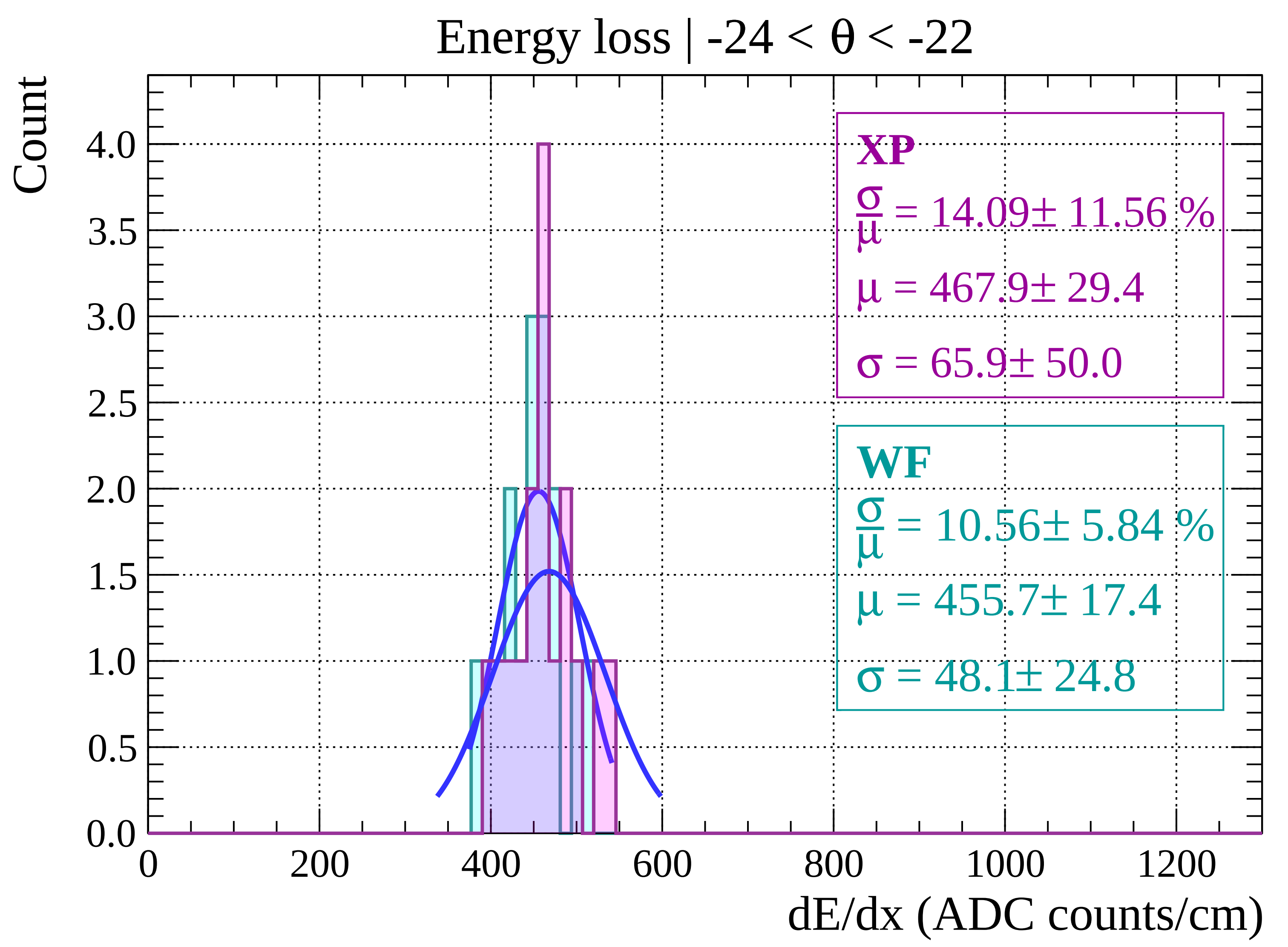
Energy loss | -30 < θ < -28
Count
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 0.06± 0.45 %
μ = 422.2± 1.4
σ = 0.3± 1.9
WF
σ/μ = 0.06± 0.50 %
μ = 409.2± 1.9
σ = 0.3± 2.0



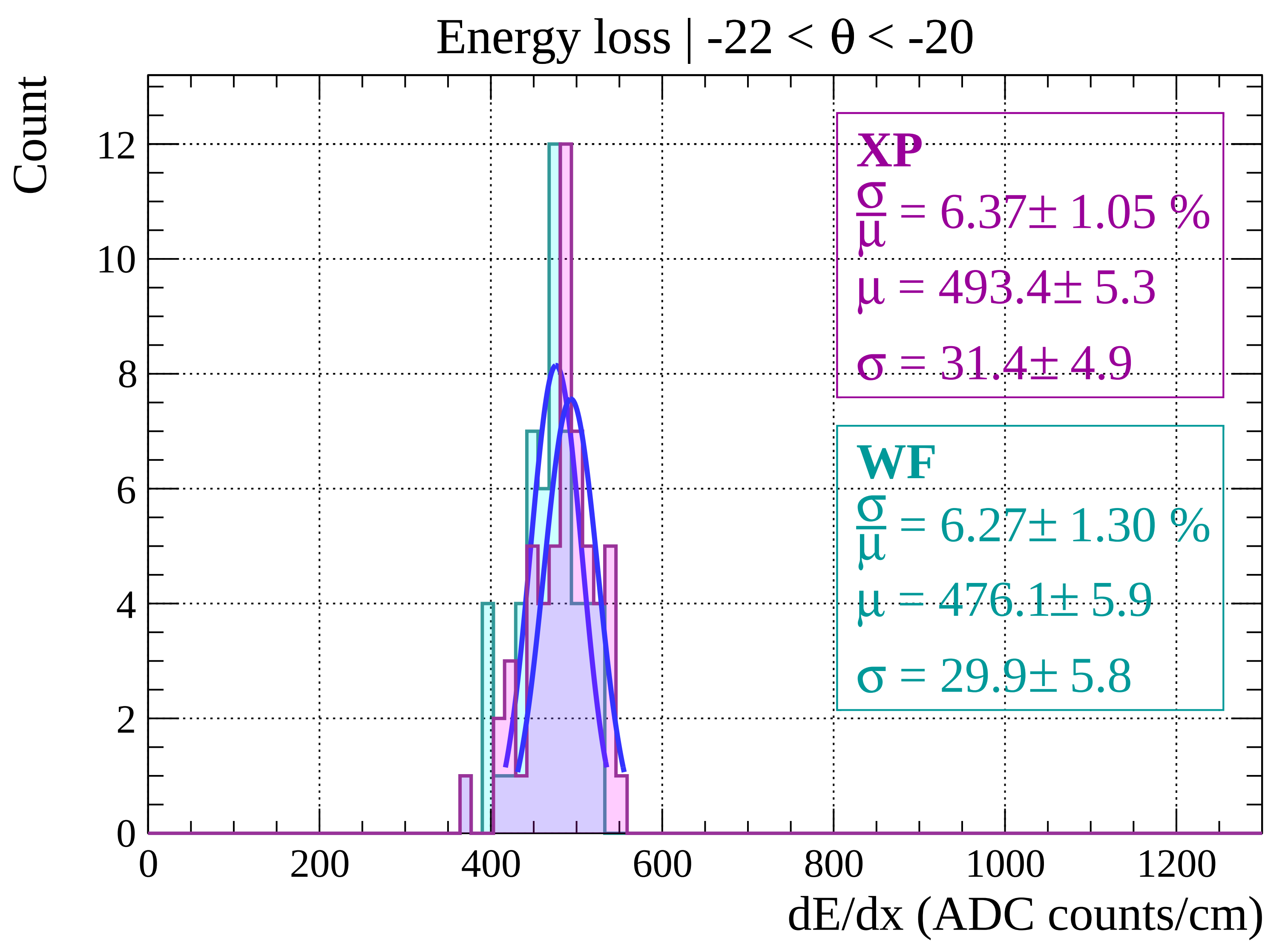
Energy loss | -28 < θ < -26
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



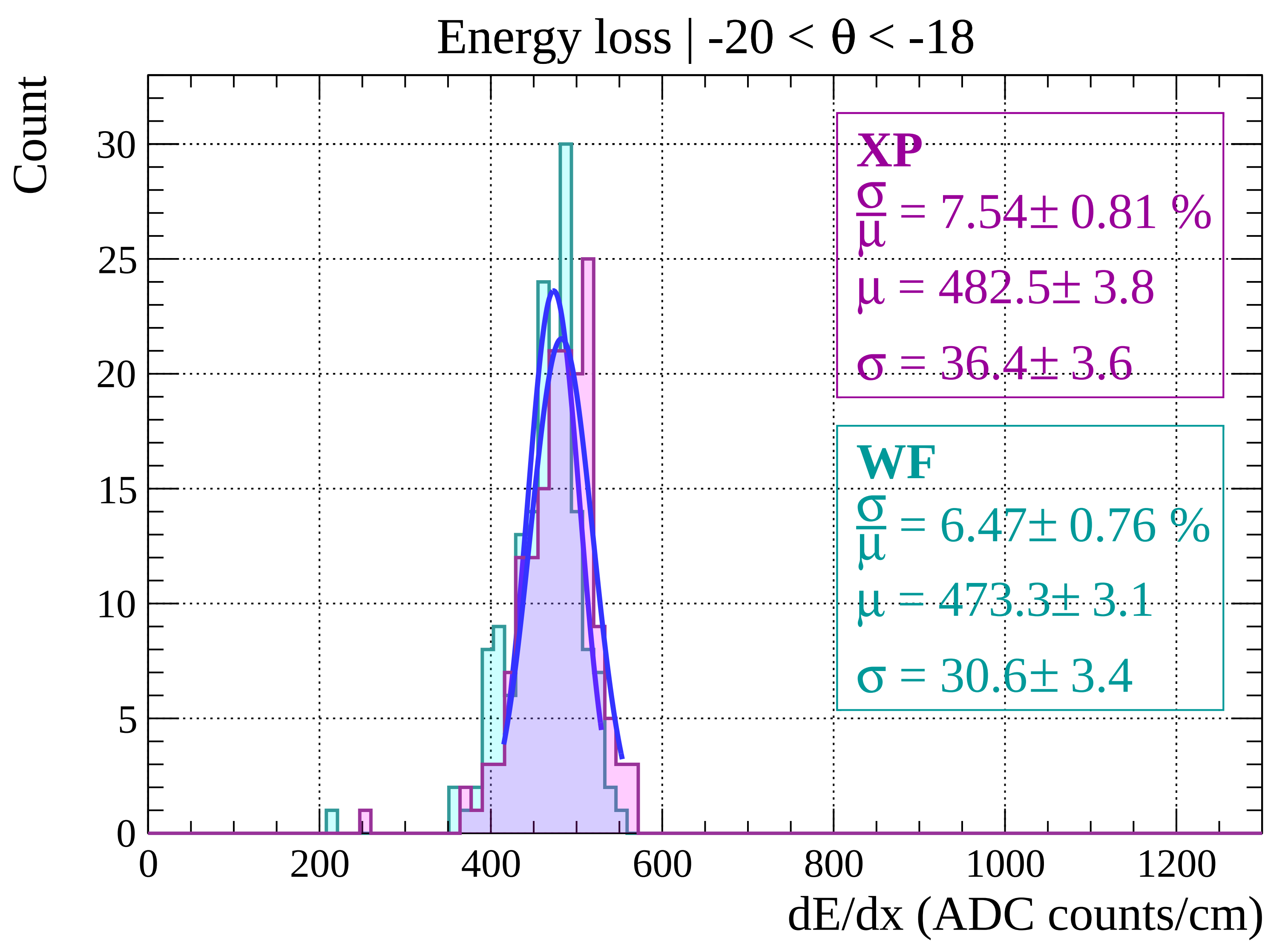
Energy loss | -26 < θ < -24
Count
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 5.42± 5.39 %
μ = 461.5± 12.0
σ = 25.0± 24.2
WF
$\frac{\sigma}{\mu}$ = 9.78± 6.26 %
μ = 429.7± 18.3
σ = 42.0± 25.1



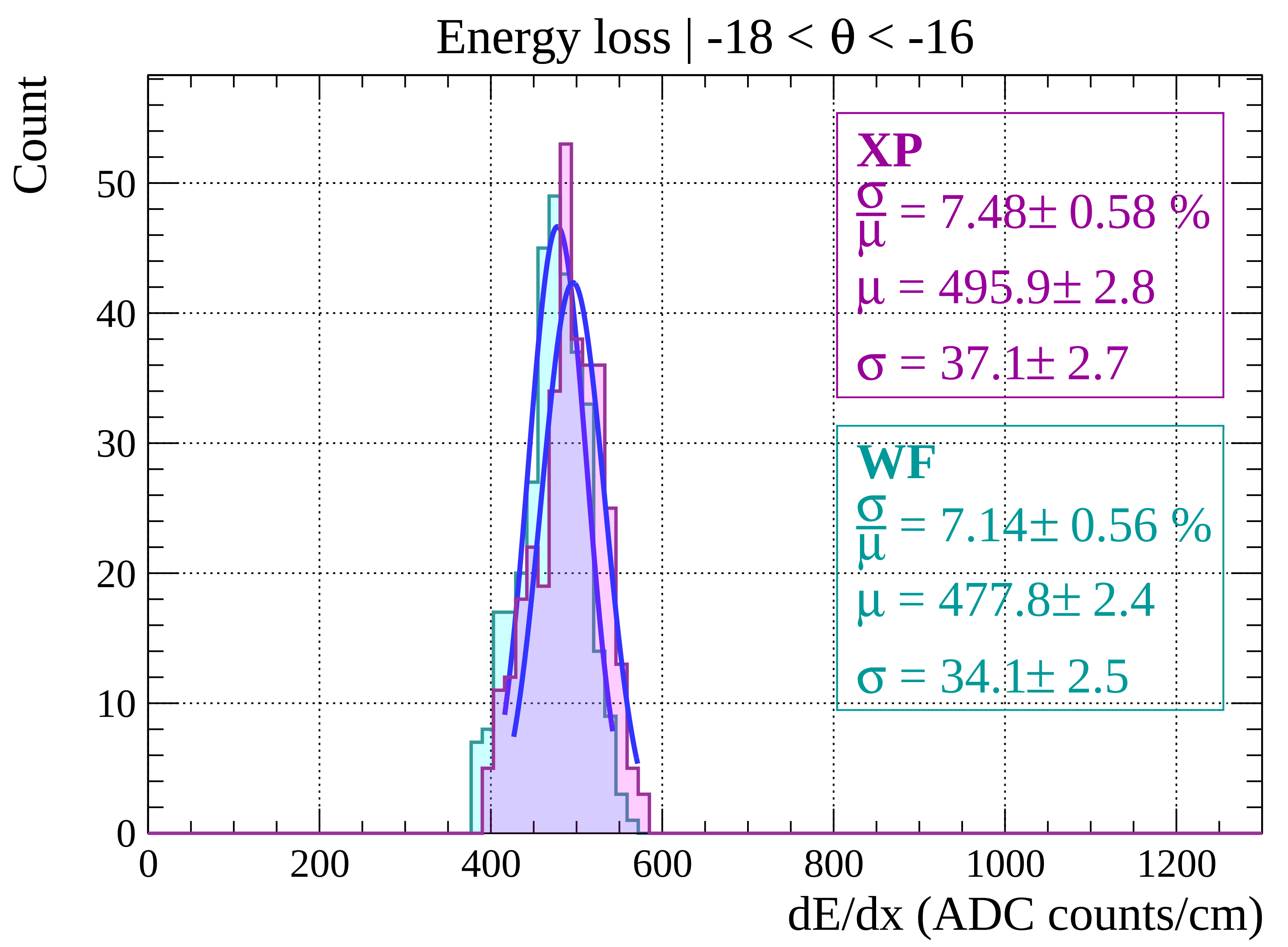
Energy loss | -24 < θ < -22
Count
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 14.09± 11.56 %
μ = 467.9± 29.4
σ = 65.9± 50.0
WF
σ/μ = 10.56± 5.84 %
μ = 455.7± 17.4
σ = 48.1± 24.8

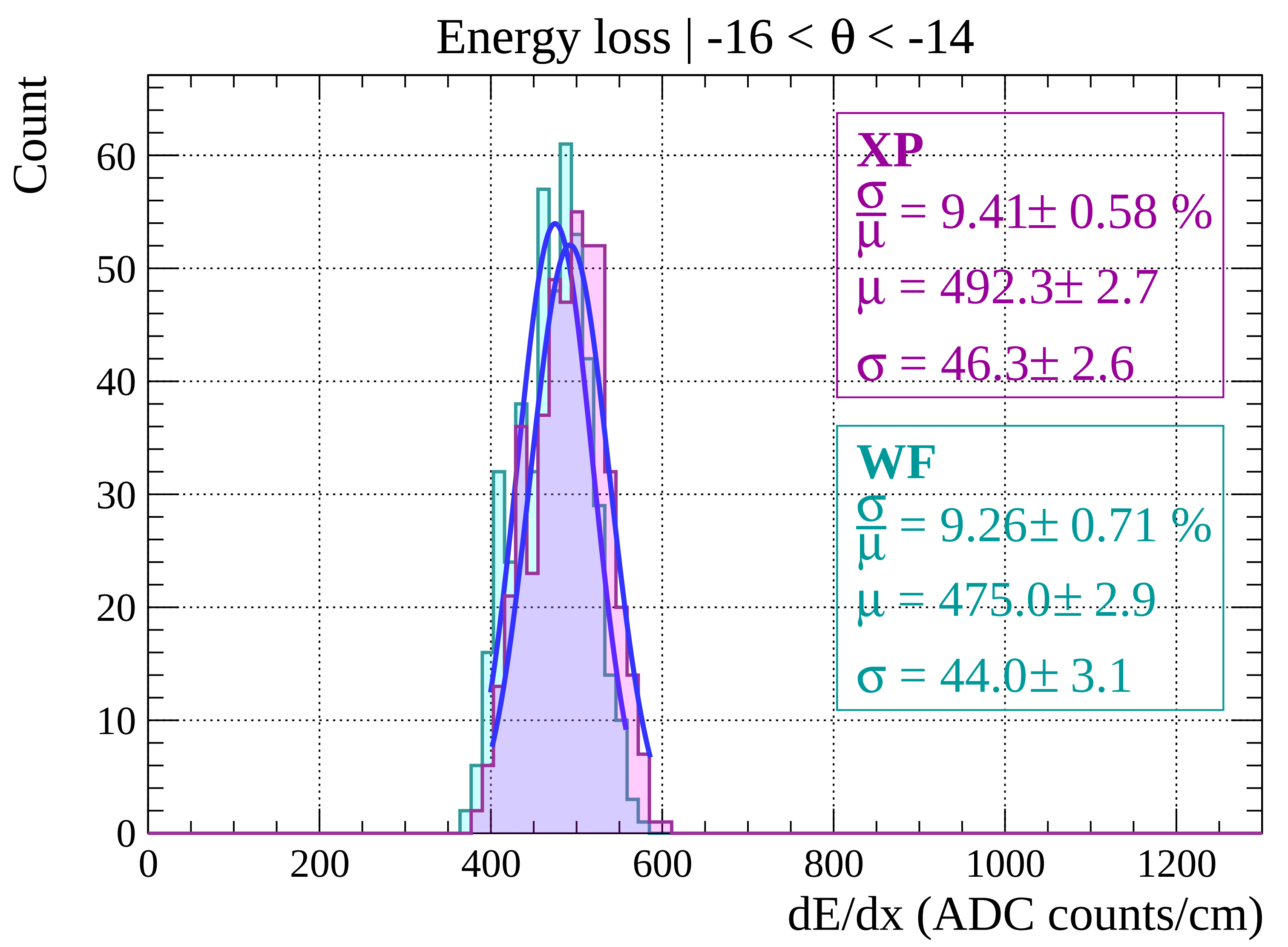


Energy loss | -22 < θ < -20
Count
dE/dx (ADC counts/cm)
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200
XP
σ/μ = 6.37± 1.05 %
μ = 493.4± 5.3
σ = 31.4± 4.9
WF
σ/μ = 6.27± 1.30 %
μ = 476.1± 5.9
σ = 29.9± 5.8

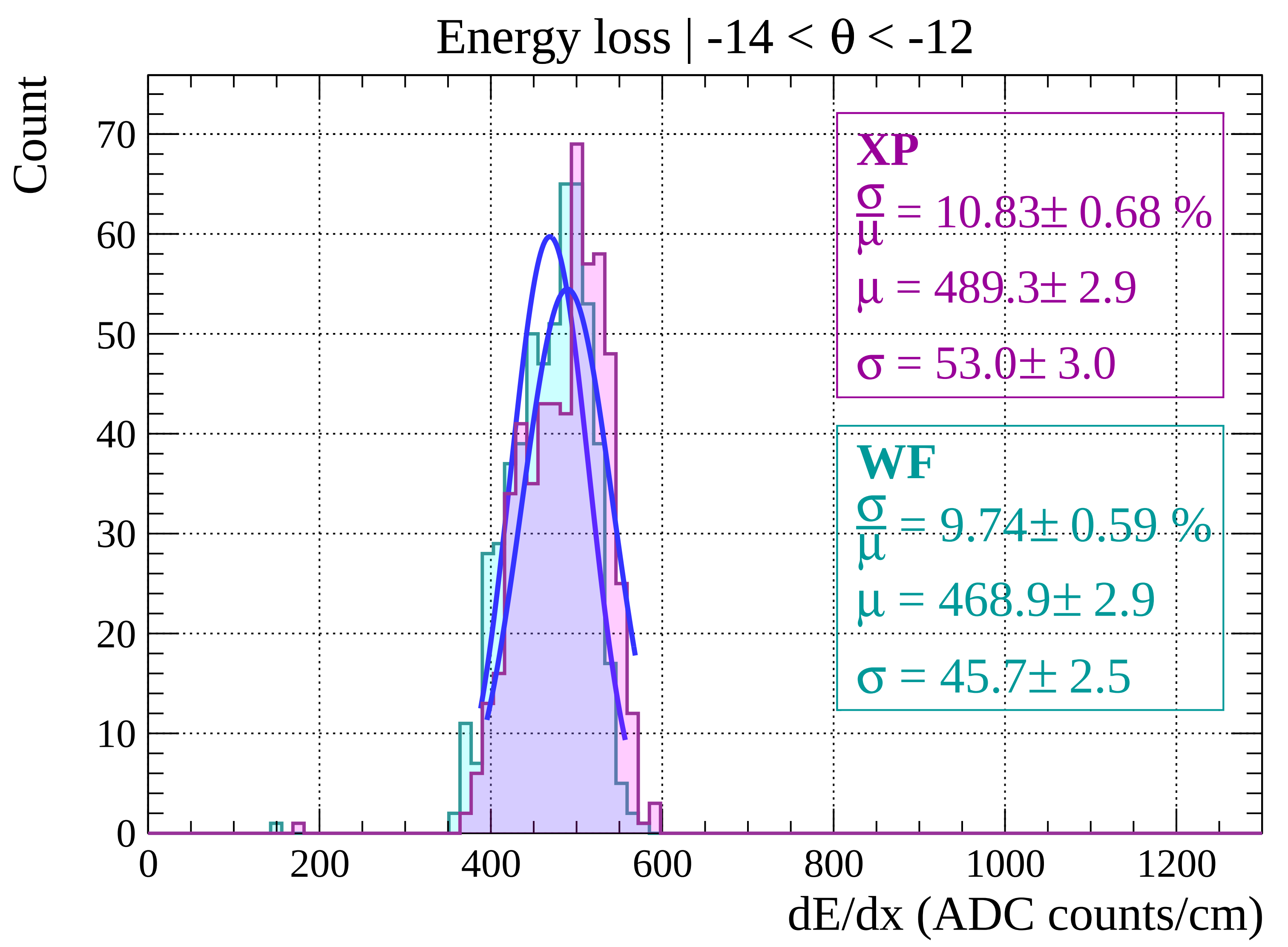


Energy loss | -20 < θ < -18
Count
30
25
20
15
10
5
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 7.54± 0.81 %
μ = 482.5± 3.8
σ = 36.4± 3.6
WF
σ/μ = 6.47± 0.76 %
μ = 473.3± 3.1
σ = 30.6± 3.4

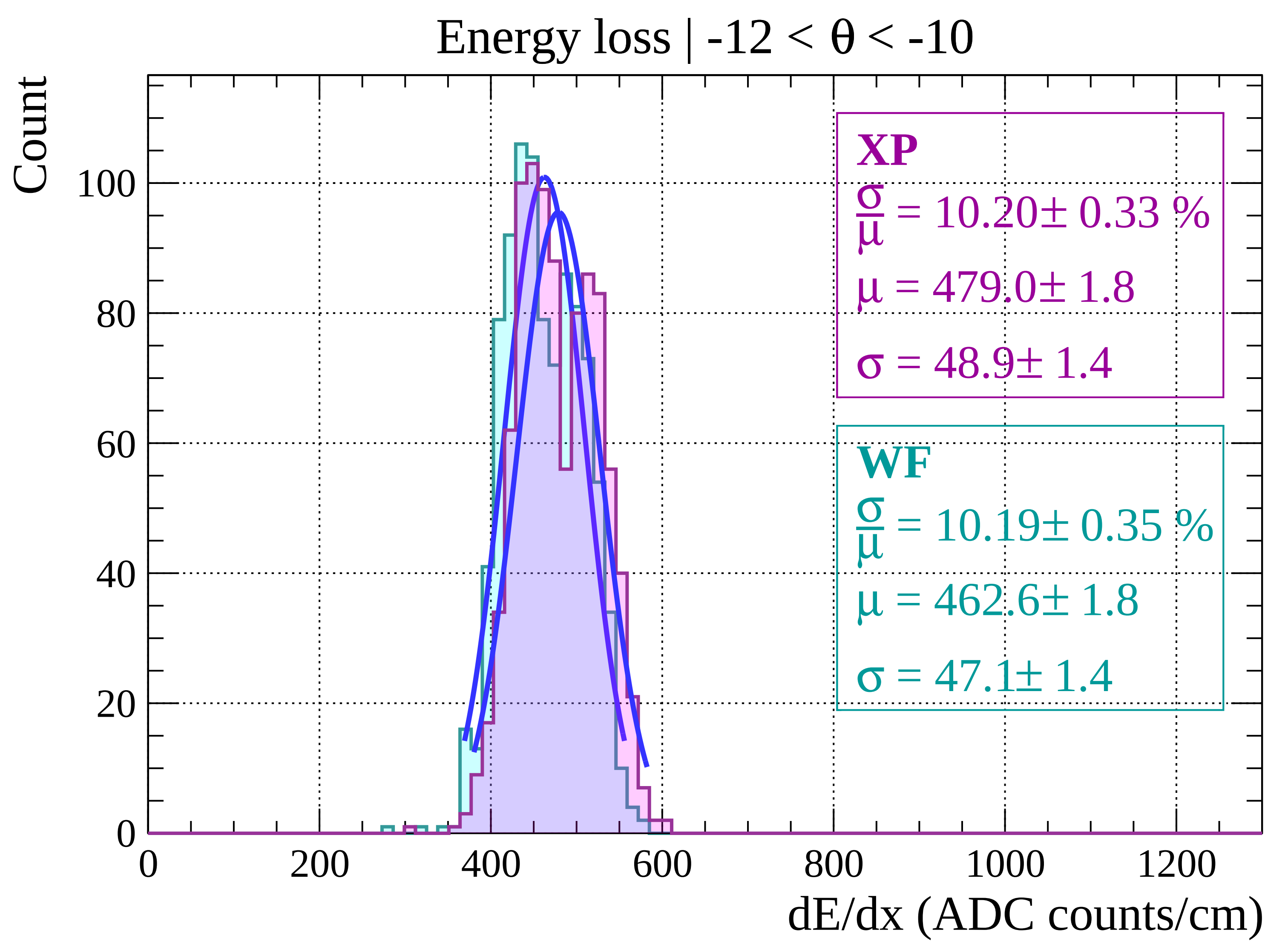

Energy loss | -18 < θ < -16
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
0
200
400
600
800
1000
1200
XP
σ/μ = 7.48± 0.58 %
μ = 495.9± 2.8
σ = 37.1± 2.7
WF
σ/μ = 7.14± 0.56 %
μ = 477.8± 2.4
σ = 34.1± 2.5



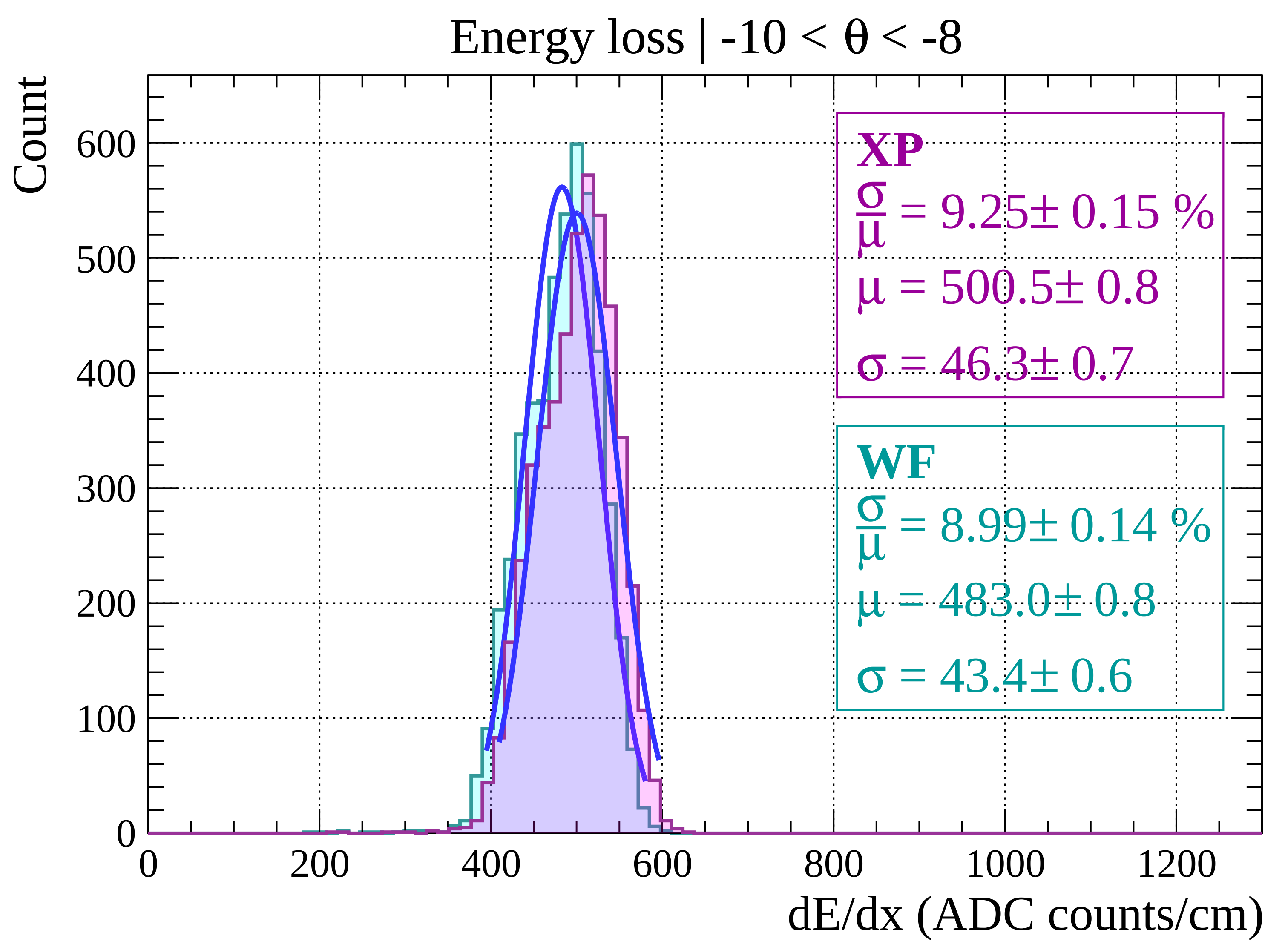
Energy loss | -16 < θ < -14
Count
0
10
20
30
40
50
60
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 9.41± 0.58 %
μ = 492.3± 2.7
σ = 46.3± 2.6
WF
σ/μ = 9.26± 0.71 %
μ = 475.0± 2.9
σ = 44.0± 3.1



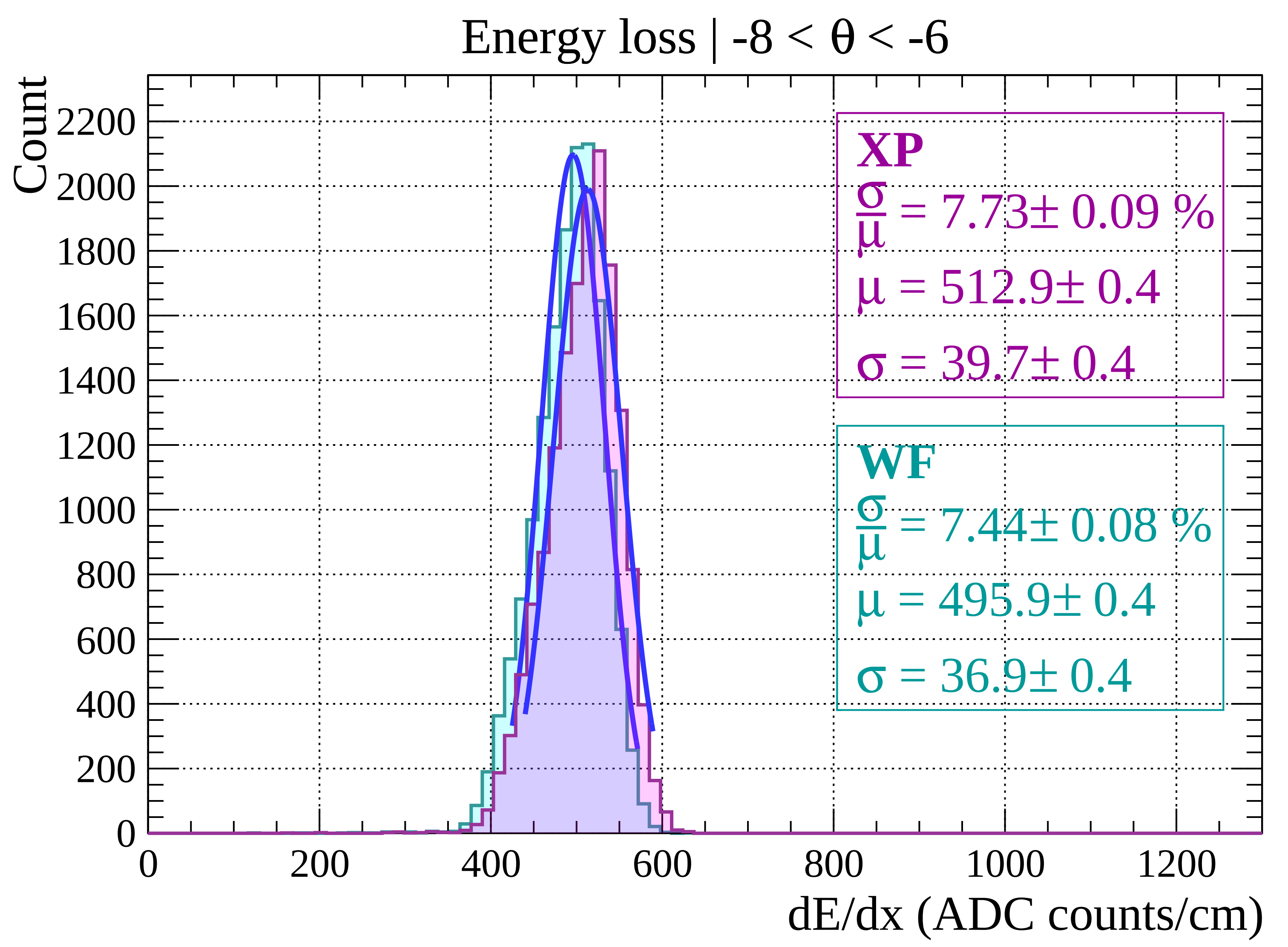
Energy loss | -14 < θ < -12
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 10.83± 0.68 %
μ = 489.3± 2.9
σ = 53.0± 3.0
WF
σ/μ = 9.74± 0.59 %
μ = 468.9± 2.9
σ = 45.7± 2.5



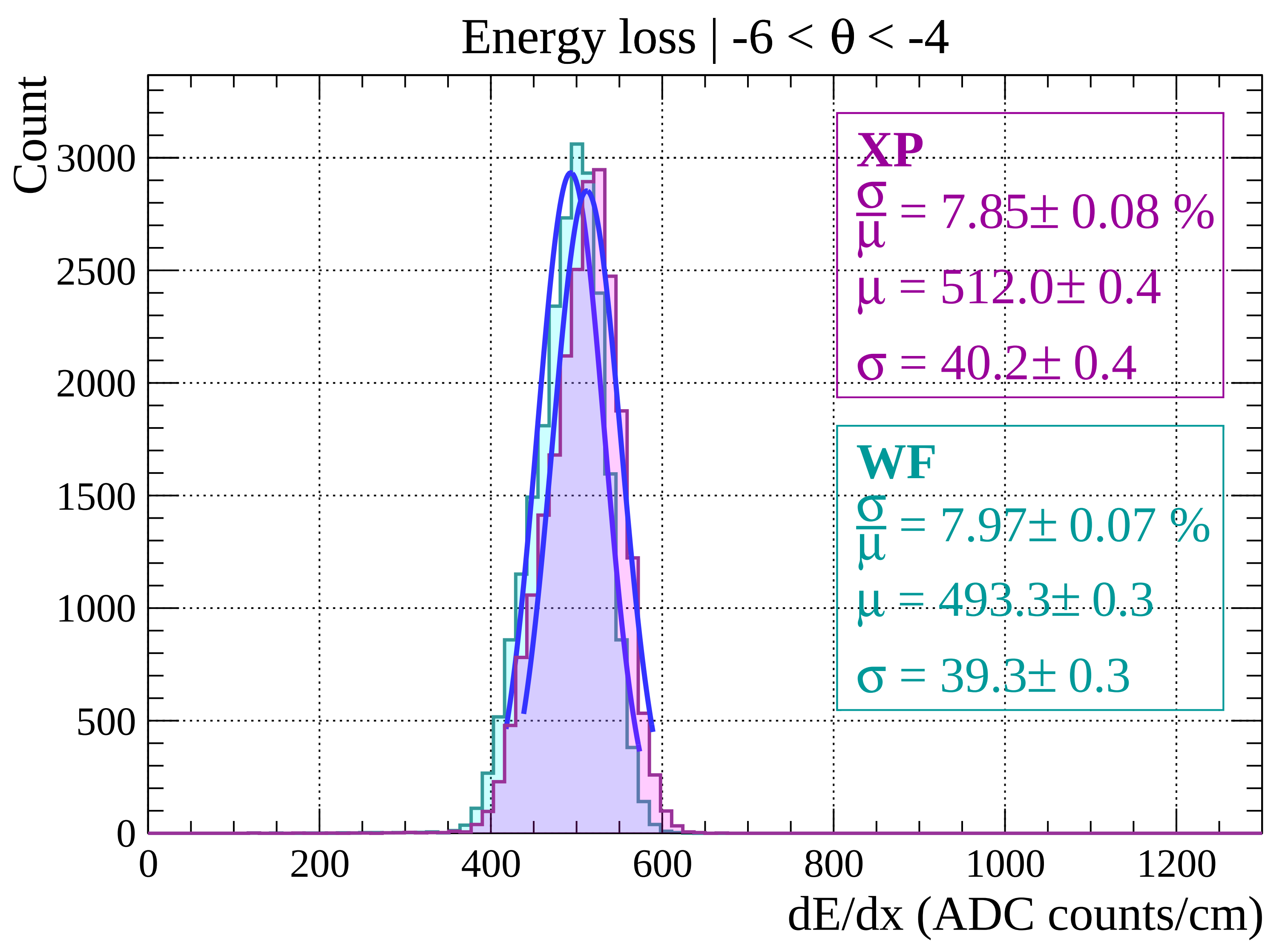
Energy loss | -12 < θ < -10
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP
σ/μ = 10.20± 0.33 %
μ = 479.0± 1.8
σ = 48.9± 1.4
WF
σ/μ = 10.19± 0.35 %
μ = 462.6± 1.8
σ = 47.1± 1.4



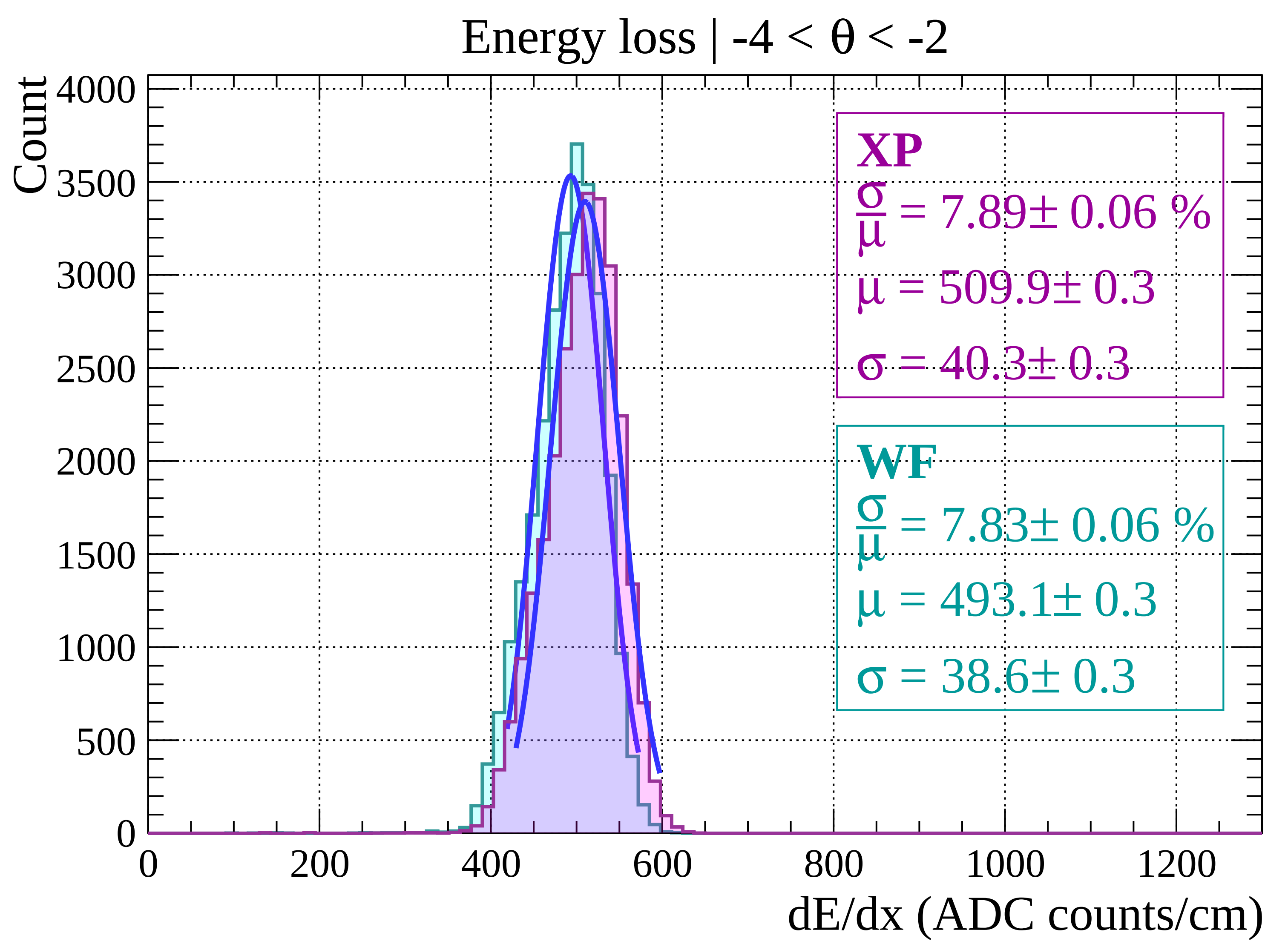
Energy loss | -10 < θ < -8
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
XP
σ/μ = 9.25± 0.15 %
μ = 500.5± 0.8
σ = 46.3± 0.7
WF
σ/μ = 8.99± 0.14 %
μ = 483.0± 0.8
σ = 43.4± 0.6



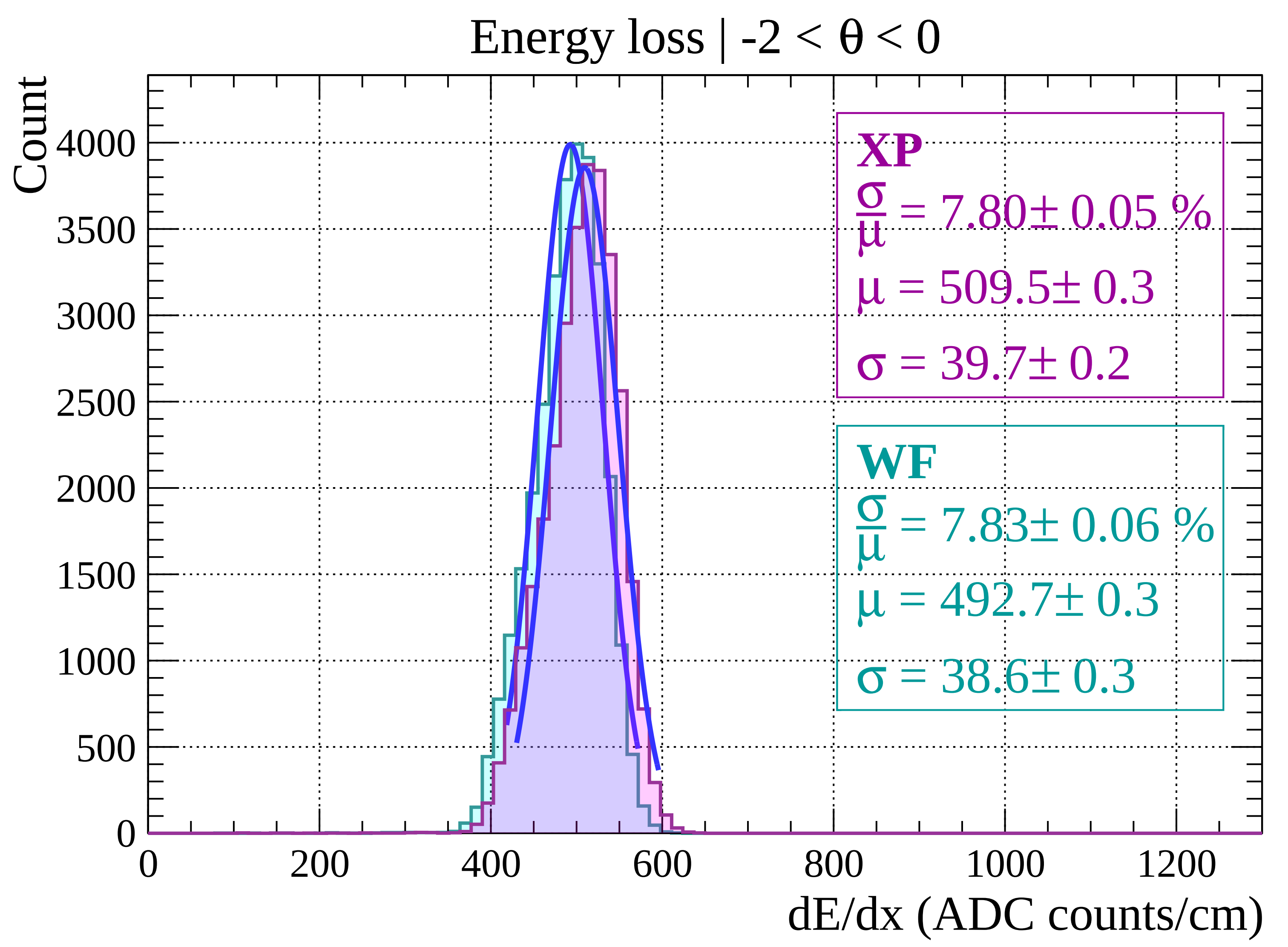
Energy loss | -8 < θ < -6
Count
0
200
400
600
800
1000
1200
1400
1600
1800
2000
2200
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 7.73± 0.09 %
μ = 512.9± 0.4
σ = 39.7± 0.4
WF
σ/μ = 7.44± 0.08 %
μ = 495.9± 0.4
σ = 36.9± 0.4



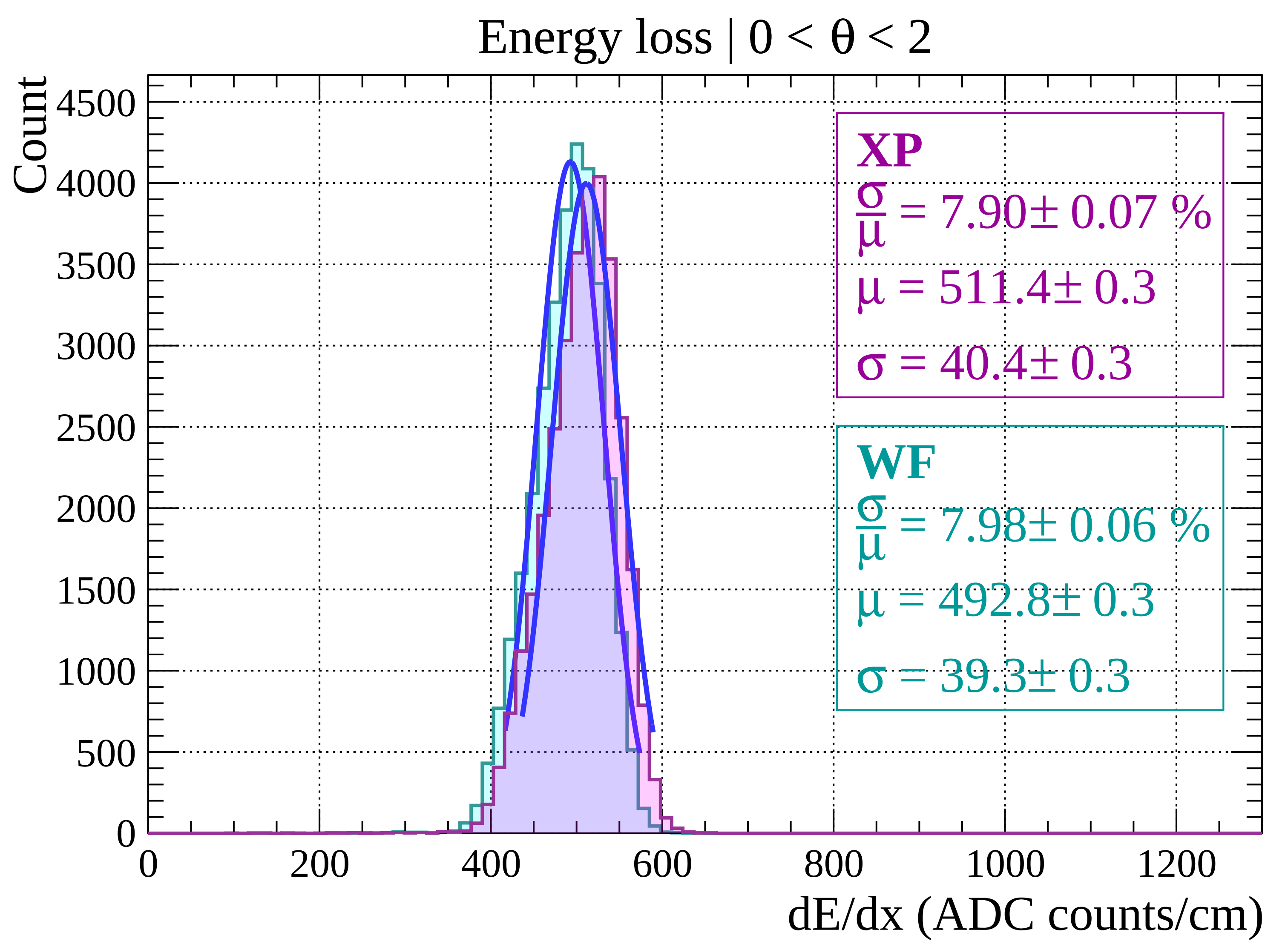
Energy loss | -6 < θ < -4
Count
3000
2500
2000
1500
1000
500
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 7.85± 0.08 %
μ = 512.0± 0.4
σ = 40.2± 0.4
WF
σ/μ = 7.97± 0.07 %
μ = 493.3± 0.3
σ = 39.3± 0.3



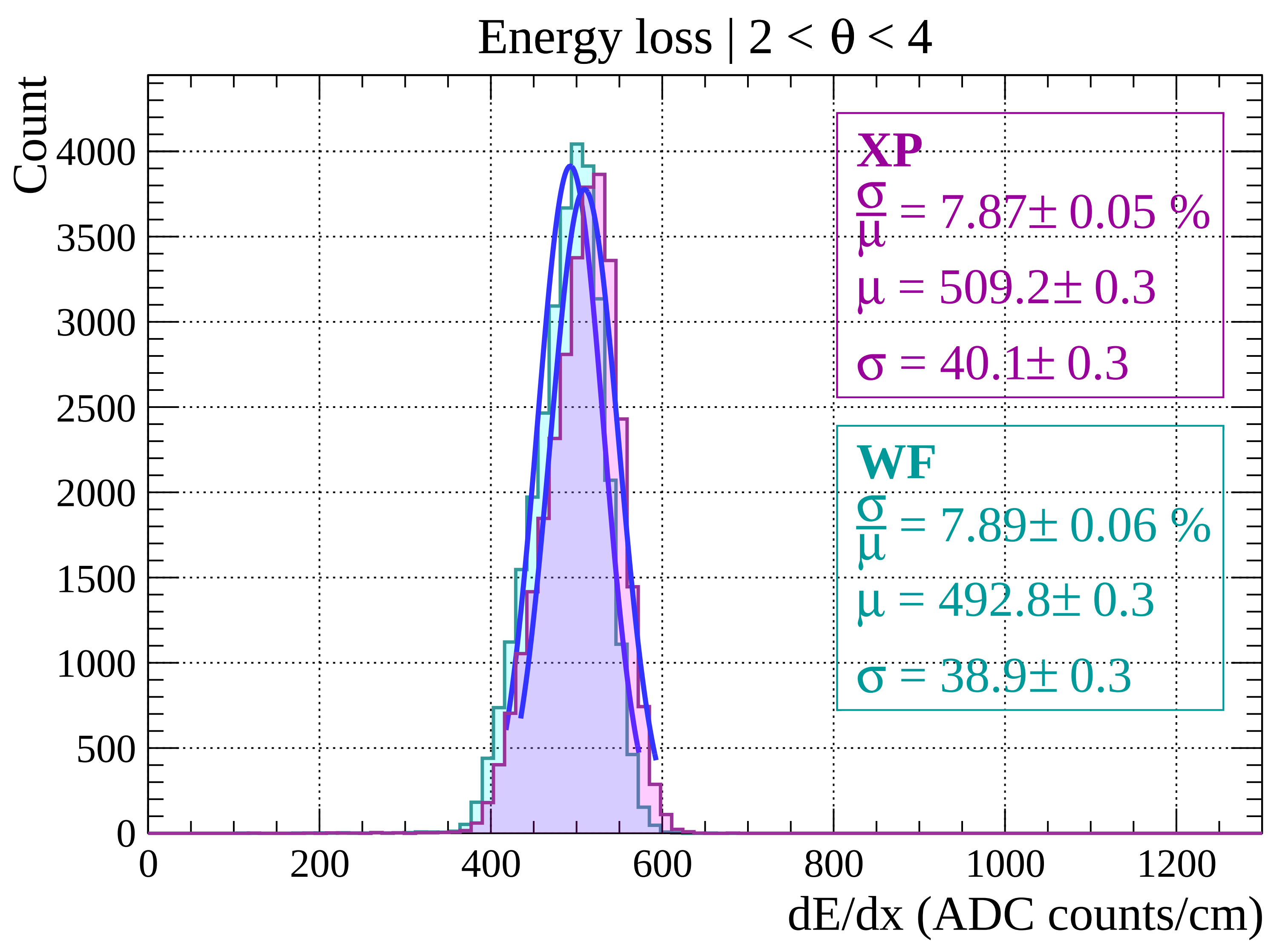
Energy loss | -4 < θ < -2
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 7.89± 0.06 %
μ = 509.9± 0.3
σ = 40.3± 0.3
WF
σ/μ = 7.83± 0.06 %
μ = 493.1± 0.3
σ = 38.6± 0.3



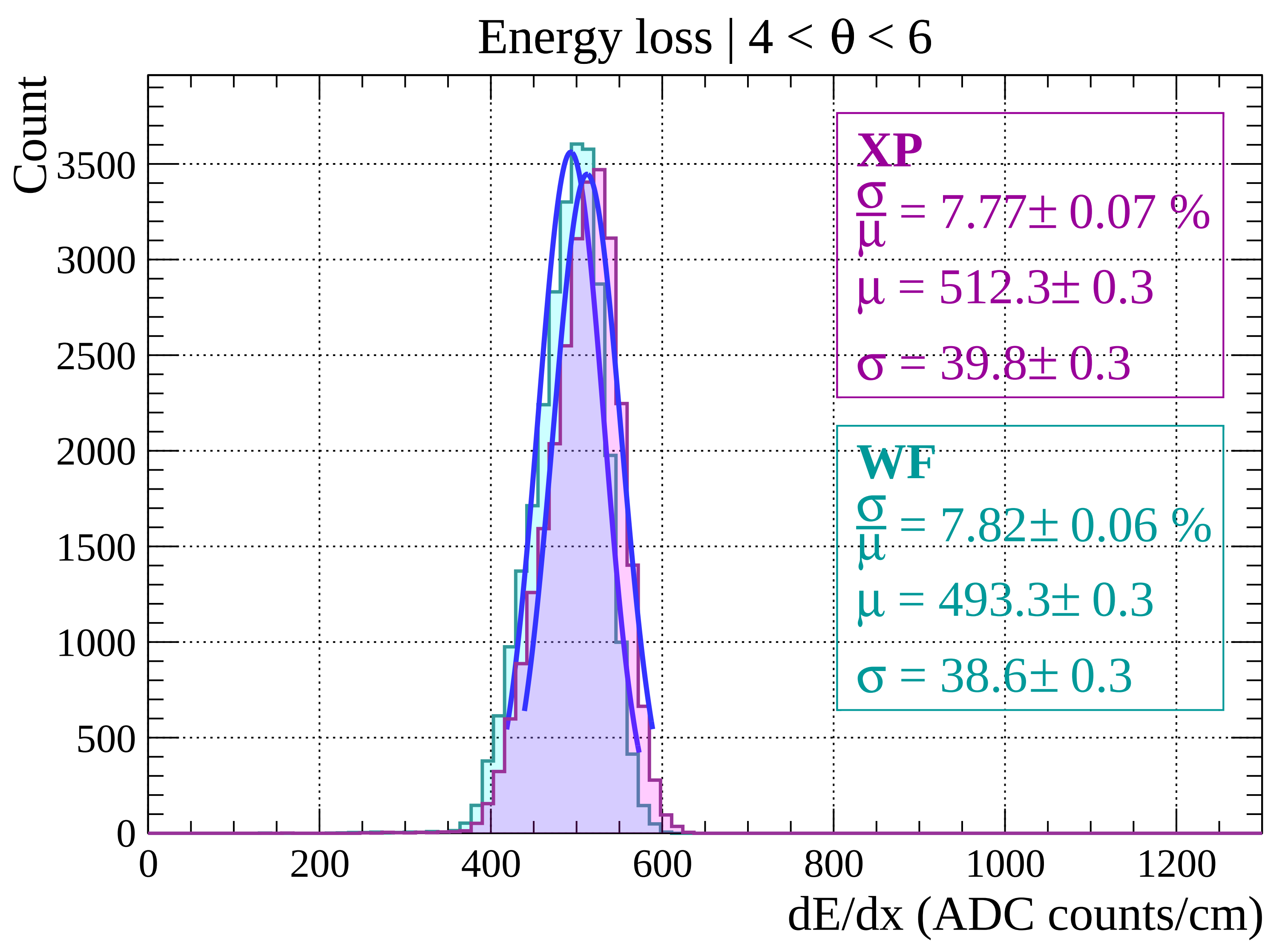
Energy loss | -2 < θ < 0
Count
4000
3500
3000
2500
2000
1500
1000
500
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 7.80± 0.05 %
μ = 509.5± 0.3
σ = 39.7± 0.2
WF
$\frac{\sigma}{\mu}$ = 7.83± 0.06 %
μ = 492.7± 0.3
σ = 38.6± 0.3



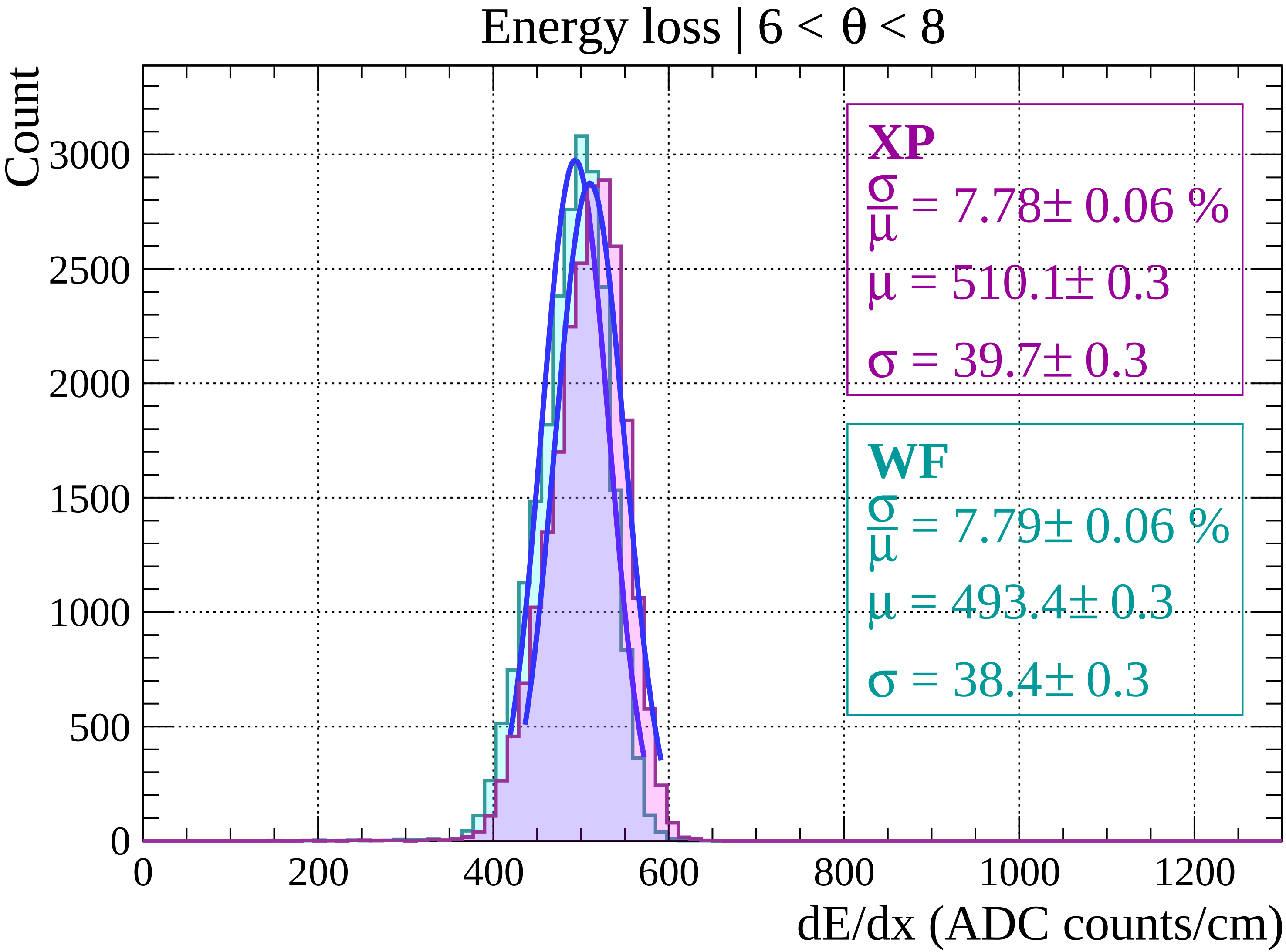
Energy loss | 0 < θ < 2
Count
4500
4000
3500
3000
2500
2000
1500
1000
500
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 7.90± 0.07 %
μ = 511.4± 0.3
σ = 40.4± 0.3
WF
σ/μ = 7.98± 0.06 %
μ = 492.8± 0.3
σ = 39.3± 0.3



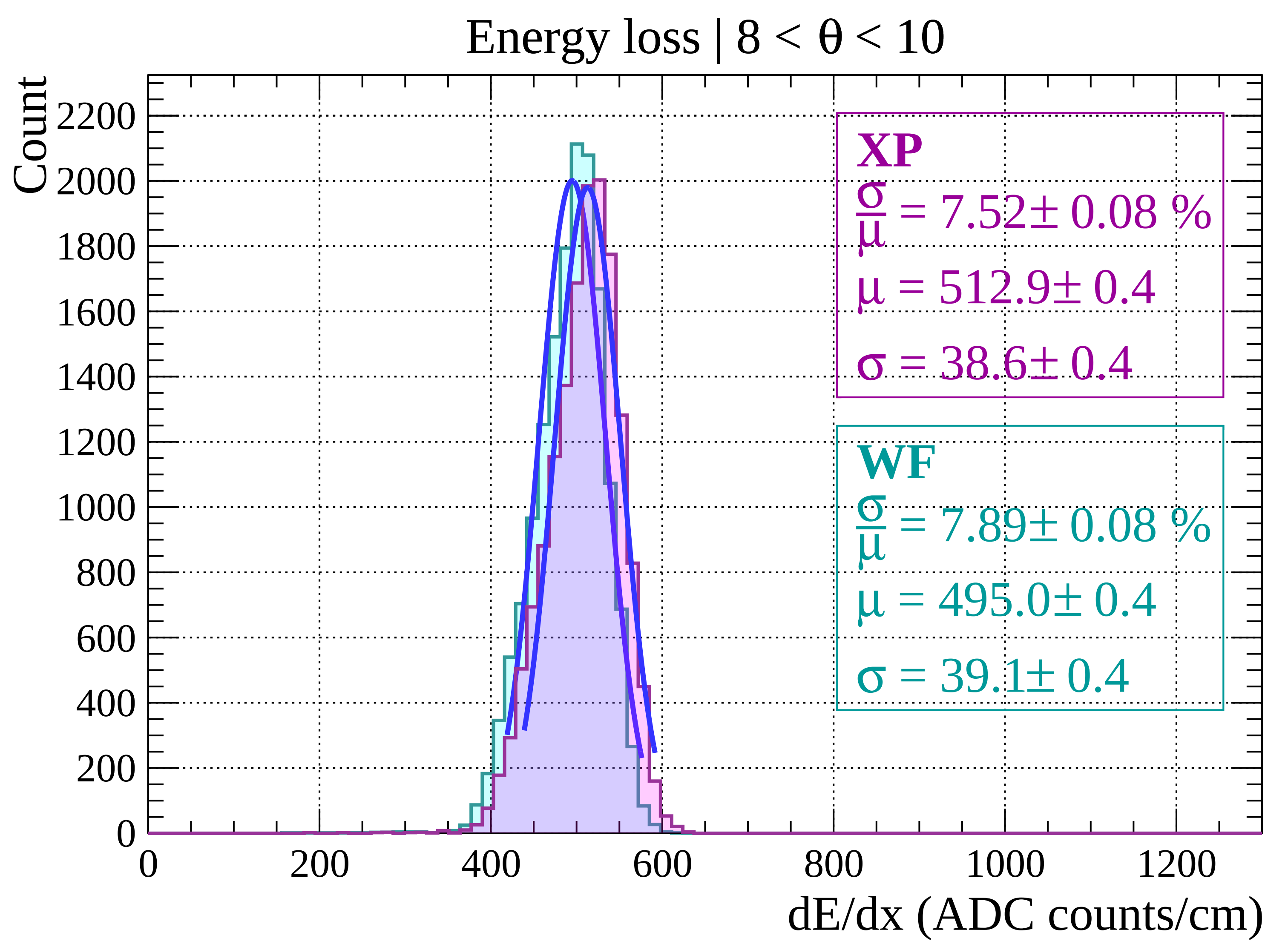
Energy loss | 2 < θ < 4
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
500
1000
1500
2000
2500
3000
3500
4000
XP
σ/μ = 7.87± 0.05 %
μ = 509.2± 0.3
σ = 40.1± 0.3
WF
σ/μ = 7.89± 0.06 %
μ = 492.8± 0.3
σ = 38.9± 0.3



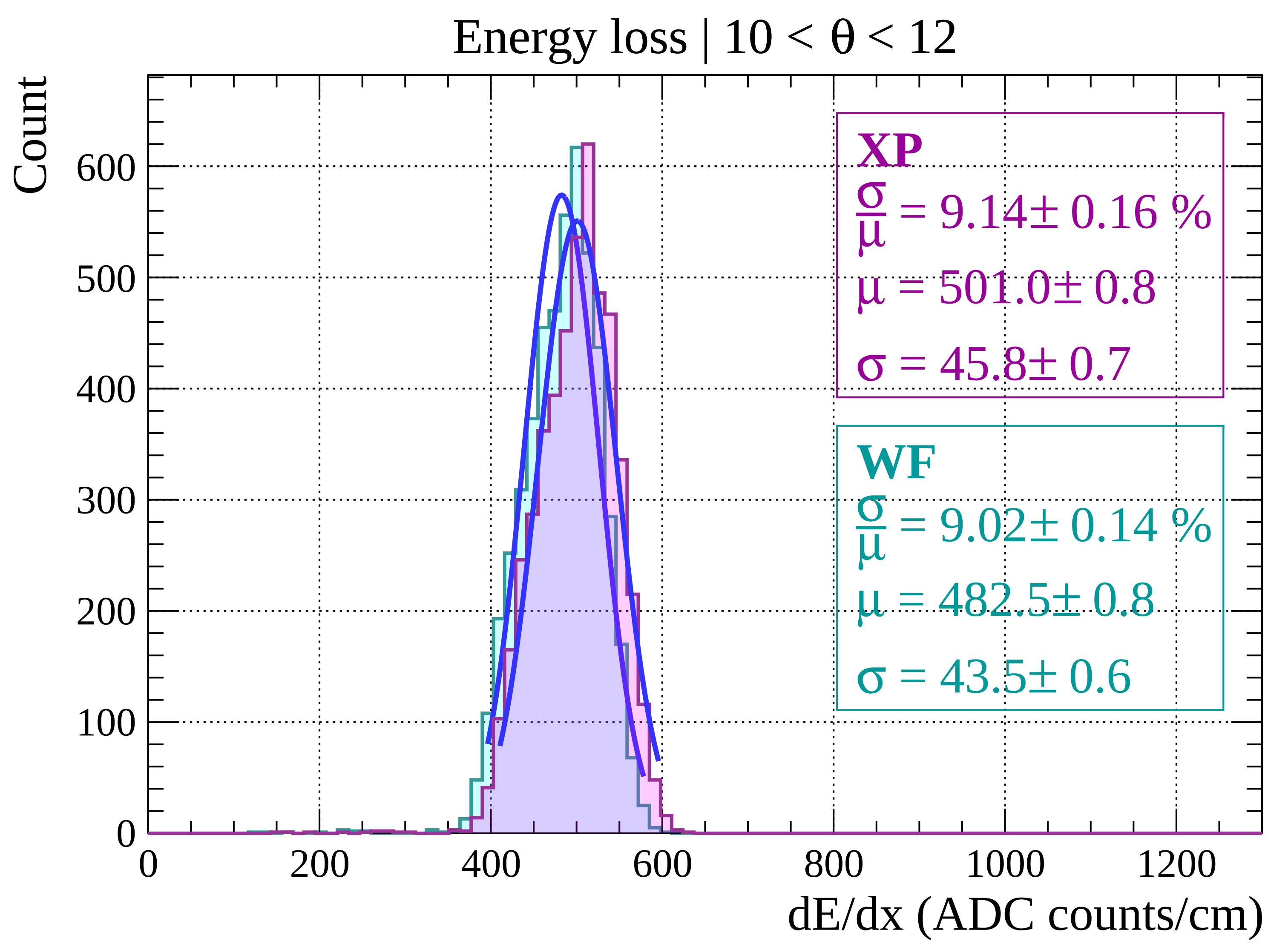
Energy loss | 4 < θ < 6
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
500
1000
1500
2000
2500
3000
3500
XP
$\frac{\sigma}{\mu}$ = 7.77± 0.07 %
μ = 512.3± 0.3
σ = 39.8± 0.3
WF
$\frac{\sigma}{\mu}$ = 7.82± 0.06 %
μ = 493.3± 0.3
σ = 38.6± 0.3



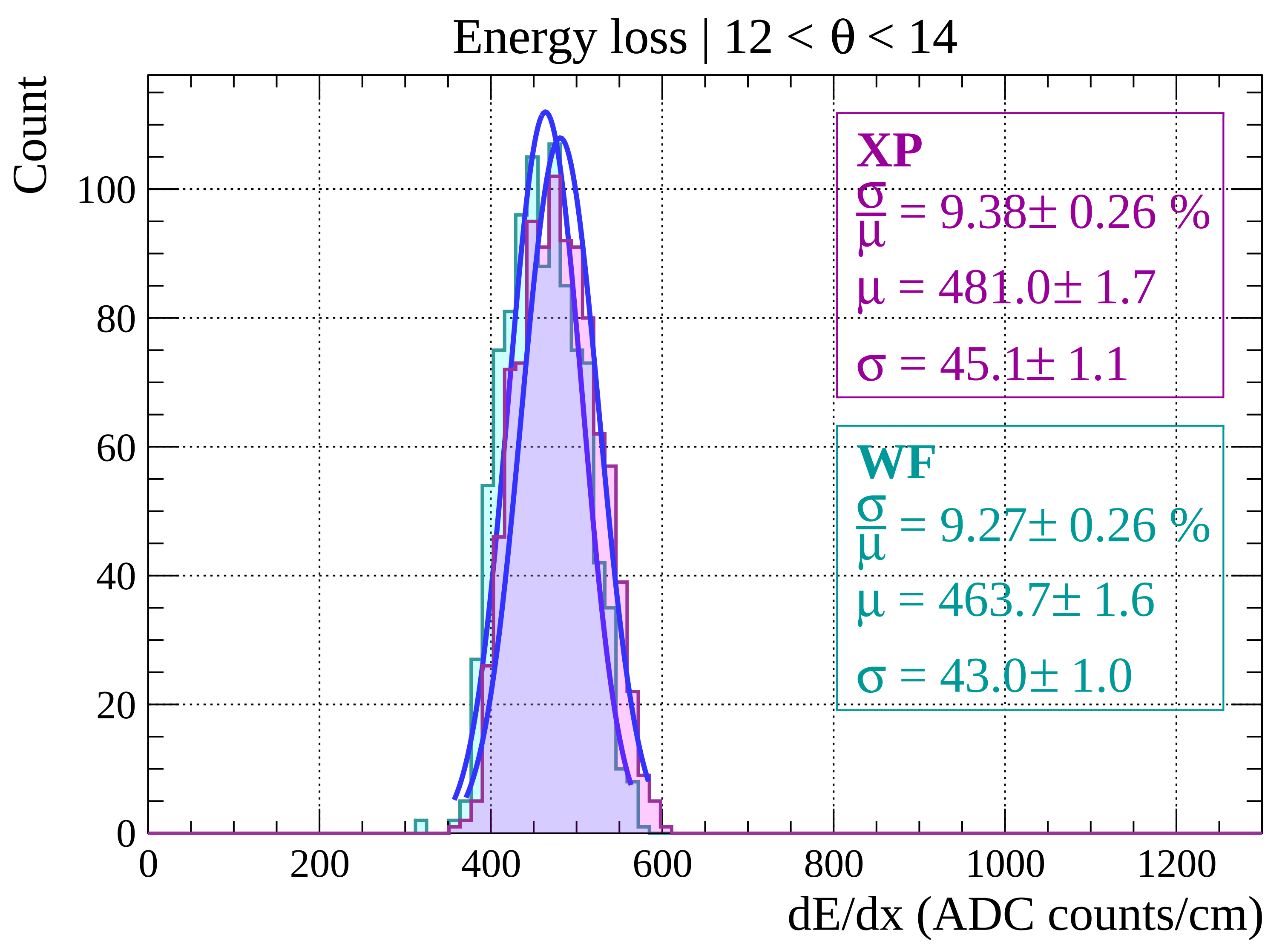
Energy loss | 6 < θ < 8
Count
3000
2500
2000
1500
1000
500
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 7.78± 0.06 %
μ = 510.1± 0.3
σ = 39.7± 0.3
WF
σ/μ = 7.79± 0.06 %
μ = 493.4± 0.3
σ = 38.4± 0.3



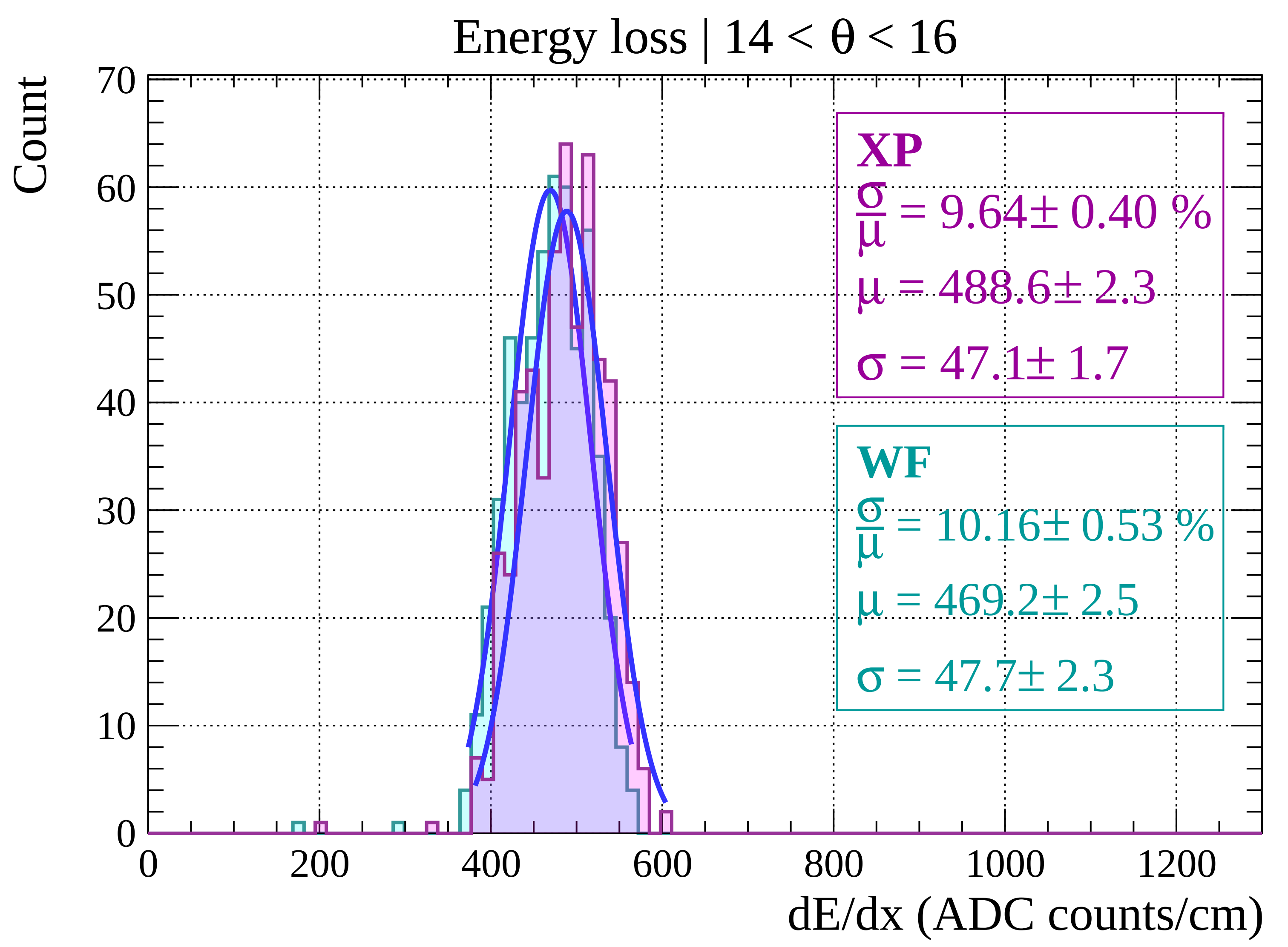
Energy loss | 8 < θ < 10
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1400
1600
1800
2000
2200
0
200
400
600
800
1000
1200
XP
σ/μ = 7.52± 0.08 %
μ = 512.9± 0.4
σ = 38.6± 0.4
WF
σ/μ = 7.89± 0.08 %
μ = 495.0± 0.4
σ = 39.1± 0.4



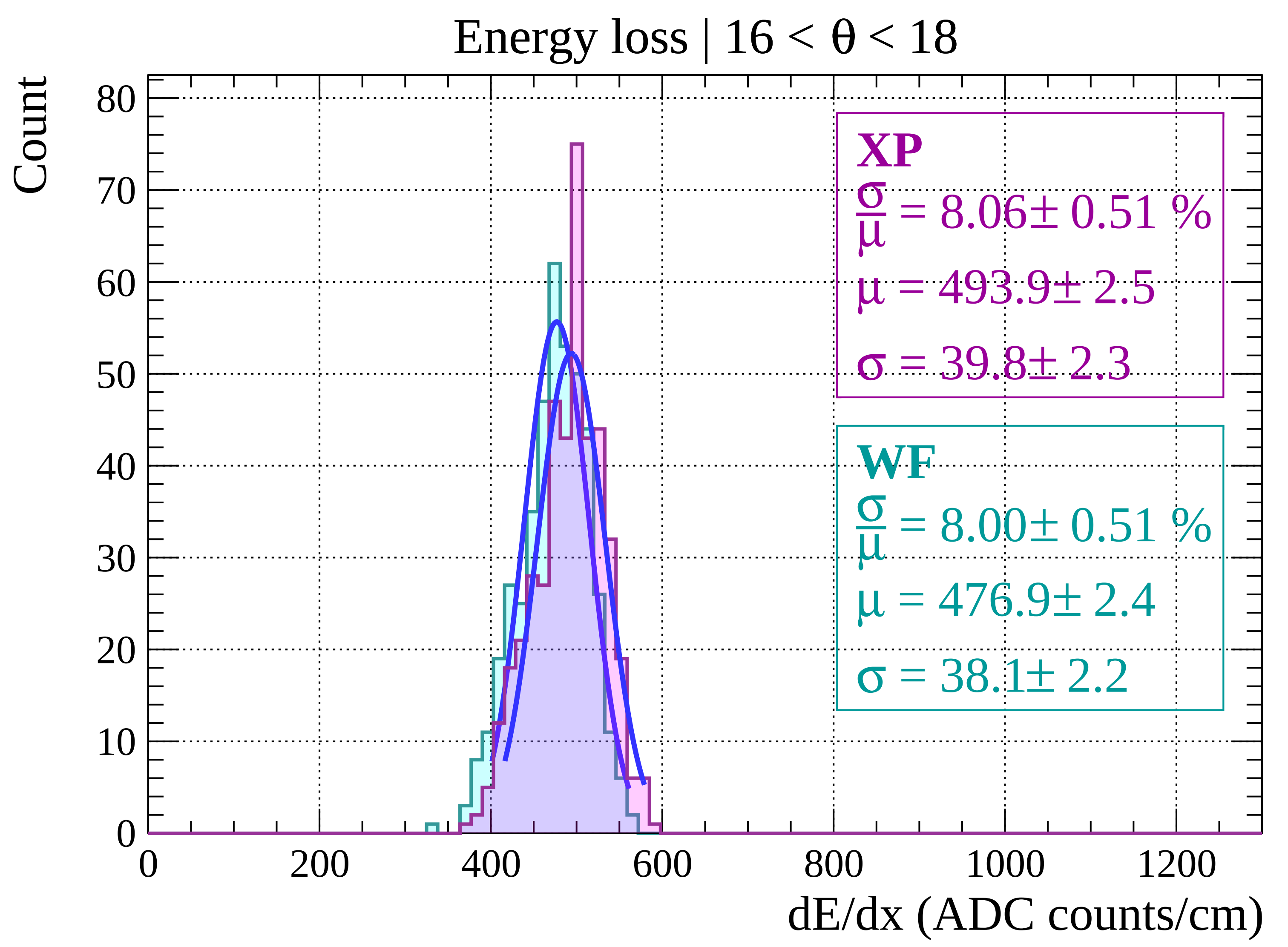
Energy loss | 10 < θ < 12
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
XP
σ/μ = 9.14± 0.16 %
μ = 501.0± 0.8
σ = 45.8± 0.7
WF
σ/μ = 9.02± 0.14 %
μ = 482.5± 0.8
σ = 43.5± 0.6



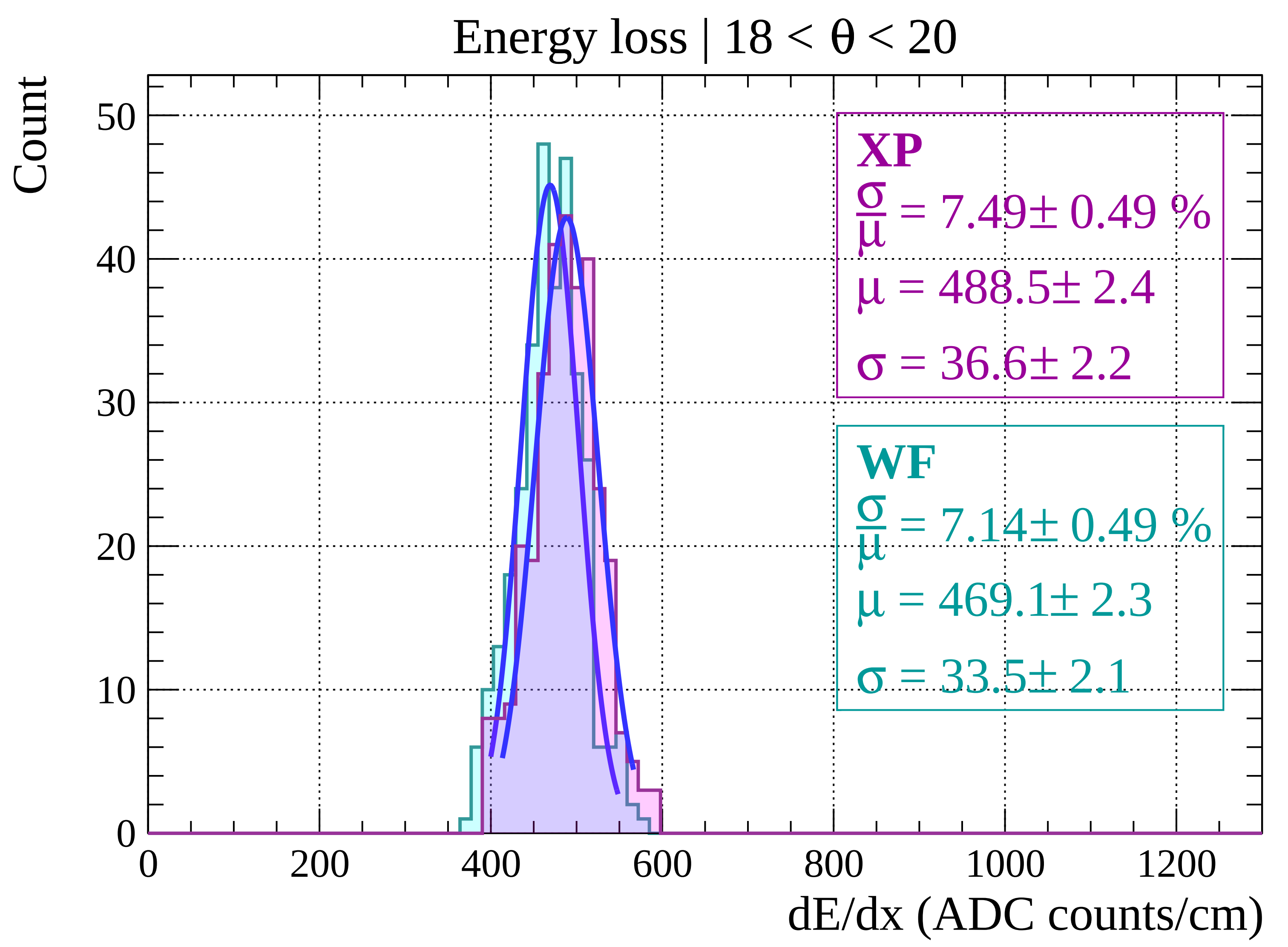
Energy loss | 12 < θ < 14
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP
σ/μ = 9.38± 0.26 %
μ = 481.0± 1.7
σ = 45.1± 1.1
WF
σ/μ = 9.27± 0.26 %
μ = 463.7± 1.6
σ = 43.0± 1.0

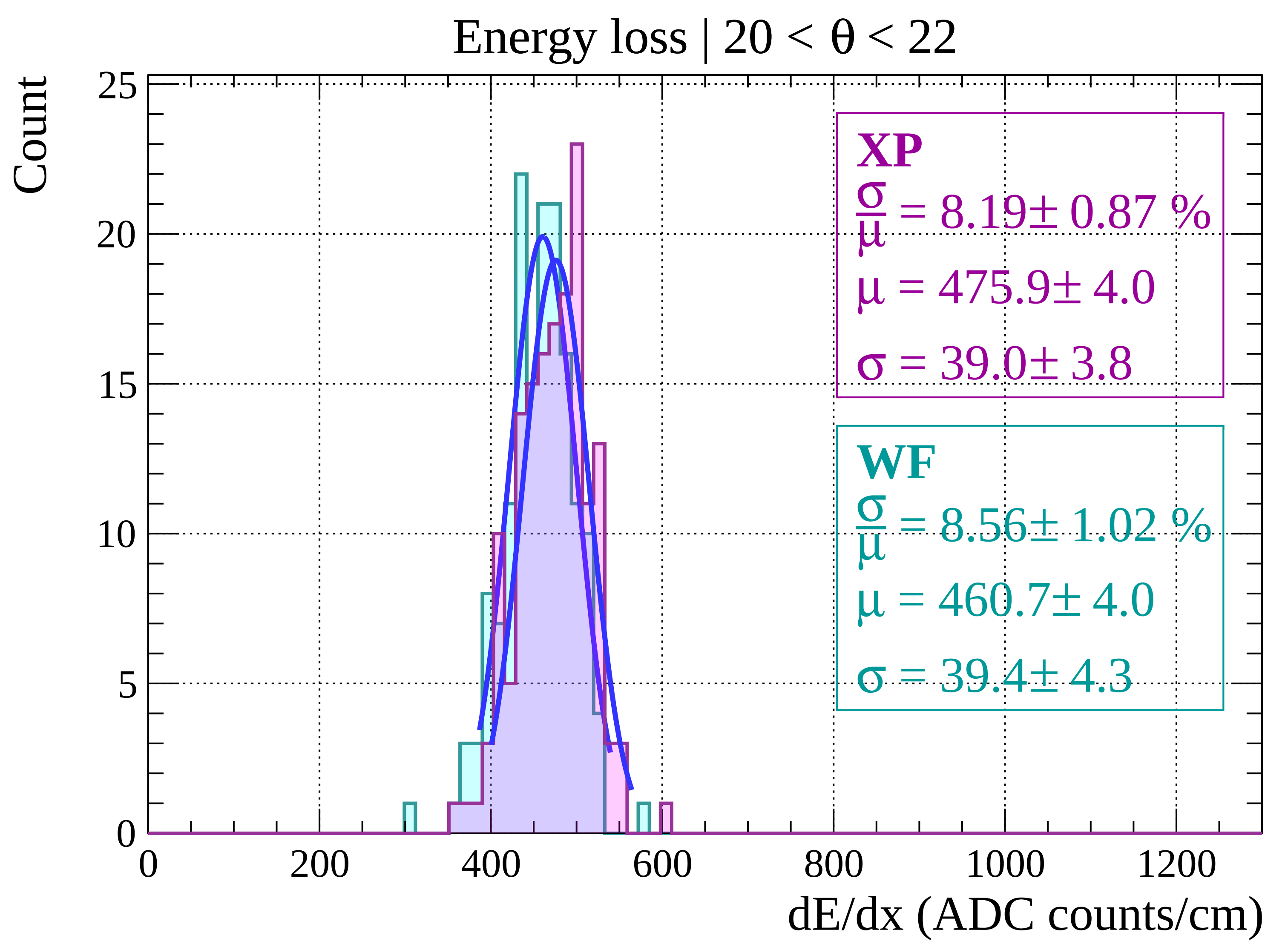


Energy loss | 14 < θ < 16
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
60
70
XP
σ/μ = 9.64± 0.40 %
μ = 488.6± 2.3
σ = 47.1± 1.7
WF
σ/μ = 10.16± 0.53 %
μ = 469.2± 2.5
σ = 47.7± 2.3

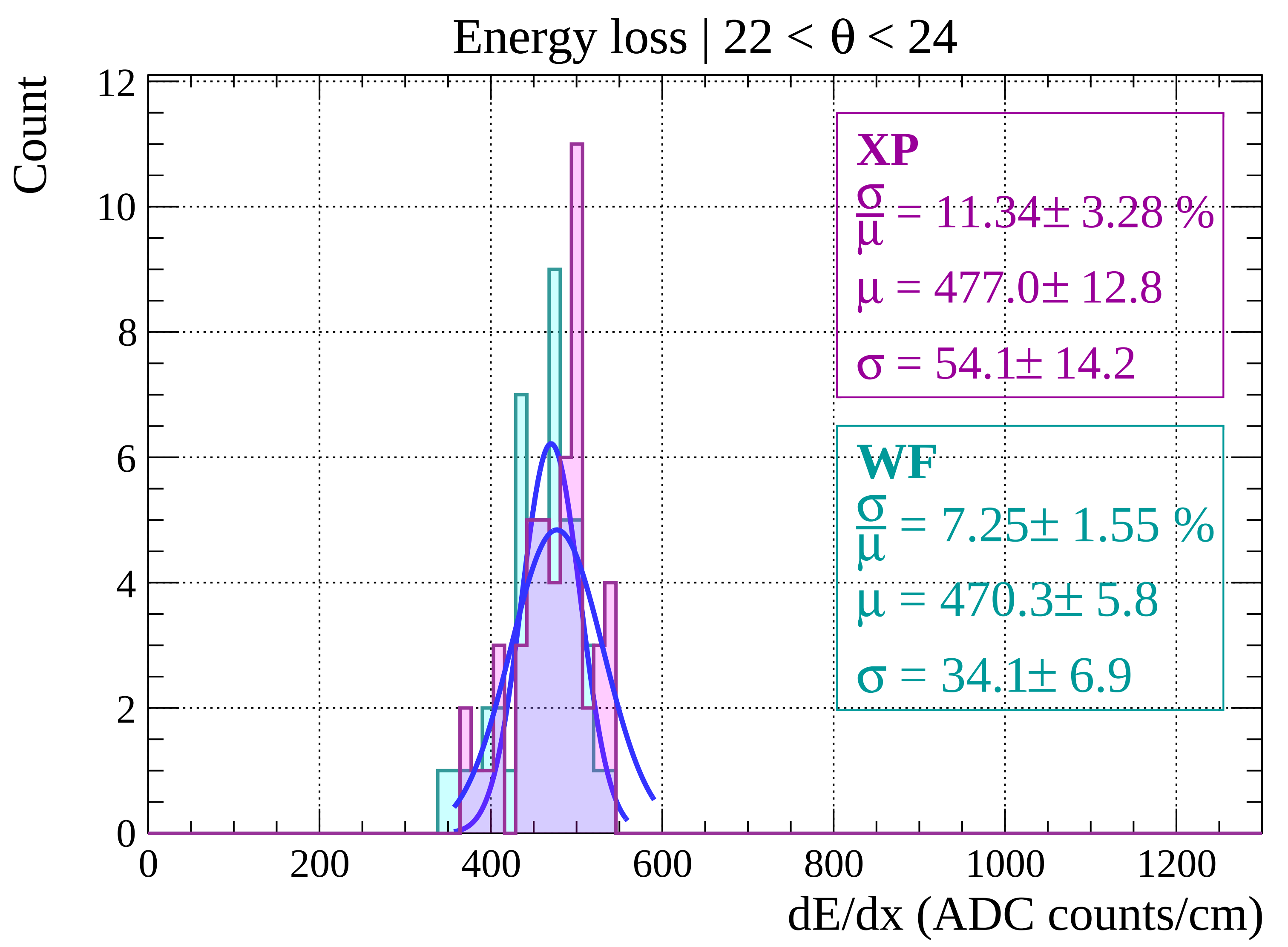


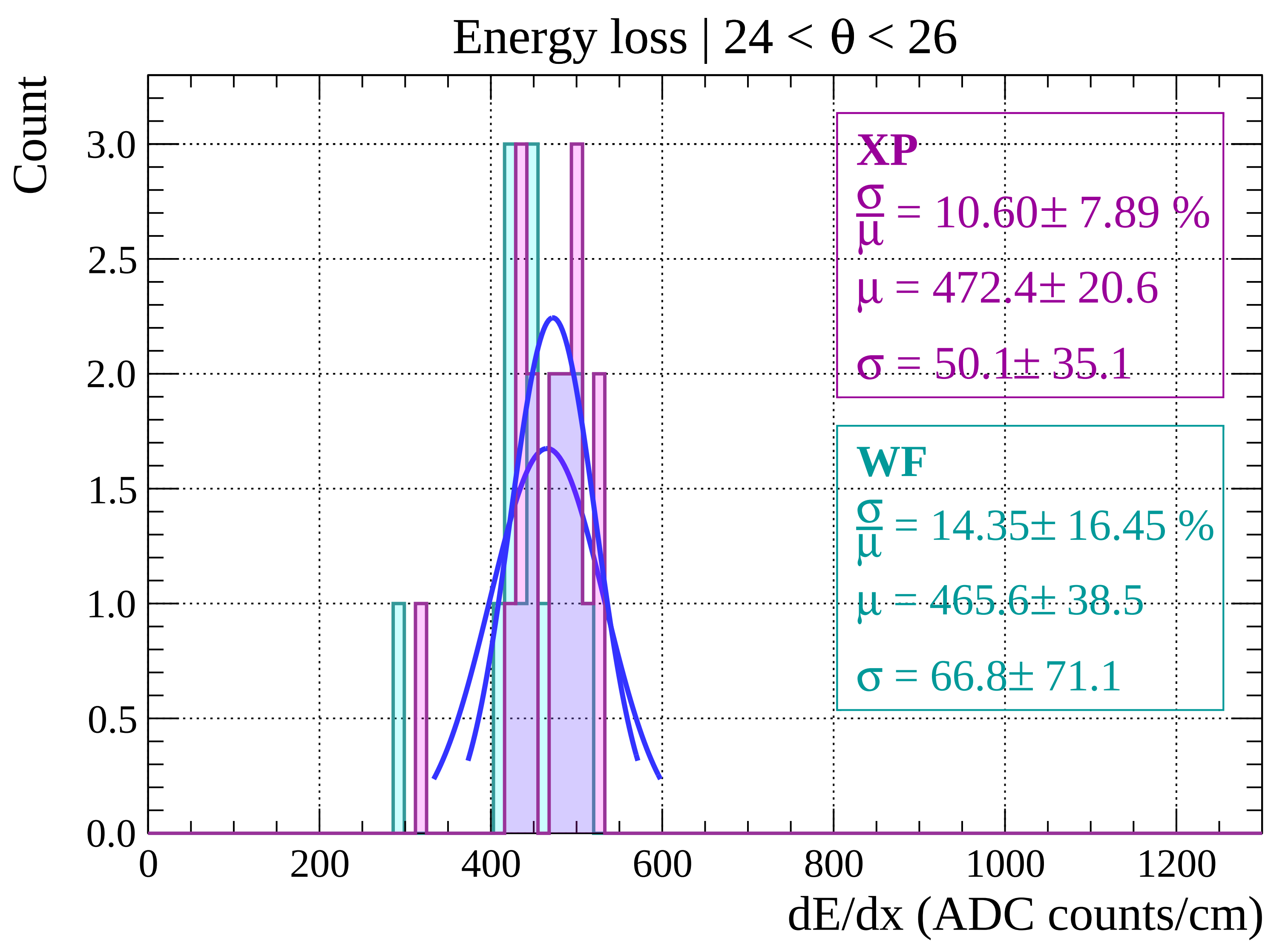
Energy loss | 16 < θ < 18
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 8.06± 0.51 %
μ = 493.9± 2.5
σ = 39.8± 2.3
WF
σ/μ = 8.00± 0.51 %
μ = 476.9± 2.4
σ = 38.1± 2.2


Energy loss | 18 < θ < 20
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 7.49± 0.49 %
μ = 488.5± 2.4
σ = 36.6± 2.2
WF
$\frac{\sigma}{\mu}$ = 7.14± 0.49 %
μ = 469.1± 2.3
σ = 33.5± 2.1

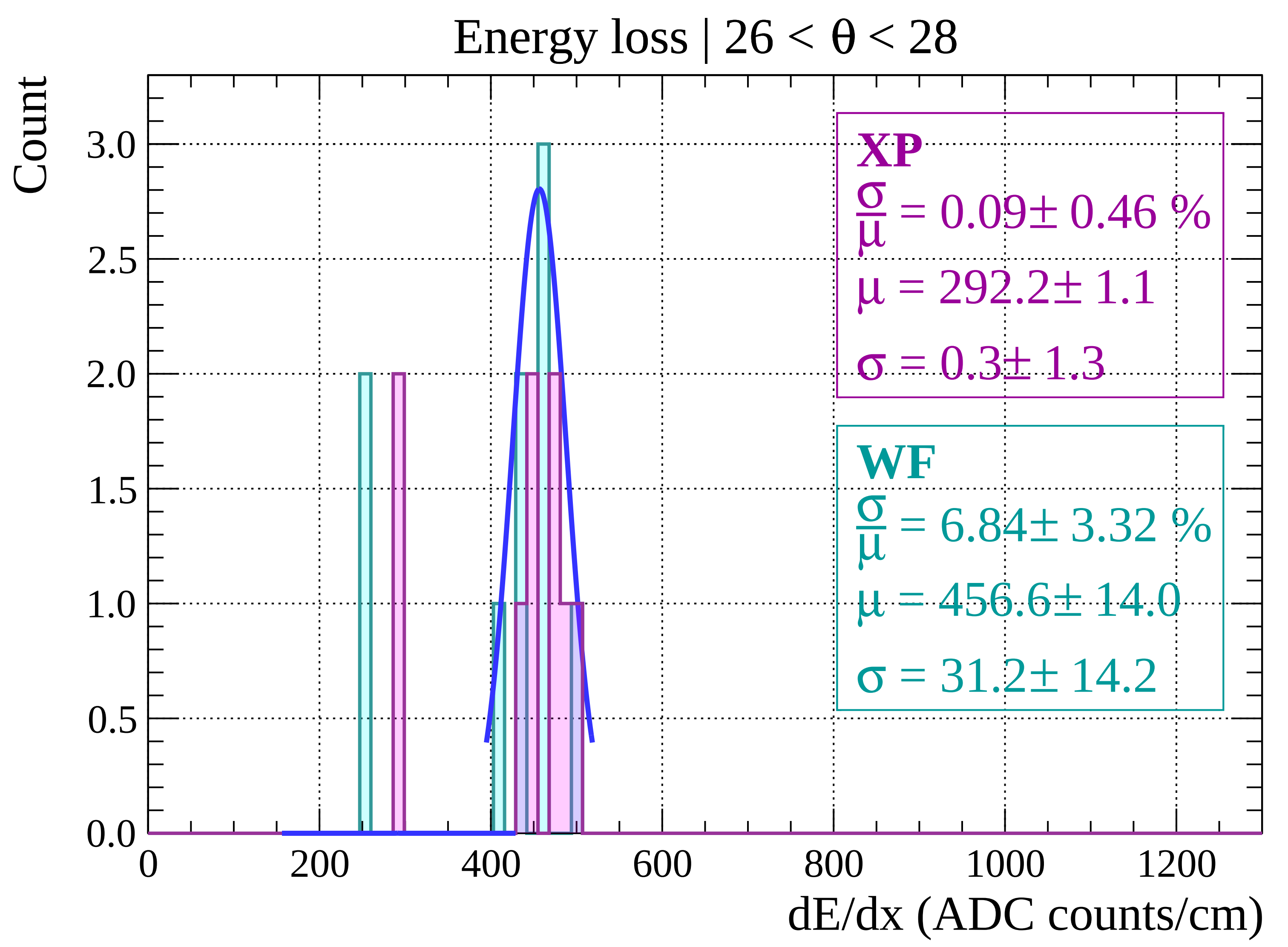


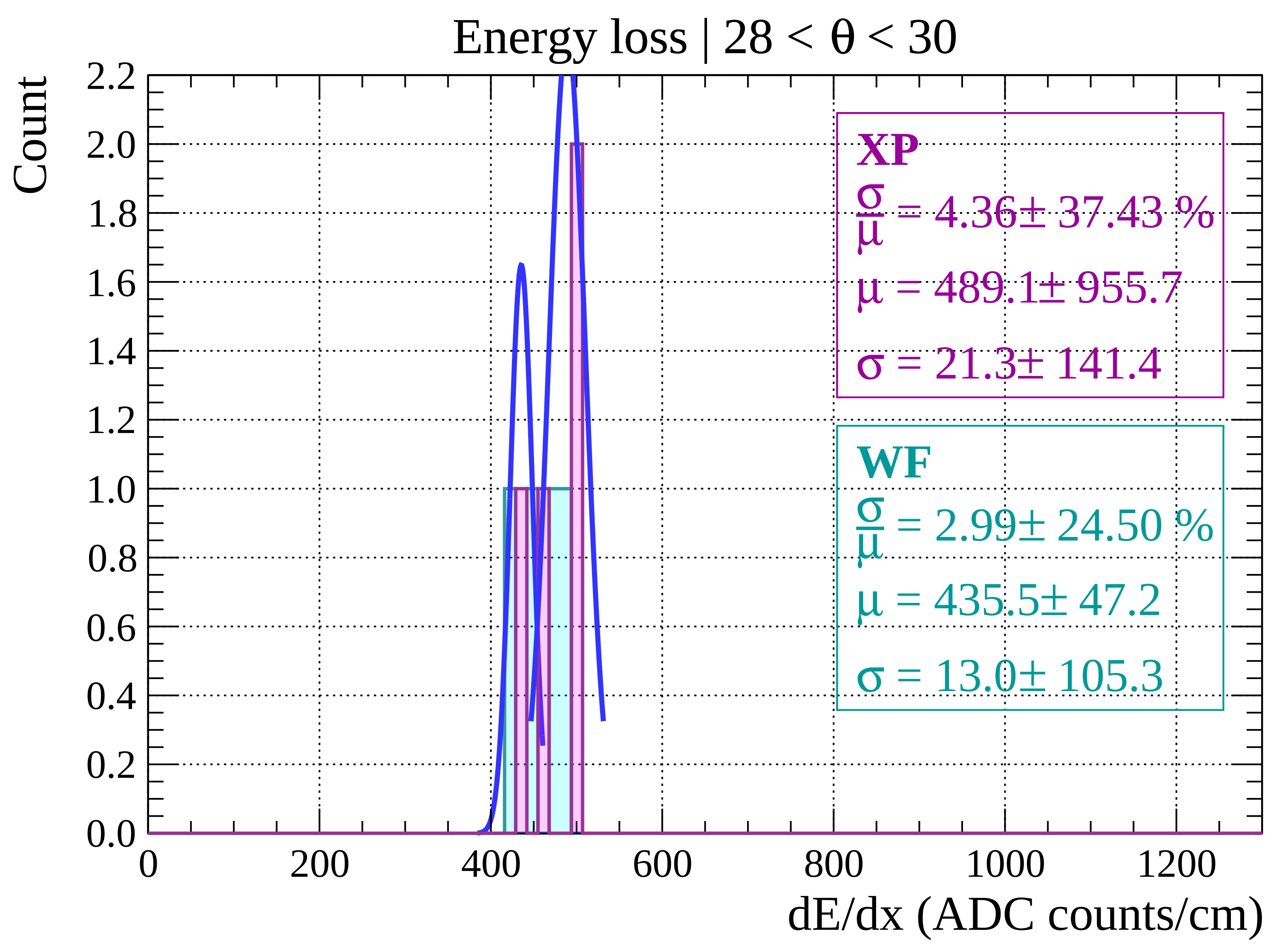
Energy loss | 20 < θ < 22
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
5
10
15
20
25
XP
σ/μ = 8.19± 0.87 %
μ = 475.9± 4.0
σ = 39.0± 3.8
WF
σ/μ = 8.56± 1.02 %
μ = 460.7± 4.0
σ = 39.4± 4.3

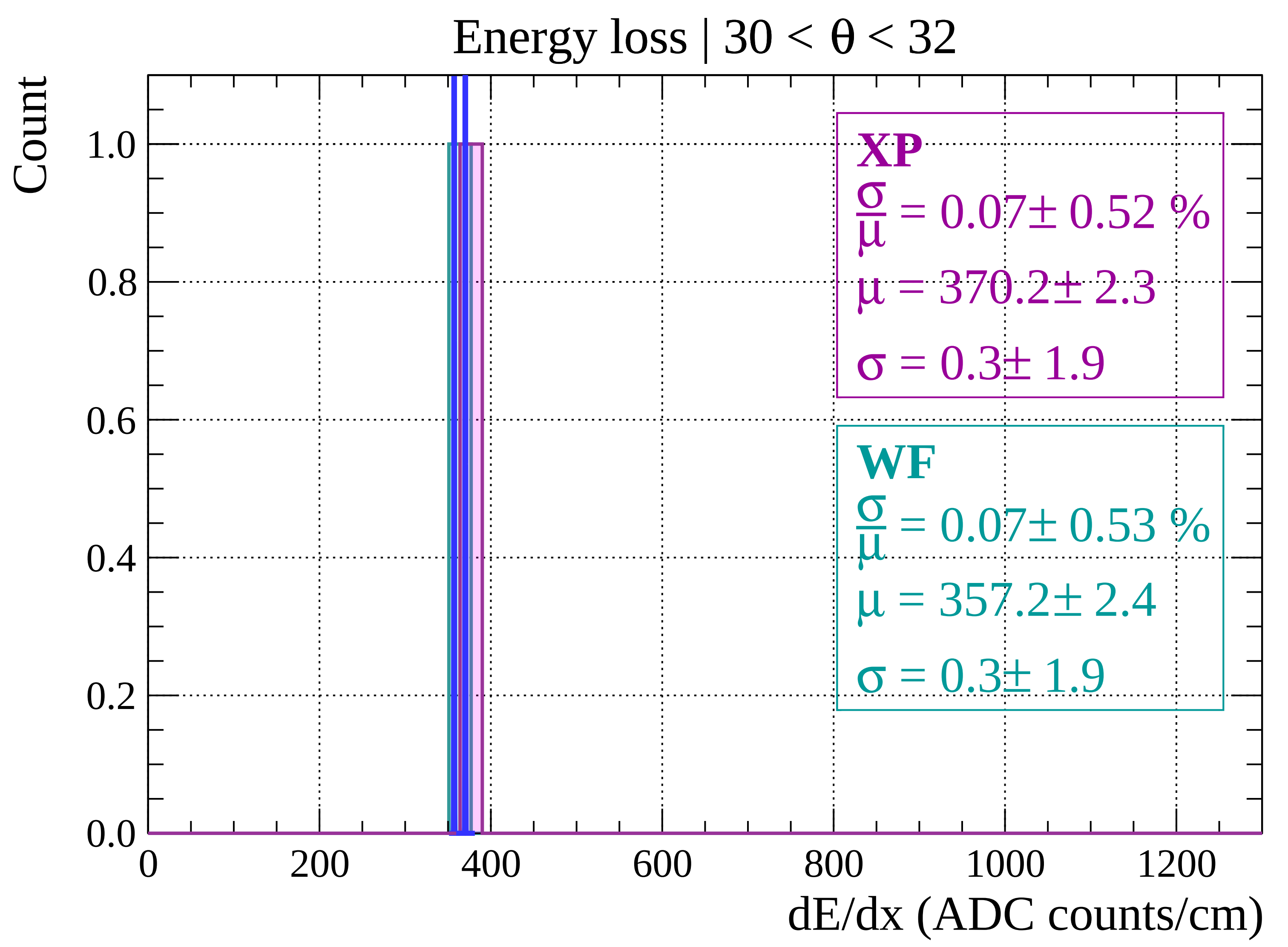

Energy loss | 22 < θ < 24
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 11.34± 3.28 %
μ = 477.0± 12.8
σ = 54.1± 14.2
WF
σ/μ = 7.25± 1.55 %
μ = 470.3± 5.8
σ = 34.1± 6.9
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200

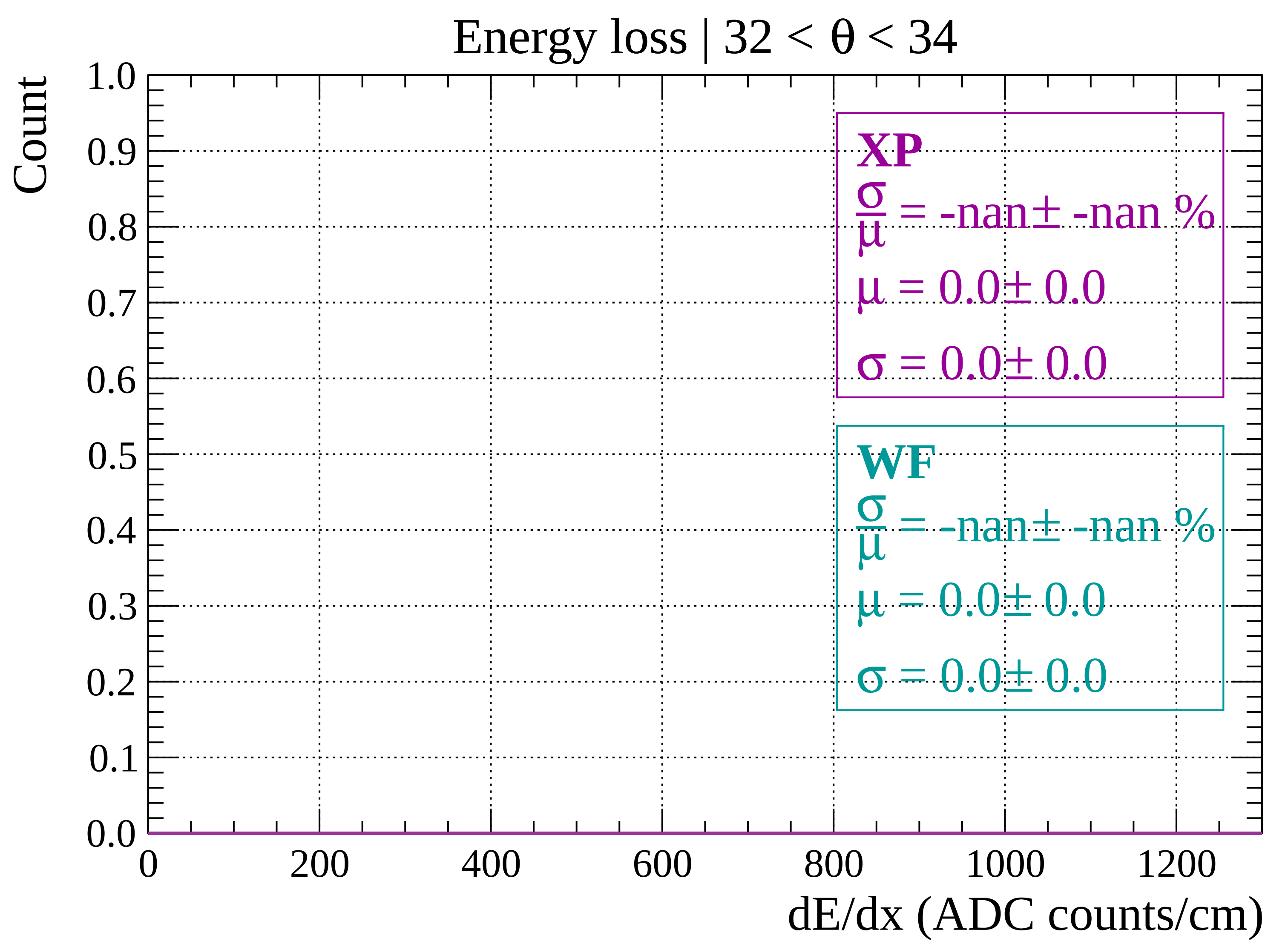


Energy loss | 24 < θ < 26
Count
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 10.60± 7.89 %
μ = 472.4± 20.6
σ = 50.1± 35.1
WF
σ/μ = 14.35± 16.45 %
μ = 465.6± 38.5
σ = 66.8± 71.1

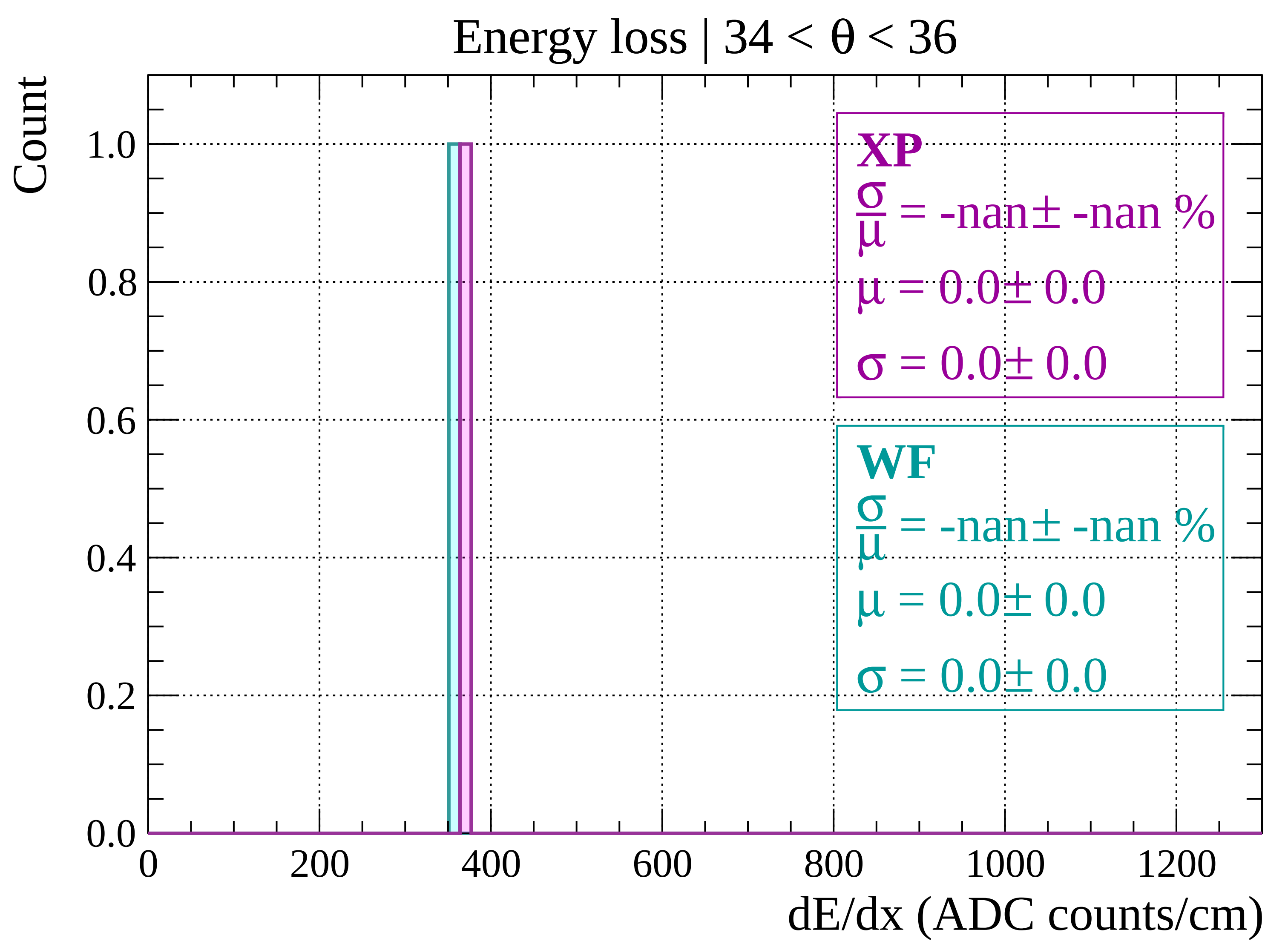

Energy loss | 26 < θ < 28
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 0.09± 0.46 %
μ = 292.2± 1.1
σ = 0.3± 1.3
WF
σ/μ = 6.84± 3.32 %
μ = 456.6± 14.0
σ = 31.2± 14.2
0.0
0.5
1.0
1.5
2.0
2.5
3.0
0
200
400
600
800
1000
1200


Energy loss | 28 < θ < 30
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 4.36± 37.43 %
μ = 489.1± 955.7
σ = 21.3± 141.4
WF
$\frac{\sigma}{\mu}$ = 2.99± 24.50 %
μ = 435.5± 47.2
σ = 13.0± 105.3
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
2.2
0
200
400
600
800
1000
1200

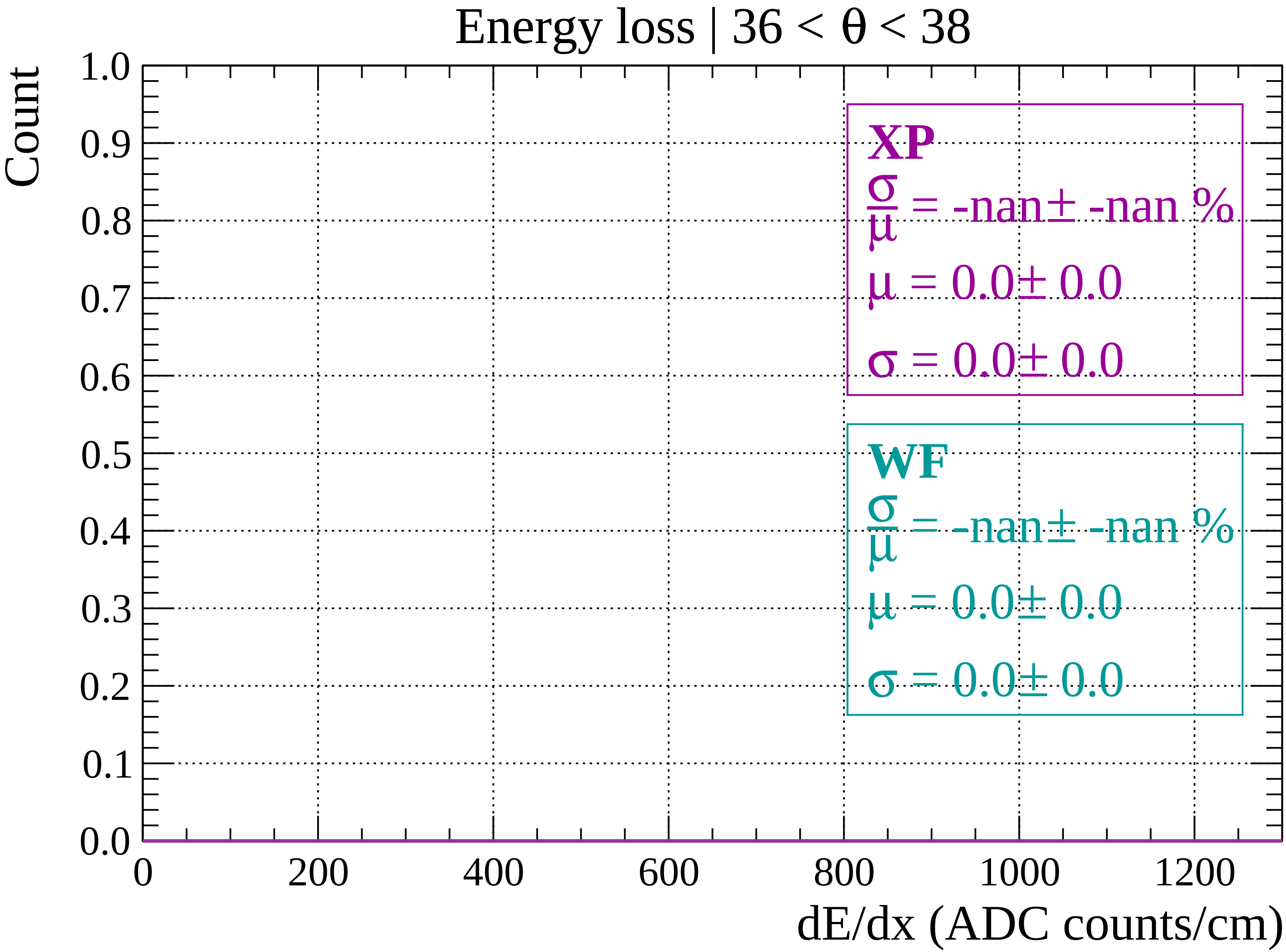

Energy loss | 30 < θ < 32
Count
dE/dx (ADC counts/cm)
0.0
0.2
0.4
0.6
0.8
1.0
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 0.07± 0.52 %
μ = 370.2± 2.3
σ = 0.3± 1.9
WF
$\frac{\sigma}{\mu}$ = 0.07± 0.53 %
μ = 357.2± 2.4
σ = 0.3± 1.9



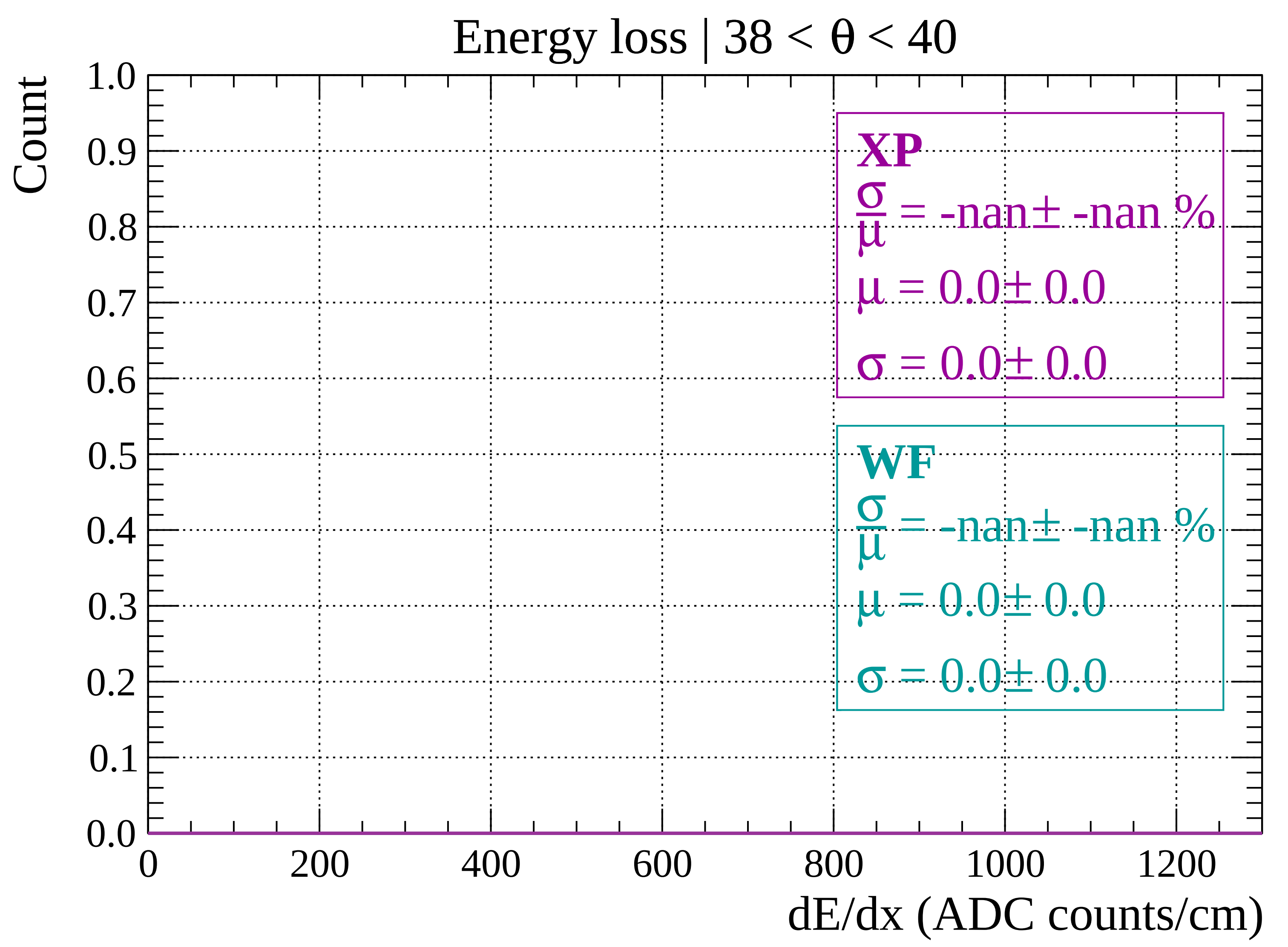
Energy loss | 32 < θ < 34
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



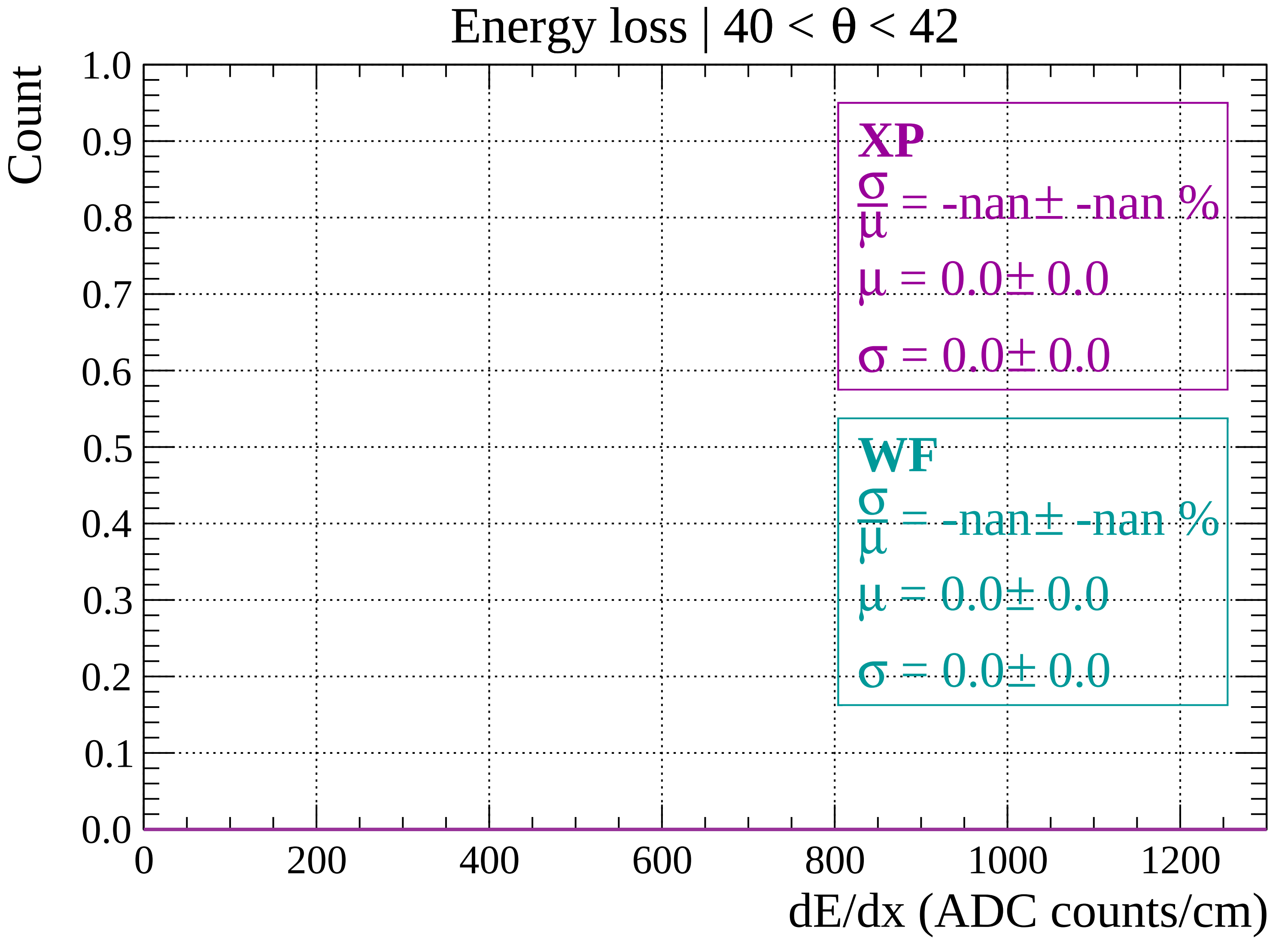
Energy loss | 34 < θ < 36
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



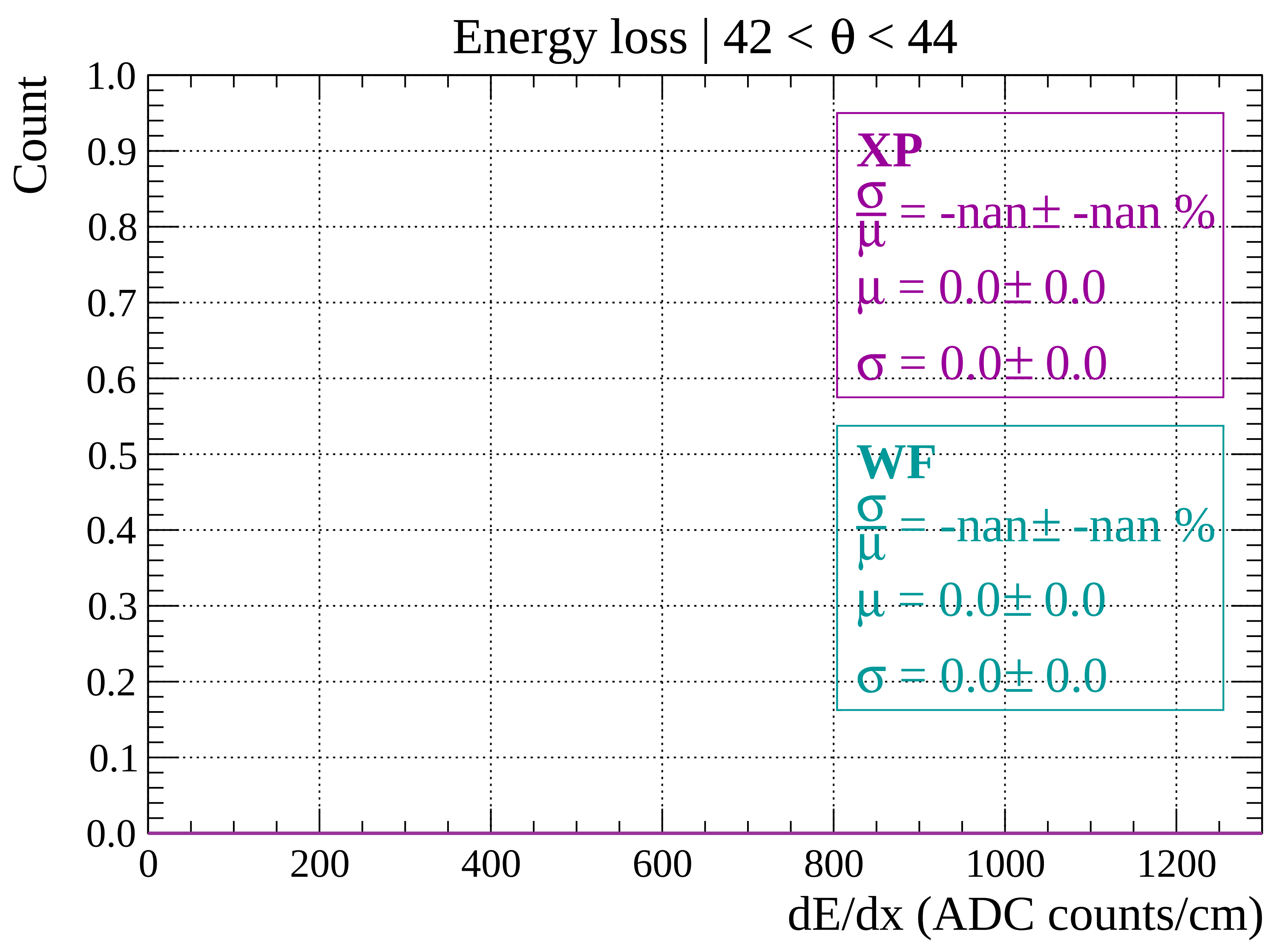
Energy loss | 36 < θ < 38
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

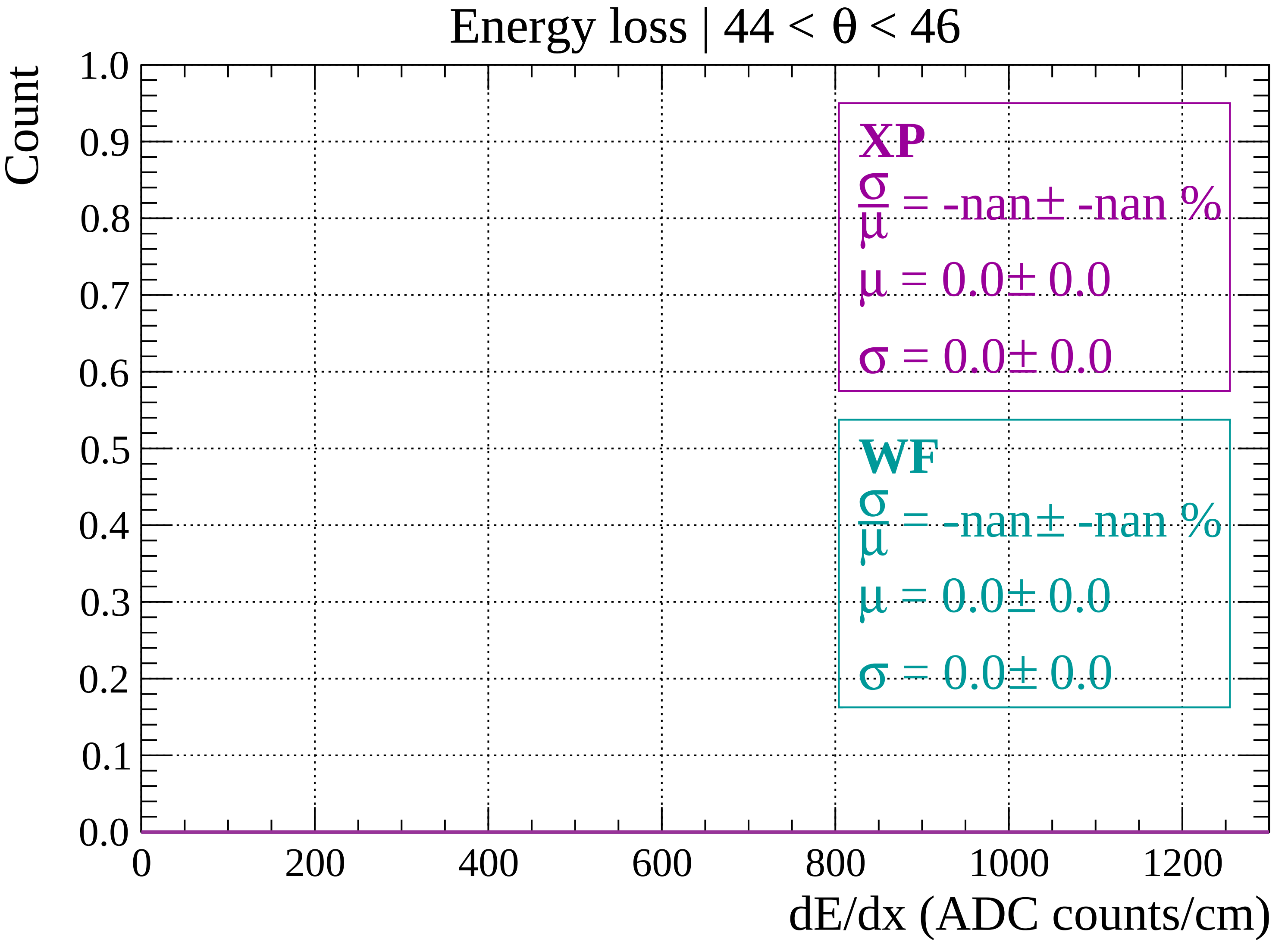


Energy loss | 38 < θ < 40
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

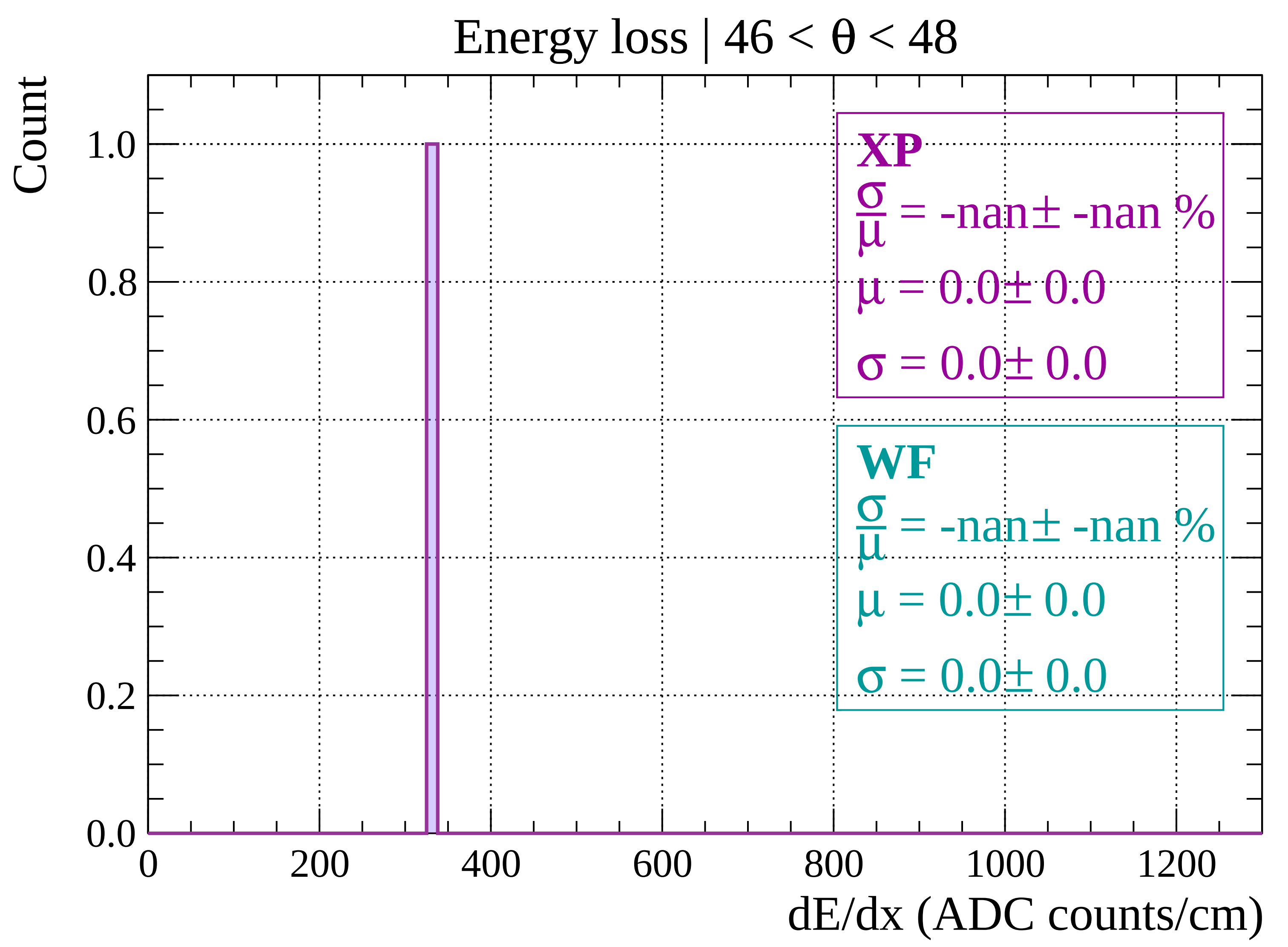


Energy loss | 40 < θ < 42
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

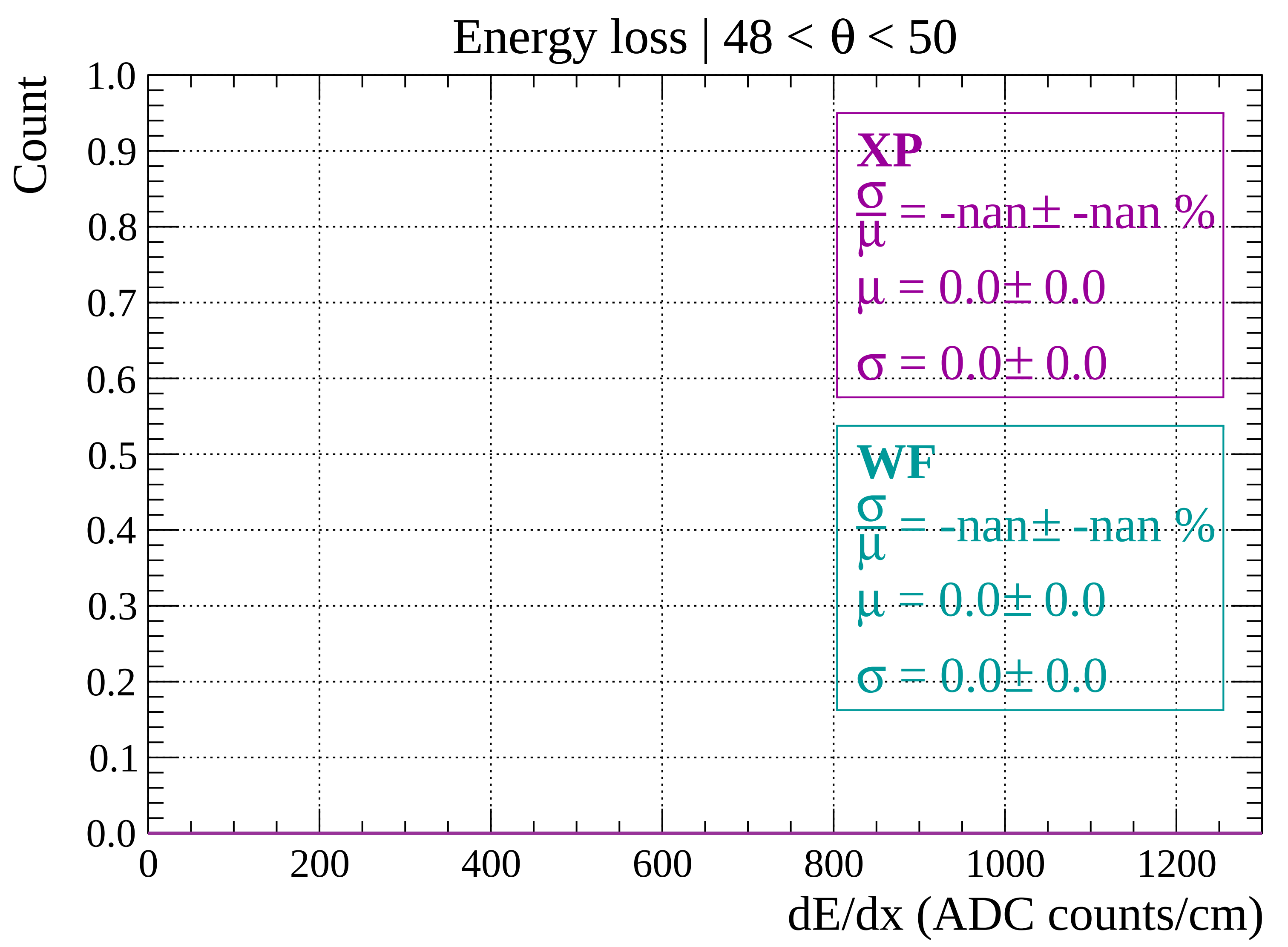

Energy loss | 42 < θ < 44
Count
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200



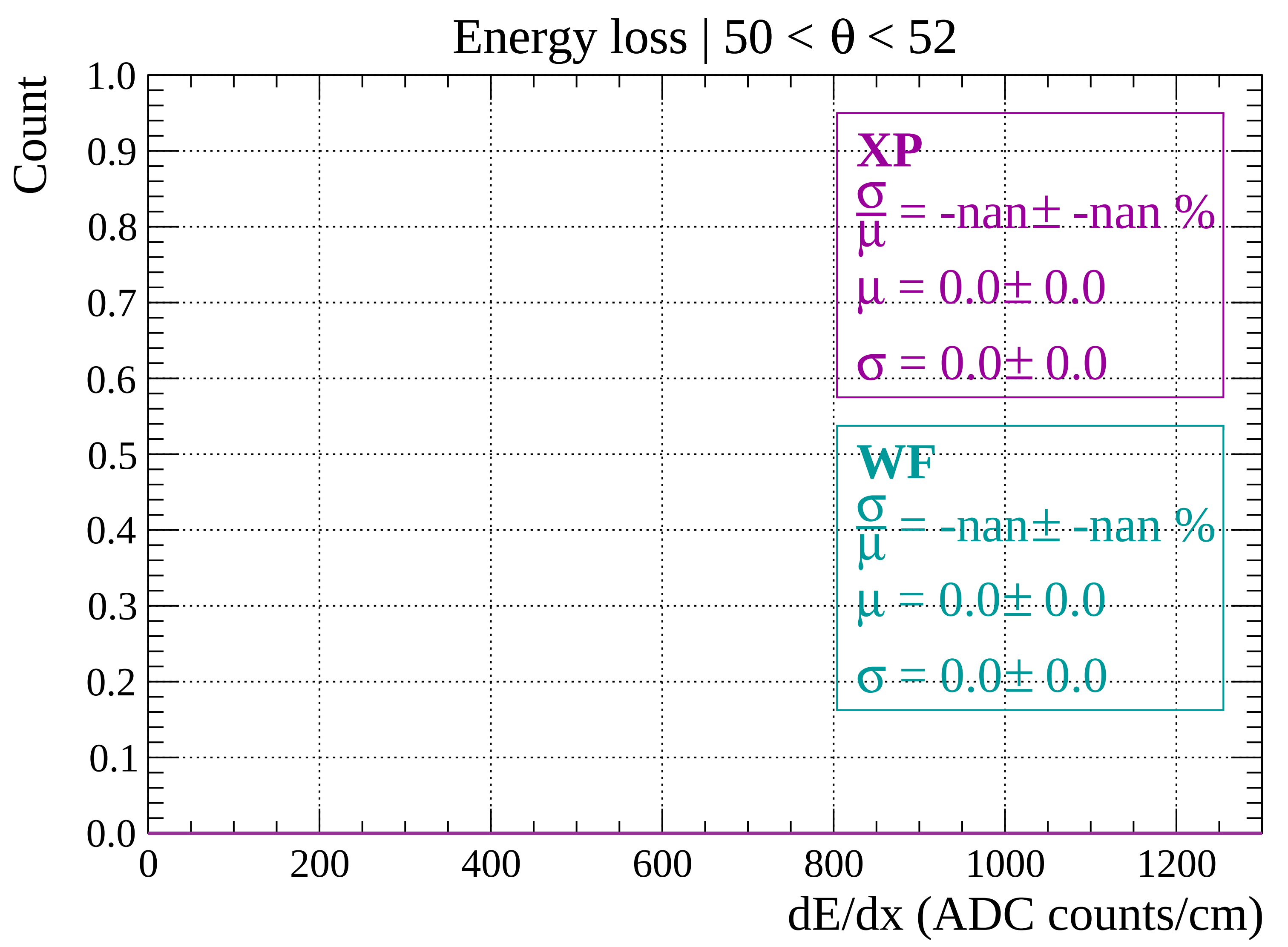
Energy loss | 44 < θ < 46
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



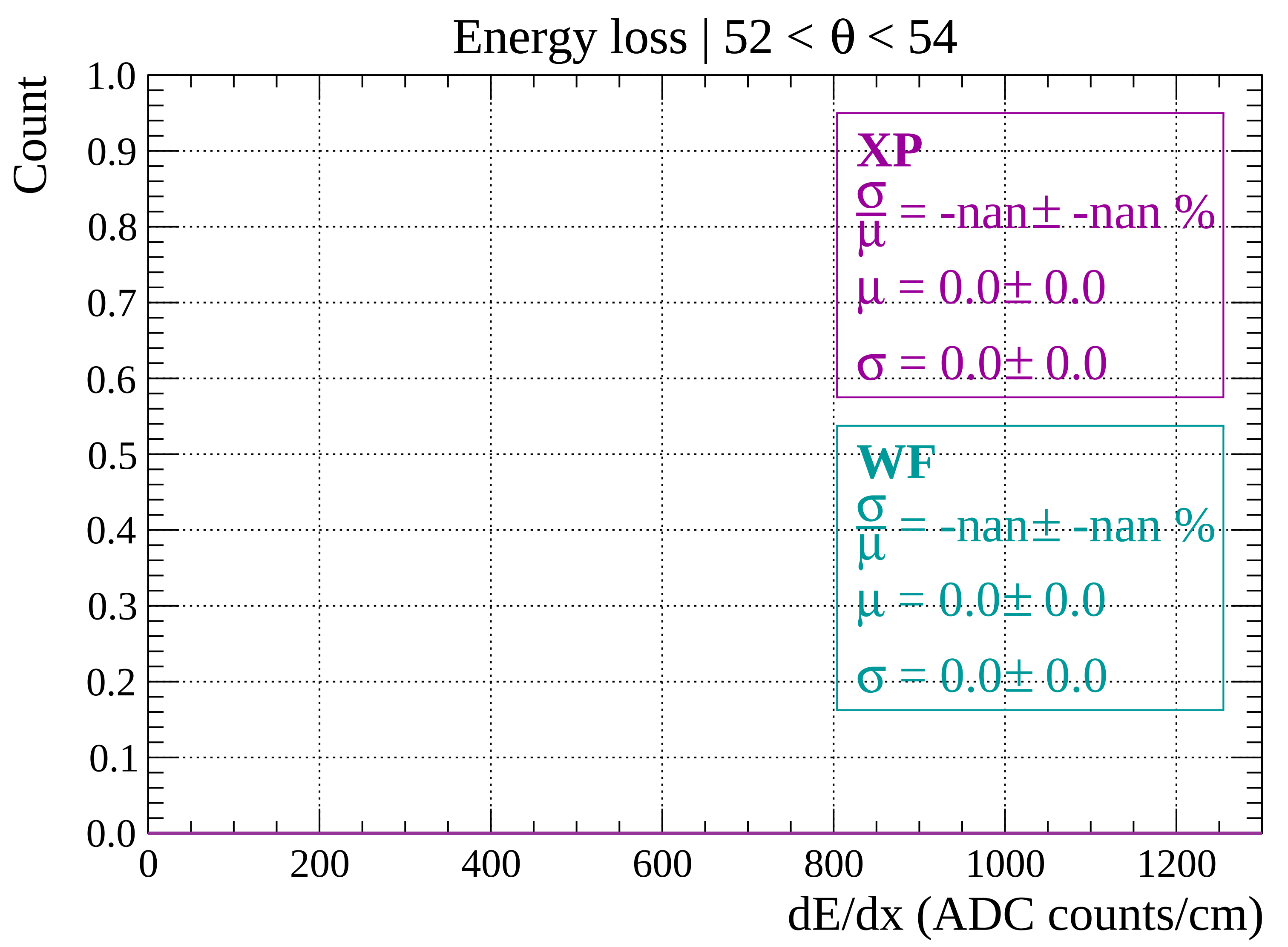
Energy loss | 46 < θ < 48
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



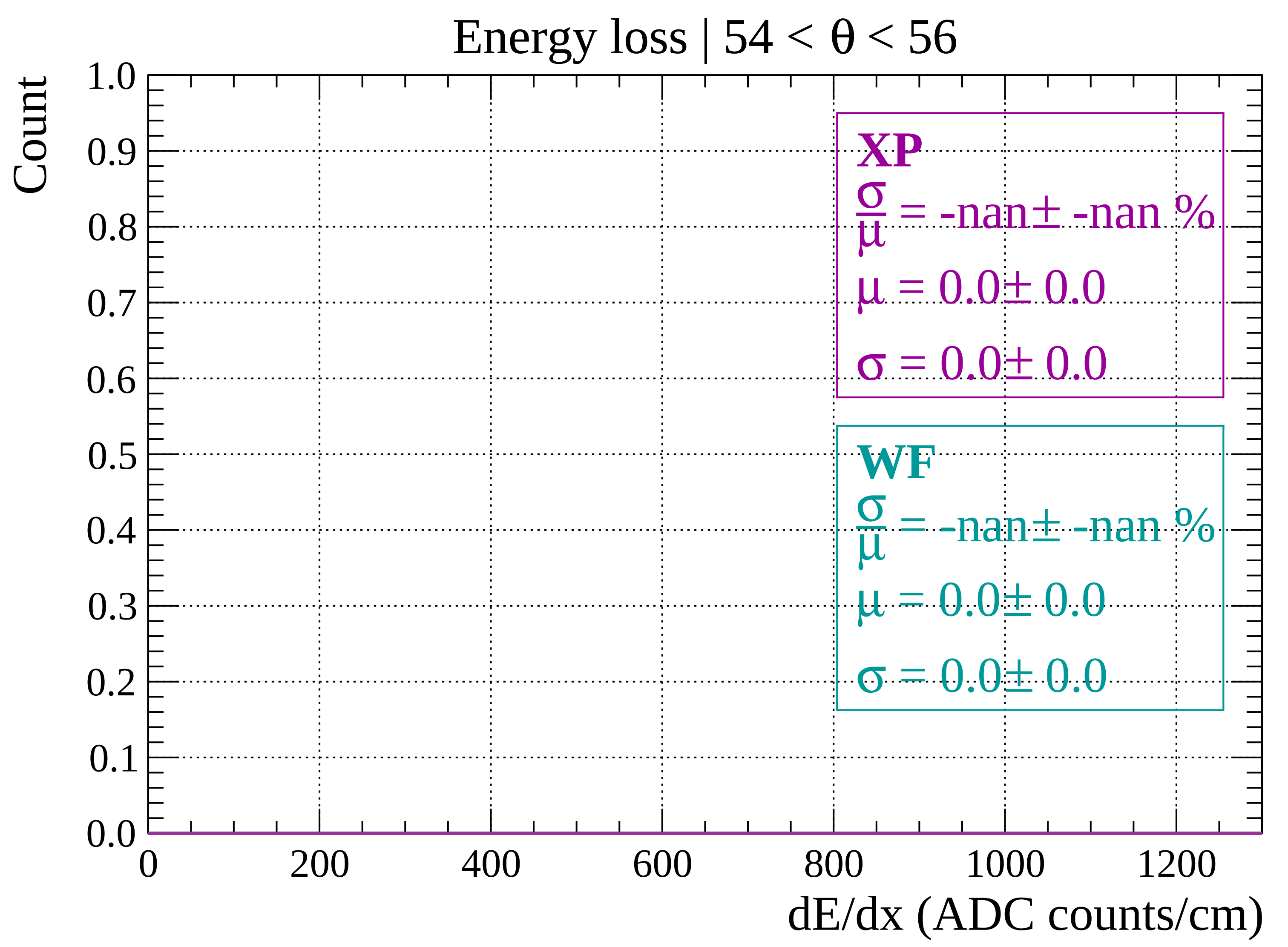
Energy loss | 48 < θ < 50
Count
dE/dx (ADC counts/cm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

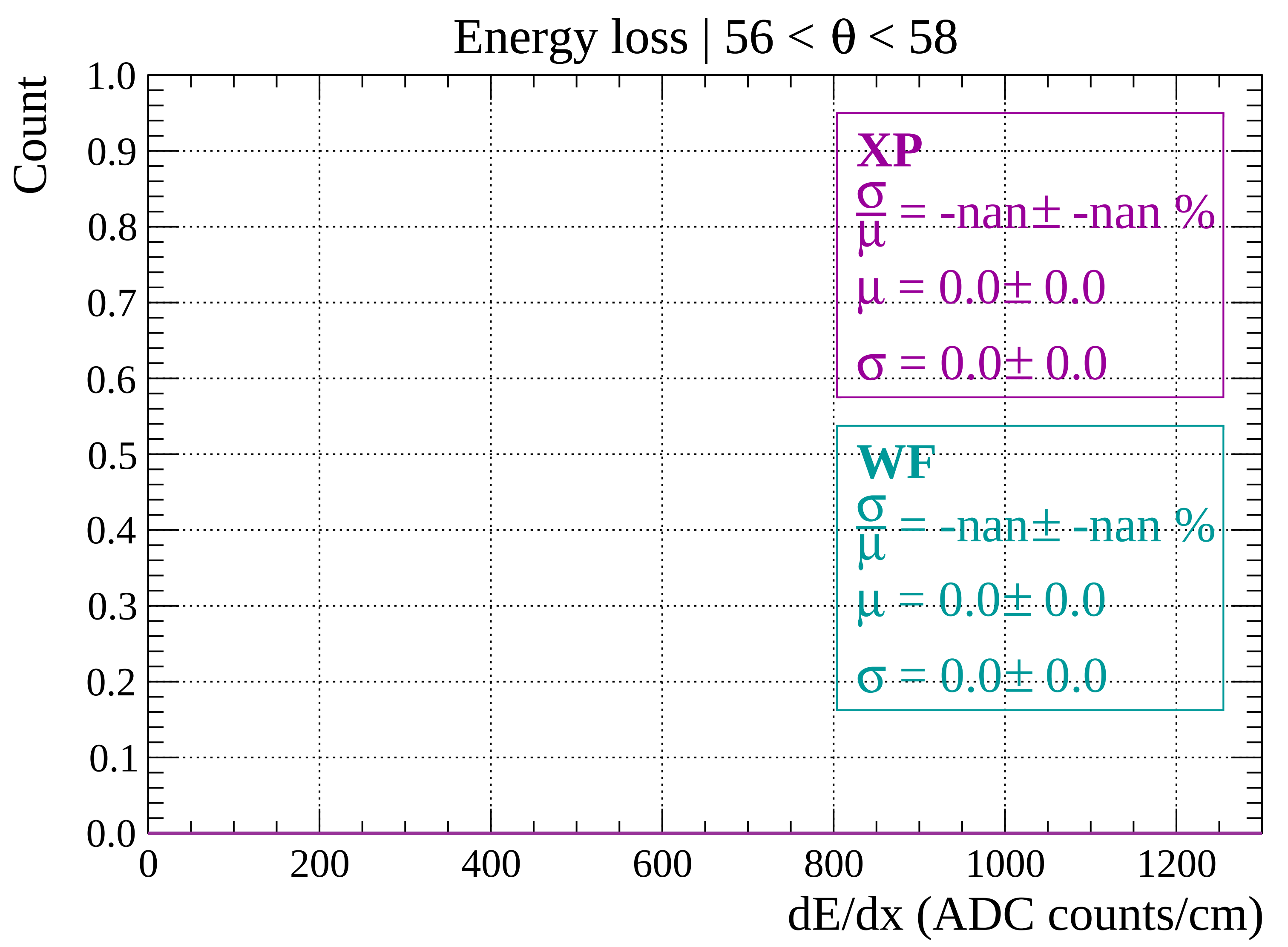


Energy loss | 50 < θ < 52
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

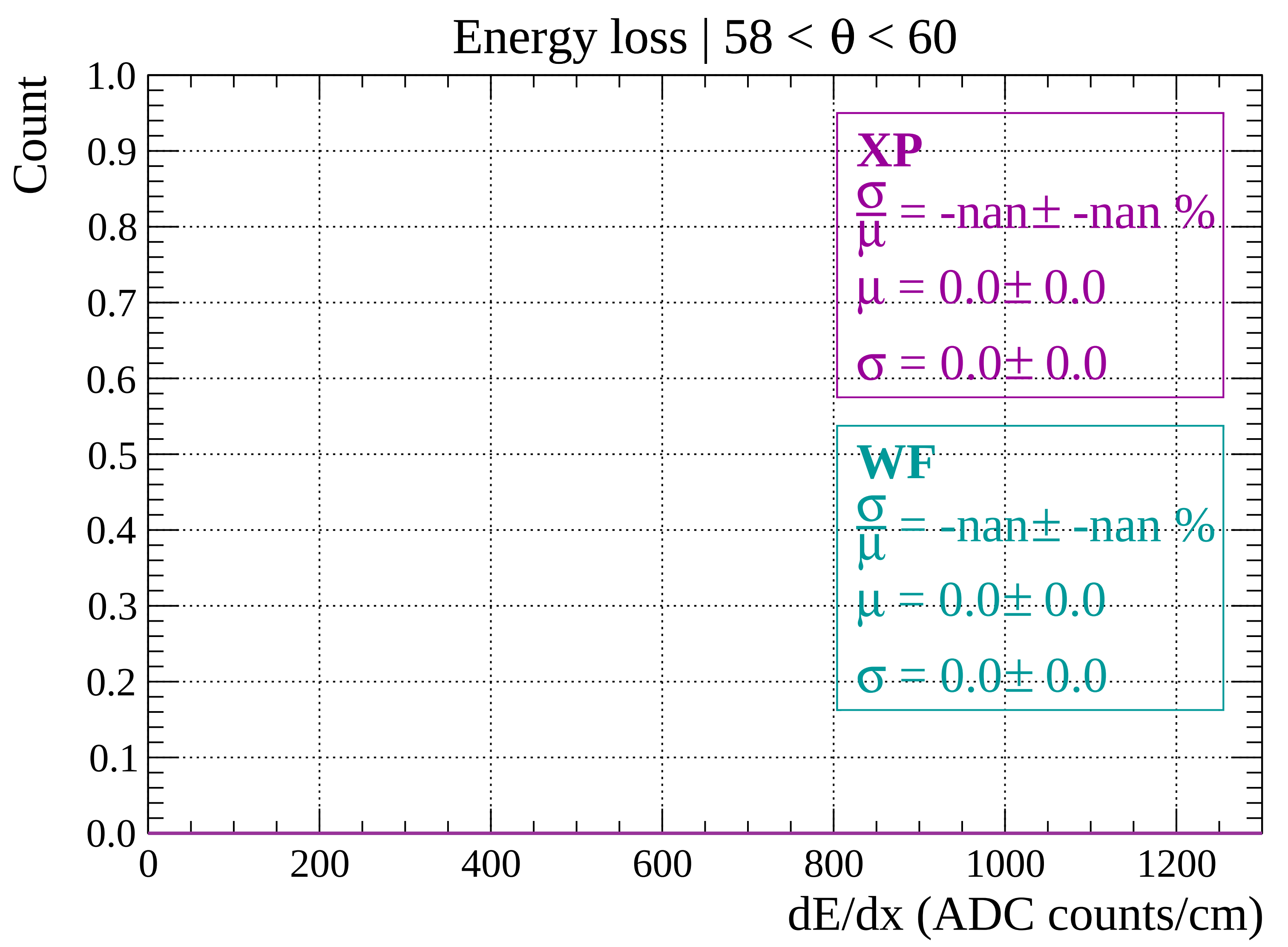


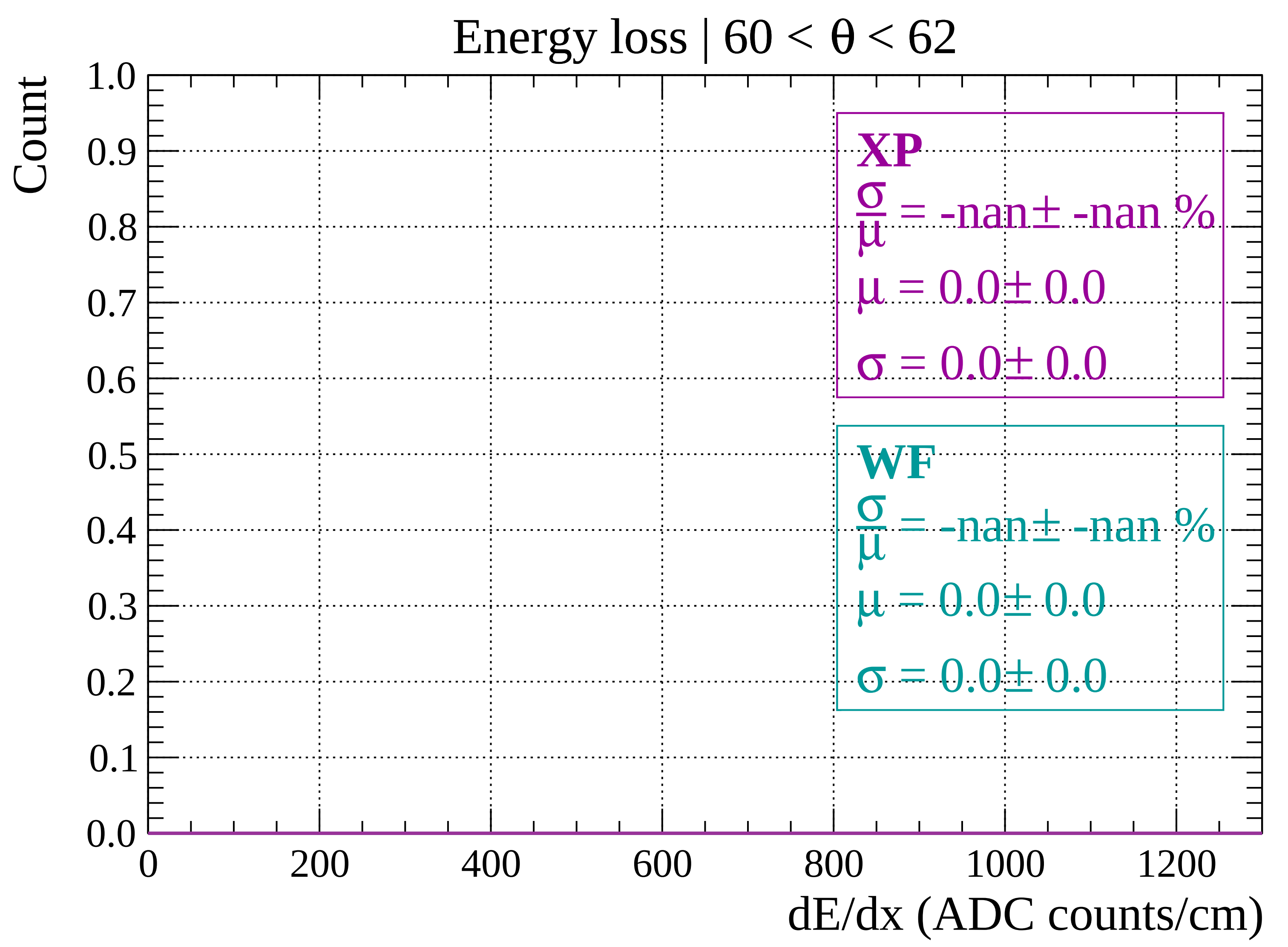
Energy loss | 52 < θ < 54
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0


Energy loss | 54 < θ < 56
Count
dE/dx (ADC counts/cm)
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

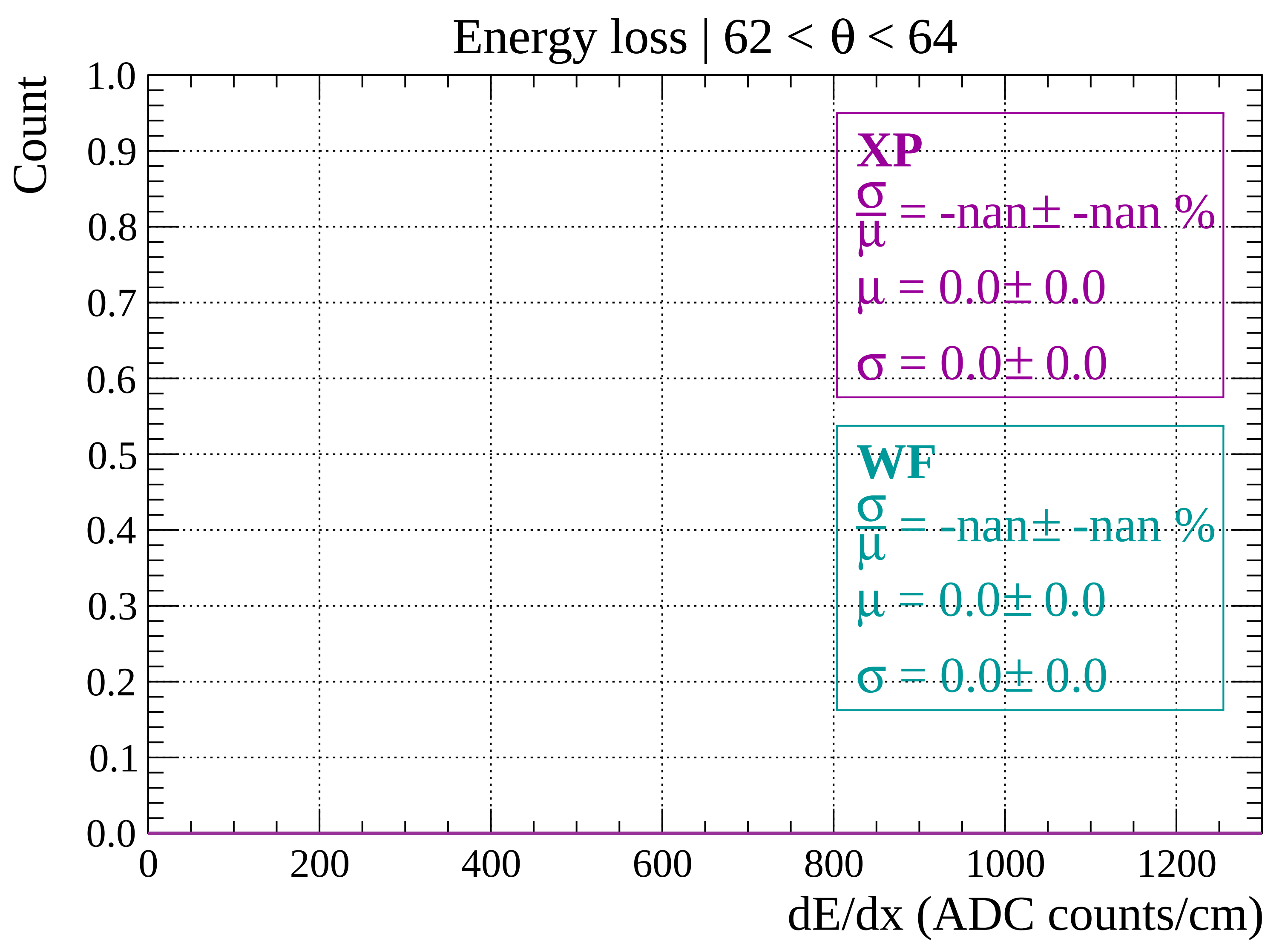


Energy loss | 56 < θ < 58
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

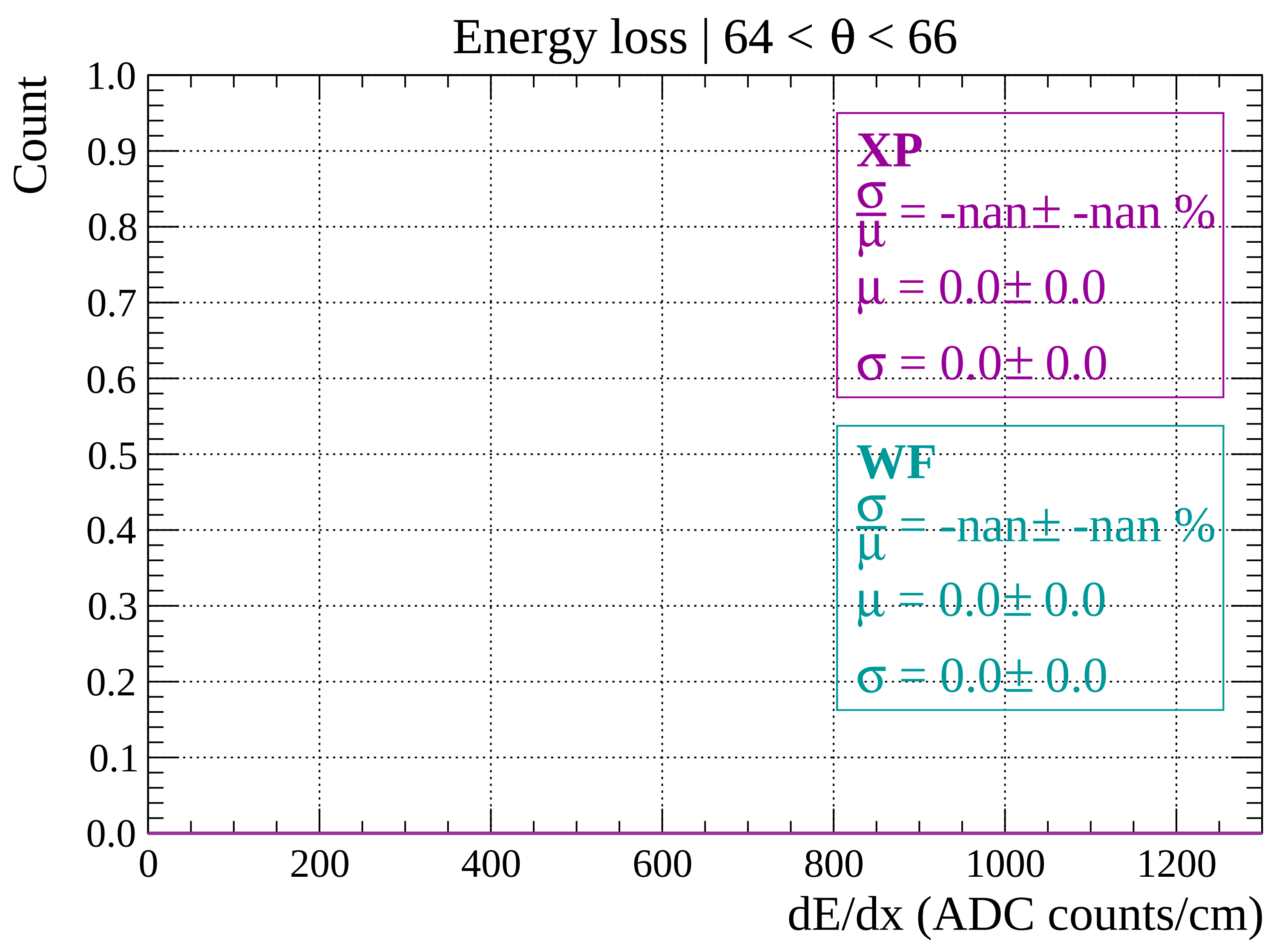

Energy loss | 58 < θ < 60
Count
dE/dx (ADC counts/cm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



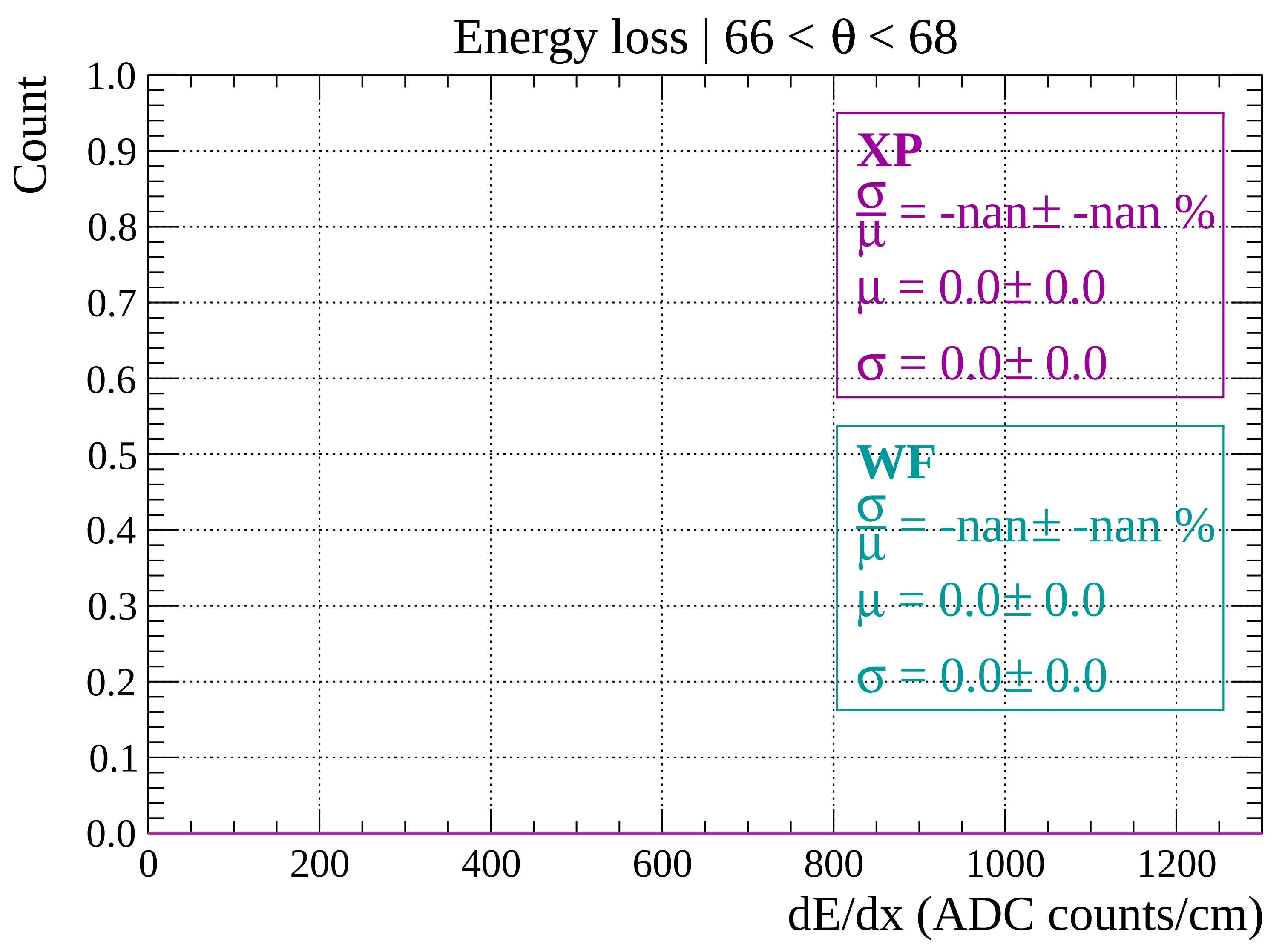
Energy loss | 60 < θ < 62
Count
dE/dx (ADC counts/cm)
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

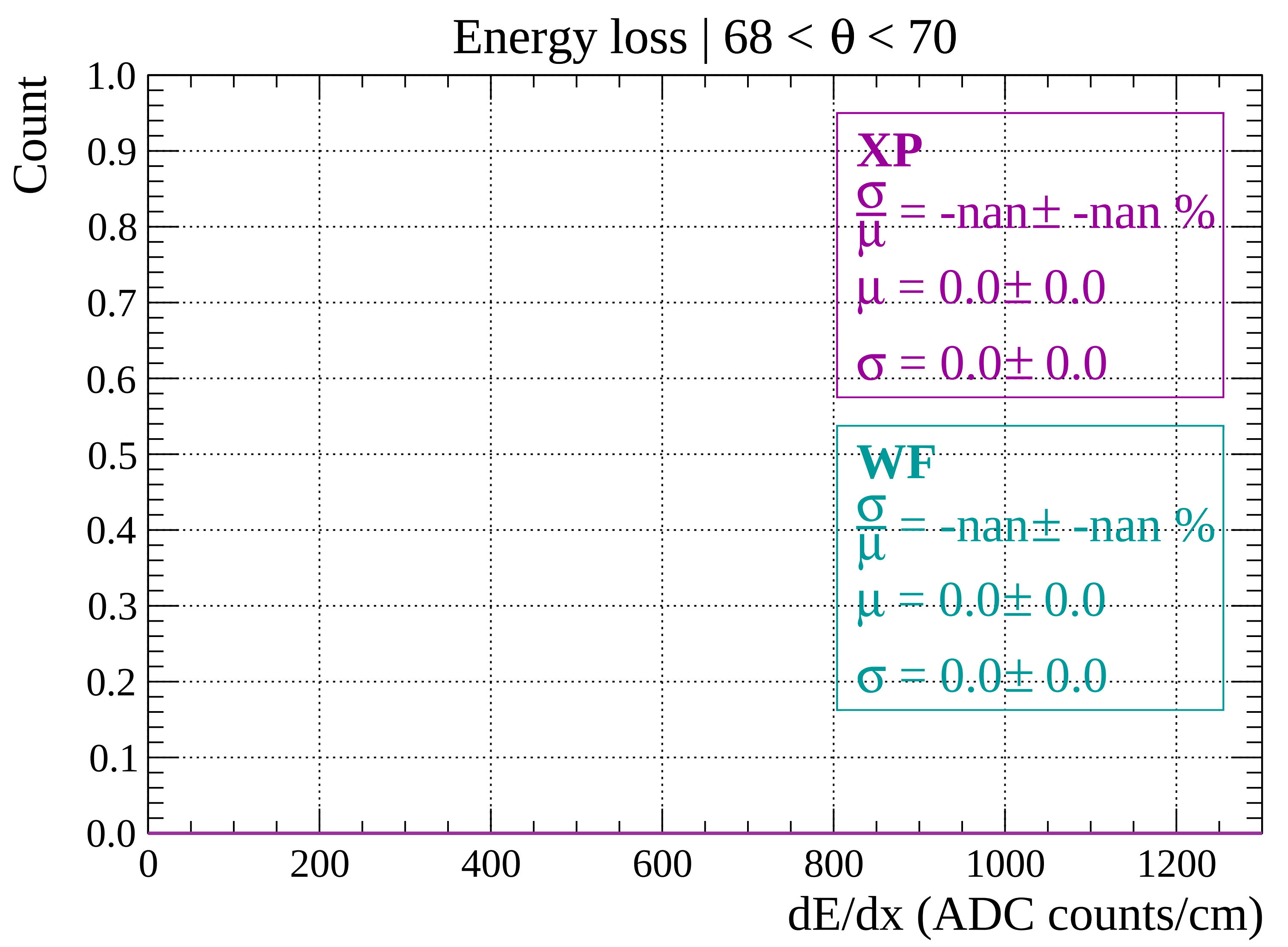


Energy loss | 62 < θ < 64
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

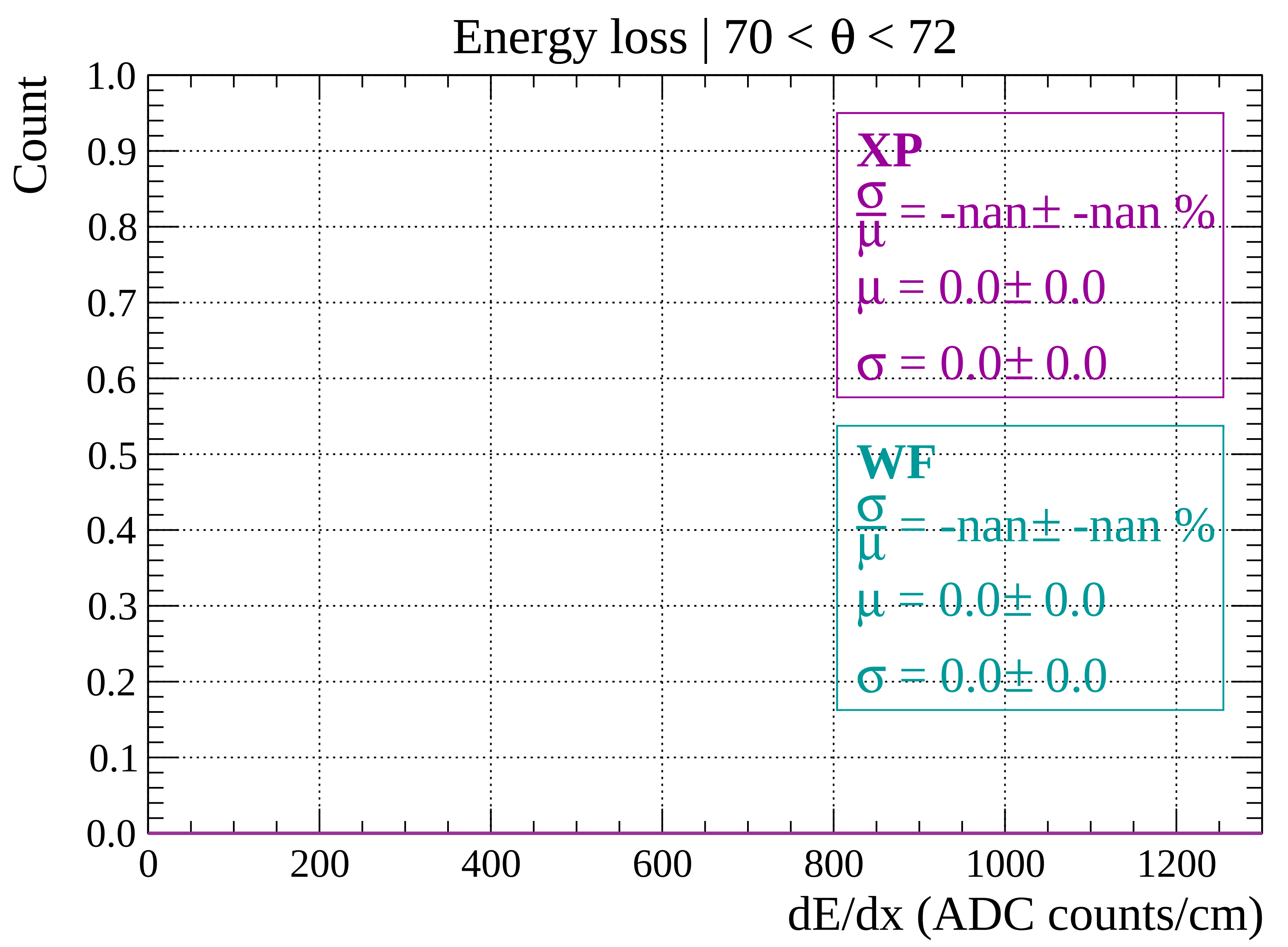


Energy loss | 64 < θ < 66
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

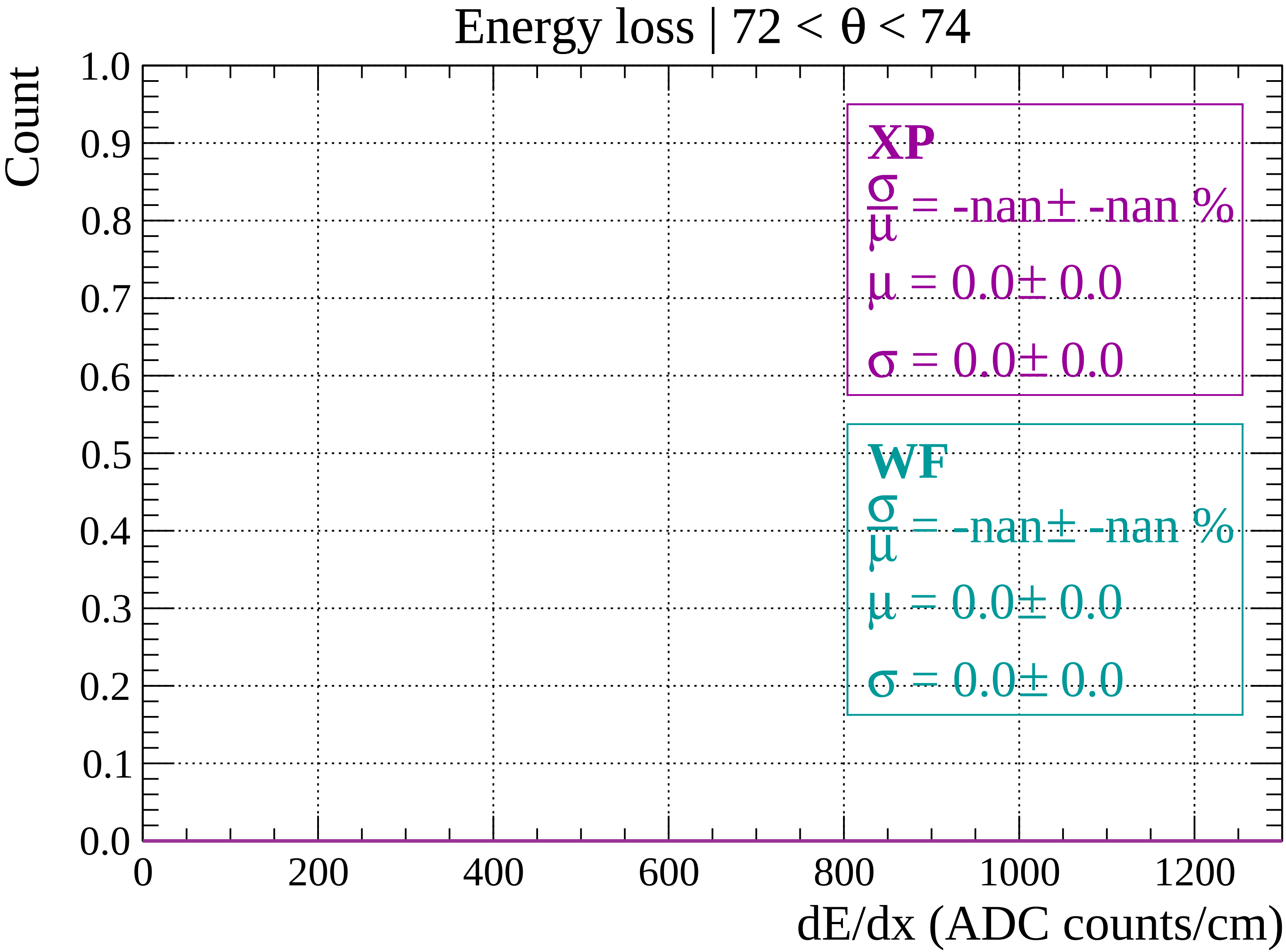

Energy loss | 66 < θ < 68
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



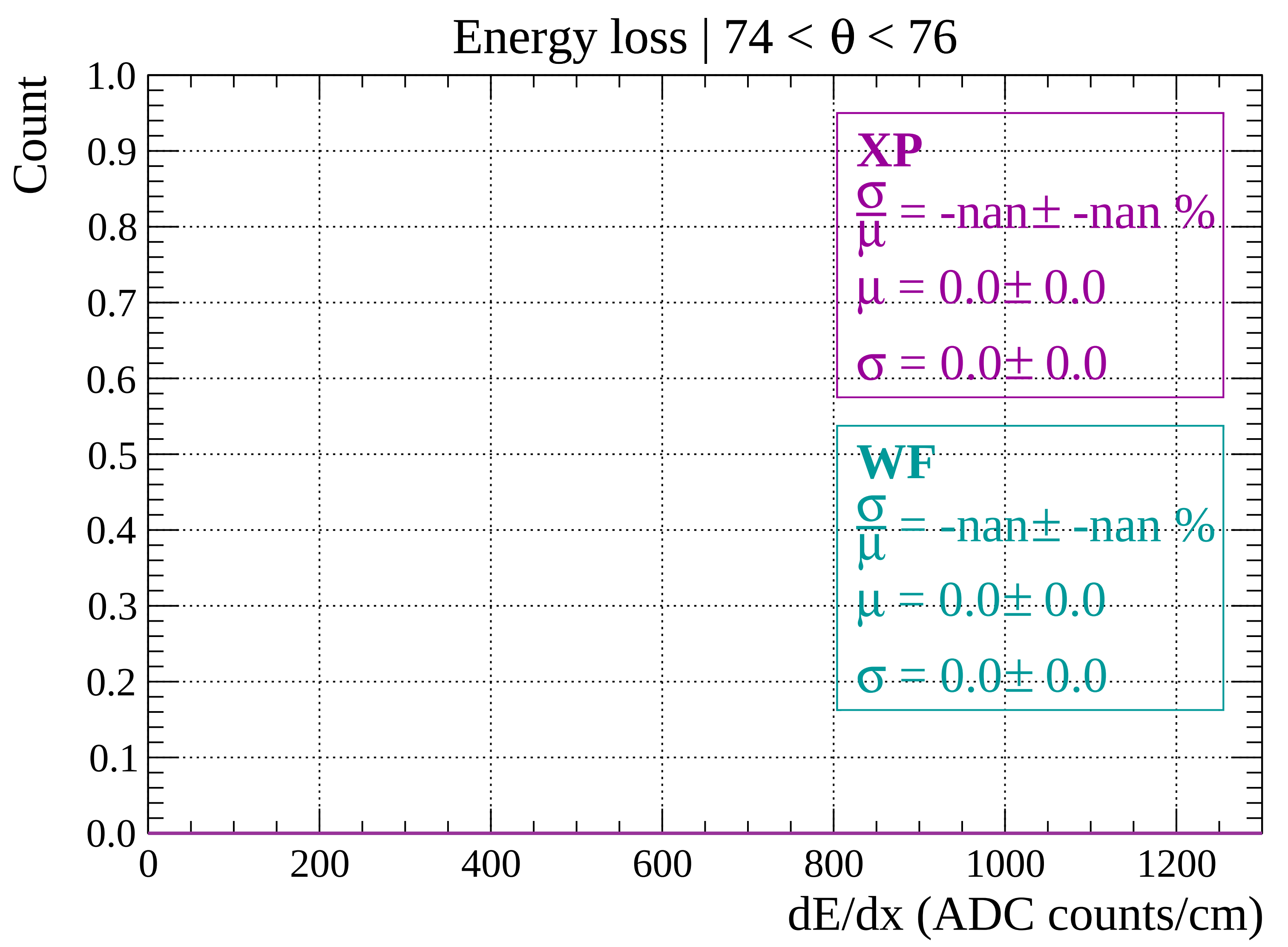
Energy loss | 68 < θ < 70
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

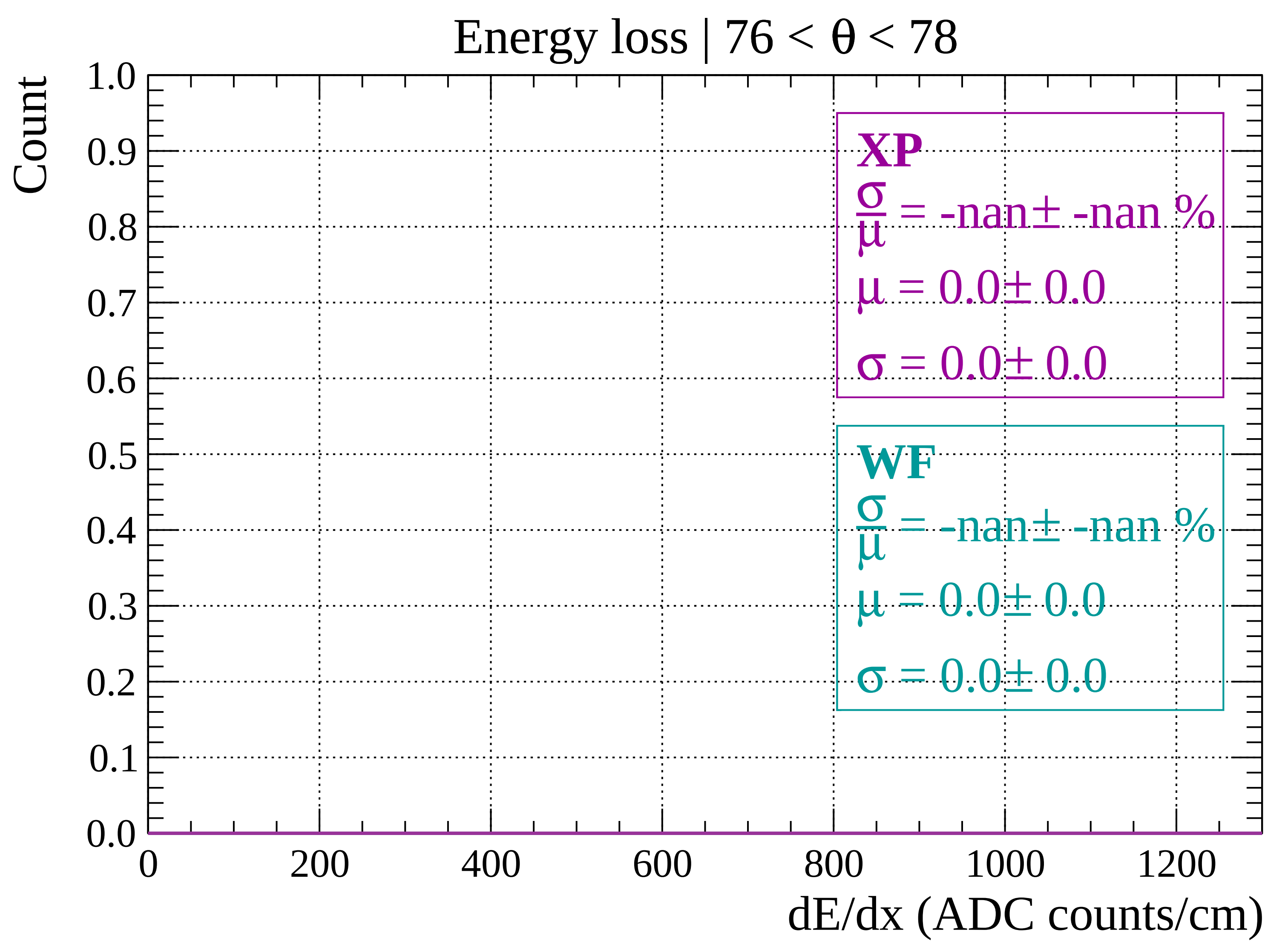


Energy loss | 70 < θ < 72
Count
dE/dx (ADC counts/cm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

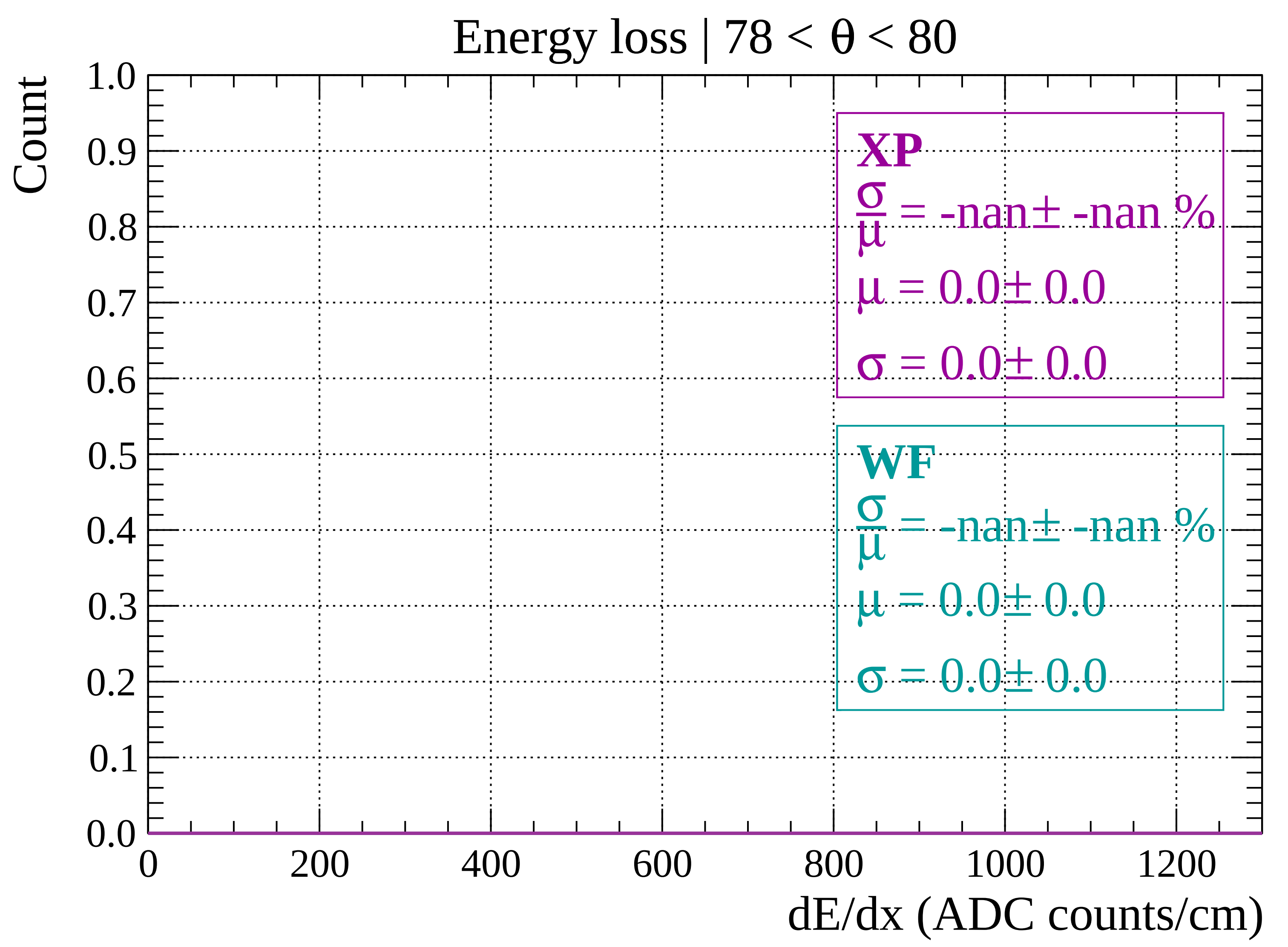


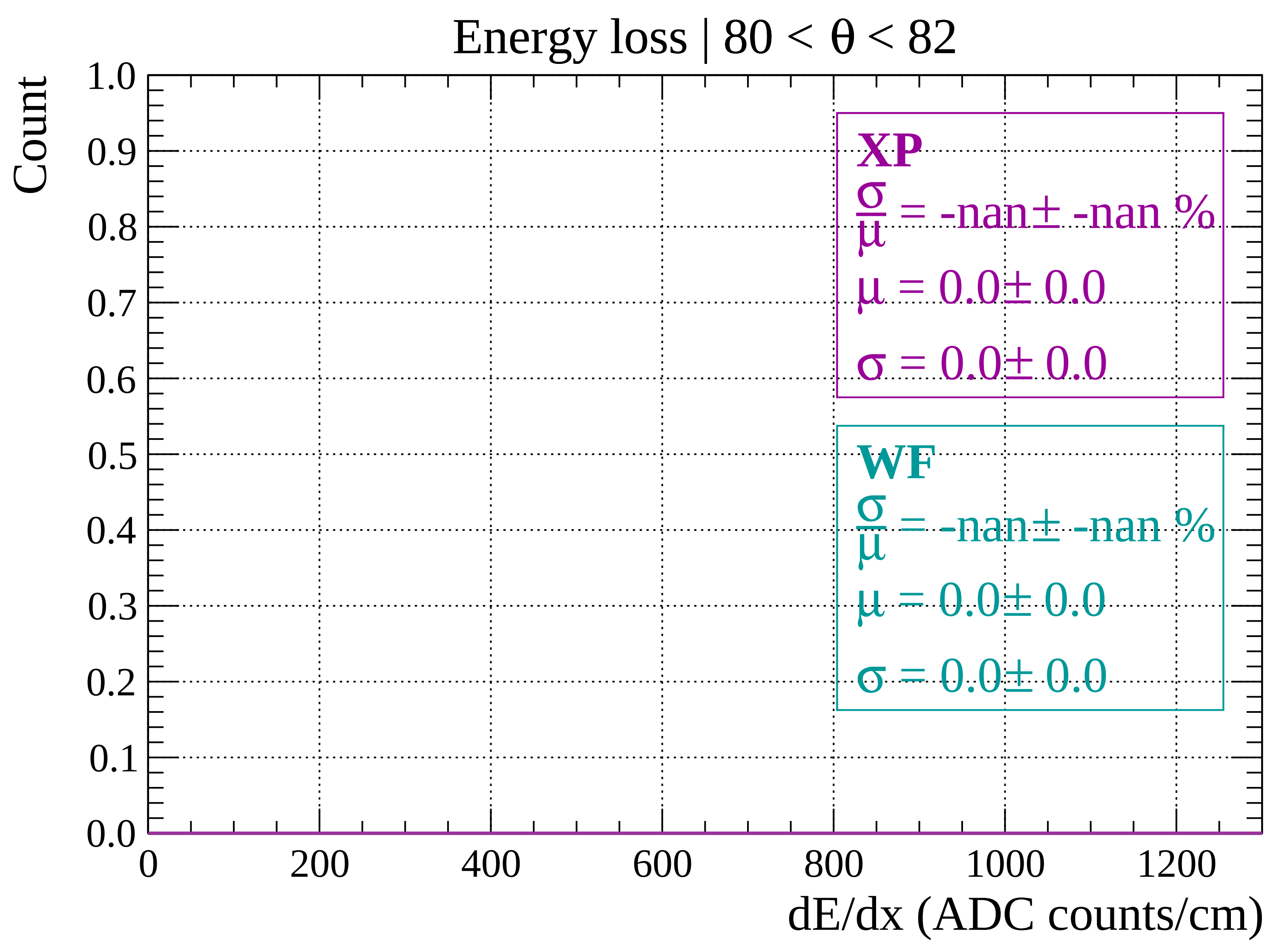
Energy loss | 72 < θ < 74
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

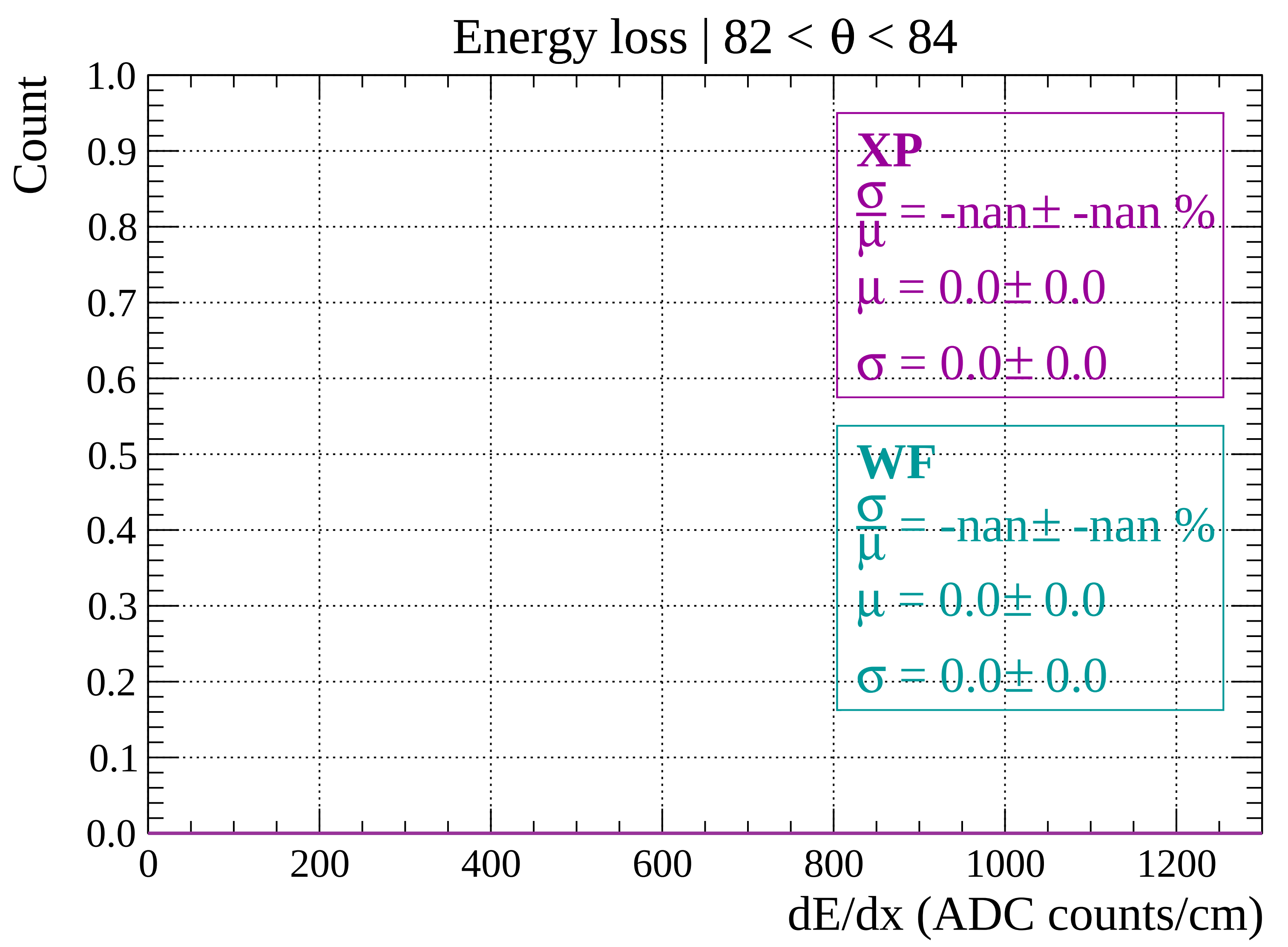

Energy loss | 74 < θ < 76
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



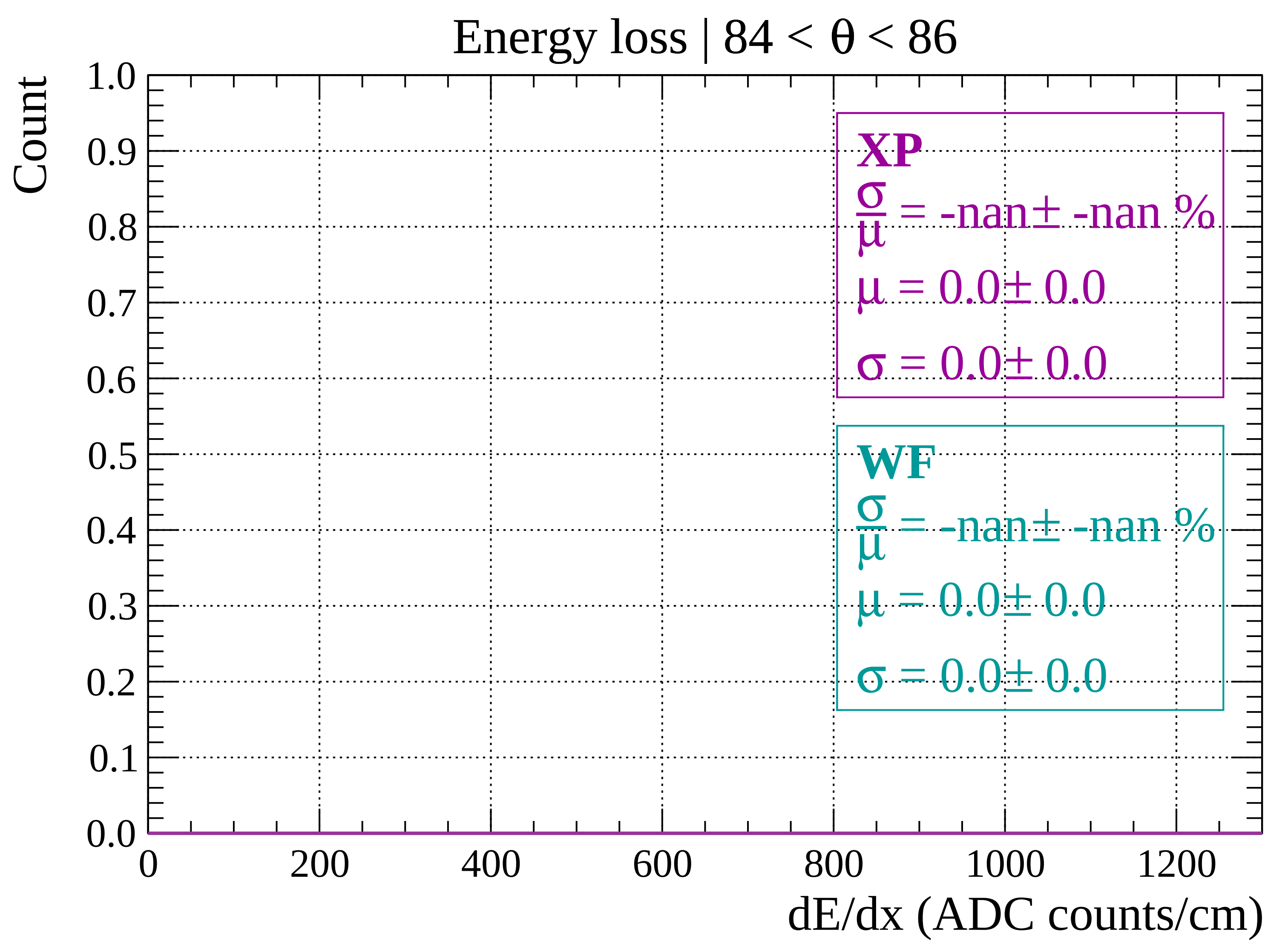
Energy loss | 76 < θ < 78
Count
dE/dx (ADC counts/cm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0


Energy loss | 78 < θ < 80
Count
dE/dx (ADC counts/cm)
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

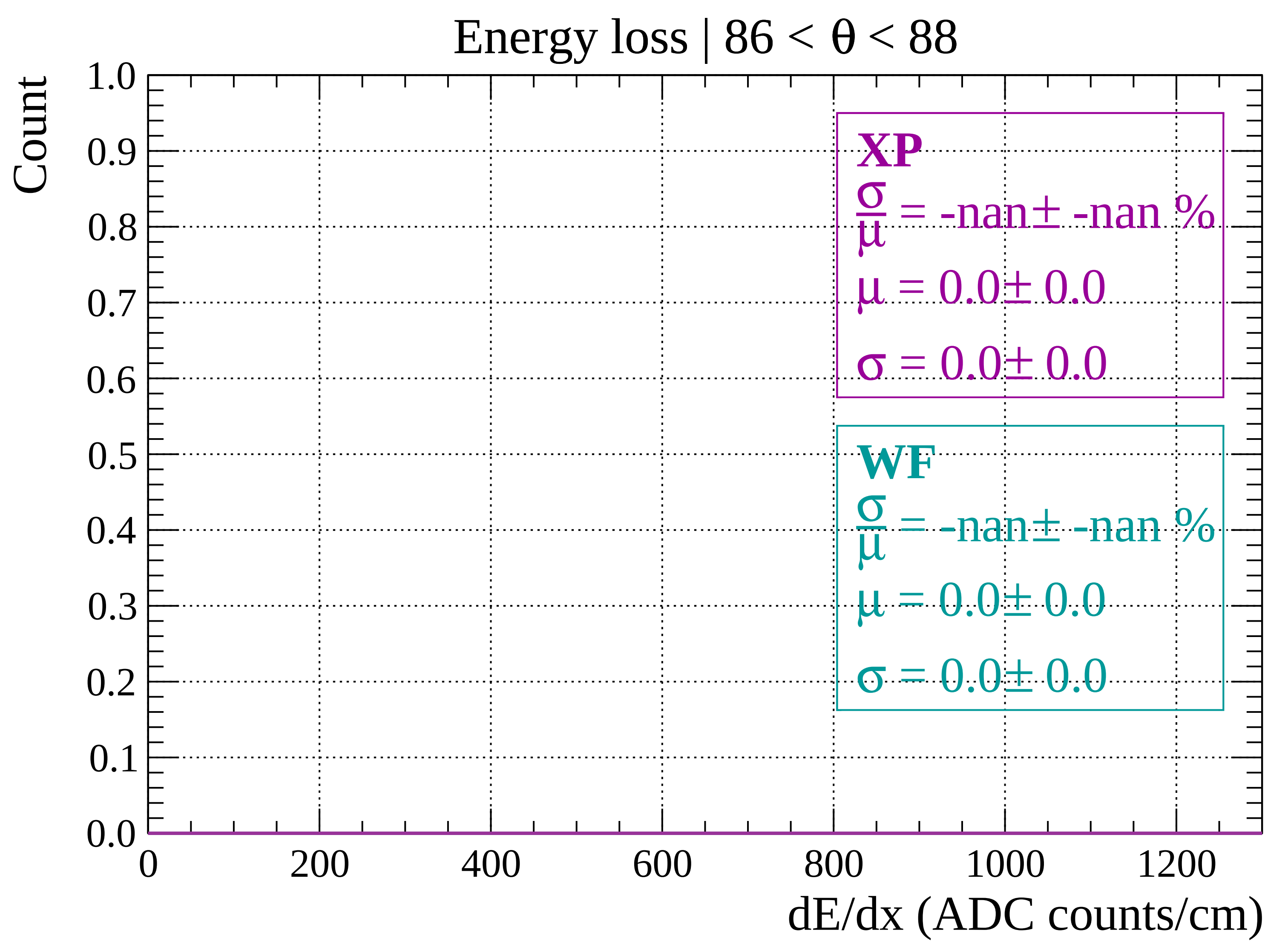

Energy loss | 80 < θ < 82
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



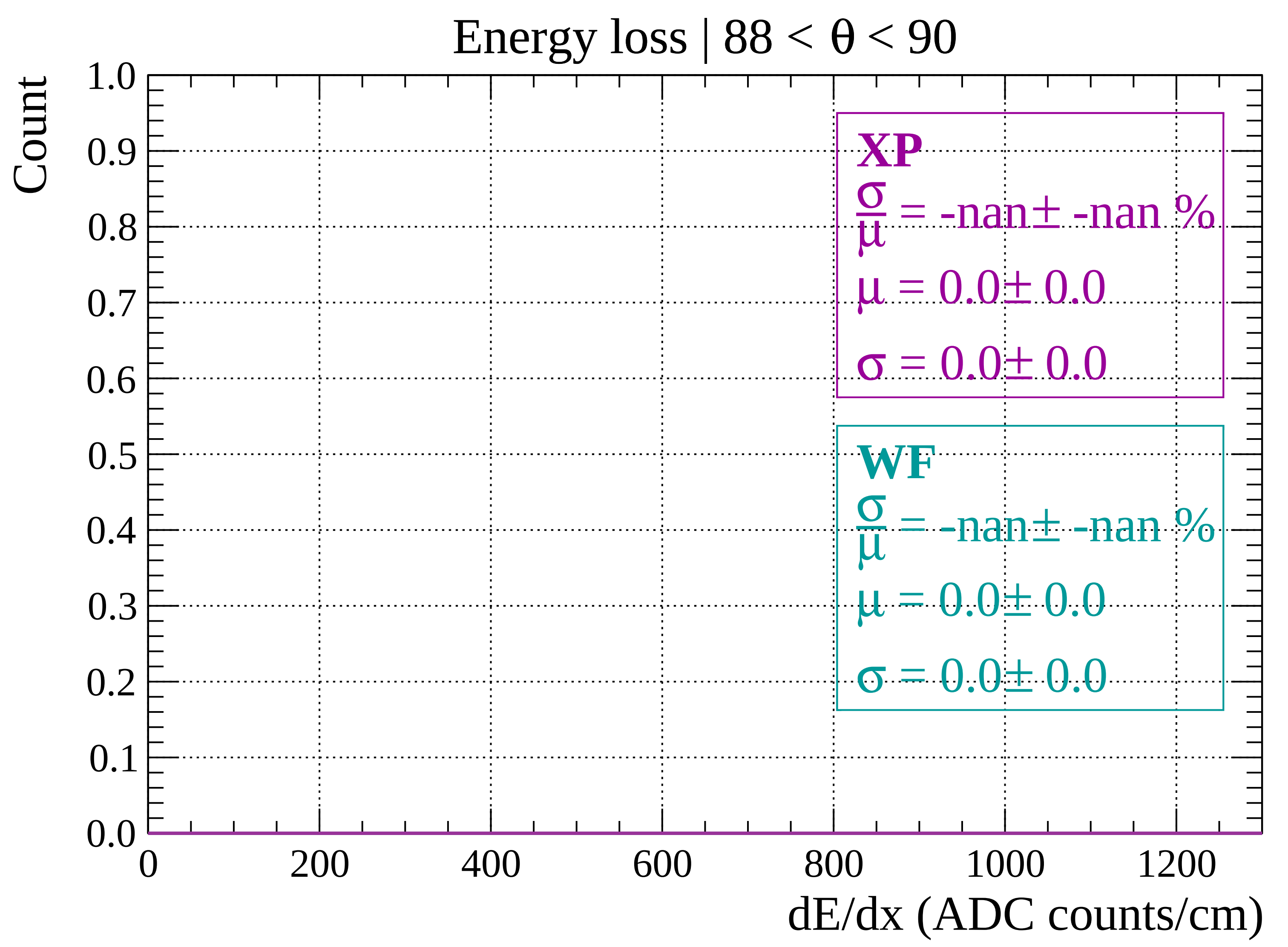
Energy loss | 82 < θ < 84
Count
dE/dx (ADC counts/cm)
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

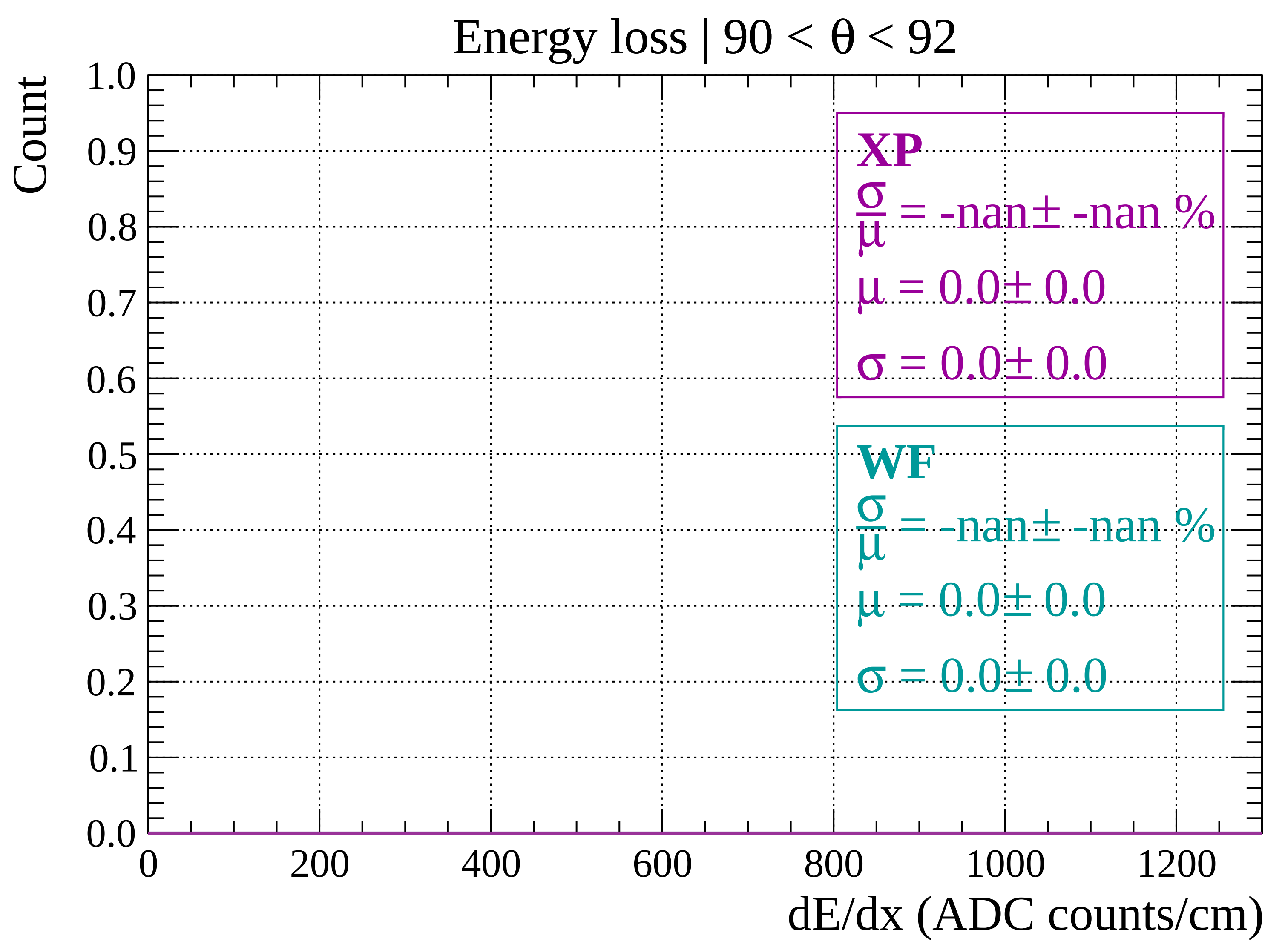


Energy loss | 84 < θ < 86
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

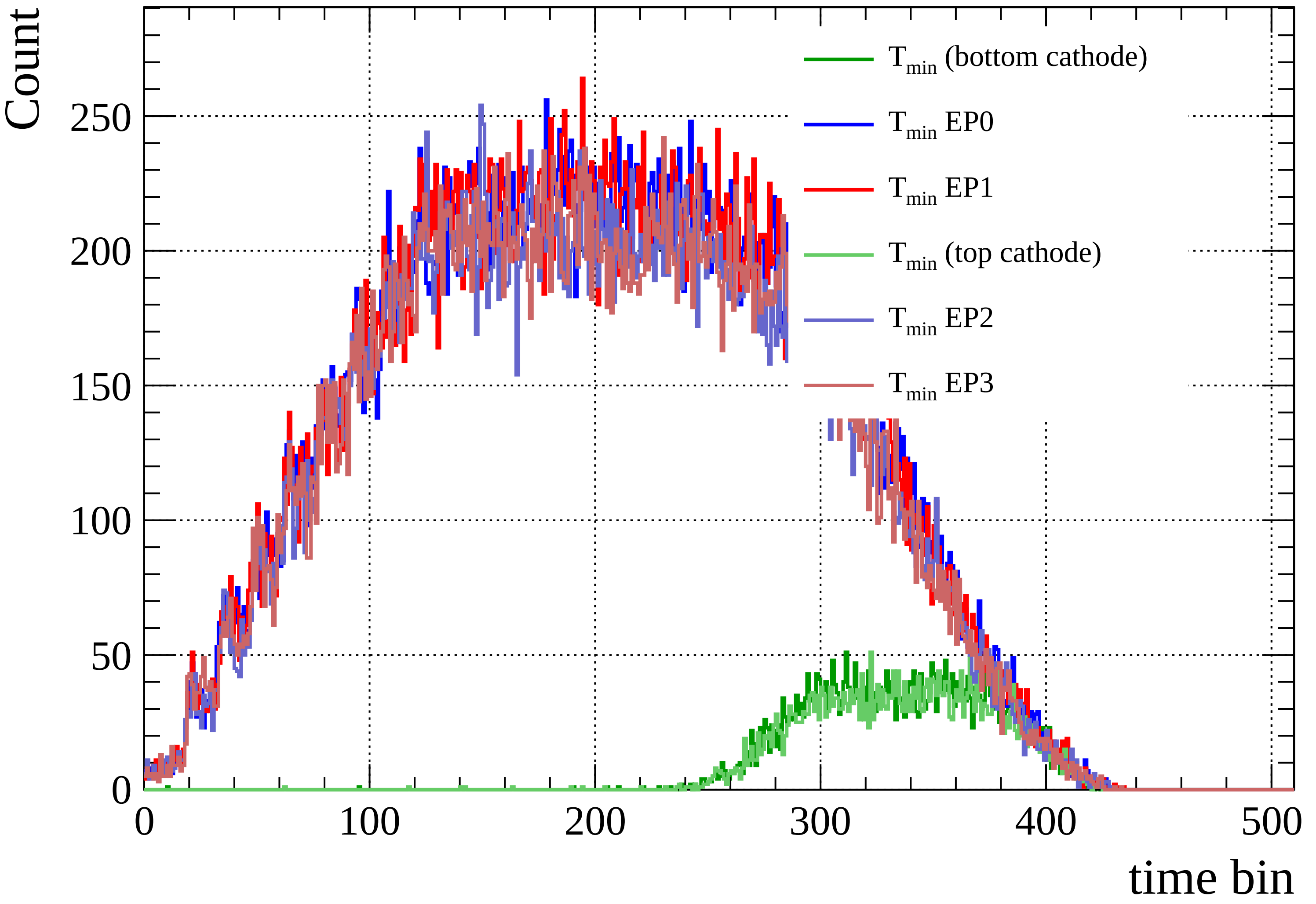


Energy loss | 86 < θ < 88
Count
dE/dx (ADC counts/cm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

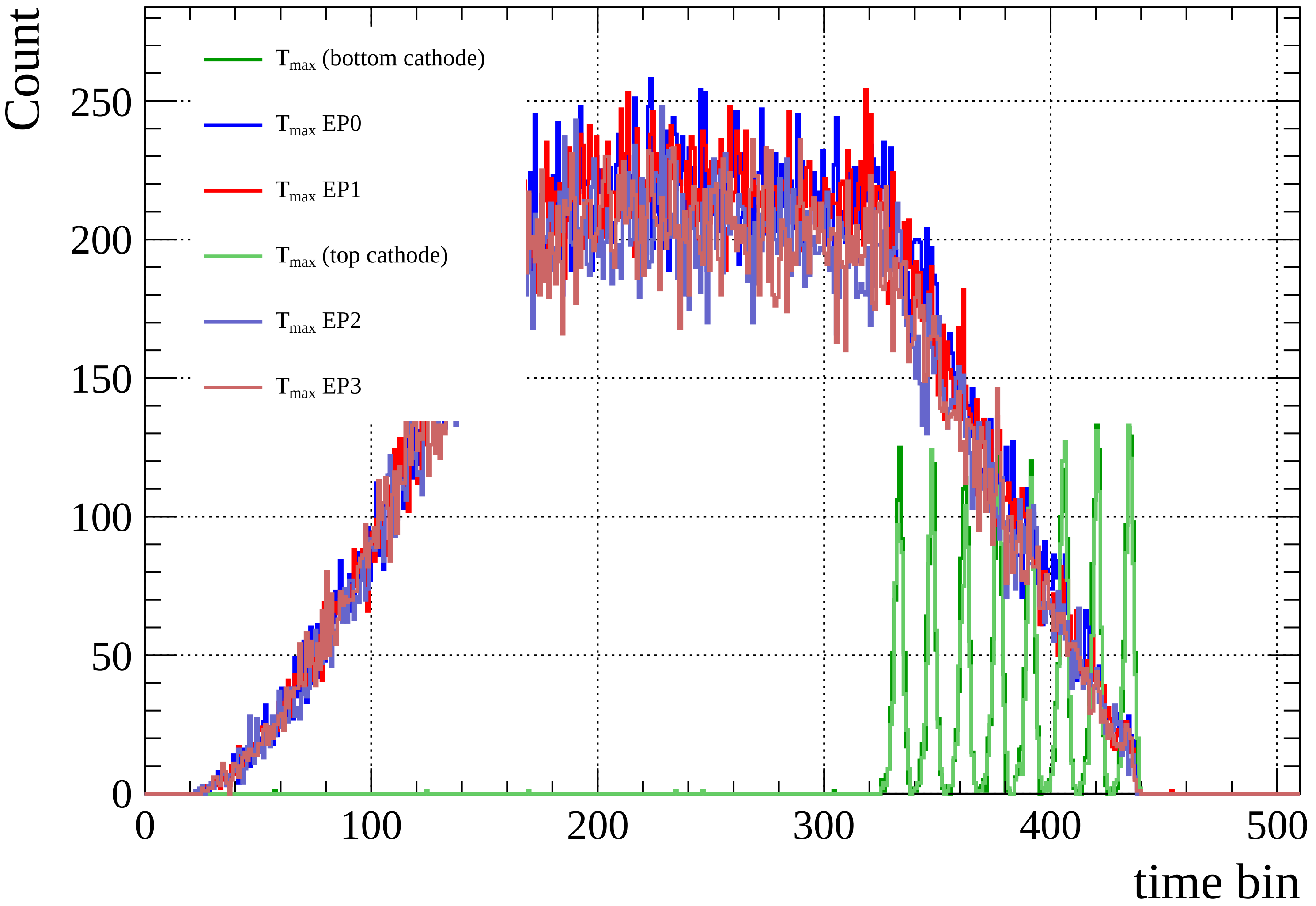

Energy loss | 88 < θ < 90
Count
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



Energy loss | 90 < θ < 92
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



Count
time bin
0
50
100
150
200
250
0
100
200
300
400
500
T_{min} (bottom cathode)
T_{min} EP0
T_{min} EP1
T_{min} (top cathode)
T_{min} EP2
T_{min} EP3



Count
T_{max} (bottom cathode)
T_{max} EP0
T_{max} EP1
T_{max} (top cathode)
T_{max} EP2
T_{max} EP3
0
50
100
150
200
250
0
100
200
300
400
500
time bin